

LECTURE NOTES

FALL 2025

Group Theory

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Part II

Lie Groups and Lie Algebras

Chapter 13

Introduction to Lie groups and Lie algebras

Chapter 14

Some global properties of some Lie groups

14.1 $SU(2)$ and $SO(3)$

14.2 $Sl(2, \mathbb{C})$ and the Lorentz group $SO(3, 1)$

14.3 The Möbius group, $Gl(2, \mathbb{C})$ and $Sl(2, \mathbb{C})$

Chapter 15

General compact simple Lie algebras (LA's)

- 15.1 The Cartan-Weyl basis, weights and roots
- 15.2 Simple roots, fundamental weights, and Dynkin diagrams
- 15.3 General form of a simple LA in the Cartan-Weyl basis
- 15.4 Symmetric LA's, regular subalgebras and extended Dynkin diagrams

Chapter 16

$SU(N)$

The matrices defining the group $SU(N)$ are $N \times N$ unitary matrices with unit determinant, $U^\dagger U = I$ and $\det U = 1$. Because the entries of these matrices are bounded by unity, $|U_j^i| \leq 1$, the group parameters form a compact domain, which is the meaning of the statement “ $SU(N)$ is compact”. A unitary matrix can always be written as the exponent of an antihermitian matrix. We discussed this for $SU(2)$ in detail. If the determinant of the unitary matrix is unity, the antihermitian matrices are traceless. A unitary matrix with unit determinant depends on $N^2 - 1$ real parameters. (For example, $SU(2)$ depends on 3 real parameters.) Thus the $N^2 - 1$ generators T_a^{def} of $SU(N)$ in the defining $(N \times N)$ representation are **antihermitian** traceless $N \times N$ matrices. A basis is given by matrices of the form

$$T_a^{\text{def}} : \quad -\frac{1}{2} \begin{pmatrix} \ddots & & & 1 \\ & \ddots & & \\ & & \ddots & \\ -1 & & & \ddots \end{pmatrix}, \quad -\frac{1}{2} \begin{pmatrix} \ddots & & & i \\ & \ddots & & \\ & & \ddots & \\ i & & & \ddots \end{pmatrix}, \quad -\frac{1}{2} \begin{pmatrix} 0 & & & \\ & i & & \\ & & -i & \\ & & & 0 \end{pmatrix} \quad (16.1)$$

$\frac{1}{2}N(N-1)$ matrices $\frac{1}{2}N(N-1)$ matrices $N-1$ matrices

One usually orthogonalizes and normalizes them to $\text{tr } T_a^{\text{def}} T_b^{\text{def}} = -\frac{1}{2} \delta_{ab}$. Then the diagonal matrices must be modified, see (16.2). For $SU(2)$ this is the set $T_a^{\text{def}} = -\frac{i}{2} \sigma_a$ with σ_a the three Pauli matrices, and for $SU(3)$ this is the set $T_a^{\text{def}} = -\frac{i}{2} \lambda_a$ with λ_a “the eight Gell-Mann matrices”. The rank of $SU(N)$ is the maximal number of commuting generators T_a^{def} which is $N - 1$. Thus, the rank of $SU(2)$ is 1, and the rank of $SU(3)$ is 2.

16.1 Roots, simple roots, weights and fundamental reps

To study the representations of the Lie algebra we go to the Cartan-Weyl basis (a different real form¹, with complex combinations of the T_a such that all generators in the defining representation become real and traceless, with real structure constants). The generators of $SU(N)$ in the Cartan-Weyl basis generate the noncompact group $SL(N, R)$. One divides the generators into a maximal set of commuting hermitian generators, called the Cartan generators, and further pairs of raising and lowering generators. As Cartan generators we take the following set of diagonal **hermitian** commuting $N \times N$ matrices, obtained by multiplying the last set of matrices in (16.1) by a factor i and taking real linear combinations

$$\begin{aligned} H_I^{\text{def}} &= \frac{1}{2} \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & 0 & \\ & & & \ddots \\ & & & & 0 \end{pmatrix}; \quad H_{II}^{\text{def}} = \frac{1}{\sqrt{12}} \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & -2 & \\ & & & 0 \\ & & & & \ddots \end{pmatrix}; \\ H_{III}^{\text{def}} &= \frac{1}{\sqrt{24}} \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & -3 \\ & & & & \ddots \end{pmatrix}; \cdots H_{N-1}^{\text{def}} = \frac{1}{\sqrt{2N(N-1)}} \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \\ & & & & -(N-1) \end{pmatrix}. \end{aligned} \quad (16.2)$$

The choice of H_I^{def} is natural but the choice of H_{II}^{def} etc is motivated by the fact that they give a diagonal metric $g_{JK}^{\text{def}} = \text{tr } H_J^{\text{def}} H_K^{\text{def}} = \frac{1}{2} \delta_{JK}$. As mentioned in the introduction, the number of Cartan generators is called the **rank** of the Lie algebra. So $SU(N)$ has rank $N - 1$.

The states of the defining representation² correspond to the N -component vectors $\vec{e}^1 = (1, 0, \dots, 0)$, $\vec{e}^2 = (0, 1, 0, \dots)$, $\dots$, $\vec{e}^N = (0, \dots, 0, 1)$, and the **weights of the defining representation** are the eigenvalues of H_J on these states

$$H_J^{\text{def}} \vec{e}^j = \mu_{\text{def}, J}^j \vec{e}^j = (\vec{\mu}_{\text{def}}^j)_J \vec{e}^j, \quad j = 1, \dots, N; \quad J = 1, \dots, N - 1. \quad (16.3)$$

¹In fact, the Cartan-Weyl basis corresponds to the maximally noncompact form. If one multiplies a compact generator by i , it becomes a noncompact generator (more precisely: the generator of a noncompact group). One can take all generators compact (as in (16.1)), but one cannot take all generators noncompact and still have real structure constants. The maximal number N_{nc} of noncompact generators minus the corresponding number N_c of compact generators satisfies $N_{nc} - N_c = r$ where r is the rank of the Lie algebra. The generators in the Cartan-Weyl basis for the defining rep of $SU(N)$ are all real matrices, see (16.2) and (16.5). So they generate $SL(N, R)$. For $SL(N, R)$ the $(N - 1)$ hermitian Cartan generators in (16.2) are noncompact, and the $\frac{1}{2}N(N - 1)$ antihermitian generators $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ are compact (they generate the Lie algebra of $SO(N)$). The remaining $\frac{1}{2}N(N - 1)$ hermitian generators $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ are noncompact. Hence $N_{nc} - N_c = (N - 1 + \frac{1}{2}N(N - 1) - \frac{1}{2}N(N - 1)) = N - 1$, which is indeed the rank of $SU(N)$.

²The relation between states and tensors will be discussed later in more detail. The states corresponding to the vectors $\vec{e}^j$ are denoted by $|\vec{\mu}^j, \text{def}\rangle$, or shorter by $|j\rangle$.

Note that there are $N - 1$ generators but N eigenvectors. The eigenvectors are the following N vectors, each with $N - 1$ components,

$$\begin{aligned}
\vec{\mu}_{\text{def}}^1 &= \left(\frac{1}{2}, \frac{1}{\sqrt{12}}, \frac{1}{\sqrt{24}}, \dots, \frac{1}{\sqrt{2(N-1)N}} \right) \\
\vec{\mu}_{\text{def}}^2 &= \left(-\frac{1}{2}, \frac{1}{\sqrt{12}}, \frac{1}{\sqrt{24}}, \dots, \frac{1}{\sqrt{2(N-1)N}} \right) \\
\vec{\mu}_{\text{def}}^3 &= \left(0, -\frac{2}{\sqrt{12}}, \frac{1}{\sqrt{24}}, \dots, \frac{1}{\sqrt{2(N-1)N}} \right) \\
\vec{\mu}_{\text{def}}^4 &= \left(0, 0, -\frac{3}{\sqrt{24}}, \dots, \frac{1}{\sqrt{2(N-1)N}} \right) \\
&\vdots \\
\vec{\mu}_{\text{def}}^{N-1} &= \left(0, \dots, 0, -\frac{(N-2)}{\sqrt{2(N-2)(N-1)}}, \frac{1}{\sqrt{2(N-1)N}} \right) \\
\vec{\mu}_{\text{def}}^N &= \left(0, \dots, 0, -\frac{N-1}{\sqrt{2(N-1)N}} \right).
\end{aligned} \tag{16.4}$$

Because the Cartan generators H_J are traceless, the sum of the entries in each column (the sum $\sum_{j=1}^{N-1} \mu_J^j$ for fixed J) vanishes. It follows that the sum of all weight vectors of the defining rep vanishes. With the definition that weight vectors (and the differences of two weight vectors) are positive if their **last** nonvanishing entry is positive³, we see that $\vec{\mu}_{\text{def}}^1 > \vec{\mu}_{\text{def}}^2 > \dots > \vec{\mu}_{\text{def}}^N$. We shall use the notation that indices I, J run from 1 to $N - 1$, while indices i, j run from 1 to N . So the weights of the defining rep are $\mu_{\text{def}, J}^j$. (We write the quantifier “def” whenever it helps to avoid confusion.)

The raising and lowering operators are eigenvectors of the H_I in the sense that $[H_I, E_\alpha^+] = \alpha_I E_\alpha^+$ and $[H_I, E_\alpha^-] = -\alpha_I E_\alpha^-$. Then the α_I are the components of the **root vectors** $\vec{\alpha}$ (usually called roots). For $SU(N)$ the E_α^+ and E_α^- are chosen as follows

$$\begin{aligned}
\text{raising operators: } E_{jk}^{\text{def}} &= \begin{pmatrix} & 1 \\ 0 & \end{pmatrix} \quad \text{if } j < k \\
\text{lowering operators: } E_{jk}^{\text{def}} &= \begin{pmatrix} & 0 \\ 1 & \end{pmatrix} \quad \text{if } j > k \\
(E_{jk}^{\text{def}})^l{}_m &= \delta_j^l \delta_{km}.
\end{aligned} \tag{16.5}$$

From $[H_I, E_{jk}] = \alpha_I^{jk} E_{jk}$ we then read off the root vectors $\vec{\alpha}^{jk}$ with components α_I^{jk}

$$\alpha_I^{jk} = \left[(H_I^{\text{def}})^j{}_j - (H_I^{\text{def}})^k{}_k \right] = \mu_{\text{def}, I}^j - \mu_{\text{def}, I}^k, \tag{16.6}$$

³One could also require that the first nonvanishing entry is positive, but this leads to awkward-looking weight diagrams.

or in vector notation

$$\vec{\alpha}^{jk} = \vec{\mu}_{\text{def}}^j - \vec{\mu}_{\text{def}}^k. \quad (16.7)$$

We discuss the $[E, E]$ commutators later.

It is now clear that the positive roots $\vec{\alpha}^{jk}$ have $j < k$. **Simple roots** are positive roots that cannot be written as the sum of two positive roots. The simple roots are then $\vec{\alpha}^{12}, \vec{\alpha}^{23}, \dots, \vec{\alpha}^{N-1,N}$. Other positive roots, for example, $\vec{\alpha}^{13} = \vec{\alpha}^{12} + \vec{\alpha}^{23}$, are not simple because they are a sum of positive roots. Relabelling these simple roots as follows

$$\vec{\alpha}^{12} = \vec{\alpha}^I, \quad \vec{\alpha}^{23} = \vec{\alpha}^{II}, \quad \dots \quad \vec{\alpha}^{N-1,N} = \vec{\alpha}^{N-1}, \quad (16.8)$$

we find the following explicit expressions for them

$$\begin{aligned} \vec{\alpha}^I &= (1, 0, 0, 0, \dots, 0, 0, 0) \equiv \alpha_J^I \quad \text{where } \alpha_J^I = (\vec{\alpha}^I)_J \text{ with } J = 1, \dots, N-1 \\ \vec{\alpha}^{II} &= \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0, 0, \dots, 0, 0, 0\right) \equiv \alpha_J^{II} \\ \vec{\alpha}^{III} &= \left(0, -\frac{1}{\sqrt{3}}, \frac{4}{\sqrt{24}}, 0, \dots, 0, 0, 0\right) \\ &\vdots \\ \vec{\alpha}^{N-2} &= \left(0, 0, 0, 0, \dots, -\frac{(N-3)}{\sqrt{2(N-3)(N-2)}}, \frac{N-1}{\sqrt{2(N-2)(N-1)}}, 0\right) \\ \vec{\alpha}^{N-1} &= \left(0, 0, 0, 0, \dots, 0, -\frac{N-2}{\sqrt{2(N-2)(N-1)}}, \frac{N}{\sqrt{2(N-1)N}}\right) \equiv \alpha_J^{N-1}. \end{aligned} \quad (16.9)$$

Note that the components α_K^J form a square $(N-1) \times (N-1)$ matrix. Each of the $\vec{\alpha}^J$ has only two nonvanishing entries. They satisfy the following inner products $(\vec{\alpha}^J \cdot \vec{\alpha}^K) \equiv \alpha_M^J g^{MN} \alpha_N^K$. Using the metric $g_{\text{def}}^{JK} = 2\delta^{JK}$, which is the inverse of g_{IJ}^{def} introduced before, we find

$$(\vec{\alpha}^J \cdot \vec{\alpha}^J) = 2; \quad (\vec{\alpha}^J \cdot \vec{\alpha}^{J+1}) = -1; \quad (\vec{\alpha}^J \cdot \vec{\alpha}^L) = 0 \text{ if } |J - L| > 1. \quad (16.10)$$

(Easy to check for $J = 1$ and $L \geq 2$.)

Dynkin has introduced so-called **Dynkin diagrams** for simple Lie algebras. Each simple root corresponds to a little open circle, and two simple roots α and β are connected by lines (rungs) as follows

$$\text{if } \cos^2 \theta_{\alpha\beta} = \frac{(\alpha \cdot \beta)(\alpha \cdot \beta)}{(\alpha \cdot \alpha)(\beta \cdot \beta)} = \begin{cases} 0 & \text{then one writes } \bigcirc \quad \bigcirc \quad \theta_{\alpha\beta} = 90^\circ \\ \frac{1}{4} & \text{then one writes } \bigcirc \text{---} \bigcirc \quad \theta_{\alpha\beta} = 120^\circ \\ \frac{1}{2} & \text{then one writes } \bigcirc \text{=}\bullet \quad \theta_{\alpha\beta} = 135^\circ \\ \frac{3}{4} & \text{then one writes } \bigcirc \text{=}\!\!=\!\!=\bullet \quad \theta_{\alpha\beta} = 150^\circ \end{cases} \quad (16.11)$$

(If $\cos^2 \theta_{\alpha\beta}$ equals $\frac{1}{2}$ or $\frac{3}{4}$, one finds that the lengths of all roots take on two different values, and

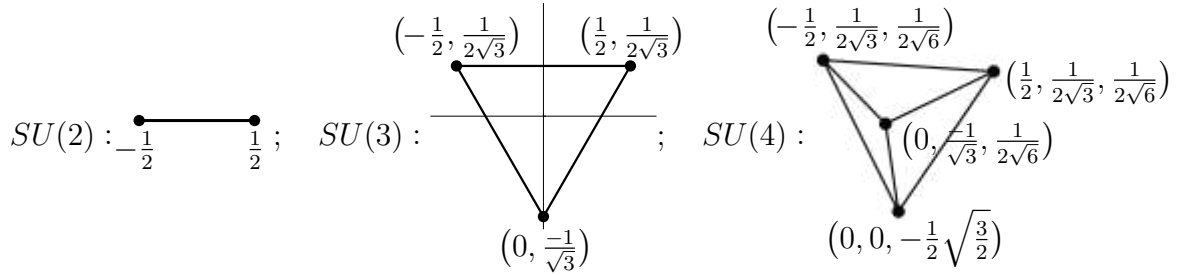
the convention is to fill in the little open circles of the shorter roots.) Since for $SU(N)$ two simple roots α^J and α^{J+1} have $\frac{(\alpha \cdot \beta)^2}{(\alpha \cdot \alpha)(\beta \cdot \beta)} = \frac{1}{4}$, the Dynkin diagram for $SU(N)$ is as follows

$$\bigcirc_1 - \bigcirc_2 - \bigcirc_3 \cdots - \bigcirc_{N-1} \quad (16.12)$$

One may also calculate with some effort the inner products between two weights⁴ of the defining representation

$$(\vec{\mu}_{\text{def}}^j \cdot \vec{\mu}_{\text{def}}^j) = \frac{N-1}{N}; \quad (\vec{\mu}_{\text{def}}^j \cdot \vec{\mu}_{\text{def}}^k) = -\frac{1}{N} \quad \text{for all } j \neq k. \quad (16.13)$$

The latter result can be understood geometrically by noting that the vectors $\vec{\mu}_{\text{def}}^j$ span a regular simplex in N dimensions: for $SU(2)$ a line segment, for $SU(3)$ an equilateral triangle, for $SU(4)$ one lifts the triangle of $SU(3)$ up along the positive 3^{rd} axis and adds a weight on the negative 3^{rd} axis yielding a regular tetrahedron, and similarly for $N \geq 4$. Using (16.13), one can reobtain (16.10).



Next we determine the **fundamental weights** $\vec{\mu}_{FW}^K$. They are orthogonal to the simple roots and normalized to

$$\frac{2\vec{\alpha}^J \cdot \vec{\mu}_{FW}^K}{\vec{\alpha}^J \cdot \vec{\alpha}^J} = \delta_J^K. \quad (16.14)$$

Note that with our index conventions these relations are “covariant”: the left- and right-hand sides have the same number of upper and lower indices J and K . Note also that the scale of the metric g_{IJ}^{def} cancels in (16.14). One may check that the fundamental weights are given by

$$\vec{\mu}_{FW}^J = \sum_{k=1}^J \vec{\mu}_{\text{def}}^k. \quad (16.15)$$

To check this, use $\vec{\alpha}^J = \vec{\mu}_{\text{def}}^j - \vec{\mu}_{\text{def}}^{j+1}$ and (16.13).

The fundamental weights (we should write fundamental weight vectors, but we follow common practice and call them fundamental weights) are the highest weights of some particular representations. The first fundamental weight is $\vec{\mu}_{\text{def}}^1$, and this is the highest weight of the defining representation whose weights were given in (16.4). The linear vector space on which the generators in the defining representation act is spanned by the states $|\vec{\mu}^k, \text{def}\rangle$ with weights $H_J |\vec{\mu}^k, \text{def}\rangle = \mu_{\text{def}, J}^k |\vec{\mu}^k, \text{def}\rangle$. We denote these states by $|_k\rangle$; the reason why we write the index k down will be explained later. We next consider the two-index antisymmetric tensor represen-

⁴One should call the vectors $\vec{\mu}^j$ weight vectors instead of weights, but the terminology weights is common.

tation. Its states span a linear vector space of dimension $\frac{1}{2}N(N-1)$, and a basis is given by $|12\rangle, |13\rangle, \dots, |1N\rangle, |23\rangle, |24\rangle, |2N\rangle, \dots, |N-1,N\rangle$. One can write these states as tensor products $|i\rangle \otimes |j\rangle - |j\rangle \otimes |i\rangle \equiv |ij\rangle$, whose corresponding tensors are $t^{ij} = -t^{ji}$. The Cartan generators H_I^{AT} in the antisymmetric tensor representation act on the states $|ij\rangle = |i\rangle \otimes |j\rangle$ as follows

$$H_I^{AT}|ij\rangle = \left[(H_I^{\text{def}})^i_i + (H_I^{\text{def}})^j_j \right] |ij\rangle = (\mu_I^{\text{def},i} + \mu_I^{\text{def},j})|ij\rangle. \quad (16.16)$$

So the weights of these states are $\vec{\mu}_{\text{def}}^i + \vec{\mu}_{\text{def}}^j$ for $i \neq j$. The state with the highest weight is $|12\rangle$ whose weight is $\vec{\mu}_{\text{def}}^1 + \vec{\mu}_{\text{def}}^2$. This is the second fundamental weight. Going on in this way we see that the fundamental weights are all weights of the highest component of vector states $|j\rangle$, 2-index antisymmetric tensor states $|ij\rangle, \dots$, up to $(N-1)$ -index antisymmetric tensor states.

We finally come back to the commutation relations of the raising and lowering operators E_{ij} . For any simple Lie algebra in the Cartan-Weyl basis one denotes the raising generators by E_α where $\vec{\alpha}$ is a positive root, and the lowering generators by $E_{-\alpha}$. **In any representation we define $\mathbf{E}_{-\alpha} = \mathbf{E}_\alpha^\dagger$.** From the explicit expressions for these generators in the defining representation in (16.5) we easily derive

$$[E_\alpha^{\text{def}}, E_{-\alpha}^{\text{def}}] = [E_{jk}^{\text{def}}, E_{kj}^{\text{def}}] = \begin{pmatrix} 0 & & & & & & & & \\ & \ddots & & & & & & & \\ & & 0 & & & & & & \\ & & & 1 & & & & & \\ & & & & 0 & & & & \\ & & & & & \ddots & & & \\ & & & & & & 0 & & \\ & & & & & & & -1 & \\ & & & & & & & & 0 \\ & & & & & & & & & \ddots \\ & & & & & & & & & & 0 \end{pmatrix} \quad \text{for } j < k. \quad (16.17)$$

For any positive root $\vec{\alpha}$ the matrix $[E_\alpha^{\text{def}}, E_{-\alpha}^{\text{def}}]$ commutes with each of the H_J^{def} . This is obvious from (16.17) but one can also prove this algebraically as follows

$$\begin{aligned} [H_J^{\text{def}}, [E_\alpha^{\text{def}}, E_{-\alpha}^{\text{def}}]] &= [[H_J^{\text{def}}, E_\alpha^{\text{def}}], E_{-\alpha}^{\text{def}}] + [E_\alpha^{\text{def}}, [H_J^{\text{def}}, E_{-\alpha}^{\text{def}}]] \\ &= \alpha_J [E_\alpha^{\text{def}}, E_{-\alpha}^{\text{def}}] + [E_\alpha^{\text{def}}, -\alpha_J E_{-\alpha}^{\text{def}}] = 0 \quad (\text{"one pulls } H_J^{\text{def}} \text{ through"}) \end{aligned} \quad (16.18)$$

Hence the commutator $[E_\alpha^{\text{def}}, E_{-\alpha}^{\text{def}}]$ can only be proportional to the generators without roots, the Cartan generators H_J ,

$$[E_\alpha^{\text{def}}, E_{-\alpha}^{\text{def}}] = \beta^J H_J^{\text{def}}. \quad (16.19)$$

To determine β^J we multiply by H_I^{def} and take the trace. On the right-hand side we obtain

$$\beta^J \text{tr} H_I^{\text{def}} H_J^{\text{def}} = \beta^J g_{IJ}^{\text{def}}. \quad (16.20)$$

Since the trace is cyclic, the left-hand side can be rewritten as follows

$$\text{tr} H_I^{\text{def}} [E_\alpha^{\text{def}}, E_{-\alpha}^{\text{def}}] = \text{tr} [H_I^{\text{def}}, E_\alpha^{\text{def}}] E_{-\alpha}^{\text{def}} = \alpha_I \text{tr} E_\alpha^{\text{def}} E_{-\alpha}^{\text{def}} \equiv \alpha_I g_{\alpha, -\alpha}^{\text{def}} \quad (16.21)$$

where we defined the metric g_{ab}^{def} as

$$g_{ab}^{\text{def}} \equiv \text{tr} X_a^{\text{def}} X_b^{\text{def}} \quad (16.22)$$

for any generators X_a^{def} . In the case at hand X_a is the set H_J , E_α and $E_{-\alpha}$. The metric has then the following block form

$$g_{ab}^{\text{def}} = \left(\begin{array}{c|ccc} \frac{1}{2}\delta_{IJ} & & & 0 \\ \hline & 0 & g_{\alpha, -\alpha}^{\text{def}} & \\ & g_{\alpha, -\alpha}^{\text{def}} & 0 & \\ 0 & & & \begin{array}{cc} 0 & g_{\beta, -\beta}^{\text{def}} \\ g_{\beta, -\beta}^{\text{def}} & 0 \end{array} \end{array} \right) \quad (16.23)$$

where we used that $\text{tr} H_I E_\alpha = 0$ since $H_I E_\alpha$ is an off-diagonal matrix. Similarly, $g_{\alpha\beta}^{\text{def}} = \text{tr} E_\alpha^{\text{def}} E_\beta^{\text{def}} = 0$ for $\beta \neq -\alpha$ because the matrices $E_{ij}^{\text{def}} E_{kl}^{\text{def}}$ for $ij \neq kl$ are off-diagonal. For a simple (and even for a semisimple) Lie algebra the metric g_{IJ}^{def} is always invertible as we shall prove. Denoting the inverse by g_{def}^{IJ} , we find

$$\beta^J = g_{\text{def}}^{JI} \alpha_I g_{\alpha, -\alpha}^{\text{def}}. \quad (16.24)$$

Hence we find the following covariant result

$$[E_\alpha^{\text{def}}, E_{-\alpha}^{\text{def}}] = (\alpha_I g_{\text{def}}^{IJ} H_J^{\text{def}}) g_{\alpha, -\alpha}^{\text{def}} \quad (16.25)$$

where for $SU(N)$ $E_\alpha^{\text{def}} = E_{jk}^{\text{def}}$ with $j < k$. One sometimes chooses at this point a particular normalization such that, for example, $g_{\alpha, -\alpha} = 1$, but we shall not impose any normalization. Then (16.25) is scale covariant: the scale factors from the two terms with the metric in this relation cancel, and the two E 's scale the same way as α and H . It is easy to check this relation for $j = 1, k = 2$ or $j = 1, k = 3$ or $j = N - 1, k = N$.

The remaining commutation relations are $[E_{jk}^{\text{def}}, E_{lm}^{\text{def}}]$ for $jk \neq ml$. One finds

$$[E_{jk}^{\text{def}}, E_{lm}^{\text{def}}] = \delta_{kl} E_{jm}^{\text{def}} - \delta_{mj} E_{lk}^{\text{def}} \quad (\text{note } lk \text{ instead of } kl). \quad (16.26)$$

They follow directly from the expression for E_{jk}^{def} in (16.5). The structure is $[E_{\vec{\alpha}}, E_{\vec{\beta}}] = f_{\vec{\alpha}, \vec{\beta}}^{\vec{\gamma}} E_{\vec{\gamma}}$, where $f_{\vec{\alpha}, \vec{\beta}}^{\vec{\gamma}}$ is only nonvanishing if $\vec{\gamma} = \vec{\alpha} + \vec{\beta}$ and if $\vec{\alpha} + \vec{\beta}$ is a root. (It follows from (16.26) that if $\vec{\alpha} + \vec{\beta}$ is a root, $f_{\vec{\alpha}, \vec{\beta}}^{\vec{\gamma}}$ does not vanish.) One sometimes writes this as $[E_{\vec{\alpha}}, E_{\vec{\beta}}] = N_{\vec{\alpha}, \vec{\beta}} E_{\vec{\alpha} + \vec{\beta}}$.

We can now write the complete set of commutation relations for any simple Lie algebra. We

derived it for the defining rep of $SU(N)$, but it should hold for any rep

$$\begin{aligned} [H_I, H_J] &= 0; & [H_I, E_{\pm\alpha}] &= \pm\alpha_I E_{\pm\alpha}; \\ [E_\alpha, E_{-\alpha}] &= (\alpha_I g^{IJ} H_J) g_{\alpha, -\alpha}; \\ [E_\alpha, E_\beta] &= f_{\alpha\beta}^\gamma E_\gamma \quad \text{with } f_{\alpha\beta}^\gamma \neq 0 \text{ if } \gamma \text{ is a root.} \end{aligned} \tag{16.27}$$

One remark is in order: as we shall later prove, the metric g_{IJ}^{def} for the defining rep, and a corresponding matrix g_{IJ}^R for any other irrep R are proportional. Hence $g_{\text{def}}^{IJ} g_{\alpha, -\alpha}^{\text{def}} = g_R^{IJ} g_{\alpha, -\alpha}^R = g^{IJ} g_{\alpha, -\alpha}$ is independent of R , and this was used in (16.27).

16.2 The adjoint rep and complex conjugate irreps

“The roots are the non-zero weights of the adjoint representation.” We now explain what is meant by this sentence. We can introduce states $|X_\alpha\rangle = \{|H_I\rangle, |E_\alpha\rangle, |E_{-\alpha}\rangle\}$ as the states of the adjoint rep, and define how they transform

$$X_b |X_a\rangle = |[X_b, X_a]\rangle = f_{ba}^c |X_c\rangle. \tag{16.28}$$

So we use the commutation relations to define how the states of the adjoint representation transform. There are as many states as generators, and since the weights of any representation are the eigenvalues of the states with respect to the Cartan generators H_I , we find that the nonvanishing weights of the adjoint rep are the roots

$$H_I |E_\alpha\rangle = |[H_I, E_\alpha]\rangle = |\alpha_I E_\alpha\rangle = \alpha_I |E_\alpha\rangle. \tag{16.29}$$

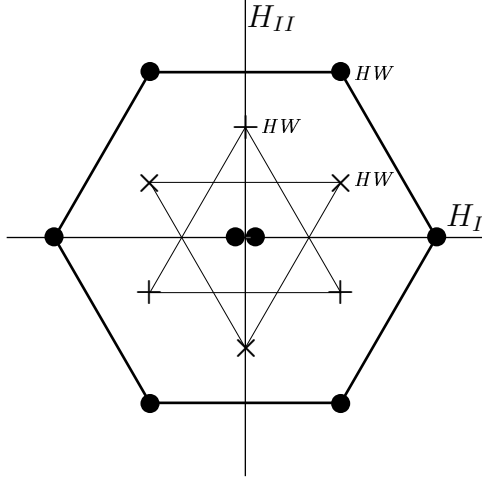
There are also vanishing weights of the adjoint rep, namely as many as there are Cartan generators and thus as many as the rank of the Lie algebra

$$H_I |H_J\rangle = |[H_I, H_J]\rangle = 0 |H_J\rangle. \tag{16.30}$$

For $SU(N)$ we found that the roots are given by $\vec{\alpha}^{jk} = \vec{\mu}_{\text{def}}^j - \vec{\mu}_{\text{def}}^k$. Thus the highest weight of the adjoint rep is given by $\vec{\mu}_{\text{def}}^1 - \vec{\mu}_{\text{def}}^N$. Now the sum of all weight vectors of the defining irrep vanishes, $\sum_{j=1}^N \vec{\mu}_{\text{def}}^j = 0$ as we have shown, and the fundamental weight vectors $\vec{\mu}_{FW}^J$ were given by $\vec{\mu}_{FW}^J = \sum_{k=1}^J \vec{\mu}_{\text{def}}^k$. Hence, $-\vec{\mu}_{\text{def}}^N = \vec{\mu}_{FW}^{N-1}$, and thus

$$\vec{\mu}_{HW}^{\text{adj}} = \vec{\mu}_{FW}^1 + \vec{\mu}_{FW}^{N-1}. \tag{16.31}$$

For $SU(3)$ the weights of the adjoint irrep and the two fundamental irreps are as shown in figure 16.1. The weights of the adjoint rep are the six roots and two weights at the origin (the weights of the Cartan generators), and the second fundamental weight is $\vec{\mu}_1^{\text{def}} + \vec{\mu}_2^{\text{def}} = -\vec{\mu}_3^{\text{def}}$. From



- × : $\vec{\mu}_{FW}^1 = \vec{\mu}_{\text{def}}^{(1)} = (\frac{1}{2}, \frac{1}{2\sqrt{3}})$
- + : $\vec{\mu}_{FW}^2 = \vec{\mu}_{\text{def}}^1 + \vec{\mu}_{\text{def}}^2 = -\vec{\mu}_{\text{def}}^{(3)} = (0, \frac{1}{\sqrt{3}})$
- : adjoint rep.

Figure 16.1: Three important irreps of $SU(3)$. Their highest weights are indicated by HW. According to our definition of positivity of weights, the highest weight vectors of irreps of $SU(3)$ lie as far “north” as possible.

this diagram we see that the weights of the second fundamental irrep are minus the weights of the first fundamental irrep. In particular, the highest weight of the second fundamental irrep, $(0, \frac{1}{\sqrt{3}})$, is minus the lowest weight of the defining irrep. These observations are explained by introducing the concept of a complex-conjugate representation. If T_a^R are the matrices which represent the generators of some irrep R of a Lie algebra, also the complex-conjugate matrices $(T_a^R)^* \equiv T_a^{R*}$ form a representation of the same Lie algebra, because the structure constants are real

$$\begin{aligned} ([T_a, T_b])^* &= (f_{ab}^c T_c)^* \\ [T_a^*, T_b^*] &= f_{ab}^c T_c^* \end{aligned} \tag{16.32}$$

The Cartan generators in any rep of a simple Lie algebra such as $SU(N)$ are obtained from the antihermitian generators of the corresponding compact Lie algebra by multiplication with i (for example $H_I = iT_3$ for $SU(2)$).

$$H_I^{R*} = iT_I^{R*} = i(-T_I^R) = -iT_I^R = -H_I^R. \tag{16.33}$$

We used that for $SU(N)$ the antihermitian generators T_I^{def} in the defining rep, and therefore also in any other rep, are imaginary and diagonal. Thus **the weights of a complex-conjugate rep R^* of $SU(N)$ are minus the weights of the rep R** . Later, when we introduce the concepts of real, pseudoreal, and complex irreps, we shall discuss in which cases the weight diagrams of a rep R and the complex-conjugate rep R^* are equal.

16.3 States and tensors

For $SU(2)$ one may consider states $|j, m\rangle$ on which the generators $J_3, J_+ = (J_1 + iJ_2)$ and $J_- = (J_1 - iJ_2)$ of the Cartan-Weyl basis act as $J_3|j, m\rangle = m|j, m\rangle$ and $J_+|j, m\rangle = \sqrt{(j+m+1)(j-m)}|j, m+1\rangle$ and $J_-|j, m\rangle = \sqrt{(j-m+1)(j+m)}|j, m-1\rangle$. (On the compact basis with hermitian generators $J_i = i(T_i)$ one has $[J_i, J_j] = i\epsilon_{ijk}J_k$.) For arbitrary Lie algebras we introduced before states $|T_b\rangle$ of the adjoint representation by defining how the generators act on these states

$$T_a|T_b\rangle = |[T_a, T_b]\rangle = f_{ab}^c|T_c\rangle. \quad (16.34)$$

Thus generators act on states just as they act (commute) on other generators, and this is reminiscent of the regular representation of finite groups, where group elements act on the set of all group elements, but here we are in the Lie algebra, not the Lie group. Let us apply these results to $SU(N)$. In the Cartan-Weyl basis there are r Cartan generators H_J with $J = 1, \dots, r$

$$H_J|H_K\rangle = 0, \quad H_J|E_{\pm\alpha}\rangle = |[H_J, E_{\pm\alpha}]\rangle = \pm\alpha_J|E_{\pm\alpha}\rangle. \quad (16.35)$$

We also introduce states $|\vec{\mu}^j, R\rangle$ for an arbitrary representation R , where j labels the weight vectors of the irrep R

$$H_I|\vec{\mu}^j, R\rangle = \mu_I^{j(R)}|\vec{\mu}^j, R\rangle \quad (16.36)$$

If a state is nondegenerate⁵, we define $E_{\pm\vec{\alpha}}|\vec{\mu}, R\rangle = N_{\pm\vec{\alpha}, \vec{\mu}}|\vec{\mu} \pm \vec{\alpha}, R\rangle$, where $N_{\pm\vec{\alpha}, \vec{\mu}}$ is a normalization factor.

However, in many cases in physics we are more accustomed to transforming tensors instead of states under symmetries. For example, the coordinates x^μ transform as a vector under finite Lorentz transformations, $(x')^\mu = L^\mu_\nu x^\nu$, and under infinitesimal Lorentz transformations we find to first order in L^μ_ν that $\delta x^\mu = L^\mu_\nu x^\nu$ where $L^\mu_\nu = \delta^\mu_\nu + l^\mu_\nu$ and where by definition $\delta x^\mu = (x')^\mu - x^\mu$. We now introduce tensors for a general simple Lie algebra.

The connection between states and tensors is given (for physicists at least) by the bra and ket formalism of Dirac for quantum mechanics. Given ket-states $|j\rangle$ we define bra-states $\langle^i|$ by $\langle^i|j\rangle = \delta^i_j$. Using completeness of the states, $\sum_i |i\rangle\langle^i| = I$, we find that a state $|i\rangle$ transforms as follows under the generators $\hat{T}_a$ (omitting summation symbols; the hat indicates that we view T_a as an operator, not a matrix) of a representation R

$$\hat{T}_a|i\rangle = |j\rangle\langle^j|\hat{T}_a|i\rangle = |j\rangle(T_a^R)^j_i, \quad (16.37)$$

where we denoted the matrix element $\langle^j|\hat{T}_a|i\rangle$ by $(T_a^R)^j_i$. Note that we wrote the indices of the kets downstairs, as $|i\rangle$ and those of the bras upstairs, as $\langle^j|$. Indices are then contracted in the usual way, with one up and one down.

Let us first check that this gives a representation of the Lie algebra. We shall check this first

⁵If the state is degenerate, we first choose a state which is an eigenvector of $E_{-\vec{\alpha}}E_{\vec{\alpha}}$.

for group elements because then the notation is simpler. We have

$$|i\rangle' = \hat{g}|i\rangle = |j\rangle\langle^j\hat{g}|i\rangle = |j\rangle M^R(g)^j_i. \quad (16.38)$$

Then

$$\hat{g}_1\hat{g}_2|i\rangle = \hat{g}_1(\hat{g}_2|i\rangle) = \hat{g}_1|j\rangle M^R(g_2)^j_i = |k\rangle M^R(g_1)^k_j M^R(g_2)^j_i = |k\rangle M^R(g_1g_2)^k_i = (\hat{g}_1\hat{g}_2)|i\rangle. \quad (16.39)$$

(The operators $\hat{g}_1$ act on the state $|j\rangle$ in the linear combination $|j\rangle M(g_2)^j_i$, not on the index i ; this is the passive interpretation of symmetries, and natural from our physical point of view.) So the matrices $M(g)^i_j$ form a representation of the Lie group⁶. Let us now derive the corresponding result for Lie algebras. We write the matrix $M(g)$ as $M(g) = e^{\lambda^a T_a}$. For g near the unit matrix, $M(g) = I + \lambda^a T_a$, we get

$$\delta(\lambda)|i\rangle = |j\rangle(\lambda^a T_a^R)^j_i, \quad (16.40)$$

which agrees with the expression in (16.37) we started from.

Let us also check that this agrees with our earlier definition of the transformation law for states of the adjoint representation

$$\hat{T}_a|T_b\rangle = |[T_a, T_b]\rangle = f_{ab}^c|T_c\rangle. \quad (16.41)$$

We identify $|b\rangle$ with $|T_b\rangle$ and $(T_a^{\text{adj}})^c_b = f_{ab}^c$ according to (16.34). Then we get

$$\hat{T}_a|T_b\rangle = |T_c\rangle(T_a^{\text{adj}})^c_b = f_{ab}^c|T_c\rangle \quad (16.42)$$

which agrees with (16.37). So the law $T_a|i\rangle = |j\rangle(T_a)^j_i$ also agrees with our definition for $T_a|T_b\rangle$.

We used in (16.41) that the matrices $(T_a^{\text{adj}})^c_b = f_{ab}^c$ form a representation. Let us prove this. We must show that

$$\begin{aligned} (T_a^{\text{adj}})^p_q (T_b^{\text{adj}})^q_r - (T_b^{\text{adj}})^p_q (T_a^{\text{adj}})^q_r &= f_{ab}^c (T_c^{\text{adj}})^p_r \\ f_{aq}^p f_{br}^q - f_{bq}^p f_{ar}^q &= f_{ab}^c f_{cr}^p. \end{aligned} \quad (16.43)$$

Using the antisymmetry $f_{ab}^c = -f_{ba}^c$, this expression can be rewritten as the Jacobi identity

$$f_{rb}^q f_{qa}^p + f_{ar}^q f_{qb}^p + f_{ba}^c f_{cr}^p = 0. \quad (16.44)$$

Hence the matrices $(T_a^{\text{adj}})^c_b = f_{ab}^c$ form indeed a representation.

We want now to introduce tensors, and define how they transform. Given a state $|i\rangle$ we construct the state $|v\rangle = v^j|j\rangle$ where v^j will be the tensor corresponding to the state $|j\rangle$. There are now two ways to derive the transformation law of tensors from the transformation law of the corresponding state. The results of these two methods differ by an overall minus sign, and we shall use that method which gives the usual sign. However, we could equally well use the other method,

⁶We have a transformation group: the group acts on the carrier space of the states $|i\rangle$.

and add as overall minus sign afterwards. Given $|v\rangle$ we define $\delta|v\rangle$ by how the states $|_j\rangle$ transform

$$\delta(\lambda)|v\rangle = v^j \delta(\lambda)|_j\rangle = |_k\rangle \lambda^a (T_a)^k_j v^j. \quad (16.45)$$

We next move the matrix $\lambda^a T_a$ from acting on the state $|_k\rangle$ to acting on the tensor v^j

$$v^j \delta(\lambda)|_j\rangle = (\delta(\lambda)v^j)|_j\rangle \quad (16.46)$$

and this defines then how v^j transform. The result is

$$\boxed{\delta v^j = \lambda^a (T_a)^j_k v^k.} \quad (16.47)$$

The other way of defining δv^j goes as follows. One requires that $|v\rangle = v^j |_j\rangle$ is **in**variant if one both transforms v^j and $|_j\rangle$. (This is reminiscent of the approach with forms where the one-form $v \equiv v_\lambda dx^\lambda$ is invariant under general coordinate transformations.) Obviously, the two methods differ by an overall sign.

In addition to kets $|_j\rangle$ we needed bras $\langle^j|$ with upper indices j . We defined the inner product $\langle^j|_k\rangle = \delta^j_k$ and now we require that the inner product $\langle^j|_k\rangle$ is invariant under symmetry (group) transformations. Since we know how the $|_j\rangle$ transform, this tells us how the $\langle^j|$ transform.

$$\delta\langle^j| = -(\lambda^a T_a)^j_k \langle^k| \quad (16.48)$$

Again we introduce vectors w_j in $w = w_j \langle^j|$ and following the same steps as for v^j , we deduce how w_j transforms

$$\delta w_j = -w_k (\lambda^a T_a)^k_j \quad (16.49)$$

For unitary representations (to which we do not restrict ourselves) with antihermitian generators T_a

$$(T_a)^k_j = -(T_a^*)_j^k \quad (16.50)$$

where T_a^* denotes the complex conjugate of the matrix T_a , we obtain

$$\delta w_j = (T_a^*)_j^k w_k. \quad (16.51)$$

So w_j transforms in the complex conjugate representation of v^j in (16.47).

Let us now take direct products of m states $|^i\rangle$ and n states $|_j\rangle$

$$|^i_1\rangle \otimes \cdots |^i_m\rangle \otimes |_{j_1}\rangle \otimes \cdots |_{j_n}\rangle. \quad (16.52)$$

More generally, we consider states $|^{i_1 \cdots i_m}_{j_1 \cdots j_n}\rangle$. Again we can contract with the components of a corresponding tensor

$$|v\rangle = |^{i_1 \cdots i_m}_{j_1 \cdots j_n}\rangle v^{j_1 \cdots j_n}_{i_1 \cdots i_m} \quad (16.53)$$

and following the same steps as for v^j and w_j , we find the transformation property of mixed tensors, with n contravariant (upper) indices and m covariant (lower) indices

$$\begin{aligned}\delta_a v_{i_1 \dots i_l \dots i_m}^{j_1 \dots j_l \dots j_n} &= (T_a)^{j_l}_{k} v_{i_1 \dots i_l \dots i_m}^{j_1 \dots k \dots j_n} - v_{i_1 \dots i_l \dots i_m}^{j_1 \dots j_n} (T_a)^k_{i_l} \quad (\text{general, summation over } k) \\ &= (T_a)^{j_l}_{k} v_{i_1 \dots i_l \dots i_m}^{j_1 \dots k \dots j_n} + (T_a^*)_{i_l}^{k} v_{i_1 \dots i_l \dots i_m}^{j_1 \dots j_n} \quad (\text{for unitary reps})\end{aligned}\tag{16.54}$$

We have now tensors at our disposal, and we shall use tensors instead of states to study the representations of simple compact Lie algebras.

Given how the tensor components transform, we can find a subset of those components which transforms into itself, and which is irreducible (does not contain a subset of this subset which transforms only into itself). For $SU(3)$ the highest-weight state is $|1^3 \dots 1^3\rangle$. (We saw that the highest-weight state of the defining representation was the state $|1\rangle$ and for its complex conjugate rep the highest weight state was $|\bar{3}\rangle$.) This corresponds to a tensor whose only nonvanishing component has these indices

$$v_{i_1 \dots i_m}^{j_1 \dots j_n} = \delta_1^{j_1} \dots \delta_1^{j_n} \delta_{i_1}^3 \dots \delta_{i_m}^3 .\tag{16.55}$$

We now **construct** the lower components of this tensor by acting with the lowering operators. Note that the highest-weight component in (16.55) is

- symmetric in the upper indices $j_1, \dots, j_n$,
- symmetric in the lower indices $i_1, \dots, i_m$,
- traceless: $\delta_{j_1}^{i_1} v_{i_1 \dots i_m}^{j_1 \dots j_n} = \delta_3^1 \dots = 0$.

These properties are preserved by the transformation laws. We now first apply these results to $SU(3)$ to see how one constructs complete irreps.

16.4 Tensor analysis for $SU(3)$

The discussion of tensors in $SU(3)$ makes use of a special property of $SU(3)$, namely antisymmetric tensors can at most have two indices. As antisymmetric tensor with 3 indices can be expressed in terms of a scalar as $t^{[ijk]} = \epsilon^{ijk}s$ or $t_{[ijk]} = \epsilon_{ijk}s$. Similarly, an antisymmetric tensor $t^{[ij]}$ can be written in terms of a vector w_k as $t^{[ij]} = \epsilon^{ijk}w_k$, and $t_{[ij]} = \epsilon_{ijk}v^k$. Then the most general tensor can be written as $t_{pqr\dots}^{ijk\dots}$. For $SU(N)$ one has more possible index structures, and the analysis of tensors in $SU(N)$ will make use of Young tableaux. Since $SU(3)$ plays an important role in physics, we shall first discuss its tensor analysis, and afterwards the more general Young tableaux.

For $SU(3)$ the highest weight component of the symmetric traceless tensor t_{kl}^{ij} (so $t_{kl}^{ij} = t_{kl}^{ji} = t_{lk}^{ij}$

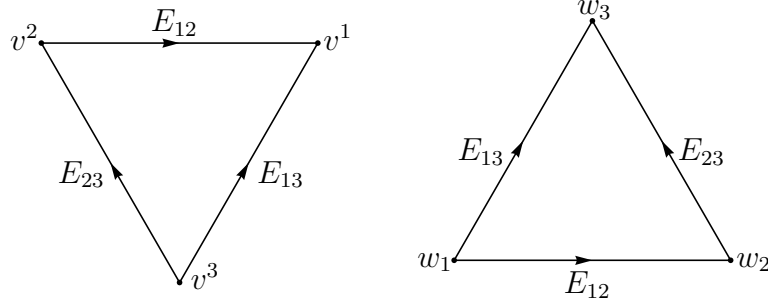
and $\delta t_{il}^{ij} = 0$) is t_{33}^{11} , and transforms as follows

$$\begin{aligned} X_a t_{33}^{11} &= (X_a)^1_k t_{33}^{k1} + (X_a)^1_k t_{33}^{1k} - t_{k3}^{11} (X_a)^k_3 - t_{3k}^{11} (X_a)^k_3 \quad (\text{sum over } k) \\ &= [2(X_a)^1_1 - 2(X_a)^3_3] t_{33}^{11} + (X_a)^1_2 (t_{33}^{21} + t_{33}^{12}) \\ &\quad + (X_a)^1_3 (t_{33}^{31} + t_{33}^{13} - t_{13}^{11} - t_{31}^{11}) - (X_a)^2_3 (t_{23}^{11} + t_{32}^{11}). \end{aligned} \quad (16.56)$$

where X_a is either the set $H_I, E_\alpha, E_{-\alpha}$ or the set T_a . Hence t_{33}^{11} transforms only into itself and the combinations $t_{33}^{12} + t_{33}^{21}$, $t_{23}^{11} + t_{32}^{11}$, $t_{13}^{11} + t_{31}^{11} - t_{33}^{13} - t_{33}^{31}$. It is clear that only combinations of tensor components appear that are symmetric in the upper indices and separately symmetric in lower indices. Moreover, these combinations of tensor components

$$t_{33}^{11}; \quad t_{33}^{12} + t_{33}^{21}; \quad t_{23}^{11} + t_{32}^{11}; \quad t_{13}^{11} + t_{31}^{11} - t_{33}^{13} - t_{33}^{31} \quad (16.57)$$

are also traceless. (To spell this out, t_{33}^{11} is a tensor $t_{i_1 i_2}^{j_1 j_2} = \delta_{i_1}^3 \delta_{i_2}^3 \delta_{j_1}^1 \delta_{j_2}^1$ and $\delta_{j_1}^{i_1} v_{i_1 i_2}^{j_1 j_2} = \delta_3^1 \delta_{i_2}^3 \delta_1^{j_2} = 0$.) We can now vary each of these combinations, and find again only symmetric traceless combinations on the right-hand side. A quick way to obtain these combinations is to use that E_{12} transforms $v^2 \rightarrow v^1$ and $w_1 \rightarrow -w_2$, while E_{13} maps $v^3 \rightarrow v^1$ and $w_1 \rightarrow -w_3$, and finally E_{23} maps $v^3 \rightarrow v^2$ and $w_2 \rightarrow -w_3$.



These pictures illustrate that E_{12} is a raising operator. Similarly one finds that E_2 is a lowering operator with the same pictures but reversed arrows. Acting on the combination in (16.57), one finds again combinations that are symmetric and traceless.

Consider next a tensor v_j^i in $SU(3)$ with $n = 1$ and $m = 1$ (a (1,1) tensor). We can take out its trace

$$s_j^i = t_j^i - \frac{1}{3} \delta_j^i t_k^k \quad (16.58)$$

and the components of s_j^i will transform only into each other but not into other components of t_j^i . For example

$$\begin{aligned} \delta_a s_3^1 &= \delta_a t_3^1 = (T_a)^1_k t_3^k - t_k^1 (T_a)^k_3 \\ &= (T_a)^1_1 s_3^1 + (T_a)^1_2 s_3^2 + (T_a)^1_3 (t_3^3 - t_1^1) - s_2^1 (T_a)^2_3 - s_3^1 (T_a)^3_3 \end{aligned} \quad (16.59)$$

and $t_1^1 - t_3^3 = s_1^1 - s_3^3$. One can understand what the operation of taking a trace means, by noting

that δ_j^i , viewed as a particular tensor v_j^i , is an **invariant tensor**: it does not transform

$$\delta_a t_j^i = (T_a)^i_k t_j^k - t_k^i (T_a)^k_j = (T_a)^i_k \delta_j^k - \delta_k^i (T_a)^k_j = 0. \quad (16.60)$$

Contracting v^i and w_j yields a scalar (an invariant)

$$\delta_a(v^i w_i) = (\delta_a v^i) w_i + v^i (\delta_a w_i) = ((T_a)^i_k v^k) w_i + v^i (-w_k (T_a)^k_i) = 0. \quad (16.61)$$

Hence $\delta_j^i t_i^j$ is a scalar, and $\delta_i^k t_{kl}^{ij} \equiv s_l^j$ transforms as an (1,1) tensor. In particular, when we wrote

$$s_j^i = t_j^i - \frac{1}{3} \delta_j^i (\delta_l^k t_k^l) \quad (16.62)$$

the t_j^i and δ_j^i on the right-hand side are tensors, and $\delta_l^k t_k^l$ is a scalar. The difference of tensors is a tensor, hence $t_j^i - \frac{1}{3} \delta_j^i (\delta_l^k t_k^l)$ is a tensor, and its components (a subset of all t_j^i !) transform into each other.

We consider a few more applications of the tensor formalism to $SU(3)$. First consider the decomposition

$$\begin{aligned} u^i v^j &= \frac{1}{2} (u^i v^j + u^j v^i) + \frac{1}{2} (u^i v^j - u^j v^i) \\ &= \frac{1}{2} (u^i v^j + u^j v^i) + \frac{1}{2} \epsilon^{ijk} w_k; \quad w_k = \epsilon_{klm} u^l v^m. \end{aligned} \quad (16.63)$$

We used $\frac{1}{2} \epsilon^{kij} \epsilon_{klm} u^l v^m = \frac{1}{2} (\delta_l^i \delta_m^j - \delta_m^i \delta_l^j) u^l v^m = \frac{1}{2} (u^i v^j - u^j v^i)$. The $(u^i v^j + u^j v^i)$ forms the **6** of $SU(3)$, and w_k forms a **3*** of $SU(3)$. (If one transforms the u^l and v^m in $\epsilon_{klm} u^l v^m$, then w_k transforms as the **3*** because ϵ_{klm} is an invariant tensor. For the same reason $\epsilon^{ijk} w_k$ transforms like an antisymmetric tensor $t^{ij} = -t^{ji}$.) Thus

$$\mathbf{3} \times \mathbf{3} = \mathbf{6} + \mathbf{3}^*. \quad (16.64)$$

Next $\mathbf{3} \times \mathbf{3}^*$. We get

$$u^i v_j = (u^i v_j - \frac{1}{3} \delta_j^i u \cdot v) + \frac{1}{3} \delta_j^i u \cdot v \quad (16.65)$$

which corresponds to

$$\mathbf{3} \times \mathbf{3}^* = \mathbf{8} + \mathbf{1}. \quad (16.66)$$

More generally: $t_j^i = (\tau)_j^i + \frac{1}{3} \delta_j^i s$ where τ_j^i and s are irreducible tensors, the **8** and **1** of $SU(3)$.

Consider $\mathbf{3} \times \mathbf{8}$. The **3** corresponds to u^i , and the **8** to v_j^i . Note that v_j^i is traceless, $\delta_i^j v_j^i = 0$.

Then we decompose the product $u^i v_k^j$ as follows

$$\begin{aligned}
u^i v_k^j &= \frac{1}{2} \left(u^i v_k^j + u^j v_k^i \right) + \frac{1}{2} \left(u^i v_k^j - u^j v_k^i \right) \\
&= \frac{1}{2} \left[u^i v_k^j + u^j v_k^i - \alpha \left(\delta_k^i u \cdot v^j + \delta_k^j u \cdot v^i \right) \right] + \frac{1}{2} \left[u^i v_k^j - u^j v_k^i - \beta \left(\delta_k^i u \cdot v^j - \delta_k^j u \cdot v^i \right) \right] \\
&\quad + \frac{1}{2} (\alpha + \beta) \delta_k^i u \cdot v^j + \frac{1}{2} (\alpha - \beta) \delta_k^j u \cdot v^i,
\end{aligned} \tag{16.67}$$

where $u \cdot v^j$ denotes $u^k v_k^j$, and α, β are constants which we now fix. The expressions inside square brackets are traceless for $1 - \alpha(N + 1) = 0$ and $1 - \beta(N - 1) = 0$ which for $N = 3$ yields $\alpha = \frac{1}{4}$ and $\beta = \frac{1}{2}$. Thus we have broken up the reducible tensor $u^i v_k^j$ into a symmetric traceless tensor s_k^{ij} , an antisymmetric traceless tensor a_k^{ij} , and a trace part which depends only on a vector (on $(u \cdot v)^j$). The number of components of s_k^{ij} is $\frac{1}{2}N(N + 1) \times N$ for a tensor $s^{ij}w_k$, minus N for the trace $s^{il}w_l$, so $\frac{1}{2}N^2(N + 1) - N = \mathbf{15}$ for $SU(3)$. Similarly, the number of components of a_k^{ij} is $\frac{1}{2}N(N - 1) \times N$ for a tensor $a^{ij}w_k$, minus N for the trace, giving $\frac{1}{2}N^2(N - 1) - N = 6$ components for a_k^{ij} . However, we cannot claim that a_k^{ij} is the **6** of $SU(3)$ because we already defined the **6** as a symmetric tensor $u^i v^j + u^j v^i$, see before. In fact, a_k^{ij} transforms like the **6*** because $a^{ij} = \epsilon^{ijk} b_k$, and $b_k w_l = \frac{1}{2} (b_k w_l + b_l w_k) + \frac{1}{2} (b_k w_l - b_l w_k)$. The tensor $\frac{1}{2} (b_k w_l + b_l w_k)$ transforms like the complex conjugate of $\frac{1}{2} (u^k v^l + u^l v^k)$, hence as **6***, while $(b_k w_l - b_l w_k) = \epsilon_{klm} y^m$ transforms like **3** of $SU(3)$, and this **3** is the trace we subtracted in a_k^{ij} . Thus, finally

$$\begin{array}{c} \mathbf{3} \end{array} \times \begin{array}{c} \mathbf{8} \end{array} = \begin{array}{c} \mathbf{15} \end{array} + \begin{array}{c} \mathbf{6^*} \end{array} + \begin{array}{c} \mathbf{3} \end{array}$$

$$\begin{array}{|c|} \hline \square \\ \hline \end{array} \otimes \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \oplus \left(\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} = \begin{array}{|c|} \hline \square \\ \hline \end{array} \right) \tag{16.68}$$

16.5 The Cartan-Weyl basis and other real forms for $SU(N)$

The Cartan-Weyl basis of a simple Lie algebra is the real noncompact form which has the least number of compact generators. The number D of noncompact generators minus the number of compact generators is smaller or equal to the rank of Lie algebra. For the Cartan-Weyl basis this number is equal to the rank.

For $SU(N)$ the real forms of its Lie group are the following

- 1) $SU(N)$ which is the compact form with $D = -(N^2 - 1)$. For example, $D = -3$ for $SU(2)$ and $D = -8$ for $SU(3)$.
- 2) $SU(p, q)$ which preserve the bilinear $x^\dagger \eta y$ with $\eta = \begin{pmatrix} I_{p \times p} & 0 \\ 0 & -I_{q \times q} \end{pmatrix}$. The generators in the defining representation are traceless and have the form $M = \begin{pmatrix} A_1 & B \\ B^\dagger & A_2 \end{pmatrix}$ where A_1 and A_2 are antihermitian satisfying $\text{tr } A_1 + \text{tr } A_2 = 0$, and B is an arbitrary complex matrix. One obtains $D = 2pq - (p^2 + q^2 - 1) = -(p - q)^2 + 1$. For example, $D = 1$ for $SU(1, 1)$ and $D = 0$ for $SU(2, 1)$.

- 3) $SL(N, R)$, the real $N \times N$ matrices with determinant one. For them $D = \frac{1}{2}N(N+1) - 1 - \frac{1}{2}N(N-1) = N-1$. For example, $D = 1$ for $SL(2, R)$ and $D = 2$ for $SU(3)$.
- 4) $SU(2N)^*$ whose defining representation consists of the complex $2N \times 2N$ matrices that are pseudoreal with determinant one

$$M^* = \Omega M \Omega^{-1}; \quad \Omega = \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}; \quad \det M = 1 \quad (16.69)$$

The generators satisfy the relations $m^* = \Omega m \Omega^{-1}$ and $\text{tr } m = 0$. They are complex $2N \times 2N$ matrices of the form

$$m = \begin{pmatrix} Z_1 & Z_2 \\ -Z_2^* & Z_1^* \end{pmatrix} \quad \text{tr } Z_1 + \text{tr } Z_1^* = 0 \quad (16.70)$$

where Z_1, Z_2 are complex. There are $(2N)^2 - 1$ generators. For low N there are isomorphisms: $SU(2)^* \simeq SU(2)$ and $SU(4)^* \simeq SO(5, 1)$. If one writes the generators of $SU(2N)^*$ as real or pure imaginary, symmetric (s) or antisymmetric (a), block-diagonal or block off-diagonal matrices, one finds the following set of matrices, below which we indicate whether they are antihermitian (compact, c) or hermitian (noncompact, nc)

$$\begin{array}{cccc} \begin{pmatrix} 0 & s_1 \\ -s_1 & 0 \end{pmatrix}, & \begin{pmatrix} 0 & a_1 \\ -a_1 & 0 \end{pmatrix}, & \begin{pmatrix} 0 & is_2 \\ is_2 & 0 \end{pmatrix}, & \begin{pmatrix} 0 & ia_2 \\ ia_2 & 0 \end{pmatrix}, \\ c & nc & c & nc \\ \begin{pmatrix} s_3 & 0 \\ 0 & s_3 \end{pmatrix}, & \begin{pmatrix} a_3 & 0 \\ 0 & a_3 \end{pmatrix}, & \begin{pmatrix} is_4 & 0 \\ 0 & -is_4 \end{pmatrix}, & \begin{pmatrix} ia_4 & 0 \\ 0 & -ia_4 \end{pmatrix} \\ nc & c & c & nc \end{array} \quad (16.71)$$

The number D of noncompact generators minus the number of compact generators is

$$\begin{aligned} D &= N(N-1) - N(N+1) + \left[\frac{1}{2}N(N+1) - 1 \right] - \frac{1}{2}N(N-1) - \frac{1}{2}N(N+1) + \frac{1}{2}N(N-1) \\ &= -2N - 1. \end{aligned} \quad (16.72)$$

For example, $D = -3$ for $SU(2)^*$ and $D = -5$ for $SU(4)^*$. As a check one may verify that $D = -2N - 1$ also for $SU(2)$ and $SO(5, 1)$.

Comparing the results for the four real forms, it is clear that $SL(N, R)$ yields the Cartan-Weyl basis for $SU(N)$. This was our starting point in section 16.1.

Chapter 17

$SO(2N)$

We consider next $SO(2N)$ in general, and use $SO(16)$ as an example. The group elements of the defining rep of $SO(2N)$, or more precisely $SO(2N, R)$, are $2N \times 2N$ real orthogonal matrices ($O^T O = I$) with unit determinant ($\det O = 1$). Then the generators in the defining representation of $SO(2N, R)$ are real antisymmetric $2N \times 2N$ matrices. A basis is given by matrices with only two nonzero entries, in the i -th row and j -th column, or the j -th row and i -th column,

$$M_{ij}^{\text{def}} = \begin{pmatrix} & 1 \\ -1 & \end{pmatrix} \quad \text{for } i < j; \quad \left(M_{ij}^{\text{def}}\right)^k{}_l = \delta_i^k \delta_{jl} - \delta_j^k \delta_{il}. \quad (17.1)$$

These $\frac{1}{2}(2N)(2N-1)$ matrices form a basis for the Lie algebra. For $i > j$ we define $M_{ij}^{\text{def}} = -M_{ji}^{\text{def}}$.

$$M_{ij}^{\text{def}} = \begin{pmatrix} & -1 \\ 1 & \end{pmatrix} \quad \text{for } i > j \quad (17.2)$$

Then in all cases $\left(M_{ij}^{\text{def}}\right)^k{}_l = \delta_i^k \delta_{jl} - \delta_j^k \delta_{il}$. They are real, antisymmetric, and thus antihermitian, and satisfy¹

$$[M_{ij}^{\text{def}}, M_{kl}^{\text{def}}] = \delta_{jk} M_{il}^{\text{def}} - \delta_{jl} M_{ik}^{\text{def}} - \delta_{ik} M_{jl}^{\text{def}} + \delta_{il} M_{jk}^{\text{def}} \quad i, j, k, l = 1, \dots, 2N \quad (17.3)$$

As the name indicates: the defining representation defines the Lie algebra, so any other matrix representation must also satisfy (17.3).

¹One way to derive this result is to decompose M_{ij} into the raising and lowering operators $E_{ij} = \begin{pmatrix} & 1 \\ & \end{pmatrix}$ for $i < j$ and $E_{ij} = \begin{pmatrix} & \\ 1 & \end{pmatrix}$ for $i > j$ of $SU(N)$ as $M_{ij} = E_{ij} - E_{ji}$ for $i < j$, and then to use the commutation relation $[E_{ij}, E_{kl}] = \delta_{jk} E_{il} - \delta_{il} E_{kj}$ of $SU(N)$. (Note the ordering kj instead of jk in the second term.)

17.1 Tensor representations of $SO(2N)$

To obtain the Cartan-Weyl basis for this Lie algebra we need Cartan generators, and raising and lowering generators. For the Cartan generators in the defining representation we choose the following maximal set of **hermitian** commuting generators H_I with $I = 1, \dots, 8$ with 2×2 block matrices along the diagonal with only two nonzero entries

$$H_1^{\text{def}} = -i \begin{pmatrix} 0 & 1 & & \\ -1 & 0 & & \\ & & \ddots & \end{pmatrix}; H_2^{\text{def}} = -i \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ & & 0 & 1 \\ & & -1 & 0 \\ & & & \ddots \end{pmatrix} \dots;$$

$$H_8^{\text{def}} = -i \begin{pmatrix} & & & \\ & & & \\ & & 0 & 1 \\ & & -1 & 0 \end{pmatrix} \text{ so } (H_J^{\text{def}})^i_j \text{ with } \begin{matrix} J = 1, \dots, N \\ i, j = 1, \dots, 2N. \end{matrix} \quad (17.4)$$

The factors $-i$ are needed for hermiticity. Clearly, there are 8 Cartan generators for $SO(16)$, hence the rank of $SO(2N)$ is N . Obviously $[H_I^{\text{def}}, H_J^{\text{def}}] = 0$, and $H_I^{\text{def}} = -iM_{2I-1, 2I}^{\text{def}}$ for $I = 1, 2, \dots, N$. We have chosen to normalize to $\text{tr}(H_I^{\text{def}} H_J^{\text{def}}) \equiv g_{IJ}^{\text{def}} = 2\delta_{IJ}$.

For later use we mention that the Lie algebra of $SO(p, q)$ with $p + q = 16$ has a basis with M_{ij}^{def} antisymmetric as in (17.1) for the diagonal blocks $1 \leq i < j \leq p$ or for $p + 1 \leq i < j \leq 16$, but for the off-diagonal blocks with $1 \leq i \leq p$ and $p + 1 \leq j \leq 16$, or $p + 1 \leq i \leq 16$ and $1 \leq j \leq p$, M_{ij}^{def} is symmetric

$$M_{ij}^{\text{def}} = \left(\begin{array}{cc|cc} \ddots & 1 & & \\ & \ddots & & \\ -1 & & & \\ \hline & & \ddots & \end{array} \right); \quad M_{ij}^{\text{def}} = \left(\begin{array}{c|c} & 1 \\ \hline 1 & \end{array} \right); \quad M_{ij}^{\text{def}} = \left(\begin{array}{c|c} & \\ \hline & -1 \end{array} \right) \quad (17.5)$$

An example is the Lorentz group $SO(3, 1)$. The commutation relations of $SO(p, q)$ are then obtained from (17.3) by replacing δ_{ij} by $\eta_{ij} = (+1, \dots, +1, -1, \dots, -1)$ with p plus signs and q minus signs. We do not discuss $SO(p, q)$ at this moment any further.

A matrix representation R of $SO(16)$ is an n -dimensional linear vector space on which $n \times n$ matrices $M_{ij}^{(R)}$ act which satisfy the same commutation relations as the 16×16 matrices M_{ij} in (17.3). In this n -dimensional linear vector space we choose a basis of states $|\vec{\mu}^k, R\rangle$ where $k = 1, \dots, n$ labels the states, which are eigenstates of the Cartan generators

$$H_I |\vec{\mu}^k, R\rangle = \mu_{R,I}^k |\vec{\mu}^k, R\rangle; \quad \mu_{R,I}^k \text{ real and } I = 1, \dots, 8. \quad (17.6)$$

The set of 8 real numbers μ_I^k for $I = 1, \dots, 8$ and fixed k are called the **weights** of the state k .

Together these 8 numbers form the weight vector $\vec{\mu}_R^k$ of the state. A representation is given by the weight vectors of all states. One calls a whole weight vector sometimes “the weight” of the corresponding state.

The carrier space of the defining representation is the linear vector space of 16-component vectors. So we have designated the 16×16 matrices in (17.1) by M_{ij}^{def} . The states on which H_I^{def} are diagonal correspond to the 16-component vectors $(1, i, 0, \dots, 0)$, $(1, -i, 0, \dots, 0)$, $(0, 0, 1, i, 0, \dots, 0)$, $(0, 0, 1, -i, 0, \dots, 0)$, etc. There are 16 such states, which is indicated by writing **16**, and the weights for these 16 basis states are the 8-component vectors $(1, 0, \dots, 0)$, $(-1, 0, \dots, 0)$, $(0, 1, 0, \dots, 0)$, $(0, -1, \dots, 0)$, etc.

$$\vec{\mu}_{\text{def}}^j = (0, \dots, 0, \pm 1, 0, \dots, 0) \quad (\text{defining representation, } j = 1, \dots, 2N) \quad (17.7)$$

A basis for the adjoint representation of $SO(2N)$ consists of states $|M_{jk}\rangle$ with $j < k$, so its dimension is $\frac{1}{2}16 \times 15 = 120$. They transform by definition as follows

$$M_{ij}|M_{kl}\rangle = |M_{ij}, M_{kl}\rangle = \delta_{jk}|M_{il}\rangle + 3 \text{ terms as in (17.3)} \quad (17.8)$$

So we view the second matrix M_{kl} in $[M_{ij}, M_{kl}]$ as a state $|M_{kl}\rangle$ which transforms under M_{ij} according to the Lie algebra. Given a Lie algebra with $[T_a, T_b] = f_{ab}^c T_c$, we can obtain the matrices of the adjoint representation $T_b(\text{adj})$ with indices $(T_b(\text{adj}))_a^c$ from the structure constants as follows:² $(T_b(\text{adj}))_a^c = f_{ab}^c$. The $[T, T] = fT$ commutation relations become then the Jacobi identity. In our case T_a corresponds to M_{ij} . The 120×120 matrices $(M_{ij})_{kl}^{mn} = f_{kl, ij}^{mn}$ will be denoted by $M_{ij}(\text{adj})$ or M_{ij}^{adj} .

In the Cartan-Weyl basis, the number of states is the same, but the states are now $|H_J\rangle$ and $|E_{JK}^{\eta\eta'}\rangle$. The $E_{JK}^{\eta\eta'}$ are the raising and lowering generators, whose matrix representation in the defining representation will be given in (17.10). The nonvanishing weights of the adjoint representation in the Cartan-Weyl basis are called **roots**. (The weights of the Cartan generators are vanishing, namely the vectors $(0, \dots, 0)$). To find the roots, we need to find the eigenvectors of H_I .

To obtain the raising and lowering generators E , we use the defining irrep in (17.1), and find those complex linear combinations of M_{ij}^{def} which are eigenvectors of H_I^{def} , meaning $[H_I^{\text{def}}, E_\alpha^{\text{def}}] = \alpha_I E_\alpha^{\text{def}}$. The eigenvectors of H_I^{def} in the defining representation are given³ by the matrices (=

²If the indices of T_b^{adj} are $(T_b^{\text{adj}})_a^c$ then $(T_b^{\text{adj}})_a^c = f_{bc}^a$.

³To find E_{JK} one may begin with $M_{2J-1, 2K-1}$. The commutator with H_J maps it to the state $M_{2J, 2K-1}$, and the commutator with H_K maps it to $M_{2J-1, 2K}$. The state $M_{2J, 2K-1}$ is mapped by H_K to $M_{2J, 2K}$, as is the state $M_{2J-1, 2K}$ by H_J . Acting with H_J or H_K on $M_{2J, 2K}$ gives back the states $M_{2J-1, 2K}$ and $M_{2J, 2K-1}$, so we have a closed system. Taking an arbitrary linear combination of these four states, and requiring it to be an eigenvector of the H_I , yields E_{JK} .

states, see above) $E_{JK}^{\eta\eta'}(\text{def})$ where $J < K$ and $J, K = 1, \dots, 8$ while $\eta = \pm 1$ and $\eta' = \pm 1$

$$E_{JK}^{\eta\eta'}(\text{def}) = \frac{1}{2} \left[M_{2J-1, 2K-1}^{\text{def}} + i\eta M_{2J, 2K-1}^{\text{def}} + i\eta' M_{2J-1, 2K}^{\text{def}} - \eta\eta' M_{2J, 2K}^{\text{def}} \right] \quad \text{with } J, K = 1, \dots, N \quad (17.9)$$

We extend the definition of these matrices to $J > K$ by requiring antisymmetry: $E_{JK}^{\eta\eta'}(\text{def}) = -E_{KJ}^{\eta\eta'}(\text{def})$ for $J > K$. (Since the $E_{JK}^{\eta\eta'}$ with $J > K$ are linearly dependent on the $E_{JK}^{\eta\eta'}$ with $J < K$, we shall mostly consider $E_{JK}^{\eta\eta'}(\text{def})$ with $J < K$.) They have the following form (we give the derivation below)

$$\begin{aligned} E_{12}^{\eta\eta'}(\text{def}) &= \frac{1}{2} \begin{pmatrix} (0) & \begin{pmatrix} 1 & i\eta' \\ i\eta & -\eta\eta' \end{pmatrix} & \cdots \\ \begin{pmatrix} -1 & -i\eta \\ -i\eta' & \eta\eta' \end{pmatrix} & \ddots & \\ \vdots & & \end{pmatrix}_{16 \times 16} = \frac{1}{2} \begin{pmatrix} (0) & \begin{pmatrix} 1 \\ i\eta \end{pmatrix} \begin{pmatrix} 1 & i\eta' \end{pmatrix} & \cdots \\ \begin{pmatrix} -1 \\ -i\eta' \end{pmatrix} \begin{pmatrix} 1 & i\eta \end{pmatrix} & \ddots & \\ \vdots & & \end{pmatrix} \\ E_{18}^{\eta\eta'}(\text{def}) &= \frac{1}{2} \begin{pmatrix} (0) & \cdots & \begin{pmatrix} 1 \\ i\eta \end{pmatrix} \begin{pmatrix} 1 & i\eta' \end{pmatrix} \\ \vdots & & \\ \begin{pmatrix} -1 \\ -i\eta' \end{pmatrix} \begin{pmatrix} 1 & i\eta \end{pmatrix} & \cdots & \vdots \end{pmatrix}_{16 \times 16} \\ \vdots & \\ E_{78}^{\eta\eta'}(\text{def}) &= \frac{1}{2} \begin{pmatrix} \vdots & \vdots & \vdots & \vdots \\ \cdots & \cdots & (0) & \begin{pmatrix} 1 \\ i\eta \end{pmatrix} \begin{pmatrix} 1 & i\eta' \end{pmatrix} \\ \cdots & \cdots & \begin{pmatrix} -1 \\ -i\eta' \end{pmatrix} \begin{pmatrix} 1 & i\eta \end{pmatrix} & (0) \end{pmatrix}_{16 \times 16} \end{aligned} \quad (17.10)$$

Writing the 2×2 matrices as a product of the two eigenvalues of H_j makes the evaluation of $[H_J, E_{IJ}^{\eta\eta'}]$ obvious. Also the evaluation of $[E_{IJ}^{\eta\eta'}, E_{KL}^{\zeta\zeta'}]$ will be easy because it only involves inner products such as $\begin{pmatrix} 1 & i\eta \end{pmatrix} \begin{pmatrix} 1 \\ i\zeta \end{pmatrix} = 1 - \eta\zeta$. All these matrices are antisymmetric and complex. The factor $\frac{1}{2}$ in front is needed to obtain later the standard form $[E_\alpha, E_{-\alpha}] = \alpha_J g^{JK} H_K g_{\alpha, -\alpha}$. At the end of this chapter, we shall show that we can cast them by a similarity transformation into a real form with both antisymmetric and symmetric matrices as in (17.5). One cannot multiply them by a factor i because then the structure constants $[E, E] = fE$ would become imaginary. If $J > K$

we use the antisymmetry mentioned before, for example

$$E_{21}^{\eta\eta'}(\text{def}) = \frac{1}{2} \begin{pmatrix} (0) & \begin{pmatrix} -1 & -i\eta \\ -i\eta' & \eta\eta' \end{pmatrix} & \cdots \\ \begin{pmatrix} 1 & i\eta' \\ i\eta & -\eta\eta' \end{pmatrix} & (0) & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \quad (17.11)$$

The set $E_{JK}^{\eta\eta'}$ with $J < K$ is the complete set of raising and lowering operators; the split into raising and lowering operators will be discussed below. There are $\frac{1}{2} \times 8 \times 7$ blocks above the diagonal (7 blocks in the first row above the diagonal, 6 blocks in the second row, $\dots$, 1 block in the seventh row) and each block can take on 4 different values ($\eta = \pm 1, \eta' = \pm 1$). So the number of matrices $E_{JK}^{\eta\eta'}(\text{def})$ (which is the number of roots) is $\frac{1}{2} 8 \times 7 \times 4 = 112$. Together with the 8 Cartan generators this is indeed the number of 120 generators of $SO(16)$. The counting for $SO(2N)$ is similar.

The $H_I^{\text{def}} = -iM_{2I-1,2I}^{\text{def}}$ act on the $E_{JK}^{\eta\eta'}(\text{def})$ as follows

$$[H_I^{\text{def}}, E_{JK}^{\eta\eta'}(\text{def})] = \eta\delta_{IJ}E_{JK}^{\eta\eta'}(\text{def}) + \eta'\delta_{IK}E_{JK}^{\eta\eta'}(\text{def}) \quad (17.12)$$

This is straightforward to check if one uses the eigenvector form of the block matrices in (17.10), and uses $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ i\eta \end{pmatrix} = i\eta \begin{pmatrix} 1 \\ i\eta \end{pmatrix}$ and $\begin{pmatrix} 1 & i\eta' \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ -1 \end{pmatrix} = -i\eta' \begin{pmatrix} 1 & i\eta' \end{pmatrix}$. For example, if $I = J$ one finds in the KJ sector

$$H_J E_{JK}^{\eta\eta'} = (-i) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \frac{1}{2} \begin{pmatrix} 1 \\ i\eta \end{pmatrix} \begin{pmatrix} 1 & i\eta' \end{pmatrix} = \eta E_{JK}^{\eta\eta'} \quad (17.13)$$

while if $I = K$ one finds in the KJ sector

$$-E_{JK}^{\eta\eta'} H_K = \frac{1}{2} \begin{pmatrix} 1 \\ i\eta' \end{pmatrix} \begin{pmatrix} 1 & i\eta \end{pmatrix} (-i) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \eta E_{JK}^{\eta\eta'} \quad (17.14)$$

Together these two contributions from the JK sector and the KJ sector yield the full term $\eta\delta_{IJ}E_{JK}^{\eta\eta'}(\text{def})$ in (17.13).

Root vectors are usually denoted by $\vec{\alpha}$. Thus the root vector corresponding to the state $E_{JK}^{\eta\eta'}(\text{def})$ with $J < K$ is given by $\vec{\alpha}^{JK\eta\eta'} = (0, \dots, 0, \eta, 0, \dots, 0, \eta', 0, \dots, 0)$, with entry η at position J and entry η' at position K . We define that a root of $SO(2N)$ is positive if its **first** nonvanishing component is positive. Thus for $\eta = +1$ the $E_{JK}^{\eta\eta'}(\text{def})$ are raising generators, and for $\eta = -1$ they are lowering generators. The H_I^{def} are diagonal on the $E_{JK}^{\eta\eta'}(\text{def})$ by construction, and the 112 roots of $SO(16)$ are given by

$$\vec{\alpha}^{JK,\eta\eta'} = \alpha_I^{JK,\eta\eta'} = (0, \underbrace{\pm 1}_{J^{\text{th}} \text{ component}}, 0, \dots, 0, \underbrace{\pm 1}_{K^{\text{th}} \text{ component}}, 0) \quad (\text{roots}) \quad 1 \leq J < K \leq N \quad (17.15)$$

Note that the contravariant composite index $JK, \eta\eta'$ labels the root vectors while the covariant

index I labels the components of a given root vector.⁴ Since we define positivity of roots (and weights) if their **first** nonzero entry is positive, the positive roots are

$$(0, \dots, 0, 1, 0, \dots, \pm 1, 0, \dots, 0) \quad (\text{positive roots}) \quad (17.16)$$

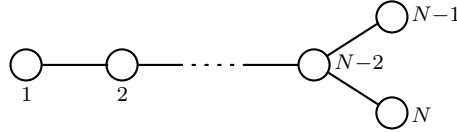
The simple roots are roots which cannot be written as the sum of two positive roots, and for $SO(2N)$ the N simple roots are

$$(1, -1, 0 \dots 0), \quad (0, 1, -1, 0 \dots 0), \quad \dots \quad (0, \dots, 0, 1, -1), \quad (0, \dots, 0, 1, 1) \quad (17.17)$$

For example, $(1, 1, 0, \dots 0)$ is not simple because it can be written as the sum of the following two positive roots $(1, 0, -1, 0, \dots, 0) + (0, 1, 1, 0, \dots, 0)$ but $(0, 0, \dots, 1, 1)$ is simple. To simplify the notation, one sometimes introduces vectors $\vec{e}^i = (0, \dots, 1, \dots, 0)$. Then the N simple roots are given by $\vec{\alpha}^J$ with $J = 1, \dots, N$ where

$$\begin{aligned} \vec{\alpha}^1 &\equiv \vec{e}^1 - \vec{e}^2, & \vec{\alpha}^2 &\equiv \vec{e}^2 - \vec{e}^3, & \dots \\ \vec{\alpha}^{N-2} &\equiv \vec{e}^{N-2} - \vec{e}^{N-1}, & \vec{\alpha}^{N-1} &\equiv \vec{e}^{N-1} - \vec{e}^N, & \vec{\alpha}^N &\equiv \vec{e}^{N-1} + \vec{e}^N. \end{aligned} \quad (17.18)$$

It is clear that with the metric $g_{\text{def}}^{KL} = \frac{1}{2}\delta^{KL}$ one has $\alpha^J \cdot \alpha^{J+1} = \alpha_K^J g_{\text{def}}^{KL} \alpha_L^{J+1} = -\frac{1}{2}$ for $J = 1, \dots, N-2$, while $\alpha^{N-1} \cdot \alpha^N = 0$. Further $\alpha^J \cdot \alpha^J = 1$ for all J , and $\alpha^J \cdot \alpha^K = 0$ if $|J - K| \geq 2$ except $\vec{\alpha}^{N-2} \cdot \vec{\alpha}^N = -\frac{1}{2}$. Thus $\frac{(\vec{\alpha}^J \cdot \vec{\alpha}^K)^2}{\vec{\alpha}^J \cdot \vec{\alpha}^J \vec{\alpha}^K \cdot \vec{\alpha}^K} = \frac{1}{4}$ if $K = J + 1$, or if $J = N - 2$ and $K = N$. Hence the Dynkin diagram for $SO(2N)$ is:



One reason we are interested in Dynkin diagrams is that they can give us a set of interesting subalgebras, as we shall discuss when we reach the exceptional Lie algebras.

The fundamental weights $\vec{\mu}_{FW}^K$ are defined by

$$2 \frac{\vec{\alpha}^J \cdot \vec{\mu}_{FW}^K}{\vec{\alpha}^J \cdot \vec{\alpha}^J} = \delta_J^K \quad (17.19)$$

Using (17.18) it is easy to check that they are given by

$$\begin{aligned} \vec{\mu}_{FW}^1 &= (1, 0 \dots 0), & \vec{\mu}_{FW}^2 &= (1, 1, 0 \dots 0), & \dots & \vec{\mu}_{FW}^{N-2} &= (1 \dots 1, 0, 0), \\ \vec{\mu}_{FW}^{N-1} &= \frac{1}{2}(1, 1, \dots, 1, -1), & \vec{\mu}_{FW}^N &= \frac{1}{2}(1, 1, \dots, 1, 1). \end{aligned} \quad (17.20)$$

⁴We could have written the E matrix as $E^{JK, \eta \eta'}$. Then $[H_I, E^{JK, \eta \eta'}] = \eta \delta_I^J E^{JK, \eta \eta'} + \eta' \delta_I^K E^{JK, \eta \eta'}$. However, to simplify the notation, we write instead $E_{JK}^{\eta \eta'}$.

The last two fundamental weights are the highest weights of the **two spinor** irreps in $2N$ dimensions. Spinor irreps only occur for $SO(N)$ groups. They are irreps of the Lie algebra, but not of the Lie group $SO(N)$; rather, they are irreps of the covering group⁵ of $SO(N)$. The familiar example is $SO(3)$ whose covering group is $SU(2)$.

The other fundamental irreps are antisymmetric tensor representations as we now show, just as for $SU(N)$. The first fundamental weight $\vec{\mu}_{FW}^1 = (1, 0, \dots, 0)$ is the highest weight of the defining representation. Its weights were $(0, 0, \pm 1, 0, \dots, 0)$, and its highest weight is indeed $(1, 0, \dots, 0)$. The corresponding states were $(1, i, 0, \dots, 0)$, $(1, -i, 0, \dots, 0)$, $\dots$, $(0, \dots, 1, i)$, $(0, \dots, 1, -i)$. One could denote these states by $|_1\rangle, |_2\rangle, \dots, |_{2N}\rangle$, but using the relation between states and tensors one usually writes that the fundamental weight $\vec{\mu}_{FW}^1$ corresponds to the vector (=defining) representation v^i . The next fundamental weight is $\vec{\mu}_{FW}^2 = (1, 1, 0, \dots, 0)$. It is the highest root vector, see (17.15), so the second fundamental weight for $SO(2N)$ is the highest weight of the adjoint representation. (It corresponds to the antisymmetric tensor representation $t^{ij} = -t^{ji}$. This tensor is dual to the states $|i\rangle |j\rangle - |j\rangle |i\rangle$, in fact to the states $|M_{ij}\rangle$ which form the adjoint representation.) Thus for $SO(2N)$ the adjoint representation is the second fundamental irrep. (For $SU(N)$ this is not the case; the highest weight of the adjoint representation of $SU(N)$ is the sum of the two fundamental weights $\vec{\mu}_{FW}^1 + \vec{\mu}_{FW}^{N-1}$.)

Let us summarize the results obtained so far. The states of the adjoint representation consist of $|E_{JK}^{\eta\eta'}\rangle$ and $|H_I\rangle$. The weights of the adjoint representation consist of the roots $\vec{\alpha} = (0, \dots, 0, \eta, 0 \dots 0, \eta', 0 \dots 0)$ and N vectors $(0, \dots, 0, \dots, 0)$, which are the weight vectors for the Cartan generators H_I . This is the complete set of all weight vectors of the adjoint representation **120**. Note that both H_I and $E_{JK}^{\eta\eta'}$ are complex linear combinations of the generators M_{ij} , but the structure constants for the Cartan-Weyl basis are still real.

One can obtain the commutator of two E matrices by straightforward but tedious calculation. For completeness we give the result of this calculation, but a much simpler derivation will be given using the spinor irreps. If only two of the indices IJ and KL are equal (so except for the case $[E_{IJ}^{\eta\eta'}, E_{IJ}^{\zeta\zeta'}]$), one finds

$$[E_{IJ}^{\eta\eta'}, E_{KL}^{\zeta\zeta'}] = \delta_{JK} \frac{(1 - \eta'\zeta)}{2} E_{IL}^{\eta\zeta'} + 3 \text{ terms} \quad (17.21)$$

where the 3 terms can be obtained by imposing the antisymmetry $E_{IJ}^{\eta\eta'} = -E_{JI}^{\eta'\eta}$

$$3 \text{ terms} = -\delta_{JL} \frac{(1 - \eta'\zeta')}{2} E_{IK}^{\eta\zeta} - \delta_{IK} \frac{(1 - \eta\zeta)}{2} E_{JL}^{\eta'\zeta'} + \delta_{IL} \frac{(1 - \eta\zeta')}{2} E_{JK}^{\eta'\zeta} \quad (17.22)$$

The structure of this equation is clear: one contracts in four ways simultaneously a pair of lower indices, giving a factor δ_{JK} and the corresponding pair of upper indices, giving a factor $\frac{1}{2}(1 - \eta'\zeta)$. One may also check that the generators E on the right-hand side carry the sum of the roots of the generators E on the left-hand side. For example, if $J = K$, in the first term in (17.21) the η' and ζ

⁵A covering group is simply connected, but $SO(N)$ is not simply connected. The groups $SU(N)$ and $USp(2N)$ are simply connected, hence only $SO(N)$ has nontrivial covering groups.

must be opposite due to factor $(1 - \eta'\zeta)$. Then η' and ζ cancel in the sum of weights, so that one is left with $E_{IL}^{\eta\zeta'}$ which only depends on η and ζ' . So the commutator in (17.21) has the structure $[E_\alpha, E_\beta] = N_{\alpha\beta}E_{\alpha+\beta}$.

On the other hand if KL equals IJ one finds

$$[E_{IJ}^{\eta\eta'}, E_{IJ}^{\zeta\zeta'}] = -\frac{(1 - \eta'\zeta')}{2} \frac{(\eta - \zeta)}{2} H_I - \frac{(1 - \eta\zeta)}{2} \frac{(\eta' - \zeta')}{2} H_J \quad (17.23)$$

This commutator is only nonvanishing if the two E matrices have opposite roots; if one root corresponds to (η, η') the other root corresponds to $(-\eta, -\eta')$. In this case the commutator can be written as

$$[E_{IJ}^{\eta\eta'}, E_{IJ}^{-\eta-\eta'}] = -\eta H_I - \eta' H_J \quad (\text{for } I < J). \quad (17.24)$$

The raising operators have $\eta = +1$ according to (17.12). Because the raising operators $E_{JK}^{+\eta'}$ in the defining representation of (17.10) are related to $E_{JK}^{-\eta'}$ by $(E_{JK}^{+\eta'})^\dagger = -E_{JK}^{-\eta'}$ (more generally, $(E_{JK}^{\eta\eta'})^\dagger = -E_{JK}^{-\eta, -\eta'}$), the right-hand side of (17.24) contains minus signs. In general one defines the lowering operators $E_{-\alpha}$ by $(E_\alpha)^\dagger = E_{-\alpha}$. Then the raising and lowering generators are given by

$$E_\alpha = E_{JK}^{+\eta'}, \quad E_{-\alpha} = (E_\alpha)^\dagger = -E_{JK}^{-\eta, -\eta'} \quad (17.25)$$

Then $[E_\alpha, E_{-\alpha}]$ yields $\eta H_I + \eta' H_J$.

The Killing metric g_{IJ} is usually defined as the trace over two generators in the adjoint representation. However, the same trace in any other representation differs only by an overall constant, as we shall prove later. Since it is easier to compute the trace over matrices in the defining $2N \times 2N$ matrix representation, we call that trace the Killing metric of the defining rep. Then the Killing metric of the defining rep in the Cartan-Weyl basis is given by⁶

$$g_{IJ}^{\text{def}} = \text{tr} H_I^{\text{def}} H_J^{\text{def}} = 2\delta_{IJ}; \quad g_{\alpha, -\alpha}^{\text{def}} = \text{tr} E_\alpha^{\text{def}} E_{-\alpha}^{\text{def}} = -\text{tr} E_{JK}^{+\eta'} E_{JK}^{-\eta, -\eta'} = 2 \quad (17.26)$$

and thus (17.24) can be rewritten as

$$[E_\alpha, E_{-\alpha}] = (\alpha^I H_I + \alpha^J H_J) g_{\alpha, -\alpha}, \quad (17.27)$$

where E_α denotes $E_{IJ}^{\eta\eta'}$, and $\alpha_I = \eta$ and $\alpha_J = \eta'$. (Recall $\alpha^I = g_{\text{def}}^{IJ} \alpha_J = \frac{1}{2} \alpha_I$.)

The complete set of commutation relations of the generators H_I and $E_\alpha = E_{JK}^{\eta\eta'}$ of $SO(2N)$ in

⁶The evaluation of $g_{\alpha, -\alpha}$ by using the explicit expression for $E_\alpha = E_{IJ}^{+\eta, \eta'}$ in (17.10) and $E_{-\alpha} = -E_{IJ}^{-\eta, -\eta'}$ is tedious. It is easier to use (17.9) and $\text{tr} M_{ij}^{\text{def}} M_{ij}^{\text{def}} = -2$ for fixed i and j . It is even easier to use the spinor representation as we shall see.

the Cartan-Weyl basis has the generic form of any simple Lie algebra

$$\begin{aligned}
[H_I, H_J] &= 0 \\
[H_I, E_{\pm\alpha}] &= \pm\alpha_I E_{\pm\alpha} \quad (\text{where } \alpha_I = \eta \text{ and } \eta') \\
[E_\alpha, E_\beta] &= N_{\alpha\beta} E_{\alpha+\beta} \quad \text{if } \alpha + \beta \text{ is a root} \\
[E_\alpha, E_{-\alpha}] &= \alpha_I g^{IJ} H_J g_{\alpha, -\alpha} \quad (\text{sum over } I, J)
\end{aligned} \tag{17.28}$$

where α_J are the components of the positive root vectors $\vec{\alpha}$ in (17.16), and we recall that the raising operators are $E_\alpha = E_{JK}^{+\eta'}$ for $J < K$, and the lowering operators are $E_{-\alpha} = (E_\alpha)^\dagger = -E_{JK}^{-\eta'}$ for $J < K$. Although (17.28) was derived from the defining representation, it should be satisfied by any other matrix representation.

17.2 The two spinor irreps of $SO(2N)$

To construct the weights of the two spinor representations of $SO(16)$, we begin by defining a linear vector space which is a direct product of eight (or N) two-dimensional linear vector spaces. A basis is given by

$$|\pm 1\rangle \otimes |\pm 1\rangle \cdots \otimes |\pm 1\rangle \tag{17.29}$$

For $SO(16)$ there are $2^8 = 256$ states which form a basis. The Dirac matrices are given on this basis by the following⁷ direct product of eight 2×2 matrices

$$\begin{aligned}
\Gamma_1 &= \sigma_1 \otimes \mathbb{1} \otimes \cdots \otimes \mathbb{1} \\
\Gamma_2 &= \sigma_2 \otimes \mathbb{1} \otimes \cdots \otimes \mathbb{1} \\
\Gamma_3 &= \sigma_3 \otimes \sigma_1 \otimes \mathbb{1} \otimes \cdots \otimes \mathbb{1} \\
\Gamma_4 &= \sigma_3 \otimes \sigma_2 \otimes \mathbb{1} \otimes \cdots \otimes \mathbb{1} \\
&\vdots \quad \vdots \quad \vdots \\
\Gamma_i &= \sigma_3 \otimes \cdots \otimes \sigma_3 \otimes (\sigma_1 \text{ or } \sigma_2) \otimes \mathbb{1} \otimes \cdots \otimes \mathbb{1} \\
&\vdots \quad \vdots \quad \vdots \\
\Gamma_{2N} &= \sigma_3 \otimes \cdots \otimes \sigma_3 \otimes \sigma_2
\end{aligned} \tag{17.30}$$

Clearly there are 16 Dirac matrices; they are hermitian and they satisfy the Clifford algebra

$$\{\Gamma_i, \Gamma_j\} = 2\delta_{ij}I; \quad i, j = 1, \dots, 16 \tag{17.31}$$

⁷The alternative direct product of [1] which σ_3 matrices standing to the right of σ_1 and σ_2 , and unit matrices standing to the left of σ_1 and σ_2 , yields an equivalent representation because there exists only one representation of the Clifford algebra in even dimensions up to equivalence.

The antihermitian generators for the (reducible as we shall see!) spinor representation are given by

$$M_{ij}^{\text{RS}} = \frac{1}{2}\Gamma_i\Gamma_j \equiv \frac{1}{2}\Gamma_{ij} = \frac{1}{4}[\Gamma_i, \Gamma_j] \quad \text{for } i < j \quad (\text{RS} = \text{reducible spinor rep}) \quad (17.32)$$

They satisfy⁸ (17.3). The hermitian Cartan generators in the spinor representation are now

$$\begin{aligned} H_1^{\text{RS}} &= -i\frac{1}{2}\Gamma_{12} = \frac{-i}{2}\sigma_1\sigma_2 \otimes \mathbb{1} \otimes \cdots \otimes \mathbb{1} = \frac{1}{2}\sigma_3 \otimes \mathbb{1} \otimes \cdots \otimes \mathbb{1} \\ H_2^{\text{RS}} &= -i\frac{1}{2}\Gamma_{34} = \mathbb{1} \otimes \frac{-i}{2}\sigma_1\sigma_2 \otimes \mathbb{1} \cdots \otimes \mathbb{1} = \mathbb{1} \otimes \frac{1}{2}\sigma_3 \otimes \mathbb{1} \cdots \otimes \mathbb{1} \\ &\vdots \quad \quad \quad \vdots \\ H_8^{\text{RS}} &= -i\frac{1}{2}\Gamma_{15,16} = \mathbb{1} \otimes \cdots \otimes \mathbb{1} \otimes \frac{-i}{2}\sigma_1\sigma_2 = \mathbb{1} \otimes \cdots \otimes \mathbb{1} \otimes \frac{1}{2}\sigma_3 \end{aligned} \quad (17.33)$$

By acting with these operators on the states in (17.29) we find the weights of the spinor representation

$$\vec{\mu}^\alpha(\text{RS}) = (\pm\frac{1}{2}, \pm\frac{1}{2}, \dots, \pm\frac{1}{2}) \quad (\alpha = 1, \dots, 2^8 \text{ for the reducible spinor representation}) \quad (17.34)$$

These are the $2^8 = \mathbf{256}$ weight vectors for the 2^8 states of the two spinor representations. This representation is reducible; it splits into two irreducible representations denoted by s (for spinor) and c (for conjugate spinor). The spinor representation $|\mathbf{128}, s\rangle$ consists of the states with an even number of $+$ signs, while the conjugate spinor representation $|\mathbf{128}, c\rangle$ has an odd number of $+$ signs. Note that the generators $M_{ij}^{(\text{RS})} = \frac{1}{2}\Gamma_i\Gamma_j$ map states from the s -representation into states of the s -representation because only σ_1 and σ_2 flip $|+1\rangle$ into $|-1\rangle$ and vice versa, and Γ_i and Γ_j have each precisely one σ_1 or σ_2 . The same holds for the $E_{IJ}^{\eta\eta'}(\text{RS})$ as we now show.

The raising and lowering generators take on a very simple form in this spinor representation. Taking (17.9), and substituting $M_{ij}^{\text{RS}} = \frac{1}{2}\Gamma_i\Gamma_j$ and the expressions for Γ_i in (17.30), we obtain

$$E_{JK}^{\eta\eta'}(\text{RS}) = \mathbb{1} \otimes \dots \otimes \mathbb{1} \otimes \sigma^\eta \sigma_3 \otimes \sigma_3 \cdots \otimes \sigma_3 \otimes \sigma^{\eta'} \otimes \mathbb{1} \cdots \otimes \mathbb{1} \quad \text{for } J < K \quad (17.35)$$

where $\sigma^\eta = \frac{\sigma^1 + i\eta\sigma^2}{2} = \frac{\sigma^1 + i\sigma^2}{2}$ or $\frac{\sigma^1 - i\sigma^2}{2}$ as $\eta = +1$ or $\eta = -1$. In this representation the relation $[H, E] \sim E$ in (17.12) becomes obvious if one uses $\sigma_3\sigma^\eta = \eta\sigma^\eta$ and $\sigma^\eta\sigma_3 = -\eta\sigma^\eta$, while also the commutator $[E_\alpha, E_{-\alpha}] \sim \alpha^I H_I$ in (17.24) comes out easily if one uses in addition $\sigma^\eta\sigma^{-\eta} = \frac{1}{2}(1 + \eta\sigma_3)$

$$[E_{IJ}^{\eta\eta'}, -E_{IJ}^{-\eta, -\eta'}] = \eta H_I + \eta' H_J \quad (17.36)$$

Even the more complicated expression for the $[E, E]$ commutator in (17.21) becomes clear if one stares at it for a while.

Sometimes another basis for the Γ_i is more useful. In this basis the Γ_{ij} have the form of two blocks on the diagonal, and one block gives the s representation, while the other block gives the c

⁸To evaluate $[\frac{1}{2}\Gamma_{ij}, \frac{1}{2}\Gamma_{kl}]$ one may use the identity $[AB, CD] = A\{B, C\}D - \{A, C\}BD + CA\{B, D\} - C\{A, D\}B$ and then apply (17.31). One pulls first C to the left of AB , and then D to the left of AB , using the identity $[AB, C] = A\{B, C\} - \{A, C\}B$.

representation. Given Dirac matrices γ_i in one lower dimension, $d = 2N - 1$, we construct the $2N$ Dirac matrices for $d = 2N$ as follows

$$\begin{aligned}\Gamma_i &= \gamma_i \otimes \sigma_2 = \begin{pmatrix} 0 & -i\gamma_i \\ i\gamma_i & 0 \end{pmatrix}_{256 \times 256} \quad i = 1, \dots, 15 \\ \Gamma_{16} &= I \otimes \sigma_1 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}_{256 \times 256}\end{aligned}\quad (17.37)$$

They clearly are hermitian, satisfy the Clifford algebra, and are block off-diagonal.

For example, in $d = 4$ one can use the Pauli matrices $\sigma^1, \sigma^2, \sigma^3$ of $d = 3$, and obtain for the Dirac matrices $\Gamma^4, \Gamma^1, \Gamma^2, \Gamma^3$ in $d = 4$ Euclidean space

$$\Gamma^\mu = \begin{pmatrix} 0 & -i\sigma^\mu \\ i\bar{\sigma}^\mu & 0 \end{pmatrix}; \quad \sigma^\mu \equiv \{iI, \sigma^1, \sigma^2, \sigma^3\} \quad ; \quad \mu = 4, 1, 2, 3. \quad (17.38)$$

Here Γ_4 is hermitian, $\Gamma_4 = \Gamma^4$, and $\Gamma_4^2 = I$.

In Minkowski space one obtains a representation by removing the factor i from Γ^4 , and one obtains the representation for $\Gamma^0, \Gamma^1, \Gamma^2, \Gamma^3$ often used for supersymmetry

$$\Gamma^\mu = \begin{pmatrix} 0 & -i\sigma^\mu \\ i\bar{\sigma}^\mu & 0 \end{pmatrix}; \quad \sigma^\mu = (I, \sigma^1, \sigma^2, \sigma^3) \quad ; \quad \mu = 0, 1, 2, 3. \quad (17.39)$$

Now Γ^0 is antihermitian, and $\Gamma_0 = -\Gamma^0$ while $(\Gamma_0)^2 = -I$.

Because the Dirac matrices are block off-diagonal, the generators of $SO(2N)$ are block-diagonal

$$\begin{aligned}M_{ij}^{(\text{spin})} &= \frac{1}{2} \begin{pmatrix} \gamma_{ij} & 0 \\ 0 & \gamma_{ij} \end{pmatrix} \quad \text{for } 1 \leq i < j \leq 2N - 1 \\ M_{i,2N}^{(\text{spin})} &= \frac{1}{2} \begin{pmatrix} -i\gamma_i & 0 \\ 0 & i\gamma_i \end{pmatrix} \quad \text{for } 1 \leq i \leq 2N - 1\end{aligned}\quad (17.40)$$

Here $\gamma_{ij} = \gamma_i \gamma_j$ with $i \neq j$. These matrices are clearly anti-hermitian. The two irreducible spinor representations are now manifest: for the s -representation we take the upper blocks, and for the c representation the lower blocks

$$\begin{aligned}s \text{ rep : } & \frac{1}{2} \{ \gamma_{ij}, -i\gamma_i \} \text{ spinor irrep} \\ c \text{ rep : } & \frac{1}{2} \{ \gamma_{ij}, +i\gamma_i \} \text{ conjugate spinor irrep}\end{aligned}\quad (17.41)$$

Each of them satisfies the $SO(2N)$ algebra in (17.3). The s rep and c rep are inequivalent. If they were equivalent, one would have the relations $S\gamma_{ij}S^{-1} = \gamma_{ij}$ and $Si\gamma_iS^{-1} = -i\gamma_i$ for some matrix

S . However, the relation $S\gamma_i S^{-1} = -\gamma_i$ is inconsistent in odd dimension.⁹ If we know the reality properties of the γ_i , we can deduce the reality properties of the s and c spinor representations of $SO(2N)$. We postpone this analysis till we have discussed also $SO(2N+1)$.

17.3 The Cartan-Weyl basis and other real forms for $SO(2N)$

For $SU(N)$ the Cartan-Weyl (CW) basis is given by the Lie algebra of $Sl(N, R)$. Its defining representation are real traceless $N \times N$ matrices, and it is the maximally noncompact real form of $SU(N)$. However, for $SO(2N)$ we found that the CW basis is defined in terms of $2N \times 2N$ matrices by the matrices H and E which are complex. There are 3 real forms of $SO(2N)$:

- 1) the compact $SO(2N, R)$,
- 2) the noncompact $SO(p, q; R)$ with $p + q = 2N$, and
- 3) the noncompact $SO(2N)^*$. ($SO(2)^* \simeq SO(2)$, $SO(6)^* \simeq SU(3, 1)$, $SO(4)^* \simeq SU(2) \times Sl(2, R)$, while $SO(8)^* \simeq SO(6, 2)$.) $SO(2N)^*$ is defined as the group which preserves $x^T y$ and $x^\dagger \Omega y$ with $\Omega = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$. The generators of the defining rep are $2N \times 2N$ complex antisymmetric matrices ($M^T + M = 0$) which are pseudoreal instead of real ($M^\dagger \Omega + \Omega M = 0$ implies $M^* = \Omega M \Omega^{-1}$ for antisymmetric M). They generate a subgroup of $SO(2N, C)$ with the same number of generators as $SO(2N, R)$, and are of the form $\begin{pmatrix} A & H \\ -H^T & A^* \end{pmatrix}$ where A is complex antisymmetric, while H is hermitian.

If one defines the degree D of noncompactness of a Lie algebra as the difference of the number N_{nc} of noncompact generators and the number N_c of compact generators, then the degree of noncompactness is at most the rank r of the Lie algebra

$$D \equiv N_{nc} - N_c \leq r. \quad (17.42)$$

The CW basis has $D = r$, so it is the maximally noncompact real form. For $SO(p, q; R)$ with $p + q = 2N$, one finds from (17.5) that $D(SO(p, q; R)) = N - \frac{1}{2}(p - q)^2$. For $SO(2N)^*$ one finds $D(SO(2N)^*) = -N$. It is thus clear that $SO(N, N; R)$ should be the CW basis for $SO(2N, R)$. Since $SO(N, N; R)$ is defined in terms of real $2N \times 2N$ matrices, the conclusion is that there should exist a similarity transformation which maps the matrices H_J and $E_{JK}^{\eta\eta'}$ to matrices of the form in (17.1) for the block-diagonal part, and (17.5) for the block off-diagonal part. The problem we wish to solve is to find this similarity transformation.

⁹In odd $2N - 1$ dimensions, one of the $2N - 1$ Dirac matrices is proportional to the product of all $2N - 2$ others. Denoting it by γ_{2N-1} , it satisfies $S\gamma_{2N-1}S^{-1} = +\gamma_{2N-1}$ because one gets $2N - 2$ factors -1 if each γ_i satisfies $S\gamma_i S^{-1} = \pm\gamma_i$. This means that the two irreps of the Dirac matrices γ_i and $-\gamma_i$ in odd dimensions are inequivalent, a well-known result. Thus the fact that there are two inequivalent spinor irreps of $SO(2N)$ in even dimensions is a consequence of the existence of two inequivalent irreps of the Dirac matrices in odd dimensions.

Inspection of the E matrices in (17.9) reveals that if one numbers the rows and columns from 1 to $2N$, then the even-odd and odd-even entries are imaginary, but the even-even and odd-odd entries are real. The same holds for the H matrices in (17.3). One way to proceed would be to renumber the indices as follows: $1, 3, \dots, 2N-1$ followed by $2, 4, \dots, 2N$. Then the two diagonal $N \times N$ blocks would be real, and the two off-diagonal $N \times N$ blocks would be imaginary, and one could make an ansatz for the similarity matrix J in terms of four $N \times N$ blocks of the same form. However, by playing a bit around, one finds a rather simple matrix S already for the matrices H and E in their original form, namely

$$S = \begin{pmatrix} 1 & -i & & & \\ & 1 & & & \\ & & -i & & \\ & & & 1 & \\ \emptyset & & & & \ddots & \\ & \emptyset & & & & 1 & -i \end{pmatrix} \quad (17.43)$$

Direct evaluation reveals that all $2N \times 2N$ matrices SES^{-1} and SHS^{-1} are real, and on the renumbered basis they have indeed the form of (17.1) and (17.5). To illustrate this, consider $SH_{J=1}S^{-1}$ for $SO(4)$. The calculations boils down to

$$\begin{pmatrix} 1 & -i & & \\ \emptyset & & 1 & -i \end{pmatrix} \begin{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} & (\emptyset) \\ (\emptyset) & (\emptyset) \end{pmatrix} \begin{pmatrix} 1 & i & & \\ \emptyset & & 1 & i \end{pmatrix} = \begin{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & (\emptyset) \\ (\emptyset) & (\emptyset) \end{pmatrix} \quad (17.44)$$

which becomes of the form in (17.5) after renumbering rows and columns as discussed above

$$\begin{pmatrix} \emptyset & \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \\ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} & \emptyset \end{pmatrix} \quad (17.45)$$

Similarly $E_{12} = \begin{pmatrix} \emptyset & \begin{pmatrix} 1 & i\eta' \\ i\eta & -\eta\eta' \end{pmatrix} \\ \begin{pmatrix} -1 & -i\eta \\ -i\eta' & \eta\eta' \end{pmatrix} & \emptyset \end{pmatrix}$ becomes $\begin{pmatrix} \emptyset & \begin{pmatrix} 1 & -\eta' \\ \eta & -\eta\eta' \end{pmatrix} \\ \begin{pmatrix} -1 & \eta \\ -\eta' & \eta\eta' \end{pmatrix} & \emptyset \end{pmatrix}$ which becomes in the renumbered basis of the form in (17.1) and (17.5),

$$\begin{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} & \begin{pmatrix} 0 & -\eta' \\ \eta & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & \eta \\ -\eta' & 0 \end{pmatrix} & \begin{pmatrix} 0 & -\eta\eta' \\ \eta\eta' & 0 \end{pmatrix} \end{pmatrix} \quad (17.46)$$

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Chapter 18

$SO(2N + 1)$

By $SO(2N + 1)$ it is meant the compact group $SO(2N + 1, R)$, defined by the set of $(2N + 1) \times (2N + 1)$ real orthogonal matrices ($O^T O = I$) with unit determinant ($\det O = 1$). The generators T_a^{def} of the defining (vector) representation are real antisymmetric (hence antihermitian) matrices, for which a basis is given by the following matrices

$$(M_{ij}^{\text{def}})^k_l = \begin{pmatrix} \ddots & & 1 \\ & \ddots & \\ -1 & & \ddots \end{pmatrix}_{(2N+1) \times (2N+1)} = \delta_i^k \delta_{jl} - \delta_j^k \delta_{il} \quad (i, j, k, l = 1, \dots, 2N + 1), \quad (18.1)$$

The number of these generators is $\frac{1}{2}(2N + 1)(2N) = N(2N + 1)$. We define M_{ij} with $i > j$ by $M_{ij} = -M_{ji}$. Then the Lie algebra has the same form as for $SO(2N)$

$$[M_{ij}, M_{kl}] = \delta_{jk} M_{il} - \delta_{jl} M_{ik} - \delta_{ik} M_{jl} + \delta_{il} M_{jk}. \quad (18.2)$$

18.1 Tensor representations of $SO(2N + 1)$

The Cartan-Weyl basis is the maximally noncompact form. As Cartan generators we take again 2×2 blocks along the diagonal and add a factor $-i$ to obtain hermitian generators

$$H_J^{\text{def}} = -i M_{2J-1, 2J}^{\text{def}}; \quad g_{JK}^{\text{def}} = \text{tr}(H_J^{\text{def}} H_K^{\text{def}}) = 2\delta_{JK}; \quad J, K = 1, \dots, N$$

$$H_1^{\text{def}} = \begin{pmatrix} 0 & -i & & \\ i & 0 & & \\ & & \ddots & \\ & & & \emptyset \end{pmatrix} \cdots; H_N^{\text{def}} = \begin{pmatrix} \emptyset & & & 0 \\ & \ddots & & \vdots \\ & & 0 & -i \\ & & i & 0 \\ 0 & \cdots & & 0 \end{pmatrix}_{(2N+1) \times (2N+1)} \quad (18.3)$$

Note that the last row and column of these $(2N + 1) \times (2N + 1)$ matrices contain only zeros. Since there are N Cartan generators the rank of $SO(2N + 1)$ is N .

These matrices act in the space of $(2N + 1)$ component vectors (the states of the defining representation), and the eigenvectors consist of the $2N$ eigenvectors of $SO(2N)$ (but with an extra

zero in the last component) together with the $(2N + 1)$ th vector $(0, 0, \dots, 0, 1)$. Then the weights of the defining representation are as follows

$$\begin{aligned} \vec{\mu}_{\text{def}}^j &= (0, \dots, 0, \pm 1, \dots, 0) \text{ for } j \leq 2N : 2N \text{ weight vectors} \\ \vec{\mu}_{\text{def}}^{2N+1} &= \underbrace{(0, \dots, 0, 0, \dots, 0)}_{N \text{ components}} : \text{ the } (2N + 1)\text{th weight vector.} \end{aligned} \quad (18.4)$$

For example, the weight diagram of the defining (=vector) representation of $SO(5)$ looks as shown in figure 18.1.

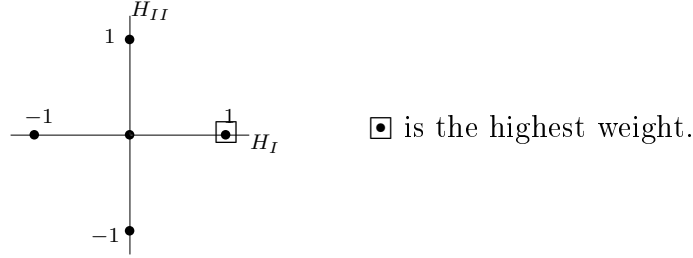


Figure 18.1: Weight diagram of the vector representation of $SO(5)$. A weight, or difference of weights is again positive if its first nonvanishing component is positive.

The roots $\vec{\alpha}$ are, as always, defined by $[H_J, E_{\vec{\alpha}}] = \alpha_J E_{\vec{\alpha}}$, where the $E_{\vec{\alpha}}$ are defined by this equation. Most of the E matrices we already know: those of $SO(2N)$ but with an extra column and row with zeros

$$E_{JK}^{\eta\eta'}(\text{def}) = \frac{1}{2} \begin{pmatrix} \ddots & \begin{pmatrix} 1 & i\eta' \\ i\eta & -\eta\eta' \end{pmatrix} & \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} \\ \begin{pmatrix} -1 & -i\eta \\ -i\eta' & \eta\eta' \end{pmatrix} & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix}_{(2N+1) \times (2N+1)} = \frac{1}{2} \begin{pmatrix} \ddots & \begin{pmatrix} 1 \\ i\eta \end{pmatrix} \begin{pmatrix} 1 & i\eta \end{pmatrix} & \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} \\ -\begin{pmatrix} 1 \\ i\eta' \end{pmatrix} \begin{pmatrix} 1 & i\eta \end{pmatrix} & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix}, \quad (18.5)$$

where $1 \leq J < K \leq N$, $\eta = \pm 1$, $\eta' = \pm 1$. There are $4 \times \frac{1}{2}N(N-1) = 2N(N-1)$ of these matrices and those with $\eta = +1$ are raising generators. The corresponding roots are

$$\underbrace{(0, 0, \eta, 0, \dots, 0, \eta', 0)}_{N \text{ components}}, \quad 4 \times \frac{1}{2}N(N-1) = 2N(N-1) \text{ roots.} \quad (18.6)$$

But there are further E matrices, with entries only in the last row and column.

$$E_J^{\eta}(\text{def}) = \frac{1}{\sqrt{2}} \begin{pmatrix} \emptyset & \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \\ 0 & 0 & -1 & -i\eta & 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \left(M_{2J-1, 2N+1} + i\eta M_{2J, 2N+1} \right), \quad (18.7)$$

where $J = 1, \dots, N$ and $\eta = \pm 1$. The factor $\frac{1}{\sqrt{2}}$ is needed to give the factor $\frac{1}{2}$ in $E_{JK}^{\eta\eta'}$, see below.

It is easy to determine the corresponding roots by evaluating $[H_J^{(\text{def})}, E_K^\eta(\text{def})]$

$$[H_J^{(\text{def})}, E_K^\eta(\text{def})] = \eta \delta_{JK} E_K^\eta(\text{def}). \quad (18.8)$$

Hence the new roots are $\eta \delta_{JK}$, or in vector notation

$$\underbrace{(0, 0, \eta, 0, \dots, 0)}_{N \text{ components}}, \quad 2N \text{ roots.} \quad (18.9)$$

Together with the N Cartan generators one indeed obtains $2N^2 + N = N(2N + 1) = \frac{1}{2}(2N)(2N - 1)$ generators. For example, the root diagram for $SO(5)$ looks as shown in figure 18.2. One often calls this the root diagram, but we added the two weights for H_1 and H_2 , so this is strictly speaking not the root diagram but the weight diagram of the adjoint rep.

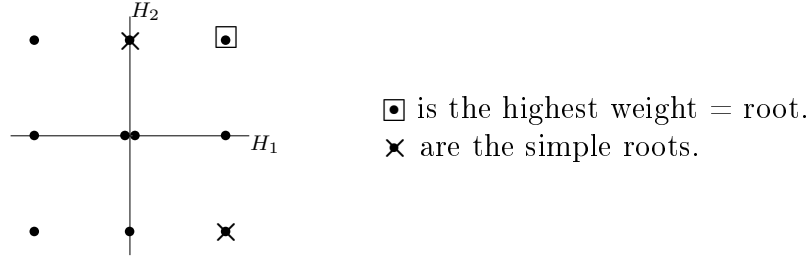


Figure 18.2: Weight diagram of the adjoint representation of $SO(5)$.

The matrices $E_J^\eta(\text{def})$ are in some sense the square roots of the matrices $E_{JK}^{\eta\eta'}(\text{def})$ since

$$[E_J^\eta(\text{def}), E_K^{\eta'}(\text{def})] = -E_{JK}^{\eta\eta'}(\text{def}) \quad \text{for } J < K \quad (18.10)$$

In fact, one could have started with first determining E_J^η , and subsequently obtaining $E_{JK}^{\eta\eta'}$ from this commutator. To obtain this relation with the proper $\frac{1}{2}$ of (17.10) one needs the normalization factor $\frac{1}{\sqrt{2}}$ in front of E_J^η .

If $J = K$ in (18.10), we find $[E_J^+, E_J^-] = -H_J$. However, as in the case of $E_{jk}^{\eta\eta'}$, E_j^- is not the lowering operator since $(E_j^\eta)^\dagger = -E_j^{-\eta}$, hence $(E_j^+)^\dagger = -E_j^-$. Thus $-E_j^-$ is the lowering generator, and we obtain

$$[E_J^+, (E_J^+)^\dagger] = [E_J^+, -E_J^-] = H_J. \quad (18.11)$$

Having obtained all $2N^2$ roots, we go through the usual steps. We define positivity of a root or weight if its first nonvanishing component is positive. Then the N^2 positive roots are

$$\begin{aligned} (0, \dots, 0, 1, 0, \dots, \eta', 0) & \quad N(N - 1) \text{ roots} \\ (0, \dots, 0, 1, 0, \dots, 0) & \quad N \text{ roots.} \end{aligned} \quad (18.12)$$

The N simple roots (each having N components) and the raising operators corresponding to these

simple roots are

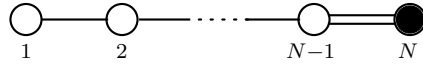
$$\begin{aligned}
(1, -1, 0, \dots, 0) &= \vec{\alpha}^I &\leftrightarrow& E_{12}^{+-} \\
(0, 1, -1, \dots, 0) &= \vec{\alpha}^{II} &\leftrightarrow& E_{23}^{+-} \\
&\vdots && \vdots \\
(0, \dots, 0, 1, -1) &= \vec{\alpha}^{N-1} &\leftrightarrow& E_{N-1,N}^{+-} \\
(0, \dots, 0, 0, 1) &= \vec{\alpha}^N &\leftrightarrow& E_N^+
\end{aligned} \tag{18.13}$$

The first $N-1$ simple roots are the same as for $SO(2N)$, but the last one is different; it was $\vec{\alpha}^N = (0, \dots, 0, 1, 1)$ for $SO(2N)$ but in $SO(2N+1)$ this is not a simple root because $(0, \dots, 0, 1, 1) = (0, 0, \dots, 0, 1, 0) + (0, 0, \dots, 0, 0, 1)$. Instead $(0, \dots, 0, 1)$ is now the last simple root.

The simple roots determine the Dynkin diagram, but note that $\vec{\alpha}^N$ is shorter than the other simple roots. It has only a nonvanishing inner product with $\vec{\alpha}^{N-1}$

$$\frac{(\vec{\alpha}^{N-1} \cdot \vec{\alpha}^N)^2}{(\vec{\alpha}^{N-1} \cdot \vec{\alpha}^{N-1})(\vec{\alpha}^N \cdot \vec{\alpha}^N)} = \frac{1}{2} = \cos^2 \theta_{\alpha^{N-1}, \alpha^N} \Rightarrow \theta_{\alpha^{N-1}, \alpha^N} = 135^\circ \tag{18.14}$$

and we find for the Dynkin diagram:



The solid dot denotes the shorter root. In particular one obtains for $SO(3)$: $\bigcirc$ (or $\bullet$); $SO(5)$: $\bigcirc=\bullet$; $SO(7)$: $\bigcirc-\bigcirc=\bullet$. The Dynkin diagram for $SU(2)$ is $\bigcirc$, which reflects that the Lie algebras of $SO(3)$ and $SU(2)$ are isomorphic. In a similar way the Lie algebras of $SO(5)$ and $USp(4)$ are related as we shall discuss when we reach $USp(2N)$ since the Dynkin diagram for $SO(5)$ is $\bigcirc=\bullet$ while for $USp(4)$ it is $\bullet=\bigcirc$.

Next we determine the fundamental weights $\vec{\mu}_{FW}^K$. There should be N of them and they are determined by

$$2 \frac{\vec{\alpha}^J \cdot \vec{\mu}_{FW}^K}{\vec{\alpha}^J \cdot \vec{\alpha}^J} = \delta_J^K \tag{18.15}$$

Writing $\vec{\alpha}^I = \vec{e}^1 - \vec{e}^2$, $\vec{\alpha}^{II} = \vec{e}^2 - \vec{e}^3$, $\dots$, $\vec{\alpha}^{N-1} = \vec{e}^{N-1} - \vec{e}^N$ and $\vec{\alpha}^N = \vec{e}^N$ where $\vec{e}^J$ are unit vectors, it is not too difficult to find the $\vec{\mu}_{FW}^K$:

$$\left. \begin{aligned}
\vec{\mu}_{FW}^I &= \vec{e}^1 = (1, 0, \dots, 0) && \text{defining (vector) irrep} \\
\vec{\mu}_{FW}^{II} &= \vec{e}^1 + \vec{e}^2 = (1, 1, 0, \dots, 0) && \text{adjoint irrep} \\
\vec{\mu}_{FW}^{III} &= \vec{e}^1 + \vec{e}^2 + \vec{e}^3 = (1, 1, 1, 0, \dots, 0) \\
&\vdots \\
\vec{\mu}_{FW}^{N-1} &= \vec{e}^1 + \vec{e}^2 + \dots + \vec{e}^{N-1} = (1, 1, \dots, 1, 0)
\end{aligned} \right\} \text{AT irreps}$$

$$\vec{\mu}_{FW}^N = \frac{1}{2} \left(\vec{e}^1 + \vec{e}^2 + \cdots + \vec{e}^N \right) = \left(\frac{1}{2}, \frac{1}{2}, \dots, \frac{1}{2} \right) \quad : \text{ spinor irrep} \quad (18.16)$$

The first fundamental weight $(1, 0, \dots, 0)$ is the highest weight of the defining (vector) irrep, see (18.4). The second fundamental weight, $(1, 1, 0, \dots, 0)$, is the highest root, see (18.12), so it is the highest weight of the adjoint representation. The rest of the fundamental weights, except for the last one, are all highest weights of antisymmetric tensor irreps with states $|i_1\rangle \otimes |i_2\rangle \otimes \cdots \otimes |i_N\rangle$ totally antisymmetric in all $i_1, \dots, i_N$. However, the last fundamental weight is the highest weight of the spinor irrep of $SO(2N+1)$. We discuss it in detail below, but first we note that **there is only one spinor irrep for $SO(2N+1)$** . Recall that for $SO(2N)$ there were two spinor irreps: the s -spinor irrep and the c -spinor irrep.

18.2 The spinor irrep of $SO(2N+1)$

As in the case of $SO(2N)$ we introduce spinor space with states

$$\underbrace{|\pm 1\rangle \otimes |\pm 1\rangle \otimes \cdots \otimes |\pm 1\rangle}_{N \text{ direct products}} : \quad 2^N \text{ states.} \quad (18.17)$$

There are now $2N+1$ Dirac matrices Γ_j . The first $2N$ Dirac matrices are the same as for $SO(2N)$, but Γ_{2N+1} is new¹

$$\begin{aligned} \Gamma_j &= \sigma_3 \otimes \cdots \otimes \sigma_3 \otimes (\sigma_1 \text{ or } \sigma_2) \otimes \mathbb{1} \otimes \cdots \otimes \mathbb{1} \quad j = 1, \dots, 2N \\ \Gamma_{2N+1} &= \sigma_3 \otimes \cdots \otimes \sigma_3 \otimes \sigma_3 \otimes \sigma_3 \otimes \cdots \otimes \sigma_3. \end{aligned} \quad (18.18)$$

So Γ_{2N+1} contains only σ_3 's. Because $\{\sigma_1, \sigma_2\} = 0$, $\{\sigma_3, \sigma_1\} = 0$ and $\{\sigma_3, \sigma_2\} = 0$, it is easy to see that the Clifford algebra is satisfied

$$\{\Gamma_j, \Gamma_k\} = 2\delta_{jk} \quad j, k = 1, \dots, 2N+1. \quad (18.19)$$

The generators of $SO(2N+1)$ in the spinor irrep are given by $M_{ij}^{(\text{spin})} = \frac{1}{2}\Gamma_i\Gamma_j$ for $i < j$. This rep is irreducible, so we write $M_{ij}^{(\text{spin})}$ instead of M_{ij}^{RS} (we used RS in the $SO(2N)$ case to denote the reducible spinor rep). It is straightforward to check that these matrices satisfy the Lie algebra of $SO(2N+1)$

$$[M_{ij}^{(\text{spin})}, M_{kl}^{(\text{spin})}] = \delta_{jk}M_{il}^{(\text{spin})} - \delta_{jl}M_{ik}^{(\text{spin})} - \delta_{ik}M_{jl}^{(\text{spin})} + \delta_{il}M_{jk}^{(\text{spin})} \quad (18.20)$$

We recall that to evaluate the commutators it is useful to write the commutation relations in terms

¹We could also have defined $\Gamma_j = \mathbb{1} \otimes \cdots \otimes \mathbb{1} \otimes (\sigma_1 \text{ or } \sigma_2) \otimes \sigma_3 \otimes \cdots \otimes \sigma_3$ for $j = 1, \dots, 2N$ and Γ_{2N+1} as in (18.18). This is the choice of Georgi, but with our choice the number of σ_3 in Γ_j increases as j increases, culminating with Γ_{2N+1} .

of anticommutators

$$[AB, CD] = [AB, C]D + C[AB, D] = A\{B, C\}D - \{A, C\}BD + CA\{B, D\} - C\{A, D\}B. \quad (18.21)$$

This is an identity for any A, B, C and D . Using the Clifford algebra for the anticommutators, we recover the $SO(2N+1)$ Lie algebra.

To study the transformation properties of spinors in more detail, we evaluate the generators of $SO(2N+1)$ in the spinor irrep explicitly. The M_{ij}^{spin} are as for $SO(2N)$ but the $M_{j,2N+1}^{\text{spin}}$ are new.

$$\begin{aligned} M_{jk}^{(\text{spin})} &= \frac{1}{2}\Gamma_j\Gamma_k = \frac{1}{2}\mathbb{1} \otimes \cdots \otimes \mathbb{1} \otimes (\sigma_1 \text{ or } \sigma_2)\sigma_3 \otimes \sigma_3 \otimes \cdots \otimes \sigma_3 \otimes (\sigma_1 \text{ or } \sigma_2) \otimes \mathbb{1} \otimes \cdots \otimes \mathbb{1} \\ &\quad j,k=1,\dots,2N \\ M_{j,2N+1}^{(\text{spin})} &= \frac{1}{2}\Gamma_j\Gamma_{2N+1} = \frac{1}{2}\mathbb{1} \otimes \cdots \otimes \mathbb{1} \otimes (\sigma_1 \text{ or } \sigma_2)\sigma_3 \otimes \sigma_3 \otimes \cdots \otimes \sigma_3 \end{aligned} \quad (18.22)$$

These are the generators of the compact group $SO(2N+1)$, but for the study of the representations we need the CW basis, so we construct the Cartan generators $H_J^{(\text{spin})} = -iM_{2J-1,2J}^{(\text{spin})}$ and the raising and lowering generators $E_J^\eta(\text{spin}) = \frac{1}{\sqrt{2}} \left(M_{2J-1,N+1}^{(\text{spin})} + i\eta M_{2J,N+1}^{(\text{spin})} \right)$:

$$\begin{aligned} H_J^{(\text{spin})} &= -i\frac{1}{2}\Gamma_{2J-1}\Gamma_{2J} = \mathbb{1} \otimes \cdots \otimes \mathbb{1} \otimes \frac{1}{2}\sigma_3 \otimes \mathbb{1} \otimes \cdots \otimes \mathbb{1} \quad (\text{as for } SO(2N)) \\ E_J^\eta(\text{spin}) &= \frac{1}{\sqrt{2}}\mathbb{1} \otimes \cdots \otimes \mathbb{1} \otimes \sigma^\eta\sigma_3 \otimes \sigma_3 \otimes \cdots \otimes \sigma_3 \end{aligned} \quad (18.23)$$

We can then obtain the raising and lowering generators $E_{JK}^{\eta\eta'}$ from the commutators of two E_J^η generators as in (18.10) but the calculation is much simpler for the present case of the spinor irreps

$$E_{JK}^{\eta\eta'}(\text{spin}) = - \left[E_J^\eta(\text{spin}), E_K^{\eta'}(\text{spin}) \right] = \mathbb{1} \otimes \cdots \otimes \mathbb{1} \otimes \sigma^\eta\sigma_3 \otimes \sigma_3 \otimes \cdots \otimes \sigma_3 \otimes \sigma^{\eta'} \otimes \mathbb{1} \otimes \cdots \otimes \mathbb{1}. \quad (18.24)$$

This agrees with (17.35). We used the relation $\sigma_3\sigma^{\eta'}\sigma_3 = -\sigma^{\eta'}$. We recall the definitions $\sigma^\eta = \frac{1}{2}(\sigma_1 + i\eta\sigma_2)$ and $\sigma^{\eta'} = \frac{1}{2}(\sigma_1 + i\eta'\sigma_2)$. They satisfy $\sigma^3\sigma^\eta = \eta\sigma^\eta = -\sigma^\eta\sigma_3$ and $\sigma^3\sigma^{\eta'} = \eta'\sigma^{\eta'} = -\sigma^{\eta'}\sigma_3$. The last relation can be used to evaluate the commutators $[E_{JK}^{\eta\eta'}(\text{spin}), E_L^{\eta''}(\text{spin})]$ and $[E_{JK}^{\eta\eta'}(\text{spin}), E_{LM}^{\eta''\eta'''}(\text{spin})]$, straightforward calculations which we omit since we do not need the results.

We can now prove our earlier claim that $\vec{\mu}_{FW}^N = (\frac{1}{2}, \dots, \frac{1}{2})$ is the highest weight of the spinor irrep. The weights of the 2^N states $|\pm 1\rangle \otimes \cdots \otimes |\pm 1\rangle$ of the spinor irrep are clearly

$$\underbrace{(\pm\frac{1}{2}, \pm\frac{1}{2}, \dots, \pm\frac{1}{2})}_N : \text{ the } 2^N \text{ weight vectors of the spinor irrep of } SO(2N+1) \quad (18.25)$$

because the $\frac{1}{2}\sigma_3$ in $H_J^{(\text{spin})}$ in (18.23) has eigenvalues $\pm\frac{1}{2}$. The highest weight is $(\frac{1}{2}, \dots, \frac{1}{2})$, which indeed agrees with $\vec{\mu}_{FW}^N$ in (18.16). For $SO(5)$ one finds the weight diagram for the spinor representation as shown in figure 18.3.

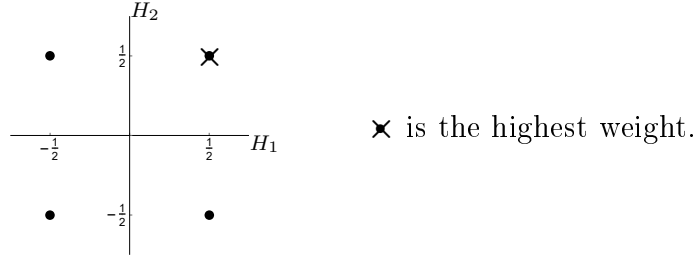


Figure 18.3: Weight diagram of the spinor representation of $SO(5)$.

18.3 The Cartan-Weyl basis and other real forms for $SO(2N + 1)$

The last issue concerns the Cartan-Weyl (CW) basis of $SO(2N + 1)$. For $SO(2N)$ we found that the CW basis yielded the Lie algebra of $SO(N, N)$. Since there are only 2 real forms of $SO(2N + 1)$, namely the compact Lie algebra $SO(2N, R)$ and the noncompact $SO(p, q; R)$ with $p + q = 2N + 1$, while the degree D of non compactness of $SO(p, q; R)$ equals $D = N + \frac{1}{2} - \frac{1}{2}(p - q)^2$, where $p + q = 2N + 1$, it is clear that the maximal noncompact form of $SO(2N + 1)$ is obtained for $p - q = \pm 1$, hence it is $SO(N + 1, N; R)$ or (equivalently) $SO(N, N + 1; R)$. This should be the CW basis. To check this we evaluate the same kind of similarity transformation as for $SO(2N, R)$, but with a diagonal matrix S with a $(2N + 1)$ th entry $+1$.

$$S = \begin{pmatrix} 1 & & & & \\ & -i & & & \\ & & \ddots & & \\ & & & 1 & \\ & & & & -i & 1 \end{pmatrix} \quad (18.26)$$

For the matrices $E_{JK}^{\eta\eta'}$ we already proved that $SE_{JK}^{\eta\eta'}S^{-1}$ is of the form of the generators of $SO(N, N; R)$, and nothing changes if one adds a $(2N + 1)$ th row and column of vanishing entries. For E_J^η one finds for $SO(5)$ by direct computation

$$SE_2^\eta S^{-1} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & & & & \\ & -i & & & \\ & & 1 & & \\ & & & -i & \\ & & & & 1 \end{pmatrix} \begin{pmatrix} 0 & & & & \\ 0 & 0 & & & \\ & 1 & & & \\ i\eta & & & & \\ 0 & & & & \end{pmatrix} \begin{pmatrix} 1 & & & & \\ & i & & & \\ & & 1 & & \\ & & & i & \\ & & & & 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \emptyset & & 0 \\ & \emptyset & 1 \\ 0 & 0 & -1 & \eta & 0 \end{pmatrix} \quad (18.27)$$

and renumbering rows and columns as 24|135 we obtain a matrix of the form in (18.4)

$$\begin{array}{c} 2 \\ 4 \\ 1 \\ 3 \\ 5 \end{array} \left(\begin{array}{cc|ccc} 2 & 4 & 1 & 3 & 5 \\ \hline & \emptyset & & \emptyset & 0 \\ & & & & \eta \\ \hline & \emptyset & 0 & 0 & 0 \\ & & 0 & 0 & 1 \\ 0 & \eta & 0 & -1 & 0 \end{array} \right) \quad (18.28)$$

This shows that $SO(N, N + 1; R)$ is the CW real form of $SO(2N + 1)$.

Chapter 19

$USp(2N)$

The group $Sp(2N, C)$ is defined as those complex transformations $x'^\alpha = \mathcal{M}^\alpha_\beta x^\beta$ which leave the bilinear form $y^\alpha \Omega_{\alpha\beta} x^\beta$ invariant where $\Omega_{\alpha\beta}$ is real and antisymmetric¹, $\Omega_{\alpha\beta} = -\Omega_{\beta\alpha}$. We shall use $\Omega_{\alpha\beta} = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$ but one can also use other real antisymmetric Ω 's. The group $USp(2N)$ is the intersection of $Sp(2N, C)$ and $SU(2N)$. (In fact, the generators of $Sp(2N, C)$ are traceless,² so we could also use $U(2N)$ instead of $SU(2N)$ in the definition of $USp(2N)$.) Thus the group $USp(2N)$ is compact. It is also simple, connected, and simply connected. On the other hand, the group $Sp(2N, R)$ consists of the real transformations $x'^\alpha = \mathcal{M}^\alpha_\beta x^\beta$ which leave $x^\alpha \Omega_{\alpha\beta} y^\beta$ invariant with real $\mathcal{M}^\alpha_\beta$, x and y . This group $Sp(2N, R)$ is noncompact, and as we shall see, it is the Cartan-Weyl basis for $USp(2N)$, the maximally noncompact real form of $USp(2N)$.

The group $USp(2N)$ is $\dots$ as the group of canonical transformations of bosonic operators a^i and $a_i^\dagger$. Defining a linear infinitesimal transformations by $\hat{\alpha}^i = a^i + \phi^i_k a^k + \psi^{ik} a_k^\dagger$ this transformation is canonical ($[\hat{\phi}^i, \hat{\psi}_j^\dagger] = \delta_j^i$) if $A = \begin{pmatrix} \phi & \psi \\ \psi^* & \phi^* \end{pmatrix}$ in $\begin{pmatrix} \hat{\phi} \\ \hat{\phi}^\dagger \end{pmatrix} = \begin{pmatrix} \phi \\ \phi^\dagger \end{pmatrix} + A \begin{pmatrix} \phi \\ \phi^\dagger \end{pmatrix}$ satisfies $A^T J + J A = 0$ where $J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$. Similarly, the linear canonical transformation of a fermionic oscillators form the group $SO(N, N)$, and canonical transformations which also mix bosonic and fermionic oscillations form the superalgebra $OSp(N, N|2N, R)$ [1].

Expanding the group elements $\mathcal{M}$ as $\mathcal{M} = I + M + \dots$, the generators M of $USp(2N)$ in the

¹A general $2N \times 2N$ complex antisymmetric tensor T can always be cast in the form of $\Omega' = \begin{pmatrix} 0 & 1 & & \\ -1 & 0 & & \\ & & \ddots & \\ & & & 0 & 1 \\ & & & -1 & 0 \end{pmatrix}$

by $S^T U^T T U S$ where U is unitary and S a real diagonal matrix. By a further basis transformation one reaches the form $\Omega = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$.

²To prove that the generators of $USp(2N)$ are traceless, we only have to look at (19.4). But also the generators of $Sp(2N, C)$ or $Sp(2N, R)$ are traceless, hence the condition $\det M = 1$ need not be added, it is automatically satisfied. Let us show that $\det M = 1$ for any matrix M satisfying $M^T \Omega M = \Omega$. Since $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ can be decomposed as $M = \begin{pmatrix} A & 0 \\ C & I \end{pmatrix} \begin{pmatrix} I & A^{-1}B \\ 0 & D - CA^{-1}B \end{pmatrix} = \begin{pmatrix} I & B \\ 0 & D \end{pmatrix} \begin{pmatrix} A - BD^{-1}C & 0 \\ D^{-1}C & I \end{pmatrix}$, it follows that $\det M = \det A \det(D - CA^{-1}B) = \det A^T \det(D - CA^{-1}B) = \det(A^T D - A^T C A^{-1} B)$. From $M^T \Omega M = \Omega$ we learn that $C^T A = A^T C$, $D^T B = B^T D$, $-C^T B + A^T D = I$, and $D^T A - B^T C = I$. Then $\det M = \det(A^T D - C^T A A^{-1} B) = \det A^T D - C^T B = \det I = 1$.

defining representation are the $2N \times 2N$ complex matrices satisfying

$$\begin{aligned} M^T \Omega + \Omega M &= 0 \text{ or } (M^T)_{\alpha}^{\beta} \Omega_{\beta\gamma} + \Omega_{\alpha\beta} M^{\beta}_{\gamma} = 0 \text{ for } Sp(2N, C) \\ M^{\dagger} &= -M \text{ or } (M^{\dagger})^{\alpha}_{\beta} = -M^{\alpha}_{\beta} \text{ for } SU(2N), \end{aligned} \quad (19.1)$$

where $(M^T)_{\alpha}^{\beta} = M^{\beta}_{\alpha}$ and $(M^{\dagger})^{\alpha}_{\beta} = (M^{\beta}_{\alpha})^*$. Writing M in block form as $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$, we get

$$\begin{pmatrix} A^T & C^T \\ B^T & D^T \end{pmatrix} \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} + \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = 0 \Rightarrow \begin{aligned} C^T &= C; B^T = B \\ D &= -A^T \end{aligned} \quad (19.2)$$

$$\begin{pmatrix} A^{\dagger} & C^{\dagger} \\ B^{\dagger} & D^{\dagger} \end{pmatrix} + \begin{pmatrix} A & B \\ C & D \end{pmatrix} = 0 \Rightarrow \begin{aligned} A^{\dagger} + A &= 0; D^{\dagger} + D = 0 \\ C^{\dagger} + B &= 0; B^{\dagger} + C = 0 \end{aligned} \quad (19.3)$$

The generators of $USp(2N)$ have then the following form

$$M = \begin{pmatrix} A_r + iS_3 & S_2 + iS_1 \\ -S_2 + iS_1 & A_r - iS_3 \end{pmatrix}_{2N \times 2N} \quad \text{with} \quad \begin{aligned} A_r &= \text{real antisymmetric} \\ S_j &= \text{real symmetric} \end{aligned} \quad (19.4)$$

If one writes this as $M = A_r \otimes I_{2 \times 2} + S_j \otimes i\tau_j$, where τ_j are the Pauli matrices, it is clear that M is antihermitian, but also the equation $M^T \Omega + \Omega M = 0$ is easily checked if one writes Ω as $\Omega = I \otimes i\tau_2$ and uses the relation $\tau_j^T \tau_2 + \tau_2 \tau_j = 0$.

19.1 Tensor representations of $USp(2N)$

To analyze the representations, we need the Cartan-Weyl basis. As we show in section 19.4, the CW form of $USp(2N)$ is the group $Sp(2N, R)$. It contains only real matrices and can be obtained by dropping the i 's in (19.4). Then the generators of $USp(2N)$ in the Cartan-Weyl basis become the generators of $Sp(2N, R)$

$$M = \begin{pmatrix} A_r + S_3 & S_1 + S_2 \\ S_1 - S_2 & A_r - S_3 \end{pmatrix}. \quad (19.5)$$

The $\frac{1}{2}N(N-1)$ generators with A_r and the $\frac{1}{2}N(N+1)$ generators with S_2 are still compact (antihermitian), but the $\frac{1}{2}N(N+1)$ generators with S_1 and the $\frac{1}{2}N(N+1)$ generators with S_3 are noncompact (hermitian). The total number of generators is $2N^2 + N$. The number of noncompact generators minus the number of compact generators is N , which is the rank of $USp(2N)$ and $Sp(2N, R)$. This confirms that (19.5) is the maximally noncompact form of the symplectic group. The matrices M in (19.5) still satisfy $M^T \Omega + \Omega M = 0$ which implies that ΩM is symmetric, and indeed $\Omega M = \begin{pmatrix} S_1 - S_2 & A_r - S_3 \\ -A_r - S_3 & -S_1 - S_2 \end{pmatrix}$ are the most general $2N \times 2N$ real symmetric matrices.

The equation $M^T \Omega + \Omega M = 0$ implies that Ω is an invariant tensor of the defining irrep of

$Sp(2N, R)$. If we define $\delta v^\alpha = M^\alpha_\beta v^\beta$ and $\delta w_\alpha = -w_\beta M^\beta_\alpha$, then

$$\delta\Omega_{\alpha\beta} = -\Omega_{\alpha\gamma}M^\gamma_\beta - \Omega_{\gamma\beta}M^\gamma_\alpha = -\left(\Omega M + M^T\Omega\right)_{\alpha\beta} = 0. \quad (19.6)$$

Also δ^α_β is an invariant tensor of $Sp(2N, R)$ because its variation vanishes: $\delta\delta^\alpha_\beta = M^\alpha_{\alpha'}\delta^{\alpha'}_\beta - \delta^\alpha_{\beta'}M^{\beta'}_\beta = M^\alpha_\beta - M^\alpha_\beta = 0$. In fact, one defines the transformation law of covariant vectors $\delta v_a = -v_\beta M^\beta_\alpha$ such that δ^α_β be an invariant tensor (or: in order that $v_\beta w^\beta$ be invariant). We shall use $\Omega_{\alpha\beta} = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$ and $\Omega^{\beta\gamma} = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$ to raise and lower α, β, γ indices³. A tensor remains a tensor when some of its indices have been raised or lowered with invariant tensors.

A basis for the generators of $Sp(2N, R)$ in terms of the invariant tensors δ^α_β and $\Omega_{\alpha\beta}$ as follows⁴

$$(M_{\alpha\beta})^\gamma_\delta = \delta^\gamma_\alpha \Omega_{\beta\delta} + \delta^\gamma_\beta \Omega_{\alpha\delta} \quad \text{with } \alpha \leq \beta \quad (19.7)$$

These matrices are real and they also satisfy the symplectic condition because $(\Omega M_{\alpha\beta})_{\gamma\delta} = \Omega_{\gamma\alpha}\Omega_{\beta\delta} + \Omega_{\gamma\beta}\Omega_{\alpha\delta}$ is symmetric in $\gamma\delta$. But $M_{\alpha\beta}$ is also symmetric in $\alpha\beta$, hence the number of these matrices is $\frac{1}{2}2N(2N+1)$, which is the number of generators of $Sp(2N, R)$. Hence the $M_{\alpha\beta}$ form a complete basis for $Sp(2N, R)$ (which looks similar to the usual basis for $SO(N)$: $(M_{ij})^k_l = \delta_i^k\delta_{jl} - \delta_j^k\delta_{il}$). Note that the matrix elements $(M_{\alpha\beta})^\gamma_\delta$ are made out of the invariant tensors $\Omega_{\beta\gamma}$ and δ^γ_α , so $(M_{\alpha\beta})^\gamma_\delta$ is a 4-index invariant tensor of $Sp(2N, R)$.⁵ Each matrix $M_{\alpha\beta}$ is traceless and real, and has two nonzero entries for $\alpha < \beta$, and one nonzero entry for $\alpha = \beta$. These

³We use the northwest-southeast convention both for raising and for lowering indices: $\Omega^{\alpha\beta}v_\beta = v^\alpha$ and $v^\beta\Omega_{\beta\alpha} = v_\alpha$. Using these rules to raise both indices of $\Omega_{\alpha\beta}$, it follows that $\Omega^{\alpha\beta}\Omega_{\gamma\beta} = \delta^\alpha_\gamma$, so $\Omega^{\alpha\beta}$ and $\Omega_{\alpha\beta}$ are numerically equal, but $\Omega^{\alpha\beta}\Omega_{\beta\gamma} = -(I)^\alpha_\gamma$.

⁴From $M^T\Omega + \Omega M = 0$ it follows that $\Omega^{-1}M^T + M\Omega^{-1} = 0$. Then $M\Omega^{-1}$ is a symmetric matrix S . A basis in S is given by $(S_{IJ})^\alpha_\beta = \delta_I^\alpha\delta_{J\beta} + \delta_J^\alpha\delta_{I\beta}$. Then $M = S_{IJ}\Omega$ which is (19.7).

⁵One can at this point start constructing a tensor calculus for tensors of $Sp(2N, R)$. One begins with contravariant vectors v^α which transform per definition as $\delta v^\alpha = N^\alpha_\beta v^\beta$ with $N^\alpha_\beta = \frac{1}{2}s^{pq}(M_{pq})^\alpha_\beta$ with real symmetric parameters s^{pq} and $(M_{pq})^\alpha_\beta$ defined in (19.7). Next one introduces covariant vectors w_α which transform such that $v^\alpha w_\alpha$ is invariant; the result is $\delta w_\alpha = -w_\beta N^\beta_\alpha = -w_\beta \frac{1}{2}s^{pq}(M_{pq})^\beta_\alpha$. Indices are raised and lowered by $\Omega^{\alpha\beta}$ and $\Omega_{\alpha\beta}$ with northwest-southeast contractions: $v_\alpha = v^\beta\Omega_{\beta\alpha}$; $w^\alpha = \Omega^{\alpha\beta}w_\beta$. Then $v^\alpha = \Omega^{\alpha\beta}v_\beta = \Omega^{\alpha\beta}(v^\gamma\Omega_{\gamma\beta})$, hence for consistency $\Omega^{\alpha\beta}\Omega_{\gamma\beta} = \delta^\alpha_\gamma$. This defines $\Omega^{\alpha\beta}$ in terms of $\Omega_{\alpha\beta}$: $\Omega_{\alpha\beta} = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$; $\Omega^{\alpha\beta} = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$. As a particular case, one may verify that $\Omega^{\alpha\beta} = \Omega^{\alpha\alpha'}\Omega^{\beta\beta'}\Omega_{\alpha'\beta'}$.

Next one shows that $v_\alpha = v^\beta\Omega_{\beta\alpha}$ transforms like w_α : $\delta v_\alpha = -v_\beta \frac{1}{2}s^{pq}(M_{pq})^\beta_\alpha$ equals $\delta v^\beta\Omega_{\beta\alpha} = (\frac{1}{2}s^{pq}(M_{pq})^\beta_\gamma v^\gamma)\Omega_{\beta\alpha}$. This equality holds since $(M_{pq})_{\alpha\beta} \equiv (M_{pq})^\gamma_\beta\Omega_{\gamma\alpha}$ satisfy $(M_{pq})_{\alpha\beta} = (M_{pq})_{\beta\alpha}$. Using these rules one may verify that $(M_{pq})_{\alpha\beta} = \Omega_{p\alpha}\Omega_{q\beta} + \Omega_{q\alpha}\Omega_{p\beta}$ and $(M_{pq})^{\alpha\beta} = (\delta_p^\alpha\delta_q^\beta + \delta_p^\beta\delta_q^\alpha)$.

The statement that $\Omega_{\alpha\beta}$, $\Omega^{\alpha\beta}$ and δ^α_α are invariant tensors of $Sp(2N, R)$ means

$$\begin{aligned} \delta\Omega_{\alpha\beta} &= -\Omega_{\alpha'\beta}N^{\alpha'}_\alpha - \Omega_{\alpha\beta'}N^{\beta'}_\beta = 0 \\ \delta\Omega^{\alpha\beta} &= N^\alpha_{\alpha'}\Omega^{\alpha'\beta} + N^\beta_{\beta'}\Omega^{\alpha\beta'} = 0 \\ \delta\delta^\alpha_\alpha &= N^\beta_{\beta'}\delta^{\beta'}_\alpha - \delta^\beta_{\alpha'}N^{\alpha'}_\alpha = 0. \end{aligned}$$

With these preliminaries we can now be more explicit about the statement that $(M_{\alpha\beta})^\gamma_\delta$ is an invariant tensor: it means the following

$$0 = \delta(M_{\alpha\beta})^\gamma_\delta = -(M_{\alpha'\beta})^\gamma_\delta N^{\alpha'}_\alpha - (M_{\alpha\beta'})^\gamma_\delta N^{\beta'}_\beta + N^\gamma_{\gamma'}(M_{\alpha\beta})^{\gamma'}_\delta - (M_{\alpha\beta})^\gamma_{\delta'}N^{\delta'}_{\delta}.$$

matrices satisfy $M^T \Omega + \Omega M = 0$ but not all of them are antihermitian. So they do not correspond to the matrices in (19.4), but rather to the matrices in (19.5) with the factors i deleted. These are the generators in the (maximally noncompact) Cartan-Weyl basis. They generate $Sp(2N, R)$. They are still invariant tensors, because (19.6) clearly remains true if one deletes a factor of i in M . (To avoid confusion, note that δ_{jl} is an invariant tensor of $SO(N)$ but not of $Sp(2N, C)$ or $SU(N)$). Using (19.7), one obtains for the Lie algebra of $Sp(2N, R)$

$$[M_{\alpha\beta}, M_{\gamma\delta}] = \Omega_{\beta\gamma} M_{\alpha\delta} + \Omega_{\alpha\gamma} M_{\beta\delta} + \Omega_{\beta\delta} M_{\alpha\gamma} + \Omega_{\alpha\delta} M_{\beta\gamma}. \quad (19.8)$$

This looks like the $SO(N)$ algebra except for the Ω 's and the signs. The Lie algebra of $USp(2N)$ has a more complicated form, which follows from (19.4).⁶

The dimension of $USp(2N)$ (and thus also the dimension of $Sp(2N, R)$) is the dimension of the $SO(N)$ subgroup generated by the generators A_r , and the sum of the number of generators S_j

$$\dim USp(2N) = \frac{1}{2}N(N-1) + 3 \cdot \frac{1}{2}N(N+1) = \frac{1}{2}2N(2N+1). \quad (19.9)$$

which is also the number of independent components of $M_{\alpha\beta}$ because $M_{\alpha\beta}$ is symmetric. Note that this is also the dimension of $SO(2N+1)$. In fact, $USp(2N)$ and $Sp(2N, R)$ resemble $SO(2N+1)$ in many ways.

We now construct the Cartan generators and the raising and lowering generators of the defining representation in the Cartan-Weyl basis. The Cartan generators are those of $SU(N)$ but doubled and with extra minus signs. The last one, H_N , is due to the unit matrix in S_3 in (19.5)

$$\begin{aligned} H_1^{\text{def}} &= \frac{1}{2} \begin{pmatrix} 1 & & & & & \\ & -1 & & & & \\ & & 0 & & & \\ & & & \ddots & & \\ & & & & -1 & \\ & & & & & 1 \\ & & & & & & 0 \\ & & & & & & & \ddots \end{pmatrix}; H_2^{\text{def}} = \frac{1}{\sqrt{12}} \begin{pmatrix} 1 & & & & & \\ & 1 & & & & \\ & & -2 & & & \\ & & & 0 & & \\ & & & & \ddots & \\ & & & & & -1 \\ & & & & & & -1 \\ & & & & & & & 2 \\ & & & & & & & & 0 \\ & & & & & & & & & \ddots \end{pmatrix}; \dots \\ H_{N-1}^{\text{def}} &= \frac{1}{\sqrt{2(N-1)N}} \begin{pmatrix} 1 & & & & & \\ & 1 & & & & \\ & & 1 & & & \\ & & & \ddots & & \\ & & & & -(N-1) & \\ & & & & & -1 \\ & & & & & & -1 \\ & & & & & & & -1 \\ & & & & & & & & \ddots \\ & & & & & & & & & N-1 \end{pmatrix}; H_N^{\text{def}} = \frac{1}{\sqrt{2N}} \begin{pmatrix} 1 & & & & & \\ & 1 & & & & \\ & & 1 & & & \\ & & & \ddots & & \\ & & & & 1 & \\ & & & & & -1 \\ & & & & & & -1 \\ & & & & & & & -1 \\ & & & & & & & & \ddots \\ & & & & & & & & & -1 \end{pmatrix}. \end{aligned} \quad (19.10)$$

They clearly are traceless, and satisfy $\text{tr } H_J^{\text{def}} H_K^{\text{def}} \equiv g_{JK}^{\text{def}} = \delta_{JK}$.

With the explicit form of the H_J we can construct the weights of the defining irrep. The

⁶Using $M^{(i)} = A_r^{(i)} \otimes I + S_j^{(i)} \otimes i\tau_j$, one finds $[M^{(1)}, M^{(2)}] = A_r^{(12)} \otimes I + S_j^{(12)} \otimes i\tau_j$ with $A_r^{(12)} = [A_r^{(1)}, A_r^{(2)}] - \sum_j [S_j^{(1)}, S_j^{(2)}]$ and $S_j^{(12)} = [A_r^{(1)}, S_j^{(2)}] + [S_j^{(1)}, A_r^{(2)}] - \epsilon_{jkl} \{S_k^{(1)}, S_l^{(2)}\}$.

dimension of the linear vector space for the defining irrep is $2N$, but there are only N Cartan generators, hence we get $2N$ weight vectors, where each vector has N components. We denote these weights by $\vec{\nu}^j$ with $1 \leq j \leq 2N$ (we shall later find their relation to the weights $\vec{\mu}_{\text{def}}^j$ of the defining irrep of $SU(N)$)

$$\begin{aligned}
& \left. \begin{array}{l} \text{Mostly} \\ \text{the } \mathbf{N} \\ \text{of } SU(N) : \\ \vec{\nu}^j = (\vec{\mu}^j, \frac{1}{\sqrt{2N}}) \\ \text{for } 1 \leq j \leq N \end{array} \right\} \left\{ \begin{array}{l} \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = |1\rangle \text{ has } \vec{\nu}^1 = \left(\frac{1}{2}, \frac{1}{\sqrt{12}}, \dots, \frac{1}{\sqrt{2(N-1)N}}, \frac{1}{\sqrt{2N}} \right) \\ \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = |2\rangle \text{ has } \vec{\nu}^2 = \left(-\frac{1}{2}, \frac{1}{\sqrt{12}}, \dots, \frac{1}{\sqrt{2(N-1)N}}, \frac{1}{\sqrt{2N}} \right) \\ \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} = |3\rangle \text{ has } \vec{\nu}^3 = \left(0, -\frac{2}{\sqrt{12}}, \dots, \frac{1}{\sqrt{2(N-1)N}}, \frac{1}{\sqrt{2N}} \right) \\ \vdots \\ \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} = |_{N-1}\rangle \text{ has } \vec{\nu}^{N-1} = \left(0, 0, \dots, 0, -\frac{(N-2)}{\sqrt{2(N-2)(N-1)}}, \frac{1}{\sqrt{2(N-1)N}}, \frac{1}{\sqrt{2N}} \right) \\ \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} = |_N\rangle \text{ has } \vec{\nu}^N = \left(0, 0, \dots, 0, -\frac{(N-1)}{\sqrt{2(N-1)N}}, \frac{1}{\sqrt{2N}} \right) \end{array} \right. \\
& \left. \begin{array}{l} \vec{\nu}^{j+N} = -\vec{\nu}^j \\ \text{for } 1 \leq j \leq N \end{array} \right\} \left\{ \begin{array}{l} \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} = |_{N+1}\rangle \text{ has } \vec{\nu}^{N+1} = \left(-\frac{1}{2}, -\frac{1}{\sqrt{12}}, \dots, -\frac{1}{\sqrt{2N}} \right) \\ \begin{pmatrix} 0 \\ 1 \\ 1 \\ \vdots \\ 0 \end{pmatrix} = |_{N+2}\rangle \text{ has } \vec{\nu}^{N+2} = \left(\frac{1}{2}, -\frac{1}{\sqrt{12}}, \dots, -\frac{1}{\sqrt{2N}} \right) \\ \vdots \\ \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = |_{2N}\rangle \text{ has } \vec{\nu}^{2N} = \left(0, 0, \dots, 0, \frac{N-1}{\sqrt{2(N-1)N}}, \frac{-1}{\sqrt{2N}} \right). \end{array} \right. \quad (19.11)
\end{aligned}$$

So the components of the ν 's are ν_j^j with $j = 1, \dots, 2N$ and $J = 1, \dots, N$. Clearly $\vec{\nu}^{j+N} = -\vec{\nu}^j$ for $j = 1, \dots, N$. It is also clear that if we define positivity of the weights as positivity of the **last** nonvanishing component (just as we did for $SU(N)$; note that $USp(2N)$ is a subgroup of $SU(2N)$), then the highest weight of the defining irrep is $\vec{\nu}^1$

$$\vec{\mu}_{HW}(\text{def}) = \vec{\nu}^1 = \left(\frac{1}{2}, \frac{1}{2\sqrt{3}}, \dots, \frac{1}{\sqrt{2(N-1)N}}, \frac{1}{\sqrt{2N}} \right). \quad (19.12)$$

To find the roots of $Sp(2N, R)$, recall that in the Cartan-Weyl basis the generators are given

by

$$M = A_r \otimes I + S_3 \otimes \tau_3 + S_2 \otimes i\tau_2 + S_1 \otimes \tau_1 = \begin{pmatrix} A_r + S_3 & S_1 + S_2 \\ S_1 - S_2 & A_r - S_3 \end{pmatrix} = \begin{pmatrix} P & S_I \\ S_{II} & -P^T \end{pmatrix} \quad (19.13)$$

Here P is an arbitrary real $N \times N$ matrix, and S_I and S_{II} are arbitrary symmetric real $N \times N$ matrices. Then the raising and lowering operators are those of $SU(N)$ in the diagonal blocks, and new ones in the off-diagonal blocks

$$\begin{aligned} E_{JK} &= \left(\begin{array}{c|c} \begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix} & \\ \hline \begin{smallmatrix} 0 & 0 \\ -1 & 0 \end{smallmatrix} \end{array} \right); & E_{KJ} &= \left(\begin{array}{c|c} \begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix} & \\ \hline \begin{smallmatrix} 0 & -1 \\ 0 & 0 \end{smallmatrix} \end{array} \right) & \#E\text{'s} &= 2 \cdot \frac{1}{2}N(N-1). \\ F_{JK} &= \left(\begin{array}{c|c} & \begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix} \\ \hline & \end{array} \right); & F_{KJ} &= \left(\begin{array}{c|c} & \\ \hline \begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix} \end{array} \right) & \#F\text{'s} &= 2 \cdot \frac{1}{2}N(N-1). \\ G_K^+ &= \left(\begin{array}{c|c} & \begin{smallmatrix} 0 & 1 \\ & 0 \end{smallmatrix} \\ \hline & \end{array} \right); & G_K^- &= \left(\begin{array}{c|c} & \\ \hline \begin{smallmatrix} 0 & 1 \\ & 0 \end{smallmatrix} \end{array} \right) & \#G\text{'s} &= 2 \cdot N. \end{aligned} \quad (19.14)$$

With the N Cartan generators, this is indeed the total number of $\frac{1}{2}(2N)(2N+1)$ generators of $USp(2N)$ and $Sp(2N, R)$.

We can now read off the roots from the general formula $[H_J, E_\alpha] = \alpha_J E_\alpha$ and idem for F_α and G_α . There are ‘‘E-roots’’ which are essentially the roots of $SU(N)$, and ‘‘F-roots’’ and ‘‘G-roots’’ which are new. Denoting the weights of the defining representation of $SU(N)$ by $\vec{\mu}_{\text{def}}^j$ we get

$$\begin{aligned} \pm \vec{\alpha}^{JK}(E) &= \pm(\vec{\nu}^J - \vec{\nu}^K) = \pm(\vec{\mu}^J - \vec{\mu}^K, 0) \text{ with } 1 \leq J < K \leq N: 2 \cdot \frac{1}{2}N(N-1) \text{ roots.} \\ \pm \vec{\alpha}^{JK}(F) &= \pm(\vec{\nu}^J + \vec{\nu}^K) = \pm\left(\vec{\mu}^J + \vec{\mu}^K, \sqrt{\frac{2}{N}}\right) \text{ with } 1 \leq J < K \leq N: 2 \cdot \frac{1}{2}N(N-1) \text{ roots.} \\ \pm \vec{\alpha}^{K\pm}(G) &= \pm 2\vec{\nu}^K = \pm\left(2\vec{\mu}^K, \sqrt{\frac{2}{N}}\right) \text{ with } 1 \leq K \leq N: 2 \cdot N \text{ roots.} \end{aligned} \quad (19.15)$$

The E -generators are block-diagonal and they commute with H_N , but the F -generators and G -generators are block off-diagonal and do not commute with H_N . In total we get the correct number of $2N^2$ roots.

The positive roots are

$$\left(\vec{\mu}^J - \vec{\mu}^K, 0\right) \text{ for } 1 \leq J < K \leq N; \quad \left(\vec{\mu}^J + \vec{\mu}^K, \sqrt{\frac{2}{N}}\right) \text{ for } 1 \leq J < K \leq N. \quad (19.16)$$

(The case $J = K$ accounts for the generators G_K .) The simple roots are then essentially the $N-1$ simple roots of $SU(N)$ plus an extra simple root

$$\vec{\alpha}^I = (\vec{\mu}^1 - \vec{\mu}^2, 0); \quad \vec{\alpha}^{II} = (\vec{\mu}^2 - \vec{\mu}^3, 0); \quad \dots; \quad \vec{\alpha}^{N-1} = (\vec{\mu}^{N-1} - \vec{\mu}^N, 0); \quad \vec{\alpha}^N = \left(2\vec{\mu}^N, \sqrt{\frac{2}{N}}\right). \quad (19.17)$$

The simple root $\vec{\alpha}^N$ is the least positive G -root; other positive G -roots are not simple, for example $\left(\vec{\mu}^1 + \vec{\mu}^1, \sqrt{\frac{2}{N}}\right) = (\vec{\mu}^1 - \vec{\mu}^2, 0) + \left(\vec{\mu}^1 + \vec{\mu}^2, \sqrt{\frac{2}{N}}\right)$.

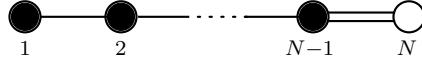
To construct the Dynkin diagram we need the inner products of the N simple roots. Using the metric $g_{\text{def}}^{IJ}(Sp(2N, R)) = \delta^{IJ}$ instead of $g_{\text{def}}^{IJ}(SL(N, R)) = 2\delta^{IJ}$ which we used for $SU(N)$, the inner products of $\vec{\mu}^j$'s are a factor 2 smaller than we found in $SU(N)$

$$\left(\vec{\mu}^j \cdot \vec{\mu}^j\right) = \frac{N-1}{2N}; \quad \vec{\mu}^j \cdot \vec{\mu}^k = -\frac{1}{2N} \text{ for } j \neq k. \quad (19.18)$$

Then we get for $Sp(2N, R)$

$$\begin{aligned} \vec{\alpha}^1 \cdot \vec{\alpha}^2 &= -\frac{1}{2}, & \vec{\alpha}^1 \cdot \vec{\alpha}^3 &= 0, \dots, & \vec{\alpha}^{N-1} \cdot \vec{\alpha}^N &= -1 \\ \vec{\alpha}^j \cdot \vec{\alpha}^j &= 1 \text{ for } 1 \leq j < N; & \vec{\alpha}^N \cdot \vec{\alpha}^N &= 2. \end{aligned} \quad (19.19)$$

This yields the following Dynkin Diagram



where the solid circles correspond to shorter roots.

To construct the fundamental weights we first write down an explicit expression for the simple roots. The first $N-1$ simple roots are essentially (*i.e.* up to the last entry which is zero) those of $SU(N)$

$$\begin{aligned} \text{The simple roots of } SU(N) \text{ with an extra zero} & \left\{ \begin{array}{l} \vec{\alpha}^1 = \vec{\nu}^1 - \vec{\nu}^2 = (1, 0, \dots, 0) \\ \vec{\alpha}^2 = \vec{\nu}^2 - \vec{\nu}^3 = \left(-\frac{1}{2}, \frac{3}{\sqrt{12}}, 0, \dots, 0\right) \\ \vec{\alpha}^3 = \vec{\nu}^3 - \vec{\nu}^4 = \left(0, -\frac{2}{\sqrt{12}}, \frac{4}{\sqrt{24}}, 0, \dots, 0\right) \\ \vdots \\ \vec{\alpha}_{N-1} = \vec{\nu}^{N-1} - \vec{\nu}^N = \left(0, \dots, -\frac{N-2}{\sqrt{2(N-2)(N-1)}}, \frac{N}{\sqrt{2(N-1)N}}, 0\right) \\ \vec{\alpha}_N = 2\vec{\nu}^N = \left(2\vec{\mu}^N, \sqrt{\frac{2}{N}}\right) = \left(0, \dots, 0, -\sqrt{\frac{2(N-1)}{N}}, \sqrt{\frac{2}{N}}\right). \end{array} \right. \quad (19.20) \end{aligned}$$

Using that the $N-1$ fundamental weights of $SU(N)$ are given by

$$\vec{\mu}_{FW}^J(SU(N)) = \sum_{k=1}^J \vec{\mu}_{def}^k \quad (J = 1, \dots, N-1) \quad (19.21)$$

we can construct the fundamental weights of $USp(2N)$. By looking at the explicit expressions for the weights $\vec{\nu}^J$ of the defining irrep of $USp(2N)$, we see that the first $N-1$ of them are orthogonal

to $\vec{\alpha}^N$. Thus the first $N - 1$ fundamental weights of $USp(2N)$ are essentially those of $SU(N)$

$$\vec{\mu}_{FW}^J(USp(2N)) = \sum_{k=1}^J \vec{\nu}^k = \left(\vec{\mu}_{FW}^J(SU(N)), J \frac{1}{\sqrt{2N}} \right) \text{ for } 1 \leq J \leq N - 1. \quad (19.22)$$

The inner product of these $\vec{\mu}_{FW}^J(USp(2N))$ for $J \leq N - 1$ with the first $N - 1$ $\vec{\alpha}$'s is of course correct because it is the same as for $SU(N)$. But also the inner product with the simple root $\vec{\alpha}^N$ is correct: it vanishes because the last two entries of $\sum_{k=1}^J \vec{\nu}^k = \left(\sum_{k=1}^J \vec{\mu}^k, J \frac{1}{\sqrt{2N}} \right)$ are $(*, *, \dots, *, J \frac{1}{\sqrt{2(N-1)N}}, J \frac{1}{\sqrt{2N}})$.

Of the last one, $\vec{\mu}_{FW}^N(USp(2N))$, clearly only the last entry is nonvanishing because $\sum_{k=1}^N \vec{\mu}_{\text{def}}^k = 0$

$$\vec{\mu}_{FW}^N(USp(2N)) = \left(0, \dots, \sqrt{\frac{N}{2}} \right). \quad (19.23)$$

It is clearly orthogonal to the first $N - 1$ $\vec{\alpha}_J$ in (19.20), and also $\vec{\alpha}^N$. $\vec{\mu}_{FW}^N$ has the correct value 1.

Hence the N fundamental weights of $USp(2N)$ are given by

$$\vec{\mu}_{FW}^J(USp(2N)) = \left(\sum_{j=1}^J \vec{\mu}_{\text{def}}^j, J \frac{1}{\sqrt{2N}} \right) \text{ with } 1 \leq J \leq N. \quad (19.24)$$

This formula is also correct for $J = N$ because $\sum_{j=1}^N \vec{\mu}_{\text{def}}^j = 0$. The tensors which correspond to the N fundamental weights are antisymmetric tensors, for the same reasons as for $SU(N)$. The highest root follows from (19.16) and is $\left(2\vec{\mu}_{\text{def}}^1, \sqrt{\frac{2}{N}} \right)$, hence the adjoint representation corresponds to a symmetric tensor $t^{\alpha\beta} = t^{\beta\alpha}$.

19.2 $Sp(4, R)$ ($N = 2$)

As an example we work out the case $Sp(4, R)$. There are 10 generators. In the defining (vector) representation all matrices are 4×4 . The 2 Cartan generators are

$$H_1 = \frac{1}{2} \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & 1 \end{pmatrix}; \quad H_2 = \frac{1}{2} \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}. \quad (19.25)$$

So the weights of the defining irrep **4** are

$$\begin{aligned} \vec{\nu}^1 &= \left(\frac{1}{2}, \frac{1}{2} \right) && \leftarrow \text{highest weight} \\ \vec{\nu}^2 &= \left(-\frac{1}{2}, \frac{1}{2} \right) \\ \vec{\nu}^3 &= \left(-\frac{1}{2}, -\frac{1}{2} \right) \\ \vec{\nu}^4 &= \left(\frac{1}{2}, -\frac{1}{2} \right). \end{aligned} \quad (19.26)$$

Of course, $\vec{\nu}^3 = -\vec{\nu}^1$ and $\vec{\nu}^4 = -\vec{\nu}^2$.

The 4 raising and 4 lowering generators are (we only display nonvanishing entries)

$$\begin{aligned} E_{12} &= \left(\begin{array}{c|c} \begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix} & \\ \hline \begin{smallmatrix} 0 & 0 \\ -1 & 0 \end{smallmatrix} \end{array} \right); & E_{21} &= \left(\begin{array}{c|c} \begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix} & \\ \hline \begin{smallmatrix} 0 & -1 \\ 0 & 0 \end{smallmatrix} \end{array} \right); \\ G_1^+ &= \left(\begin{array}{c|c} & \begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix} \\ \hline & \end{array} \right); & G_2^+ &= \left(\begin{array}{c|c} & \begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix} \\ \hline & \end{array} \right); & F_{12} &= \left(\begin{array}{c|c} & \begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix} \\ \hline & \end{array} \right); \\ G_1^- &= \left(\begin{array}{c|c} & \\ \hline \begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix} \end{array} \right); & G_2^- &= \left(\begin{array}{c|c} & \\ \hline \begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix} \end{array} \right); & F_{21} &= \left(\begin{array}{c|c} & \\ \hline \begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix} \end{array} \right). \end{aligned} \quad (19.27)$$

The 8 roots are $\pm(\vec{\nu}^1 - \vec{\nu}^2)$ and $\pm(\vec{\nu}^i + \vec{\nu}^j)$ with $1 \leq i, j \leq 2$ and $\vec{\nu}^1 = (\frac{1}{2}, \frac{1}{2})$ while $\vec{\nu}^2 = (-\frac{1}{2}, \frac{1}{2})$.

$$\begin{aligned} \vec{\alpha}(E_{12}) &= (1, 0); & \vec{\alpha}(E_{21}) &= (-1, 0); \\ \vec{\alpha}(G_1^+) &= (1, 1); & \vec{\alpha}(G_2^+) &= (-1, 1); & \vec{\alpha}(F_{12}) &= (0, 1); \\ \vec{\alpha}(G_1^-) &= (-1, -1); & \vec{\alpha}(G_2^-) &= (1, -1); & \vec{\alpha}(F_{21}) &= (0, -1). \end{aligned} \quad (19.28)$$

The 4 positive roots are $(1, 0); (1, 1); (-1, 1); (0, 1)$. The 2 simple roots are

$$\vec{\alpha}^I = (1, 0), \quad \vec{\alpha}^{II} = (-1, 1). \quad (19.29)$$

The positive root $(0, 1)$ is not simple because $(0, 1) = \vec{\alpha}^I + \vec{\alpha}^{II}$. Also $(1, 1)$ is not simple because $(1, 1) = (1, 0) + (0, 1) = 2\vec{\alpha}^I + \vec{\alpha}^{II}$. (Simple means: not the sum of **two** positive roots.)

The two fundamental weights are

$$\vec{\mu}_{FW}^I = \left(\frac{1}{2}, \frac{1}{2}\right), \quad \vec{\mu}_{FW}^{II} = (0, 1). \quad (19.30)$$

Clearly the first fundamental weight is the highest weight of the defining rep, see (19.26). We now consider other reps.

Weights of $USp(4)$. If we look at the 8 roots in (19.28), the one with the H.W. is $\vec{\alpha} = (1, 1)$. This is equal to $2\vec{\mu}_{FW}^I$, so the adjoint rep correspond to a symmetric tensor $t^{\alpha\beta} = t^{\beta\alpha}$ with $\alpha, \beta = 1, \dots, 4$. Consider next antisymmetric tensors. The 5 components of the antisymmetric symplectic-traceless tensor $t^{\alpha\beta} - \frac{1}{4}\Omega^{\alpha\beta}(\Omega_{\gamma\delta}t^{\gamma\delta})$, with their weights written below them, are

$$\begin{array}{ccccc} t^{12} & \frac{1}{2}(t^{13} - t^{24}) & t^{14} & t^{23} & t^{34} \\ (0,1) & (0,0) & (1,0) & (-1,0) & (0,-1) \end{array} \quad (19.31)$$

Clearly the highest weight of this irrep is $(0, 1)$ which is the second fundamental weight.

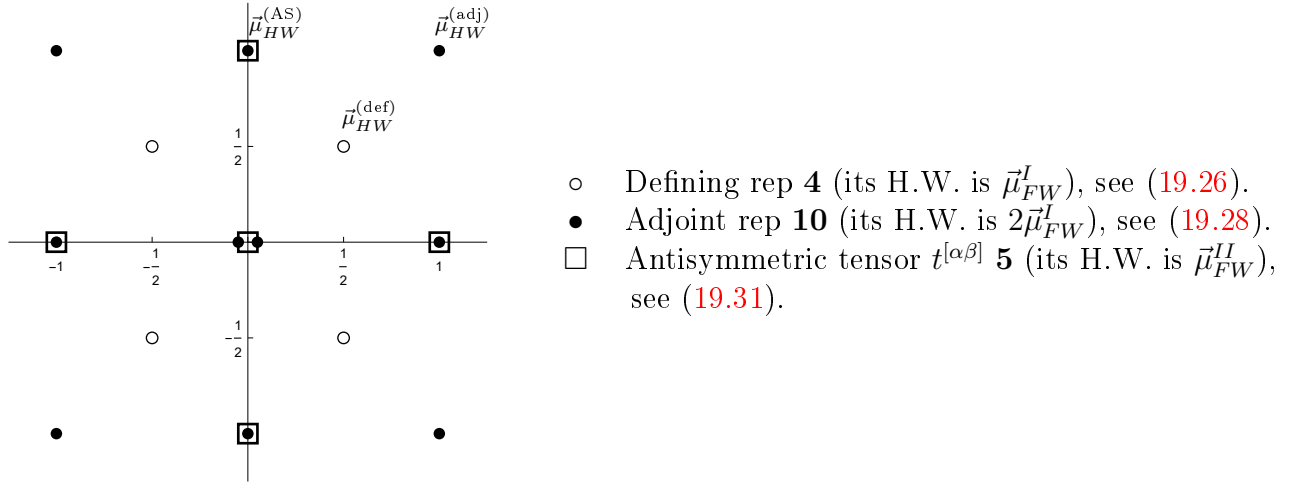


Figure 19.1: Weight diagrams of the **4**, **5** and **10** irreps of $USp(4)$.

Young tableaux. The first few irreps of $USp(4)$ have the following Young tableaux

$$\square, \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array}, \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \quad (19.32)$$

4, 5, 10

In simple cases one can deduce the Clebsch-Gordan decomposition of products of irreps by matching dimensions. For $SU(N)$ we have presented the general rules, but we are not familiar with similar rules for $USp(2N)$. A good strategy is to start with the corresponding results for $SU(N)$ and then to use the invariant tensors to subtract sub-reps, and finally match the dimensions of the irreps on the left- and right-hand side of a product of two irreps. Furthermore, for $USp(4)$ we can use the isomorphism $USp(4) \simeq SO(5)$ to check the results obtained for $USp(4)$. Here are some examples for $USp(4)$. Note that the Young tableaux for $USp(2N)$ have at most N boxes in a column, just like $SO(2N)$ and $SO(2N+1)$.

$$\begin{array}{rcl} \square & \otimes & \square = \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \oplus \bullet \\ \mathbf{4} & \times & \mathbf{4} = \mathbf{10} + \mathbf{5} + \mathbf{1} \end{array}$$

$$\begin{array}{rcl} \square & \otimes & \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} = \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \oplus \square \\ \mathbf{4} & \times & \mathbf{5} = \mathbf{16} + \mathbf{4} \end{array}$$

$$\begin{array}{rcl} \square & \otimes & \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \oplus \square \\ \mathbf{4} & \times & \mathbf{10} = \mathbf{20} + \mathbf{16} + \mathbf{4} \end{array}$$

We used $\Omega_{\alpha\beta}$ to subtract $t^{\alpha\beta}\Omega_{\alpha\beta} = \mathbf{1}$ and $t^{\alpha\beta}{}_{\gamma}\Omega_{\alpha\beta} = \mathbf{4}$. We can check this last result by writing the corresponding decomposition for $SO(5)$

$$SO(5): \quad \begin{array}{rcl} \times & \otimes & \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} = \begin{array}{|c|} \hline \times \\ \hline \times \\ \hline \end{array} \oplus \begin{array}{|c|} \hline \times \\ \hline \end{array} \oplus \times \\ \mathbf{4} & \times & \mathbf{10} = \mathbf{20} + \mathbf{16} + \mathbf{4} \end{array} \quad (19.33)$$

The **4** is the spinor irrep of $SO(5)$, and the **10** is an antisymmetric tensor. The dimensions of these spinorial irreps of $SO(5)$ can be obtained from chapter 20, but in this case we can see directly how the spinor-tensor $t_{\mu\nu}^\alpha = -t_{\nu\mu}^\alpha$ decomposes: it contains $\gamma^\mu\gamma^\nu t_{\mu\nu}^\alpha = \mathbf{4}$ and $\gamma^\mu t_{\mu\nu}^\alpha = t_\nu^\alpha = \mathbf{20} - \mathbf{4} = \mathbf{16}$.

We give some further decompositions of $USp(4)$, **deduced** by matching dimensions. As a check, one can look up the dimensions of irreps of $USp(2N)$ also given in chapter 20.

$$\begin{array}{ccccccc}
\begin{array}{|c|} \hline \square \\ \hline \end{array} & \otimes & \begin{array}{|c|} \hline \square \\ \hline \end{array} & = & \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} & \oplus & \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} & \oplus & \bullet \\
\mathbf{5} & \times & \mathbf{5} & = & \mathbf{14} & + & \mathbf{10} & + & \mathbf{1} \\
\\
\begin{array}{|c|} \hline \square \\ \hline \end{array} & \otimes & \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} & = & \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} & \oplus & \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} & \oplus & \begin{array}{|c|} \hline \square \\ \hline \end{array} \\
\mathbf{5} & \times & \mathbf{10} & = & \mathbf{35'} & + & \mathbf{10} & + & \mathbf{5} \\
\\
\begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} & \otimes & \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} & = & \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \end{array} & \oplus & \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} & \oplus & \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} & \oplus & \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} & \oplus & \begin{array}{|c|} \hline \square \\ \hline \end{array} & \oplus & \bullet \\
\mathbf{10} & \times & \mathbf{10} & = & \mathbf{35} & + & \mathbf{35'} & + & \mathbf{14} & + & \mathbf{10} & + & \mathbf{5} & + & \mathbf{1}
\end{array}$$

In $SU(4)$ $\begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array}$ is **45**, but in $USp(4)$ it decomposes into **35' + 10**. Likewise, the $\begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} = \mathbf{20}$ of $SU(4)$ decomposes into **14 + 5 + 1**. One can again check all these results for $USp(4)$ by comparing with the corresponding results for $SO(5)$.

19.3 Reality properties of $USp(2N)$ irreps

The vector (defining) irrep of $USp(4)$ transforms as: $\delta v^\alpha = M^\alpha_\beta v^\beta$ with $M = A_r \otimes I + \vec{S} \otimes i\vec{\tau}$ (see (19.4)). The complex-conjugate irrep can be written as follows

$$\begin{aligned}
M^* &= A_r \otimes I - S_1 \otimes i\tau_1 + S_2 \otimes i\tau_2 - S_3 \otimes i\tau_3 \\
&= (I \otimes \tau_2) (A_r \otimes I + S_1 \otimes i\tau_1 + S_2 \otimes i\tau_2 + S_3 \otimes i\tau_3) (I \otimes \tau_2) \\
&= CMC^{-1} \text{ with } C = I \otimes \tau_2.
\end{aligned} \tag{19.34}$$

So M is pseudoreal because $I \otimes \tau_2$ is antisymmetric. For example $USp(2)$ has generators $(i\tau_1, i\tau_2, i\tau_3)$, see (19.4), and $(i\tau_1 + i\tau_2 + i\tau_3)^* \tau_2 = i\tau_2 (i\tau_1 + i\tau_2 + i\tau_3) i\tau_2$. In fact $USp(2) \simeq SU(2)$ so also the **2** of $SU(2)$ is pseudoreal. We shall simplify the notation and write τ_2 instead of $I \otimes \tau_2$.

Theorem: $\tau_2(v^\alpha)^*$ transforms like v^α .

Proof: $\delta \tau_2(v^\alpha)^* = \tau_2(M^\alpha_\beta v^\beta)^* = \tau_2(M^\alpha_\beta)^* (\tau_2 v^{*\beta}) = M^\alpha_\beta (\tau_2 v^*)$. So, just as for $SO(N)$ but unlike $SU(N)$, the defining representation and its conjugate (with an upper index and a lower index) are equivalent.

Theorem: Direct products with an even (odd) number of vector irreps are real (pseudoreal).

Proof: The direct product of $\tau_2 \otimes \tau_2$ (or an even number of τ_2 's) is symmetric, hence these reps

(and irreps) are real.

Comment 1. The symmetric tensor $t^{\alpha\beta} = t^{\beta\alpha}$ of $Sp(4)$ has 10 states, and we identified it with the adjoint rep because it also has 10 states. But the matrices M in the Cartan-Weyl basis were not symmetric matrices

$$(M)^\gamma_\delta = \begin{pmatrix} A_r + S_3 & S_1 + S_2 \\ S_1 - S_2 & A_r - S_3 \end{pmatrix}^\gamma_\delta \text{ is not symmetric in } \gamma\delta. \quad (19.35)$$

However, we could obtain the adjoint rep by acting on states $|M_{\alpha\beta}\rangle$ as $\delta|M_{\alpha\beta}\rangle = [[\lambda^{\gamma\delta} M_{\gamma\delta}, M_{\alpha\beta}]]\rangle$, but $M_{\alpha\beta}$ is not symmetric. However, the matrices $M^\gamma_\delta \Omega^{\beta\delta} \equiv M^{\gamma\beta}$ are symmetric (and real), so $M^{\gamma\beta}$ corresponds to $t^{\alpha\beta} = t^{\beta\alpha}$ above

$$M\Omega = \begin{pmatrix} S_1 + S_2 & -A_r - S_3 \\ A_r - S_3 & -S_1 + S_2 \end{pmatrix} \text{ is symmetric.} \quad (19.36)$$

(We used $\Omega^{\alpha\beta} = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$ although this is not essential for the symmetry.)

Comment 2. Given a highest weight of some representation, we can use the lowering operators of the simple roots and/or Weyl reflections to construct the whole irrep. Often it is enough to know its dimension, and that one most easily gets from tensor methods.

19.4 The Cartan-Weyl basis and other real forms for $USp(2N)$

We finally prove that the Cartan-Weyl (CW) basis of $USp(2N)$ is $Sp(2N, R)$. As in the cases of $SU(N)$, $SO(2N)$ and $SO(2N+1)$ we begin by listing all real forms of $USp(2N)$, and then we shall identify the CW basis.

There are only 3 real forms for $USp(2N)$

- 1) $USp(2N)$ itself. Introducing again D = the number of noncompact generators minus the number of compact generators, we find from (19.4) $D = -N(2N+1)$.
- 2) $Sp(2N, R)$, the real symplectic matrices. We have discussed this group above. In this case $D = N$, as we derived below (19.5).
- 3) $USp(2p, 2q)$ with $2p+2q=2N$. This group leaves invariant the two forms $x^T \Omega y$ and $x^\dagger H y$ where $H = \begin{pmatrix} I_p & & \\ & -I_q & \\ & & I_p & \\ & & & -I_q \end{pmatrix}$ with I_p and I_q the $p \times p$ and $q \times q$ unit matrices, respectively. So it is a subgroup of $Sp(2N, C)$ but it is the intersection of $Sp(2N, C)$ with $SU(2p, 2q)$ with

$p + q = N$ instead of $SU(N)$. Assuming block form, the generators are

$$M = \begin{pmatrix} A_1 & Z_{12} & S_1 & Z_{14} \\ Z_{12}^\dagger & A_2 & Z_{14}^T & S_2 \\ -S_1^\dagger & Z_{14}^* & -A_1^T & -Z_{12}^* \\ Z_{14}^\dagger & -S_2^\dagger & -Z_{12}^T & -A_2^T \end{pmatrix} \quad (19.37)$$

where A_1, A_2 are antihermitian, S_1, S_2 complex symmetric, and Z_{12}, Z_{14} complex. One finds $D = -2(p - q)^2$.

Clearly, the CW basis is $Sp(2N, R)$ since it is the only real form for which D is equal to the rank N .

Bibliography

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Chapter 20

Dimensions of irreps of the classical groups

20.1 $SU(N)$

20.2 $SO(2N)$

The irreducible representations (irreps) of $SO(2N)$ for tensors are denoted by Young tableaux as usual, but the spinor irreps are denoted by adding a dot in each box. For example

$$\begin{array}{cc} \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} & \begin{array}{|c|} \hline \bullet \\ \hline \bullet \\ \hline \end{array} \\ t^{\mu\nu} = -t^{\nu\mu} & t_A^{\mu\nu} = -t_A^{\nu\mu} \end{array}$$

where the spinor index runs from $A = 1$ to $A = 2^{N-1}$.

We label the rows for $SO(2N)$ from bottom to top by the row number r_i where $r_i = 0, \dots, N-1$. The number of boxes in each row is λ_i , with $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N$ from bottom to top. Then we add the λ_i to r_i for tensor irreps, or $\lambda_i + \frac{1}{2}$ to r_i for spinor irreps. (The dots in the Young tableaux corresponds to these $\frac{1}{2}$.) This gives a second sequence of row numbers

$$\begin{aligned} R_i &= (0 + \lambda_1, \dots, N - 1 + \lambda_N) \text{ for tensor irreps.} \\ &= (0 + \lambda_1 + \tfrac{1}{2}, \dots, N - 1 + \lambda_N + \tfrac{1}{2}) \text{ for spinor irreps.} \end{aligned}$$

The degeneracy formula (the dimension of the representation) is then

$$\dim = \frac{\prod \text{Sums} \prod \text{Differences}}{\prod \text{sums} \prod \text{differences}}. \quad (20.1)$$

where Sums (Differences) are all $R_j + R_i$ ($R_j - R_i$), and sums (differences) are all $r_j + r_i$ ($r_j - r_i$), with always $j > i$.

We now give some examples for $SO(6)$. Of course, $SO(6)$ is isomorphic to $SU(4)$, so one can also get these results from the “factors over hooks” rule for unitary groups, and this yields a nice

check.

Antisymmetric tensor:

$$\begin{array}{rclcl}
 r_i & +\lambda_i = R_i & & & \\
 2 & \begin{array}{|c|} \hline \square \\ \hline \end{array} +1 = 3 & \text{Sums} = (5, 3, 2) & \text{Differences} = (1, 3, 2) & \\
 1 & \begin{array}{|c|} \hline \square \\ \hline \end{array} +1 = 2 & \text{sums} = (3, 2, 1) & \text{differences} = (1, 2, 1) & \\
 0 & +0 = 0 & & & \\
 \dim \left(\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \right) = \frac{(5 \cdot 3 \cdot 2) \cdot (1 \cdot 3 \cdot 2)}{(3 \cdot 2 \cdot 1) \cdot (1 \cdot 2 \cdot 1)} = \mathbf{15}.
 \end{array}$$

Indeed, an antisymmetric tensor $t^{\mu\nu} = -t^{\nu\mu}$ has $\binom{6}{2} = 15$ states. It is the adjoint representation of $SO(6)$, and corresponds to $\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \\ \hline \square & \\ \hline \end{array}$ in $SU(4)$ which is the adjoint representation of $SU(4)$.

Antisymmetric tensor-spinor:

$$\begin{array}{rclcl}
 r_i & +\lambda_i = R_i & & & \\
 2 & \begin{array}{|c|} \hline \bullet \\ \hline \end{array} +1 = 3 + \frac{1}{2} & \text{Sums} = (6, 4, 3) & \text{Differences} = (1, 3, 2) & \\
 1 & \begin{array}{|c|} \hline \bullet \\ \hline \end{array} +1 = 2 + \frac{1}{2} & \text{sums} = (3, 2, 1) & \text{differences} = (1, 2, 1) & \\
 0 & +0 = 0 + \frac{1}{2} & & & \\
 \dim \left(\begin{array}{|c|} \hline \bullet \\ \hline \square \\ \hline \end{array} \right) = \frac{(6 \cdot 4 \cdot 3) \cdot (1 \cdot 3 \cdot 2)}{(3 \cdot 2 \cdot 1) \cdot (1 \cdot 2 \cdot 1)} = \mathbf{36}.
 \end{array}$$

Indeed, the tensor-spinor $t_A^{\mu\nu} = -t_A^{\nu\mu}$ has $\binom{6}{2} \cdot 4 = 60$ states, but the gamma-traceless irreducibility condition $\gamma_\mu t_A^{\mu\nu} = 0$ subtracts $6 \times 4 = 24$ states, leaving 36 states. In $SU(4)$, we obtain this irrep as follows

$$\begin{array}{ccccccc}
 \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \\ \hline \square & \\ \hline \end{array} & \otimes & \begin{array}{|c|} \hline \square \\ \hline \end{array} & = & \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & & \\ \hline \square & & \\ \hline \end{array} & \oplus & \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \square & \\ \hline \end{array} & \oplus & \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \\ \hline \square & \\ \hline \square & \\ \hline \end{array} \\
 \mathbf{15} & \times & \mathbf{4} & = & \mathbf{36} & + & \mathbf{20} & + & \mathbf{4}
 \end{array}$$

Symmetric tensor:

$$\begin{array}{rclcl}
r_i & & +\lambda_i = R_i & & \\
2 & \begin{array}{|c|c|} \hline & \\ \hline \end{array} & +2 = 4 & \text{Sums} = (5, 4, 1) & \text{Differences} = (3, 4, 1) \\
1 & & +0 = 1 & \text{sums} = (3, 2, 1) & \text{differences} = (1, 2, 1) \\
0 & & +0 = 0 & & \\
\dim(\begin{array}{|c|c|} \hline & \\ \hline \end{array}) & = & \frac{(5 \cdot 4 \cdot 1) \cdot (3 \cdot 4 \cdot 1)}{(3 \cdot 2 \cdot 1) \cdot (1 \cdot 2 \cdot 1)} = \mathbf{20}.
\end{array}$$

Indeed, a general symmetric tensor $t^{\mu\nu}$ has $\frac{1}{2} \times 6 \times 7 = 21$ components, but the trace removes 1 component.

Symmetric tensor-spinor:

$$\begin{array}{rclcl}
r_i & & +\lambda_i = R_i & & \\
2 & \begin{array}{|c|c|} \hline \bullet & \bullet \\ \hline \end{array} & +2 = 4 + \frac{1}{2} & \text{Sums} = (6, 5, 2) & \text{Differences} = (3, 4, 1) \\
1 & & +0 = 1 + \frac{1}{2} & \text{sums} = (3, 2, 1) & \text{differences} = (1, 2, 1) \\
0 & & +0 = 0 + \frac{1}{2} & & \\
\dim(\begin{array}{|c|c|} \hline \bullet & \bullet \\ \hline \end{array}) & = & \frac{(6 \cdot 5 \cdot 2) \cdot (3 \cdot 4 \cdot 1)}{(3 \cdot 2 \cdot 1) \cdot (1 \cdot 2 \cdot 1)} = \mathbf{60}.
\end{array}$$

A general symmetric tensor $t^{\mu\nu}$ with a spinor index has $21 \times 4 = 84$ components but gamma-tracelessness removes $6 \times 4 = 24$ components, leaving 60 components.

Another symmetric tensor:

$$\begin{array}{rclcl}
r_i & & +\lambda_i = R_i & & \\
2 & \begin{array}{|c|c|c|} \hline & & \\ \hline \end{array} & +3 = 5 & \text{Sums} = (6, 5, 1) & \text{Differences} = (4, 5, 1) \\
1 & & +0 = 1 & \text{sums} = (3, 2, 1) & \text{differences} = (1, 2, 1) \\
0 & & +0 = 0 & & \\
\dim(\begin{array}{|c|c|c|} \hline & & \\ \hline \end{array}) & = & \frac{(6 \cdot 5 \cdot 1) \cdot (4 \cdot 5 \cdot 1)}{(3 \cdot 2 \cdot 1) \cdot (1 \cdot 2 \cdot 1)} = \mathbf{50}.
\end{array}$$

Indeed, a symmetric tensor $t^{\mu\nu\rho}$ has $\binom{n+k-1}{k} = \binom{6+3-1}{3} = 56$ components, but one must subtract 6 for the trace.

Tensor with mixed symmetry:

$$\begin{array}{rclcl}
 r_i & +\lambda_i = R_i & & & \\
 2 & \begin{array}{|c|c|} \hline & \\ \hline \end{array} & +2 = 4 & \text{Sums} = (6, 4, 2) & \text{Differences} = (2, 4, 2) \\
 1 & \begin{array}{|c|} \hline \\ \hline \end{array} & +1 = 2 & \text{sums} = (3, 2, 1) & \text{differences} = (1, 2, 1) \\
 0 & & +0 = 0 & & \\
 & \dim \left(\begin{array}{|c|c|} \hline & \\ \hline \end{array} \right) = \frac{(6 \cdot 4 \cdot 2) \cdot (2 \cdot 4 \cdot 2)}{(3 \cdot 2 \cdot 1) \cdot (1 \cdot 2 \cdot 1)} = \mathbf{64}. & & &
 \end{array}$$

We can check (or prove) this by taking tensor products. In $SO(6)$ we find the **64** as follows

$$\begin{array}{ccccccc}
 \begin{array}{|c|c|} \hline & \\ \hline \end{array} & \otimes & \begin{array}{|c|} \hline \\ \hline \end{array} & = & \begin{array}{|c|c|c|} \hline & & \\ \hline \end{array} & \oplus & \begin{array}{|c|} \hline \\ \hline \end{array} & \oplus & \begin{array}{|c|c|} \hline & \\ \hline \end{array} \\
 \mathbf{20} & \times & \mathbf{6} & = & \mathbf{50} & + & \mathbf{6} & + & \mathbf{64}
 \end{array}$$

where we used that a totally symmetric tensor in $SO(6)$ with 3 indices has **56** components, but the trace removes **6**. We can also obtain the **64** in another way

$$\begin{array}{ccccccc}
 \begin{array}{|c|} \hline \\ \hline \end{array} & \otimes & \begin{array}{|c|} \hline \\ \hline \end{array} & = & \begin{array}{|c|} \hline \\ \hline \end{array} & \oplus & \begin{array}{|c|} \hline \\ \hline \end{array} & \oplus & \begin{array}{|c|c|} \hline & \\ \hline \end{array} \\
 \mathbf{15} & \times & \mathbf{6} & = & \mathbf{20} & + & \mathbf{6} & + & \mathbf{64}
 \end{array}$$

A more difficult example with one representation appearing twice in the expansion of a direct product is as follows

$$\begin{array}{ccccccc}
 \begin{array}{|c|} \hline \\ \hline \end{array} & \otimes & \begin{array}{|c|c|c|c|} \hline & & & \\ \hline \end{array} & = & \begin{array}{|c|c|c|c|c|} \hline & & & & \\ \hline \end{array} & \oplus & \begin{array}{|c|c|c|c|} \hline & & & \\ \hline \end{array} & \oplus & \begin{array}{|c|c|c|c|} \hline & & & \\ \hline \end{array} & \oplus & \begin{array}{|c|c|c|c|} \hline & & & \\ \hline \end{array} \\
 \mathbf{6} & \times & \mathbf{384} & = & \mathbf{735} & + & \mathbf{729} & + & \mathbf{280} & + & \mathbf{280} \\
 & & & & & \oplus & \begin{array}{|c|c|c|} \hline & & \\ \hline \end{array} & \oplus & \begin{array}{|c|c|c|c|} \hline & & & \\ \hline \end{array} & & \\
 & & & & & + & \mathbf{175} & + & \mathbf{105}
 \end{array}$$

$$\begin{array}{ccccccc}
 \begin{array}{|c|c|} \hline & \\ \hline \end{array} & \otimes & \begin{array}{|c|c|} \hline & \\ \hline \end{array} & = & \begin{array}{|c|c|c|c|} \hline & & & \\ \hline \end{array} & \oplus & \begin{array}{|c|c|c|} \hline & & \\ \hline \end{array} & \oplus & \begin{array}{|c|c|} \hline & \\ \hline \end{array} & \oplus & \begin{array}{|c|c|} \hline & \\ \hline \end{array} & \oplus & \begin{array}{|c|c|} \hline & \\ \hline \end{array} & \oplus & \begin{array}{|c|} \hline \\ \hline \end{array} & \oplus & \bullet \\
 \mathbf{20} & \times & \mathbf{20} & = & \mathbf{105} & + & \mathbf{175} & + & \mathbf{84} & + & \mathbf{20} & + & \mathbf{15} & + & \mathbf{1}
 \end{array}$$

Tensor-spinor with mixed symmetry:

$$\begin{array}{rcl}
 r_i & & +\lambda_i = R_i \\
 2 & \begin{array}{|c|c|} \hline \bullet & \bullet \\ \hline \end{array} & +2 = 4 + \frac{1}{2} \quad \text{Sums} = (7, 5, 3) \quad \text{Differences} = (2, 4, 2) \\
 1 & \begin{array}{|c|} \hline \bullet \\ \hline \end{array} & +1 = 2 + \frac{1}{2} \quad \text{sums} = (3, 2, 1) \quad \text{differences} = (1, 2, 1) \\
 0 & & +0 = 0 + \frac{1}{2} \\
 & & \dim \left(\begin{array}{|c|c|} \hline \bullet & \bullet \\ \hline \bullet & \bullet \\ \hline \end{array} \right) = \frac{(7 \cdot 5 \cdot 3) \cdot (2 \cdot 4 \cdot 2)}{(3 \cdot 2 \cdot 1) \cdot (1 \cdot 2 \cdot 1)} = \mathbf{140}.
 \end{array}$$

Since $t_A^{\mu\nu\rho}$ has $64 \times 4 = 256$ states, the condition of gamma-tracelessness subtracts 116 states.

Another tensor-spinor with mixed symmetry:

$$\begin{array}{rcl}
 r_i & & +\lambda_i = R_i \\
 2 & \begin{array}{|c|c|c|} \hline \bullet & \bullet & \bullet \\ \hline \end{array} & +3 = 5 + \frac{1}{2} \quad \text{Sums} = (8, 6, 3) \quad \text{Differences} = (3, 5, 2) \\
 1 & \begin{array}{|c|} \hline \bullet \\ \hline \end{array} & +1 = 2 + \frac{1}{2} \quad \text{sums} = (3, 2, 1) \quad \text{differences} = (1, 2, 1) \\
 0 & & +0 = 0 + \frac{1}{2} \\
 & & \dim \left(\begin{array}{|c|c|c|} \hline \bullet & \bullet & \bullet \\ \hline \bullet & \bullet & \bullet \\ \hline \end{array} \right) = \frac{(8 \cdot 6 \cdot 3) \cdot (3 \cdot 5 \cdot 2)}{(3 \cdot 2 \cdot 1) \cdot (1 \cdot 2 \cdot 1)} = \mathbf{360}.
 \end{array}$$

Another symmetric tensor-spinor:

$$\begin{array}{rcl}
 r_i & & +\lambda_i = R_i \\
 2 & \begin{array}{|c|c|c|} \hline \bullet & \bullet & \bullet \\ \hline \end{array} & +3 = 5 + \frac{1}{2} \quad \text{Sums} = (7, 6, 2) \quad \text{Differences} = (4, 5, 1) \\
 1 & & +0 = 1 + \frac{1}{2} \quad \text{sums} = (3, 2, 1) \quad \text{differences} = (1, 2, 1) \\
 0 & & +0 = 0 + \frac{1}{2} \\
 & & \dim \left(\begin{array}{|c|c|c|} \hline \bullet & \bullet & \bullet \\ \hline \bullet & \bullet & \bullet \\ \hline \bullet & \bullet & \bullet \\ \hline \end{array} \right) = \frac{(7 \cdot 6 \cdot 2) \cdot (4 \cdot 5 \cdot 1)}{(3 \cdot 2 \cdot 1) \cdot (1 \cdot 2 \cdot 1)} = \mathbf{140}.
 \end{array}$$

As a check we note that a general symmetric tensor-spinor $t_A^{\mu\nu\rho}$ has 56×4 components, but the condition of gamma-tracelessness removes the 21×4 components of $t_A^{\nu\rho}$, leaving $(56 - 21) \times 4 = 35 \times 4 = 140$ components. For a symmetric tensor gamma-tracelessness implies ordinary tracelessness, so we did not begin with a symmetric traceless tensor $t^{\mu\nu\rho}$ with 50 components.

20.3 $SO(2N+1)$

Again we discuss tensors first, then spinors, but the formula for the dimension of a Young tableau has now two extra factors, $\prod R_i$ in the numerator and $\prod r_i$ in the denominator. Recall that the r_i now run from $\frac{1}{2}$ to $\frac{2N-1}{2}$. The formula for the dimensions is

$$\dim = \frac{\prod R_i \prod \text{Sums} \prod \text{Differences}}{\prod r_i \prod \text{sums} \prod \text{differences}}. \quad (20.2)$$

We give the formulas for $SO(2N+1)$, but we check the formula for $SO(5)$.

Tensors:

$$\begin{array}{ccc}
 r_i & & +\lambda_i = R_i \\
 \frac{2N-1}{2} & \begin{array}{|c|c|} \hline & \\ \hline \end{array} & +\lambda_N = R_N \\
 \vdots & \begin{array}{|c|} \hline \\ \hline \end{array} & \vdots \\
 \frac{3}{2} & \begin{array}{|c|} \hline \\ \hline \end{array} & +\lambda_2 = R_2 \\
 \frac{1}{2} & & +\lambda_1 = R_1
 \end{array}
 \quad \lambda_i = \text{number of boxes in row } i.$$

Examples: $\square$ and $\begin{array}{|c|} \hline \square \\ \hline \end{array}$ in $SO(5)$ (of course $\dim \square = \mathbf{5}$ and $\dim \begin{array}{|c|} \hline \square \\ \hline \end{array} = \mathbf{10}$)

$$\begin{array}{ccc}
 \frac{3}{2} & \begin{array}{|c|} \hline \square \\ \hline \end{array} & +1 = \frac{5}{2} \\
 \frac{1}{2} & & +0 = \frac{1}{2}
 \end{array}
 \quad \dim(\square) = \frac{\frac{5}{2} \cdot \frac{1}{2} \cdot 3 \cdot 2}{\frac{3}{2} \cdot \frac{1}{2} \cdot 2 \cdot 1} = \mathbf{5}, \text{ correct.}$$

$$\begin{array}{ccc}
 \frac{3}{2} & \begin{array}{|c|} \hline \square \\ \hline \end{array} & +1 = \frac{5}{2} \\
 \frac{1}{2} & \begin{array}{|c|} \hline \square \\ \hline \end{array} & +1 = \frac{3}{2}
 \end{array}
 \quad \dim\left(\begin{array}{|c|} \hline \square \\ \hline \end{array}\right) = \frac{\frac{5}{2} \cdot \frac{3}{2} \cdot 4 \cdot 1}{\frac{3}{2} \cdot \frac{1}{2} \cdot 2 \cdot 1} = \mathbf{10}, \text{ which is correct.}$$

Spinors:

$$\begin{array}{ccc}
 r_i & & +\lambda_i + \frac{1}{2} = R_i \\
 \frac{2N-1}{2} & \begin{array}{|c|c|} \hline & \\ \hline \end{array} & +\lambda_N + \frac{1}{2} = \lambda_N + N \\
 \vdots & \begin{array}{|c|c|} \hline & \\ \hline \end{array} & \vdots \\
 & \begin{array}{|c|} \hline \\ \hline \end{array} & \\
 \frac{1}{2} & & +\lambda_1 + \frac{1}{2} = \lambda_1 + 1
 \end{array}$$

Examples: For spinor $\bullet$, we should get **4**. Using the rules we get

$$\begin{array}{rcl} \frac{3}{2} & \bullet + 0 + \frac{1}{2} & = 2 \\ \frac{1}{2} & + 0 + \frac{1}{2} & = 1 \end{array} \quad \dim(\bullet) = \frac{2 \cdot 1 \cdot 3 \cdot 1}{\frac{3}{2} \cdot \frac{1}{2} \cdot 2 \cdot 1} = \mathbf{4}, \text{ correct.}$$

For a vector-spinor $\square\bullet$, we get

$$\begin{array}{rcl} \frac{3}{2} & \square\bullet + 1 + \frac{1}{2} & = 3 \\ \frac{1}{2} & + 0 + \frac{1}{2} & = 1 \end{array} \quad \dim(\square\bullet) = \frac{3 \cdot 1 \cdot 4 \cdot 2}{\frac{3}{2} \cdot \frac{1}{2} \cdot 2 \cdot 1} = \mathbf{16}.$$

Indeed, a vector-spinor ψ_μ^α has $5 \times 4 = 20$ components but the “gamma-trace” $\gamma^\mu \psi_\mu^\alpha = \tilde{\psi}^\alpha$ has 4 components, so the irreducible rep has $20 - 4 = 16$ components.

20.4 $USp(2N)$

Number the rows r_i from 1 to N , starting below. λ_k is number of boxes in row k .

$R_k = k + \lambda_k$; Sums = $R_k + R_l$; Differences = $R_k - R_l$ for $k > l$.

$$\dim = \frac{\prod R_k \prod \text{Sums} \prod \text{Differences}}{1! 3! \cdots (2N-1)!}. \quad (20.3)$$

Example 1: $USp(2) (= SU(2))$.

$$\begin{array}{rcl} r_k & + \lambda_k = R_k & \dim(\square\square\square\square) = \frac{5}{1} = \mathbf{5}. \\ 1 & \square\square\square\square + 4 = 5 & \text{Indeed, spin } j = 2 \text{ of } SU(2). \end{array}$$

Example 2: $USp(4) (= SO(5))$.

Vector v^μ :

$$\begin{array}{rcl} 2 & \square + 1 = 3 & \dim(\square) = \frac{3 \cdot 1 \cdot 4 \cdot 2}{1! 3!} = \mathbf{4}, \text{ (spinor irrep of } SO(5)). \\ 1 & + 0 = 1 & \end{array}$$

Symmetric tensor $t^{\mu\nu} = t^{\nu\mu}$:

$$\begin{array}{rcl} 2 & \square\square + 2 = 4 & \dim(\square\square) = \frac{4 \cdot 1 \cdot 5 \cdot 3}{1! 3!} = \mathbf{10}, \text{ (} \square \text{ of } SO(5)). \\ 1 & + 0 = 1 & \end{array}$$

Antisymmetric Tensor $t^{\mu\nu} = -t^{\nu\mu}$ minus symplectic trace:

$$\begin{array}{rcl} 2 & \square + 1 = 3 & \dim(\square) = \frac{3 \cdot 2 \cdot 5 \cdot 1}{1! 3!} = \mathbf{5}, \text{ (} \square \text{ of } SO(5)). \\ 1 & \square + 1 = 2 & \end{array}$$

Example 3: $USp(6)$.

Antisymmetric symplectic-traceless tensor:

$$\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} \text{ has } \dim \left(\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} \right) = \mathbf{14} \text{ since } \frac{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 1 \cdot 1 \cdot 2}{1!3!5!} = 14.$$

Indeed, $t^{\mu\nu\rho} + \frac{1}{4} (\Omega^{\mu\nu} \Omega_{\alpha\beta} t^{\alpha\beta\rho} + 2 \text{ cyclic terms})$: $\binom{6}{3} - 6 = \mathbf{14}$.

Example 4: $USp(8)$.

$$\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} \text{ has } \dim \left(\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} \right) = \mathbf{42} \quad \left(\binom{8}{4} - \binom{8}{2} = 42 \right).$$

Chapter 21

$SU(N) \subset SO(2N)$

We have seen that the generators M of the defining representation of $SO(2N)$ in the compact (unitary) basis have the form

$$M^{def} = \left\{ \begin{pmatrix} A_s^{(1)} & 0 \\ 0 & A_s^{(2)} \end{pmatrix}, \begin{pmatrix} 0 & S \\ -S & 0 \end{pmatrix}, \begin{pmatrix} 0 & A_s^{(3)} \\ A_s^{(3)} & 0 \end{pmatrix} \right\} \quad (21.1)$$

where $A_s^{(j)}$ for $j = 1, 2, 3$ are real antisymmetric matrices, and S are real symmetric matrices. The total number of generators is $3 \times \frac{1}{2}N(N-1) + \frac{1}{2}N(N+1) = 2N^2 - N = \frac{1}{2}(2N)(2N-1)$ which is indeed the number of generators of $SO(2N)$. The generators of $SU(N)$ in the defining representation have the form $M^{def} = A_s + iS$ with traceless S . Their Lie algebra has the form

$$[A_s^{(1)} + iS^{(1)}, A_s^{(2)} + iS^{(2)}] = A_s^{(12)} + iS^{(12)} \quad (21.2)$$

where

$$\begin{aligned} A_s^{(12)} &= [A_s^{(1)}, A_s^{(2)}] - [S^{(1)}, S^{(2)}] = \text{antisymmetric} \\ S^{(12)} &= [A_s^{(1)}, S^{(2)}] - [A_s^{(2)}, S^{(1)}] = \text{symmetric, traceless} \end{aligned} \quad (21.3)$$

It is easy to find a subgroup $SU(N)$ inside the compact group $SO(2N)$: the generators $\begin{pmatrix} A_s & 0 \\ 0 & A_s \end{pmatrix}$ and $\begin{pmatrix} 0 & S \\ -S & 0 \end{pmatrix}$ satisfy the same Lie algebra as A_s and iS in (21.2). One can write them as $A_s \otimes I_2$ and $S \otimes i\tau^2$, and the matrices $i\tau^2$ play the role of i in $A + iS$. Adding the trace to S replaces $SU(N)$ by $U(N)$ in this discussion.

However, in many applications involving spinors, one is interested in $SO(2N)$ as “holonomy group” (under round-transport of a vector along a closed loop, it is rotated such that its length stays fixed. This gives a holonomy group $SO(2N)$ for vectors. Similarly one can round-transport a spinor, and obtain a corresponding rotation of the spinor, with the same holonomy group as for the vector). If one makes further suitable restrictions, only a subgroup $SU(N)$ or $U(N)$ remains. For these applications it is convenient to construct generators of an $SU(N)$ subgroup inside $SO(2N)$ from Dirac matrices. This construction proceeds as follows.

Let $\{\Gamma_i, \Gamma_j\} = 2\delta_{ij}$ with $i, j = 1, \dots, 2N$. Then $\frac{1}{2}\Gamma_{ij}$ are the generators of $SO(2N)$ in the reducible spinor rep. Construct N creation and N annihilation operators defined by

$$\begin{aligned} A_J^\dagger &= \frac{1}{2} (\Gamma_{2j-1} + i\Gamma_{2j}), \quad A^J = \frac{1}{2} (\Gamma_{2j-1} - i\Gamma_{2j}) \\ \{A^J, A_K^\dagger\} &= \delta_K^J; \quad \{A^J, A^K\} = \{A_J^\dagger, A_K^\dagger\} = 0. \end{aligned} \quad (21.4)$$

Given matrices $(T_a)^j_k$ of a Lie algebra with N -dimensional irrep, then $\hat{T}_a = A_J^\dagger (T_a)^J_K A^K$ are operators which satisfy the same Lie algebra as the matrices T_a . This is clear from the following straightforward calculation

$$\begin{aligned} [\hat{T}_a, \hat{T}_b] &= A_J^\dagger (T_a)^J_K \{A^K, A_L^\dagger\} (T_b)^L_M A^M - A_L^\dagger (T_b)^L_M (T_a)^J_K \{A_J^\dagger, A^M\} A^K \\ &= A_J^\dagger (T_a T_b)^J_M A^M - A_L^\dagger (T_b T_a)^L_K A^K \\ &= A_J^\dagger [T_a, T_b]^J_M A^M \\ &= f_{ab}^c A_J^\dagger (T_c)^J_M A^M = f_{ab}^c \hat{T}_c. \end{aligned} \quad (21.5)$$

If one takes for $(T_a)^J_K$ the matrices of $SU(N)$ in the defining representation in the Cartan-Weyl basis (either off-diagonal matrices with only $+1$ in the JK th entry, or diagonal matrices with 1 and -1 in diagonal entries), one obtains the operators $A_J^\dagger A^K$ for $J \neq K$ or $A_J^\dagger A^J - A_{J+1}^\dagger A^{J+1}$ (no sum over J). For the former we find

$$\begin{aligned} A_J^\dagger A^K &= \frac{1}{4} (\Gamma_{2j-1} + i\Gamma_{2j}) (\Gamma_{2k-1} - i\Gamma_{2k}) \\ &= \frac{1}{4} (\Gamma_{2j-1}\Gamma_{2k-1} + i\Gamma_{2j}\Gamma_{2k-1} - i\Gamma_{2j-1}\Gamma_{2k} + \Gamma_{2j}\Gamma_{2k}) \\ &= \frac{1}{2}\delta_J^K + \frac{1}{4} (\Gamma_{2j-1,2k-1} + i\Gamma_{2j,2k-1} - i\Gamma_{2j-1,2k} + \Gamma_{2j,2k}) \\ &= \frac{1}{2} \left[\delta_J^K + M_{2j-1,2k-1}^{spin} + iM_{2j,2k-1}^{spin} - iM_{2j-1,2k}^{spin} + M_{2j,2k}^{spin} \right]. \end{aligned} \quad (21.6)$$

In the commutation relations

$$[A_J^\dagger A^K, A_L^\dagger A^M] = \delta_L^K A_J^\dagger A^M - \delta_J^M A_L^\dagger A^K \quad (21.7)$$

all constant terms (the terms with δ_J^K in (21.6)) cancel. But note that

$$\frac{1}{2} \left(M_{2j-1,2k-1}^{def} + iM_{2j,2k-1}^{def} - iM_{2j-1,2k}^{def} + M_{2j,2k}^{def} \right) \quad (21.8)$$

for $j < k$ are just the raising operators $E_{JK}^{\eta\eta'}$ with $\eta = +1$ and $\eta' = -1$ of the defining representation of $SO(2N)$ in the Cartan-Weyl basis (and for $j > k$ the lowering operators $-E_{JK}^{+-} = (E_{JK}^{+-})^\dagger$, see (17.25)). The Cartan generators $A_J^\dagger A^J - A_{J+1}^\dagger A^{J+1}$ are equal to

$$\left(-iM_{2j-1,2j}^{spin} \right) - \left(-iM_{2j+1,2j+2}^{spin} \right). \quad (21.9)$$

So the operators $A_J^\dagger A^K - \frac{1}{N} \delta_J^K A_L^\dagger A^L$ (summation over L is understood) generate $SU(N)$, while $A^J A^K$ and $A_J^\dagger A_K^\dagger$ are further operators which together with $A_K^\dagger A^K$ generate the full $SO(2N)$.

If one transforms from the Cartan-Weyl basis to the compact basis, both for $SO(2N)$ and for $SU(N)$ by removing the factors i , one recovers the result with which we began, namely that $SU(N)$ is (isomorphic to, by identifying i with $i\tau^2$) a subgroup of $SO(2N)$.

We can define a vacuum $|0\rangle$ that is annihilated by all annihilation operators A^J with $J = 1, \dots, N$. Then acting on $|0\rangle$ with the creation operators $A_J^\dagger$ gives states with none, one, two, $\dots, N$ creation operators (not more because the $A_J^\dagger$ anticommute). The states with an even number of $A_J^\dagger$ transform under $SO(2N)$ into each other (but not into states with an odd number of $A_J^\dagger$) because the $SO(2N)$ generators are quadratic in Γ_j , and each Γ_j is a sum of an annihilation and a creation operator. The $SU(N)$ generators just permute all states with a given number of $A_J^\dagger$ operators, while the rest of the generators of $SO(2N)$ maps between states with different numbers of $A_J^\dagger$ generators. Because all generators of $SO(2N)$ (and thus also $SU(N)$) have an even number of A and $A^\dagger$ operators, the states with an even number of $A_J^\dagger$ form an irreducible rep of $SO(2N)$, the s -spinor irrep. The states with an odd number of $A_J^\dagger$ form the c -spinor irrep.

We can extend $SU(N)$ to $U(N)$ by adding a $U(1)$ generator which commutes with all $SU(N)$ generators. The generator which commutes with all $A_J^\dagger A^K$ is

$$S = \sum_{J=1}^N A_J^\dagger A^J = \sum_{J=1}^N \left(\frac{1}{2} - iM_{2j-1,2j} \right). \quad (21.10)$$

It is clearly hermitian (or antihermitian if one multiplies by i), and since

$$[S, A_K^\dagger] = A_K^\dagger; \quad [S, A^J] = -A^J \quad (21.11)$$

it is clear that S counts how many $A_J^\dagger$ a given state has: $S = m$ for a state with m operators $A_J^\dagger$.

Chapter 22

The Exceptional Lie algebras

G_2, F_4, E_6, E_7, E_8

The total set of simple Lie groups G consists of the 4 classical families and the exceptional Lie groups. There are 5 exceptional Lie groups, namely G_2, F_4, E_6, E_7 and E_8 , and we choose in each of the 5 cases a basis for the generators T_a of the Lie algebra (LA) as a set of generators H_i of a suitable subgroup H (suitable meaning: well-known, maximal, and more properties to be discussed) and the rest of the generators denoted by K_α (and sometimes called coset generators)

$$T_a = \{H_i, K_\alpha\}. \quad (22.1)$$

We require that the split is reductive ($[H, K] \subset K$), but it will not be symmetric ($[K, K] \subset H$) in the case of G_2 . A maximally noncompact version of an algebra always yields a symmetric decomposition, and has the property that the number of noncompact generators minus the number of compact generators is equal to its rank ($8 - 6 = 2$ for G_2). Such a decomposition is also called a Cartan decomposition. For some of the 5 Lie algebras (E_7 and E_8) we shall use such a Cartan decomposition. For F_4 and E_6 we use a basis which is not a Cartan decomposition but still symmetric, and for G_2 , as already mentioned, we use a nonsymmetric decomposition (and hence not a Cartan decomposition). For all LA's we shall use a basis where the generators are divided into two sets. One set contains the Cartan generators C_I which form a maximal abelian hermitian subalgebra of G . The remaining generators $E_{\vec{\alpha}}$ satisfy eigenvalue equations of the form $[C_I, E_{\vec{\alpha}}] = \alpha_I E_{\vec{\alpha}}$ where the $\vec{\alpha}$ with components α_I are the roots. We shall use as Cartan generators of G those of the subgroup H ; the subalgebras for which this is possible are called maximal regular subalgebras. We denote the Cartan generators by H_I , so the H_I are a commuting subset of the H_i . Here is the list of the groups G and subgroups H we have chosen

$$\begin{aligned} G_2 &\supset SU(3); & F_4 &\supset SO(9); \\ E_6 &\supset SU(6) \otimes SU(2); & E_7 &\supset SU(8); & E_8 &\supset SO(16). \end{aligned} \quad (22.2)$$

Our strategy is as follows. In each case we choose a low-dimensional irreducible representation (irrep) R of G , and decompose it into several irreps r of H . We also decompose the group parameters Λ^a of G into group parameters λ^i for H_i and further parameters σ^α called coset parameters for K_α . The main idea is to write all transformations rules of the tensors and spinors of the irreps r with the parameters λ^i and σ^α in a manifestly H -covariant way. This is possible because due to reductivity the σ^α are tensors of H . (When we say, for example, that the parameters σ^α transform as **3** under $SU(3)$, we mean that we have the following commutator for two infinitesimal transformations $[\delta(\sigma), \delta(\lambda)] = \delta(\sigma')$ with $\sigma'^\alpha = \lambda^\alpha_\beta \sigma^\beta$.) Then, if the transformation rules yield a commutator algebra that closes, we have proven the existence of the corresponding LA¹. Moreover, the composition laws of the parameters give us the commutation relations of the generators of the Lie algebra. We do not prove that these are the only LA's in addition to the four classical families. (For a proof of this statement, see the textbook on Group Theory by H. Georgi.) Here is a list of the representations and decompositions we use

G/H	$R \rightarrow r_1 + r_2 + \dots$	$\Lambda^a \rightarrow \{\lambda^i, \sigma^\alpha\}$
$G_2/SU(3)$	7 $\rightarrow \{x^\alpha(\mathbf{3}), \bar{x}_\alpha(\mathbf{3}^*), y(\mathbf{1})\}$	14 $\rightarrow \{\lambda^\alpha_\beta(\mathbf{8}), \sigma^\alpha(\mathbf{3}), \bar{\sigma}_\alpha(\mathbf{3}^*)\}$
$F_4/SO(9)$	26 $\rightarrow \{v^i(\mathbf{9}), \psi^\alpha(\mathbf{16}), s(\mathbf{1})\}$	52 $\rightarrow \{\Lambda^i_j(\mathbf{36}), \lambda^\alpha(\mathbf{16})\}$
$E_6/USp(8)$	27 $\rightarrow z^{\alpha\beta} = -z^{\beta\alpha}(\mathbf{27})$	78 $\rightarrow \{\Lambda^{\alpha\beta} = \Lambda^{\beta\alpha}(\mathbf{36}), \Sigma[[\alpha\beta\gamma\delta]](\mathbf{42})\}$
$E_7/SU(8)$	56 $\rightarrow \{x^{ij}(\mathbf{28}), \bar{x}_{ij}(\mathbf{28}^*)\}$	133 $\rightarrow \{\Lambda^i_j(\mathbf{63}), \Sigma^{[ijkl]}(\mathbf{70})\}$
$E_8/SO(16)$	248 $\rightarrow \{\phi^{ij} = -\phi^{ji}(\mathbf{120}), \psi^\alpha(\mathbf{128})\}$	248 $\rightarrow \{\Lambda^{ij} = -\Lambda^{ji}(\mathbf{120}), \lambda^\alpha(\mathbf{128})\}$

Table 22.1: The list of cosets, irreps R of G , the decomposition of R into irreps r of H , and the decomposition of the group parameters Λ^a of G into group parameters λ^i of H and coset parameters σ^α . The symbol $[]$ denotes antisymmetrization and $[[]]$ denotes anti-symmetrization and subtraction of a symplectic trace.

We have denoted the irreps R of G and the group parameters Λ^a by their dimensions. For example, the defining representation (rep) of G_2 is the **7** and it decomposes under $SU(3)$ into the defining **3** of $SU(3)$, its complex conjugate **3**^{*} and a singlet **1**. Furthermore, there are 14 parameters for G_2 , and they decompose under $SU(3)$ into the 8 real group parameters of $SU(3)$, and further 3 complex (6 real) parameters σ^α and $\bar{\sigma}_\alpha$ that transform as **3** and **3**^{*} under $SU(3)$.

We shall also discuss other choices of subgroups H , but we are only interested in the maximal regular subgroups. A regular subalgebra H of G has roots which are a subset of the roots of G , and Cartan generators which are a linear combination of the Cartan generators of G . If it is maximal, the rank of H is equal to the rank of G . For a maximal regular subalgebra H of a group G the roots of G are the sum of the roots and weights of H . One can obtain all maximal regular subalgebras of a Lie group G by first constructing the so-called extended Dynkin diagram, and then removing one node from it. The extended Dynkin diagram is obtained from the ordinary

¹The transformations rules have the generic form $\delta x^\alpha = \Lambda^a (T_a)^\alpha_\beta x^\beta$, from which one can extract the matrices T_a which form the representation R of G . We shall not determine these matrices, but work with δx^α . Similarly, from the commutator algebra one can extract the structure constants, and hence an explicit matrix representation of the adjoint rep. Again we shall not determine these matrices.

Dynkin diagram by adding minus the highest root $-\vec{\theta}$ to the rest of the simple roots, and then connecting the nodes according to the same rules as for ordinary Dynkin diagrams.

22.1a Definition and closure of G_2 transformations

We now work out the details for G_2 . If we can prove that the commutator of two G_2 -transformation is again a G_2 transformation (closure), we shall have proven that G_2 exists.

Input. G_2 has 14 generators and the fundamental irrep is **7**. All irreps of G_2 are real. We choose the compact form of G_2 , and $SU(3)$ as maximal subgroup; this is not a Cartan decomposition, see the comment below, but $SU(3)$ is so well-known that we prefer to use it as H . Under $SU(3)$ we have the following decompositions:

$$\begin{aligned} \mathbf{14} &\rightarrow \mathbf{8} + \mathbf{3} + \mathbf{3}^* & (\lambda^\alpha_\beta, \sigma^\alpha, \bar{\sigma}_\alpha) \\ \mathbf{7} &\rightarrow \mathbf{3} + \mathbf{3}^* + \mathbf{1} & (x^\alpha, \bar{x}_\alpha, y) \end{aligned} \quad (22.3)$$

Using this input, we can then write all transformation rules in a manifestly $SU(3)$ -covariant form [1]

$$\begin{aligned} \delta x^\alpha &= \lambda^\alpha_\beta x^\beta + \frac{1}{\sqrt{2}} \epsilon^{\alpha\beta\gamma} \bar{\sigma}_\beta \bar{x}_\gamma + \sigma^\alpha y \\ \delta \bar{x}_\alpha &= \bar{\lambda}_\alpha^\beta \bar{x}_\beta + \frac{1}{\sqrt{2}} \epsilon_{\alpha\beta\gamma} \sigma^\beta x^\gamma + \bar{\sigma}_\alpha y \\ \delta y &= a (\bar{\sigma}_\alpha x^\alpha + \sigma^\alpha \bar{x}_\alpha) \text{ with real } a \text{ (} a = -1 \text{)}. \end{aligned} \quad (22.4)$$

By rescaling the parameters σ^α and the scalar y , we can fix all constants in δx^α (and $\delta \bar{x}_\alpha$). The reality of y leaves only the combination $\bar{\sigma}_\alpha x^\alpha + \sigma^\alpha \bar{x}_\alpha$ in δy . So the whole Lie algebra (LA) has only one free real parameter left, a . The factors $\frac{1}{\sqrt{2}}$ in (22.4) will simplify the results for the commutators. We defined $\bar{x}_\alpha = (x^\alpha)^*$, $\bar{\sigma}_\alpha = (\sigma^\alpha)^*$ and $\bar{\lambda}_\alpha^\beta = (\lambda^\alpha_\beta)^*$. The $SU(3)$ parameters λ^α_β are the group parameters λ^i of $SU(3)$ (with $i = 1, \dots, 8$) multiplied by the $SU(3)$ generators T_i ($i = 1, \dots, 8$) in the defining (**3**) rep.

$$\lambda^\alpha_\beta = \lambda^i (T_i)^\alpha_\beta, \quad T_i \text{ are the 8 "Gell-Mann" matrices} \quad (22.5)$$

The **7** contains 7 real components: $x^\alpha + \bar{x}_\alpha$, $\frac{1}{i}(x^\alpha - \bar{x}_\alpha)$ and y . The **14** real group parameters of G_2 are λ^i , $\sigma^\alpha + \bar{\sigma}_\alpha$, $\frac{1}{i}(\sigma^\alpha - \bar{\sigma}_\alpha)$. The antihermiticity of the generators of $SU(3)$ in the defining representation implies² $(\lambda^\beta_\alpha)^* = -\lambda^\alpha_\beta$, and using the notation $(\lambda^\beta_\alpha)^* = \bar{\lambda}_\beta^\alpha$, we get $\bar{\lambda}_\beta^\alpha = -\lambda^\alpha_\beta$. Of course the S of $SU(3)$ implies that the generators are traceless, $\lambda^\alpha_\alpha = 0$. Finally, the ϵ -symbols $\epsilon^{\alpha\beta\gamma}$ and $\epsilon_{\alpha\beta\gamma}$ are numerically equal ($\epsilon^{123} = \epsilon_{123} = 1$) and they are invariant tensors of $SU(3)$. This concludes the definition of the symbols in (22.4). All transformation rules are manifestly $SU(3)$ covariant.

²Antihermiticity implies $(\lambda^{T*})^\alpha_\beta = -\lambda^\alpha_\beta$. We define for any matrix $(M)^\alpha_\beta$ that $(M^T)^\alpha_\beta = M^\alpha_\beta$ and $(M^\alpha_\beta)^* = (M^*)^\alpha_\beta$. Then, using $(\lambda^{T*})^\alpha_\beta = ((\lambda^T)^*)^\alpha_\beta = (\lambda^T_\alpha^\beta)^* = (\lambda^\beta_\alpha)^*$, we obtain $(\lambda^\beta_\alpha)^* = -\lambda^\alpha_\beta$.

The proof that G_2 is a LA is that the commutators of all symmetry transformations “close”, namely they should be a sum of symmetry transformations with composite parameters λ', σ' and λ'', σ''

$$[\delta(\lambda_2), \delta(\lambda_1)] = \delta(\lambda' = [\lambda_1, \lambda_2]) \quad \text{on } x^\alpha, \bar{x}_\alpha, y \quad (22.6)$$

$$[\delta(\lambda), \delta(\sigma)] = \delta(\sigma'^\alpha = -\lambda^\alpha_\beta \sigma^\beta) \quad \text{on } x^\alpha, \bar{x}_\alpha, y, \quad \text{see (22.15)} \quad (22.7)$$

$$[\delta(\sigma_2), \delta(\sigma_1)] = \delta(\lambda'') + \delta(\sigma'') \quad \text{on } x^\alpha, \bar{x}_\alpha, y, \quad \text{see (22.17) and (22.23)} \quad (22.8)$$

The first two commutators express closure of the subalgebra $SU(3)$ and reductivity; they are “trivially” satisfied because they follow from the manifest $SU(3)$ -covariance of the transformation laws. The last commutator is not trivial at all; if it is satisfied, one has a proof that G_2 exists, and, as we shall see, it only closes for a particular value of a .

Now we prove (22.6), (22.7) and (22.8). First (22.6).

$$\begin{aligned} [\delta(\lambda_2), \delta(\lambda_1)]x^\alpha &= \delta(\lambda_2)(\lambda_1^\alpha_\beta x^\beta) - 1 \leftrightarrow 2 \\ &= \lambda_1^\alpha_\beta (\lambda_2^\beta_\gamma x^\gamma) - 1 \leftrightarrow 2 = [\lambda_1, \lambda_2]^\alpha_\gamma x^\gamma \\ &= \delta([\lambda_1, \lambda_2])x^\alpha. \end{aligned} \quad (22.9)$$

We used the passive approach³: $\delta(\lambda_2)(\lambda_1^\alpha_\beta x^\beta) = \lambda_1^\alpha_\beta \delta(\lambda_2)x^\beta$. Since $[\lambda_1, \lambda_2]^\alpha_\beta = \lambda_1^i \lambda_2^j [T_i, T_j]^\alpha_\beta = \lambda_1^i \lambda_2^j f_{ij}{}^k (T_k)^\alpha_\beta = (\lambda')^k (T_k)^\alpha_\beta$ with $(\lambda')^k = \lambda_1^i \lambda_2^j f_{ij}{}^k$, the λ -transformations close on x^α (of course, since $SU(3)$ is a subgroup). For $\bar{x}_\alpha$ one finds the same composition law of the parameters

$$\begin{aligned} [\delta(\lambda_2), \delta(\lambda_1)]\bar{x}_\alpha &= \bar{\lambda}_{1\alpha}{}^\beta (\bar{\lambda}_{2\beta}{}^\gamma \bar{x}_\gamma) - 1 \leftrightarrow 2 \\ &= (\lambda_1^\alpha_\beta \lambda_2^\beta_\gamma)^* \bar{x}_\gamma - 1 \leftrightarrow 2 \\ &= \delta([\lambda_1, \lambda_2])\bar{x}_\alpha. \end{aligned} \quad (22.10)$$

For y one has $\delta(\lambda)y = 0$, so $[\delta(\lambda_1), \delta(\lambda_2)]y = \delta([\lambda_1, \lambda_2])y = 0$ is trivially satisfied. Thus (22.6) is uniformly satisfied.

Next we turn to (22.7). Here we shall need “the Schouten identity”

$$\lambda^\alpha_\beta \epsilon^{\beta\gamma\delta} = \lambda^\gamma_\beta \epsilon^{\beta\alpha\delta} + \lambda^\delta_\beta \epsilon^{\beta\gamma\alpha}, \quad (22.11)$$

which can be written in simpler form as $\lambda^{[\alpha}_\beta \epsilon^{\beta\gamma\delta]} = 0$ (since $\lambda^\beta_\beta = 0$). It is an identity since any expression with 4 totally antisymmetrized indices which can only take on 3 values, must vanish.

³In the active approach $\delta(\lambda_2)(\lambda_1^\alpha_\beta x^\beta) = \lambda_2^\alpha_\beta \lambda_1^\beta_\gamma x^\gamma$, which yields minus the results of the passive approach.

First on x^α

$$\begin{aligned} [\delta(\lambda), \delta(\sigma)]x^\alpha &= \delta(\lambda) \left[\frac{1}{\sqrt{2}} \epsilon^{\alpha\beta\gamma} \bar{\sigma}_\beta \bar{x}_\gamma + \sigma^\alpha y \right] - \delta(\sigma) [\lambda^\alpha_\beta x^\beta] \\ &= \frac{1}{\sqrt{2}} \epsilon^{\alpha\beta\gamma} \bar{\sigma}_\beta (\bar{\lambda}_\gamma^\delta \bar{x}_\delta) - \lambda^\alpha_\beta \left(\frac{1}{\sqrt{2}} \epsilon^{\beta\gamma\delta} \bar{\sigma}_\gamma \bar{x}_\delta + \sigma^\beta y \right) \end{aligned} \quad (22.12)$$

The last term, $-\lambda^\alpha_\beta \sigma^\beta y$, can be written as $\sigma'^\alpha y$ with $\sigma'^\alpha = -\lambda^\alpha_\beta \sigma^\beta$. This suggests that the sum of the first two terms (the terms with $\epsilon^{\alpha\beta\gamma}$ and $\epsilon^{\beta\gamma\delta}$) can be written as $\frac{1}{\sqrt{2}} \epsilon^{\alpha\beta\gamma} \bar{\sigma}'_\beta \bar{x}_\gamma$ with the same σ' . Let us work this out. We want to prove (using $\lambda^\alpha_\beta = -\bar{\lambda}_\beta^\alpha$) that

$$\frac{1}{\sqrt{2}} \epsilon^{\alpha\beta\gamma} \bar{\sigma}_\beta \bar{\lambda}_\gamma^\delta \bar{x}_\delta - \frac{1}{\sqrt{2}} \epsilon^{\beta\gamma\delta} \lambda^\alpha_\beta \bar{\sigma}_\gamma \bar{x}_\delta = \frac{1}{\sqrt{2}} \epsilon^{\alpha\beta\delta} (-\bar{\lambda}_\beta^\gamma \bar{\sigma}_\gamma) \bar{x}_\delta \quad (22.13)$$

This can be rewritten after relabeling $\bar{\sigma}$ and extracting $\frac{1}{\sqrt{2}} \bar{\sigma}_\beta$ and $\bar{x}_\delta$ as follows

$$\epsilon^{\alpha\beta\gamma} \bar{\lambda}_\gamma^\delta + \epsilon^{\gamma\beta\delta} \bar{\lambda}_\gamma^\alpha + \epsilon^{\alpha\gamma\delta} \bar{\lambda}_\gamma^\beta = 0 \quad (22.14)$$

which is the Schouten identity.

For $\bar{x}_\alpha$ one finds similarly that $[\delta(\lambda), \delta(\sigma)]\bar{x}_\alpha = \delta(\sigma')\bar{x}_\alpha$ with the same $\sigma'^\alpha = -\lambda^\alpha_\beta \sigma^\beta$. For y one finds

$$\begin{aligned} [\delta(\lambda), \delta(\sigma)]y &= \delta(\lambda) a (\bar{\sigma}_\alpha x^\alpha + \sigma^\alpha \bar{x}_\alpha) \\ &= a \left(\bar{\sigma}_\alpha \lambda^\alpha_\beta x^\beta + \sigma^\alpha \bar{\lambda}_\alpha^\beta \bar{x}_\beta \right) \\ &= \delta(\sigma')y \quad \text{with } \sigma'^\beta = \sigma^\alpha \bar{\lambda}_\alpha^\beta = -\lambda^\beta_\alpha \sigma^\alpha. \end{aligned} \quad (22.15)$$

So (22.7) is satisfied on all coordinates x^α , $\bar{x}_\alpha$ and y , and the parameter a is not yet fixed. These results are expected as they follow from the manifest $SU(3)$ covariance of the transformation rules.

Finally we come to the really nontrivial computation, the commutator $[\delta(\sigma_2), \delta(\sigma_1)]$. Here we begin by acting on y , which will only give us the expression for σ''^α because y is invariant under $\delta(\lambda)$. Having fixed σ''^α we shall then turn to x^α where $[\delta(\sigma_2), \delta(\sigma_1)]x^\alpha$ will give us λ''^α_β as well.

For the commutator of two σ transformations of y we find

$$[\delta(\sigma_2), \delta(\sigma_1)]y = a \bar{\sigma}_{1\alpha} \left(\frac{1}{\sqrt{2}} \epsilon^{\alpha\beta\gamma} \bar{\sigma}_{2\beta} \bar{x}_\gamma + \sigma_2^\alpha y \right) + a \sigma_1^\alpha \left(\frac{1}{\sqrt{2}} \epsilon_{\alpha\beta\gamma} \sigma_2^\beta x^\gamma + \bar{\sigma}_{2\alpha} y \right) - 1 \leftrightarrow 2 \quad (22.16)$$

The terms with y cancel, and the terms with $\bar{x}_\gamma$ give us σ''^γ

$$\sigma''^\alpha = \sqrt{2} \epsilon^{\alpha\beta\gamma} \bar{\sigma}_{1\beta} \bar{\sigma}_{2\gamma}. \quad (22.17)$$

The terms with x^γ confirm the result for σ''^α .

It all depends now on the commutator $[\delta(\sigma_2), \delta(\sigma_1)]x^\alpha$. If it closes (if it is a sum of a λ transformation with a new composite parameter λ'' , and a σ -transformation with the composite

parameter σ'' given above), the Lie algebra (G_2) exists. We find

$$[\delta(\sigma_2), \delta(\sigma_1)]x^\alpha = \frac{1}{\sqrt{2}}\epsilon^{\alpha\beta\gamma}\bar{\sigma}_{1\beta}\left(\frac{1}{\sqrt{2}}\epsilon_{\gamma\delta\epsilon}\sigma_2^\delta x^\epsilon + \bar{\sigma}_{2,\gamma}y\right) + \sigma_1^\alpha a\left(\bar{\sigma}_{2\beta}x^\beta + \sigma_2^\beta \bar{x}_\beta\right) - 1 \leftrightarrow 2 \quad (22.18)$$

The terms with y are as follows

$$2 \times \frac{1}{\sqrt{2}}\epsilon^{\alpha\beta\gamma}\bar{\sigma}_{1\beta}\bar{\sigma}_{2\gamma}y, \quad (22.19)$$

and this agrees with (22.17). The terms with $\bar{x}_\gamma$ are given by

$$a\left(\sigma_1^\alpha\sigma_2^\beta\bar{x}_\beta - \sigma_2^\alpha\sigma_1^\beta\bar{x}_\beta\right), \quad (22.20)$$

and they also agree with (22.17) because they can be rewritten as follows

$$(\sigma_1^\gamma\sigma_2^\alpha - \sigma_2^\gamma\sigma_1^\alpha)\bar{x}_\gamma = \epsilon^{\alpha\beta\gamma}\left(\epsilon_{\beta\delta\epsilon}\sigma_1^\delta\sigma_2^\epsilon\right)\bar{x}_\gamma = \frac{1}{\sqrt{2}}\epsilon^{\alpha\beta\gamma}\sigma''_{\beta}\bar{x}_\gamma \quad (22.21)$$

provided $a = -1$. Finally we analyze the terms with x^ϵ . They will yield λ'' . They are given by

$$\left(\frac{1}{2}\epsilon^{\alpha\beta\gamma}\epsilon_{\gamma\delta\epsilon}\bar{\sigma}_{1\beta}\sigma_2^\delta + a\sigma_1^\alpha\bar{\sigma}_{2\epsilon}\right)x^\epsilon - 1 \leftrightarrow 2 = \left(\frac{1}{2}\bar{\sigma}_{1\epsilon}\sigma_2^\alpha - \frac{1}{2}\delta_\epsilon^\alpha\bar{\sigma}_1 \cdot \sigma_2 + a\sigma_1^\alpha\bar{\sigma}_{2\epsilon}\right)x^\epsilon - 1 \leftrightarrow 2 \equiv \lambda''^\alpha_\epsilon x^\epsilon \quad (22.22)$$

with

$$\lambda''^\alpha_\epsilon = -\frac{1}{2}\delta_\epsilon^\alpha(\bar{\sigma}_1 \cdot \sigma_2 - \bar{\sigma}_2 \cdot \sigma_1) + \left(a - \frac{1}{2}\right)(\sigma_1^\alpha\bar{\sigma}_{2\epsilon} - \sigma_2^\alpha\bar{\sigma}_{1\epsilon}). \quad (22.23)$$

In order that $\lambda''^\alpha_\epsilon$ be an $SU(3)$ parameter, it should satisfy $((\lambda'')^\alpha_\alpha)^* = -(\lambda'')^\epsilon_\alpha$ and $(\lambda'')^\alpha_\alpha = 0$. These two properties are satisfied, the latter if $a = -1$ to cancel the trace $\delta^\alpha_\alpha = 3$, which agrees with the value of a found before. This concludes the proof that G_2 exists.

Comment. The decomposition of the generators of G_2 into the **8** of $SU(3)$ and the remaining **6** is not a symmetric decomposition (because of the term with $\delta(\sigma''^\alpha)$ in (22.8)). The decomposition into the subgroup $SU(3)$ and six coset generators is also not a Cartan decomposition because the subgroup $SU(3)$ does not have six generators. For G_2 a Cartan decomposition is, for example, into the generators of the maximal subgroup $SU(2) \otimes SU(2)$ and the **(2, 4)** coset generators. The **7** decomposes under $SU(2) \otimes SU(2)$ into **(1, 3)+ (2, 2)**. One can find the maximal regular subalgebras of G_2 from its extended Dynkin diagram (to be discussed next)

$$G'_2 = \bullet \text{---} \bullet \text{---} \bullet; \quad G_2 = \bullet \text{---} \bullet \text{---} \bullet \quad (22.24)$$

Deleting one dot, one finds the maximal (rank-2) regular subalgebras $SU(3)$, $SU(2) \otimes SU(2)$ and of course G_2 itself. We could have proven the existence of G_2 by choosing $SU(2) \otimes SU(2)$ as subgroup, but $SU(3)$ is simpler to deal with.

22.1b Roots of G_2

We now construct the simple roots and fundamental representations of G_2 . We take as Cartan generators the same as for $SU(3)$, but now in the **7** of G_2

$$H_I = \frac{1}{2} \begin{pmatrix} 1 & & & & & & \\ & -1 & & & & & \\ & & 0 & & & & \\ & & & -1 & & & \\ & & & & 1 & & \\ \emptyset & & & & & 0 & \\ & & & & & & 0 \end{pmatrix}, \quad H_{II} = \frac{1}{\sqrt{12}} \begin{pmatrix} 1 & & & & & & \\ & 1 & & & & & \\ & & -2 & & & & \\ & & & -1 & & & \\ & & & & -1 & & \\ \emptyset & & & & & 2 & \\ & & & & & & 0 \end{pmatrix}. \quad (22.25)$$

They are normalized to $\text{tr } H_I H_J = \delta_{IJ}$. Then the 7 weight vectors of the defining representation are given by

$$\begin{aligned} \vec{\mu}^{(1)} &= \left(\frac{1}{2}, \frac{1}{\sqrt{12}} \right), & \vec{\mu}^{(2)} &= \left(-\frac{1}{2}, \frac{1}{\sqrt{12}} \right), & \vec{\mu}^{(3)} &= \left(0, -\frac{1}{\sqrt{3}} \right), \\ \vec{\mu}^{(4)} &= \left(-\frac{1}{2}, -\frac{1}{\sqrt{12}} \right), & \vec{\mu}^{(5)} &= \left(\frac{1}{2}, -\frac{1}{\sqrt{12}} \right), & \vec{\mu}^{(6)} &= \left(0, \frac{1}{\sqrt{3}} \right), & \vec{\mu}^{(7)} &= (0, 0). \end{aligned} \quad (22.26)$$

We define positivity of weights as positivity of their **last nonvanishing** component (as we did for $SU(N)$). Then the weight diagram of the defining irrep of G_2 is the sum of the weight diagrams of **3**, **3*** and **1** of $SU(3)$, as shown in figure 22.1.

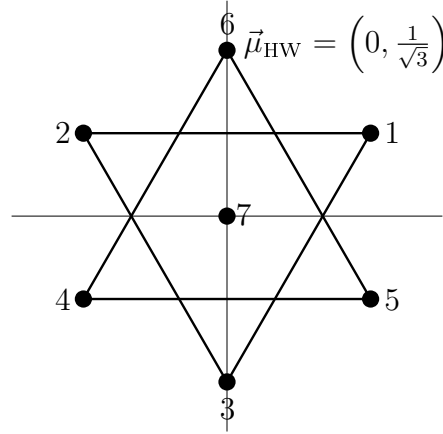


Figure 22.1: The weight diagram of the defining irrep of G_2 . The highest weight is denoted by HW.

From the transformation rules we see that the adjoint rep of G_2 decomposes under $SU(3)$ into the adjoint rep of $SU(3)$ and the **3** and **3*** reps of $SU(3)$. Since for a maximal regular subalgebra the roots of G are the sum of the roots of H and the weights of the irreps $r_1, r_2, \dots$, the 12 roots of G_2 consist of (i) those of $SU(3)$, $\vec{\alpha}^{ij} = \pm(\vec{\mu}^{(i)} - \vec{\mu}^{(j)})$, and (ii) the weights of the **3** and **3*** of $SU(3)$, $\pm\vec{\mu}^{(i)}$. They are shown in figure 22.2.

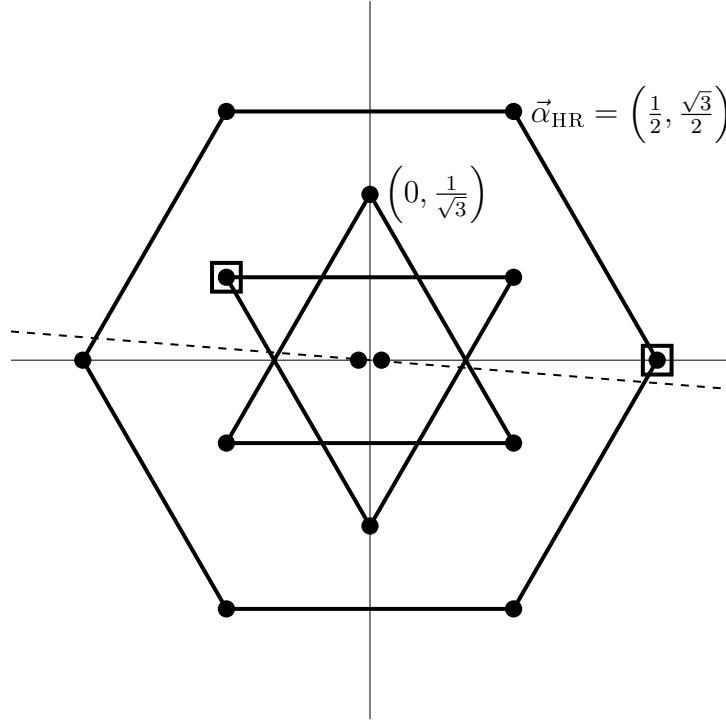


Figure 22.2: The 12 roots and the two vanishing weights at the origin of G_2 . The positive roots are those above the dashed line, the simple roots are denoted by a surrounding square, and the highest root is labelled HR. A weight vector or root is positive if its last nonvanishing component is positive, the same definition as for $SU(N)$. The first simple root is $\vec{\alpha}^I = (1, 0)$, and it is the same for G_2 as for $SU(3)$. But the second simple root is different for G_2 and $SU(3)$. For G_2 it is $\vec{\alpha}^{II} = (-\frac{1}{2}, \frac{1}{2\sqrt{3}})$ while for $SU(3)$ it is $\vec{\alpha}^{II} = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$.

The simple roots are

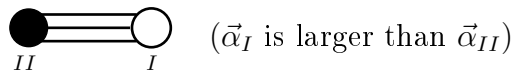
$$\vec{\alpha}^I = (1, 0), \quad \vec{\alpha}^{II} = \left(-\frac{1}{2}, \frac{1}{\sqrt{12}}\right), \quad (22.27)$$

For $SU(3)$ the second simple root is $\vec{\alpha}^{II} = \vec{\mu}^2 - \vec{\mu}^3$, but for G_2 $\vec{\mu}^2 - \vec{\mu}^3$ is not a simple root because it is the sum of the two positive roots $\vec{\mu}^2$ and $-\vec{\mu}^3$. Rather it is $\vec{\mu}^2 = (-\frac{1}{2}, \frac{1}{2\sqrt{3}})$. The fundamental weights are

$$\vec{\mu}_{FW}^I = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \quad \vec{\mu}_{FW}^{II} = \left(0, \frac{1}{\sqrt{3}}\right). \quad (22.28)$$

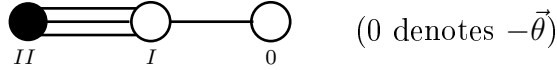
Clearly, the first fundamental weight is the highest root of the adjoint representation, and the second fundamental weight is the highest weight of the complex conjugate of the defining representation.

Since $\cos \phi_{I,II} = \frac{\vec{\alpha}^I \cdot \vec{\alpha}^{II}}{|\vec{\alpha}^I| |\vec{\alpha}^{II}|} = -\frac{\sqrt{3}}{2}$, the angle between the two simple roots is 150° . Hence the Dynkin diagram is



An extended Dynkin diagram is obtained by adding minus the highest root $-\vec{\theta}$ to the simple roots, and following the same rules as for the construction of an ordinary Dynkin diagram. For G_2 we

have $-\vec{\theta} = \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$ and $\frac{(-\vec{\theta}, \vec{\alpha}^I)}{|\vec{\theta}||\vec{\alpha}^I|} = -\frac{1}{2}$ while $\frac{(-\vec{\theta}, \vec{\alpha}^{II})}{|\vec{\theta}||\vec{\alpha}^{II}|} = 0$. This yields the extended Dynkin diagram



A general irrep of G_2 is determined by its Dynkin labels (m, n) . The label for the defining representation is $(0, 1)$ and that for the adjoint representation is $(1, 0)$. The Clebsch-Gordan decomposition series of the direct product of irreps of G_2 can be written in terms of Young tableaux or Dynkin labels. For example, the symmetric and antisymmetric parts of the direct product of two defining irreps of G_2 decompose as follows

$$\begin{aligned}
(\mathbf{7} \times \mathbf{7})_s &= \mathbf{27} + \mathbf{1} \\
(\square \otimes \square)_s &= \square\square \oplus \bullet \\
((0, 1) \otimes (0, 1))_s &= (0, 2) \oplus (0, 0) \\
(\mathbf{7} \times \mathbf{7})_a &= \mathbf{14} + \mathbf{7} \\
(\square \otimes \square)_a &= \begin{array}{|c|} \hline \square \\ \hline \end{array} \oplus \square \\
((0, 1) \otimes (0, 1))_a &= (1, 0) \oplus (0, 1) \\
\mathbf{7} \times \mathbf{7} &= \mathbf{27} + \mathbf{14} + \mathbf{7} + \mathbf{1} \\
\square \otimes \square &= \square\square \oplus \begin{array}{|c|} \hline \square \\ \hline \end{array} \oplus \square \oplus \bullet \\
(0, 1) \otimes (0, 1) &= (0, 2) \oplus (1, 0) \oplus (0, 1) \oplus (0, 0)
\end{aligned} \tag{22.29}$$

$(0, 2)$ must occur since its highest weight is the sum of the highest weight of the two reps $(0, 1)$. (The rep $(\mathbf{7} \times \mathbf{7})_s$ is reducible into $\mathbf{27} + \mathbf{1}$ because there is a symmetric invariant tensor S_{IJ} of G_2 . This follows from the reality of the $\mathbf{7}$, because then $T_A^{(7)*} = S T_A^{(7)} S^{-1}$ with symmetric S . In this case the symmetric invariant tensor S_{IJ} is δ_{IJ} .⁴) Since $\mathbf{7}$ appears in the decomposition of $(\mathbf{7} \times \mathbf{7})_a$: there must exist an invariant tensor $g_{abc} = -g_{bac}$ with 3 indices in the defining representation.

Weyl (1926) has obtained a formula for the dimensions of all irreps of any simple Lie group [2]. If Λ is the highest weight of an irrep R , and $g = \frac{1}{2} \sum \alpha^+$ is half the sum of all positive roots, then the dimension of the irrep is given by

$$\dim R = \prod_{\alpha^+} \frac{(\Lambda + g, \alpha^+)}{(g, \alpha^+)} \tag{22.30}$$

where α^+ are all positive roots. For example, for G_2 we saw that the positive roots are $\vec{\mu}^i - \vec{\mu}^j$ with $i < j$ and $\vec{\mu}^1$, $\vec{\mu}^2$ and $-\vec{\mu}^3$. Then $g = \left(\frac{1}{2}, \frac{5}{2\sqrt{3}}\right)$ and for the $\mathbf{7}$ one has $\Lambda = \left(0, \frac{1}{\sqrt{3}}\right)$. Then $\Lambda + g = \left(\frac{1}{2}, \frac{7}{2\sqrt{3}}\right)$ and one finds $\dim R = 1 \times \frac{3}{2} \times \frac{4}{3} \times \frac{5}{4} \times 2 \times \frac{7}{5} = \mathbf{7}$, which is the correct result. The

⁴Another argument is that one always finds a singlet in the decomposition of a direct product of a representation R with its complex conjugate R^* , and the $\mathbf{7}$ is a real representation.

first few irreps of G_2 are as follows (see Slansky's tables)

$$\begin{aligned} (0, 1) = \mathbf{7}, \quad (0, 2) = \mathbf{27}, \quad (0, 3) = \mathbf{77}', \quad (0, 4) = \mathbf{182}, \quad (2, 1) = \mathbf{286}, \\ (1, 0) = \mathbf{14}, \quad (1, 1) = \mathbf{64}, \quad (2, 0) = \mathbf{77}, \quad (1, 2) = \mathbf{189}. \end{aligned} \quad (22.31)$$

There are no simple rules for the Clebsch-Gordan (CG) decomposition of products of irreps of exceptional groups, but by matching dimensions and making some simple observations one can construct the CG series, at least for the lowest irreps.

Consider for example the following Clebsch-Gordan decomposition for G_2

$$\begin{aligned} \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \otimes \square &= \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \oplus \square \\ \mathbf{14} \times \mathbf{7} &= \mathbf{27} + \mathbf{64} + \mathbf{7} \\ (1, 0) \otimes (0, 1) &= (0, 2) \oplus (1, 1) \oplus (0, 1) \end{aligned} \quad (22.32)$$

The **64** must occur because its highest weight is the sum of the HW of the **14** and the **7**. By analogy with $SU(N)$, one might (erroneously) have expected that for G_2

$$\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \otimes \square = \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \quad (\text{wrong}) \quad (22.33)$$

but Young tableaux with more than two boxes in a column are not allowed in G_2 . On the other hand, starting from the tensor product $t^{[\alpha\beta]}r^\gamma$, one can certainly antisymmetrize in α, β, γ which would correspond to $\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array}$ in G_2 . The highest weight of this representation is $\vec{\mu}_6 + \vec{\mu}_1 + \vec{\mu}_2 = (0, \frac{2}{\sqrt{3}})$ of the defining rep, and this is also the highest weight $2\vec{\mu}_6$ of $\begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array}$. This explains the presence of $\begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array}$ in (22.32). Comparing dimensions, it is clear that one needs a **7** in (22.32).

To confirm the dimension of $\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array}$ we consider another Clebsch-Gordan decomposition

$$\begin{aligned} \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \otimes \square &= \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \oplus \square \\ \mathbf{27} \times \mathbf{7} &= \mathbf{77}' + \mathbf{64} + \mathbf{27} + \mathbf{14} + \mathbf{7} \\ (0, 2) \otimes (0, 1) &= (0, 3) \oplus (1, 1) \oplus (0, 2) \oplus (1, 0) \oplus (0, 1) \end{aligned} \quad (22.34)$$

The **77'** occurs since its HW is the sum of the HW of **27** and **7**. The dimensions add up correctly, confirming that $\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array}$ has dimension 64. One could decompose all these irreps of G_2 into irreps of $SU(3)$, and then check the result by using the Clebsch-Gordan rules for $SU(3)$.

Another exercise is the decomposition of $\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \otimes \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array}$. It simplifies the analysis if one first decom-

poses this product into symmetric and antisymmetric parts

$$\begin{aligned} (\mathbf{14} \times \mathbf{14})_s &= \mathbf{105} = \mathbf{77} + \mathbf{27} + \mathbf{1} = (2,0) \oplus (0,2) \oplus (0,0) \\ \left(\begin{array}{|c|} \hline \square \\ \hline \end{array} \otimes \begin{array}{|c|} \hline \square \\ \hline \end{array} \right)_s &= \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \oplus \bullet \end{aligned} \quad (22.35)$$

$$\begin{aligned} (\mathbf{14} \times \mathbf{14})_a &= \mathbf{91} = \mathbf{77}' + \mathbf{14} = (0,3) \oplus (1,0) \\ \left(\begin{array}{|c|} \hline \square \\ \hline \end{array} \otimes \begin{array}{|c|} \hline \square \\ \hline \end{array} \right)_a &= \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|} \hline \square \\ \hline \end{array} \end{aligned} \quad (22.36)$$

The presence of $(2,0)$ in $((1,0) \otimes (1,0))_s$ is expected from addition of angular momenta: the highest weights of a product of irreps is the sum of the highest weights of the irreps. The presence of a singlet in $((1,0) \otimes (1,0))_s$ is also expected: one always finds a singlet (scalar) in the product $R \otimes R^*$ and $R = \mathbf{14}$ is real, so $\mathbf{14} \times \mathbf{14}$ contains a scalar. That the scalar appears in the symmetric part of the product follows from the reality of $\mathbf{14}$. (Reality implies $T^* = STS^{-1}$ with symmetric S . Here S is the unit matrix.) The presence of $\mathbf{14}$ in $(\mathbf{14} \times \mathbf{14})_a$ is also expected: the structure constants are totally antisymmetric, and always couple two adjoint reps to a third adjoint rep. The presence of $(0,3)$ and $(0,2)$ in $\mathbf{14} \times \mathbf{14}$ can be deduced from the weight diagram given in figure 22.3.

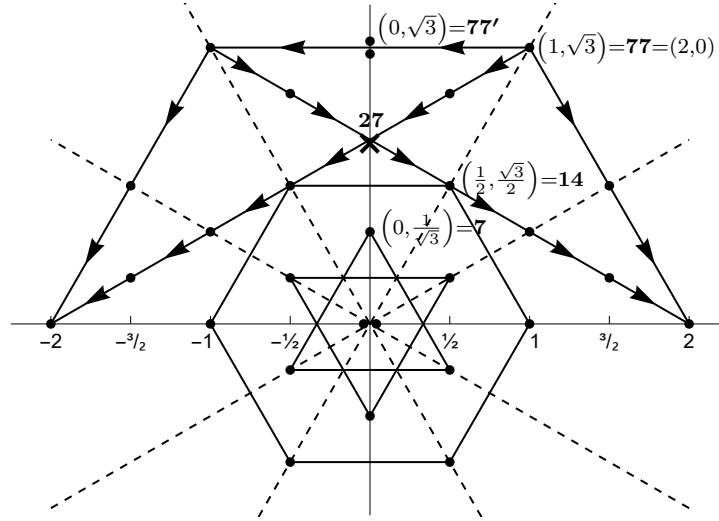


Figure 22.3: The beginning of the weight diagram of the irrep $\mathbf{77}$ with Dynkin labels $(2,0)$. Its highest weight $(1, \sqrt{3})$ is twice the highest weight of the adjoint rep $\mathbf{14}$. Constructing the $\vec{\alpha}^I$ -chain, one finds a state at $(0, \sqrt{3})$. This state is degenerate with another state with this weight vector, but which is the highest weight of a second $\mathbf{77}'$. The dotted lines are the hyper“surfaces” w.r.t. which weight diagrams of G_2 are invariant under Weyl reflections. The $\times$ denotes weight vector $(0, \frac{2}{\sqrt{3}})$ which is the highest weight of the $\mathbf{27} = (0,2)$. The weight vector $(\frac{1}{2}, \frac{\sqrt{3}}{2})$ is the highest weight of the adjoint $\mathbf{14}$ and also the first fundamental weight. The weight vector $(0, \frac{1}{\sqrt{3}})$ is the highest weight of the defining rep $\mathbf{7}$ and also the second fundamental weight.

The group G_2 plays an important role in particle physics because it is the subgroup of the Euclidean rotation group $SO(7)$ in seven dimensions which keeps one real spinor fixed.

22.2a Definition and closure of F_4 transformations

Next, we consider the group F_4 . It has 52 generators, the defining representation is **26** and $SO(9)$ is a maximal regular subalgebra. Spinors ψ^α of $SO(9)$ have 16 components and $SO(9)$ has 36 generators, so we expect that under $SO(9)$ the **26** and the **52** decompose as follows:

$$\begin{aligned}\mathbf{26} &\rightarrow \mathbf{9} + \mathbf{16} + \mathbf{1} \\ \mathbf{52} &\rightarrow \mathbf{36} + \mathbf{16}.\end{aligned}\tag{22.37}$$

So the **26** consists of a vector v^i , a commuting spinor ψ^α and a scalar s , all of which are real. Furthermore, the parameters of F_4 decompose into the parameters Λ^i_j of the adjoint rep **36** of $SO(9)$ and a real commuting spinor λ^α . We define F_4 by its action on the **26**. Manifest $SO(9)$ covariance only allows the following transformation rules [1]

$$\begin{aligned}\delta v^i &= \Lambda^i_j v^j + \lambda^\alpha \Gamma_{\alpha\beta}^i \psi^\beta \\ \delta \psi^\alpha &= \frac{1}{4} \Lambda^{ij} (\Gamma_{ij})^\alpha_\beta \psi^\beta + v^i (\Gamma_i)^\alpha_\beta \lambda^\beta + s \lambda^\alpha \\ \delta s &= a \lambda^\alpha \psi_\alpha \quad (a = 3, \text{ see below}).\end{aligned}\tag{22.38}$$

By suitable rescaling of fields and parameters we can achieve that only a has to be fixed. Orthogonal groups leave $y^i \delta_{ij} x^j$ invariant, hence $\delta_{ik} \Lambda^k_j + \Lambda^k_i \delta_{kj} = 0$. Then $\Lambda_{ij} = -\Lambda_{ji}$ where $\Lambda_{ij} = \delta_{ik} \Lambda^k_j$ are the 36 real parameters of $SO(9)$, and λ^α are the 16 extra real parameters. Since the Dirac matrices (with $\{\Gamma_i, \Gamma_j\} = 2\delta_{ij}$) in nine Euclidean dimensions can be taken to be real and symmetric 16×16 matrices, the **26** is real. In fact, the charge conjugation matrix is the unit matrix since $C\Gamma^i = \Gamma^{i,T}C$ for $C = I$. (In odd dimensions one has either $C_-\Gamma^i = -\Gamma^{i,T}C_-$ or $C_+\Gamma^i = +\Gamma^{i,T}C_+$, but not both possibilities. Here, clearly one has the $+$ sign.) Because $C = I$, we may raise and lower spinor indices α, β at will, and thus the matrices $\Gamma_{\alpha\beta}^i$ and $(\Gamma^i)^\alpha_\beta$ in (22.38) are numerically the same.

To show that (22.38) defines a Lie algebra, we must show that the commutators close. Of course the subalgebra of $SO(9)$ closes, $[\delta(\Lambda_2), \delta(\Lambda_1)] = \delta([\Lambda_1, \Lambda_2])$. Also $[\delta(\Lambda), \delta(\lambda)]$ is, as expected, equal to $\delta(\lambda' = -\frac{1}{4} \Lambda^{jk} \Gamma_{jk} \lambda)$. The crucial question is whether $[\delta(\lambda_2), \delta(\lambda_1)]$ closes. On s and v^i one easily establishes that this commutator is equal to an $SO(9)$ rotation with composite parameter $\Lambda'_{ij} = 2\lambda_1 \Gamma_{ij} \lambda_2$. (Recall that the spinorial parameters λ^α are commuting.) On ψ a Fierz rearrangement yields

$$[\delta(\lambda_2), \delta(\lambda_1)]\psi = \frac{1}{16} \left(\lambda_2 \Gamma^i O^I \Gamma_i \lambda_1 + a \lambda_2 O^I \lambda_1 \right) O_I \psi - (1 \leftrightarrow 2) \tag{22.39}$$

without overall minus sign because the spinors λ are commuting, and O^I denotes the complete set $I, \Gamma^i, \Gamma^{ij}, \Gamma^{ijk}, \Gamma^{ijkl}$ (256 matrices in all). Moreover $\Gamma^i O^I \Gamma_i$ is proportional to O^I , so only $O^I \sim \Gamma^{jkl}$ and $O^I \sim \Gamma^{jk}$ contribute (the rest, I and Γ^i and Γ^{ijkl} are symmetric matrices). The contribution of Γ^{jkl} cancels if one chooses $a = 3$, and then the contribution of Γ^{jk} yields an $SO(9)$ rotation with parameter $\Lambda'_{ij} = -2\lambda_2 \Gamma_{ij} \lambda_1$. Hence, (22.38) defines a Lie algebra, namely F_4 .

The decomposition of the generators of F_4 into generators of $SO(9)$ and coset generators is not a Cartan decomposition because that would require 24 subgroup generators and 28 coset generators; however, it still is a symmetric decomposition (because two λ transformations only produce an $SO(9)$ rotation). Hence, not every symmetric decomposition is a Cartan decomposition. (The reverse is, however, true: every Cartan decomposition is a symmetric decomposition.) Of course, 24 is the number of generators of $SU(5)$, but 28 is not the sum of the dimensions of a set of irreducible representations of $SU(5)$ (except if one has many $SU(5)$ singlets). So $SU(5)$ is not a subgroup of F_4 . In fact, we may use the extended Dynkin diagram F'_4 (see below) to find all maximal regular subalgebras of F_4 ,

$$F'_4 = \bullet \text{---} \bullet \text{---} \circ \text{---} \circ \text{---} \circ; \quad F_4 = \bullet \text{---} \bullet \text{---} \circ \text{---} \circ \quad (22.40)$$

Deleting any point gives the following set of regular subalgebras of F_4

$$SO(9); \quad SU(2) \otimes SU(4); \quad SU(3) \otimes SU(3); \quad USp(6) \otimes SU(2). \quad (22.41)$$

All except $SU(2) \otimes SU(4)$ are maximal regular subalgebras.⁵ The last one yields a Cartan decomposition ($USp(6) \otimes SU(2)$ has $21 + 3 = 24$ generators), and the **28**-dimensional coset is indeed a representation of $USp(6) \otimes SU(2)$, namely **28** = (**14**, **2**) (where **14** is the antisymmetric symplectic-traceless representation $t^{ij} = -t^{ji}$ of $USp(6)$). However, $SO(9)$ is easier to work with, and for that reason we choose $H = SO(9)$.

22.2b Roots of F_4

We take as Cartan generators those of $SO(9)$. The **26** weights of the defining representation of F_4 are those of the vector, spinor, and singlet irreps of $SO(9)$, all 4-component vectors.

$$\begin{aligned} \vec{\mu}(v) &= (0, \pm 1, 0, 0), (0, 0, 0, 0) & (9 \text{ weights}) \\ \vec{\mu}(\psi) &= (\pm \tfrac{1}{2}, \pm \tfrac{1}{2}, \pm \tfrac{1}{2}, \pm \tfrac{1}{2}) & (16 \text{ weights}) \\ \vec{\mu}(s) &= (0, 0, 0, 0) & (1 \text{ weight}) \end{aligned} \quad (22.42)$$

A weight vector $\vec{\mu}$ is positive if its first nonvanishing component is positive (as for $SO(N)$). The highest weight is

$$\vec{\mu}_{HW}^{\text{def}} = (1, 0, 0, 0). \quad (22.43)$$

⁵Actually, it is known that for the exceptional groups there are five regular subalgebras obtained in this way which are not maximal, and one of these exceptions is for F_4 where $SU(2) \otimes SU(4)$ is contained in $SO(9)$. (Recall that the $SU(2)$ Lie algebra is isomorphic to that of $SO(3)$, and the $SU(4)$ Lie algebra is isomorphic to that of $SO(6)$.)

The $52 - 4 = 48$ roots of F_4 consist of the 32 roots of $SO(9)$, together with the 16 weights of the spinor irrep λ^α of $SO(9)$

$$\begin{aligned} \text{roots of } SO(9) : (\pm 1, 0, \pm 1, 0), (0, \pm 1, 0, 0) & \quad (24 + 8 = 32 \text{ roots}) \\ \text{weights of } \lambda^\alpha : (\pm \frac{1}{2}, \pm \frac{1}{2}, \pm \frac{1}{2}, \pm \frac{1}{2}) & \quad (16 \text{ roots}) \end{aligned} \quad (22.44)$$

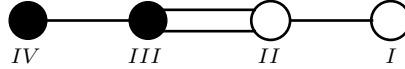
The highest root of F_4 is clearly $(1, 1, 0, 0)$. The 24 positive roots are

$$\begin{aligned} (0, 1, \pm 1, 0), (0, 1, 0, 0) & \quad (12 + 4) = 16 \text{ positive roots} \\ (\frac{1}{2}, \pm \frac{1}{2}, \pm \frac{1}{2}, \pm \frac{1}{2}) & \quad 8 \text{ positive roots} \end{aligned} \quad (22.45)$$

The 4 simple roots are⁶

$$\vec{\alpha}^I = (0, 1, -1, 0), \quad \vec{\alpha}^{II} = (0, 0, 1, -1), \quad \vec{\alpha}^{III} = (0, 0, 0, 1), \quad \vec{\alpha}^{IV} = (\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}). \quad (22.46)$$

with Dynkin diagram

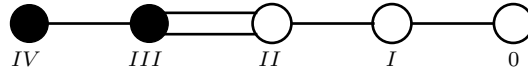


where solid dots denote the shorter roots. The four fundamental weights can now be computed.

$$\begin{aligned} \vec{\mu}^I &= (1, 1, 0, 0) & \text{highest weight of adjoint rep } \mathbf{52} \\ \vec{\mu}^{II} &= (2, 1, 1, 0) & \text{highest weight of } (\text{adj} \otimes \text{adj})_a \mathbf{1274} \\ \vec{\mu}^{III} &= (\frac{3}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}) & \text{contains } \boxtimes \text{ of } SO(9) \text{ (see below) } \mathbf{273} \\ \vec{\mu}^{IV} &= (1, 0, 0, 0) & \text{highest weight of defining rep } \mathbf{26} \end{aligned} \quad (22.47)$$

The symbol $\times$ denotes the spinor irrep, and $\bullet$ will indicate the singlet irrep.

The highest root is $\vec{\theta} = (1, 1, 0, 0)$. Adding $-\vec{\theta}$ to the simple roots yields the extended Dynkin diagram, which we used in (22.40),



The subscript $_0$ denotes the contribution from $\vec{\theta}$. From this diagram we obtained the maximal regular subalgebras in (22.41).

We shall now try to identify a few of the lowest irreps in terms of Young tableaux for F_4 , but no general theory of Young tableaux for F_4 seems to exist, so we use inspired guess work. The first few irreps with their F_4 Young tableaux, Dynkin labels, dimensions, and decomposition into

⁶The root $(1, -1, 0, 0)$ is a simple root in $SO(9)$, but it is not simple in F_4 as it is equal to $(\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}) + (\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2})$. For the same reason, $(1, 1, 0, 0)$, $(1, 0, \pm 1, 0)$, $(1, 0, 0, \pm 1)$ are not simple. Also the roots $(0, 1, 1, 0) = (0, 1, 0, -1) + (0, 0, 1, 1)$ and $(0, 1, 0, \pm 1) = (0, 1, -1, 0) + (0, 0, 1, \pm 1)$ and $(0, 0, 1, 1) = (0, 0, 1, 0) + (0, 0, 0, 1)$ are not simple. None of the spinor weights give simple roots except $(\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2})$. For example $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}) = (\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{1}{2}) + (0, 1, 0, 0)$. That leaves as candidates for simple roots: $(0, 1, -1, 0)$, $(0, 0, 1, -1)$, $(0, 0, 0, 1)$, $(\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2})$. Since there should be 4 simple roots, we got them all.

$SO(9)$ irreps are given in table 22.2. The Young tableaux for F_4 which are shown in the table are suggested by the branching rule $\mathbf{26} \times \mathbf{26} = \mathbf{1}_s + \mathbf{26}_s + \mathbf{52}_a + \mathbf{273}_a + \mathbf{324}_s$ (see [3]). However, it is not clear whether, and if so which, Young tableaux correspond to the other irreps.

F_4 tableau	Dynkin labels	Dim	Decomposition into $SO(9)$ irreps
$\square$	$(0, 0, 0, 1)$	26	defining rep, $\mathbf{9} + \mathbf{16} + \mathbf{1} = \square \oplus \times \oplus \bullet$
?	$(1, 0, 0, 0)$	52	adjoint rep, $\mathbf{36} + \mathbf{16} = \begin{array}{ c } \hline \square \\ \hline \end{array} \oplus \times$
$\begin{array}{ c } \hline \square \\ \hline \end{array}$	$(0, 0, 1, 0)$	273	$\mathbf{9} + \mathbf{16} + \mathbf{36} + \mathbf{84} + \mathbf{128} = \square \oplus \times \oplus \begin{array}{ c } \hline \square \\ \hline \end{array} \oplus \begin{array}{ c } \hline \square \\ \hline \end{array} \oplus \begin{array}{ c } \hline \square \\ \hline \end{array} \oplus \begin{array}{ c } \hline \square \\ \hline \end{array}$
$\square \square$	$(0, 0, 0, 2)$	324	$\mathbf{1} + \mathbf{9} + \mathbf{16} + \mathbf{44} + \mathbf{126} + \mathbf{128}$ $= \bullet \oplus \square \oplus \times \oplus \square \square \oplus \begin{array}{ c } \hline \square \\ \hline \end{array} \oplus \begin{array}{ c } \hline \square \\ \hline \end{array} \oplus \begin{array}{ c } \hline \square \\ \hline \end{array}$
	$(1, 0, 0, 1)$	1053	$\mathbf{16} + \mathbf{36} + \mathbf{84} + \mathbf{126} + \mathbf{128} + \mathbf{231} + \mathbf{432}$
	$(2, 0, 0, 0)$	1053'	$\mathbf{126} + \mathbf{432} + \mathbf{495}$
	$(0, 1, 0, 0)$	1274	$\mathbf{36} + \mathbf{84} + \mathbf{128} + \mathbf{432} + \mathbf{594}$

Table 22.2: The lowest irreps of F_4 . Shown are the Young tableaux (conjectured), the Dynkin labels, the dimensions and the decomposition of these irreps into irreps of subgroup $SO(9)$. The $\times$ denotes the spinor irrep and $\bullet$ denotes singlets.

The irrep **273** is obtained by taking the antisymmetric part of the product of two defining irreps and subtracting the adjoint rep ($\frac{1}{2} \times 26 \times 25 - 52 = 273$). It has the following F_4 and $SO(9)$ content (shaded boxes are **26** of F_4 , ordinary boxes are **9** of $SO(9)$)

$$\begin{aligned}
\begin{array}{|c|} \hline \blacksquare \\ \hline \end{array} &= (\blacksquare \otimes \blacksquare)_a \ominus \text{adj of } F_4 \\
\mathbf{273} &= \mathbf{325} - \mathbf{52} \\
&= [(\square \oplus \times \oplus \bullet) \otimes (\square \oplus \times \oplus \bullet)]_a \ominus \begin{array}{|c|} \hline \square \\ \hline \end{array} \oplus \times \\
&= \left[\begin{array}{|c|} \hline \square \\ \hline \end{array} \oplus \left(\begin{array}{|c|} \hline \square \\ \hline \end{array} \oplus \begin{array}{|c|} \hline \square \\ \hline \end{array} \right) \oplus (\boxtimes \oplus \times) \oplus \square \oplus \times \right]_a \ominus \begin{array}{|c|} \hline \square \\ \hline \end{array} \oplus \times \\
&= \begin{array}{|c|} \hline \square \\ \hline \end{array} \oplus \begin{array}{|c|} \hline \square \\ \hline \end{array} \oplus \square \oplus \boxtimes \oplus \times \\
&= \mathbf{84} + \mathbf{36} + \mathbf{9} + \mathbf{128} + \mathbf{16}
\end{aligned} \tag{22.48}$$

The irrep **324** is obtained by taking the symmetric product of two defining irreps of F_4 and subtracting a defining irrep of F_4 and a singlet

$$\begin{aligned}
\blacksquare \blacksquare &= (\blacksquare \otimes \blacksquare)_s \ominus \blacksquare \ominus \bullet \\
\mathbf{324} &= \mathbf{351} - \mathbf{26} - \mathbf{1}
\end{aligned} \tag{22.49}$$

The decomposition of the right-hand side into irreps of $SO(9)$ ⁷ is given in [3].

The group F_4 has, so far at least, not played an important role in physics. But it exists, so there may be surprises.

22.3a Definition and closure of E_6 transformations

The Lie algebra of E_6 has rank 6, as the notation indicates, and 78 generators. Its lowest-dimensional irreducible representations (irreps) are **27**, **78**, and **351** [3]. We shall take the fundamental irrep **27** as the carrier space on which the generators act. For our purposes we also need a subalgebra of E_6 whose properties are well-known. In the Kaluza-Klein reduction of $d = 11$ supergravity to $d = 5$, a coset manifold $E_6/USp(8)$ appears, but $USp(8)$ is not suitable for our purpose because its rank is 4 instead of 6. We need a maximal regular subalgebra of E_6 . A regular subalgebra is an algebra whose Cartan generators are linear combinations of the Cartan generators of the original algebra and whose rank is the same as the rank of the original algebra.

There are eight maximal subalgebras of E_6 , three of which are regular and denoted by (R) (see [3, table 15])

$$\begin{array}{llll} SU(6) \otimes SU(2)(R); & USp(8); & SO(10) \otimes U(1)(R); & SU(3) \\ SU(3) \otimes SU(3) \otimes SU(3)(R); & F_4; & G_2; & G_2 \otimes SU(3) \end{array} \quad (22.50)$$

All are semi-simple except $SO(10) \otimes U(1)$ which has the abelian ideal $U(1)$. The fundamental irrep **27** of E_6 decomposes as follows under these subalgebras

$$\begin{array}{ll} SU(6) \otimes SU(2) & : \mathbf{27} \rightarrow (\mathbf{6}^*, \mathbf{2}) + (\mathbf{15}, \mathbf{1}) \\ SU(3) \otimes SU(3) \otimes SU(3) & : \mathbf{27} \rightarrow (\mathbf{3}^*, \mathbf{3}, \mathbf{1}) + (\mathbf{3}, \mathbf{1}, \mathbf{3}) + (\mathbf{1}, \mathbf{3}^*, \mathbf{3}^*) \\ SO(10) \otimes U(1) & : \mathbf{27} \rightarrow \mathbf{16}_1 + \mathbf{10}_{-2} + \mathbf{1}_4 \\ USp(8) & : \mathbf{27} \rightarrow \mathbf{27} \\ F_4 & : \mathbf{27} \rightarrow \mathbf{26} + \mathbf{1} \\ SU(3) & : \mathbf{27} \rightarrow \mathbf{27} \\ G_2 & : \mathbf{27} \rightarrow \mathbf{27} \\ SU(3) \otimes G_2 & : \mathbf{27} \rightarrow (\mathbf{6}^*, \mathbf{1}) + (\mathbf{3}, \mathbf{7}) \end{array} \quad (22.51)$$

(Recall that the lowest-dimensional irreps of G_2 are **7**, **14**, **27**, and those of F_4 are **26**, **52**, **273**).

⁷Table 38 of [3] gives the following irreps of $SO(9)$: **9** = (1000), **16** = (0001), **44** = (2000), **126** = (0002) and **128** = (1001). The fundamental irreps of $SO(9)$ are: t_μ , $t_{\mu\nu}$, $t_{\mu\nu\rho}$, ψ^α . The tensor $t_{\mu\nu\rho\sigma}$ (**126**) is the highest irrep in the decomposition of $\psi^\alpha \otimes \psi^\beta = \mathbf{16} \times \mathbf{16}$.

The adjoint irrep **78** has the following decomposition (see [3, table 49]):

$$\begin{aligned}
SU(6) \otimes SU(2) &: \mathbf{78} \rightarrow (\mathbf{35}, \mathbf{1}) + (\mathbf{1}, \mathbf{3}) + (\mathbf{20}, \mathbf{2}) \\
SO(10) \otimes U(1) &: \mathbf{78} \rightarrow \mathbf{45}_0 + \mathbf{16}_{-3} + \mathbf{16}^*_3 + \mathbf{1}_0 \\
USp(8) &: \mathbf{78} \rightarrow \mathbf{36} + \mathbf{42} \\
F_4 &: \mathbf{78} \rightarrow \mathbf{52} + \mathbf{26}
\end{aligned} \tag{22.52}$$

We choose $SU(6) \otimes SU(2)$ as subalgebra because it is the simplest maximal regular semisimple subalgebra, but afterwards we shall discuss how the subalgebra $USp(8)$ appears in supergravity. (The subgroup $SU(6) \otimes SU(2)$ does not yield a Cartan decomposition because the number of compact generators is $35 + 3 = 38$ instead of 36.)

The carrier space **27** decomposes under $SU(6) \otimes SU(2)$ into an antisymmetric tensor $v^{ij} = -v^{ji}$ corresponding to the irrep $(\mathbf{15}, \mathbf{1})$, and a vector-spinor t_i^α corresponding to the irrep $(\mathbf{6}^*, \mathbf{2})$ where $i, j = 1, \dots, 6$ and $\alpha = 1, 2$ (see [3, table 49]).

$$\mathbf{27} \xrightarrow{SU(6) \otimes SU(2)} (\mathbf{15}, \mathbf{1})_{v^{ij}} \oplus (\mathbf{6}^*, \mathbf{2})_{t_i^\alpha} \tag{22.53}$$

The adjoint irrep **78** of E_6 decomposes under the subalgebra $SU(6) \otimes SU(2)$ into the adjoint irreps of $SU(6)$ and $SU(2)$, and extra generators with parameters $\mu^{\alpha,ijk}$ which are totally antisymmetric in ijk (see [3, table 49])

$$\mathbf{78} \xrightarrow{SU(6) \otimes SU(2)} (\mathbf{35}, \mathbf{1})_{\lambda^i_j} \oplus (\mathbf{1}, \mathbf{3})_{\sigma^\alpha_\beta} \oplus (\mathbf{20}, \mathbf{2})_{\mu^{\alpha,ijk}} \tag{22.54}$$

There is only one irrep of dimension **20** for $SU(6)$, and it corresponds to the Young tableau with 3 boxes in one column. In order that the μ parameters contain **40** real parameters we need a reality condition on them which we shall discuss later. The matrices λ^i_j and σ^α_β are antihermitian matrices of the fundamental irreps of $SU(6)$ and $SU(2)$, respectively, so they contribute 35 and 3 real parameters.

We must now write down the transformation rules of v^{ij} and t_i^α with parameters λ^i_j , σ^α_β , and $\mu^{\alpha,ijk}$. Imposing $SU(6)$ and $SU(2)$ covariance severely restricts the possible terms, and we find [4]

$$\begin{aligned}
\delta v^{ij} &= (\lambda^i_{i'} v^{i'j} + \lambda^j_{j'} v^{ij'}) + \mu_\alpha^{ijk} t_k^\alpha \\
\delta t_i^\alpha &= (\bar{\lambda}_i^j t_j^\alpha + \sigma^\alpha_\beta t_i^\beta) + \epsilon_{ijklmn} \mu^{\alpha,jkl} v^{mn}
\end{aligned}$$

After rescaling t_i^α , v^{ij} and $\mu^{\alpha,ijk}$ there are no free parameters left. The matrix $\bar{\lambda}_i^j$ is the complex conjugate of λ^i_j , and $SU(2)$ indices α and β are raised and lowered with the $SU(2)$ invariant tensor $\epsilon_{\alpha\beta} = -\epsilon_{\beta\alpha} = \epsilon^{\alpha\beta} = -\epsilon^{\beta\alpha}$

$$\bar{\lambda}_i^j = (\lambda^i_j)^*; \quad \mu_\alpha^{ijk} = \mu^{\beta,ijk} \epsilon_{\beta\alpha} \tag{22.55}$$

We must now show that the commutator of two E_6 transformation is again an E_6 transformation. It is obvious that two $SU(6)$ or two $SU(2)$ transformations give again an $SU(6)$ or $SU(2)$

transformation, respectively, and also the commutator of an $SU(6)$ or $SU(2)$ transformation with a μ transformation yields back a μ transformation, with composite parameters μ_c

$$\begin{aligned} [\delta(\mu), \delta(\lambda)] &= \delta((\mu^{\alpha,ijk})' = \lambda^i_{i'} \mu^{\alpha,i'jk} + 2 \text{ terms}) \\ [\delta(\sigma), \delta(\mu)] &= \delta(\mu_c^{\alpha,ijk} = \sigma^\alpha_\beta \mu^{\beta,ijk}) \end{aligned} \quad (22.56)$$

These results follow from the manifest $SU(6)$ and $SU(2)$ covariance. Less obvious is that two μ transformations produce an $SU(6)$ and an $SU(2)$ transformation. If we can show this, we have shown that (22.55) defines a Lie algebra, and we identify this Lie algebra with E_6 .

The commutator of two μ -transformations acting on v^{ij} must produce the composite λ parameter, and acting on t_i^α we can then determine the composite σ parameter. On v^{ij} we find

$$[\delta(\mu_2), \delta(\mu_1)] v^{ij} = \mu_1^{\beta,ijk} \epsilon_{\beta\alpha} \left(\epsilon_{kj'k'l'mn} \mu_2^{\alpha,j'k'l'} v^{mn} \right) - 1 \leftrightarrow 2 \quad (22.57)$$

To obtain an $SU(6)$ transformation with $\lambda_c^i v^{ij}$ we must move the index j onto v^{mn} . We use the ‘‘Schouten identity’’ (the fact that if one totally antisymmetrizes 7 indices which can take on only 6 values, one gets zero) to rewrite the expression in (22.57) as a sum of 6 other expressions, obtained by exchanging the contravariant index j with the indices $kj'k'l'mn$, one at a time. This yields

$$2\mu_1^{ijk} \mu_2^{j'k'l'} v^{mn} = 3\mu_1^{ij'k} \mu_2^{jk'l'} v^{mn} + 2\mu_1^{imk} \mu_2^{j'k'l'} v^{jn} \quad (22.58)$$

We have suppressed the antisymmetry in $1 \leftrightarrow 2$, both ϵ tensors and the spinor indices. The last term has already the form of an $SU(6)$ transformation with composite parameter $\lambda_c^i{}_n$. The first term on the right-hand side of (22.58) is symmetric in i, j . To show this, use the antisymmetry in $1 \leftrightarrow 2$, the symmetry under exchange of $j'k$ with $k'l'$ (due to $\epsilon_{kj'k'l'mn}$), and the antisymmetry of $\epsilon_{\beta\alpha}$. Thus, by taking the part of (22.58) antisymmetric in i, j we find that two μ transformations of v^{ij} produce a λ -transformation with composite parameter

$$\lambda_c^i{}_n = -\frac{1}{2} \mu_1^{\beta,imk} \epsilon_{\beta\alpha} \epsilon_{kj'k'l'mn} \mu_2^{\alpha,j'k'l'} - 1 \leftrightarrow 2 \quad (22.59)$$

(The minus sign is due to replacing v^{jn} by $-v^{kj}$, and the factor $\frac{1}{2}$ is needed to produce the two λ terms in δv^{ij}).

Next we consider the commutator of two μ transformations on t_i^α . The composite $SU(6)$ parameter is already fixed, so this should give us the composite $SU(2)$ parameter. We find

$$[\delta(\mu_2), \delta(\mu_1)] t_i^\alpha = \epsilon_{ijklmn} \mu_1^{\alpha,jkl} \left(\mu_2^{\gamma,mns} \epsilon_{\gamma\beta} t_s^\beta \right) - 1 \leftrightarrow 2 \quad (22.60)$$

To produce both an $SU(6)$ term with $\bar{\lambda}_{c,i}{}^s t_s^\alpha$ and an $SU(2)$ term with $\sigma_c^\alpha{}_\beta t_i^\beta$ we first apply the Schouten identity in 7 dimensions; afterwards we shall use the Schouten identity in 3 dimensions. We obtain for the right-hand side of (22.60) after exchanging the covariant index s with the indices

of the $SU(6)$ ϵ tensor

$$\mu_1^{\alpha,jkl} \mu_2^{\gamma,mns} \epsilon_{\gamma\beta} \left(\epsilon_{sjklmn} t_i^\beta + 3\epsilon_{isklmn} t_j^\beta + 2\epsilon_{ijklsn} t_m^\beta \right) - 1 \leftrightarrow 2 \quad (22.61)$$

The first term should become the $SU(2)$ transformation because it contains t_i^β , while the two last terms should yield the $SU(6)$ transformation because they contain the index i in the composite parameter. However, for an $SU(6)$ transformation we need t^α instead of t^β , and thus we use the Schouten identity in 3 dimensions to move the spinor index α onto t

$$\begin{aligned} \mu_1^\alpha \mu_2^\gamma \epsilon_{\gamma\beta} t^\beta &= -\mu_1^\gamma \mu_2^\beta \epsilon_{\gamma\beta} t^\alpha - \mu_1^\beta \mu_2^\alpha \epsilon_{\gamma\beta} t^\gamma \\ \mu_1^\alpha \mu_2^\gamma \epsilon_{\gamma\beta} t^\beta &= -\frac{1}{2} \left(\mu_1^\gamma \mu_2^\beta \epsilon_{\gamma\beta} \right) t^\alpha \end{aligned} \quad (22.62)$$

Substituting this identity into (22.61) yields

$$\left(\mu_1^{\alpha,jk} \mu_{2,\beta}^{mns} \epsilon_{sjklmn} - 1 \leftrightarrow 2 \right) t_i^\beta + \left(-\frac{1}{2} \mu_{1,\beta}^{skl} \mu_2^{\beta,mnj} \epsilon_{ijklmn} - 1 \leftrightarrow 2 \right) t_s^\alpha \quad (22.63)$$

(The last term in (22.61) cancels the one-but-last term partly: $-\frac{3}{2} + 1 = -\frac{1}{2}$). The first term in parentheses defines $\sigma_{c,\beta}^\alpha$, while the second term is indeed $\bar{\lambda}_{c,i}^s$ as we now show.

To prove this we need a reality condition on $\mu^{\alpha,jk}$. A natural definition is

$$\left(\mu^{\alpha,jk} \right)^* = \frac{1}{3!} \epsilon_{ijk i' j' k'} \epsilon_{\alpha\alpha'} \mu^{\alpha', i' j' k'} \quad (22.64)$$

This condition is consistent: taking the complex conjugate of this relation and then inserting the relation yields

$$\mu^{\alpha,jk} = \frac{1}{3!} \epsilon^{ijk i' j' k'} \epsilon^{\alpha\alpha'} \left(\frac{1}{3!} \epsilon_{i' j' k' i'' j'' k''} \epsilon_{\alpha' \alpha''} \right) \mu^{\alpha'', i'' j'' k''} \quad (22.65)$$

and the product of the four ϵ -tensors yields two minus signs which cancel each other. It is now straightforward to prove the following equality

$$\begin{aligned} \left(\lambda_c^i \right)^* &= -\frac{1}{2} \left(\mu_1^{\beta, imk} \epsilon_{\beta\alpha} \epsilon_{kj' k' l' mn} \mu_2^{\alpha, j' k' l'} - 1 \leftrightarrow 2 \right)^* \quad \text{from (22.59)} \\ &= -\frac{1}{2} \left(\mu_1^{\gamma, skl} \epsilon_{\gamma\beta} \mu_2^{\beta, mnj} \epsilon_{ijklmn} - 1 \leftrightarrow 2 \right) \quad \text{from (22.63)} \end{aligned}$$

Thus the commutator of two μ transformations also closes on t_i^α , and thus we have proven the existence of a Lie algebra E_6 .

Let us now comment on how the group E_6 and the subgroup $USp(8)$ appear in supergravity. If one dimensionally reduces the fields g_{MN} , A_{MNP} (totally antisymmetric tensor) and ψ_M of $d = 11$ supergravity to $d = 5$ on a 6-torus, with M, N eleven-dimensional, μ, ν five-dimensional and α, β six-dimensional indices, one finds the field content in $d = 5$ as given in table 22.3. (Both the $d = 11$ supergravity and the $d = 5$ supergravity are in Minkowski space, and in both cases Majorana spinors exist.)

$d = 11$ fields	$d = 5$ fields	Number of fields
$g_{\mu\nu}$	spin 2, graviton	1
$g_{\mu\alpha}$	graviphotons (vectors)	6
$A_{\mu\alpha\beta}$	vectors	15
$A_{\mu\nu\alpha}$	dual to vectors	6
$g_{\alpha\beta}$	graviscalars	21
$A_{\mu\nu\rho}$	dual to scalar	1
$A_{\alpha\beta\gamma}$	scalars	20
ψ_μ	$\frac{32}{4}$ gravitinos	8 spin $\frac{3}{2}$
ψ_α	$6 \times \frac{32}{4}$ real spin $\frac{1}{2}$	48 spin $\frac{1}{2}$

Table 22.3:

The vectors should form an irrep of a rigid symmetry group G , the spin $\frac{3}{2}$ and $\frac{1}{2}$ fields should form an irrep of a local subgroup H , and the scalars should parametrize the coset manifold G/H [4]. So $\dim G - \dim H = 42$, and H should have the irreps **8** and **48**. This suggests $H = USp(8)$ because the dimension of the 3-index antisymmetric tensor representation is $\binom{8}{3} - 8 = 48$. Then $\dim H = 36$, so $\dim G = 78$. This yields $G = E_6$. Confirmation is obtained from the vectors since **27** is indeed an irrep of G .

There is yet further evidence for E_6 and $USp(8)$. The decomposition of G into H and coset generators should be maximally noncompact: the number of noncompact generators minus the number of compact generators should be the rank of the algebra (a Cartan decomposition). This is the case for E_6 and $USp(8)$ ($78=36+42$ and $42-36=6$). Furthermore, one can choose a “unitary” gauge in which all H parts of G group elements are set to zero. In this unitary gauge only a diagonal rigid H symmetry remains, and now all particles with a given spin should form multiplets of H . Indeed, for spin 1 the **27** is not only an irrep of E_6 , but also the $\begin{smallmatrix} \square \\ \square \end{smallmatrix}$ of $USp(8)$, and **42** is the $\begin{smallmatrix} \square \\ \square \\ \square \end{smallmatrix}$ of $USp(8)$.

Since the **27** of E_6 is irreducible under $USp(8)$, we can take as carrier space the tensor $z^{\alpha\beta} = -z^{\beta\alpha}$ with $\alpha = 1, \dots, 8$ which is symplectic traceless: $z^{\alpha\beta}\Omega_{\alpha\beta} = 0$ with $\Omega^{\alpha\beta}\Omega_{\gamma\beta} = \delta_\gamma^\alpha$ and $\Omega = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}_{8 \times 8}$. Furthermore, the adjoint irrep **78** splits under $USp(8)$ into the adjoint irrep **36** of $USp(8)$ and a **42** irrep of $USp(8)$. The latter corresponds to a 4-index totally antisymmetric tensor $\Sigma^{\alpha\beta\gamma\sigma}$ which is symplectic-traceless: $\binom{8}{4} - \frac{1}{2} \cdot 8 \cdot 7 = \mathbf{42}$. We can then write in manifestly $USp(8)$ covariant form

$$\delta z^{\alpha\beta} = \left(\Lambda^\alpha_\gamma z^{\gamma\beta} + \Lambda^\beta_\gamma z^{\alpha\gamma} \right) + \Sigma^{\alpha\beta}_{\gamma\delta} z^{\gamma\delta} \quad (22.66)$$

The 8×8 $USp(8)$ matrices Λ^α_γ are antihermitian and symplectic

$$\bar{\Lambda} = -\Lambda^T; \quad \Lambda^T \Omega + \Omega \Lambda = 0 \quad (22.67)$$

With indices these equations read

$$(\bar{\Lambda})_{\alpha}^{\beta} \equiv (\Lambda^{\alpha}_{\beta})^* = -(\Lambda^T)_{\alpha}^{\beta} \equiv -\Lambda^{\beta}_{\alpha} \quad (22.68)$$

$$(\Lambda^T)_{\alpha}^{\beta} \Omega_{\beta\gamma} + \Omega_{\alpha\beta} \Lambda^{\beta}_{\gamma} = 0 \Rightarrow \Lambda_{\gamma\alpha} = \Lambda_{\alpha\gamma} \quad (22.69)$$

Hence the matrices for the defining representation of the Lie algebra of $USp(8)$ satisfy

$$(\bar{\Lambda})_{\alpha}^{\beta} = -\Lambda_{\alpha}^{\beta} \quad \text{where } \Lambda_{\alpha}^{\beta} = \Lambda^{\alpha'}_{\beta'} \Omega_{\alpha'\alpha} \Omega^{\beta\beta'}. \quad (22.70)$$

The symmetry condition leaves $\frac{1}{2} \cdot 8 \cdot 9 = 36$ complex parameters in Λ^{α}_{β} , but the antihermiticity condition is a reality condition which leaves 36 real parameters. Furthermore, $\Sigma_{\alpha\beta\gamma\delta} \equiv \Sigma^{\alpha'\beta'}_{\gamma\delta} \Omega_{\alpha'\alpha} \Omega_{\beta'\beta}$ is totally antisymmetric, traceless with respect to $\Omega^{\alpha\beta}$. In order that also $\Sigma^{\alpha\beta\gamma\delta}$ contains 42 real parameters we impose the reality condition

$$(\Sigma^{\alpha\beta\gamma\delta})^* = \Sigma^{\alpha'\beta'\gamma'\delta'} \Omega_{\alpha'\alpha} \dots \Omega_{\delta'\delta} \quad (22.71)$$

One can also use the 8-index antisymmetric tensor $\epsilon_{\alpha\beta\gamma\delta\alpha'\beta'\gamma'\delta'}$ to impose a reality condition. This is equivalent because $\epsilon_{\alpha\beta\gamma\delta\alpha'\beta'\gamma'\delta'} = \Omega_{[\alpha\alpha'} \Omega_{\beta\beta'} \Omega_{\gamma\gamma'} \Omega_{\delta\delta']}$.

We expect now that the commutator of two Σ transformations produces a $USp(8)$ transformation with a composite parameters $\Lambda_{\epsilon'}$. We get $[\delta(\Sigma_2), \delta(\Sigma_1)] z^{\alpha\beta} = (\Sigma_1^{\alpha\beta}_{\gamma\delta} \Sigma_2^{\gamma\delta}_{\epsilon\zeta}) z^{\epsilon\zeta} - 1 \leftrightarrow 2$. We refer to [5] for the rest of the proof. The fact that E_6 and $USp(8)$ appear in 5 dimensions was first noted in [4].

22.3b Roots of E_6

We shall now go through the usual steps: we construct the weights of the defining irrep **27**, the positive roots of E_6 , and then the simple roots of E_6 . Finally we shall construct the fundamental weights of E_6 and the extended Dynkin diagram.

To obtain the weights of the **27** of E_6 and the roots of E_6 , it is useful to first obtain the weights of the defining irreps of $SU(6)$ and $SU(2)$. The Cartan generators of $SU(6)$ and $SU(2)$ together form the Cartan generators H_I of E_6 . This implies that the subalgebra is maximal and regular. We shall fix their normalization such that the trace $\text{tr}_{27} H_I H_J$ over the **27** dimensional defining irrep of E_6 is proportional to δ_{ij} (with $i, j = 1, \dots, 6$). Then we shall obtain the roots of E_6 by adding the $SU(6)$ and $SU(2)$ weights of the **(20, 2)** to the roots of $SU(6)$ and $SU(2)$.

The Cartan generators of $SU(6)$ and $SU(2)$ in the defining irrep of each are given by

$$\begin{aligned}
H_1 &= \frac{1}{2} \begin{pmatrix} 1 & & & & \\ & -1 & & & \\ & & 0 & & \\ & & & 0 & \\ & & & & 0 \end{pmatrix}, & H_2 &= \frac{1}{\sqrt{12}} \begin{pmatrix} 1 & & & & \\ & 1 & & & \\ & & -2 & & \\ & & & 0 & \\ & & & & 0 \end{pmatrix} \\
H_3 &= \frac{1}{\sqrt{24}} \begin{pmatrix} 1 & & & & \\ & 1 & & & \\ & & 1 & & \\ & & & -3 & \\ & & & & 0 \end{pmatrix}, & H_4 &= \frac{1}{\sqrt{40}} \begin{pmatrix} 1 & & & & \\ & 1 & & & \\ & & 1 & & \\ & & & 1 & \\ & & & & -4 \end{pmatrix} \\
H_5 &= \frac{1}{\sqrt{60}} \begin{pmatrix} 1 & & & & \\ & 1 & & & \\ & & 1 & & \\ & & & 1 & \\ & & & & 1 \end{pmatrix}, & H_6 &= \frac{1}{2} \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}
\end{aligned} \tag{22.72}$$

The first five matrices and the last matrix are normalized to $\text{tr} H_i H_j = \frac{1}{2} \delta_{ij}$. The defining irrep **27** of E_6 contains the tensors v^{ij} for $(\mathbf{15}, \mathbf{1})$ and t_i^α for $(\mathbf{6}^*, \mathbf{2})$. The trace of H_1^2 and H_6^2 over this 27-dimensional space decomposes then in a trace over a 15-dimensional space and a trace over a 12-dimensional space, but the latter trace factorizes into a 6-dimensional trace times a 2-dimensional trace

$$\begin{aligned}
\text{tr}_{27} H_1^2 &= \sum_{1 \leq i < j \leq 6} (\mu_1^i + \mu_1^j)^2 + 2 \sum_{1 \leq i \leq 6} (\mu_1^i)^2 = \frac{1}{2} \sum_{i,j} (\mu_1^i + \mu_1^j)^2 \\
&= \sum_{i,j} (\mu_1^i)^2 = 6 \text{tr} (H_1^{\text{def}})^2 = 6 \cdot \frac{1}{2} = 3
\end{aligned} \tag{22.73}$$

$$\text{tr}_{27} H_6^2 = 6 \left(\frac{1}{4} + \frac{1}{4} \right) = 3 \tag{22.74}$$

Hence the normalizations of the Cartan generators H_j for E_6 with $1 \leq j \leq 6$ satisfy $\text{tr} H_i H_j = 3 \delta_{ij}$.

The roots of E_6 are the roots of $SU(6)$, the roots of $SU(2)$ and the weights of the $(\mathbf{20}, \mathbf{2})$ multiplet. The components of the latter correspond to the tensor $\mu^{\alpha,ijk}$, and its weights follow

from the weights of the defining irrep of $SU(6)$ and $SU(2)$. These are given by

$$\left. \begin{aligned} \vec{\mu}^1 &= \left(\frac{1}{2}, \frac{1}{\sqrt{12}}, \frac{1}{\sqrt{24}}, \frac{1}{\sqrt{40}}, \frac{1}{\sqrt{60}} \right); & \vec{\mu}^4 &= \left(0, 0, \frac{-3}{\sqrt{24}}, \frac{1}{\sqrt{40}}, \frac{1}{\sqrt{60}} \right); \\ \vec{\mu}^2 &= \left(-\frac{1}{2}, \frac{1}{\sqrt{12}}, \frac{1}{\sqrt{24}}, \frac{1}{\sqrt{40}}, \frac{1}{\sqrt{60}} \right); & \vec{\mu}^5 &= \left(0, 0, 0, -\frac{4}{\sqrt{40}}, \frac{1}{\sqrt{60}} \right); \\ \vec{\mu}^3 &= \left(0, \frac{-2}{\sqrt{12}}, \frac{1}{\sqrt{24}}, \frac{1}{\sqrt{40}}, \frac{1}{\sqrt{60}} \right); & \vec{\mu}^6 &= \left(0, 0, 0, 0, \frac{-5}{\sqrt{60}} \right); \end{aligned} \right\} \text{ for } SU(6) \quad (22.75)$$

$$\vec{\mu}^1 = \left(\frac{1}{2} \right); \quad \vec{\mu}^2 = \left(-\frac{1}{2} \right); \quad \text{for } SU(2).$$

The simple roots of $SU(6)$ and $SU(2)$ (5-vectors and 1-vectors), denoted by $\vec{\alpha}$ and $\vec{\beta}$, are also roots of E_6 (6-vectors)

$$\begin{aligned} \vec{\alpha}^{12} &\equiv \vec{\alpha}^1 = (1, 0, 0, 0, 0, 0); & \vec{\alpha}^{45} &\equiv \vec{\alpha}^4 = \left(0, 0, \frac{-3}{\sqrt{24}}, \frac{5}{\sqrt{40}}, 0, 0 \right) \\ \vec{\alpha}^{23} &\equiv \vec{\alpha}^2 = \left(-\frac{1}{2}, \frac{3}{\sqrt{12}}, 0, 0, 0, 0 \right); & \vec{\alpha}^{56} &\equiv \vec{\alpha}^5 = \left(0, 0, 0, \frac{-4}{\sqrt{40}}, \frac{6}{\sqrt{60}}, 0 \right) \\ \vec{\alpha}^{34} &\equiv \vec{\alpha}^3 = \left(0, \frac{-2}{\sqrt{12}}, \frac{4}{\sqrt{24}}, 0, 0, 0 \right); & \vec{\beta}^{12} &= (0, 0, 0, 0, 0, 1) \end{aligned} \quad (22.76)$$

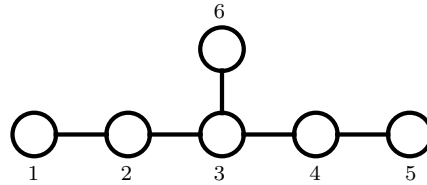
The roots of the generators corresponding to $\mu^{\alpha,ijk}$ are the following 6-component vectors

$$\vec{\alpha}(\pm, ijk) = \left(\vec{\mu}^i + \vec{\mu}^j + \vec{\mu}^k, \pm \frac{1}{2} \right) \quad \text{with } i < j < k. \quad (22.77)$$

It is clear that the root $(\vec{\mu}^1 + \vec{\mu}^2 + \vec{\mu}^4, \frac{1}{2})$ is not a simple root, because it is the sum of the two positive roots $(\vec{\mu}^1 + \vec{\mu}^3 + \vec{\mu}^4, \frac{1}{2})$ and $\vec{\alpha}^{23} = (\vec{\mu}^2 - \vec{\mu}^3, 0)$. Also $(\vec{\mu}^1 + \vec{\mu}^2 + \vec{\mu}^3, \frac{1}{2})$ is not simple because it is equal to $(\vec{\mu}^1 + \vec{\mu}^2 + \vec{\mu}^4, \frac{1}{2})$ and $(\vec{\mu}^3 - \vec{\mu}^4, 0)$. Only $(\vec{\mu}^4 + \vec{\mu}^5 + \vec{\mu}^6, \frac{1}{2})$ cannot be written as a sum of two positive roots. We denote it by $\vec{\alpha}^6$. Thus the 6 simple roots of E_6 are the 5 simple roots of $SU(6)$ given in (22.76), and the weight $\vec{\alpha}^6$

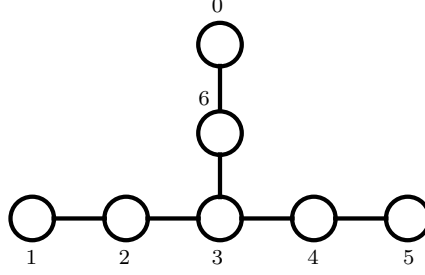
$$\vec{\alpha}^6 = \left(\vec{\mu}^4 + \vec{\mu}^5 + \vec{\mu}^6, \frac{1}{2} \right) = \left(0, 0, -\frac{3}{\sqrt{24}}, -\frac{3}{\sqrt{40}}, -\frac{3}{\sqrt{60}}, \frac{1}{2} \right) \quad (22.78)$$

Since $\vec{\alpha}^6$ has only a non-vanishing inner product with $\vec{\alpha}^{34}$ which is $-\frac{1}{2}$, and the length of all simple roots is unity, the Dynkin diagram of E_6 is given by



To obtain the extended Dynkin diagram for E_6 , we must add minus the root with the highest weight to the simple roots. This is $-\vec{\beta}^{12} = (0, 0, 0, 0, 0, -1)$, and it has only a nonvanishing inner

product with $\vec{\alpha}^6$. The extended Dynkin diagram is thus



Finally we construct the fundamental weights $\vec{\mu}^j$. They are defined by $\frac{2\vec{\mu}^i \cdot \vec{\alpha}^j}{\vec{\alpha}^j \cdot \vec{\alpha}^j} = \delta^{ij}$. One finds by straightforward but tedious algebra

$$\begin{aligned} \vec{\mu}^6 &= (0, 0, 0, 0, 0, 1); & \vec{\mu}^3 &= \left(0, 0, \sqrt{\frac{3}{8}}, \sqrt{\frac{9}{40}}, \sqrt{\frac{6}{40}}, \frac{3}{2}\right); \\ \vec{\mu}^5 &= \left(0, 0, 0, 0, \sqrt{\frac{5}{12}}, \frac{1}{2}\right); & \vec{\mu}^2 &= \left(0, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{10}}, \frac{1}{\sqrt{15}}, \frac{1}{5}\right); \\ \vec{\mu}^4 &= \left(0, 0, 0, \sqrt{\frac{2}{5}}, \sqrt{\frac{4}{15}}, 1\right); & \vec{\mu}^1 &= \left(\frac{1}{2}, \frac{1}{\sqrt{12}}, \frac{1}{\sqrt{24}}, \frac{1}{\sqrt{40}}, \frac{1}{\sqrt{60}}, \frac{1}{2}\right); \end{aligned} \quad (22.79)$$

For convenience we record again the simple roots

$$\begin{aligned} \vec{\alpha}^1 &= (1, 0, 0, 0, 0, 0); & \vec{\alpha}^4 &= \left(0, 0, -\sqrt{\frac{3}{8}}, \sqrt{\frac{5}{8}}, 0, 0\right) \\ \vec{\alpha}^2 &= \left(-\frac{1}{2}, \sqrt{\frac{3}{4}}, 0, 0, 0, 0\right); & \vec{\alpha}^5 &= \left(0, 0, 0, -\sqrt{\frac{2}{5}}, \sqrt{\frac{3}{5}}, 0\right) \\ \vec{\alpha}^3 &= \left(0, -\sqrt{\frac{1}{3}}, \sqrt{\frac{2}{3}}, 0, 0, 0\right); & \vec{\alpha}^6 &= \left(0, 0, -\frac{3}{\sqrt{24}}, -\frac{3}{\sqrt{40}}, -\frac{3}{\sqrt{60}}, \frac{1}{2}\right) \end{aligned} \quad (22.80)$$

We identify $\vec{\mu}^1$ with the highest weight of the defining **27** irrep of E_6 , which is $(\vec{\mu}^1, \frac{1}{2})$, and $\vec{\mu}^6$ with the highest root of the adjoint representation, which is $\vec{\beta}^{12} = (\vec{0}, 1)$. Finally $\vec{\mu}^5$ is the highest weight of the **27***, because the lowest weight of the **27** is $(\vec{\mu}^6, -\frac{1}{2}) = (0, 0, 0, 0, -\sqrt{\frac{5}{12}}, -\frac{1}{2})$.

The group E_6 has been used extensively in Grand Unified Theories, and also in supergravity as discussed above.

22.4a Definition and closure of E_7 transformations

The exceptional Lie algebra E_7 has rank 7, and 133 generators. Its lowest-dimensional irreps are **56**, **133** and **912**. Its maximal subgroups (see [3, table 58]) are

$$\begin{array}{cccc} E_6 \otimes U(1)(R) & SU(6) \otimes SU(3)(R) & SU(3) & SU(2) \otimes F_4 \\ SU(8)(R) & SU(2) & SU(2) \otimes G_2 & G_2 \otimes USp(6) \\ SO(12) \otimes SU(2)(R) & SU(2) & SU(2) \otimes SU(2) & \end{array} \quad (22.81)$$

From the extended Dynkin diagram (to be discussed) we read off the 3 maximal regular semisimple Lie subgroups.

$$\begin{array}{c}
 \begin{array}{ccccccc}
 & & & 7 & & & \\
 & & & \bigcirc & & & \\
 & & & | & & & \\
 \bigcirc & - & \bigcirc & - & \bigcirc & - & \bigcirc \\
 0 & & 1 & & 2 & & 3 & & 4 & & 5 & & 6
 \end{array} \\
 SU(2) \otimes SO(12) \\
 SU(3) \otimes SU(6) \\
 \text{not } SU(4) \otimes SU(4) \otimes SU(2) \\
 SU(8)
 \end{array} \tag{22.82}$$

The subgroup $SU(4) \otimes SU(4) \otimes SU(2)$ is one of the five cases where the subgroup obtained by removing a dot from the extended Dynkin diagram is not maximal (in E_8 , there are 3 exceptions, and in F_4 also one exception): 3 removed is a subgroup of the group which one obtains by removing 1 or 5 ($SU(4) \otimes SU(4) \otimes SU(2)$ is a subalgebra of $SO(12) \otimes SU(2)$).

There is also one maximal regular nonsemisimple Lie subalgebra: $U(1) \otimes E_6$ (obtained by removing the 6 from the Dynkin diagram of E_7).

A natural choice for H is clearly $SU(8)$. This is also the subalgebra which appears if one dimensionally reduces $d = 11$ supergravity to $d = 4$ [6]. One finds then $G = E_7$ and $H = SU(8)$.

Under $SU(8)$ the carrier space (fundamental irrep) **56** and the adjoint representation **133** decompose as follows

$$\begin{aligned}
 \mathbf{56} &\rightarrow \mathbf{28} + \mathbf{28}^* \\
 \mathbf{133} &\rightarrow \mathbf{63} + \mathbf{70}.
 \end{aligned} \tag{22.83}$$

The **28** of $SU(8)$ corresponds to the Young tableau $\begin{array}{|c|} \hline \square \\ \hline \end{array}$, and the **70** corresponds to $\begin{array}{|c|} \hline \square \\ \square \\ \hline \end{array}$. (The **70** satisfies a reality condition, see below, and for this reason it contributes 70 real parameters.) The **63** is of course the adjoint irrep of $SU(8)$. The decomposition of E_7 into $SU(8)$ and coset generators is a Cartan decomposition, $\dim(G/H) - \dim H = \text{rank } G$, namely $70 - 63 = 7$. Thus we obtain a symmetric algebra: the commutator of the two coset transformation will yield a subgroup transformation.

It is straightforward to write down the transformation rules which give the transformation rules of E_7 on the carrier space $\mathbf{56} = \mathbf{28} + \mathbf{28}^*$. (If we only allow real group parameters, the **56** remains irreducible under $SU(8)$.) Imposing $SU(8)$ covariance one finds [6]

$$\begin{aligned}
 \delta z^{ij} &= (\Lambda_k^i z^{kj} + \Lambda_k^j z^{ik}) + \frac{1}{4!} \epsilon^{ijklmnop} \Sigma_{mnop} \bar{z}_{kl} \\
 \delta \bar{z}_{ij} &= (\bar{\Lambda}_i^k \bar{z}_{kj} + \bar{\Lambda}_j^k \bar{z}_{ik}) + \Sigma_{ijkl} z^{kl}
 \end{aligned} \tag{22.84}$$

We defined $(\Lambda_k^i)^* = \bar{\Lambda}_i^k$ and $(z^{ij})^* = \bar{z}_{ij}$. In order that $(\delta z^{ij})^* = \delta \bar{z}_{ij}$, we must impose a reality

condition on totally antisymmetric Σ_{ijkl} , namely selfduality

$$(\Sigma_{ijkl})^* = \frac{1}{4!} \epsilon^{ijklmnop} \Sigma_{mnop} \quad (22.85)$$

The anti-hermiticity of the 8×8 $SU(8)$ matrices Λ implies that $\bar{\Lambda}_i^j = -\Lambda_i^j$. The Λ_k^i are also traceless $\Lambda_i^i = 0$. We shall see that this is a noncompact form of E_7 : the Λ_j^i are 63 compact generators, and the 70 generators Σ_{ijkl} are noncompact. We need reality conditions on Λ_j^i and Σ_{ijkl} in order to end up with 63 real and 70 real parameters. This definition of E_7 in terms of $SU(8)$ resembles the definition of G_2 in terms of $SU(3)$, but $E_7/SU(8)$ yields a Cartan decomposition.

Let us now prove that (22.84) defines a closed symmetric Lie algebra. From the $SU(8)$ covariance we obtain closure of two Λ transformations, or one Λ and one Σ transformation. So let us evaluate the commutator of two Σ transformation on z^{ij} . One finds

$$\left[\delta \left(\Sigma^{(2)} \right), \delta \left(\Sigma^{(1)} \right) \right] z^{ij} = \frac{1}{4} \epsilon^{ij \underline{kl} mnop} \Sigma_{\underline{mnop}}^{(1)} \Sigma_{\underline{klrs}}^{(2)} z^{\underline{rs}} - 1 \leftrightarrow 2 \quad (22.86)$$

We have underlined contracted indices. There is an identity which casts the right-hand side into an $SU(8)$ transformation of z^{ij} , thereby proving closure and symmetry

$$\epsilon^{ij \underline{kl} mnop} \Sigma_{\underline{mnop}}^{(1)} \Sigma_{\underline{klrs}}^{(2)} = \frac{1}{14} \left(\delta_r^i E^j_s - \delta_r^j E^i_s - \delta_s^i E^j_r + \delta_s^j E^i_r \right) \quad (22.87)$$

where E^i_r has one more contraction

$$E^i_r = \epsilon^{i \underline{kl} mnop} \Sigma_{\underline{mnop}}^{(1)} \Sigma_{r \underline{tkl}}^{(2)} \quad (22.88)$$

For further details of E_7 , see [6].

22.4b Roots of E_7

To find the roots of E_7 , we need to first choose the 7 Cartan generators. For these we take of course the 7 Cartan generators H_i of $SU(8)$. Then the roots of E_7 are the 56 roots of $SU(8)$ which we know, and the weights of the **70** under $SU(8)$. The latter follow from the weights of the defining representation.

The Cartan generators H_i of $SU(8)$ are 8×8 diagonal matrices whose entries yield the weights $\vec{\mu}^i$ ($i = 1, \dots, 8$) of the defining irrep of $SU(8)$. Specifically, if we normalize the H_i such that $\text{tr } H_i H_j = \frac{1}{2} \delta_{ij}$ we obtain (suppressing the subscripts def for notational clarity)

$$\begin{aligned} \vec{\mu}_1 &= \left(\frac{1}{2}, \frac{1}{\sqrt{12}}, \frac{1}{\sqrt{24}}, \frac{1}{\sqrt{40}}, \frac{1}{\sqrt{60}}, \frac{1}{\sqrt{84}}, \frac{1}{\sqrt{112}} \right) \\ \vec{\mu}_2 &= \left(-\frac{1}{2}, \frac{1}{\sqrt{12}}, \frac{1}{\sqrt{24}}, \frac{1}{\sqrt{40}}, \frac{1}{\sqrt{60}}, \frac{1}{\sqrt{84}}, \frac{1}{\sqrt{112}} \right) \end{aligned}$$

$$\begin{aligned}
\vec{\mu}_3 &= \left(0, \frac{-2}{\sqrt{12}}, \frac{1}{\sqrt{24}}, \frac{1}{\sqrt{40}}, \frac{1}{\sqrt{60}}, \frac{1}{\sqrt{84}}, \frac{1}{\sqrt{112}}\right) \\
\vec{\mu}_4 &= \left(0, 0, \frac{-3}{\sqrt{24}}, \frac{1}{\sqrt{40}}, \frac{1}{\sqrt{60}}, \frac{1}{\sqrt{84}}, \frac{1}{\sqrt{112}}\right) \\
\vec{\mu}_5 &= \left(0, 0, 0, \frac{-4}{\sqrt{40}}, \frac{1}{\sqrt{60}}, \frac{1}{\sqrt{84}}, \frac{1}{\sqrt{112}}\right) \\
\vec{\mu}_6 &= \left(0, 0, 0, 0, \frac{-5}{\sqrt{60}}, \frac{1}{\sqrt{84}}, \frac{1}{\sqrt{112}}\right) \\
\vec{\mu}_7 &= \left(0, 0, 0, 0, 0, \frac{-6}{\sqrt{84}}, \frac{1}{\sqrt{112}}\right) \\
\vec{\mu}_8 &= \left(0, 0, 0, 0, 0, 0, \frac{-7}{\sqrt{112}}\right)
\end{aligned} \tag{22.89}$$

The 28 positive roots of $SU(8)$ are $\vec{\mu}^i - \vec{\mu}^j$ with $1 \leq i < j \leq 8$. The 7 simple roots of $SU(8)$ are

$$\begin{aligned}
\vec{\alpha}^{12} \equiv \vec{\alpha}^1 &= (1, 0, 0, 0, 0, 0, 0); & \vec{\alpha}^{45} \equiv \vec{\alpha}^4 &= \left(0, 0, \frac{-3}{\sqrt{24}}, \frac{5}{\sqrt{40}}, 0, 0, 0\right) \\
\vec{\alpha}^{23} \equiv \vec{\alpha}^2 &= \left(-\frac{1}{2}, \frac{3}{\sqrt{12}}, 0, 0, 0, 0, 0\right); & \vec{\alpha}^{56} \equiv \vec{\alpha}^5 &= \left(0, 0, 0, \frac{-4}{\sqrt{40}}, \frac{6}{\sqrt{60}}, 0, 0\right) \\
\vec{\alpha}^{34} \equiv \vec{\alpha}^3 &= \left(0, \frac{-2}{\sqrt{12}}, \frac{4}{\sqrt{24}}, 0, 0, 0, 0\right); & \vec{\alpha}^{67} \equiv \vec{\alpha}^6 &= \left(0, 0, 0, 0, \frac{-5}{\sqrt{60}}, \frac{7}{\sqrt{84}}, 0\right) \\
& & \vec{\alpha}^{78} &= \left(0, 0, 0, 0, 0, \frac{-6}{\sqrt{84}}, \frac{8}{\sqrt{112}}\right)
\end{aligned} \tag{22.90}$$

The 70 weights of the generators T^{ijkl} associated to the parameters Σ_{ijkl} are

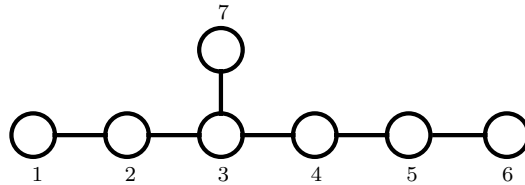
$$\vec{\mu}(\Sigma) = (\vec{\mu}^i + \vec{\mu}^j + \vec{\mu}^k + \vec{\mu}^l) \quad \text{with } 1 \leq i < j < k < l \leq 8. \tag{22.91}$$

We must now find the simple roots of E_7 . From our experience with E_6 we expect that the last of the simple roots of $SU(8)$ must be replaced by a Σ weight. As we shall shortly explain, this Σ weight is

$$\vec{\alpha}^7 = \vec{\mu}^4 + \vec{\mu}^5 + \vec{\mu}^6 + \vec{\mu}^7 = \left(0, 0, \frac{-3}{\sqrt{24}}, \frac{-3}{\sqrt{40}}, \frac{-3}{\sqrt{60}}, \frac{-3}{\sqrt{84}}, \frac{4}{\sqrt{112}}\right). \tag{22.92}$$

The root $\vec{\alpha}^{78}$ is a simple root of $SU(8)$ but not of E_7 , as we shall show.

All 7 simple roots of E_7 have the same length, and $\vec{\alpha}^7$ has only a nonvanishing inner product with $\vec{\alpha}^3$, yielding $\frac{\vec{\alpha}^7 \cdot \vec{\alpha}^3}{|\vec{\alpha}^3||\vec{\alpha}^7|} = -\frac{1}{2}$. Hence the Dynkin diagram is

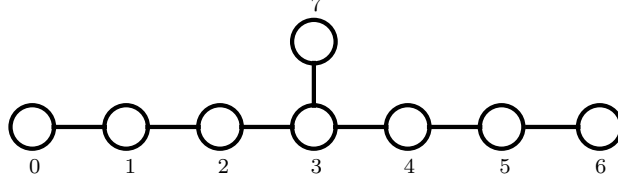


To obtain the extended Dynkin diagram of E_7 we must add the negative of the root of E_7 with

highest weight. This is

$$-\vec{\theta} = -(\vec{\mu}^1 - \vec{\mu}^8) = -\left(\frac{1}{2}, \frac{1}{\sqrt{12}}, \frac{1}{\sqrt{24}}, \frac{1}{\sqrt{40}}, \frac{1}{\sqrt{60}}, \frac{1}{\sqrt{84}}, \frac{8}{\sqrt{112}}\right) \quad (22.93)$$

Also, this vector has length unity, and only a nonvanishing inner product with $\vec{\alpha}^1$ whose value is $-\frac{1}{2}$. The extended Dynkin diagram becomes then



From this diagram we obtained the subgroups in (22.82).

We now discuss the construction of the simple roots and fundamental representations of E_7 . The roots of E_7 are the roots of $SU(8)$ and further the 70 weights of the 4-index antisymmetric tensor representation were obtained before

$$\begin{aligned} & \pm (\vec{\mu}^{(i)} - \vec{\mu}^{(j)}) \quad \text{with } 1 \leq i < j \leq 8 \quad (56 \text{ roots of } SU(8)) \\ & \vec{\mu}^{(i)} + \vec{\mu}^{(j)} + \vec{\mu}^{(k)} + \vec{\mu}^{(l)} \quad \text{with } 1 \leq i < j < k < l \leq 8 \quad (70 \text{ weights of } t^{ijkl}) \end{aligned} \quad (22.94)$$

The 70 weights split into 35 weights $\vec{\mu}^{(i)} + \vec{\mu}^{(j)} + \vec{\mu}^{(k)} + \vec{\mu}^{(l)}$ where i, j, k, l run from 1 till 7, and 35 more weights $\vec{\mu}^{(i)} + \vec{\mu}^{(j)} + \vec{\mu}^{(k)} + \vec{\mu}^{(8)}$ where i, j, k run from 1 till 7. The latter 35 weights can also be written as⁸ $-(\vec{\mu}^{(i)} + \vec{\mu}^{(j)} + \vec{\mu}^{(k)} + \vec{\mu}^{(l)})$ with $1 \leq i < j < k < l \leq 7$ since $\vec{\mu}^{(8)} = -\vec{\mu}^{(1)} - \vec{\mu}^{(2)} - \dots - \vec{\mu}^{(7)}$.

The positive roots of E_7 are the 28 roots $\vec{\mu}^{(i)} - \vec{\mu}^{(j)}$ of $SU(8)$ with $i < j$ and the 35 weights $\vec{\mu}^{(i)} + \vec{\mu}^{(j)} + \vec{\mu}^{(k)} + \vec{\mu}^{(l)}$ with $1 \leq i < j < k < l \leq 7$. Then the simple roots are those of $SU(8)$ except the last one which can be written as sum of two positive weights of E_7

$$\vec{\alpha}^{78} = (\vec{\mu}^{(1)} + \vec{\mu}^{(2)} + \vec{\mu}^{(3)} + \vec{\mu}^{(7)}) + (\vec{\mu}^{(4)} + \vec{\mu}^{(5)} + \vec{\mu}^{(6)} + \vec{\mu}^{(7)}) \quad (22.95)$$

Instead of $\vec{\alpha}^{78}$ we need the lowest but still positive tensor weight of E_7 . This is $\vec{\mu}^{(4)} + \vec{\mu}^{(5)} + \vec{\mu}^{(6)} + \vec{\mu}^{(7)}$. For example $\vec{\mu}^1 + \vec{\mu}^2 + \vec{\mu}^3 + \vec{\mu}^4$ is not simple (it is $(\vec{\mu}^1 + \vec{\mu}^2 + \vec{\mu}^4 + \vec{\mu}^5) + (\vec{\mu}^3 - \vec{\mu}^4) + (\vec{\mu}^4 - \vec{\mu}^5)$) and even $\vec{\mu}^{(3)} + \vec{\mu}^{(5)} + \vec{\mu}^{(6)} + \vec{\mu}^{(7)}$ is not simple (it is $(\vec{\mu}^3 + \vec{\mu}^4) + (\vec{\mu}^4 + \vec{\mu}^5 + \vec{\mu}^6 + \vec{\mu}^7)$). So the set of 7

⁸A more symmetric way to rewrite the definition of the 70 weights is as follows

$$\frac{1}{2} (\pm \vec{\mu}^{(1)} \pm \vec{\mu}^{(2)} \pm \vec{\mu}^{(3)} \pm \vec{\mu}^{(4)} \pm \vec{\mu}^{(5)} \pm \vec{\mu}^{(6)} \pm \vec{\mu}^{(7)} \pm \vec{\mu}^{(8)}) \quad \text{with four } + \text{ and four } - \text{ signs.}$$

Since the sum of all 8 weights $\vec{\mu}^{(i)}$ of the defining representation of $SU(8)$ vanishes, this definition is equivalent to the previous one.

simple roots of E_7 is as follows

$$\begin{aligned}\vec{\alpha}^1 &= \vec{\alpha}^{12}, & \vec{\alpha}^2 &= \vec{\alpha}^{23}, & \vec{\alpha}^3 &= \vec{\alpha}^{34}, & \vec{\alpha}^4 &= \vec{\alpha}^{45}, & \vec{\alpha}^5 &= \vec{\alpha}^{56}, & \vec{\alpha}^6 &= \vec{\alpha}^{67}, \\ \vec{\alpha}^7 &= \vec{\mu}^{(4)} + \vec{\mu}^{(5)} + \vec{\mu}^{(6)} + \vec{\mu}^{(7)}.\end{aligned}\tag{22.96}$$

With metric $g^{IJ} = 2\delta^{IJ}$ and $\vec{\alpha}^{ij} = \vec{\mu}^i - \vec{\mu}^j$, $\vec{\mu}^{(j)} \cdot \vec{\mu}^{(j)} = \frac{N-1}{N}$ and $\vec{\mu}^{(j)} \cdot \vec{\mu}^{(k)} = -\frac{1}{N}$ for $j \neq k$, one obtains the Dynkin diagram shown before. For convenience we give the numerical values of the 7 simple roots of E_7

$$\begin{aligned}\vec{\alpha}^1 &= (1, 0, 0, 0, 0, 0, 0) \\ \vec{\alpha}^2 &= \left(-\frac{1}{2}, \frac{3}{\sqrt{2}}, 0, 0, 0, 0, 0\right) \\ \vec{\alpha}^3 &= \left(0, -\frac{2}{\sqrt{12}}, \frac{4}{\sqrt{24}}, 0, 0, 0, 0\right) \\ \vec{\alpha}^4 &= \left(0, 0, -\frac{3}{\sqrt{24}}, \frac{5}{\sqrt{40}}, 0, 0, 0\right) \\ \vec{\alpha}^5 &= \left(0, 0, 0, -\frac{4}{\sqrt{40}}, \frac{6}{\sqrt{60}}, 0, 0\right) \\ \vec{\alpha}^6 &= \left(0, 0, 0, 0, -\frac{5}{\sqrt{60}}, \frac{7}{\sqrt{84}}, 0\right) \\ \vec{\alpha}^7 &= \left(0, 0, -\frac{3}{\sqrt{24}}, -\frac{3}{\sqrt{40}}, -\frac{3}{\sqrt{60}}, -\frac{3}{\sqrt{84}}, \frac{4}{\sqrt{112}}\right)\end{aligned}\tag{22.97}$$

It is clear that $\vec{\alpha}^7$ has only a nonvanishing inner product with itself and with $\vec{\alpha}^3$, $(\vec{\alpha}^7, \vec{\alpha}^7) = 1$ and $(\vec{\alpha}^7, \vec{\alpha}^3) = -\frac{1}{2}$. The construction of the Dynkin diagram is now very easy.

The fundamental weights follow from $2\frac{(\vec{\alpha}^i, \vec{\mu}_{FW}^j)}{(\vec{\alpha}^i, \vec{\alpha}^i)} = \delta^{ij}$. They can be expressed in terms of the simple roots, and the result [2, Chapter 13] reads as follows

$$\begin{aligned}\vec{\mu}_{FW}^1 &= (2\vec{\alpha}^1 + 3\vec{\alpha}^2 + 4\vec{\alpha}^3 + 3\vec{\alpha}^4 + 2\vec{\alpha}^5 + \vec{\alpha}^6 + 2\vec{\alpha}^7) \\ \vec{\mu}_{FW}^2 &= (3\vec{\alpha}^1 + 6\vec{\alpha}^2 + 8\vec{\alpha}^3 + 6\vec{\alpha}^4 + 4\vec{\alpha}^5 + 2\vec{\alpha}^6 + 4\vec{\alpha}^7) \\ \vec{\mu}_{FW}^3 &= (4\vec{\alpha}^1 + 8\vec{\alpha}^2 + 12\vec{\alpha}^3 + 9\vec{\alpha}^4 + 6\vec{\alpha}^5 + 3\vec{\alpha}^6 + 6\vec{\alpha}^7) \\ \vec{\mu}_{FW}^4 &= \frac{1}{2} (6\vec{\alpha}^1 + 12\vec{\alpha}^2 + 18\vec{\alpha}^3 + 15\vec{\alpha}^4 + 10\vec{\alpha}^5 + 5\vec{\alpha}^6 + 9\vec{\alpha}^7) \\ \vec{\mu}_{FW}^5 &= (2\vec{\alpha}^1 + 4\vec{\alpha}^2 + 6\vec{\alpha}^3 + 5\vec{\alpha}^4 + 4\vec{\alpha}^5 + 2\vec{\alpha}^6 + 3\vec{\alpha}^7) \\ \vec{\mu}_{FW}^6 &= \frac{1}{2} (2\vec{\alpha}^1 + 4\vec{\alpha}^2 + 6\vec{\alpha}^3 + 5\vec{\alpha}^4 + 4\vec{\alpha}^5 + 3\vec{\alpha}^6 + 3\vec{\alpha}^7) \\ \vec{\mu}_{FW}^7 &= \frac{1}{2} (4\vec{\alpha}^1 + 8\vec{\alpha}^2 + 12\vec{\alpha}^3 + 9\vec{\alpha}^4 + 6\vec{\alpha}^5 + 3\vec{\alpha}^6 + 7\vec{\alpha}^7).\end{aligned}\tag{22.98}$$

Using⁹ $(\vec{\alpha}^j, \vec{\alpha}^j) = 1$ for $1 \leq j \leq 7$ and $(\vec{\alpha}^j, \vec{\alpha}^{j+1}) = -\frac{1}{2}$ for $1 \leq j \leq 6$ while $(\vec{\alpha}^3, \vec{\alpha}^7) = -\frac{1}{2}$, one can check that the fundamental weights $\vec{\mu}^j$ for $j = 1, \dots, 7$ satisfy $2\frac{(\vec{\alpha}^j, \vec{\mu}^k)}{(\vec{\alpha}^j, \vec{\alpha}^j)} = \delta^{jk}$.

Our last task for E_7 is to identify the 7 fundamental weights with irreps of E_7 . The adjoint rep **138** consists of the adjoint rep t^i_j of $SU(8)$ and an antisymmetric tensor t^{ijkl} of $SU(8)$. The highest weight comes from t^i_j , namely $\vec{\mu}^i - \vec{\mu}^j = \vec{\mu}^1 - \vec{\mu}^8$, which is $\vec{\mu}_{FW}^1$. The next irrep is the

⁹We use here the metric δ^{IJ} instead of $g^{IJ} = 2\delta^{IJ}$. This overall scale cancels in $2\frac{(\vec{\alpha}^i, \vec{\mu}_{FW}^j)}{(\vec{\alpha}^i, \vec{\alpha}^i)}$.

fundamental irrep **56** which decomposes into t^{ij} (**28**) and t_{ij}^* (**28**) of $SU(8)$. Its weights are $(\vec{\mu}^i + \vec{\mu}^j)$ and $-(\vec{\mu}^i + \vec{\mu}^j)$ for $i < j$, and the highest weight is $-(\vec{\mu}^7 + \vec{\mu}^8) = \vec{\mu}^1 + \vec{\mu}^2 + \vec{\mu}^3 + \vec{\mu}^4 + \vec{\mu}^5 + \vec{\mu}^6$. Hence $\vec{\mu}_{FW}^6$ is the highest weight of the defining rep **56** of E_7 . Consider next $\vec{\mu}_{FW}^7 = -2\vec{\mu}^8$. It is the highest weight of a symmetric tensor $t_{ij}^* = t_{ji}$ of $SU(8)$ with dimension **28**, but if it can be extended to an irrep of E_7 , it needs further irreps of $SU(8)$ (just as in the case of the adjoint reps of $SU(8)$ and E_7). One could use Weyl's dimension formula to calculate the dimensions of the E_7 irreps whose highest weights are $\vec{\mu}_{FW}^j$ for $j = 2, 3, 4, 5, 7$, but we stop here.

22.5a Definition and closure of E_8 transformations

The exceptional compact simple Lie group E_8 can be rather easily constructed by using the properties of $SO(16)$ which we have derived before. As we shall show, one can construct an explicit representation of E_8 in terms of the adjoint and one of the spinor representations of $SO(16)$. Thus the root system of E_8 is the sum (or rather the union) of the roots of $SO(16)$ and the weights of its spinor irrep. Since $SO(16)$ has $\frac{1}{2} \times 16 \times 15 = 120$ generators, and its spinor representations have dimensions $2^{8-1} = 128$, the defining representation is thus the **248** which is at the same time the adjoint representation. There are no spinor representations of E_8 . Let us now discuss E_8 in more detail.

The group E_8 has 248 generators, and $SO(16)$ is a maximal regular subgroup. The **248** adjoint representation of E_8 splits into the adjoint representation **120** of $SO(16)$ and one of the two spinor representations (**128_s** or **128_c**) of $SO(16)$. It does not matter which spinor representation one chooses, hence

$$\mathbf{248} \xrightarrow{SO(16)} \mathbf{120} + \mathbf{128}. \quad (22.99)$$

This is a Cartan decomposition.

We take as carrier space the adjoint representation since this is for E_8 also the defining representation, and the representation with the lowest dimension. So as carrier space we take an antisymmetric tensor $\phi^{ij} = -\phi^{ji}$ ($i, j = 1, 16$) and a chiral spinor ψ^α ($\alpha = 1, 128$). The **248** parameters of E_8 are also decomposed into the **120** parameters Λ_j^i of $SO(16)$ and **128** further parameter λ^α . It is then fairly obvious what the transformation rules are [7, Appendix 6A]

$$\begin{aligned} \delta\phi^{ij} &= \Lambda_k^i \phi^{kj} + \Lambda_k^j \phi^{ik} + \lambda^\alpha \Gamma_{\alpha\beta}^{ij} \psi^\beta & i, j, k = 1, \dots, 16 \\ \delta\psi^\alpha &= \frac{1}{4} \Lambda^{ij} (\Gamma_{ij})^\alpha_\beta \psi^\beta + \phi^{ij} (\Gamma_{ij})^\alpha_\beta \lambda^\beta. & \alpha, \beta = 1, \dots, 128 \end{aligned} \quad (22.100)$$

By rescaling one can eliminate all free constants in these terms. Clearly both ψ^α and λ^α have the same chirality. The Dirac matrices in 8 Euclidean dimensions, and therefore also in 16 dimensions, can be chosen to be real, symmetric and off-diagonal. In that representation the charge conjugation matrix $(C_+)_{\alpha\beta}$, defined by $C_+ \Gamma_i C_+^{-1} = +\Gamma_i^T$, is the unit matrix, $C_+ = I$. This explains why there is no "bar" on λ in the term $\lambda \Gamma^{ij} \psi$. So all indices i, j, α, β are raised and lowered with Kronecker

deltas. All $\phi^{ij}, \Lambda_j^\alpha, \lambda^\alpha$ and ψ^α are real.¹⁰ We take the λ^α as commuting objects (they are just parameters) and also ψ^α will be taken to be commuting.

To prove that the transformation rules define a Lie algebra, we must show that

- (i) $[\delta(\Lambda_2), \delta(\Lambda_1)] = \delta[\Lambda_1, \Lambda_2]$ which is just the statement that $SO(16)$ is a Lie subalgebra.
- (ii) $[\delta(\lambda), \delta(\Lambda)] = \delta[\lambda']$ with $\lambda' = \frac{1}{4}\Lambda^{ij}\Gamma_{ij}\lambda$ (hence $\lambda'^T = \frac{1}{4}\lambda^T\Lambda^{ij}\Gamma_{ij}$). This states that the coset generators corresponding to the parameters λ form a representation of the subgroup (a so-called reductive algebra). This is clearly true in our case.
- (iii) We should find that $[\delta(\lambda_2), \delta(\lambda_1)] = \delta[\Lambda']$ with $\Lambda'_{ij} \sim \lambda_2\Gamma_{ij}\lambda_1$ because we have a Cartan decomposition of the Lie algebra of E_8 . This is the crucial commutator; if it is satisfied we have a Lie algebra.

Consider first the last commutator on ϕ^{ij}

$$[\delta(\lambda_2), \delta(\lambda_1)]\phi^{ij} = \lambda_1^\alpha \Gamma_{\alpha\beta}^{ij} (\phi^{kl}\Gamma_{kl})^\beta{}_\gamma \lambda_2^\gamma - 1 \leftrightarrow 2 . \quad (22.101)$$

One can write the sum of both terms as the commutator of $[\Gamma^{ij}, \Gamma_{kl}]$, and this proves (iii) with $\Lambda'_{ij} = -4\lambda_1\Gamma_{ij}\lambda_2 = +4\lambda_2\Gamma_{ij}\lambda_1$ when acting on ϕ^{ij} . All that is left is to prove that (iii) also holds when acting on ψ^α . One finds

$$[\delta(\lambda_2), \delta(\lambda_1)]\psi^\alpha = (\lambda_2\Gamma^{ij}\psi)(\Gamma_{ij}\lambda_1) - 1 \leftrightarrow 2 . \quad (22.102)$$

We next make a Fierz rearrangement which couples to λ 's together

$$= + \sum_I \left(\frac{1}{256} \right) (\lambda_2 O^I \lambda_1) (\Gamma_{ij} O^I \Gamma^{ij} \psi) - 1 \leftrightarrow 2 \quad (22.103)$$

(with a plus sign since the λ 's are commuting spinors). We used that the Dirac matrices in 16 dimensions are 256×256 matrices and O^I are all 2^{16} products of Dirac matrices normalized to $\text{Tr } O^I O^J = 256 \delta^{IJ}$.

Because the λ 's are chiral, only products of an even number of Γ_i matrices contribute. Moreover, the matrices $\Gamma_c, \Gamma^i\Gamma_c, \Gamma^{ij}\Gamma_c \dots$ (where Γ_c is the chirality matrix in 16 dimensions) give the same result as $I, \Gamma^i, \Gamma^{ij} \dots$ so we only need to consider products with none, two, $\dots$, eight Dirac matrices. But $\lambda_2\lambda_1, \lambda_2\Gamma_{klmn}\lambda_1$ and $\lambda_2\Gamma_{i_1\dots i_8}\lambda_1$ are symmetric in λ_2 and λ_1 , so only $O^I \sim \Gamma_{ij}$ and $O^I \sim \Gamma^{i_1\dots i_6}$ can contribute. But now there is an amusing identity

$$\Gamma^{ij}\Gamma^{i_1\dots i_6}\Gamma_{ij} = 0 . \quad (22.104)$$

¹⁰In technical terms, both λ^α and ψ^α are Majorana spinors. The consistency condition for a Majorana spinor to exist in Euclidean space is $\bar{\lambda}_M \equiv \lambda^T C = \bar{\lambda}_D \equiv \lambda^\dagger$. Taking the complex conjugate yields $\lambda^\dagger C^* = \lambda^T$ and iteration yields $\lambda^T C C^* = \lambda^T$. The consistency condition is thus $C C^* = I$ which is satisfied by $C = C_+ = I$.

Indeed, there are $\binom{6}{2}$ choices of ij from the set $i_1 \cdots i_6$, $\binom{10}{2}$ choices if ij lie outside this same set, and 6×10 ways of choosing i from $i_1 \cdots i_6$ and j from the complement. The first two cases yield $+\Gamma^{i_1 \cdots i_6}$, while the last case yields $-\Gamma^{i_1 \cdots i_6}$. Since $\binom{6}{2} + \binom{10}{2} - 6 \times 10 = 0$, this proves the identity.

So we only need to retain the $O^I \equiv \Gamma^{ij}$. This yields for the commutator

$$\sum_{ij} \sum_I \frac{1}{256} (\lambda_2 O^I \lambda_1) (\Gamma_{ij} O^I \Gamma^{ij} \psi) - 1 \leftrightarrow 2 = 4 \left(-\frac{1}{2} \right) \frac{1}{256} \sum_{ij} \sum_{kl} (\lambda_2 \Gamma^{kl} \lambda_1) (\Gamma_{ij} \Gamma^{kl} \Gamma^{ij} \psi). \quad (22.105)$$

The factor 4 is needed to include $O^I = \Gamma^{ij} \Gamma_c$ and the terms due to $1 \leftrightarrow 2$, and the factor $-\frac{1}{2}$ is needed since $\Gamma^{12} \Gamma^{12} = -1$ and we sum over $k > l$ and $k < l$. The same arguments as used before now show that $\sum_{ij} \Gamma_{ij} \Gamma^{kl} \Gamma^{ij} = -128 \Gamma^{kl}$ (since $2\binom{2}{2} + 2\binom{14}{2} - 2 \times 2 \times 14 = 128$). Thus we obtain $(\lambda_2 \Gamma_{kl} \lambda_1) \Gamma^{kl} \psi$, which confirms the expression for $[\delta(\lambda_2), \delta(\lambda_1)]$ in (iii) also for ψ^α . This proves the existence of E_8 , and provides at the same time a simple explicit representation of the adjoint representation.

22.5b Roots of E_8

The roots of E_8 consist of the roots of $SO(16)$ and the weights of one of the spinor irreps

$$\begin{aligned} (0, \pm 1, 0, \pm 1, \dots)_8 & \quad (112 \text{ roots of } SO(16)) \\ (\pm \tfrac{1}{2}, \pm \tfrac{1}{2}, \dots, \pm \tfrac{1}{2})_8 & \quad \text{with even number of } + \text{ signs} \quad (128 \text{ weights}) \end{aligned} \quad (22.106)$$

All roots have the same length.

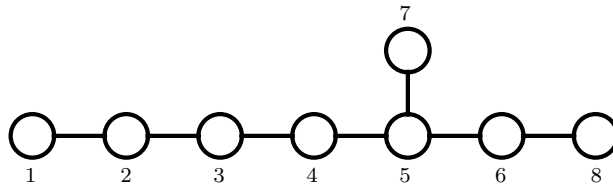
The simple roots are those of $SO(16)$, except that the first simple root $(1, -1, 0, 0, 0, 0, 0, 0)$ of $SO(16)$ is not a simple root of E_8 because

$$(1, -1, 0, 0, 0, 0, 0, 0) = \left(\tfrac{1}{2}, -\tfrac{1}{2}, -\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2} \right) + \left(\tfrac{1}{2}, -\tfrac{1}{2}, \tfrac{1}{2}, -\tfrac{1}{2}, -\tfrac{1}{2}, -\tfrac{1}{2}, -\tfrac{1}{2}, -\tfrac{1}{2} \right). \quad (22.107)$$

Instead, we need the simple spinor weight $(\frac{1}{2}, -\frac{1}{2}, \dots, -\frac{1}{2}, \frac{1}{2})$ (the lowest but still positive weight). The set of simple roots of E_8 can be conveniently written in terms of the vectors $e_j = (0, 0, \dots, 1, \dots, 0)$ as follows

$$\begin{aligned} \vec{\alpha}^1 &= e_2 - e_3, & \vec{\alpha}^2 &= e_3 - e_4, & \vec{\alpha}^3 &= e_4 - e_5, & \vec{\alpha}^4 &= e_5 - e_6, & \vec{\alpha}^5 &= e_6 - e_7 \\ \vec{\alpha}^6 &= e_7 - e_8, & \vec{\alpha}^7 &= e_7 + e_8, & \vec{\alpha}^8 &= \tfrac{1}{2}(e_1 - e_2 - \dots - e_7 + e_8). \end{aligned} \quad (22.108)$$

The Dynkin diagram is

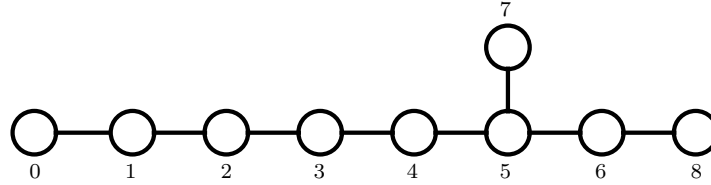


The fundamental weights of E_8 are as follows

$$\begin{aligned}
\vec{\mu}_{\text{FW}}^1 &= e_1 + e_2, & \vec{\mu}_{\text{FW}}^5 &= 5e_1 + e_2 + e_3 + e_4 + e_5 + e_6, \\
\vec{\mu}_{\text{FW}}^2 &= 2e_1 + e_2 + e_3, & \vec{\mu}_{\text{FW}}^6 &= \frac{1}{2}(7e_1 + e_2 + e_3 + e_4 + e_5 + e_6 + e_7 - e_8), \\
\vec{\mu}_{\text{FW}}^3 &= 3e_1 + e_2 + e_3 + e_4, & \vec{\mu}_{\text{FW}}^7 &= \frac{1}{2}(5e_1 + e_2 + e_3 + e_4 + e_5 + e_6 + e_7 + e_8), \\
\vec{\mu}_{\text{FW}}^4 &= 4e_1 + e_2 + e_3 + e_4 + e_5, & \vec{\mu}_{\text{FW}}^8 &= 2e_1.
\end{aligned} \tag{22.109}$$

The first seven fundamental weights were obtained by requiring that they are orthogonal to $\vec{\alpha}^8$, and $\vec{\mu}_{\text{FW}}^8$ was found by requiring orthogonality to the first seven simple roots.

To construct the extended Dynkin diagram, we add minus the highest weight $-\vec{\theta} = -e_1 - e_2$ to the simple roots. Then the extended Dynkin diagram becomes



Removing a dot from the extended Dynkin diagram yields in general maximal regular subalgebra, but for E_8 there are 3 exceptions (and for F_4 one exception, and also for E_7 one exception, see the discussions there). We obtain, starting with removing 1, then 2, etc. the following regular subalgebras

$SU(2) \times E_7$	R	$SU(3) \times SU(2) \times SU(6)$	
$SU(3) \times E_6$	R	$SU(2) \times SU(8)$	
$SU(4) \times SO(10)$		$SU(9)$	R
$SU(5) \times SU(5)$	R	$SO(16)$	R

Not maximal are $SU(4) \times SO(10) \subset SO(16)$, $SU(3) \times SU(2) \times SU(6) \subset SU(3) \times E_6$ and $SU(2) \times SU(6) \subset SU(2) \times E_7$. The maximal regular subalgebra $SO(16)$ which we used is obtained by deleting the node 7.

The highest weight of the adjoint representation is $e_1 + e_2$, which is also the first fundamental weight $\vec{\mu}_{\text{FW}}^{\text{adj}} = \vec{\mu}_{\text{FW}}^1$. Inspection of the set of fundamental weights suggests that $\vec{\mu}_{\text{FW}}^{(1)}$ is the smallest irrep; it is both the defining rep and the adjoint rep. All other irreps are much larger. (This can be proven by using Weyl's dimension formula.)

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Chapter 23

The adjoint representation, Cartan's criterion, and Weyl's theorem

Every Lie algebra (LA) has an adjoint representation. If the structure constants of the LA are given by $[T_a^{\text{def}}, T_b^{\text{def}}] = f_{ab}^c T_c^{\text{def}}$ where T_a^{def} are the generators in the representation which defines the Lie algebra, then the adjoint representation is given in terms of the structure constants by

$$(T_a^{\text{adj}})_b^c = f_{ba}^c \quad \text{or} \quad (T_a^{\text{adj}})_c^b = f_{ac}^b. \quad (23.1)$$

The T_a^{adj} denote the generators in the adjoint rep of any LA, for example the compact form of a simple LA or its CW form, or even a non-semisimple LA. The proof is based on the Jacobi identities

$$\begin{aligned} [T_a, [T_b, T_c]] + [T_b, [T_c, T_a]] + [T_c, [T_a, T_b]] &= 0 \\ f_{bc}^p f_{ap}^q + f_{ca}^p f_{bp}^q + f_{ab}^p f_{cp}^q &= 0 \end{aligned} \quad (23.2)$$

Using (23.1), the Jacobi identities can be written as follows

$$\begin{aligned} (T_b^{\text{adj}})_c^p (T_a^{\text{adj}})_p^q - (T_a^{\text{adj}})_c^p (T_b^{\text{adj}})_p^q + f_{ab}^p (T_p^{\text{adj}})_c^q &= 0 \\ [T_a^{\text{adj}}, T_b^{\text{adj}}] &= f_{ab}^p T_p^{\text{adj}} \quad \text{for } (T_a^{\text{adj}})_b^c = f_{ba}^c \end{aligned} \quad (23.3)$$

which proves the first part of (23.1).

To prove that $(T_a^{\text{adj}})_c^b = f_{ac}^b$ is also a representation, we write the Jacobi identities as follows

$$\begin{aligned} (T_a^{\text{adj}})_c^q (T_b^{\text{adj}})_p^c - (T_b^{\text{adj}})_c^q (T_a^{\text{adj}})_p^c - f_{ab}^p (T_p^{\text{adj}})_c^q &= 0 \\ [T_a^{\text{adj}}, T_b^{\text{adj}}] &= f_{ab}^p T_p^{\text{adj}} \quad \text{for } (T_a^{\text{adj}})_c^b = f_{ac}^b \end{aligned} \quad (23.4)$$

We shall later in this section prove that the matrices $(T_a^{\text{adj}})_b^c = f_{ba}^c$ and $(T_a^{\text{adj}})_c^b = f_{ac}^b$ are equivalent for semisimple LA's.

Given any LA, not necessarily even semisimple, the structure constants f_{abc} with all three

indices down are totally antisymmetric. They are defined by

$$f_{abc} = f_{ab}{}^{c'} g_{c'c}^{\text{adj}} \quad (23.5)$$

where g_{ab}^{adj} is the Killing metric for the adjoint representation $(T_a^{\text{adj}})_b{}^c = f_{ba}{}^c$

$$g_{ab}^{\text{adj}} = \text{tr } T_a^{\text{adj}} T_b^{\text{adj}} = f_{ap}{}^q f_{bq}{}^p \quad (23.6)$$

This Killing metric g_{ab}^{adj} and f_{abc} exist for any LA. The proof that f_{abc} are totally antisymmetric is as follows

$$\begin{aligned} f_{abc} &= \text{tr}[T_a^{\text{adj}}, T_b^{\text{adj}}] T_c^{\text{adj}} = \text{tr}(T_a^{\text{adj}} T_b^{\text{adj}} T_c^{\text{adj}} - T_b^{\text{adj}} T_a^{\text{adj}} T_c^{\text{adj}}) \\ &= \text{tr}\left(T_b^{\text{adj}} T_c^{\text{adj}} T_a^{\text{adj}} - T_c^{\text{adj}} T_b^{\text{adj}} T_a^{\text{adj}}\right) = \text{tr}[T_b^{\text{adj}}, T_c^{\text{adj}}] T_a^{\text{adj}} = f_{bca} . \end{aligned} \quad (23.7)$$

Hence $f_{abc} = f_{bca}$, and since $f_{ab}{}^c = -f_{ba}{}^c$, this ensures the total antisymmetry of f_{abc}

$$f_{abc} = f_{bca} = f_{cab} = -f_{bac} = -f_{acb} = -f_{cba} . \quad (23.8)$$

Cartan (1894) has proved the following important theorem

Cartan's criterion: A Lie algebra is semisimple iff (if and only if) $\det g_{ab}^{(\text{adj})} \neq 0$.

Proof: We shall prove this theorem in two steps:

A) If there exists a nontrivial abelian ideal (NAI) (if the LA is not semisimple), then $\det g_{ab}^{(\text{adj})} = 0$,

B) If $\det g_{ab}^{(\text{adj})} = 0$, then there exists a NAI.

Part A) is easy to prove. Let $X_{\lambda'}, X_{\rho'}, \dots$ denote a set of generators which form a basis for the NAI, and let a general generator be denoted by X_M . Then

$$\begin{aligned} g_{M\lambda'}^{(\text{adj})} &= \text{tr } X_M X_{\lambda'} = f_{PM}{}^Q f_{Q\lambda'}{}^P \\ &= f_{\mu'M}{}^Q f_{Q\lambda'}{}^{\mu'} \quad (\text{so } P = \mu' \text{ because } X_{\lambda'} \text{ lies in the ideal}) \\ &= f_{\mu'M}{}^{\rho'} f_{\rho'\lambda'}{}^{\mu'} \quad (\text{so } Q = \rho' \text{ because } X_{\mu'} \text{ lies in the ideal}) \\ &= 0 \quad (\text{since the ideal is abelian}). \end{aligned} \quad (23.9)$$

So the column $g_{M\lambda'}^{(\text{adj})}$ for fixed λ' vanishes. This implies that $\det g_{ab}^{(K)} = 0$.

The proof of part B) is more complicated. If $\det g_{MN}^{(\text{adj})} = 0$, there are null eigenvectors: $g_{MN} x_{(j)}^N = 0$ for all M (j labels these eigenvectors). Hence $\text{tr } X_M Y_{(j)} = 0$ where $Y_{(j)} = X_N x_{(j)}^N$. All $Y_{(j)}$ which satisfy $\text{tr } X_M Y_{(j)} = 0$ for all M span a linear vector space. They also form a Lie algebra (closure holds: $\text{tr } X_M [Y_{(j)}, Y_{(k)}] = \text{tr } [X_M, Y_{(j)}] Y_{(k)} = \text{tr } [X'_M, Y_{(k)}] = 0$ for all M). In fact, they form

an invariant Lie subalgebra. To prove this we must show that

$$[X_M, Y_{(j)}] \in Y \Rightarrow \text{tr } X_N[X_M, Y_{(j)}] = 0 \quad \text{for all } N. \quad (23.10)$$

This follows from the cyclicity of the trace

$$\text{tr}[X_M, Y_{(j)}]Y_{(k)} = \text{tr } X_M[Y_{(j)}, Y_{(k)}] = 0. \quad (23.11)$$

So the $Y_{(j)}$ form an ideal. If we could prove that this ideal is abelian, we would have proven part B). To prove that the $Y_{(j)}$ commute, one needs first to introduce the concept of Solvable Lie algebras, then to show that the $Y_{(j)}$ form a solvable Lie algebra, and finally to prove that

Theorem: a LA is semisimple iff it contains no solvable ideal.

We shall not give the proofs of these steps.

At this point we can prove for semisimple LA's the equivalence of both ways to represent the adjoint rep. in (23.1). If we use the Killing metric and its inverse (which exists as the LA is semisimple) to raise the indices c in (23.3) and lower the indices q , and replace the northwest-southeast contractions p_p by southwest-northeast contraction p^p , then one obtains (23.4).

Weyl proved an interesting result for compact Lie algebras. (A Lie group is compact if its volume evaluated with the invariant (Haar) measure is finite. The LA of a compact Lie group is called a compact Lie algebra.)

Weyl's Theorem: A Lie group (or LA) is compact if and only if its Killing metric is **negative definite**. Thus an algebraic property is equivalent to a topological property.

Example 1: For the compact group $SU(2)$ with generators $-\frac{i}{2}\sigma_k$ and structure constants $f_{ij}^k = \epsilon_{ij}^k$, we get $g_{ij}^{(adj)} = \epsilon_{ik}^l \epsilon_{jl}^k = -2\delta_{ij}$, which is indeed negative definite. But for the non-compact group $SU(1,1)$ with generators $\frac{\alpha}{2}\sigma_1, \frac{\beta}{2}\sigma_2, \frac{i\gamma}{2}\sigma_3$ with real α, β, γ and structure constants $f_{12}^3 = \frac{\alpha\beta}{\gamma}; f_{13}^2 = \frac{\alpha\gamma}{\beta}; f_{23}^1 = -\frac{\beta\gamma}{\alpha}$, we get

$$g_{ab}^{(adj)} = \begin{pmatrix} 2\alpha^2 & & \\ & 2\beta^2 & \\ & & -2\gamma^2 \end{pmatrix} \quad (23.12)$$

which is indeed not negative definite.

Here are two further useful theorems:

Theorem I: If a LA is semisimple (if $\det g_{ab}^{(K)} \neq 0$), then its adjoint rep is faithful.

Proof: If the adjoint rep were not faithful, there would exist two elements X_a and X_b of the LA such that $f_{Ma}^N = f_{Mb}^N$ for all M and N . Then $f_{b-a,M}^N = 0$, but this would imply that the Killing metric is singular:

$$g_{a-b,P}^{(K)} = f_{a-b,M}^N f_{PN}^M = 0 \quad \text{for all } P \quad (23.13)$$

(The Killing metric would be singular because a row vanishes.)

Theorem II: If a LA is simple, then its adjoint rep is irreducible.

Proof: If the adjoint rep of a LA were reducible, then there would exist a linear space B' of generators $X_{b'}$ such that $[X_a, X_{b'}] \in B'$ for all X_a in the LA. But then B' would be a proper invariant subalgebra of the LA, which is impossible because the LA was simple.

The direct product of two simple groups (or the direct sum of two LA's) is a semisimple group (or a semisimple LA). That is clear: one group is a normal subgroup of the direct product group, but this group is simple, thus nonabelian, hence the direct product group does not have an abelian invariant subgroup. But the converse is also true. For the LA's we get (a similar result holds for the groups)

Theorem: A LA is semisimple if and only if it is the direct sum of simple LA's. In formula:

$$A = A_1 \oplus A_2 \oplus \cdots \oplus A_l \quad \text{with } A_j \text{ simple LA's.} \quad (23.14)$$

Proof: If A is a simple LA, the proof is done. If there is an ideal I (an invariant subalgebra), choose a basis $v_1, \dots, v_p$ of this ideal. We shall show that $LA = I \oplus I^\perp$ where $I^\perp$ is "the orthogonal complement of I with respect to the Killing metric" (to be explained). Then $I^\perp$ is also an ideal of the LA. If $I^\perp$ is simple, the theorem is proven, if not repeat this process. After a finite number of steps the theorem is proven (since a LA has a finite number of generators).

So how is $I^\perp$ defined? Consider $g(v_i, w) \equiv \text{tr}(v_i^{\text{adj}} w^{\text{adj}})$. Then the linear vector space W of w satisfying $g(v_i, w) = 0$ satisfies $\dim LA = \dim I + \dim W$, and the w form an ideal of I .

Chapter 24

α -chains, the Weyl group, and conjugacy classes

Any irrep R of a simple LA with rank r is determined by its highest weight (HW), and this HW is a non-negative integer linear combination of the fundamental weights

$$\vec{\mu}_{HW}^{(R)} = \sum_{J=1}^r n_J \vec{\mu}_{FW}^J \quad \text{with } n_J \geq 0 \quad (24.1)$$

The highest weight vector $\vec{\mu}_{HW}^{(R)}$ corresponds to a state $|\vec{\mu}_{HW}, R\rangle$ of the irrep R . We describe in this chapter how to find all the other states. This generalizes the well-known construction in $SU(2)$ where one starts at the highest weight of a multiplet with spin j , and then obtains all other states with spin j by repeatedly acting with the lowering operator. After this demonstration how to fill out any given irrep, we consider the set of all irreps of one of the classical groups, and show that they can be grouped together into a few classes, the so-called conjugacy classes.

Consider a state in a given irrep. We shall use the lowering generators $E_{-\alpha}, E_{-\beta}, \dots$ to construct further states. For example, $E_{-\alpha}|\vec{\mu}_{HW}, R\rangle$ can be a new state (it may vanish for some α), and $E_{-\alpha}E_{-\alpha}|\vec{\mu}_{HW}, R\rangle$ could be another state, but eventually we cannot get new states (all unitary irreps of simple LA's are finite-dimensional), so

$$E_{-\alpha}^q |\vec{\mu}_{HW}, R\rangle = 0 \quad \text{for some } q > 0. \quad (24.2)$$

More generally, if we pick any state $|\vec{\mu}, R\rangle$ of an irrep R , with weight vector $\vec{\mu}$ satisfying $H_I |\vec{\mu}, R\rangle = \mu_I |\vec{\mu}, R\rangle$, we can repeatedly act on it with E_{α} and $E_{-\alpha}$, and in each case the series should terminate

$$E_{\alpha}^p |\vec{\mu}, R\rangle = 0; \quad E_{-\alpha}^q |\vec{\mu}, R\rangle = 0 \quad \text{for some } p \geq 0 \text{ and } q > 0. \quad (24.3)$$

This yields what is called an **α -chain of states**. We shall now determine these p and q .

Define

$$E_\alpha|\vec{\mu}, R\rangle = N_{\alpha,\mu}|\vec{\mu} + \vec{\alpha}, R\rangle; \quad E_{-\alpha}|\vec{\mu}, R\rangle = N_{-\alpha,\mu}|\vec{\mu} - \vec{\alpha}, R\rangle \quad (24.4)$$

Since

$$H_I E_{\pm\alpha}|\vec{\mu}, R\rangle = [H_I, E_{\pm\alpha}]|\vec{\mu}, R\rangle + E_{\pm\alpha}H_I|\vec{\mu}, R\rangle \quad (24.5)$$

one obtains

$$H_I E_{\pm\alpha}|\vec{\mu}, R\rangle = (\pm\alpha_I + \mu_I)E_{\pm\alpha}|\vec{\mu}, R\rangle \quad (24.6)$$

This implies that if $E_{\pm\alpha}|\vec{\mu}, R\rangle$ is nonvanishing, it yields a state $|\vec{\mu} \pm \vec{\alpha}, R\rangle$. It shows in particular that E_α for positive root $\vec{\alpha}$ is a raising operator (because $\vec{\mu} + \vec{\alpha}$ is more positive than $\vec{\mu}$). We now proceed as with angular momentum theory in Quantum Mechanics, by acting with $[E_\alpha, E_{-\alpha}]$ on the series of states $E_\alpha^k|\vec{\mu}, R\rangle$ and $E_{-\alpha}^l|\vec{\mu}, R\rangle$. We define bra-states as the set of states $\langle\vec{\mu}, R|$ which are orthogonal to the set of states $|\vec{\mu}, R\rangle$, so for example $\langle\vec{\mu}, R|\vec{\mu}, R\rangle = 1$ and $\langle\vec{\mu} + \vec{\alpha}, R|\vec{\mu}, R\rangle = 0$. For a given state $|\vec{\mu}, R\rangle$ we get, using $[E_\alpha, E_{-\alpha}] = \alpha_I g^{IJ} H_J g_{\alpha, -\alpha}$, the following relation¹

$$\begin{aligned} \langle\vec{\mu}, R|[E_\alpha, E_{-\alpha}]\vec{\mu}, R\rangle &= \langle\vec{\mu}, R|\alpha^I H_I g_{\alpha, -\alpha}|\vec{\mu}, R\rangle = \alpha^I \mu_I g_{\alpha, -\alpha} = \alpha \cdot \mu g_{\alpha, -\alpha} \\ &= \langle\vec{\mu}, R|E_\alpha E_{-\alpha}|\vec{\mu}, R\rangle - \langle\vec{\mu}, R|E_{-\alpha} E_\alpha|\vec{\mu}, R\rangle = |N_{-\alpha,\mu}|^2 - |N_{\alpha,\mu}|^2 \end{aligned} \quad (24.7)$$

Next we relate the $N_{-\alpha,\mu}$ to the $N_{\alpha,\mu}$ as follows

$$\langle\vec{\mu} - \vec{\alpha}, R|E_{-\alpha}|\vec{\mu}, R\rangle = N_{-\alpha,\mu} = \langle\vec{\mu} - \vec{\alpha}, R|E_\alpha^\dagger|\vec{\mu}, R\rangle = (\langle\vec{\mu}, R|E_\alpha|\vec{\mu} - \vec{\alpha}, R\rangle)^* = N_{\alpha,\mu-\alpha}^* \quad (24.8)$$

So

$$|N_{\alpha,\mu-\alpha}|^2 - |N_{\alpha,\mu}|^2 = \vec{\alpha} \cdot \vec{\mu} g_{\alpha, -\alpha}. \quad (24.9)$$

We now construct a hierarchy of such relations. We start with the state $|\vec{\mu} + p\vec{\alpha}, R\rangle$ for which $E_\alpha|\vec{\mu} + p\vec{\alpha}, R\rangle = 0$, hence $N_{\alpha,\mu+p\alpha} = 0$. This yields the following results

$$\begin{aligned} |N_{\alpha,\mu+(p-1)\alpha}|^2 - 0 &= \vec{\alpha} \cdot (\vec{\mu} + p\vec{\alpha}) g_{\alpha, -\alpha} \\ |N_{\alpha,\mu+(p-2)\alpha}|^2 - |N_{\alpha,\mu+(p-1)\alpha}|^2 &= \vec{\alpha} \cdot (\vec{\mu} + (p-1)\vec{\alpha}) g_{\alpha, -\alpha} \\ \vdots &\quad \quad \quad \vdots \\ |N_{\alpha,\mu-q\alpha}|^2 - |N_{\alpha,\mu+(q-1)\alpha}|^2 &= \vec{\alpha} \cdot (\vec{\mu} - (q-1)\vec{\alpha}) g_{\alpha, -\alpha} \\ 0 - |N_{\alpha,\mu-q\alpha}|^2 &= \vec{\alpha} \cdot (\vec{\mu} - q\vec{\alpha}) g_{\alpha, -\alpha} \end{aligned} \quad (24.10)$$

We used in the last step that $N_{\alpha,\mu-(q+1)\alpha} = N_{-\alpha,\mu-q\alpha}^* = 0$. Adding these relations, all terms on the left-hand side cancel, so we obtain, dropping the common nonvanishing factor $g_{\alpha, -\alpha}$,

$$0 = (p+q+1)\vec{\alpha} \cdot \vec{\mu} + \left(\frac{1}{2}p(p+1) - \frac{1}{2}q(q+1)\right) \vec{\alpha} \cdot \vec{\alpha} \quad (24.11)$$

¹We recall that $g^{IJ}g_{\alpha, -\alpha}$ is representation-independent, so one may choose any normalization of the Killing metric in this relation.

Since $\frac{1}{2}p(p+1) - \frac{1}{2}q(q+1)$ factorizes into $\frac{1}{2}(p-q)(p+q+1)$, and $p+q+1$ is nonzero, we obtain² “the master formula” (terminology used in Georgi)

$$\boxed{2 \frac{\vec{\alpha} \cdot \vec{\mu}}{\vec{\alpha} \cdot \vec{\alpha}} = -p + q} \quad (24.12)$$

This is true for any irrep, but it has particularly simple and strong consequences for the adjoint rep. In that case the weights $\vec{\mu}$ are the roots $\vec{\beta}$, so

$$\frac{\vec{\alpha} \cdot \vec{\beta}}{\vec{\alpha} \cdot \vec{\alpha}} = -\frac{1}{2}(p-q) \equiv \frac{1}{2}m \quad \text{for some integer } m. \quad (24.13)$$

On the other hand, we can interchange α and β . If we begin with $|\vec{\mu}, R\rangle$ and construct a β chain, and then set $\vec{\mu} = \vec{\alpha}$, we get

$$\frac{\vec{\beta} \cdot \vec{\alpha}}{\vec{\beta} \cdot \vec{\beta}} = \frac{1}{2}m'. \quad (24.14)$$

Multiplying both expressions yields

$$\frac{(\vec{\alpha} \cdot \vec{\beta})^2}{(\vec{\alpha} \cdot \vec{\alpha})(\vec{\beta} \cdot \vec{\beta})} = \frac{1}{4}mm' = (\cos \theta_{\alpha,\beta})^2 \quad (24.15)$$

where $\theta_{\alpha,\beta}$ is the angle between the root vectors $\vec{\alpha}$ and $\vec{\beta}$. Since $|\cos \theta_{\alpha,\beta}|$ is bounded by unity, there are only a few possibilities for mm' , given in table 24.1.

mm'	$\theta_{\alpha,\beta}$
0	90°
1	$60^\circ, 120^\circ$
2	$45^\circ, 135^\circ$
3	$30^\circ, 150^\circ$
4	$0^\circ, 180^\circ$

Table 24.1: The possible angles between any two root vectors.

We next consider simple roots, and prove that for them $\theta_{\alpha,\beta} \geq 90^\circ$. To show this, we begin by noting that **if α and β are simple roots, then $\vec{\alpha} - \vec{\beta}$ and $\vec{\beta} - \vec{\alpha}$ are not roots**. The proof is easy: if $\vec{\alpha} - \vec{\beta}$ is a root and $\vec{\alpha} - \vec{\beta} > 0$ then $\vec{\alpha} = \vec{\alpha} - \vec{\beta} + \vec{\beta}$ would not be simple. And if $\vec{\alpha} - \vec{\beta} < 0$, then $\vec{\beta} - \vec{\alpha} > 0$ ³ and $\vec{\beta} = \vec{\beta} - \vec{\alpha} + \vec{\alpha}$ would not be simple. Then $E_{-\alpha}|E_\beta\rangle = 0$ and $E_{-\beta}|E_\alpha\rangle = 0$. So $q = 0$ in (24.13), and $\frac{\vec{\alpha} \cdot \vec{\beta}}{\vec{\alpha} \cdot \vec{\alpha}} = -\frac{1}{2}p \leq 0$. This proves that $\theta_{\alpha,\beta} \geq 90^\circ$. So, the possible angles between

²There exists a simpler derivation of the master formula in (24.12), based on the observation that (after suitable normalizations) the generators $\vec{\alpha} \cdot \vec{H}$, E_α and $E_{-\alpha}$ form an $SU(2)$ algebra. We shall give this derivation when we discuss Weyl reflections.

³If $\vec{\alpha}$ is a root, also $-\vec{\alpha}$ is a root, and vice-versa. We proved this before.

two simple roots are 90° , 120° , 135° and 150° . (The case $\theta = 180^\circ$ is not possible for simple roots because $\vec{\beta} = -\vec{\alpha}$ would not be positive, hence not simple.)

This is enough information to introduce Dynkin diagrams (1947).⁴ Take all simple roots, and connect them with bars depending on the angles between them as follows

$$\begin{array}{ll}
\bigcirc \quad \bigcirc & 90^\circ : \text{no bar} \\
\bigcirc \text{---} \bigcirc & 120^\circ : \text{one bar} \\
\bigcirc \text{=}\bigcirc & 135^\circ : \text{two bars} \\
\bigcirc \text{=}\text{=}\bigcirc & 150^\circ : \text{three bars}
\end{array} \tag{24.16}$$

Example: for $SU(N)$ we found the following results

$$\vec{\alpha}^I \cdot \vec{\alpha}^I = 2; \quad \vec{\alpha}^J \cdot \vec{\alpha}^{J+1} = -1; \quad \vec{\alpha}^J \cdot \vec{\alpha}^K = 0 \text{ if } |J - K| \geq 2 \tag{24.17}$$

Then $\frac{\vec{\alpha}^J \cdot \vec{\alpha}^{J+1}}{\vec{\alpha}^J \cdot \vec{\alpha}^J} = -\frac{1}{2}$, hence $\theta_{\alpha^J, \alpha^{J+1}} = 120^\circ$. The Dynkin diagram for $SU(N)$ becomes

$$\bigcirc_1 \text{---} \bigcirc_2 \text{---} \bigcirc_3 \cdots \text{---} \bigcirc_{N-1} \tag{24.18}$$

We now resume our discussion of states. Consider the following set of states

$$\cdots, \quad |\vec{\mu} - 2\vec{\alpha}, R\rangle, \quad |\vec{\mu} - \vec{\alpha}, R\rangle, \quad |\vec{\mu}, R\rangle, \quad |\vec{\mu} + \vec{\alpha}, R\rangle, \quad |\vec{\mu} + 2\vec{\alpha}, R\rangle, \quad \cdots \tag{24.19}$$

This set of states forms an α -chain. We can go $p \geq 0$ steps up, and $q \geq 0$ steps down, and

$$2 \frac{\vec{\alpha} \cdot \vec{\mu}}{\vec{\alpha} \cdot \vec{\alpha}} = -p + q. \tag{24.20}$$

If one starts at the highest weight state, one gets $p = 0$ in this formula. So

$$2 \frac{\vec{\alpha} \cdot \vec{\mu}_{HW}}{\vec{\alpha} \cdot \vec{\alpha}} = q. \tag{24.21}$$

This tell us how far down we can go along an α -chain, which starts at the HW state. At any other state $|\vec{\mu}, R\rangle$, the master formula tells us the value of $-p + q$, and if one knows either p or q , then one also knows the other.

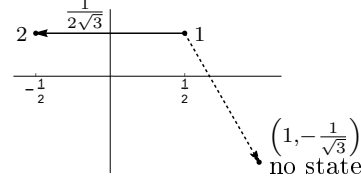
The master formula holds for any irrep and any root $\vec{\alpha}$ and any weight $\vec{\mu}$. But in order to fill out a complete representation (a complete multiplet), it is economical to only use simple roots. We now give some examples.

⁴E. B. Dynkin was a Soviet mathematician who emigrated in 1976 to the US.

Example 1: The 3 of $SU(3)$. The input is as follows

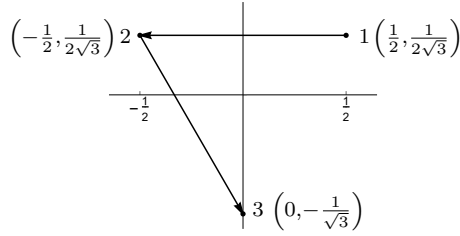
$$\vec{\mu}_{FW}^I = \vec{\mu}_{HW}^{\text{def}} = \left(\frac{1}{2}, \frac{1}{2\sqrt{3}}\right) \equiv \vec{\mu}^1; \quad \vec{\alpha}^I = (1, 0); \quad \vec{\alpha}^{II} = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \quad (24.22)$$

Step 1. The master formula gives for $\vec{\alpha}^I$ and $\vec{\alpha}^{II}$ chains starting at the highest weight of the defining rep $\vec{\mu}_{\text{def}}^1$

$$\left. \begin{aligned} \frac{2\vec{\alpha}^I \cdot \vec{\mu}_{\text{def}}^1}{\vec{\alpha}^I \cdot \vec{\alpha}^I} = 1 = q \Rightarrow q = 1 \\ \frac{2\vec{\alpha}^{II} \cdot \vec{\mu}_{\text{def}}^1}{\vec{\alpha}^{II} \cdot \vec{\alpha}^{II}} = 0 = q \Rightarrow q = 0 \end{aligned} \right\} \quad (24.23)$$


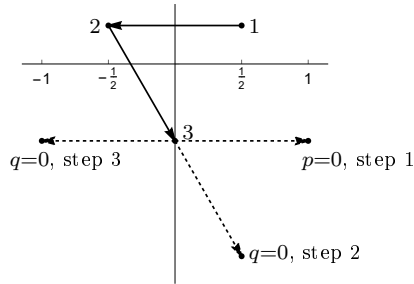
Step 2. At the state $|\vec{\mu}_{\text{def}}^1 - \vec{\alpha}^I\rangle = |\vec{\mu}_{\text{def}}^2\rangle$ with $\vec{\mu}_{\text{def}}^2 = \left(-\frac{1}{2}, \frac{1}{2\sqrt{3}}\right)$ the master formula now gives

$$\frac{2\vec{\alpha}^{II} \cdot \vec{\mu}_{\text{def}}^2}{\vec{\alpha}^{II} \cdot \vec{\alpha}^{II}} = 1 = -p + q = q \Rightarrow q = 1 \quad (p = 0 \text{ because } \vec{\mu}_{\text{def}}^1 \text{ is HW.}) \quad (24.24)$$



Step 3. At $|\vec{\mu}_{\text{def}}^1 - \vec{\alpha}^I - \vec{\alpha}^{II}\rangle = |\vec{\mu}_{\text{def}}^3\rangle$ with $\vec{\mu}_{\text{def}}^3 = \left(0, -\frac{1}{\sqrt{3}}\right)$ the master formula gives

$$\frac{2\vec{\alpha}^I \cdot \vec{\mu}_{\text{def}}^3}{\vec{\alpha}^I \cdot \vec{\alpha}^I} = 0 = -p + q = q \quad (p = 0, \text{ see step 1.}) \quad (24.25)$$



So we get the **3** of $SU(3)$. Since there is only one way to reach all states by lowering operators of simple roots, there is no degeneracy of the states.

Example 2: The 10 of $SU(3)$. Input: $\vec{\mu}_{HW}^{R=10} = 3\vec{\mu}_{FW}^I = \left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right); \vec{\alpha}^I = (1, 0); \vec{\alpha}^{II} = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right).$

Step 1.

$$\left. \begin{aligned} \frac{2\vec{\alpha}^I \cdot \vec{\mu}_{HW}}{\vec{\alpha}^I \cdot \vec{\alpha}^I} &= 3 = q \\ \frac{2\vec{\alpha}^{II} \cdot \vec{\mu}_{HW}}{\vec{\alpha}^{II} \cdot \vec{\alpha}^{II}} &= 0 = q \end{aligned} \right\} \quad \begin{array}{c} \text{Diagram: A horizontal line with points at } -2, -\frac{3}{2}, -1, -\frac{1}{2}, \frac{1}{2}, 1, \frac{3}{2}, 2. \text{ Arrows point left from } -2, -\frac{3}{2}, -1, -\frac{1}{2}, \frac{1}{2}, 1. \text{ A dashed arrow points from } \frac{3}{2} \text{ to } 2. \text{ The point } 2 \text{ is labeled 'no state'.} \end{array} \quad (24.26)$$

Step 2.

$$\left. \begin{aligned} \frac{2\vec{\alpha}^{II} \cdot (\vec{\mu}_{HW} - \vec{\alpha}^I)}{\vec{\alpha}^{II} \cdot \vec{\alpha}^{II}} &= 1 = -p + q = q \\ \frac{2\vec{\alpha}^{II} \cdot (\vec{\mu}_{HW} - 2\vec{\alpha}^I)}{\vec{\alpha}^{II} \cdot \vec{\alpha}^{II}} &= 2 = -p + q = q \\ \frac{2\vec{\alpha}^{II} \cdot (\vec{\mu}_{HW} - 3\vec{\alpha}^I)}{\vec{\alpha}^{II} \cdot \vec{\alpha}^{II}} &= 3 = -p + q = q \end{aligned} \right\} \quad \begin{array}{c} \text{Diagram: A root diagram with points labeled A, B, C, D, E, F, G, H, I. A horizontal line has points A, B, C, D. A diagonal line has points E, F, G. A vertical line has points H, I. The origin is labeled 0. Arrows point from A to B, B to C, C to D, D to E, E to F, F to G, G to H, H to I, and I to 0.} \end{array} \quad (24.27)$$

At E : Nothing to the left: $\frac{2\vec{\alpha}^I \cdot (\vec{\mu}_{HW} - 3\vec{\alpha}^I - \vec{\alpha}^{II})}{\vec{\alpha}^I \cdot \vec{\alpha}^I} = -2 = -p$.

At F : Nothing to the left.

At G : Nothing to the left and nothing below.

Now, however, there are more ways than one to reach states using lowering operators of simple roots. The states A, B, C, D, I are still unique, but for example the state(s) at the origin can be reached in two ways

$$\begin{array}{ccc} \begin{array}{c} C \longleftarrow A \\ \searrow \\ 0 \end{array} & \text{and} & \begin{array}{c} B \longleftarrow A \\ \searrow \\ 0 \longleftarrow I \end{array} \end{array} \quad (24.28)$$

There is a method to conclude without much work that states related by an operation called Weyl reflections have the same degeneracy. We next discuss these Weyl reflections. Using them, we shall conclude that all states of the **10** except the states at the origin are nondegenerate. To conclude that the state at the origin is also nondegenerate requires a more detailed analysis which we first provide.

There are two ways of creating a state at the origin of the root diagram as shown in (24.28):

$$\begin{aligned} \text{State I: } & E_{-\alpha^{II}} E_{-\alpha^I} E_{-\alpha^I} |\vec{\mu}_{HW}\rangle \\ \text{State II: } & E_{-\alpha^I} E_{-\alpha^{II}} E_{-\alpha^I} |\vec{\mu}_{HW}\rangle \end{aligned} \quad (24.29)$$

We start by rewriting state I as state II plus a commutator

$$\text{State I} = E_{-\alpha^I} E_{-\alpha^{II}} E_{-\alpha^I} |\vec{\mu}_{HW}\rangle + [E_{-\alpha^{II}}, E_{-\alpha^I}] E_{-\alpha^I} |\vec{\mu}_{HW}\rangle \quad (24.30)$$

Since $[E_{-\alpha^{II}}, E_{-\alpha^I}]$ can only be proportional to $E_{-\alpha^{II}-\alpha^I}$, and $E_{-\alpha^{II}-\alpha^I}$ commutes with $E_{-\alpha^I}$ because $-\vec{\alpha}^{II} - 2\vec{\alpha}^I$ is not a root, we can rewrite the term with the commutator as follows

$$E_{-\alpha^I}[E_{-\alpha^{II}}, E_{-\alpha^I}]|\vec{\mu}_{HW}\rangle = E_{-\alpha^I}E_{-\alpha^{II}}E_{-\alpha^I}|\vec{\mu}_{HW}\rangle - E_{-\alpha^I}E_{-\alpha^I}E_{-\alpha^{II}}|\vec{\mu}_{HW}\rangle = \text{State II} + 0. \quad (24.31)$$

We used that $E_{-\alpha^{II}}|\vec{\mu}_{HW}\rangle = 0$, as we discovered before (see step 1). Hence collecting all terms we discover that

$$\text{State I} = 2 \text{ State II} \quad (24.32)$$

This proves that there is only one state at the origin. We now turn to the Weyl reflections to study the degeneracy of the states not at the origin.

Weyl reflections. Recall the CW form of a simple LA

$$\begin{aligned} [H_I, H_J] &= 0; \quad [H_I, E_{\pm\alpha}] = \pm\alpha_I E_{\pm\alpha}; \\ [E_{\alpha}, E_{-\alpha}] &= \alpha_I g^{IJ} H_J g_{\alpha, -\alpha}; \quad [E_{\alpha}, E_{\beta}] = N_{\alpha, \beta} E_{\alpha+\beta} \quad \text{if } \beta \neq -\alpha. \end{aligned} \quad (24.33)$$

Defining

$$\begin{aligned} J_3^{\alpha} &= \frac{\alpha_I g^{IJ} H_J}{\vec{\alpha} \cdot \vec{\alpha}} \equiv \frac{\alpha \cdot H}{\alpha \cdot \alpha} \\ J_+^{\alpha} &= \frac{E_{\alpha}}{\sqrt{\alpha \cdot \alpha g_{\alpha, -\alpha}}}, \quad J_-^{\alpha} = \frac{E_{-\alpha}}{\sqrt{\alpha \cdot \alpha g_{\alpha, -\alpha}}} \end{aligned} \quad (24.34)$$

it follows that $J_3^{\alpha}, J_+^{\alpha}, J_-^{\alpha}$ form an $SU(1, 1)$ algebra

$$[J_3^{\alpha}, J_{\pm}^{\alpha}] = \pm J_{\pm}^{\alpha}; \quad [J_+^{\alpha}, J_-^{\alpha}] = J_3^{\alpha} \quad (24.35)$$

Consider now an α -chain of states in a representation R starting at weight $\vec{\mu}$. Acting with J_3^{α} and $J_{\pm}^{\alpha}$ on these states, they form a representation of the $SU(1, 1)$ algebra. Such representations have spin j , meaning that $J^2 = j(j+1)$ and J_3 runs from $-j$ to $+j$. But the eigenvalues of J_3^{α} on the states $|\vec{\mu} + k\vec{\alpha}, R\rangle$ of the α -chain are

$$J_3^{\alpha}|\vec{\mu} + k\vec{\alpha}, R\rangle = \frac{\vec{\alpha} \cdot (\vec{\mu} + k\vec{\alpha})}{\vec{\alpha} \cdot \vec{\alpha}}. \quad (24.36)$$

At the top of the α -chain we get

$$\frac{\vec{\alpha} \cdot (\vec{\mu} + p\vec{\alpha})}{\vec{\alpha} \cdot \vec{\alpha}} = j \quad (24.37)$$

and at the bottom we get

$$\frac{\vec{\alpha} \cdot (\vec{\mu} - q\vec{\alpha})}{\vec{\alpha} \cdot \vec{\alpha}} = -j \quad (24.38)$$

Adding both equations yields

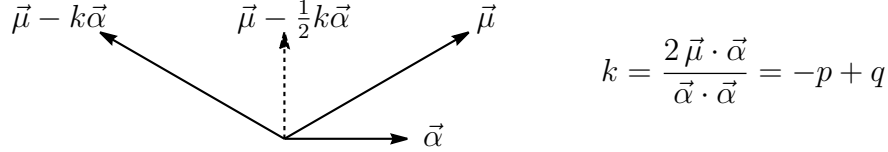
$$2 \frac{\vec{\alpha} \cdot \vec{\mu}}{\vec{\alpha} \cdot \vec{\alpha}} = -p + q \quad (24.39)$$

This is the master formula, derived in a simpler way than before. But taking the difference yields $j = \frac{p+q}{2}$. The middle of the α -chain occurs at

$$(\vec{\mu} + p\vec{\alpha}) - j\vec{\alpha} = \vec{\mu} + \frac{1}{2}(p - q)\vec{\alpha} = \vec{\mu} - \frac{\vec{\alpha} \cdot \vec{\mu}}{\vec{\alpha} \cdot \vec{\alpha}}\vec{\alpha}. \quad (24.40)$$

This result has an interesting geometrical interpretation.

For each root $\vec{\alpha}$ and state $|\vec{\mu}, R\rangle$ with weight $\vec{\mu}$, there is an α -chain, and the states (weights) on this chain are symmetric with respect to a hyperplane orthogonal to $\vec{\alpha}$.



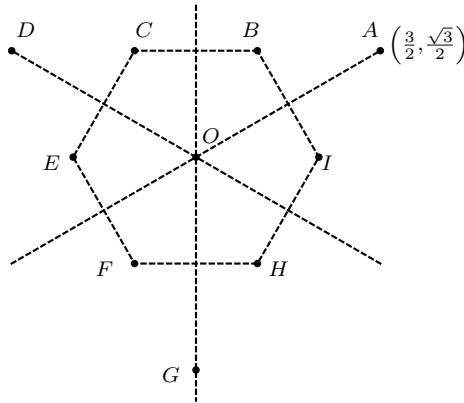
Given a weight (state) $\vec{\mu}$, also $\vec{\mu} - k\vec{\alpha}$ is a state

$$\vec{\mu} - 2\frac{\vec{\alpha} \cdot \vec{\mu}}{\vec{\alpha} \cdot \vec{\alpha}} \quad \text{is a weight.} \quad (24.41)$$

(If j is half-integer, there is no state in the hyperplane at $\vec{\mu} - \frac{1}{2}k\vec{\alpha}$.) Thus, given a root $\vec{\alpha}$ and a state $|\vec{\mu}, R\rangle$, we can **use a Weyl reflection to obtain another state with the same degeneracy**.

One can use any root $\vec{\alpha}$ to perform a Weyl reflection on all states of an irrep R . The set of all these reflections for all roots forms a finite group, called the Weyl group of the LA. It can act on any irrep.

We can now complete the filling out of the **10** multiplet of $SU(3)$.



The states I, B, C, E, F, H have the same weights as the roots of $SU(3)$. The 3 hyperplanes orthogonal to the roots are indicated by dotted lines. Starting with the state at A , consider the simple root OI . It yields a hyperplane (a line in this case) containing O and G , and reflection about this line yields the state D . Consider next the simple root OC ; it yields a hyperplane containing O and A , and the state D is reflected to G . Since A was nondegenerate, also D and G are nondegenerate.

Next consider the state at B . It is nondegenerate because it can only be reached in one way, by the lowering operator $E_{-\alpha^I}$. Successive Weyl reflections show that the states B, C, E, F, H, I are all also nondegenerate. The state at the origin is invariant under Weyl reflections, and to show that it, too, is nondegenerate, we needed a more detailed analysis.

The Weyl group is generated by the simple Weyl reflections (reflections with respect to hyperplanes that are orthogonal to the simple roots). Products of simple Weyl reflections generate Weyl reflections for all other roots, and products of an even number of Weyl reflections yield rotations. For the case of $SU(3)$ with the two simple Weyl reflections s_1 (about the line OG) and s_2 (about the line OA), the Weyl group thus constructed has the following 6 group elements

$$W = \{1, s_1, s_2, s_1 s_2, s_2 s_1, s_1 s_2 s_1\} \quad (24.42)$$

The group element $s_1 s_2$ yields a rotation over $\frac{2\pi}{3}$, and $s_2 s_1$ yields a rotation over $-\frac{2\pi}{3}$, while $s_1 s_2 s_1 = s_2 s_1 s_2$ yields a Weyl reflection corresponding to the root $\vec{\alpha}^I + \vec{\alpha}^{II}$ (the highest weight). The Weyl group for $SU(3)$ is the dihedral finite group D_3 .

Weyl groups.⁵ We now determine the Weyl group for each of the four families of classical groups. We begin with $SU(N)$.

$SU(N)$ Recall the following results for the weights $\vec{\mu}^i$ of the defining rep and the roots of $SU(N)$

$$\begin{aligned} \vec{\mu}^i \cdot \vec{\mu}^j &= \delta^{ij} - \frac{1}{N} \quad \text{see (16.13); } \vec{\mu}^1 + \dots + \vec{\mu}^N = 0 \\ \vec{\alpha}^{ij} &= \pm(\vec{\mu}^i - \vec{\mu}^j); \quad (\vec{\alpha}^{ij})^2 = (\vec{\mu}^i)^2 + (\vec{\mu}^j)^2 - 2\vec{\mu}^i \cdot \vec{\mu}^j = 2\left(1 - \frac{1}{N}\right) + \frac{2}{N} = 2. \end{aligned} \quad (24.43)$$

The simple roots are $\vec{\alpha}^I = \vec{\mu}^I - \vec{\mu}^{I+1}$ for $I = 1, \dots, N-1$.

A Weyl reflection of a weight $\vec{\mu}^k$ wrt a root $\vec{\alpha}^{ij}$ is given by (see (24.41))

$$R(\vec{\alpha}^{ij})\vec{\mu}^k = \vec{\mu}^k - \frac{2\vec{\alpha}^{ij} \cdot \vec{\mu}^k}{(\vec{\alpha}^{ij})^2} \vec{\alpha}^{ij} \quad \text{with } R(\vec{\alpha}^{ij}) = I - 2\frac{\vec{\alpha}^{ij} \vec{\alpha}^{ij,T}}{(\vec{\alpha}^{ij})^2} = R(-\vec{\alpha}^{ij}) \quad (24.44)$$

For $k \neq i, k \neq j$ one has $\vec{\mu}^k \cdot \vec{\alpha}^{ij} = 0$ hence

$$R(\vec{\alpha}^{ij})\vec{\mu}^k = R(-\vec{\alpha}^{ij})\vec{\mu}^k = \vec{\mu}^k. \quad (24.45)$$

For $\vec{\mu}^i$ and $\vec{\mu}^j$ one obtains

$$\begin{aligned} R(\vec{\alpha}^{ij})\vec{\mu}^i &= \vec{\mu}^i - 2\frac{\vec{\alpha}^{ij} \cdot \vec{\mu}^i}{(\vec{\alpha}^{ij})^2} \vec{\alpha}^{ij} = \vec{\mu}^i - ((\vec{\mu}^i)^2 - \vec{\mu}^j \cdot \vec{\mu}^i) \vec{\alpha}_{ij} = \vec{\mu}^i - \vec{\alpha}^{ij} = \vec{\mu}^j \\ R(\vec{\alpha}^{ij})\vec{\mu}^j &= \vec{\mu}^j + \vec{\alpha}^{ij} = \vec{\mu}^i. \end{aligned} \quad (24.46)$$

⁵We thank Luis Alvarez-Gaumé for help in writing this section.

Thus the reflection operators $R(\vec{\alpha}^{ij})$ act as transpositions on the pairs $(\vec{\mu}^i, \vec{\mu}^j)$ and keep the other weights $\vec{\mu}^k$ fixed. The complete set of $\binom{N}{2}$ transpositions generate the finite group S_N . Hence the Weyl group for $SU(N)$ is given by S_N

$$\mathcal{W}(SU(N)) = S_N. \quad (24.47)$$

This agrees with (24.42) since $D_3 = S_3$.

$SO(2N)$ We recall that the roots of $SO(2N)$ are given by $(0, \dots, \pm 1, \dots, \pm 1, 0)$. We introduce a basis of N orthonormal vectors $\vec{e}^1, \dots, \vec{e}^N$ such that $\vec{e}^i \cdot \vec{e}^j = \delta^{ij}$. Then the positive roots are $\vec{\alpha}_{\pm}^{ij} = \vec{e}^i \pm \vec{e}^j$ for $i < j$, and the simple roots are $\vec{e}^1 - \vec{e}^2, \dots, \vec{e}^{N-1} - \vec{e}^N, \vec{e}^{N-1} + \vec{e}^N$. (We may restrict the analysis to positive roots since $R(-\vec{\alpha}^{ij}) = R(+\vec{\alpha}^{ij})$.) The action of the Weyl group on the basis $\vec{e}^1, \dots, \vec{e}^N$ is easy to compute. Since $\vec{e}^k$ is orthogonal to $\vec{\alpha}_{\pm}^{ij}$ for $k \neq i, k \neq j$, we have

$$R(\vec{\alpha}_{\pm}^{ij})\vec{e}^k = \vec{e}^k \quad \text{for } k \neq i, k \neq j. \quad (24.48)$$

On the other hand

$$\begin{aligned} R(\vec{\alpha}_{-}^{ij})\vec{e}^i &= \vec{e}^i - 2\frac{\vec{\alpha}_{-}^{ij} \cdot \vec{e}^i}{(\alpha_{-}^{ij})^2}\vec{e}^i = \vec{e}^i - \vec{\alpha}_{-}^{ij} = \vec{e}^j \\ R(\vec{\alpha}_{-}^{ij})\vec{e}^j &= \vec{e}^j + \vec{\alpha}_{-}^{ij} = \vec{e}^i. \end{aligned} \quad (24.49)$$

These are the same results as for $SU(N)$, reflecting that $SU(N) \subset SO(2N)$. (In $SU(N)$ we used $\vec{\mu}^i$ instead of $\vec{e}^i$, satisfying $\vec{\mu}^i \cdot \vec{\mu}^j = \delta^{ij} - \frac{1}{N}$ instead of $\vec{e}^i \cdot \vec{e}^j = \delta^{ij}$, but the $-\frac{1}{N}$ cancel in the inner products $\vec{\alpha}^{ij} \cdot \vec{\mu}^j$ and $\vec{\alpha}^{ij} \cdot \vec{\alpha}^{ij}$.)

For the remaining roots $\vec{\alpha}_{+}^{ij}$ we find

$$\begin{aligned} R(\vec{\alpha}_{+}^{ij})\vec{e}^i &= \vec{e}^i - 2\frac{\vec{\alpha}_{+}^{ij} \cdot \vec{e}^i}{(\alpha_{+}^{ij})^2}\vec{\alpha}_{+}^{ij} = \vec{e}^i - \vec{\alpha}_{+}^{ij} = -\vec{e}^j \\ R(\vec{\alpha}_{+}^{ij})\vec{e}^j &= \vec{e}^j - \vec{\alpha}_{+}^{ij} = -\vec{e}^i. \end{aligned} \quad (24.50)$$

Instead of using $R(\vec{\alpha}_{+}^{ij})$ as generators of the Weyl group, we choose the following operators

$$T^{ij} = R(\vec{\alpha}_{+}^{ij})R(\vec{\alpha}_{-}^{ij}) \quad \text{with} \quad T^{ij} \begin{pmatrix} \vec{e}^i \\ \vec{e}^j \end{pmatrix} = \begin{pmatrix} -\vec{e}^i \\ -\vec{e}^j \end{pmatrix}. \quad (24.51)$$

So the generators of the Weyl group for $SO(2N)$ are $R(\vec{\alpha}_{+}^{ij})$ and T^{ij} . One can evaluate $R(\vec{\alpha}_{+}^{ij})R(\vec{\alpha}_{-}^{ij})$ to obtain an operator expression for T^{ij} , but it is clear by inspection what it is

$$T^{ij} = 1 - 2\frac{\vec{e}^i \vec{e}^i, T}{(\vec{e}^i)^2} - 2\frac{\vec{e}^j \vec{e}^j, T}{(\vec{e}^j)^2} \quad \text{for } i < j \quad (24.52)$$

It is also clear that it factorizes as follows

$$T = R(e^i)R(e^j) \quad \text{with } R(e^i) = 1 - 2\frac{\vec{e}^i \vec{e}^{i,T}}{(\vec{e}^i)^2} \quad (24.53)$$

One has $R(e^i)e^i = -e^i$ but $R(e^i)e^k = e^k$ for $k \neq i$ because $e^{i,T} \cdot e^j = 0$ for $i \neq j$.

We first consider the group generated by $R(\vec{\alpha}_{-}^{ij})$ and $R(e^i)$. The $R(e^i)$ satisfy

$$R(e^i)R(e^i) = I, \quad R(e^i)R(e^j) = R(e_j)R(e_i) \text{ for } i \neq j \quad (24.54)$$

So the $R(e_i)$ generate N commuting groups Z_2 , denoted by $(Z_2)^N$. They form an invariant subgroup

$$\begin{aligned} R(\vec{\alpha}_{-}^{ij})R(e^i)R(\vec{\alpha}_{-}^{ij}) &= R(\vec{\alpha}_{-}^{ij}) \left(1 - 2\frac{\vec{e}^i \vec{e}^{i,T}}{(\vec{e}^i)^2} \right) R(\vec{\alpha}_{-}^{ij}) \\ &= 1 - 2 \left(\frac{R(\vec{\alpha}_{-}^{ij})e^i}{(\vec{e}^i)^2} \right) \left(\frac{e^{i,T}R(\vec{\alpha}_{-}^{ij})}{(\vec{e}^i)^2} \right) \\ &= 1 - 2\frac{e^j e^{j,T}}{(\vec{e}^j)^2} = R(e^j) \end{aligned} \quad (24.55)$$

where we used $R(\vec{\alpha}_{-}^{ij})\vec{e}^i = \vec{e}^j$ and $\vec{e}^{i,T}R(\vec{\alpha}_{-}^{ij}) = \vec{e}^{i,T} - (e^i \cdot \alpha^{ij})\alpha^{ij,T} = e^{j,T}$. Thus the group generated by $R(\vec{\alpha}_{-}^{ij})$ and $R(e^i)$ is the semi-direct product $S_N \rtimes (Z_2)^N$. However, the Weyl group is generated by the $R(\vec{\alpha}_{-}^{ij})$ and the $R(e^i)R(e^j)$. So the Weyl group is the group of all products of $R(\vec{\alpha}_{-}^{ij})$ and all **even** number of products of $R(e^i)$. We denote this group by

$$\mathcal{W}(SO(2N)) = S_N \rtimes (Z_2)_+^N. \quad (24.56)$$

As an example consider $SO(5)$ with $\mathcal{W}(SO(5)) = S_2 \rtimes (Z_2)^2$. Denoting the reflections wrt the simple roots $\vec{\alpha}^I = (1, -1)$ and $\vec{\alpha}^{II} = (0, 1)$ by σ_1 and σ_2 , one finds that σ_1 and σ_2 act on the coordinates x and y as $\sigma_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}$ and $\sigma_2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ -y \end{pmatrix}$. Hence σ_I and σ_{II} can be represented by Pauli matrices $\sigma_I = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $\sigma_{II} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Taking products of σ_I and σ_{II} one finds the Weyl group $\mathcal{W}(SO(5)) = D_4$ where $D_4 = \{a, b | a^4 = e, b^2 = e, bab^{-1} = a^{-1}\}$. This does not look like $S_2 \rtimes (Z_2)^2$, but these two groups are the same if one sets $a = \sigma_I \sigma_{II}$ and $b = \sigma_{II}$ because then one can write the Weyl group as follows $(e, ab) \rtimes [(e, a^2b) \otimes (e, b)]$ (use that a^2 commutes with b).

$SO(2N+1)$ The roots are $\pm e^i \pm e^j$ and $\pm e^i$. The Weyl reflection wrt the roots $\pm e^i$ are just the $R(e^i)$ we constructed for $SO(2N)$. Hence now the Weyl group contains products of any number of $R(e^i)$, and the restriction that the number of $R(e^i)$ has to be even, no longer holds. We conclude

$$\mathcal{W}(SO(2N+1)) = S_N \rtimes (Z_2)^N. \quad (24.57)$$

$USp(2N)$ The positive roots are the N -component vectors $(\vec{\mu}^i - \vec{\mu}^j, 0)$ for $i < j$ and $(\vec{\mu}^i + \vec{\mu}^j, \sqrt{\frac{2}{N}})$ for $i \leq j$, see (18.16), where the $(N-1)$ -component vectors $\vec{\mu}^i - \vec{\mu}^j$ are the positive roots of $SU(N)$. One can again introduce operators $R(\vec{\alpha}_{-}^{ij})$ and $R(e^i)$, and the analysis proceeds as in the case of $SO(2N+1)$. Hence

$$\mathcal{W}(USp(2N)) = S_N \rtimes (Z_2)^N. \quad (24.58)$$

Conjugacy classes. For each of the classical groups, the set of all irreps is infinite, but can be divided into a finite set of conjugacy classes. Each conjugacy class is infinite, and two irreps in the same conjugacy class have weights that are, in a sense to be explained, equivalent. Consider a simple Lie algebra of rank n . States of an irrep are labeled by their weights: vectors in the n -dimensional weight space. The set of all possible weights of all possible representations forms a lattice, called the **weight lattice** Λ_W . Another lattice, the **root lattice** Λ_R , is formed by the integral linear combinations of the weights of the adjoint representation (the roots). Two representations are said to belong to the same conjugacy class if their weights differ by a vector on the root lattice.

For $SO(2N)$ there are 4 conjugacy classes: the class which includes the defining (vector) irrep, the class with the adjoint irrep, the class with the spinor (s) irrep, and the class with the conjugate spinor (c) irrep. We first check that the weights of these four irreps indeed belong to different conjugacy classes. Afterwards we prove that there are precisely 4 classes.

Inside a given irrep, the weight vectors differ by a vector on the root lattice. So it is sufficient to consider only one weight vector of each irrep. The following are weight vectors of the 4 irreps of $SO(2N)$ mentioned above.

$$\begin{aligned} (1, \dots, 0, 0, 0, \dots, 0) &: \text{defining rep.} \\ (1, 1, \dots, 0, 0, 0, \dots, 0, 0, \dots, 0) &: \text{adjoint rep.} \\ (\frac{1}{2}, \frac{1}{2}, \dots, \frac{1}{2}, \frac{1}{2}) &: \text{s-spinor irrep.} \\ (\frac{1}{2}, \frac{1}{2}, \dots, \frac{1}{2}, -\frac{1}{2}) &: \text{c-spinor irrep.} \end{aligned} \quad (24.59)$$

- The difference of the two spinor weights is $(0, 0, \dots, 1)$ which does not lie on the root lattice.
- The differences between the first two and the last two weight vectors contains half-integral entries, so these differences do not lie on the root lattice.
- The difference between the first two weights is $(0, 1, 0, 0)$. This difference does not lie on the root lattice.

Thus there are at least 4 classes for $SO(2N)$. We shall later prove that there are no more. For example, the highest weight of the symmetric tensor $t^{\mu\nu} = t^{\nu\mu}$ is $\vec{\mu}_{def}^1 + \vec{\mu}_{def}^1 = 2\vec{\mu}_{FW}^1 = (2, 0, \dots, 0)$ which lies on Λ_R since $(2, 0, \dots, 0)$ is the sum of $(1, -1, 0, \dots, 0)$ and $(1, +1, 0, \dots, 0)$. So this

irrep does not give another conjugacy class, but it belongs to the class which contains the adjoint rep.

Next we consider $SO(2N+1)$. We claim that there are 2 conjugacy classes. Some weights of three of the fundamental irreps of $SO(2N+1)$ are as follows:

$$\begin{aligned} (1, 0, \dots, 0) : & \quad \text{defining (vector) rep.} \\ (1, 0, \dots, 0) : & \quad \text{adjoint rep.} \\ (\tfrac{1}{2}, \tfrac{1}{2}, \dots, \tfrac{1}{2}, \tfrac{1}{2}) : & \quad \text{spinor irrep.} \end{aligned} \tag{24.60}$$

Clearly the spinor irrep belongs to a different conjugacy class as the other two irreps. But the other two irreps belong to the same conjugacy class because the weight vectors of the defining rep are a subset of the roots. So for $SO(2N+1)$ there are (at least, but in fact precisely) 2 conjugacy classes which one might call the tensor class and the spinor class. The highest weight of $t^{\mu\nu} = t^{\nu\mu}$ is $(2, 0, \dots, 0) = 2\vec{\mu}_{FW}^1$ which belongs to the first class (the tensor class).

We consider $SU(N)$. We claim that for $SU(N)$ there are N conjugacy classes: $N-1$ classes containing the $N-1$ fundamental irreps, and the class with the adjoint rep. First we prove that for $SU(N)$ each of the $N-1$ fundamental irreps belongs to a different conjugacy class and then that the fundamental weights do not lie on the root lattice. The highest weight vectors $\vec{\mu}_{FW}^J$ with $J \leq N-1$ are given by $\vec{\mu}_{FW}^J = \sum_{k=1}^J \vec{\mu}_{def}^k$. The root lattice Λ_R is spanned by the $N-1$ simple roots $\vec{\alpha}^J = \vec{\mu}_{def}^J - \vec{\mu}_{def}^{J+1}$. We claim that the difference of the highest weights of two different fundamental irreps $\vec{\mu}_{FW}^J - \vec{\mu}_{FW}^K = \sum_{k=J+1}^K \vec{\mu}_{def}^k$ does not lie on Λ_R . To prove this claim, note that the $N-1$ vectors $\vec{\mu}_{def}^k$ with $k = 1, \dots, N-1$ form a basis; on this basis the coefficients of $\vec{\mu}_{FW}^J - \vec{\mu}_{FW}^K$ are nonnegative integers. Then the equations to be proved are as follows

$$\sum_{k=J}^K \vec{\mu}_{def}^k \neq \sum_{l=1}^{N-1} n_l (\vec{\mu}_{def}^l - \vec{\mu}_{def}^{l+1}) \quad \text{for } J = 1, \dots, N-1; \quad J \leq K \leq N \tag{24.61}$$

The cases with $J = 1$ prove that the fundamental weights do not lie on the root lattice, while the cases $J > 1$ show that two different fundamental irreps do not belong to the same conjugacy class. If $j = J$ we substitute $\vec{\mu}_{def}^N = -\vec{\mu}_{def}^1 - \dots - \vec{\mu}_{def}^{N-1}$ to obtain the following $N-1$ equations for the nonnegative integers n_l

$$\begin{aligned} O = (n_1 + n_{N-1} - 1, n_2 - n_1 + n_{N-1} - 1, \dots, n_K - n_{K-1} + n_{N-1} - 1, n_{K+1} - n_K + n_{N-1}, \\ \dots, n_{N-1} - n_{N-2} + n_{N-1}) \end{aligned} \tag{24.62}$$

Solving as $n_{N-2} = 2n_{N-1}$, $n_{N-3} = 3n_{N-1}$, $\dots$, $n_{K-1} = (N-K+1)n_{N-1} - 1$, $\dots$ we arrive at the equation $n_1 + n_{N-1} - 1 = 0$ with $n_1 = (N-1)n_{N-1} - K + 1$. This would imply that $K = Nn_{N-1}$, a contradiction. For the case with $J > 1$ one should proceed similarly. We consider again as an example the tensor $t^{\mu\nu} = t^{\nu\mu}$. It has $2\vec{\mu}_{FW}^1$ as highest weight, so it belongs to the first conjugacy class.

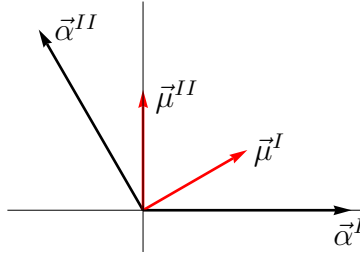
Finally we consider the Lie algebra of $USp(2N)$. For $USp(2N)$ there are 2 conjugacy classes. The highest weight of some of the irreps of $USp(2N)$ are as follows:

$$\begin{aligned}
\left(\vec{\mu}_{def}^1, \frac{1}{\sqrt{2N}}\right) &: \text{defining (vector) rep.} \\
\left(\vec{\mu}_{def}^{(1)} + \vec{\mu}_{def}^{(2)}, \frac{2}{\sqrt{2N}}\right) &: \text{AT } (t^{\mu\nu} = -t^{\nu\mu}) \text{ rep.} \\
\left(2\vec{\mu}_{def}^{(1)}, \frac{2}{\sqrt{2N}}\right) &: \text{symmetric tensor } (t^{\mu\nu} = t^{\nu\mu}) = \text{adjoint rep.}
\end{aligned} \tag{24.63}$$

The N simple roots are

$$(\vec{\mu}_{def}^J - \vec{\mu}_{def}^{J+1}, 0) \text{ for } 1 \leq J \leq N-1, \text{ and } \left(2\vec{\mu}^N, \sqrt{\frac{2}{N}}\right). \tag{24.64}$$

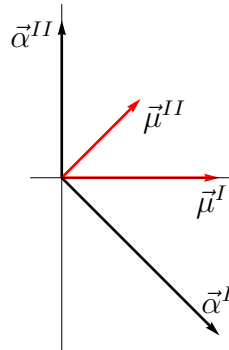
$SU(3)$.



$$\begin{aligned}
\vec{\alpha}^I &= (1, 0) \\
\vec{\alpha}^{II} &= \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \\
\vec{\mu}_{FW}^I &= \left(\frac{1}{2}, \frac{1}{2\sqrt{3}}\right) \\
\vec{\mu}_{FW}^{II} &= \left(0, \frac{1}{\sqrt{3}}\right)
\end{aligned}
\quad \frac{\text{vol } \Lambda_W^{(0)} = \frac{1}{2\sqrt{3}}}{\text{vol } \Lambda_R^{(0)} = \frac{1}{2}\sqrt{3}} = \frac{1}{3}. \tag{24.65}$$

For $SU(N)$ product is always $(\frac{1}{2})^2$ since $\frac{\vec{\alpha} \cdot \vec{\mu}}{\vec{\alpha} \cdot \vec{\alpha}} = 1$ and all roots have same length.

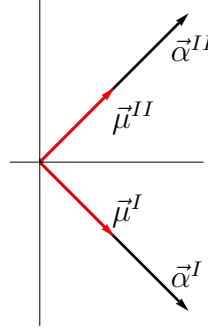
$SO(5)$.



$$\begin{aligned}
\vec{\alpha}^I &= (1, -1) \\
\vec{\alpha}^{II} &= (0, 1) \\
\vec{\mu}_{FW}^I &= (1, 0) \\
\vec{\mu}_{FW}^{II} &= \left(\frac{1}{2}, \frac{1}{2}\right)
\end{aligned}
\quad \frac{\text{vol } \Lambda_W^{(0)} = \frac{1}{2}}{\text{vol } \Lambda_R^{(0)} = 1} = \frac{1}{2}. \tag{24.66}$$

For $SO(2N+1)$ the product is $\frac{1}{2}$ instead of $\frac{1}{4}$ since simple roots have different length.

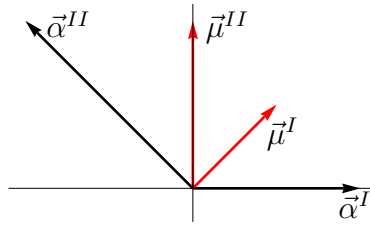
$SO(4)$.



$$\begin{aligned}
 \vec{\alpha}^I &= (1, -1) \\
 \vec{\alpha}^{II} &= (1, 1) \\
 \vec{\mu}_{FW}^I &= \left(\frac{1}{2}, -\frac{1}{2}\right) \\
 \vec{\mu}_{FW}^{II} &= \left(\frac{1}{2}, \frac{1}{2}\right)
 \end{aligned}
 \quad
 \frac{\text{vol } \Lambda_W^{(0)} = \frac{1}{4}}{\text{vol } \Lambda_R^{(0)} = 1} = \frac{1}{4}. \quad (24.67)$$

For $SO(N)$ the product is always $\frac{1}{4}$.

$USp(4)$.



$$\begin{aligned}
 \vec{\alpha}^I &= (1, 0) \\
 \vec{\alpha}^{II} &= (-1, 1) \\
 \vec{\mu}_{FW}^I &= \left(\frac{1}{2}, \frac{1}{2}\right) \\
 \vec{\mu}_{FW}^{II} &= (0, 1)
 \end{aligned}
 \quad
 \frac{\text{vol } \Lambda_W^{(0)} = \frac{1}{2}}{\text{vol } \Lambda_R^{(0)} = 1} = \frac{1}{2}. \quad (24.68)$$

The number of conjugacy classes for $SU(3)$, $SO(5)$, $SO(4)$ and $USp(4)$ obtained as the ratio of the volumes of the unit cell of the root lattice $\Lambda_R^{(0)}$ and the weight lattice $\Lambda_W^{(0)}$.

The number of conjugacy classes is equal to the ratio of the volume of the unit cell $\Lambda_R^{(0)}$ of the root lattice divided by the volume of the unit cell of the weight lattice $\Lambda_W^{(0)}$. This proves that the number of equivalence classes of $SO(2N)$, $SO(2N+1)$ and $SU(N)$ is 4, 2, N , respectively, and 2 for $USp(2N)$. So there are 2 conjugacy classes for $USp(2N)$, and the weight diagram for $USp(4)$ in chapter 19, figure 19.1, shows that the adjoint (symmetric tensor) and the AT-tensor irreps belong to the same conjugacy class, and the defining irrep to the other class. This suggests that irreps constructed from an even number of vector irreps form one class, and those with an odd number of vector irreps form another class. Indeed, the difference of a weight of an irrep due to an even number of vector irreps and the weight of an irrep due to an odd number of vector irreps has an odd number of irreps, while the adjoint irrep is even.

Chapter 25

Casimir operators, Killing metrics, and Dynkin indices

Given a semisimple Lie group G with Lie algebra $\mathcal{L}\mathcal{A}$, the Killing metric for the adjoint representation is by definition nonsingular. Hence its inverse g_{adj}^{ab} exists. One can then construct the so-called **quadratic Casimir operator** for a representation R

$$C_2(R) \sim g_{\text{adj}}^{ab} T_a^{(R)} T_b^{(R)} \quad (25.1)$$

We have written $\sim$ because we shall want to choose a convenient normalization. To discuss this normalization, we go back to g_{ab}^{adj} and consider what we shall call the Killing metric in a representation R

$$g_{ab}(R) = \text{tr } T_a^{(R)} T_b^{(R)} \quad (25.2)$$

For R equal to the adjoint representation $g_{ab}(\text{adj})$ is equal to g_{ab}^{adj} , but for different R the Killing metrics are still the same up to a constant

$$g_{ab}(R) = c(R) g_{ab}^{\text{adj}} \quad (25.3)$$

The proof will be given later.

For all simple compact $\mathcal{L}\mathcal{A}$'s we have chosen the generators in the defining representation such that

$$\text{tr } T_a^{\text{def}} T_b^{\text{def}} = -\delta_{ab} T(\text{def}). \quad (25.4)$$

The minus sign is included in order that $T(\text{def})$ is positive if the T_a^{def} are antihermitian. Then in any representation one has

$$\text{tr } T_a^{(R)} T_b^{(R)} = -\delta_{ab} T(R). \quad (25.5)$$

The $T(R)$ are called **Dynkin indices**. Clearly, $c(R)$ in (25.3) is equal to $\frac{T(R)}{T(\text{adj})}$.

The metric $g_{ab}(R)$ for any R is an invariant tensor: it does not change if one transforms it as

a tensor

$$\delta g_{ab}(R) = \lambda^c (T_c^{\text{adj}})_a^{a'} g_{a'b}^{(R)} + \lambda^c (T_c^{\text{adj}})_b^{b'} g_{ab'}^{(R)} = 0 \quad (25.6)$$

The proof follows from the observation that

$$\text{tr } T_a^{(R)} T_b^{(R)} = \text{tr} \left(g^{(R),-1} T_a^{(R)} g^{(R)} g^{(R),-1} T_b^{(R)} g^{(R)} \right) \quad (25.7)$$

Expanding $g^{(R)} = I + \lambda^c T_c^{(R)} + \dots$, one obtains to linear order in λ

$$g^{(R),-1} T_a^{(R)} g^{(R)} = T_a + f_{ac}{}^b \lambda^c T_b^{(R)} \quad (25.8)$$

and substitution into (25.7) proves (25.6).

If $g_{ab}(R)$ is an invariant tensor, and if $g_{ab}(R) = -T(R)\delta_{ab}$, then also δ_{ab} is an invariant tensor. We shall use δ_{ab} and δ^{ab} to lower and raise indices $a, b, \dots$ of the adjoint representation. Using this, we fix the normalization of the quadratic Casimir operator in a representation R by defining

$$C_2(R) = -\delta^{ab} T_a^{(R)} T_b^{(R)} \quad (25.9)$$

The operator $C_2(R)$ has an important property: it commutes with all generators $T_a^{(R)}$. The proof is straightforward

$$\begin{aligned} -[C_2(R), T_c^{(R)}] &= \delta^{ab} [T_a^{(R)} T_b^{(R)}, T_c^{(R)}] = \delta^{ab} T_a^{(R)} [T_b^{(R)}, T_c^{(R)}] + \delta^{ab} [T_a^{(R)}, T_c^{(R)}] T_b^{(R)} \\ &= \delta^{ab} T_a^{(R)} f_{bc}{}^p T_p^{(R)} + \delta^{ab} f_{ac}{}^p T_p^{(R)} T_b^{(R)} = \delta^{ab} f_{bc}{}^p \left(T_a^{(R)} T_p^{(R)} + T_p^{(R)} T_a^{(R)} \right) = 0 \end{aligned} \quad (25.10)$$

We used that $\delta^{ab} f_{bc}{}^p = f^a{}_c{}^p$ is antisymmetric in a, p .

Suppose the representation R is an irreducible representation. Since $C_2(R)$ commutes with all generators $T_a^{(R)}$, it commutes with all group elements in the representation R , and hence Schur's lemma tells us that $C_2(R)$ is constant when acting on all states of the irrep R .

Consider now the trace of the quadratic Casimir operator. It can be evaluated in two ways

$$\text{tr } \delta^{ab} T_a^{(R)} T_b^{(R)} = -C_2(R) \dim R = \delta^{ab} (-\delta_{ab} T(R)) = -T(R) \dim G \quad (25.11)$$

where by $\dim R$ we mean the dimension of the matrices $T_a^{(R)}$, and $\dim G$ equals the number of generators (which is also the dimensions of the matrices $(T_a^{\text{adj}})_b^c = f_{ba}{}^c$). Hence we get the important relation

$$\boxed{C_2(R) \dim R = T(R) \dim G} \quad (25.12)$$

We shall now first show how to calculate $T(R)$, and then use this relation to obtain the value of $C_2(R)$.

To compute $T(R)$ as defined by $\text{tr } T_a^{(R)} T_b^{(R)} = -\delta_{ab} T(R)$, we choose a particular generator which is as simple as possible. For $SU(N)$ groups we choose a generator A in the defining rep which is

diagonal and has imaginary entries on the diagonal. (We could further specify the values of these entries, but we shall not need this information.) Since every irrep of $SU(N)$ can be obtained from the defining (sometimes called vector or fundamental) irrep by tensoring and reducing (by taking direct products of tensors and reducing these direct products into irreps), we see that the constants $T(R)$ are **integers** (positive, negative or zero). The following rules are useful

1) For a direct sum of two irreps R and R' one has

$$T(R \oplus R') = T(R) + T(R'). \quad (25.13)$$

This follows from $A^{R \oplus R'} = \begin{pmatrix} A^{(R)} & 0 \\ 0 & A^{(R')} \end{pmatrix}$.

2) For a direct product of two irreps R and R' one has

$$T(R \otimes R') = T(R) \dim(R') + T(R') \dim(R). \quad (25.14)$$

This follows from $A^{R \otimes R'} = A^{(R)} \otimes I^{R'} + I^R \otimes A^{(R')}$, and $\text{tr } A^{(R)} = \text{tr } A^{(R')} = 0$. So $\text{tr } A^{R \otimes R'} A^{R \otimes R'}$ is equal to $(\text{tr } A^{(R)} A^{(R)}) \text{tr } I^{R'} + (\text{tr } A^{(R')} A^{(R')}) \text{tr } I^R$.

3) Under complex conjugation one has

$$T(R^*) = T(R). \quad (25.15)$$

This follows from the antihermiticity of the generators which implies $(T_a^{(R)})^* = -(T_a^{(R)})^T$, and the invariance of the trace under transpositions, $\text{tr } (T_a^{(R)})^T (T_b^{(R)})^T = \text{tr } T_a^{(R)} T_b^{(R)}$.

We shall now compute $T(R)$ for some irreps of $SU(N)$ and deduce the corresponding $C_2(R)$.

The N of $SU(N)$. $C_2(\text{def}) = T(\text{def}) \frac{\dim G}{\dim R(\text{def})} = T(\text{def}) \frac{N^2-1}{N}$. For $SU(N)$ we choose the normalization $\text{tr } T_a^{\text{def}} T_b^{\text{def}} = -\frac{1}{2} \delta_{ab}$. For example, for $SU(2)$ one has $T_a^{\text{def}} = -\frac{i}{2} \sigma_a$, and $\text{tr } (-\frac{i}{2} \sigma_a) (-\frac{i}{2} \sigma_b) = -\frac{1}{2} \delta_{ab}$. For $SU(3)$, we choose as generators $T_a^{\text{def}} = -\frac{i}{2} \lambda_a$, where λ_a are the 3×3 hermitian Gell-Mann matrices. Then, we also have $T(\text{def}) = \frac{1}{2}$ for $SU(3)$. For all $SU(N)$ we then get

$$C_2(\text{def}) = \frac{1}{2} \frac{N^2 - 1}{N}. \quad (25.16)$$

The adjoint of $SU(N)$. The adjoint representation of $SU(N)$ is obtained from

$$\begin{array}{c} \begin{array}{|c|} \hline \square \\ \hline \vdots \\ \hline \square \\ \hline \end{array} \otimes \begin{array}{|c|} \hline \square \\ \hline \end{array} = \begin{array}{|c|c|} \hline \square & \square \\ \hline \vdots & \vdots \\ \hline \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|} \hline \square \\ \hline \vdots \\ \hline \square \\ \hline \end{array} \\ N^* \times N = \text{adjoint} + \mathbf{1} \end{array} \quad (25.17)$$

Hence $T(\text{adj}) = T(\mathbf{N}^* \times \mathbf{N} - \mathbf{1})$, and using the rules given above, we get

$$\begin{aligned} T(\text{adj}) &= T(\mathbf{N}) \dim(\mathbf{N}^*) + T(\mathbf{N}^*) \dim(\mathbf{N}) - T(\mathbf{1}) \\ &= NT(\text{def}) + NT(\text{def}) - 0 = 2NT(\text{def}) = N. \end{aligned} \quad (25.18)$$

Since for the adjoint rep $\dim T(\text{adj})$ is equal to $\dim G$, $T(\text{adj})$ is equal to $C_2(\text{adj})$, hence

$$C_2(\text{adj}) = 2NT(\text{def}) = N. \quad (25.19)$$

A simple Lie algebra of rank r has r generalized Casimir operators, and r generalized Dynkin indices, of which $C_2(R)$ and $T(R)$ are the first. The cubic Casimir operator (sometimes called the d -symbol) plays a role in anomalies, and is discussed in detail in chapter 29.

The general form of the Killing metric in the Cartan-Weyl basis is block diagonal:

$$c(K)g_{ab}^{(K)} = \begin{pmatrix} g_{IJ}^{\text{adj}} & & & & \\ & 0 & g_{\alpha, -\alpha}^{\text{adj}} & & \\ & g_{\alpha, -\alpha}^{\text{adj}} & 0 & & \\ & & & 0 & g_{\beta, -\beta}^{\text{adj}} \\ & & & g_{\beta, -\beta}^{\text{adj}} & 0 \\ & & & & & \ddots \end{pmatrix} \quad (25.20)$$

where $g_{IJ}^{\text{adj}} = \text{tr } H_I^{\text{adj}} H_J^{\text{adj}}$ and $g_{\alpha, -\alpha}^{\text{adj}} = \text{tr } E_{\alpha}^{\text{adj}} E_{-\alpha}^{\text{adj}}$. Hence if $\det g_{ab}^{(K)} \neq 0$, then also $\det g_{IJ}^{(K)} \neq 0$. To prove this structure, we first show that $g_{I\alpha}$ vanishes

$$c(K)g_{I\alpha}^{(K)} = \text{tr } H_I^{\text{adj}} E_{\alpha}^{\text{adj}} = f_{\beta I}^{\beta} f_{\beta \alpha}^{\beta} = 0 \quad (25.21)$$

since $f_{\beta \alpha}^{\beta} = 0$. What about $g_{\alpha \beta}^{(K)}$ for $\beta \neq -\alpha$? They vanish, too,

$$\begin{aligned} c(K)g_{\alpha \beta}^{(K)} &= \text{tr } E_{\alpha}^{\text{adj}} E_{\beta}^{\text{adj}} = f_{M\alpha}^N f_{N\beta}^M \quad (M = I \text{ or } \gamma) \\ &= f_{I\alpha}^{\alpha} f_{\alpha \beta}^I + f_{\gamma \alpha}^{\gamma + \alpha} f_{\gamma + \alpha, \beta}^{\gamma} = 0. \end{aligned} \quad (25.22)$$

We used that $f_{\alpha \beta}^I = 0$ if $\beta \neq -\alpha$, and $f_{\gamma + \alpha, \beta}^{\gamma} = 0$ if $\beta \neq -\alpha$. In particular the diagonal entries $g_{\alpha \alpha}$ vanish

$$\text{tr } E_{\alpha} E_{\alpha} = f_{M\alpha}^N f_{N\alpha}^M = f_{I\alpha}^{\alpha} f_{\alpha \alpha}^I + f_{\beta \alpha}^{\beta + \alpha} f_{\beta + \alpha, \alpha}^{\beta} = 0 \quad (25.23)$$

since $f_{\alpha \alpha}^I = 0$ and $f_{\alpha + \beta, \alpha}^{\beta} = 0$. Note that if $-\alpha$ were not a root, $g_{\alpha, -\alpha}$ would not exist, and the Killing metric $g_{ab}^{(K)}$ would be singular. Cartan's criterion would then imply that the LA is not semisimple. So, **if α is a root of a simple LA, also $-\alpha$ is a root.**

Finally, we discuss the values of $g_{\alpha, -\alpha}^{\text{adj}}$. We defined $E_{-\alpha} = E_{\alpha}^{\dagger}$, hence $g_{\alpha, -\alpha}^{\text{adj}} = \text{tr } E_{\alpha}^{\text{adj}} E_{\alpha}^{\text{adj} \dagger} > 0$. We could normalize E_{α} such that $g_{\alpha, -\alpha} = 1$; however, we prefer to leave this normalization open, and fix it for each of the simple Lie groups in the most convenient way. Then (25.20) is proven.

Uniqueness of the Killing metric. We want to prove that the Killing metrics for all representations R are proportional to each other: there is only one invariant symmetric bilinear form (with indices in the adjoint rep) for a simple Lie algebra up to an overall constant.¹

$$\begin{aligned}\mathrm{tr} T_a^{\mathrm{def}} T_b^{\mathrm{def}} &= -g_{ab}^{\mathrm{def}} = -T(\mathrm{def})\delta_{ab} \\ \mathrm{tr} T_a^R T_b^R &= -h_{ab}^R = -T(R)\delta_{ab} \quad (\text{to be proven}).\end{aligned}\tag{25.24}$$

Since invariant tensors are Clebsch-Gordan coefficients, the invariance and uniqueness of g_{ab} implies that in the decomposition of the symmetric product of two adjoint reps one finds the adjoint rep back precisely once. It is amusing to check this using Young tableaux for $SU(N)$.

To prove the uniqueness of the Killing metric, one considers the following matrix

$$\phi^a_b = (h^{R,-1})^{ac} g_{cb}^{\mathrm{def}}.\tag{25.25}$$

At the end we show that h_{ab}^R is invertible. We shall show that ϕ commutes with the generators in the adjoint irrep. Then Schur's lemma tells us that ϕ is proportional to the unit matrix, and thus that h_{ab}^R and g_{ab}^{def} are equal up to an overall constant.

To avoid too many uncontracted indices we consider Lie algebra elements $X = X^a T_a^R$ where X^a are numbers, and T_a^R generators in a representation R . Consider now the following expression

$$h(\phi([X, Y]), Z) \equiv h_{ab}^R \phi^a_c f_{pq}^c X^p Y^q Z^b.\tag{25.26}$$

We used the following definitions and notation

$$\begin{aligned}h(X, Z) &= -\mathrm{tr} XZ = h_{ab}^R X^a Z^b, & [X, Y] &= f_{pq}^c X^p Y^q T_c^R, \\ \phi(X) &= \phi^a_b X^b T_a^R, & \phi([X, Y]) &= \phi^c_d f_{pq}^d X^p Y^q T_c^R.\end{aligned}\tag{25.27}$$

Since by definition $h\phi = g^{\mathrm{def}}$, we can rewrite (25.26) as follows in terms of $X^{\mathrm{def}} = X^a T_a^{\mathrm{def}}$

$$\begin{aligned}g_{ab}^{\mathrm{def}} f_{pq}^a X^p Y^q Z^b &= -\mathrm{tr}[X^{\mathrm{def}}, Y^{\mathrm{def}}] Z^{\mathrm{def}} && (\text{trace over } T_a^{\mathrm{def}} T_b^{\mathrm{def}}) \\ &= \mathrm{tr} Y^{\mathrm{def}} [X^{\mathrm{def}}, Z^{\mathrm{def}}] && (\text{cyclicity of trace}) \\ &= -g_{ab}^{\mathrm{def}} (Y^a) (f_{pq}^b X^p Z^q) && (\text{definition of } g^{\mathrm{def}}) \\ &= -h_{ac}^R (\phi^c_b Y^a) (f_{pq}^b X^p Z^q) && (\text{definition of } \phi) \\ &= -h_{ac}^R \phi^c_b Y^b f_{pq}^a X^p Z^q && (\text{symmetry of } g_{ab}^{\mathrm{def}}) \\ &= \mathrm{tr} \phi(Y[X, Z]) && (\text{trace over } T_a^R T_b^R) \\ &= -\mathrm{tr}[X, \phi(Y)] Z && (\text{cyclicity of trace}) \\ &= h_{ab}^R (f_{pc}^a X^p \phi^c_q Y^q) (Z^b)\end{aligned}\tag{25.28}$$

¹I thank J.P. Ang who constructed this elegant proof.

Since X^a, Y^a, Z^a are arbitrary, and h is invertible, we arrive at $\phi^a{}_c f_{pq}{}^c = f_{pc}{}^a \phi^c{}_q$, or $\phi^a{}_c (T_p^{\text{adj}})^c{}_q = (T_p^{\text{adj}})^a{}_c \phi^c{}_q$, or

$$[\phi, T_p^{\text{adj}}] = 0. \quad (25.29)$$

Finally we prove that h_{ab}^R is invertible. From its definition in terms of $\text{tr } T_a^R T_b^R$ it follows that h_{ab}^R is an invariant metric, meaning

$$Y^c f_{ac}{}^{a'} h_{a'b}^R + Y^c f_{bc}{}^{b'} h_{ab'}^R = 0 \quad (25.30)$$

where Y^a are Lie group parameters. Consider now the null space of h_{ab}^R , namely those X^a which satisfy $X^a h_{ab}^R = 0$. If we contract (25.30) with X^a , the second term vanishes, hence also the first term vanishes

$$X^a Y^c f_{ac}{}^{a'} h_{a'b}^R = [X, Y]^{a'} h_{a'b}^R = 0. \quad (25.31)$$

Thus if X lies in the null space, also $[X, Y]$ lies in the null space. This means that the X 's form an ideal (an invariant subalgebra). But we assumed that we have a simple Lie algebra; then the only permitted ideals are either the empty algebra or the whole algebra. It cannot be the whole algebra because if for all X^a one has $X^a h_{ab}^R = 0$, the trace $\text{tr } T_a^R T_b^R$ would vanish for all a and b , which is impossible (for example $\text{tr } T_a T_a^\dagger = -\text{tr } T_a T_a$ is nonvanishing). Thus the null space is zero: there are no X^a with $h_{ab}^R X^a = 0$. This means that h_{ab}^R is invertible.

Chapter 26

The Haar measure

26.1 Definition and simple examples of the Haar measure

In some physical problems one studies functions such as spherical harmonics on spheres or other group manifolds or coset spaces, and one must integrate these functions. This requires a measure of integration. For example, if one integrates over $SU(2) = S_3$, one needs a definition of integration such that the result does not depend where on the sphere one picks the origin. When we studied the proofs of unitary and complete reducibility of irreps in finite group theory, we used that any function of group elements g satisfies the following equation

$$\sum_{g \in G} f(g) = \sum_{g \in G} f(g_0 g) \quad (26.1)$$

for any group element g_0 in the group G . We view this as a discrete integration over the group. The proof was almost obvious: acting with a fixed group element g_0 on all group elements of a group G leads merely to a permutation of all group elements and summing over a permuted set of elements is of course the same as summing over the original set of elements. For Lie groups we want to find a similar result: if $f(g)$ is an arbitrary function of the group elements of a Lie group, and if the group elements are parametrized by parameters $\alpha^1, \dots, \alpha^n$, then the aim is to find a measure $\mu(\alpha)$ such that

$$\int_G f(g(\alpha)) \mu(\alpha) d\alpha = \int_G f(g_0 g(\alpha)) \mu(\alpha) d\alpha \quad (26.2)$$

We call the measure which satisfies this requirement the left measure $\mu_L(\alpha)$ because we multiply with g_0 on the left. (The name left-invariant measure would be more appropriate.) The corresponding right measure $\mu_R(\alpha)$ will be discussed later.

This problem was solved for Lie groups by Hermann Weyl in 1925, 1926 for the groups $O(N)$ and $U(N)$, and extended by him and his student F. Peter in 1927 to any compact Lie group. The Hungarian mathematician Alfréd Haar extended their work to “any locally compact group with a countable open basis”. The notion of an “invariant integration” as in (26.2) was already proposed

in 1897 by Adolf Hurwitz, a student of Felix Klein. In what follows we shall consider arbitrary n -parameter Lie groups, not necessarily compact ones. The group volume $\int_G \mu(\alpha) d\alpha$ is finite if and only if the group is compact. Integration over coset manifolds will be deferred till chapter 34.

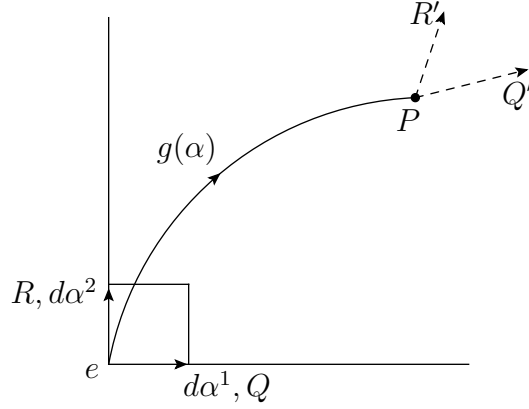


Figure 26.1:

There is a clear and elementary way to obtain a measure satisfying (26.2). It uses the concept of “sweeping out” of a little cube from the origin in group parameter space to a point P on the group manifold, and computing at P the volume of the swept-out little cube (see figure 26.1). Let the Lie group elements be parametrized by real parameters $\alpha^1, \dots, \alpha^n$, and let the origin $\alpha^1 = \dots = \alpha^n = 0$ correspond to the unit element e . Consider now little vectors at the origin of the parameter space along the coordinate axes. The corresponding n group elements are $g(d\alpha^1, 0, \dots, 0)$, $g(0, d\alpha^2, 0, \dots, 0)$, $\dots$, $g(0, \dots, 0, d\alpha^n)$. Consider next an arbitrary group element $g(\vec{\alpha}_0)$, and act on the n group elements near the origin by left multiplication (right multiplication will be discussed later). The origin is mapped to a point P with coordinates α_0^μ by the group element $g(\vec{\alpha}_0)$, but the group elements $g(0, \dots, d\alpha^j, \dots, 0)$ are mapped to group elements near P , namely, if we denote the group multiplication rule by

$$g(\alpha^\mu)g(\beta^\mu) = g(\phi^\mu(\alpha, \beta)) \quad (26.3)$$

then the group element $g(\vec{\alpha})g(0, \dots, d\alpha^j, \dots, 0)$ for given j is parametrized by $\phi^\mu(\vec{\alpha}; 0, \dots, d\alpha^j, \dots, 0)$. The parameter composition rule satisfies of course $\phi^\mu(0, 0) = 0$ and $\phi^\mu(\alpha, 0) = \alpha^\mu$ and $\phi^\mu(0, \beta) = \beta^\mu$. We shall often denote the set of parameters $\alpha^1, \dots, \alpha^n$ by α or $\vec{\alpha}$. The differences $\phi^\mu(\vec{\alpha}; 0, \dots, d\alpha^j, \dots, 0) - \alpha^\mu$ are little vectors $\vec{v}_{(1)}, \dots, \vec{v}_{(j)}, \dots, \vec{v}_{(n)}$ in parameter space at P which form a small parallelogram. These vectors have the following components

$$v_{(j)}^\mu = \left. \frac{\partial \phi^\mu(\alpha, \beta)}{\partial \beta^j} \right|_{\beta=0} d\alpha^j \quad (\text{for fixed } j = 1, \dots, n, \text{ and } \mu = 1, \dots, n). \quad (26.4)$$

If the volume in parameter space of the little cube at the origin is **by definition** given by

$$dV(0) = \prod_{j=1}^n d\alpha^j \quad (26.5)$$

then, according to basic algebra, the volume of the little parallelogram spanned by the vectors $\vec{v}_1, \dots, \vec{v}_n$ is given by the determinant of the $n \times n$ matrix whose rows are the little vectors

$$dV(\alpha) = \left(\det \frac{\partial \phi^\mu(\alpha, \beta)}{\partial \beta^j} \Big|_{\vec{\beta}=0} \right) dV(0). \quad (26.6)$$

The equation to solve is $\mu_L(0)dV(0) = \mu_L(\alpha)dV(\alpha)$. But now the problem of the measure is solved: we must use a function $\mu_L(\alpha)$ which is proportional to the inverse of $\frac{dV(\alpha)}{dV(0)}$.

$$\mu_L(\alpha) = \left(\det \frac{\partial \phi^\mu}{\partial \beta^j} \Big|_{\beta=0} \right)^{-1} = \det \left[\left(\frac{\partial \phi^\mu}{\partial \beta^j} \Big|_{\beta=0} \right)^{-1} \right]. \quad (26.7)$$

Note that this construction proves that the measure $\mu_L(\alpha)$ is unique up to an overall constant which is the value of $\mu_L(0)$.

With this measure we have obtained that the product of the volume in parameter space and the measure is the same at the origin and at the point P

$$\int_{\substack{\text{small cube} \\ \text{at origin}}} \mu(\alpha) d\alpha = \int_{\substack{\text{small paralle-} \\ \text{logram at } P}} \mu(\alpha) d\alpha \quad (26.8)$$

where the point P in parameter space corresponds to $g(\alpha_0)$. Then we also have the equality

$$\int_{\substack{\text{small cube} \\ \text{at origin}}} f(g(\alpha)) \mu(\alpha) d\alpha = \int_{\substack{\text{small paralle-} \\ \text{logram at } P}} f(g^{-1}(\alpha_0)g(\alpha)) \mu(\alpha) d\alpha \quad (26.9)$$

where $f(\alpha)$ is an arbitrary function of the group elements $g(\alpha)$, because $f(g(\alpha))$ for α in the cube at the origin has the same argument as $f(g^{-1}(\alpha_0)g(\alpha))$ for α in the parallelogram at P .

Finally we can sum over all small cubes, and get an integral over the whole group, satisfying

$$\int_G f(g(\alpha)) \mu(\alpha) d\alpha = \int_G f(g_0 g(\alpha)) \mu(\alpha) d\alpha \quad (26.10)$$

for any constant group element g_0 .

Example 1. Consider the following set of triangular matrices

$$g(\beta) = \begin{pmatrix} e^{\beta^1} & \beta^2 \\ 0 & 1 \end{pmatrix} \quad (26.11)$$

We could have written $g = \begin{pmatrix} \beta^1 & \beta^2 \\ 0 & 1 \end{pmatrix}$ but then we should require $\beta^1 \neq 0$. It is easier to use e^{β^1} to avoid this. These matrices form a group

$$g(\alpha)g(\beta) = \begin{pmatrix} e^{\alpha^1} & \alpha^2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} e^{\beta^1} & \beta^2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} e^{\alpha^1+\beta^1} & e^{\alpha^1}\beta^2 + \alpha^2 \\ 0 & 1 \end{pmatrix} \quad (26.12)$$

Clearly the parameter composition rule $\phi^\mu(\alpha, \beta)$ has the following form

$$\phi^1(\alpha, \beta) = \alpha^1 + \beta^1; \quad \phi^2(\alpha, \beta) = e^{\alpha^1}\beta^2 + \alpha^2 \quad (26.13)$$

Consider $g(\beta)$ near the origin, and $g(\alpha)$ with finite α . Then

$$\left. \frac{\partial \phi^\mu(\alpha, \beta)}{\partial \beta^j} \right|_{\beta=0} = \begin{pmatrix} 1 & 0 \\ 0 & e^{\alpha^1} \end{pmatrix}. \quad (26.14)$$

Hence $\det \frac{\partial \phi}{\partial \beta}(\beta=0) = e^{\alpha^1}$ and thus

$$\mu_L(\alpha) = e^{-\alpha^1}. \quad (26.15)$$

Note also that if we consider a group element $g(\beta)$ that maps a point P to a point Q in parameter space, and use P as base point instead of the origin, then we still get the same measure because we can go from P to Q by first going from P to the origin, and then from the origin to Q . For that reason one calls this measure the left-invariant measure. The construction shows that it is unique.

Instead of left multiplication we could also use right multiplication by $g(\alpha)$ to sweep $g(\beta)$ out from the origin

$$g(\beta)g(\alpha) = \begin{pmatrix} e^{\beta^1} & \beta^2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} e^{\alpha^1} & \alpha^2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} e^{\alpha^1+\beta^1} & e^{\beta^1}\alpha^2 + \beta^2 \\ 0 & 1 \end{pmatrix} \quad (26.16)$$

Now

$$\phi^1(\beta, \alpha) = \alpha^1 + \beta^1; \quad \phi^2(\beta, \alpha) = e^{\beta^1}\alpha^2 + \beta^2 \quad (26.17)$$

and

$$\left. \frac{\partial \phi^\mu(\beta, \alpha)}{\partial \beta^j} \right|_{\beta=0} = \begin{pmatrix} 1 & 0 \\ \alpha^2 & 1 \end{pmatrix} \quad (26.18)$$

and thus

$$\mu_R(\alpha) = 1. \quad (26.19)$$

One uses the following terminology

$$\left. \frac{\partial}{\partial \beta^j} \phi^\mu(\alpha, \beta) \right|_{\beta=0} = e_j^\mu(\alpha) \quad : \quad (\text{inverse of) left-invariant group vielbein} \quad (26.20)$$

$$\left. \frac{\partial}{\partial \beta^j} \phi^\mu(\beta, \alpha) \right|_{\beta=0} = f_j^\mu(\alpha) \quad : \quad (\text{inverse of) right-invariant group vielbein} \quad (26.21)$$

The matrix inverses of $e_j^\mu(\alpha)$ and $f_j^\mu(\alpha)$ are $e_\mu^j(\alpha)$ and $f_\mu^j(\alpha)$, so $e_\mu^j e_j^\nu = \delta_\mu^\nu$. Vielbeins on coset manifolds will be discussed when we discuss coset manifolds. So

$$\mu_L(\alpha) = \det e_\mu^m(\alpha); \quad \mu_R(\alpha) = \det f_\mu^m(\alpha). \quad (26.22)$$

The measure $\det e_\mu^m$ resembles the measure of General Relativity which is also written as $\det e_\mu^m$, but note that in group theory the vielbein e_μ^m is a fixed function of the group parameters, whereas in General Relativity it is an arbitrary function of the coordinates of spacetime.

In example 1, $\mu_L(\alpha)$ is different from $\mu_R(\alpha)$. A natural question to ask is: when is μ_L equal to μ_R ? This is almost always the case: $\mu_L = \mu_R$ for the following groups: compact, simple, semisimple, finite, groups for which the adjoint rep is unimodular (has determinant unity), abelian groups, the Poincaré group. In fact, the following theorem holds.

Theorem: The left-invariant and right-invariant measure on a group manifold are equal if and only if

$$\det f_m^\mu(\alpha) e_\mu^n(\alpha) = 1 \quad (26.23)$$

and this is the case if and only if the structure constants are traceless, $f_{mn}^m = 0$ (sum over m).

Proof: For simplicity of notation¹ we write group elements as $e^{x \cdot T}$ where $x \cdot T = x^\mu T_\mu$ and $dx \cdot T = dx^m T_m$. The x^μ are the group parameters (in coset theory they are the coordinates of the coset manifold). The general rule is that vectors near the origin (like dx^m) carry Latin indices $m, n, \dots$, while vectors which are not near the origin carry Greek indices $\mu, \nu, \dots$ (again just as in General Relativity). The group generators T_m or T_μ are, by definition, equal. Then the following three steps prove the theorem:

- 1) Assuming small dx^m , we have from (26.3)

$$\begin{aligned} e^{x \cdot T} e^{dx \cdot T} &= e^{(x^\mu + dx^m e_m^\mu(x)) T_\mu} \\ e^{dx \cdot T} e^{x \cdot T} &= e^{(x^\mu + dx^m f_m^\mu(x)) T_\mu} \end{aligned} \quad (26.24)$$

¹We use vielbeins on group manifolds which are usually viewed as functions of x . So we use x^μ instead of α^μ . Furthermore, it is customary to denote right-invariant vielbeins by $f_\mu^m(x)$. They should not be confused with the structure constants $f_{\mu\nu}^\rho$ which are constants and have three indices.

2) Multiplying the second of these equations by $e^{-x \cdot T}$ on the left yields

$$\begin{aligned}
e^{-x \cdot T} e^{dx \cdot T} e^{x \cdot T} &= \exp \left(e^{-x \cdot T} dx \cdot T e^{x \cdot T} \right) \quad (\text{proof: expand the right-hand side}) \\
&= \exp \left[dx \cdot T + [dx \cdot T, x \cdot T] + \frac{1}{2!} [[dx \cdot T, x \cdot T], x \cdot T] + \dots \right] \\
&= \exp \left[dx \cdot T + dx^m (f_{m\mu}{}^n x^\mu) T_n + \frac{1}{2} dx^m (f_{m\rho}{}^n x^\rho) (f_{n\sigma}{}^r x^\sigma) T_r + \dots \right] \\
&= \exp \left[dx^m (e^{f \bullet_\mu \bullet x^\mu})_m{}^n T_n \right] \tag{26.25}
\end{aligned}$$

The symbol $e^{f \bullet_\mu \bullet x^\mu}$ denotes a matrix whose entries are denoted by dots.

3) From the first equation in step 1) we obtain after left multiplication by $e^{-x \cdot T}$

$$e^{dx \cdot T} = e^{-x \cdot T} e^{(x^\mu + dx^m e_m{}^\mu) T_\mu} \tag{26.26}$$

Defining $dx^m e_m{}^\mu \equiv \Delta x^\mu$, we can rewrite this equation as follows

$$e^{\Delta x^\mu e_\mu{}^m(x) T_m} = e^{-x \cdot T} e^{(x^\mu + \Delta x^\mu) T_\mu} \tag{26.27}$$

This equation holds for any Δx^μ , and using it in (26.25) we obtain

$$\begin{aligned}
e^{-x \cdot T} (e^{dx \cdot T} e^{x \cdot T}) &= e^{-x \cdot T} e^{(x^\mu + dx^m f_m{}^\mu(x)) T_\mu} \\
&= e^{\Delta x^\mu f_\mu{}^m(x) T_m} = e^{dx^m f_m{}^\mu(x) e_\mu{}^n T_n} \tag{26.28}
\end{aligned}$$

where we defined $dx^m f_m{}^\mu(x) \equiv \Delta x_f^\mu$. Equating the exponents in (26.25) and (26.28) yields

$$\boxed{(e^{f \bullet_\mu \bullet x^\mu})_m{}^n = f_m{}^\mu e_\mu{}^n.} \tag{26.29}$$

It is now clear that the left-invariant measure $\mu_L(\alpha) = \det e_\mu{}^m(x)$ is equal to the right-invariant measure $\mu_R(\alpha) = \det f_\mu{}^m(x)$ if and only if the determinant of $f_m{}^\mu e_\mu{}^n$ is unity, and this is the case if and only if $f \bullet_\mu \bullet x^\mu$ is traceless. Since x^μ are arbitrary this shows that all groups with traceless structure constants have $\mu_L(\alpha)$ equal to $\mu_R(\alpha)$.

$$\mu_L(\alpha) = \mu_R(\alpha) \quad \text{iff } f_{m\mu}{}^m = 0 \text{ (sum over } m). \tag{26.30}$$

Most Lie groups of interest satisfy this condition.

There is another way of computing the measure which directly gives the vielbein field $e_\mu{}^m(x)$ instead of its inverse $e_m{}^\mu(x)$. That approach starts at equation (26.27) and subtract the unit matrix on the left- and right-hand side.

$$e^{\Delta x^\mu e_\mu{}^m(x) T_m} - I = e^{-x \cdot T} (e^{(x^\mu + \Delta x^\mu) T_\mu} - e^{x \cdot T}) \tag{26.31}$$

Expanding to first order in Δx yields

$$\Delta x^\mu e_\mu{}^m(x) T_m = g^{-1}(x) (g(x + \Delta x) - g(x)) \quad (26.32)$$

where we have denoted group elements by $g(x)$. Introducing the exterior derivative $\Delta = \Delta x^\mu \frac{\partial}{\partial x^\mu}$ we find the desired result

$$\boxed{g^{-1} \Delta g = \Delta x^\mu e_\mu{}^m T_m} \quad (26.33)$$

In the approach where one begins with the parameter composition rule $\phi^\mu(\alpha, \beta)$ and differentiates w.r.t. β at $\beta = 0$, one obtains the inverse vielbein field $e_m{}^\mu(x)$, see (26.20).

Example 2. The measure of $SL(2, R)$. The group elements $g(\alpha)$ of $SL(2, R)$ are parametrized as $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with real a, b, c, d satisfying $ad - bc = 1$. Eliminating d by $d = \frac{1+bc}{a}$, we have 3 independent parameters a, b, c , and the task is to construct the left-invariant measure $\mu_L(\alpha) = \mu_L(a, b, c)$ where $\alpha^1 = a$, $\alpha^2 = b$ and $\alpha^3 = c$. To this aim we consider the product $g(\alpha)g(d\beta)$ where $d\beta = da, db, dc$ are small. Since $\det g(d\beta) = 1$ and $g(d\beta) = \begin{pmatrix} 1+da & db \\ dc & 1+dd \end{pmatrix}$, we have to first order in $d\beta$ the relation $dd = -da$. Then we obtain

$$g(\alpha)g(d\beta) = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1+da & db \\ dc & 1+dd \end{pmatrix} = \begin{pmatrix} a+ada+bdc & adb+b-bda \\ c+cda+ddc & cdb+d-dda \end{pmatrix} \quad (26.34)$$

Then the parameter composition rule $g(\alpha)g(\beta) = g(\phi(\alpha, \beta))$ reads for β equal to $d\beta$

$$\begin{aligned} \phi^1(\vec{\alpha}, d\vec{\beta}) &= a + ada + bdc \\ \phi^2(\vec{\alpha}, d\vec{\beta}) &= b + adb - bda \\ \phi^3(\vec{\alpha}, d\vec{\beta}) &= c + cda + ddc \end{aligned} \quad (26.35)$$

The left-invariant inverse vielbeins are then easily obtained

$$\left. \begin{aligned} d\beta^j \frac{\partial}{\partial \beta^j} \phi^1 \Big|_{d\beta=0} &= ada + bdc \\ d\beta^j \frac{\partial}{\partial \beta^j} \phi^2 \Big|_{d\beta=0} &= adb - bda \\ d\beta^j \frac{\partial}{\partial \beta^j} \phi^3 \Big|_{d\beta=0} &= cda + ddc \end{aligned} \right\} e_m{}^\mu(a, b, c) = \begin{pmatrix} a & -b & c \\ 0 & a & 0 \\ b & 0 & d \end{pmatrix}, \quad d = \frac{1+bc}{a} \quad (26.36)$$

Straightforward evaluation of the determinant of the 3×3 matrix $e_m{}^\mu$ yields $\det e_m{}^\mu = a(ad - bc) = a$ and hence

$$\mu_L(a, b, c) = \frac{1}{a}. \quad (26.37)$$

The group volume $\int \mu_L(a, b, c) da db dc$ is clearly infinite, as expected since $SL(2, R)$ is noncompact, but we can integrate function of a, b, c that fall off sufficiently fast as a, b, c tend to infinity.

Example 3. The measure of $SU(2)$. We now repeat this calculation for $SU(2)$ because $SU(2)$ is the compact form of $SL(2, R)$, so we expect a finite group volume with a nonsingular measure. We parametrize $SU(2)$ as follows

$$g = a_4 + a_k i\tau^k \text{ with } a_4^2 + a_k^2 = 1 \text{ hence } a_4 = \pm\sqrt{1 - a_k^2} \quad (26.38)$$

The generators are $i\tau_k$ with $k = 1, 2, 3$. Recall that $SU(2) = S_3$. We have $a_4 > 0$ on the upper half-sphere and $a_4 < 0$ on the lower half-sphere. With $g(da) = 1 + da_k i\tau^k$ (there is no term with da_4 since $da_4 = \pm\sqrt{1 - (da_k)^2}$ is of higher order in da_k) we obtain

$$(a_4 + a_k i\tau^k)(1 + da_k i\tau^k) = a_4 - a_k da_k + i\tau^k(a_k + a_4 da_k - \epsilon_{klm} a_l da_m) \quad (26.39)$$

Hence

$$\phi^k(a, da) = a_k + (a_4 \delta_m^k - \epsilon_{klm} a^l) da^m \quad (26.40)$$

We find then the measure by the following steps

$$e_m^\mu(a_k) = \begin{pmatrix} a_4 & -a_3 & a_2 \\ a_3 & a_4 & -a_1 \\ -a_2 & a_1 & a_4 \end{pmatrix} \quad (26.41)$$

$$\begin{aligned} \det e_m^\mu &= a_4(a_4^2 + a_1^2) + a_3(a_3 a_4 - a_1 a_2) + a_2(a_3 a_1 + a_2 a_4) \\ &= a_4(a_4^2 + a_k^2) = a_4 = \pm\sqrt{1 - a_1^2 - a_2^2 - a_3^2} \end{aligned} \quad (26.42)$$

Hence the left-invariant measure of $SU(2)$ is given by $\mu_L = |\det e_\mu^m| = (1 - a_1^2 - a_2^2 - a_3^2)^{-\frac{1}{2}}$ and the volume of $SU(2)$ is obtained as follows

$$\begin{aligned} \text{vol } SU(2) &= \int \mu_L da_1 da_2 da_3 = \int_{S^3} \frac{1}{\sqrt{1 - a_k^2}} da^k \\ &= 2 \int_{\text{upper hemisphere}} \frac{da^k}{\sqrt{1 - a_k^2}} = 2 \iiint_{-\infty}^{\infty} \frac{da_1 da_2 da_3}{\sqrt{1 - a_k^2}} \\ &= 2 \int_0^\infty d\rho \frac{4\pi\rho^2}{\sqrt{1 - \rho^2}} = 4\pi \int_0^\infty \frac{dt\sqrt{t}}{\sqrt{1 - t}} \\ &= 4\pi B\left(\frac{3}{2}, \frac{1}{2}\right) = 4\pi \frac{\Gamma(\frac{3}{2})\Gamma(\frac{1}{2})}{\Gamma(2)} = 4\pi \cdot \frac{1}{2} \cdot \pi = 2\pi^2. \end{aligned} \quad (26.43)$$

The parameters a_k parametrize the sphere S_3 since $SU(2) = S_3$, and $a_4 > 0$ corresponds to the upper hemisphere and $a_4 < 0$ to the lower hemisphere. We integrated over the domain $a_4 \geq 0$ and multiplied the result by 2.

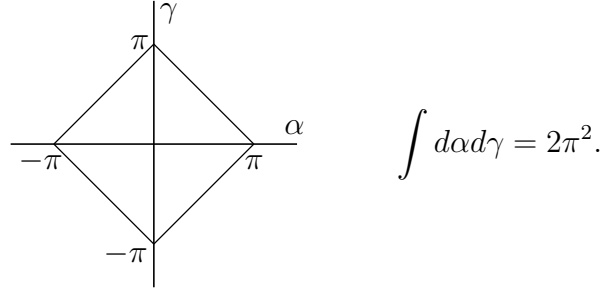
As expected, the volume is finite, and the value should not depend on the choice of coordinates,

but it does depend on the normalization of the generators. To check the coordinate invariance, we repeat the calculation for $SU(2)$ using another set of coordinates.

Example 4. The measure of $SU(2)$ with Euler angles. We write the group elements of $SU(2)$ in the spin $\frac{1}{2}$ representation as follows²:

$$\begin{aligned}
 g &= e^{\alpha T_3} e^{\beta T_1} e^{\gamma T_3}, \quad \text{with } T_j = -i\sigma_j; \quad \alpha, \beta, \gamma \text{ real} \\
 &= \begin{pmatrix} e^{-i\alpha} & 0 \\ 0 & e^{i\alpha} \end{pmatrix} \begin{pmatrix} \cos \beta & -i \sin \beta \\ -i \sin \beta & \cos \beta \end{pmatrix} \begin{pmatrix} e^{-i\gamma} & 0 \\ 0 & e^{i\gamma} \end{pmatrix} = \begin{pmatrix} \cos \beta e^{-i(\alpha+\gamma)} & -i \sin \beta e^{-i(\alpha-\gamma)} \\ -i \sin \beta e^{i(\alpha-\gamma)} & \cos \beta e^{i(\alpha+\gamma)} \end{pmatrix}
 \end{aligned} \tag{26.44}$$

The generators are $-i\sigma_j$, so the same as in example 3 up to a sign. The matrix elements of $SU(2)$ are $g = \begin{pmatrix} a & b^* \\ -b & a^* \end{pmatrix}$ with $|a|^2 + |b|^2 = 1$. Then (26.44) covers $SU(2)$ once if $-\pi \leq \alpha + \gamma \leq \pi$, $-\pi \leq \alpha - \gamma \leq \pi$, and $0 \leq \beta \leq \frac{\pi}{2}$.



Now we evaluate $g^{-1}dg$ and use (26.33) to extract e_μ^m . We use $[T_i, T_j] = 2\epsilon_{ijk}T_k$.

$$\begin{aligned}
 g^{-1}dg &= e^{-\gamma T_3} e^{-\beta T_1} e^{-\alpha T_3} d(e^{\alpha T_3} e^{\beta T_1} e^{\gamma T_3}) \\
 &= d\gamma T_3 + e^{-\gamma T_3} d\beta T_1 e^{\gamma T_3} + e^{-\gamma T_3} \left(e^{-\beta T_1} d\alpha T_3 e^{\beta T_1} \right) e^{\gamma T_3} \\
 &= d\gamma T_3 + (\cos(2\gamma)T_1 - \sin(2\gamma)T_2)d\beta + e^{-\gamma T_3} (\cos(2\beta)T_3 + \sin(2\beta)T_2) e^{\gamma T_3} d\alpha \\
 &= d\gamma T_3 + (\cos(2\gamma)T_1 - \sin(2\gamma)T_2)d\beta + (\cos(2\beta)T_3 + \sin(2\beta)\{\cos(2\gamma)T_2 + \sin(2\gamma)T_1\})d\alpha
 \end{aligned}$$

Since $g^{-1}dg = dx^\mu e_\mu^m T_m$ with $x^\mu = (\alpha, \beta, \gamma)$, we obtain

$$e_\mu^m = \begin{pmatrix} \sin(2\beta) \sin(2\gamma) & \sin(2\beta) \cos(2\gamma) & \cos(2\beta) \\ \cos(2\gamma) & -\sin(2\gamma) & 0 \\ 0 & 0 & 1 \end{pmatrix}; \quad \det e_\mu^m = -\sin(2\beta) \tag{26.45}$$

²One usually defines Euler angles with a factor $e^{\beta T_2}$ in the middle, because then for the spin $\frac{1}{2}$ irrep the matrix T_2 is real. For us this plays no role.

and thus we find that the left-invariant measure for $SU(2)$ using Euler angles is given by

$$\mu_L(\alpha, \beta, \gamma) = |\det e_\mu^m| = \sin(2\beta). \quad (26.46)$$

Hence for the volume of $SU(2)$ we now obtain

$$\text{vol } SU(2) = \int_{|\alpha+\gamma|\leq\pi} d\alpha \int_{|\alpha-\gamma|\leq\pi} d\gamma \int_0^{\frac{\pi}{2}} d\beta \sin(2\beta) = 2\pi^2 \cdot \frac{1}{2} \cdot 2 = 2\pi^2. \quad (26.47)$$

This agrees with the result obtained using Cartesian coordinates, and checks that the volume is independent of choice of coordinates.

We can check more: the measures in two different coordinate systems are related by the Jacobian for the transformation between the systems. Using the result of example 3, we should find

$$\mu_L(\alpha, \beta, \gamma) = \mu_L(a^1, a^2, a^3)J; \quad J = \det \left(\frac{\partial(a^1, a^2, a^3)}{\partial(\alpha, \beta, \gamma)} \right) \quad (26.48)$$

The calculation of the Jacobian is a little bit tedious but straightforward. Using also that $g = a_4 + a_k i \tau^k$ is equal to (26.44) we find

$$\begin{aligned} a_4 &= \cos \beta \cos(\alpha + \gamma) & a_1 &= -\sin \beta \cos(\alpha - \gamma) \\ a_3 &= -\cos \beta \sin(\alpha + \gamma) & a_2 &= -\sin \beta \sin(\alpha - \gamma). \end{aligned} \quad (26.49)$$

Then the Jacobian is given by

$$\begin{aligned} J &= \det \begin{pmatrix} \sin \beta \sin(\alpha - \gamma) & -\cos \beta \cos(\alpha - \gamma) & -\sin \beta \sin(\alpha - \gamma) \\ -\sin \beta \cos(\alpha - \gamma) & -\cos \beta \sin(\alpha - \gamma) & \sin \beta \cos(\alpha - \gamma) \\ -\cos \beta \cos(\alpha - \gamma) & \sin \beta \sin(\alpha + \gamma) & -\cos \beta \cos(\alpha + \gamma) \end{pmatrix} \\ &= 2 \sin \beta \cos^2 \beta \cos(\alpha + \gamma). \end{aligned} \quad (26.50)$$

Further, $1 - a_k^2 = 1 - \sin^2 \beta - \cos^2 \beta \sin^2(\alpha + \gamma) = \cos^2 \beta \cos^2(\alpha + \gamma)$. Hence we find that the measures are related as in (26.48)

$$\begin{aligned} \mu_L(\alpha, \beta, \gamma) &= \sin(2\beta) \\ \mu_L(a^1, a^2, a^3)J &= \frac{1}{\cos \beta \cos(\alpha + \gamma)} 2 \sin \beta \cos^2 \beta \cos(\alpha + \gamma) = \sin(2\beta). \end{aligned} \quad (26.51)$$

26.2 The Haar measure for $SU(N)$

We present now a derivation of the Haar measure for the groups $U(N)$ and $SU(N)$, and compute their volume. We follow the well-known book by Hua [1]. For another derivation based on the fact that $SU(N+1)/SU(N)$ is isomorphic to S_{2N+1} , see the textbook by Gilmore [2]. One can also

use generalized Euler angles (generalizing the expression $g = e^{i\omega_3\sigma_1}e^{i\omega_2\sigma_3}e^{i\omega_1\sigma_1}$ for $SU(2)$, see [3]).

The Haar measure for $SU(N)$ follows from that for $U(N)$ so we begin with the latter. We work at all times in the defining representation of $U(N)$: $N \times N$ unitary matrices U . Diagonalizing U by means of a unitary transformation we obtain a diagonal unitary matrix Λ

$$U = W\Lambda W^\dagger; \quad W^\dagger = W^{-1}; \quad \Lambda = \begin{pmatrix} e^{i\theta^1} & 0 & \cdots & 0 \\ 0 & e^{i\theta^2} & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & e^{i\theta^N} \end{pmatrix} \quad (26.52)$$

We discuss the range of the variables in W and Λ later. First we shall solve the easier problem of constructing the measure for functions which depend only on the θ^j such as the functions $e^{[a\text{tr}(U^4)+b\text{tr}(U^2)]}$, where U is an element of $U(N)$. This measure depends only on $\theta^1, \dots, \theta^N$. Next we shall turn to the more complicated problem of constructing the full measure on $U(N)$. We shall achieve this by writing U in terms of a hermitian matrix H

$$U = \frac{I + iH}{I - iH} \quad \text{where} \quad H = i \left(\frac{I - U}{I + U} \right). \quad (26.53)$$

The advantage is that the range of values of the entries of H is unconstrained, unlike the range of the entries of U . We begin by recalling/deriving some formulas for group vielbeins $e_\mu^m(x)$.

Consider a group element $g(x)$ for some group G with group parameters x . We shall simplify the notation by writing

$$g = e^{(x^\mu T_\mu)} = e^{(x \cdot T)} \quad (26.54)$$

but all our results remain valid for arbitrary group elements $g(x)$ that are not written as exponents. We define the left-invariant group vielbein field $e_\mu^m(x)$ by the following formula

$$g(x)g(dx) = e^{(x \cdot T)}e^{(dx \cdot T)} = e^{([x^\mu + dx^m e_m^\mu(x)]T_\mu)} \quad (26.55)$$

We follow the convention that objects near the origin of the group parameter space carry Latin indices such as dx^m , while objects far away carry Greek indices such as x^μ . This is also the convention used in General Relativity. From (26.55) we can derive another identity which will be used later

$$e^{(-x \cdot T)}e^{([x^\mu + \Delta x^\mu]T_\mu)} = e^{(dx^m T_m)} = e^{(\Delta x^\mu e_\mu^m(x)T_m)} \quad (26.56)$$

or in more general notation

$$g^{-1}(x)g(x + \Delta x) = g(\Delta x^\mu e_\mu^m(x)) \quad (26.57)$$

In this formula we may forget that Δx^μ was initially defined by $dx^m e_m^\mu(x) = \Delta x^\mu$, and take Δx^μ as an independent variable. The matrix $e_\mu^m(x)$ is the matrix inverse of $e_m^\mu(x)$, again as in General

Relativity. Using $g^{-1}(x)g(x) = I$, we subtract the unit matrix in (26.57) to obtain to first order in Δx^μ

$$g^{-1}(x) [g(x + \Delta x) - g(x)] = \Delta x^\mu e_\mu^m(x) T_m \quad (26.58)$$

Introducing the exterior derivative³

$$\Delta \equiv \Delta x^\mu \frac{\partial}{\partial x^\mu} \quad (26.59)$$

we arrive at

$$g^{-1} \Delta g = \Delta x^\mu e_\mu^m T_m \quad (26.60)$$

So the one-form $g^{-1} \Delta g$ lies in the Lie algebra. It is antihermitian since we use antihermitian generators T_m .

Consider now the following trace for $N \times N$ matrices g in $G = U(N)$:

$$\begin{aligned} (ds)^2 &= -\text{tr}[g^{-1} \Delta g g^{-1} \Delta g] \\ &= \text{tr}[(g^{-1} \Delta g)^\dagger g^{-1} \Delta g] \\ &= \text{tr}(\Delta g^\dagger \Delta g) \equiv \Delta x^\mu \Delta x^\nu g_{\mu\nu}(x) \end{aligned} \quad (26.61)$$

This formula defines the group metric $g_{\mu\nu}(x)$. Using (26.60) we find another expression for $(ds)^2$

$$\begin{aligned} (ds)^2 &= \text{tr}[(\Delta x^\mu e_\mu^m T_m)^\dagger (\Delta x^\nu e_\nu^n T_n)] \\ &= -\Delta x^\mu \Delta x^\nu e_\mu^m(x) e_\nu^n(x) \text{tr}(T_m T_n) \\ &= \Delta x^\mu \Delta x^\nu e_\mu^m(x) e_\nu^n(x) \gamma_{mn} \end{aligned} \quad (26.62)$$

where we defined

$$\gamma_{mn} \equiv -\text{tr}(T_m T_n) \quad (26.63)$$

Clearly

$$g_{\mu\nu} = e_\mu^m e_\nu^n \gamma_{mn} \implies \det(e_\mu^m) = \sqrt{\frac{\det(g_{\mu\nu})}{\det(\gamma_{mn})}} \quad (26.64)$$

Usually one takes $\det(e_\mu^m)$ as the left-invariant Haar measure μ_L on any group, but it can be multiplied by any arbitrary overall constant. We begin by choosing that constant such that the measure becomes just the square root of the group metric:

$$\mu_L(x) = \sqrt{\det(g_{\mu\nu}(x))} \quad (26.65)$$

Other normalizations will be discussed later.

Let us now compute the group metric by evaluating $\text{tr}(dU^\dagger dU)$. As mentioned before, we

³Usually one writes $g^{-1}dg$ instead of $g^{-1}\Delta g$, and defines $e^m = dx^\mu e_\mu^m$ instead of $e^m = \Delta x^\mu e_\mu^m$, but we shall also need a dx^μ later so we shall use both Δx^μ and dx^μ .

diagonalize the matrix U

$$U = W\Lambda W^\dagger; \quad W^\dagger = W^{-1}; \quad \Lambda = \begin{pmatrix} e^{i\theta^1} & 0 & \cdots & 0 \\ 0 & e^{i\theta^2} & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & e^{i\theta^N} \end{pmatrix} \quad (26.66)$$

Since any diagonal part of W cancels in this expression⁴, the effective number of real parameters on which W in (26.66) depends is $N^2 - N$. Together with the N parameters $\theta^1, \dots, \theta^N$ this yields the N^2 real parameters of U . A suitable parametrization of W (but by no means the only parametrization) is

$$W = e^{(w^a T_a)}, \quad T_a \text{ off-diagonal } N \times N \text{ antihermitian matrices} \quad (26.67)$$

However, there is a subtlety concerning the range of values of θ^j . If we permute two of the θ 's, we can undo this permutation by the action of a suitable matrix W . For example,

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} e^{i\theta^1} & 0 \\ 0 & e^{i\theta^2} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} e^{i\theta^2} & 0 \\ 0 & e^{i\theta^1} \end{pmatrix} \quad (26.68)$$

Thus, to avoid overcounting, one must restrict the ranges of values of θ^j . Therefore we impose the condition

$$2\pi > \theta^1 \geq \theta^2 \geq \dots \geq \theta^N \geq 0 \quad (26.69)$$

Given a matrix U we can of course always construct a matrix W which orders the eigenvalues of U according to (26.69), and W can always be written as in footnote 4.

Since W is unitary, $dWW^\dagger + WdW^\dagger = 0$, hence $d\mathbf{W}W^\dagger$ is **antihermitian**. We find then for dU the following expression

$$dU = dW\Lambda W^\dagger + Wd\Lambda W^\dagger + W\Lambda dW^\dagger \quad (26.70)$$

Extracting factors W and $W^\dagger$, and using that $dW^\dagger = -W^\dagger dW W^\dagger$, we find

$$\begin{aligned} dU &= W \left(W^\dagger dW \Lambda + d\Lambda - \Lambda W^\dagger dW \right) W^\dagger \\ &= W \left(d\Lambda + [W^\dagger dW, \Lambda] \right) W^\dagger \end{aligned} \quad (26.71)$$

Hence

$$(ds)^2 = \text{tr}(dU^\dagger dU) = \text{tr} \left(\left(d\Lambda + [W^\dagger dW, \Lambda] \right)^\dagger \left(d\Lambda + [W^\dagger dW, \Lambda] \right) \right) \quad (26.72)$$

⁴ Parametrizing $W = e^{(w^a T_a(\text{off-diag}))} e^{(v^a T_a(\text{diag}))}$ the dependence of U on v^a drops out.

In the trace the cross terms cancel since

$$\text{tr}\left(d\Lambda^\dagger \left[W^\dagger dW, \Lambda\right]\right) = \text{tr}\left(\left[\Lambda, d\Lambda^\dagger\right] W^\dagger dW\right) = 0 \quad (26.73)$$

Hence

$$(ds)^2 = \text{tr}\left((d\Lambda)^\dagger d\Lambda\right) + \text{tr}\left(\left[W^\dagger dW, \Lambda\right]^\dagger \left[W^\dagger dW, \Lambda\right]\right) \quad (26.74)$$

Substituting the explicit expression for Λ , and introducing indices $i, j = 1, N$ for the $N \times N$ matrices in the trace, we obtain

$$(ds)^2 = \sum_{i=1}^N d\theta^i d\theta^i + \sum_{i,j=1}^N \left[\delta\Omega_{ij}(\Lambda_j - \Lambda_i)\right]^* \left[\delta\Omega_{ij}(\Lambda_j - \Lambda_i)\right] \quad (26.75)$$

We used that for any matrix B one has

$$\text{tr}(B^\dagger B) = \sum_{i,j=1}^N B_{ij}^* B_{ij} = \sum_{i,j=1}^N |B_{ij}|^2 \quad (26.76)$$

We defined $W^\dagger dW = \delta\Omega$ and denoted the matrix elements of $\delta\Omega$ by $\delta\Omega_{ij}$. Since Λ is diagonal, the matrix elements of $[\delta\Omega, \Lambda]$ are $\delta\Omega_{ij}(\Lambda_j - \Lambda_i)$.

The crucial observation is now that $\delta\Omega_{ij}$ does not depend on θ^i but only on the w^a in (26.67). Hence

$$(ds)^2 = \sum_{i=1}^N d\theta^i d\theta^i + \sum_{i,j=1}^N \left|e^{i\theta^j} - e^{i\theta^i}\right|^2 |\delta\Omega_{ij}(w)|^2 \quad (26.77)$$

From its definition $\delta\Omega_{ij} = (W^\dagger dW)_{ij}$ it follows that $\delta\Omega_{ij}$ can be written as $dw^a V_{aij}$ where

$$V_{aij} = \left(W^\dagger \frac{\partial}{\partial w^a} W\right)_{ij} \quad (26.78)$$

and it would seem that $\det(g_{\mu\nu})$ could be easily calculated. However, there is a small complication: the range of a is over $N^2 - N$ values while the sum over i, j is over $2N^2$ values (the V_{aij} are complex). But it is easy to see that also the range over i, j can be restricted. First of all, the terms with $i = j$ cancel in (26.77) due to the factor $\left|e^{i\theta^j} - e^{i\theta^i}\right|^2$. And furthermore for $i \neq j$

$$|\delta\Omega_{ij}|^2 = |\delta\Omega_{ji}|^2 \quad (26.79)$$

because the $\delta\Omega$ are antihermitian. Hence the range of the pairs of indices i, j is over the $\frac{1}{2}N(N-1)$

values with $i < j$. We may rewrite (26.77) as follows

$$(ds)^2 = \sum_{i=1}^N d\theta^i d\theta^i + 2 \sum_{(i<j)=1}^{\frac{1}{2}N(N-1)} \left| e^{i\theta^i} - e^{i\theta^j} \right|^2 \operatorname{tr} \left| \sum_{a=1}^{N^2-N} dw^a V_{aij} \right|^2 \quad (26.80)$$

Next we double the range of the pair of indices i, j by decomposing V_{aij} into real and imaginary parts as $V_{aij} = V_{aij1} + iV_{aij2}$. Introducing the real object $V_{aij\alpha}$ which for $\alpha = 1$ is the real part of V_{aij} and for $\alpha = 2$ the imaginary part, the range of indices $ij\alpha$ of $V_{aij\alpha}$ is now from 1 to $N^2 - N$, just as the range of the index a of dw^a . We can then rewrite the second term in $(ds)^2$ as follows

$$dw^a g_{ab} dw^b = 2dw^a V_{aij\alpha} \left(\left| e^{i\theta^i} - e^{i\theta^j} \right|^2 I \right) (V_{bij\alpha})^* dw^b \quad (26.81)$$

We have omitted the symbols for the summation over a, b and $ij\alpha$ to simplify the notation. We have also introduced the $(N^2 - N) \times (N^2 - N)$ unit matrix I . Written in this form we have a product of three square matrices and we can take the determinant of the metric $g_{ab}(\theta, w)$:

$$\det(g_{ab}(\theta, w)) = 2^{N^2-N} |\det(V_{aij\alpha}(w))|^2 \prod_{i<j} \left| e^{i\theta^i} - e^{i\theta^j} \right|^4 \quad (26.82)$$

Note the fourth power which is due since the same factor $\left| e^{i\theta^i} - e^{i\theta^j} \right|^2$ appears for $\alpha = 1$ and $\alpha = 2$.

The measure for $U(N)$ is now easily obtained. We get a factor one from the part with $\sum_j d\theta^j d\theta^j$, hence

$$\mu_L = \sqrt{\det(g_{\mu\nu})} = 2^{N^2-N} |\det(V_{aij\alpha}(w))| \prod_{i<j} \left| e^{i\theta^i} - e^{i\theta^j} \right|^2 \quad (26.83)$$

Clearly, the measure factorizes into a w -dependent part and a θ -dependent part. For integrals over functions such as $f(\theta) = e^{\operatorname{tr}(U^4)}$ which only depend on θ 's, one may perform the w -integrals

$$\int f(\theta) \mu_L(w) \mu_L(\theta) d\theta^1 \dots d\theta^N dw^1 \dots dw^{N^2-N} = c \int f(\theta) \mu_L(\theta) d\theta^1 \dots d\theta^N \quad (26.84)$$

where the constant c is due to the w -integrals. Thus the measure for “gauge-invariant functions” is

$$\mu_L(\theta) = \prod_{i<j} \left| e^{i\theta^i} - e^{i\theta^j} \right|^2 \quad (26.85)$$

To obtain the corresponding measure for $SU(N)$ one inserts a delta function $\delta(\theta^1 + \dots + \theta^N)$ into the integral. Then one finds an integral of the form

$$\int f(\theta) \prod_{i<j} \left| e^{i\theta^i} - e^{i\theta^j} \right|^2 d\theta^1 \dots d\theta^{N-1} \quad (26.86)$$

and the θ^N in $f(\theta)$ and in $|e^{i\theta^i} - e^{i\theta^j}|^2$ must be replaced by $-\theta^1 - \dots - \theta^{N-1}$. For example, for $SU(3)$ one obtains

$$\int_0^{2\pi} d\theta^1 \int_0^{\theta^1} d\theta^2 f(\theta^1, \theta^2, \theta^3 = -\theta^1 - \theta^2) |e^{i\theta^1} - e^{i\theta^2}|^2 |e^{i\theta^1} - e^{-i(\theta^1+\theta^2)}|^2 |e^{i\theta^2} - e^{-i(\theta^1+\theta^2)}|^2 \quad (26.87)$$

The integral over the θ 's will be needed for the volume of the coset space $U(N)/\Lambda$. It can be evaluated by expanding the measure as follows

$$\prod_{j < k} (e^{i\theta^j} - e^{i\theta^k}) = \sum_{j_1 \dots j_N} \delta_{j_1 \dots j_N}^{1 \dots N} (-1)^P e^{[i(N-1)\theta^{j_1} + i(N-2)\theta^{j_2} + \dots + i\theta^{j_{N-1}}]} \quad (26.88)$$

There are $N!$ permutations of the indices $j_1, \dots, j_N$ on the right-hand side with a sign factor depending on whether the permutation is even or odd. For example, for $N = 3$ one obtains

$$\begin{aligned} (e^{i\theta^1} - e^{i\theta^2}) (e^{i\theta^2} - e^{i\theta^3}) (e^{i\theta^1} - e^{i\theta^3}) &= e^{i(2\theta^1+\theta^2)} + e^{i(2\theta^2+\theta^3)} + e^{i(2\theta^3+\theta^1)} \\ &\quad - e^{i(2\theta^3+\theta^2)} - e^{i(2\theta^2+\theta^1)} - e^{i(2\theta^1+\theta^3)} \end{aligned} \quad (26.89)$$

On the left-hand side one has $2^{N(N-1)/2} = 8$ terms, but on the right-hand side there are only $N! = 6$ terms. The two terms proportional to $e^{i(\theta^1+\theta^2+\theta^3)}$ cancel. If one then uses the orthogonality of the plane waves $e^{i(m_1\theta^1 + \dots + m_N\theta^N)}$, one obtains

$$\begin{aligned} &\int_0^{2\pi} d\theta^1 \int_0^{\theta^1} d\theta^2 \dots \int_0^{\theta^{N-1}} d\theta^N \prod_{(j < k)=1}^N |e^{i\theta^j} - e^{i\theta^k}|^2 \\ &= \frac{1}{N!} \int_0^{2\pi} \dots \int_0^{2\pi} d\theta^1 \dots d\theta^N \left| \sum_{j_1 \dots j_N} \delta_{j_1 \dots j_N}^{1 \dots N} e^{[i(N-1)\theta^{j_1} + i(N-2)\theta^{j_2} + \dots + i\theta^{j_{N-1}}]} \right|^2 \\ &= (2\pi)^N \end{aligned} \quad (26.90)$$

Let us now determine the full measure for the group $U(N)$. The representation of U in terms of W and Λ which we used before is not suitable because the range of the variables in W is restricted by unitarity. One would prefer variables that range from $-\infty$ to $+\infty$. Such a parametrization is suggested by the observation that a phase $e^{i\varphi}$ can be written as $(1 + ih)/(1 - ih)$ where h is a real number. As h runs from $-\infty$ to $+\infty$, the angle φ runs anti-clockwise from $-\pi$ to $-\pi$. Hence we parametrize U in terms of a hermitian matrix H as follows

$$U = \frac{I + iH}{I - iH} \quad (26.91)$$

There is a unique solution⁵ for H , namely

$$H = i \left(\frac{I - U}{I + U} \right) \quad (26.92)$$

We determine now again the group metric $g_{\mu\nu}$ by first evaluating dU and then $\text{tr}(dU^\dagger dU)$. One finds for dU

$$dU = i dH \frac{1}{I - iH} + \frac{1}{I - iH} (I + iH) i dH \frac{1}{I - iH} = 2i \frac{1}{I - iH} dH \frac{1}{I - iH} \quad (26.93)$$

Then

$$(ds)^2 = \text{tr}(dU^\dagger dU) = 4 \text{tr} \left(\frac{1}{I + H^2} dH \frac{1}{I + H^2} dH \right) \quad (26.94)$$

Introducing indices $i, j, k, l = 1, N$ for the matrices in the trace, one obtains

$$\begin{aligned} (ds)^2 &= 4 \sum_{i,j,k,l} \left(\frac{1}{I + H^2} \right)_{ij} \left(\frac{1}{I + H^2} \right)_{kl} (dH)_{jk} (dH)_{li} \\ &= 4 \sum_{i,j,k,l} \left\{ \left(\frac{1}{I + H^2} \right)_{ij} \left(\frac{1}{I + H^2} \right)_{lk}^* \right\} (dH)_{jk} (dH^*)_{il} \\ &\equiv dH_I^* M_{IJ} dH_J \end{aligned} \quad (26.95)$$

The indices I and J denote the pairs of indices il and jk , respectively, and the matrix $M_{IJ} = M_{il\ jk}$ inside the curly brackets is hermitian and the direct (Kronecker) product of the matrices $\left(\frac{1}{1+H^2} \right)_{ij}$ and $\left(\frac{1}{1+H^2} \right)_{lk}^*$. If A is an $N \times N$ matrix, B is an $M \times M$ matrix, and $A \otimes B$ denotes their direct product, then

$$\text{tr}(A \otimes B) = \text{tr}(A) \text{tr}(B); \quad \det(A \otimes B) = [\det(A)]^M [\det(B)]^N \quad (26.96)$$

There is a nice theory for such matrices that is not well-known, but one may read Muirhead [1] for more information. We now use a unitary similarity transformation to diagonalize the matrix M_{IJ} . The Jacobian for this change of variables is unity. For diagonal H^2 one finds

$$(ds)^2 = 4 \sum_{i,k} \left(\frac{1}{I + H^2} \right)_{ii} \left(\frac{1}{I + H^2} \right)_{kk}^* |dH_{ik}|^2 \quad (26.97)$$

The sum is now over diagonal terms with $i = k$, and off-diagonal terms with $i < k$ and $i > k$; the latter can be replaced by a sum over $i < k$ times a factor $2^{N(N-1)/2}$. The metric is now diagonal

⁵The 1-1 correspondence breaks down if $\det(I - iH) = 0$ or $\det(I + U) = 0$, but this happens only on a manifold of lower dimension.

in the following variables

$$h_I = \left\{ H^i_i \equiv h_i, \operatorname{Re}(H_{ik}) = h'_{ik}, \operatorname{Im}(H_{ik}) = h''_{ik} \right\} \quad \text{with } i = 1, N \quad \text{and } i < k$$

There are $N + 2 \left(\frac{1}{2} N(N-1) \right) = N^2$ integration variables h_I . For diagonal H^2 , the metric $g_{\mu\nu}$ is also diagonal. For example, for $N = 2$ the entries of $g_{\mu\nu}$ in terms of $m_{jj} = \left(\frac{1}{I+H^2} \right)_{jj}$ are given by

$$4m_{11}m_{11}; \quad 4m_{22}m_{22}; \quad 4 \cdot 2 \cdot m_{11}m_{22}; \quad 4 \cdot 2 \cdot m_{11}m_{12}; \quad (26.98)$$

Hence the determinant of the metric for diagonal H but arbitrary N reads

$$\det(g_{\mu\nu}) = 4^{N^2} 2^{N(N-1)} \left[\det \left(\frac{1}{I + H^2} \right) \right]^{2N} \quad (26.99)$$

The measure for general H is then given by

$$\begin{aligned} \int_{U(N)} &= 2^{N^2} \int_{-\infty}^{+\infty} \left[\det \left(\frac{1}{I + H^2} \right) \right]^N dH \\ dH &= 2^{N(N-1)/2} \left(\prod_{i=1}^N dh_i \right) \prod_{(j < k)=1}^N dh'_{jk} dh''_{jk} \end{aligned} \quad (26.100)$$

All variables in dH run from $-\infty$ to $+\infty$.

Up to this point the discussion was straightforward, but the evaluation of this integral requires some technical tricks. It may be useful to first outline the strategy. We shall first decompose the $N \times N$ hermitian matrix H into an $(N-1) \times (N-1)$ hermitian matrix H_1 , a complex vector v and a real scalar h as

$$H = \begin{pmatrix} H_1 & v^* \\ v^T & h \end{pmatrix} \quad (26.101)$$

We then diagonalize the matrix $I + H^2$ by means of a triangular matrix whose determinant is unity. The integral over dH then factorizes into an integral over dH_1 , dv and dh . The integrals over dv and dh are elementary, and in this way one obtains a recursion relation between the integral involving H and a similar integral involving H_1 . Iterating this recurrence relation one finds the value of the integral expressing the volume of $U(N)$ into products of gamma functions.

Consider the following integral

$$H_N(\alpha) = \int \left[\det \left(\frac{1}{I + H^2} \right) \right]^\alpha dH \quad (26.102)$$

(we shall set $\alpha = N$ later). Define

$$H = \begin{pmatrix} H_1 & v^* \\ v^T & h \end{pmatrix} \quad (26.103)$$

with h a real number, v a complex vector and H_1 a hermitian matrix. Then

$$I + H^2 = \begin{pmatrix} I + H_1^2 + v^* v^T & H_1 v^* + v^* h \\ v^T H_1 + h v^T & 1 + v^T v^* + h^2 \end{pmatrix} \quad (26.104)$$

One can diagonalize the matrix $I + H^2$ by a transformation with a triangular matrix which preserves hermiticity

$$\begin{pmatrix} I & 0 \\ -w^T A^{-1} & 1 \end{pmatrix} \begin{pmatrix} A & w^* \\ w^T & t \end{pmatrix} \begin{pmatrix} I & 0 \\ -w^T A^{-1} & 1 \end{pmatrix}^\dagger = \begin{pmatrix} A & 0 \\ 0 & t - w^T A^{-1} w^* \end{pmatrix} \quad (26.105)$$

where we used that A is hermitian. So

$$\begin{aligned} \det(I + H^2) &= \det(I + H_1^2 + v^* v^T) S \\ S &= \left[1 + v^T v^* + h^2 - \left(v^T H_1 + h v^T \right) \left(\frac{1}{I + H_1^2 + v^* v^T} \right) (H_1 v^* + v^* h) \right] \end{aligned} \quad (26.106)$$

We would like to diagonalize the matrix $I + H_1^2 + v^* v^T$. As a first step we diagonalize H_1 by a unitary similarity transformation

$$H_1 = U \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_{N-1} \end{pmatrix} U^\dagger \quad (26.107)$$

Then also $I + H_1^2$ is diagonal. We expect that we shall need to decompose $(I + H_1^2)^{-1}$ into $(I + H_1^2)^{-1/2} (I + H_1^2)^{-1/2}$. This motivates us to define

$$T = U \begin{pmatrix} \sqrt{1 + \lambda_1^2} & & 0 \\ & \ddots & \\ 0 & & \sqrt{1 + \lambda_{N-1}^2} \end{pmatrix} U^\dagger \quad (26.108)$$

Then

$$T = T^\dagger; \quad [T, H_1] = 0; \quad I + H_1^2 = T^2 \quad (26.109)$$

Consider next the matrix $A = I + H_1^2 + v^*v^T$. Define $v^* = Tu^*$ and hence $v^T = u^T T$. Then

$$\begin{aligned} I + H_1^2 + v^*v^T &= T(I + u^*u^T)T \\ \det(I + H_1^2 + v^*v^T) &= [\det(1 + H_1^2)](1 + u^T u^*) \end{aligned} \quad (26.110)$$

where we used that

$$\det(I + u^*u^T) = e^{\text{tr log}(I + u^*u^T)} = e^{\log(1 + u^T u^*)} = (1 + u^T u^*) \quad (26.111)$$

Also

$$“dv” = \prod_{j=1}^{N-1} dv'_{jN} dv''_{jN} = (\det(T))^2 du = \det(1 + H_1^2) “du” \quad (26.112)$$

The real scalar S in (26.106) can be rewritten using these definitions as follows

$$\begin{aligned} S &= 1 + u^T T T u^* + h^2 - (u^T H_1 + h u^T) \left(\frac{1}{I + u^* u^T} \right) (H_1 u^* + u^* h) \\ &\equiv a h^2 + 2 b h + c \end{aligned} \quad (26.113)$$

where

$$\begin{aligned} a &= 1 - u^T \left(\frac{1}{I + u^* u^T} \right) u^* = 1 - (u^T u^*) + (u^T u^*)^2 + \dots \\ &= \frac{1}{1 + u^T u^*} \end{aligned} \quad (26.114)$$

$$\begin{aligned} b &= \frac{1}{2} \left[-u^T H_1 \left(\frac{1}{I + u^* u^T} \right) u^* - u^T \left(\frac{1}{I + u^* u^T} \right) H_1 u^* \right] \\ &= \frac{1}{2} \left[-u^T H_1 \left\{ u^* - u^*(u^T u^*) + \dots \right\} - \left\{ u^T - (u^T u^*) u^T + \dots \right\} H_1 u^* \right] \\ &= - \left(\frac{u^T H_1 u^*}{1 + u^T u^*} \right) \end{aligned} \quad (26.115)$$

$$\begin{aligned} c &= 1 + u^T T^2 u^* - u^T H_1 \left(\frac{1}{I + u^* u^T} \right) H_1 u^* \\ &= 1 + u^T (1 + H_1^2) u^* - u^T H_1 H_1 u^* + (u^T H_1 u^*)(u^T H_1 u^*) + \dots \\ &= 1 + u^T u^* + \frac{(u^T H_1 u^*)^2}{1 + u^T u^*} \end{aligned} \quad (26.116)$$

Clearly $ac - b^2 = 1$.

Let us now return to the integral $H_N(\alpha)$. We have shown that it can be factorized as follows

$$H_N(\alpha) = 2^{N-1} \left[\int [\det(I + H_1^2)]^{1-\alpha} dH_1 \right] \left[\int "du" \frac{1}{(1 + u^T u^*)^\alpha} \right] \int_{-\infty}^{+\infty} dh (ah^2 + 2bh + c)^{-\alpha} \quad (26.117)$$

The factor 2^{N-1} is the difference between the factor $2^{N(N-1)/2}$ in dH and $2^{(N-1)(N-2)/2}$ in dH_1 . The integral over h is elementary and gives a Beta function after changing the integration variable from $t = h + \frac{b}{a}$ to $1/(1 + t^2)$

$$\int_{-\infty}^{\infty} dh (ah^2 + 2bh + c)^{-\alpha} = a^{\alpha-1} \frac{\Gamma(\alpha - \frac{1}{2}) \sqrt{\pi}}{\Gamma(\alpha)} \quad (26.118)$$

where we used that $ac - b^2 = 1$. The integral over u is also elementary after introducing polar coordinates

$$\int_{-\infty}^{\infty} "du" (1 + u^T u^*)^{-2\alpha+1} = \pi^{N-1} \frac{\Gamma(2\alpha - N)}{\Gamma(2\alpha - 1)} \quad (26.119)$$

In this way one obtains a recurrence relation

$$H_N(\alpha) = 2^{N-1} \pi^{N-\frac{1}{2}} \frac{\Gamma(\alpha - \frac{1}{2}) \Gamma(2\alpha - N)}{\Gamma(\alpha) \Gamma(2\alpha - 1)} H_{N-1}(\alpha - 1) \quad (26.120)$$

Since $H_1(\alpha - N + 1)$ is an ordinary integral, its value is easy to obtain,

$$H_1(\alpha - N + 1) = \int_{-\infty}^{\infty} \frac{dh}{[h^2 + 1]^{\alpha-N+1}} = \sqrt{\pi} \frac{\Gamma(\alpha - N + \frac{1}{2})}{\Gamma(\alpha - N + 1)} \quad (26.121)$$

and one finally finds

$$H_N(\alpha) = 2^{N(N-1)/2} \pi^{N^2/2} \prod_{j=0}^{N-1} \frac{\Gamma(\alpha - j - \frac{1}{2})}{\Gamma(\alpha - j)} \prod_{k=0}^{N-2} \frac{\Gamma(2\alpha - N - k)}{\Gamma(2\alpha - 2k - 1)} \quad (26.122)$$

For the measure of $U(N)$ in (26.100) we need $\alpha = N$. From (26.100) one obtains

$$\begin{aligned} \text{Vol}(U(N)) &= 2^{N^2} H_N(N) \\ &= 2^{N^2} 2^{N(N-1)/2} \pi^{N^2/2} \prod_{j=0}^{N-1} \frac{\Gamma(N - j - \frac{1}{2})}{\Gamma(N - j)} \prod_{k=0}^{N-2} \frac{\Gamma(N - k)}{\Gamma(2N - 2k - 1)} \end{aligned} \quad (26.123)$$

This result can be greatly simplified if one uses the doubling formula for gamma functions

$$\Gamma(x)\Gamma\left(x + \frac{1}{2}\right) = \sqrt{\pi}2^{1-2x}\Gamma(2x) \quad (26.124)$$

One finds then for the volume of the group $U(N)$:

$$\text{Vol}(U(N)) = \frac{(2\pi)^{N(N+1)/2}}{1! \cdot 2! \cdots (N-1)!} \quad (26.125)$$

For $SU(N)$, the volume is a factor 2π smaller. For example, for $SU(2)$ one finds the volume $4\pi^2$. One often defines the volume without the prefactor $2^{N(N-1)/2}$ in (26.100) and then one obtains the more familiar result $\text{Vol}(SU(2)) = 2\pi^2$ (which equals $\text{vol}(S_3)$).

A comment on the choice of coordinates. We parametrized the group elements of $U(N)$ by the matrix elements of H . However, the product of the Haar measure $[\det g_{\mu\nu}(x)]^{1/2}$ and $\prod dx^\mu$ is invariant under general coordinate transformations, hence any other parametrization would have given the same volume.

Bibliography

- [1] L. K. Hua, *Harmonic Analysis of Functions of Several Variables in the Classical Domains*, Translations of Mathematical Monographs Volume 6, American Mathematical Society 1963. In this book the measure for $SU(N)$ and other groups is derived. It uses many theorems of matrix algebra, which are derived in straightforward but somewhat old-fashioned ways. For a more modern discussion of these theorems, see for example R. J. Muirhead, *Aspects of Multivariate Statistical Theory*, Wiley 1982. For an even simpler discussion of direct (Kronecker) products of matrices, see F. A. Graybill, *Introduction to Matrices with Application in Statistics*, Wadworth 1969.
- [2] R. Gilmore, *Lie groups, Lie algebras, and some of their applications*, John Wiley, 1974. In equation (5.123) the result $\text{vol}(U(N)) = \frac{2^N \pi^{N(N+1)/2}}{1!2!\cdots(N-1)!}$ is given. Using $\text{vol}(U(1)) = 2\pi$, one obtains $\text{vol}(SU(N)) = \frac{2^{N-1} \pi^{(N-1)(N+2)/2}}{1!2!\cdots(N-1)!}$ which agrees with (26.125) without the prefactor $2^{N(N-1)/2}$. For $N = 2$ one then obtains again $\text{vol}(SU(2)) = 2\pi^2$.
- [3] E. C. G. Sudarshan and T. Tilma, *Generalized Euler angle parametrization for $SU(N)$* , [arXiv:math-ph/0205016](https://arxiv.org/abs/math-ph/0205016).

Chapter 27

Introducing Young tableaux by examples

Given the highest weight of an irreducible representation (irrep) of a simple Lie algebra, one can construct the whole multiplet of states by acting with the lowering operators associated with the simple roots, and also the Clebsch-Gordan (CG) decomposition of the tensor product of two irreps can be constructed by acting on the product of the two highest weight states. However, the procedure is laborious, and tensor methods yield the results in a much simpler and faster way. **Tensor methods can be visualized by Young tableaux.** In this section we give an introduction to Young tableaux by working out a series of simple cases. We begin with irreps of $SU(N)$ where we use the simple formulas for the dimensions of irreps and for the CG decomposition. Then we shall discuss Young tableaux for the orthogonal groups. New here are the Young tableaux for spinor irreps. Using the isomorphisms between some low $SU(N)$ and $SO(N)$ groups, we can check some of the results of Young tableaux. Finally we shall discuss Young tableaux for the symplectic group $USp(2N)$, and also here some isomorphisms allow us to make checks. Neither $SU(N)$ nor $USp(2N)$ have spinor irreps.

$SU(2)$. We begin with the simplest case, $SU(2)$. The vector (defining, fundamental) irrep is two-dimensional and it is represented by a box

$$\boxed{}_{\mathbf{2}} : \text{the spin } j = \frac{1}{2} \text{ irrep} \quad (27.1)$$

We write below Young tableaux their dimensions. From quantum mechanics we know how to add angular momenta. If $j_1 \geq j_2$ one has $j_1 \otimes j_2 = j_1 + j_2, j_1 + j_2 - 1, \dots, j_1 - j_2$. A spin j irrep is obtained by taking the product of $2j$ spin $\frac{1}{2}$ irreps and symmetrizing in all indices of the tensor

$$\boxed{}\boxed{}\boxed{}\boxed{}\boxed{} : \text{the spin } j \text{ irrep from } 2j \text{ boxes: } t^{(\alpha_1 \dots \alpha_{2j})} \quad (27.2)$$

The symbol $(\dots)$ indicates total symmetrization of the indices inside the parenthesis, and the symbol $[\dots]$ denotes total antisymmetrization. For $SO(N)$ groups we shall use the symbol $((\dots))$ to denote total symmetrization and tracelessness, and for $USp(2N)$ groups the symbol $[[\dots]]$

denotes total antisymmetrization and symplectic-tracelessness. For example in $SO(N)$ $t^{((ij))} = t^{(ij)} - \frac{1}{N}\delta^{ij}t$ with $t = \delta_{kl}t^{kl}$, and in $USp(2N)$ $t^{[[ij]]} = t^{[ij]} - \frac{1}{2N}\Omega^{ij}(\Omega_{kl}t^{kl})$ where $\Omega_{ij}\Omega^{ij} = 2N$.

Taking the product of two spin $\frac{1}{2}$ irreps yields a sum of a spin 1 and a spin 0 irrep.

$$\begin{array}{rclclcl}
\psi^\alpha \chi^\beta & = & \psi^{(\alpha} \chi^{\beta)} & + & \psi^{[\alpha} \chi^{\beta]} \\
\text{spins: } \frac{1}{2} \otimes \frac{1}{2} & = & 1 & \otimes & 0 \\
\text{dims: } \mathbf{2} \times \mathbf{2} & = & \mathbf{3} & + & \mathbf{1} \\
\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \otimes \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} & = & \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} & \oplus & \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array}
\end{array} \tag{27.3}$$

Since $\psi^{[\alpha} \chi^{\beta]} = \epsilon^{\alpha\beta}(\frac{1}{2}\epsilon_{\gamma\delta}\psi^\gamma\chi^\delta)$ and since $\epsilon_{\gamma\delta}\psi^\gamma\chi^\delta$ is a singlet (a scalar, it does not transform) because $\epsilon_{\gamma\delta}$ is an invariant tensor, we replace $\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array}$ by a dot $\bullet$.

$$\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} = \bullet \tag{27.4}$$

For the same reason we replace in $SU(N)$ a column with N boxes by a dot (we omit the dot if it is part of a Young tableaux).

$$\left. \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} \right\} N = \bullet; \quad \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} = \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \tag{27.5}$$

We can next add two spin 1 irreps, giving a sum of spin 2, 1, 0 irreps

$$\begin{array}{ccccc}
\begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} & \otimes & \begin{array}{|c|c|} \hline a & a \\ \hline \end{array} & = & \begin{array}{|c|c|c|c|} \hline \square & \square & a & a \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline \square & \square & a \\ \hline a & & \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline a & a \\ \hline \end{array} \\
\mathbf{3} & \times & \mathbf{3} & = & \mathbf{5} + \mathbf{3} + \mathbf{1}
\end{array} \tag{27.6}$$

Two boxes in one column give a singlet of $SU(2)$ and can be omitted. So for example

$$\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} = \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \quad \text{in } SU(2). \tag{27.7}$$

All these results for $SU(2)$ are familiar from quantum mechanics.

$SU(3)$. The next simplest group is $SU(3)$. Now the vector irrep is 3-dimensional, indicated by writing $\mathbf{3}$ and depicted by one box

$$\begin{array}{|c|} \hline \square \\ \hline \end{array} : v^\alpha \text{ of } SU(3). \tag{27.8}$$

Next we consider the complex conjugate of the vector v^α , which we denote by a vector w_α with a covariant index: $(v^\alpha)^* = w_\alpha$, and which corresponds to two boxes

$$\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \quad t^{\alpha\beta} = \epsilon^{\alpha\beta\gamma} w_\gamma, \quad w_\gamma = \frac{1}{2}\epsilon_{\gamma\alpha\beta} t^{\alpha\beta} \tag{27.9}$$

$\mathbf{3}^*$

We used that $\epsilon^{\alpha\beta\gamma}w_\beta = v^{\alpha\beta}$ is a two-index antisymmetric tensor $t^{[\alpha\beta]}$ in $SU(3)$. Let's do some exercises

$$\begin{array}{rclcl}
\begin{array}{|c|} \hline \square \\ \hline \end{array} & \otimes & \begin{array}{|c|} \hline \square \\ \hline \end{array} & = & \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \\
\mathbf{3}^* & \times & \mathbf{3} & = & \mathbf{8} + \mathbf{1} \\
w_\alpha & \times & v^\beta & = & w_\alpha v^\beta - \frac{1}{3}\delta_\alpha^\beta(v^\gamma w_\gamma) + \frac{1}{3}\delta_\alpha^\beta(v^\gamma w_\gamma)
\end{array} \tag{27.10}$$

Recall $\mathbf{3}^* = w_\alpha = \frac{1}{2}\epsilon_{\alpha\beta\gamma}v^{\beta\gamma}$. Next we consider the product of two tensors v^α and w^β

$$\begin{array}{rclcl}
\begin{array}{|c|} \hline \square \\ \hline \end{array} & \otimes & \begin{array}{|c|} \hline \square \\ \hline \end{array} & = & \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \\
\mathbf{3} & \times & \mathbf{3} & = & \mathbf{6} + \mathbf{3}^* \\
u^\alpha & \times & v^\beta & = & \frac{1}{2}(u^\alpha v^\beta + u^\beta v^\alpha) + \frac{1}{2}(u^\alpha v^\beta - u^\beta v^\alpha) \\
& & & = & u^{(\alpha}v^{\beta)} + u^{[\alpha}v^{\beta]}
\end{array} \tag{27.11}$$

A symmetric tensor $t^{\alpha\beta} = t^{\beta\alpha}$ of $SU(3)$ has 6 components. We go on

$$\begin{array}{rclcl}
\begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} & \otimes & \begin{array}{|c|} \hline \square \\ \hline \end{array} & = & \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \\
\mathbf{6} & \times & \mathbf{3} & = & \mathbf{10} + \mathbf{8}
\end{array} \tag{27.12}$$

A symmetric tensor with k indices in $SU(N)$ (so the indices can take on N values) has $\binom{N+k-1}{k}$ components; this explains the **10**. The **8** is the adjoint rep of $SU(3)$, its highest weight is $\vec{\mu}_{HW}(\mathbf{8}) = \vec{\mu}_{FW}^{(1)} + \vec{\mu}_{FW}^{(N-1)}$, and since $\vec{\mu}_{FW}^{(k)}$ corresponds to a k -index antisymmetric tensor, the tableaux for the **8** becomes clear. We leave as an exercise

$$\begin{array}{rclcl}
\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} & \otimes & \begin{array}{|c|} \hline \square \\ \hline \end{array} & = & \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \\
\mathbf{10} & \times & \mathbf{3} & = & \mathbf{15} + ?
\end{array} \quad \text{Clearly } \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} = \mathbf{15} \tag{27.13}$$

By matching dimensions we discover that the irreducible tensor $t_\gamma^{\alpha\beta} = \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array}$ has 15 components. That makes sense: $t^{\alpha\beta} = t^{\beta\alpha}$ has 6 components, so $t^{\alpha\beta}v_\gamma$ has 18 components, but subtracting the trace $t^{\alpha\beta}v_\beta$ removes 3 components, leaving 15 components.

Next consider the product of two irreps which each have more than one box. For example

$$\begin{array}{rclcl}
\begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} & \otimes & \begin{array}{|c|} \hline a \\ \hline b \\ \hline \end{array} & = & \left(\begin{array}{|c|c|c|} \hline \square & \square & a \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \right) \otimes \begin{array}{|c|} \hline b \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline \square & \square & a \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline \square & \square & a \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline \square & \square & b \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline \square & \square & b \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline a \\ \hline b \\ \hline \end{array} \\
\mathbf{6} & \times & \mathbf{3}^* & = & \dots = \mathbf{15} + \mathbf{3}
\end{array} \tag{27.14}$$

The rule is: add first box a in all possible ways, then box b in all possible ways, but reading from right to left and top to bottom, the number of boxes with a should always be larger or equal to the number boxes with b .

Using that $\mathbf{6}^* = (\square\square)^* = \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}$ we find

$$\begin{array}{ccccc} \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} & \otimes & \square & = & \begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix} \oplus \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} \\ \mathbf{6}^* & \times & \mathbf{3} & = & \mathbf{15}^* + \mathbf{3}^* \end{array} \quad \text{where } \left(\begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix} \right)^* = \begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix} \quad (27.15)$$

We used that $\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} = \begin{smallmatrix} \square \\ \square \end{smallmatrix} = \mathbf{3}^*$. Finally

$$\begin{array}{ccccc} \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} & \otimes & \begin{smallmatrix} a & a \end{smallmatrix} & = & \begin{smallmatrix} \square & \square & a & a \\ \square & \square & a & a \end{smallmatrix} \oplus \begin{smallmatrix} \square & \square & a \\ \square & \square & a \end{smallmatrix} \oplus \begin{smallmatrix} \square & \square & a \\ a & \square & \square \end{smallmatrix} \oplus \begin{smallmatrix} \square & \square \\ a & a \end{smallmatrix} \oplus \begin{smallmatrix} \square & \square \\ a & a \end{smallmatrix} \oplus \begin{smallmatrix} \square & \square \\ a & a \end{smallmatrix} \\ \mathbf{6}^* & \times & \mathbf{6} & = & ? + \mathbf{8} + \mathbf{1} \end{array} \quad (27.16)$$

Another rule is: no two a boxes in the same column. By matching dimensions we find out

$$\begin{smallmatrix} \square & \square & \square & \square \\ \square & \square & \square & \square \end{smallmatrix} = \mathbf{27}. \quad (27.17)$$

In tensor notation this corresponds to

$$u_{(\alpha\beta)}v^{(\gamma\delta)} = t_{\alpha\beta}^{\gamma\delta} + p\delta_{(\alpha}^{\gamma}(u \cdot v)_{\beta)}^{\delta)} + q\delta_{(\gamma}^{(\alpha}\delta_{\delta)}^{\beta)}(u \cdot v) \quad (27.18)$$

where p and q are coefficients (numbers) and $t_{\alpha\beta}^{\gamma\delta}$ is symmetric in $\alpha\beta$ and $\gamma\delta$ and traceless in $\alpha\beta$, $\alpha\delta$, $\beta\gamma$ and $\beta\delta$, and $(u \cdot v)_{\beta}^{\delta} = u_{\alpha\beta}v^{\gamma\delta}\delta_{\alpha}^{\gamma}$ while $u : v = u_{\alpha\beta}v^{\alpha\beta}$. By taking traces on both sides one may fix the coefficients p and q .

$SU(4)$. We move on to $SU(4)$. We have the following irreps of low dimensions

$$\begin{array}{ccccccccccc} \square, & \square\square, & \begin{smallmatrix} \square \\ \square \end{smallmatrix}, & \begin{smallmatrix} \square \\ \square \\ \square \end{smallmatrix}, & \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}, & \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}, & \square\square\square, & \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}, & \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}, & \dots \\ \mathbf{4} & \mathbf{10} & \mathbf{6} & \mathbf{4}^* & \mathbf{10}^* & \mathbf{15} & \mathbf{20} & \mathbf{20}' & \mathbf{20}'' \end{array} \quad (27.19)$$

The **15** is the adjoint rep, corresponding to a traceless tensor t_{β}^{α} , as follows

$$\begin{array}{ccccc} \begin{smallmatrix} \square \\ \square \\ \square \end{smallmatrix} & \otimes & \square & = & \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} \oplus \begin{smallmatrix} \square \\ \square \\ \square \end{smallmatrix} \\ \mathbf{4}^* & \times & \mathbf{4} & = & \mathbf{15} + \mathbf{1} \end{array} \quad \begin{array}{l} v^{\alpha}w_{\beta} = t_{\beta}^{\alpha} + \frac{1}{4}\delta_{\beta}^{\alpha}(v \cdot w) \\ t_{\beta}^{\alpha} = v^{\alpha}w_{\beta} - \frac{1}{4}\delta_{\beta}^{\alpha}(v \cdot w) \end{array} \quad (27.20)$$

Let us do $\mathbf{10} \times \mathbf{4}$ in two ways

$$\begin{array}{ccccc} \square\square & \otimes & \square & = & \square\square\square \oplus \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} \\ \mathbf{10} & \times & \mathbf{4} & = & \mathbf{20} + ? \end{array} \quad \text{Clearly } \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} = \mathbf{20}''^* = \left(\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} \right)^* \quad (27.21)$$

Now the other way around

$$\begin{array}{ccccccc}
\begin{array}{|c|} \hline \square \\ \hline \end{array} & \otimes & \begin{array}{|c|c|} \hline a & a \\ \hline \end{array} & = & \begin{array}{|c|c|c|} \hline & a & a \\ \hline \end{array} & \oplus & \begin{array}{|c|c|} \hline & a \\ a & \\ \hline \end{array} \oplus \begin{array}{|c|} \hline a \\ \hline \end{array} \\
\mathbf{4} & \times & \mathbf{10} & = & \mathbf{20} & + & \mathbf{20}''^*
\end{array} \tag{27.22}$$

The last diagram is to be discarded since there should not be two a -boxes in the same column.

$SO(N)$. Next we study orthogonal groups. At the level of Young tableaux for tensors we can treat $SO(2N)$ and $SO(2N+1)$ the same way, but recall that $SO(2N)$ has two spinor irreps which we denote by $\times$ and $+$, whereas $SO(2N+1)$ has only one spinor irrep which we denote by $\times$. Also Young tableaux for $SO(2N)$ and $SO(2N+1)$ are at most N boxes deep because an antisymmetric tensor with more than N boxes can be written as $\epsilon^{\mu_1 \dots \mu_{2N}}$ times a tensor with less than N boxes. A column with $2N$ boxes yields again a scalar ($t^{\mu_1 \dots \mu_{2N}}$ is invariant since it is proportional to the invariant tensor $\epsilon^{\mu_1 \dots \mu_{2N}}$). We shall use the isomorphisms between $SO(3)$ and $SU(2)$, and between $SO(6)$ and $SU(4)$, to clarify the results for $SO(N)$.

$SO(3)$. Denoting the **3** by $\square$ and the spinor irrep **2** by $\times$ we get

$$\begin{array}{ccccccc}
\square & \otimes & \square & = & (\square\square \oplus \bullet) & \oplus & \begin{array}{|c|} \hline \square \\ \hline \end{array} \\
\mathbf{3} & \times & \mathbf{3} & = & (\mathbf{5} + \mathbf{1}) & + & \mathbf{3} \\
u^\alpha & \times & v^\beta & = & (u^{((\alpha} v^{\beta))}) & + & \frac{1}{3} \delta^{\alpha\beta} u \cdot v + \frac{1}{2} \epsilon^{\alpha\beta\gamma} (\vec{u} \times \vec{v})_\gamma
\end{array} \tag{27.23}$$

Since $(\vec{u} \times \vec{v})_\gamma = \epsilon_{\gamma\alpha\beta} u^\alpha v^\beta$, the last term is equal to $u^{[\alpha} v^{\beta]}$. Here, $\begin{array}{|c|} \hline \square \\ \hline \end{array} = \square$ and $\bullet$ is the trace (a singlet). Recall that the pair of parenthesis $((\dots))$ denotes a symmetric and traceless tensor.

Now the spinor irrep. Recall $SU(2) \simeq SO(3)$ so $\square$ of $SU(2)$ is $\times$ of $SO(3)$.

$$\begin{array}{lcl}
SO(3): & \begin{array}{ccc} \square & \otimes & \times \\ \mathbf{3} & \times & \mathbf{2} \\ v^\mu & & \psi^\alpha \end{array} & = & \begin{array}{ccc} \begin{array}{|c|} \hline \times \\ \hline \end{array} & \oplus & \times \\ \mathbf{4} & + & \mathbf{2} \end{array} \\
& & & & v^\mu \psi^\alpha - \frac{1}{3} (\gamma^\mu)^\alpha_\beta v \cdot \gamma \psi^\beta & + & \frac{1}{3} (\gamma^\mu)^\alpha_\beta v \cdot \gamma \psi^\beta \\
SU(2): & \begin{array}{ccc} \square\square & \otimes & \square \\ (j=1) & \otimes & (j=\frac{1}{2}) \end{array} & = & \begin{array}{ccc} \square\square\square & \oplus & \square \\ (j=\frac{3}{2}) & \oplus & (j=\frac{1}{2}) \end{array}
\end{array} \tag{27.24}$$

The vector-spinor $v^\mu \psi^\alpha$ with $\mu = 1, 2, 3$ the vector index and $\alpha = 1, 2$ the spinor index, is made irreducible by subtracting the “gamma-trace”, namely $t^{\mu\alpha} = v^\mu \psi^\alpha - \frac{1}{3} (\gamma^\mu)^\alpha_\beta v \cdot \gamma \psi^\beta$ where $v \cdot \gamma \psi^\beta = v^\mu (\gamma_\mu)^\beta_\gamma \psi^\gamma$. The γ^μ are the three Pauli matrices. One may check that $t^{\mu\alpha}$ is gamma-traceless: $(\gamma_\mu)^\alpha_\beta t^{\mu\beta} = 0$. Another example where we compare the Young tableaux of $SO(3)$ and $SU(2)$ is as

follows

$$\begin{aligned}
SO(3) : \quad & \times \quad \otimes \quad \times \quad = \quad \square \quad \oplus \quad \bullet \\
& \mathbf{2} \quad \times \quad \mathbf{2} \quad = \quad \mathbf{3} \quad + \quad \mathbf{1} \\
SU(2) : \quad & \square \quad \otimes \quad \square \quad = \quad \square\square \quad \oplus \quad \begin{array}{|c|} \hline \square \\ \hline \end{array} \\
& (j = \tfrac{1}{2}) \quad \otimes \quad (j = \tfrac{1}{2}) \quad = \quad (j = 1) \quad \oplus \quad (j = 0)
\end{aligned} \tag{27.25}$$

$SO(6)$. Having studied $SO(3)$, consider $SO(6)$. Since $SO(6) \simeq SU(4)$ we can try to match the irreps of $SO(6)$ and $SU(4)$. The first few are easy

$$\begin{aligned}
SU(4) : \quad & \square \quad \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \quad \begin{array}{|c|} \hline \square \\ \hline \end{array} \quad \square\square \quad \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \\
& \mathbf{4} \quad \mathbf{4^*} \quad \mathbf{6} \quad \mathbf{10} \quad \mathbf{15} \quad \mathbf{20} \\
SO(6) : \quad & \times \quad + \quad \square \quad \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \quad \begin{array}{|c|} \hline \square \\ \hline \end{array} \quad \square\square
\end{aligned} \tag{27.26}$$

(For $\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array}$ of $SU(4)$ we used the formula for the dimensions of a Young tableau of $SU(N)$: $\frac{4 \times 3 \times 5 \times 4}{2 \times 3 \times 1 \times 2} = 20$.) We note a regularity: an odd number of boxes of $SU(4)$ gives spinorial irreps of $SO(6)$, while an even number gives tensorial irreps. Let us check this with a more complicated spinor irrep.

$$\begin{aligned}
SO(6) : \quad & \square \quad \otimes \quad \times \quad = \quad \boxtimes \quad \oplus \quad + \\
& \mathbf{6} \quad \times \quad \mathbf{4} \quad = \quad ? \quad + \quad \mathbf{4^*} \\
SU(4) : \quad & \begin{array}{|c|} \hline \square \\ \hline \end{array} \quad \otimes \quad \square \quad = \quad \square\square \quad \oplus \quad \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \\
& \mathbf{6} \quad \times \quad \mathbf{4} \quad = \quad \mathbf{20} \quad + \quad \mathbf{4^*}
\end{aligned} \tag{27.27}$$

Clearly $\boxtimes = \mathbf{20}$

($\dim \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} = \frac{4 \times 3 \times 5}{1 \times 3 \times 1} = 20$)

Note that we used the $+$ for $SO(6)$ to match the $\mathbf{4^*}$ in $SU(4)$. The spinorial irreps are clearly products of the lowest spinor irreps with tensor irreps. A vector-spinor $v^{\mu\alpha}$ can be decomposed into a “gamma-traceless” part $v^{\mu\alpha} - \frac{1}{6}\gamma^\mu(\gamma_\nu v^{\nu\alpha})$ and a “gamma-trace” part with $\gamma_\nu v^{\nu\alpha} = w^\alpha$. If v_μ^α is chiral, $\gamma_c v_\mu^\alpha = v_\mu^\alpha$, and $v_\mu^\alpha = w_\mu \psi^\alpha$ with ψ^α the $\times$ irrep, then $\gamma \cdot v^\alpha \equiv \gamma^\rho v_\rho^\alpha$ is antichiral, $\gamma_c \gamma \cdot v = -\gamma \cdot v$, (the $+$ irrep). Here γ_c is the chirality matrix $\gamma_c = \alpha \gamma^1 \cdots \gamma^6$ satisfying $\gamma_c^2 = I$.

Finally products of two spinor irreps

$$\begin{aligned}
& \times \quad \otimes \quad \times \quad = \quad \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} \quad \oplus \quad \square \\
& \mathbf{4} \quad \times \quad \mathbf{4} \quad = \quad \mathbf{10} \quad + \quad \mathbf{6}
\end{aligned} \tag{27.28}$$

This requires and will get much more discussion.

$USp(2N)$. Finally the symplectic groups. The basic irrep (the defining rep) is the $2N$ denoted by $\square$. A few examples for $USp(4)$

$$\begin{array}{ccccc} \square, & \square\square, & \begin{array}{|c|} \hline \square \\ \hline \end{array} & & \\ \mathbf{4} & \mathbf{10} & \mathbf{5} & & \\ t^\alpha & v^{(\alpha\beta)} & t^{[[\alpha\beta]]} & & \end{array} \quad \text{(take out the symplectic trace)} \quad (27.29)$$

Let's work out $\square \otimes \square$ for $USp(4)$

$$\begin{array}{ccccc} \square & \otimes & \square & = & \square\square \oplus \left(\begin{array}{|c|} \hline \square \\ \hline \end{array} \oplus \bullet \right) \\ \mathbf{4} & \times & \mathbf{4} & = & \mathbf{10} + (\mathbf{5} + \mathbf{1}) \end{array} \quad (27.30)$$

The $\bullet$ comes from the symplectic trace. Another case

$$\begin{array}{ccccc} \square\square & \otimes & \square & = & \square\square\square \oplus \left(\begin{array}{|c|} \hline \square\square \\ \hline \end{array} \oplus \square \right) \\ \mathbf{10} & \times & \mathbf{4} & = & \mathbf{20} + (\mathbf{16} + \mathbf{4}) \end{array} \quad (27.31)$$

The $\square$ is from the symplectic trace. Since $SO(5) \cong USp(4)$ we can do the same check as for $SO(6) \cong SU(4)$, namely identify irreps

$$\begin{array}{ccccc} SO(5) : & \square & \times & \begin{array}{|c|} \hline \square \\ \hline \end{array} & \begin{array}{|c|} \hline \times \\ \hline \end{array} \quad ? \\ & \mathbf{5} & \mathbf{4} & \mathbf{10} & \mathbf{16} \quad \mathbf{20} \\ USp(4) : & \begin{array}{|c|} \hline \square \\ \hline \end{array} & \square & \square\square & ? \quad \square\square\square \end{array} \quad (27.32)$$

The $\mathbf{16}$ of $SO(5)$ is a vector-spinor (with in general $4 \times 5 = 20$ components) made gamma-traceless (which removes a spinor with 4 components, leaving $20 - 4 = 16$ components). Since we see again that an odd number of boxes in $USp(4)$ corresponds to spinorial irreps of $SO(5)$, the $\mathbf{16}$ of $USp(4)$ must have an odd number of boxes. The case $\square\square\square$ is ruled out since it gives a $\mathbf{20}$, and $\begin{array}{|c|} \hline \square \\ \hline \end{array}$ is not allowed since only at most $N = 2$ boxes are allowed in a column. Hence the only possibility is

$$\begin{array}{|c|} \hline \square\square \\ \hline \end{array} = \mathbf{16} \quad \text{of } USp(4). \quad (27.33)$$

and indeed we found this result before, see (27.31).

Finally we must identify the $\mathbf{20}$ of $SO(5)$. It should be a spinorial irrep because $\square\square\square$ of $USp(4)$ has an odd number of boxes. It cannot be $\begin{array}{|c|} \hline \times \\ \hline \end{array}$ since that gave the $\mathbf{16}$. The next possibility is $\begin{array}{|c|} \hline \times \\ \hline \times \\ \hline \end{array}$, a tensor-spinor $t^{\mu\nu,\alpha} = -t^{\nu\mu,\alpha}$. A product of a tensor $t^{\mu\nu} = -t^{\nu\mu}$ with a spinor ψ^α gives a reducible $\mathbf{40}$, but the gamma-trace is

$$\gamma_\mu t^{\mu\nu,\alpha} = \psi^{\nu,\alpha} = \mathbf{20} = \mathbf{16} + \mathbf{4} \quad (27.34)$$

(The reducible vector-spinor $\psi^{\nu,\alpha}$ can in turn be decomposed into the two irreps $\mathbf{16} + \mathbf{4}$.) So the

dimension of the tensor-spinor $t^{\mu\nu,\alpha}$ of $SO(5)$ is $40 - (16 + 4) = 20$.

$$\begin{array}{|c|} \hline \times \\ \hline \times \\ \hline \end{array} = \mathbf{20} \quad \text{of } SO(5). \quad (27.35)$$

Comment. The formulae for the dimensions of all irreps of $SU(N)$, $SO(2N)$, $SO(2N+1)$ and $USp(2N)$ are given in chapter 20.

27.1 CG decompositions of products of spinor irreps

We now study and derive the CG series for direct products of spinorial irreps. For the $SO(2N+1)$ groups there is only one kind of fundamental spinor irreps which we denote by $\times$, but for the $SO(2N)$ groups there are two kinds, the chiral s -rep which we denote by $\times$, and the antichiral c -rep which we denote by $+$. We work through an example of each case.

$$\begin{aligned} SO(5) : & \quad \times \otimes \times \\ SO(6) : & \quad \times \otimes \times \text{ and } \times \otimes +. \end{aligned} \quad (27.36)$$

First we shall show that the product of two spinors ψ^α and χ^β of $SO(2N+1)$ can be expanded in terms of the complete set of Dirac matrices as

$$\psi^\alpha \chi^\beta = \sum_k a_k (\gamma^{\mu_1 \dots \mu_k} C^{-1})^{\alpha\beta} (\psi^T C \gamma_{\mu_1 \dots \mu_k} \chi) \quad (27.37)$$

where a_k are constants, which we shall determine. Then we consider the case of $SO(6)$.

In $SO(5)$ spinors have 4-components. The product $\psi^\alpha \chi^\beta$ of two such spinors can be viewed as a matrix $M^{\alpha\beta}$. The Dirac matrices and products thereof form a complete basis to expand any matrix. There are 16 basis elements: five γ_μ , ten $\gamma_{\mu\nu}$, and the unit element I (the matrices $\gamma_{\mu\nu\rho}$ and $\gamma_{\mu\nu\rho\sigma}$ and $\gamma_1 \dots \gamma_5$ are linearly dependent on the set of 16 matrices. For example, $\gamma_1 \dots \gamma_5$ is proportional to the unit matrix because $\gamma_5 = \gamma_1 \gamma_2 \gamma_3 \gamma_4$ and $\gamma_{\mu\nu\rho\sigma} = \epsilon_{\mu\nu\rho\sigma\tau} \gamma^\tau$, while $\gamma_{\mu\nu\rho} = -\frac{1}{2} \epsilon_{\mu\nu\rho\sigma\tau} \gamma^{\sigma\tau}$). There is one subtlety: the Dirac matrices have one index up and one index down: $(\gamma_\mu)^\alpha_\beta$ (necessary, otherwise one could not multiply two Dirac matrices). But in (27.37) the index β is up on the right hand side, so we must raise the index β ; this we do with the (inverse of) the charge conjugation matrix. For $SO(2N+1)$ it is unique (up to similarity transformations), and $C = C_-$ in $d = 3, 7$ while $C = C_+$ in $d = 5, 9$. Its index structure is as follows

$$C_{\alpha\beta} \quad \text{and} \quad (C^{-1})^{\beta\gamma} \quad (27.38)$$

Then the spinor indices in $C_\pm \gamma_\mu C_\pm^{-1} = \pm \gamma_\mu^T$ match

$$C_{\pm, \alpha\beta} (\gamma_\mu)^\beta_\gamma C_\pm^{-1, \gamma\delta} = \pm (\gamma^T)_\alpha^\delta \quad (27.39)$$

So we expand

$$\psi^\alpha \chi^\beta = c_0 C^{-1, \alpha\beta} + c_{1, \mu} (\gamma^\mu C^{-1})^{\alpha\beta} + c_{2, \mu\nu} (\gamma^{\mu\nu} C^{-1})^{\alpha\beta} \quad (27.40)$$

Since the c_k depend on the μ, ν indices, we have made that dependence explicit to avoid confusion. To find the coefficients c_k , we multiply by the inverse of the matrices and take the trace. First we multiply with $C_{\beta\alpha}$

$$\begin{aligned} C_{\beta\alpha} \psi^\alpha \chi^\beta &= (\psi^\alpha C_{\beta\alpha} \chi^\beta) = -\chi^\beta C_{\beta\alpha} \psi^\alpha = -\bar{\chi} \psi \\ &= c_0 \text{tr}(C C^{-1}) + c_1 \text{tr}(C \gamma^\mu C^{-1}) + c_2 \text{tr}(C \gamma^{\mu\nu} C^{-1}) = c_0 \times 4 \end{aligned} \quad (27.41)$$

We used that $\text{tr}(C \gamma^\mu C^{-1}) = \text{tr} \gamma^\mu = 0$, and idem $\text{tr} \gamma^{\mu\nu} = 0$. So

$$c_0 = -\frac{1}{4} (\chi^\beta C_{\beta\alpha} \psi^\alpha). \quad (27.42)$$

The minus sign is produced if ψ^α and χ^β are anticommuting. (Sometimes one uses spinors which are commuting, then no minus sign is present.) Similarly one finds

$$(C \gamma_\nu)_{\beta\alpha} \psi^\alpha \chi^\beta = -(\chi^T C \gamma_\nu \psi) = c_{1, \mu} \text{tr}(C \gamma_\nu \gamma^\mu C^{-1}) = 4 c_{1, \nu} \quad (27.43)$$

Finally

$$(C \gamma_{\mu\nu})_{\beta\alpha} \psi^\alpha \chi^\beta = -(\chi^T C \gamma_{\mu\nu} \psi) = c_{2, \rho\sigma} \text{tr}(C \gamma_{\mu\nu} \gamma^{\rho\sigma} C^{-1}) = c_{2, \rho\sigma} \text{tr}(\gamma_{\mu\nu} \gamma^{\rho\sigma}) \quad (27.44)$$

The trace $\text{tr}(\gamma_{\mu\nu} \gamma^{\rho\sigma})$ is only then nonvanishing if $\mu = \rho, \nu = \sigma$ or $\mu = \sigma, \nu = \rho$. If $\mu = \rho, \nu = \sigma$ we get $\gamma_{\mu\nu} \gamma^{\mu\nu}$ (no sum) $= -I$. Hence

$$c_{2, \rho\sigma} \text{tr}(\gamma_{\mu\nu} \gamma^{\rho\sigma}) = 4 \times (-2) c_{2, \mu\nu} \quad (27.45)$$

All in all we get

$$\begin{aligned} SO(5) : \quad \times \otimes \times &= \bullet \quad \oplus \quad \square \quad \oplus \quad \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \\ \psi^\alpha \chi^\beta &= -\frac{1}{4} \left[(C^{-1})^{\alpha\beta} (\chi^T C \psi) + (\gamma^\mu C^{-1})^{\alpha\beta} (\chi^T C \gamma_\mu \psi) - \frac{1}{2} (\gamma^{\mu\nu} C^{-1})^{\alpha\beta} (\chi^T C \gamma_{\mu\nu} \psi) \right] \\ \mathbf{4} \times \mathbf{4} &= \mathbf{1} \quad + \quad \mathbf{5} \quad + \quad \mathbf{10} \end{aligned} \quad (27.46)$$

Often one writes $\bar{\chi}$ for $\chi^T C$ where $\bar{\chi}$ is the Majorana conjugate of χ . Although we evaluated $\times \otimes \times$ for $SO(5)$, it can be done for any $SO(2N+1)$ in this way.

Next we consider in $SO(6)$ the Clebsch-Gordan expansion of two chiral spinors, or one chiral and one antichiral spinor. (We call a spinor in the s -rep a chiral spinor, and a spinor in the c -rep an antichiral spinor. That follows from the fact that the chirality matrix γ_c has the form $\gamma_c = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}$ in the representation of the Dirac matrices that we use.)

First consider $\psi^\alpha \chi^\beta$ where $\alpha, \beta = 1, \dots, 4$ are indices of the s -rep of $SO(6)$. We can again

expand on a complete basis of Dirac matrices. An arbitrary (nonchiral) spinor in 6 dimensions has 8 components, so the Dirac matrices (and products thereof) are 8×8 matrices which form the following basis

$$I, \gamma^\mu, \gamma^{\mu\nu}, \gamma^{\mu\nu\rho}, \gamma^{\mu\nu\rho\sigma}, \gamma^{\mu\nu\rho\sigma\tau}, \gamma_c \quad (27.47)$$

$$8 \times 8 = 64 = 1 + 6 + 15 + 20 + 15 + 6 + 1.$$

However, the spinors ψ^α and χ^β are chiral, and as a result only half this set of matrices will contribute because the Dirac matrices are block-offdiagonal (see (27.50)).

$$\begin{pmatrix} * & 0 \\ \chi & \end{pmatrix} \begin{pmatrix} 0 & * \\ * & 0 \end{pmatrix} \begin{pmatrix} * \\ 0 \end{pmatrix} = 0; \quad \begin{pmatrix} * & 0 \\ \chi & \end{pmatrix} \begin{pmatrix} 0 & * \\ * & 0 \end{pmatrix} \begin{pmatrix} 0 & * \\ * & 0 \end{pmatrix} \begin{pmatrix} * \\ 0 \end{pmatrix} \neq 0 \quad (27.48)$$

one γ matrix
 ψ
two γ matrices

One might conclude that only the set with an even number of Dirac matrices contributes, but there is a subtlety: we did not yet include the charge conjugation matrix. So the issue is: which of the following expressions is nonvanishing

$$\begin{pmatrix} * & 0 \\ \chi & \end{pmatrix} C \begin{pmatrix} * \\ 0 \end{pmatrix} \quad \text{OR} \quad \begin{pmatrix} * & 0 \\ \chi & \end{pmatrix} C \gamma_\mu \begin{pmatrix} * \\ 0 \end{pmatrix} \quad (27.49)$$

It clearly depends on C : if C is block-diagonal then an even number of Dirac matrices contribute to the expression, but if C is block-offdiagonal, the terms with an odd number of Dirac matrices contribute. The block-structure of C depends on the dimensions; we consider the case of 6 dimensions.

So we must first find out whether C_+ and/or C_- are block-diagonal or block-offdiagonal in 6 dimensions. There might now seem to be 3 ways to solve this question

- a) Use the fact (often mentioned in class) that in Euclidean $d = 7$ a completely imaginary set of Dirac matrices exists. Then drop one of them, and one ends up with a set of Dirac matrices in $d = 6$. In even d , the irrep of Dirac matrices is unique. However, the 6-dimensional matrices thus obtained would not be block-offdiagonal (unless we picked them in 7 dimensions in a careful way). We could try to find a similarity transformation which casts them in block form but that is tedious.
- b) Use any representation of the Dirac matrices in $d = 5 + 0$, and tensor with Pauli matrices to get $d = 6 + 0$ Dirac matrices:

$$\gamma^{(6)} = \gamma^{(5)} \otimes \sigma_2; \quad I \otimes \sigma_1 \quad (27.50)$$

Then we can construct $C_\pm^{(6)}$ from $C_\pm^{(5)}$.

- c) Use the expression for Dirac matrices as a direct product of (in this case) three two-dimensional spaces.

We use b). As charge conjugation matrices in $d = 6$ we assume a direct product

$$C^{(6)} = C^{(5)} \otimes B^{(2)} \quad (27.51)$$

where $B^{(2)}$ are matrices to be determined (they will be related to the charge conjugation matrices $C_+^{(2)}$ and $C_-^{(2)}$ in $d = 2 + 0$). The $C^{(5)}$ satisfy $C^{(5)}\gamma^\mu C^{(5),-1} = +\gamma^{\mu,T}$ with $\mu = 1, \dots, 5$. So in $d = 5 + 0$, $C = C_+$. We then get two cases to study

$$C_+^{(6)} = C^{(5)} \otimes B_+^{(2)} \quad \text{and} \quad C_-^{(6)} = C^{(5)} \otimes B_-^{(2)} \quad (27.52)$$

In each case we get two conditions on $B^{(2)}$.

$$1) \ C_\pm^{(6)}(\gamma^\mu \otimes \sigma_2)C_\pm^{(6),-1} = C^{(5)}\gamma^\mu C^{(5),-1} \otimes B_\pm^{(2)}\sigma_2 B_\pm^{(2),-1} = \pm(\gamma^\mu \otimes \sigma_2)^T = \pm\gamma^{\mu,T} \otimes (-\sigma_2).$$

$$2) \ C_\pm^{(6)}(I \otimes \sigma_1)C_\pm^{(6),-1} = C^{(5)}IC^{(5),-1} \otimes B_\pm^{(2)}\sigma_1 B_\pm^{(2),-1} = \pm(I \otimes \sigma_1)^T = \pm I \otimes \sigma_1.$$

So we get the following conditions on $B_\pm^{(2)}$:

$$C_+^{(6)} = C^{(5)} \otimes B_+^{(2)} : \begin{array}{l} B_+^{(2)}\sigma_2 B_+^{(2),-1} = -\sigma_2 \\ B_+^{(2)}\sigma_1 B_+^{(2),-1} = \sigma_1 \end{array} ; \quad C_-^{(6)} = C^{(5)} \otimes B_-^{(2)} : \begin{array}{l} B_-^{(2)}\sigma_2 B_-^{(2),-1} = \sigma_2 \\ B_-^{(2)}\sigma_1 B_-^{(2),-1} = -\sigma_1 \end{array} \quad (27.53)$$

Clearly $B_+^{(2)} = \sigma_1$ and $B_-^{(2)} = -i\sigma_2$. (The $-i$ is added to make $B_-^{(2)}$ real but that is not essential.) Both are block-offdiagonal (as expected, $C_- = C_+\Gamma_{\text{chiral}}$ and $\Gamma_{\text{chiral}} = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} = I \otimes \sigma_3$ is block-diagonal).

$$C_+^{(6)} = \begin{pmatrix} 0 & C^{(5)} \\ C^{(5)} & 0 \end{pmatrix}; \quad C_-^{(6)} = \begin{pmatrix} 0 & -C^{(5)} \\ C^{(5)} & 0 \end{pmatrix} \quad (27.54)$$

(In odd dimensions C has both indices down, as $C_{\alpha\beta}$, but in even dimensions we write a spinor as $\Psi = \begin{pmatrix} \psi^\alpha \\ \bar{\chi}_\alpha \end{pmatrix}$ and then C has a more complicated index structure which depends on whether it is block-diagonal or block-offdiagonal. We discuss this below, but for the time-being we ignore this subtlety and write indices as indices for nonchiral spinors in the same way as in odd dimensions.)

Now we get

$$\times \otimes \times = \psi^\alpha \chi^\beta = \sum_{k=\text{odd}} a_k (\gamma^{\mu_1 \dots \mu_k} C^{-1})^{\alpha\beta} (\chi^T C \gamma_{\mu_1 \dots \mu_k} \psi) \quad (27.55)$$

with an odd number of Dirac matrices: γ^μ and $\gamma^{\mu\nu\rho}$, 6 + 10 matrices.

$$\begin{array}{llll} SO(6) : \quad \times \otimes \times = & \square & \oplus & \begin{array}{c} \square \\ \square \\ \square \end{array} \\ \psi^\alpha \chi^\beta = & -\frac{1}{4}(\gamma^\mu C^{-1})^{\alpha\beta} (\chi^T C \gamma_\mu \psi) & + & \frac{1}{24}(\gamma^{\mu\nu\rho} C^{-1})^{\alpha\beta} (\chi^T C \gamma_{\mu\nu\rho} \psi) \\ \mathbf{4} \times \mathbf{4} = & \mathbf{6} & + & \mathbf{10} \end{array} \quad (27.56)$$

We used that for chiral spinors the contribution from $\gamma_{\mu_1 \dots \mu_k}$ is the same as that from $\gamma_{\mu_1 \dots \mu_{6-k}}$, so we evaluated the contributions from γ_μ and $\gamma_{\mu\nu\rho}$ and multiplied the result by a factor 2. (There are

20 matrices $\gamma^{\mu\nu\rho}$ but in $d = 6$ only half of them are independent if they act on chiral or antichiral spinors. For example γ^{123} gives the same result as γ^{456} .) This agrees with the fact that the s -irrep of $SO(6)$ corresponds to the defining rep **4** of $SU(4)$, and in $SU(4)$ we have

$$SU(4) : \quad \mathbf{4} \times \mathbf{4} = \square \otimes \square = \square\square \oplus \begin{array}{|c|} \hline \square \\ \hline \end{array} = \mathbf{10} + \mathbf{6}. \quad (27.57)$$

Finally $\times \otimes +$. Now we get

$$\psi^\alpha \chi_{\dot{\beta}} = \sum_{k=\text{even}} a_k (\gamma^{\mu_1 \dots \mu_k} C^{-1})^{\alpha\beta} (\chi^T C \gamma_{\mu_1 \dots \mu_k} \psi) \quad (27.58)$$

with an even number of Dirac matrices: I and $\gamma^{\mu\nu}$, $1 + 15$ matrices. So for $SO(6)$ we get

$$\begin{aligned} SO(6) : \quad \times \otimes + &= \bullet \oplus \begin{array}{|c|} \hline \square \\ \hline \end{array} \\ \psi^\alpha \chi_{\dot{\beta}} &= -\frac{1}{4} (C^{-1})^\alpha_{\dot{\beta}} (\chi^T C \psi) + \frac{1}{8} (\gamma^{\mu\nu} C^{-1})^\alpha_{\dot{\beta}} (\chi^T C \gamma_{\mu\nu} \psi) \\ \mathbf{4}_s \times \mathbf{4}_c &= \mathbf{1} + \mathbf{15} \end{aligned} \quad (27.59)$$

Again the contributions from $\gamma_{\mu\nu\rho\sigma}$ and γ_{123456} are the same as those of $\gamma_{\mu\nu}$ and I , so we evaluated only the latter and multiplied the result by 2. In $SU(4)$ the same CG expansion yields

$$SU(4) : \quad \mathbf{4} \times \mathbf{4}^* = \square \otimes \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} = \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} = \mathbf{1} + \mathbf{15}. \quad (27.60)$$

This confirms also the expansion of $\times \otimes +$.

Let us now come back to the issue of the indices of $C^{(6)}$ and the Dirac matrices. We are led to the analogue of what are called in $d = 4$ two-component spinors, or dotted and undotted spinors, or van der Waerden formalism. Namely, when we decompose an 8-component nonchiral spinor Ψ into a chiral part of ψ^α and an antichiral part $\chi_{\dot{\alpha}}$ as

$$\Psi^A = \begin{pmatrix} \psi^\alpha \\ \chi_{\dot{\alpha}} \end{pmatrix} \quad \begin{array}{l} \alpha = 1, \dots, 4 \\ \dot{\alpha} = 1, \dots, 4 \end{array} \quad (27.61)$$

then the structure of the Dirac matrices Γ_M in $d = 6$ given in (27.50) is as follows

$$\Gamma_M = \begin{pmatrix} 0 & -i\sigma_M^{\alpha\dot{\beta}} \\ i\bar{\sigma}_{M,\dot{\alpha}\beta} & 0 \end{pmatrix}_{8 \times 8} \quad \begin{array}{l} \sigma_M = \{\sigma_i, \sigma_6\} = \{\gamma_i^{(5)}, iI\} \\ \bar{\sigma}_M = \{\bar{\sigma}_i, \bar{\sigma}_6\} = \{\gamma_i^{(5)}, -iI\} \end{array} \quad (27.62)$$

where $M = 1, \dots, 6$, $\alpha = 1, \dots, 4$ and $\dot{\beta} = 1, \dots, 4$. This structure is familiar from $d = 4$, but it holds in any even dimension. In particular, the matrix σ_M are often called generalized Pauli matrices. Then the indices of the charge conjugation matrices are fixed by the relation $C_\pm^{(6)} \Gamma_M = \pm \Gamma_M^T C_\pm^{(6)}$, or, easier, by consistency of the indices in the CG decomposition in (27.56)

and (27.59).

$$C_{\pm}^{(6)} = \begin{pmatrix} 0 & C_{\alpha}^{(5)\dot{\beta}} \\ \pm C^{(5)\dot{\alpha}}_{\beta} & 0 \end{pmatrix} \quad (27.63)$$

The matrices $C_{\alpha}^{(5)\dot{\beta}}$ and $C^{(5)\dot{\alpha}}_{\beta}$ are both numerically equal to the charge conjugation matrix $C^{(5)}$ in $d = 5$; only the indices have been written up or down to achieve consistency in $d = 6$.

Finally we note that in the CG decomposition it makes no difference whether one uses $C_+^{(6)}$ or $C_-^{(6)}$ since the minus sign in $C_-^{(6)}$ cancels (C_- appears in two places, in terms such as $(\gamma^{\mu}C^{-1})^{\alpha\beta}$ and in terms such as $\psi^T C \gamma_{\mu} \psi$). In fact, one can rewrite the CG decomposition given above in “two-component” spinor notation, using “Pauli” matrices instead of Dirac matrices. Then one really needs to put the dotted and undotted indices in the correct positions.

27.2 Spinor calculus and invariant spinor-tensors

The Clebsch-Gordan decompositions of the product of two spinor irreps, for example the relation

$$\begin{aligned} SO(5) : \times \otimes \times = & \quad \mathbf{1} \quad + \quad \mathbf{5} \quad + \quad \mathbf{10} \\ \psi^{\alpha} \otimes \chi^{\beta} = & -\frac{1}{4} \left[(C^{-1})^{\alpha\beta} (\chi^T C \psi) + (\gamma^{\mu} C^{-1})^{\alpha\beta} (\chi^T C \gamma_{\mu} \psi) - \frac{1}{2} (\gamma^{\mu\nu} C^{-1})^{\alpha\beta} (\chi^T C \gamma_{\mu\nu} \psi) \right] \end{aligned} \quad (27.64)$$

are $SO(5)$ -covariant by which is meant that the left-hand side and the right-hand side transform the same way under $SO(5)$ transformations of ψ and χ . But if one transforms ψ and χ , the left-hand side clearly transforms as a bispinor $\phi^{\alpha\beta}$, whereas on the right-hand side the transformation of ψ and χ does not seem to be that of a bispinor. Actually, the right-hand side also transforms as a bispinor because

- (i) transforming ψ and χ in $\chi^T C \psi$ and $\chi^T C \gamma_{\mu} \psi$ and $\chi^T C \gamma_{\mu\nu} \psi$, these expressions transform as a scalar, a vector, and a tensor, respectively
- (ii) transforming the index μ of $(\gamma^{\mu} C^{-1})^{\alpha\beta}$ as a vector, this expression transforms as a bispinor because $(\gamma^{\mu} C^{-1})^{\alpha\beta}$ is an invariant tensor as we now discuss. Similarly, transforming the indices μ, ν of $(\gamma^{\mu\nu} C^{-1})^{\alpha\beta}$ as a tensor, the expression $(\gamma^{\mu\nu} C^{-1})^{\alpha\beta}$ transforms as a bispinor. And finally, transforming $(C^{-1})^{\alpha\beta}$ as a bispinor, it does not change: $(C^{-1})^{\alpha\beta}$ is also an invariant tensor. Thus the decomposition in (27.64) is covariant.

Actually, for $SO(2N + 1)$ the following are invariant spinor-tensors: $(\gamma^{\mu})^{\alpha}_{\beta}$, $C_{\alpha\beta}$, $(C^{-1})^{\alpha\beta}$ and products thereof. We shall first prove this, and then analyze the case of $SO(2N)$ where the generalized Pauli matrices $(\sigma_M)^{\alpha\dot{\beta}}$, $\bar{\sigma}_{M,\dot{\alpha}\beta}$ and the charge conjugation matrices C_+ and C_- are also invariant tensors.

$SO(2N + 1)$. Consider in $2N + 1$ dimensions the Dirac matrices $(\gamma^{\mu})^{\alpha}_{\beta}$. They must have one spinor index up and the other spinor index down in order that one can multiply two Dirac matrices

$((\gamma^\mu)^\alpha_\beta (\gamma^\nu)^\beta_\gamma = (\gamma^\mu \gamma^\nu)^\alpha_\gamma)$. A spinor χ^α has the spinor index α up in order that γ^μ can act on χ $((\gamma^\mu)^\alpha_\beta \chi^\beta)$, and it transforms as follows under $SO(2N+1)$ transformations

$$\delta\chi^\alpha = \frac{1}{2}\lambda^{mn}\frac{1}{2}(\gamma_{mn})^\alpha_\beta\chi^\beta = \frac{1}{4}\lambda^{mn}(\gamma_{mn})^\alpha_b\chi^b \quad (27.65)$$

where $\lambda^{mn} = -\lambda^{nm}$ and $\gamma_{mn} = \frac{1}{2}(\gamma_m\gamma_n - \gamma_n\gamma_m)$ because, as we have shown, $M_{mn}^{spin} = \frac{1}{2}\gamma_{mn}$ satisfy the $SO(2N+1)$ Lie algebra. The factor $\frac{1}{2}$ in front of (27.65) is needed to avoid double-counting the cases $m > n$ and $m < n$. One can define a spinor with an index down by using the (unique in odd dimensions) charge conjugation matrix $C_{\alpha\beta}$ as follows

$$\begin{aligned} \chi^\beta C_{\beta\alpha} &= \chi_\alpha = C_{\alpha\beta}^T \chi^\beta = \sigma C_{\alpha\beta} \chi^\beta \\ C^T &= \sigma C \quad (C_{\beta\alpha} = \sigma C_{\alpha\beta}) \quad \sigma = \pm 1 \end{aligned} \quad (27.66)$$

The reason of this definition of χ_α (instead of $\chi_\alpha = C_{\alpha\beta}\chi^\beta$) is our wish to agree with standard conventions in $d=5$ (and later in $d=4$). We claim that χ_α transforms as follows

$$\delta\chi_\alpha = -\frac{1}{4}\chi_\gamma\lambda^{mn}(\gamma_{mn})^\gamma_\alpha \quad (27.67)$$

which implies that $\chi_\alpha\psi^\alpha$ is invariant. To derive this result for $\delta\chi_\alpha$ from $\delta\chi^\alpha$ in (27.65) we need

$$C_\pm\gamma^\mu C_\pm^{-1} = \pm\gamma^{\mu,T} \text{ or } C_{\pm,\alpha\beta}(\gamma^\mu)^\beta_\gamma = \pm(\gamma^{\mu,T})_\alpha^\delta (C_\pm)_{\delta\gamma} \quad (27.68)$$

(with either the $+$ or the $-$ spin in odd dimensions, not both) as well as $C^T = \sigma C$ where $\sigma = +1$ or -1 depending on the dimension $2N+1$. Then we get

$$\begin{aligned} \delta\psi_\alpha &= \sigma C_{\alpha\beta} \frac{1}{4}\lambda^{mn}(\gamma_{mn})^\beta_\gamma \psi^\gamma \quad (\text{transpose}) \\ &= \sigma \frac{1}{4}\lambda^{mn} \psi^\gamma (\gamma_n^T)_{\gamma}{}^\beta (\gamma_m^T)_{\beta}{}^\delta C_{\delta\alpha}^T \quad (\text{use } C^T = \sigma C) \\ &= \frac{1}{4}\lambda^{mn} \psi^\gamma C_{\gamma\delta} (\gamma_n)^\delta_\beta (\gamma_m)^\beta_\alpha \quad (\text{twice a } - \text{ sign}) \\ &= \frac{1}{4}\lambda^{mn} (\sigma C_{\delta\gamma} \psi^\gamma) (\gamma_{nm})^\delta_\alpha \quad (\text{since } \sigma C = C^T) \\ &= -\frac{1}{4}\lambda^{mn} \psi_\delta (\gamma_{mn})^\delta_\alpha \quad (\text{since } \gamma_{nm} = -\gamma_{mn}) \end{aligned} \quad (27.69)$$

We are now ready to prove that $(\gamma^\mu)^\alpha_\beta$ is one invariant tensor. We use the rules for $\delta\chi^\alpha$ and $\delta\psi_\alpha$ to transform the spinor indices, and transfer the index μ as a vector ($\delta v^\mu = \lambda^\mu_\nu v^\nu$)

$$\begin{aligned} \delta(\gamma^\mu)^\alpha_\beta &= \frac{1}{4}\lambda^{mn}(\gamma_{mn})^\alpha_{\alpha'} (\gamma^\mu)^{\alpha'}_\beta - \frac{1}{4}\lambda^{mn} \gamma^{\mu,\alpha}_{\beta'} (\gamma_{mn})^{\beta'}_\beta + \lambda^\mu_\nu (\gamma^\nu)^a_\beta \\ &= \frac{1}{4}\lambda^{mn} [\gamma_{mn}, \gamma^\mu]^\alpha_\beta + \lambda^\mu_\nu (\gamma^\nu)^a_\beta \end{aligned} \quad (27.70)$$

Using¹ $[\gamma_{mn}, \gamma^\mu] = 2\delta_n^\mu \gamma_m - 2\delta_m^\mu \gamma_n$ we obtain the desired result

$$\delta(\gamma^\mu)^\alpha{}_\beta = \frac{1}{4}\lambda^{mn}(4\delta_n^\mu \gamma_m)^\alpha{}_\beta + \lambda^\mu{}_\nu(\gamma^\nu)^\alpha{}_\beta = 0. \quad (27.71)$$

It is also straightforward to show that the charge conjugation matrix $C_{\alpha\beta}$, (and its inverse $C^{-1,\alpha\beta}$) are invariant tensors. From (27.67) it follows that we have to show that the following expression vanishes

$$\delta C_{\alpha\beta} = C_{\alpha'\beta}(-\frac{1}{4}\lambda^{mn})(\gamma_{mn})^{\alpha'}{}_\alpha + C_{\alpha\beta'}(-\frac{1}{4}\lambda^{mn})(\gamma_{mn})^{\beta'}{}_\beta \quad (27.72)$$

In matrix notation we get the combination $C\gamma_{mn} + (\gamma_{mn})^T C$, and the combination $C\gamma_{mn}C^{-1} + (\gamma_{mn})^T$ vanishes since $C\gamma_{mn}C^{-1} = \gamma_{[m}^T \gamma_{n]}^T = (\gamma_{nm})^T = -(\gamma_{mn})^T$. Thus $C_{\alpha\beta}$ is an invariant tensor. Also $\delta_\alpha{}^\beta$ is an invariant tensor. Products of invariant tensors are invariant tensors, so the matrices $C^{-1,\alpha\beta}$, $(\gamma^\mu C^{-1})^{\alpha\beta}$ and $(\gamma^{\mu\nu} C^{-1})^{\alpha\beta}$ in (27.64) are also invariant tensors.

Let us now demonstrate the covariance of (27.64) in detail. Transforming χ and ψ in $\chi^T C \gamma_\mu \psi$ as

$$\delta\chi_\alpha = -\chi_\beta \frac{1}{4}\lambda^{mn}(\gamma_{mn})^\beta{}_\alpha \quad \text{and} \quad \delta\psi^\beta = \frac{1}{4}\lambda^{mn}(\gamma_{mn})^\beta{}_\delta \psi^\delta \quad (27.73)$$

one obtains

$$\begin{aligned} \delta(\chi^T C \gamma_\mu \psi) &= (\delta\chi_\alpha)(\gamma^\mu)^\alpha{}_\beta \psi^\beta + \chi_\alpha(\gamma^\mu)^\alpha{}_\beta \delta\psi^\beta \\ &= \chi_\alpha[\gamma_\mu, \frac{1}{4}\lambda^{mn}\gamma_{mn}]^\alpha{}_\beta \psi^\beta = \lambda_\mu{}^n \chi^T C \gamma_n \psi. \end{aligned} \quad (27.74)$$

Acting with the matrix $\lambda_\mu{}^n$ on the factor $(\gamma^\mu C^{-1})^{\alpha\beta}$ in (27.64) we obtain

$$-\lambda_\mu{}^n(\gamma^\mu C^{-1})^{\alpha\beta}, \quad (27.75)$$

and using that transforming all three indices μ, α, β of $(\gamma^\mu C^{-1})^{\alpha\beta}$ leaves it invariant, we see that $-\lambda_\mu{}^n(\gamma^\mu C^{-1})^{\alpha\beta}$ is equal to the transformation of a bispinor $\phi^{\alpha\beta}$. The other two terms are handled the same way. Thus (27.64) is indeed a covariant equation.

In 4 dimensions one usually follows conventions in which $\psi^\alpha \chi_\alpha = -\psi_\alpha \chi^\alpha$ with $\alpha = 1, 2$. In $2N + 1$ dimensions we find instead, $\chi_\alpha = \chi^\beta C_{\beta\alpha} = \sigma C_{\alpha\beta} \chi^\beta$,

$$\begin{aligned} \psi^\alpha \chi_\alpha &= \psi^\alpha \sigma C_{\alpha\beta} \chi^\beta = -\chi^\beta \psi^\alpha \sigma C_{\alpha\beta} \quad (\text{if } \psi \text{ and } \chi \text{ are anticommuting}) \\ &= -\sigma \chi^\beta \psi_\beta = \sigma \psi_\alpha \chi^\alpha \end{aligned} \quad (27.76)$$

Note that it follows from our definition $\chi^\beta C_{\beta\alpha} = \chi_\alpha$ that $\chi^\beta = \chi_\alpha C^{-1,\alpha\beta}$. We would like to follow uniformly a northwest-southeast contraction convention ($\chi^\alpha \psi_\alpha$). The law $\chi_\alpha = \chi^\beta C_{\beta\alpha}$ follows this convention, but for χ^β we would like to use a matrix $C^{\beta\alpha}$ such that $\chi^\beta = C^{\beta\alpha} \chi_\alpha$. The definition

¹Use $[\gamma_{mn}, \gamma^\mu] = [\gamma_m \gamma_n - \delta_{mn}, \gamma^\mu] = [\gamma_m \gamma_n, \gamma^\mu] = \gamma_m \{\gamma_n, \gamma^\mu\} - \{\gamma_m, \gamma^\mu\} \gamma_n$.

of the matrix $C^{\beta\alpha}$ follows from the equality of $\chi^\beta = \chi_\alpha C^{-1,\alpha\beta}$ and $\chi^\beta = C^{\beta\alpha} \chi_\alpha$

$$C^{\beta\alpha} = (C^{-1})^{\alpha\beta} = \sigma(C^{-1})^{\beta\alpha}. \quad (27.77)$$

With the northwest-southeast contraction convention to lower both indices of $C^{\alpha\beta}$ with $C_{\gamma\delta}$ we find $C_{\alpha\beta}$

$$C^{\alpha'\beta'} C_{\alpha'\alpha} C_{\beta'\beta} = \sigma C^{-1,\alpha'\beta'} C_{\alpha'\alpha} C_{\beta'\beta} = \sigma \delta_{\beta}^{\alpha'} C_{\alpha'\alpha} = \sigma C_{\beta\alpha} = C_{\alpha\beta}. \quad (27.78)$$

We shall exclusively use $C_{\alpha\beta}$ and $C^{\alpha\beta}$ to raise and lower indices in $d = 2N + 1$, but note that $C_{\alpha\beta} C^{\beta\gamma} = C_{\alpha\beta} \sigma C^{-1,\beta\gamma} = \sigma \delta_{\alpha}^{\gamma}$. A more illuminating way to write this result is $C_{\alpha\beta} C^{\gamma\beta} = \delta_{\alpha}^{\gamma}$, which parallels the relation $\epsilon_{\alpha\beta} \epsilon^{\gamma\beta} = \delta_{\alpha}^{\gamma}$ of four dimensions.

$SO(2N)$. For $SO(2N)$ the analysis is more subtle. An arbitrary nonchiral spinor ψ is decomposed into a spinors ψ^α and $\chi_{\dot{\beta}}$ as follows

$$\Psi = \begin{pmatrix} \psi^\alpha \\ \chi_{\dot{\beta}} \end{pmatrix} \quad \alpha, \dot{\beta} = 1, \dots, 2^{N-1}. \quad (27.79)$$

A useful representation of the Dirac matrices Γ_M is obtained by setting $\Gamma_m = \gamma_m \otimes \sigma_2$ and $\Gamma_{2N} = I \otimes \sigma_1$, where γ_m are Dirac matrices in $2N - 1$ dimensions. Since the product of all $2N - 1$ Dirac matrices γ_m is proportional to the unit matrix (recall that γ_{2N-1} is proportional to the product of all other Dirac matrices), the chirality matrix Γ_c in $2N$ dimensions is proportional to $I \otimes \sigma_3$, and we take it exactly equal, $\Gamma_c = I \otimes \sigma_3$. Then ψ^α is chiral and $\chi_{\dot{\beta}}$ is antichiral. The Dirac matrices can be expressed in terms of generalized Pauli matrices as follows

$$\begin{aligned} \Gamma_M &= \begin{pmatrix} 0 & -i\sigma_M^{\alpha\dot{\beta}} \\ i\bar{\sigma}_{M,\dot{\alpha}\beta} & 0 \end{pmatrix} \quad \begin{aligned} \sigma_M &= \{\gamma_m, iI\} \\ \bar{\sigma}_M &= \{\gamma_m, -iI\} \end{aligned} \quad M = 1, \dots, 2N \\ \Gamma_m &= \begin{pmatrix} 0 & -i\gamma_m \\ i\gamma_m & 0 \end{pmatrix} \quad m = 1, \dots, 2N - 1; \quad \Gamma_{2N} = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} \end{aligned} \quad (27.80)$$

The $SO(2N)$ generators $M_{MN}^{\text{spin}} = \frac{1}{2}\Gamma_{MN} = \frac{1}{2}\Gamma_{[M}\Gamma_{N]}$ are block-diagonal and have the following form

$$M_{mn} = \begin{pmatrix} \frac{1}{2}(\sigma_{mn})^{\alpha\beta} & 0 \\ 0 & \frac{1}{2}(\bar{\sigma}_{mn})_{\dot{\alpha}\dot{\beta}} \end{pmatrix}; \quad M_{m,2N} = \begin{pmatrix} -\frac{i}{2}(\sigma_m)^{\alpha\beta} & 0 \\ 0 & \frac{i}{2}(\bar{\sigma}_m)_{\dot{\alpha}\dot{\beta}} \end{pmatrix} \quad (27.81)$$

where $M, N = 1, \dots, 2N$ and $m, n = 1, \dots, 2N - 1$ while

$$\begin{aligned}
(\sigma_{mn})^\alpha{}_\beta &= \frac{1}{2} \left[(\sigma_m)^{\alpha\dot{\gamma}} \bar{\sigma}_{n,\dot{\gamma}\beta} - (\sigma_n)^{\alpha\dot{\gamma}} \bar{\sigma}_{m,\dot{\gamma}\beta} \right] \\
(\bar{\sigma}_{mn})_{\dot{\alpha}}{}^{\dot{\beta}} &= \frac{1}{2} \left[\bar{\sigma}_{m,\dot{\alpha}\gamma} (\sigma_n)^{\gamma\dot{\beta}} - \bar{\sigma}_{n,\dot{\alpha}\gamma} (\sigma_m)^{\gamma\dot{\beta}} \right] \\
-i(\sigma_m)^\alpha{}_\beta &= -i(\sigma_m)^{\alpha\dot{\gamma}} \delta_{\dot{\gamma}\beta} \\
i(\bar{\sigma}_m)_{\dot{\alpha}}{}^{\dot{\beta}} &= i\bar{\sigma}_{m,\dot{\alpha}\gamma} \delta^{\gamma\dot{\beta}}
\end{aligned} \tag{27.82}$$

All σ_m and $\bar{\sigma}_m$ are numerically equal to γ_m , but σ_m has indices $\sigma_m^{\alpha\dot{\beta}}$ while $\bar{\sigma}_m$ has indices $\sigma_{m,\dot{\alpha}\beta}$. The $\frac{1}{2}\sigma_{MN} = \frac{1}{2}(\sigma_{mn}, -i\sigma_m)$ and $\frac{1}{2}\bar{\sigma}_{MN} = \frac{1}{2}(\bar{\sigma}_{mn}, i\bar{\sigma}_m)$ satisfy the $SO(2N)$ Lie algebra (but note the position of the indices $\dot{\gamma}$ in σ_{mn} and γ in $\bar{\sigma}_{mn}$). The problem is now to show that $(\sigma_M)^{\alpha\dot{\beta}}$, $\bar{\sigma}_{M,\dot{\alpha}\beta}$, $(\sigma_{MN})^\alpha{}_\beta$ and $(\bar{\sigma}_{MN})_{\dot{\alpha}}{}^{\dot{\beta}}$ are all invariant spinor-tensors, as are the charge conjugation matrices C_+ and C_- . For the latter, however, we must be aware of a subtlety: they can be either block-diagonal or block-off-diagonal, and these blocks carry different indices in these two cases. There is a straightforward way to derive the results by starting with the corresponding results for the Dirac matrices and then decomposing into blocks. The latter are just as in $2N + 1$ dimensions, and by decomposing Γ_M as in (27.80) we can deduce the results for σ_M and $\bar{\sigma}_M$. Similarly from $C_\pm \Gamma_M C_\pm^{-1} = \pm \Gamma_M^T$ we can deduce corresponding results for the blocks which constitute C_+ and C_- . As a check we shall decompose the reducible spinor invariant $\Psi^T C_\pm X$ into chiral and antichiral parts, and check the invariance of the resulting two terms under $SO(2N)$ transformations. This checks that the charge conjugation matrices are invariant spinor-tensors. We repeat that all these results are deduced, there are no new definitions or ambiguities about the position of spinor indices. At the end we shall compare with the special case of $d = 3 + 1$ which is of course needed in much of $d = 3 + 1$ physics.

We shall express the charge conjugation matrices $C_\pm^{(2N)}$ in $d = 2N$ in terms of the (unique) charge conjugation matrix $C^{(2N-1)}$ in $d = 2N - 1$, and need the following results for the latter

$$\begin{aligned}
d = 3 : C &= C_- & C^T &= -C \\
d = 5 : C &= C_+ & C^T &= -C \\
d = 7 : C &= C_- & C^T &= C \\
d = 9 : C &= C_+ & C^T &= C \\
d = 11 : C &= C_- & C^T &= -C \\
&\vdots & &\vdots
\end{aligned} \tag{27.83}$$

These results are easily obtained by taking particular representations of the Dirac matrices in each dimension. For example, in $d = 7 + 0$ there exists a purely imaginary representation of the Dirac matrices, so $C = C_- = I$ in that case.

$d = 2N = 4n$. The charge conjugation matrices $C_{\pm}$ are block-diagonal and $C_+ = C_- \Gamma_c$

$$C_{\pm}^{(4n)} = \begin{pmatrix} C_{\alpha\beta}^{(4n-1)} & 0 \\ 0 & \pm C^{(4n-1),\dot{\alpha}\dot{\beta}} \end{pmatrix}; \quad \Psi = \begin{pmatrix} \psi^{\alpha} \\ \chi_{\dot{\alpha}} \end{pmatrix}; \quad X = \begin{pmatrix} \lambda^{\alpha} \\ \zeta_{\dot{\alpha}} \end{pmatrix} \quad (27.84)$$

Consider

$$\Psi^T C_{\pm}^{(4n)} X = \psi^{\alpha} C_{\alpha\beta}^{(4n-1)} \lambda^{\beta} \pm \chi_{\dot{\alpha}} C^{(4n-1),\dot{\alpha}\dot{\beta}} \zeta_{\dot{\beta}} \quad (27.85)$$

The position of indices on $C_{\pm}$ follow from the position of indices on the spinors. Invariance of the first of these two terms under $SO(2N)$ transformations requires

$$\begin{aligned} \psi^{\gamma} \frac{1}{4} \lambda^{MN} (\sigma_{MN}^T)_{\gamma}^{\alpha} C_{\alpha\beta}^{(4n-1)} \lambda^{\beta} + \psi^{\alpha} C_{\alpha\beta}^{(4n-1)} \frac{1}{4} \lambda^{MN} (\sigma_{MN})^{\beta}_{\gamma} \lambda^{\gamma} &= 0 \\ \text{or } C_{\alpha\beta}^{(4n-1)} (\sigma_{MN})^{\beta}_{\gamma} + (\sigma_{MN}^T)_{\alpha}^{\beta} C_{\beta\gamma}^{(4n-1)} &= 0 \end{aligned} \quad (27.86)$$

For $\sigma_{MN} = \sigma_{mn} = \gamma_{mn}$ this relation reduces to $C^{(4n-1)} \gamma_{mn} + \gamma_{mn}^T C^{(4n-1)}$ in $d = 4n - 1$ dimensions which always holds, whether $C^{(4n-1)} = C_+^{(4n-1)}$ or $C^{(4n-1)} = C_-^{(4n-1)}$. However, for $\sigma_{MN} = \sigma_{m,2N} = -i\gamma_m$, this relation reduces to

$$C_{\alpha\beta}^{(4n-1)} (-i\gamma_m)^{\beta}_{\gamma} + (-i\gamma_m^T)_{\alpha}^{\beta} C_{\beta\gamma}^{(4n-1)} = 0 \quad (27.87)$$

and this relation is only true since $C^{(4n-1)} = C_-^{(4n-1)}$ in $d = 3, 7$.

$d = 2N = 4n + 2$. Now the charge conjugation matrices are block off-diagonal

$$C_{\pm}^{(4n+2)} = \begin{pmatrix} 0 & \pm C_{\alpha}^{(4n+1)\dot{\beta}} \\ C^{(4n+1)\dot{\alpha}}_{\beta} & 0 \end{pmatrix} \quad (27.88)$$

and

$$\Psi^T C_{\pm}^{(4n+2)} X = \chi_{\dot{\alpha}} C^{(4n+1)\dot{\alpha}}_{\beta} \lambda^{\beta} \pm \psi^{\alpha} C_{\alpha}^{(4n+1)\dot{\beta}} \zeta_{\dot{\beta}} \quad (27.89)$$

The position of indices on $C_{\pm}$ follows from the position of indices on the spinors. The invariance of the first term under $SO(2N)$ transformation requires

$$\begin{aligned} \chi_{\dot{\beta}} \frac{1}{4} \lambda^{MN} (\bar{\sigma}_{MN}^T)^{\dot{\beta}}_{\dot{\alpha}} C^{(4n+1)\dot{\alpha}}_{\gamma} \lambda^{\gamma} + \chi_{\dot{\alpha}} C^{(4n+1)\dot{\alpha}}_{\beta} \frac{1}{4} \lambda^{MN} (\sigma_{MN})^{\beta}_{\gamma} \lambda^{\gamma} &= 0 \\ \text{or } C^{(4n+1)\dot{\alpha}}_{\beta} (\sigma_{MN})^{\beta}_{\gamma} + (\bar{\sigma}_{MN}^T)^{\dot{\alpha}}_{\dot{\beta}} C^{(4n+1)\dot{\beta}}_{\gamma} &= 0 \end{aligned} \quad (27.90)$$

Again we consider the two cases $\sigma_{MN} = \sigma_{mn} = \gamma_{mn}$ and $\sigma_{MN} = \sigma_{m,2N} = -i\gamma_m$ separately. In the former case we must prove the relation

$$C^{(4n+1)} \gamma_{mn} + \gamma_{mn}^T C^{(4n+1)} = 0 \quad (27.91)$$

which is again always true. But in the case $\sigma_{MN} = -i\gamma_m$ we get

$$C^{(4n+1)}(-i\gamma_m) + (i\gamma_m^T)C^{(4n+1)} = 0 \quad (27.92)$$

and this is only true since $C^{(4n+1)} = C_+^{(4n+1)}$.

Using these results it is now straightforward to check that $\sigma_M^{\alpha\dot{\beta}}$ and $\bar{\sigma}_{M,\dot{\alpha}\beta}$ are invariant spinor-tensors. Also the blocks into which the charge conjugation matrices $C_{\pm}^{(2N)}$ decompose, are invariant tensors. It follows that also the Clebsch-Gordan decompositions of products of spinor irreps of $SO(2N)$ such as in (27.56) and (27.59) are covariant equations under $SO(2N)$ transformations. Note that the matrices $C_{\alpha\beta}^{(4n)}$, $C^{(4n),\dot{\alpha}\dot{\beta}}$, $C^{(4n+2),\dot{\alpha}}_{\beta}$ and $C_{\alpha}^{(4n+2),\dot{\beta}}$ are numerically equal to the charge conjugation matrix on $(4n-1)$ and $(4n+1)$ dimensions, but they are invariant spinor-tensors of the full group $SO(4n)$ and $SO(4n+2)$, respectively.

We can now rewrite the Clebsch-Gordan decomposition of the product of two spinor irreps in terms of generalized Pauli matrices instead of Dirac matrices. For example, in $d=6$ (27.56) becomes

$$\begin{aligned} \psi^{\alpha}\chi^{\beta} = & -\frac{1}{4}\left(\sigma^{\mu}\alpha\dot{\beta}(C^{(5),-1})_{\dot{\beta}}^{\beta}\right)\left(\chi^{\alpha}C_{\alpha}^{(5),\dot{\beta}}\bar{\sigma}_{\mu,\dot{\beta}\gamma}\psi^{\gamma}\right) \\ & + \frac{1}{24}\left(\sigma^{[\mu,\alpha\dot{\beta}}\bar{\sigma}_{\dot{\beta}\gamma}^{\nu}\sigma^{\rho],\gamma\dot{\delta}}(C^{(5),-1})_{\dot{\delta}}^{\beta}\right)\left(\chi^{\alpha}C_{\alpha}^{(5),\dot{\beta}}\bar{\sigma}_{\mu,\dot{\beta}\gamma}\sigma_{\nu}^{\gamma\dot{\delta}}\bar{\sigma}_{\rho,\dot{\delta}\beta}\psi^{\beta}\right) \end{aligned} \quad (27.93)$$

or suppressing contracted indices

$$\begin{aligned} \times \otimes \times = & \quad \square \quad \oplus \quad \begin{array}{c} \square \\ \square \\ \square \end{array} \\ \psi^{\alpha}\chi^{\beta} = & -\frac{1}{4}(\sigma^{\mu}C^{(5),-1})^{\alpha\beta}(\chi^TC^{(5)}\bar{\sigma}_{\mu}\psi) + \frac{1}{24}(\sigma^{[\mu}\bar{\sigma}^{\nu}\sigma^{\rho]}C^{(5),-1})^{\alpha\beta}(\chi^TC^{(5)}\bar{\sigma}_{\mu}\sigma_{\nu}\bar{\sigma}_{\rho}\psi) \end{aligned} \quad (27.94)$$

Products of Dirac matrices such as $\gamma_{\mu}\gamma_{\nu}\gamma_{\rho}$ are replaced by $\sigma_{\mu}\bar{\sigma}_{\nu}\sigma_{\rho}$ or $\bar{\sigma}_{\mu}\sigma_{\nu}\bar{\sigma}_{\rho}$, depending on the spinor they act on. As even simpler notation is obtained by denoting $(\sigma^{\mu}C^{(5),-1})^{\alpha\beta}$ by $\sigma^{\mu,\alpha\beta}$, in which case one obtains

$$\psi^{\alpha}\chi^{\beta} = -\frac{1}{4}(\sigma^{\mu})^{\alpha\beta}(\chi^TC^{(5)}\bar{\sigma}_{\mu}\psi) + \frac{1}{24}(\sigma^{[\mu}\bar{\sigma}^{\nu}\sigma^{\rho]})^{\alpha\beta}(\chi^T\bar{\sigma}_{\mu}\sigma_{\nu}\bar{\sigma}_{\rho}\psi). \quad (27.95)$$

Chapter 28

Decomposing a tensor with Young symmetrizers

Given a Young tableau (really a Young pattern, see below) of, for example, $SU(N)$, we know how to compute the dimension of the irreducible representation (irrep) to which it corresponds (the “factors over hooks” formula). However, one sometimes wants to decompose a reducible tensor into irreps. The simplest case is an arbitrary tensor $T_{\mu\nu}$ with two indices; it can be decomposed into a symmetric part and an antisymmetric part, and both are irreps of $SU(N)$

$$T^{\mu\nu} = \frac{1}{2} (T^{\mu\nu} + T^{\nu\mu}) + \frac{1}{2} (T^{\mu\nu} - T^{\nu\mu}) . \quad (28.1)$$

This agrees of course with the rules for the Clebsch-Gordan decomposition of the product of two fundamental irreps

$$\square \otimes \square = \square\square \oplus \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \quad (28.2)$$

but (28.1) is more specific: it tells how the indices of the irreps are related to the indices of the reducible tensor $T^{\mu\nu}$. If one tries to do the same decomposition for an arbitrary tensor $T^{\mu\nu\rho}$ with three indices, one encounters already some problems. Clearly the totally symmetric part and the totally antisymmetric part are irreps, but in addition one expects to need some irreps with mixed symmetry, and then one encounters the problem of how many such mixed irreps occur. From the Clebsch-Gordan rules for $SU(N)$ one finds

$$\square \otimes \square \otimes \square = \left(\square\square \oplus \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \right) \otimes \square = \square\square\square \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} \quad (28.3)$$

and thus one expects two irreps with mixed symmetry, but the problem is how to write them with indices: to express the irreps as particular linear combinations of the components of $T^{\mu\nu\rho}$. We now describe how to achieve this for $SU(N)$. For other groups, one must further decompose the irreps of $SU(N)$ by taking ordinary traces for $SO(N)$ or symplectic traces for $USp(2N)$.

The rules for $SU(N)$ are easy to formulate. Consider a “**Young pattern**” which is a Young

tableau without numbers in it. For a k -index tensor $T_{\mu_1 \dots \mu_k}$ one needs all Young patterns with k boxes. For example, for $k = 3$, the Young patterns are

$$\begin{array}{|c|c|c|}, & \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array}, & \begin{array}{|c|} \hline \\ \hline \\ \hline \end{array} \end{array} \quad (28.4)$$

Next put the numbers $1, 2, \dots, k$ in the boxes in all ways such that

- the numbers increase in each row if one reads from left to right,
- the numbers also increase in each column if one reads from top to bottom.

Young patterns where the numbers are written down in any way, not necessarily satisfying these restrictions, are called **Young tableaux**, while if the restrictions are satisfied, they are called **standard Young tableaux**. For $k = 3$ and the Young pattern $\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array}$ one has $6(= 3!)$ Young tableaux of which only two are standard Young tableaux.

$$\underbrace{\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline \end{array}}_{\text{Standard Young tableaux}} \quad \underbrace{\begin{array}{|c|c|} \hline 2 & 1 \\ \hline 3 & \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline 3 & 1 \\ \hline 2 & \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline 3 & 2 \\ \hline 1 & \\ \hline \end{array}}_{\text{Young tableaux}} \quad (28.5)$$

Given a tensor $T_{\mu_1 \dots \mu_k}$ and a standard Young tableau,¹ one first symmetrizes the indices of $T_{\mu_1 \dots \mu_k}$ as indicated by each row, and then one antisymmetrizes the indices as indicated by each column. This will produce an irrep of $SU(N)$. For example, for $k = 3$ and the standard Young tableau $\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array}$ one obtains the following irrep

$$\hat{T}_{\mu_1 \mu_2 \mu_3}^{\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array}, I} = (T_{\mu_1 \mu_2 \mu_3} + T_{\mu_2 \mu_1 \mu_3}) - (T_{\mu_3 \mu_2 \mu_1} + T_{\mu_2 \mu_3 \mu_1}). \quad (28.6)$$

and from $\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline \end{array}$ one finds the other irrep in (28.3) with mixed symmetry

$$\hat{T}_{\mu_1 \mu_2 \mu_3}^{\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array}, II} = (T_{\mu_1 \mu_2 \mu_3} + T_{\mu_3 \mu_2 \mu_1}) - (T_{\mu_2 \mu_1 \mu_3} + T_{\mu_3 \mu_1 \mu_2}). \quad (28.7)$$

Each of these two irreps corresponds to the **8**, but they both are needed in the decomposition of $T_{\mu_1 \mu_2 \mu_3}$ into irreps. This decomposition reads as follows

$$T_{\mu_1 \mu_2 \mu_3} = \frac{1}{6} T_{\mu_1 \mu_2 \mu_3}^{\begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array}} + \frac{1}{3} T_{\mu_1 \mu_2 \mu_3}^{\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array}, I} + \frac{1}{3} T_{\mu_1 \mu_2 \mu_3}^{\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array}, II} + \frac{1}{6} T_{\mu_1 \mu_2 \mu_3}^{\begin{array}{|c|} \hline \\ \hline \\ \hline \end{array}} \quad (28.8)$$

where

$$T_{\mu_1 \mu_2 \mu_3}^{\begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array}} = T_{\mu_1 \mu_2 \mu_3} + T_{\mu_2 \mu_3 \mu_1} + T_{\mu_3 \mu_1 \mu_2} + T_{\mu_1 \mu_3 \mu_2} + T_{\mu_3 \mu_2 \mu_1} + T_{\mu_2 \mu_1 \mu_3} \quad (28.9)$$

and

$$T_{\mu_1 \mu_2 \mu_3}^{\begin{array}{|c|} \hline \\ \hline \\ \hline \end{array}} = T_{\mu_1 \mu_2 \mu_3} + T_{\mu_2 \mu_3 \mu_1} + T_{\mu_3 \mu_1 \mu_2} - T_{\mu_1 \mu_3 \mu_2} - T_{\mu_3 \mu_2 \mu_1} - T_{\mu_2 \mu_1 \mu_3}. \quad (28.10)$$

¹To simplify the notation, we write the indices on tensors as $T_{\mu_1 \dots \mu_k}$ instead of $T^{\mu_1 \dots \mu_k}$, but this is only a change in notation. The tensors $T_{\mu_1 \dots \mu_k}$ are still represented by Young tableaux, for example v_μ still corresponds to $\begin{array}{|c|} \hline \\ \hline \end{array}$.

It is straightforward to check this decomposition of $T_{\mu\nu\rho}$. The dimensions of the irreps are, of course, given by the factors over hooks rule:

$$N^3 = \frac{1}{6}N(N+1)(N+2) + 2 \times \frac{1}{3}N(N^2-1) + \frac{1}{6}N(N-1)(N-2) \quad (28.11)$$

Note that (28.7) cannot be obtained from (28.6) by interchanging the indices μ_2 and μ_3 in (28.6) because in both cases we start with the same term $T_{\mu_1\mu_2\mu_3}$.

One can define an inner product² for the components of $T_{\mu\nu\rho}$ as follows

$$(T_{\mu\nu\rho}, T_{\alpha\beta\gamma}) = \delta_{\mu\alpha}\delta_{\nu\beta}\delta_{\rho\gamma} \quad (28.12)$$

According to this inner product, $T^{\square\square\square}$ and $T^{\square}$ are orthonormal to all irreps, but $T^{\square I}$ and $T^{\square II}$ are not orthogonal to each other

$$(T^{\square I}, T^{\square II}) = -1 \quad (28.13)$$

However, one can take linear combination that are orthogonal, for example as follows

$$T^{\square I} + T^{\square II} = \beta T^{\square\square\square} + \left((1-\beta)T^{\square I} + T^{\square II}\right) \quad (28.14)$$

For $\beta = \frac{3}{4}$ the first and second tensor on the right-hand side are orthogonal. It is easier to use the basis on the left-hand side.

The theory behind these rules goes back to Young and Frobenius, and to the representation theory of the symmetric groups $S(n)$. The elements of $S(n)$ for given n can be written as products of cycles, such as (125)(37). Given a standard Young tableau T , one constructs the Young symmetrizer P (and the Young antisymmetrizer Q) as products of (anti)symmetrizers for each row (column), and the Young operator Y as $Y = cQP$, where c is a constant. (We shall presently discuss the tensors one gets if one uses PQ instead of QP .) Then YT is the irrep. For example

$$\begin{aligned} Y\left(\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 \\ \hline \end{array}\right) &= \frac{1}{3}(e - (13))(e + (12)) \quad \text{gives (28.6)} \\ Y\left(\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 \\ \hline \end{array}\right) &= \frac{1}{3}(e - (12))(e + (13)) \quad \text{gives (28.7)} \end{aligned} \quad (28.15)$$

The coefficients $\frac{1}{3}$ ensure that $Y^2 = Y$. One can also prove (and / or check) that the Y 's for different standard Young tableaux are orthogonal. For example

$$Y\left(\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 \\ \hline \end{array}\right) Y\left(\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 \\ \hline \end{array}\right) = \frac{1}{9}(e - (13))(e + (12))(e - (12))(e + (13)) = 0 \quad (28.16)$$

clearly vanishes because $(e + (12))(e - (12)) = 0$.

Let us now give the result for an arbitrary 4-index tensor $T_{\mu_1\mu_2\mu_3\mu_4}$. There are 10 standard

²An inner product (x, y) must satisfy four conditions: 1) positive definiteness: $(x, x) \geq 0$ and $(x, x) = 0$ iff $x = 0$, 2) linearity $(x, y + \alpha z) = (x, y) + \alpha(x, z)$ and 3) hermiticity $(x, y) = (y, x)^*$.

Young tableaux

$$\begin{array}{c} \boxed{1} \\ \boxed{2} \\ \boxed{3} \\ \boxed{4} \end{array}; \quad \begin{array}{cc} \boxed{1} & \boxed{2} \\ \boxed{3} & \\ \boxed{4} & \end{array}, \quad \begin{array}{cc} \boxed{1} & \boxed{3} \\ \boxed{2} & \\ \boxed{4} & \end{array}, \quad \begin{array}{cc} \boxed{1} & \boxed{4} \\ \boxed{2} & \\ \boxed{3} & \end{array}; \quad \begin{array}{cc} \boxed{1} & \boxed{2} \\ \boxed{3} & \boxed{4} \end{array}, \quad \begin{array}{cc} \boxed{1} & \boxed{3} \\ \boxed{2} & \boxed{4} \end{array}; \\
 \begin{array}{ccc} \boxed{1} & \boxed{2} & \boxed{3} \\ \boxed{4} & & \end{array}, \quad \begin{array}{ccc} \boxed{1} & \boxed{2} & \boxed{4} \\ \boxed{3} & & \end{array}, \quad \begin{array}{ccc} \boxed{1} & \boxed{3} & \boxed{4} \\ \boxed{2} & & \end{array}; \quad \begin{array}{cccc} \boxed{1} & \boxed{2} & \boxed{3} & \boxed{4} \end{array}
 \end{array} \quad (28.17)$$

So a general tensor $T_{\mu_1\mu_2\mu_3\mu_4}$ decomposes into 10 irreps of $SU(N)$. (For low N , there may be fewer irreps; for example, for $SU(3)$ the totally antisymmetric irrep $\begin{array}{c} \boxed{} \\ \boxed{} \\ \boxed{} \end{array}$ vanishes.) The Clebsch-Gordan rules confirm that one gets 10 irreps

$$\begin{array}{c} \boxed{} \\ \boxed{} \\ \boxed{} \\ \boxed{} \end{array} \otimes \begin{array}{c} \boxed{} \\ \boxed{} \\ \boxed{} \\ \boxed{} \end{array} \otimes \begin{array}{c} \boxed{} \\ \boxed{} \\ \boxed{} \\ \boxed{} \end{array} \otimes \begin{array}{c} \boxed{} \\ \boxed{} \\ \boxed{} \\ \boxed{} \end{array} = \left(\begin{array}{cc} \boxed{} & \boxed{} \\ \boxed{} & \boxed{} \end{array} \oplus \begin{array}{c} \boxed{} \\ \boxed{} \end{array} \right) \otimes \begin{array}{c} \boxed{} \\ \boxed{} \\ \boxed{} \\ \boxed{} \end{array} = \\
 \left(\begin{array}{cccc} \boxed{} & \boxed{} & \boxed{} & \boxed{} \end{array} \oplus 2 \begin{array}{cc} \boxed{} & \boxed{} \\ \boxed{} & \boxed{} \end{array} \oplus \begin{array}{c} \boxed{} \\ \boxed{} \\ \boxed{} \end{array} \right) \otimes \begin{array}{c} \boxed{} \\ \boxed{} \\ \boxed{} \\ \boxed{} \end{array} = \begin{array}{cccc} \boxed{} & \boxed{} & \boxed{} & \boxed{} \end{array} \oplus 3 \begin{array}{ccc} \boxed{} & \boxed{} & \boxed{} \\ \boxed{} & & \end{array} \oplus 2 \begin{array}{cc} \boxed{} & \boxed{} \\ \boxed{} & \boxed{} \end{array} \oplus 3 \begin{array}{cc} \boxed{} & \boxed{} \\ \boxed{} & \boxed{} \end{array} \oplus \begin{array}{c} \boxed{} \\ \boxed{} \\ \boxed{} \end{array}. \quad (28.18)$$

To decompose $T_{\mu_1\mu_2\mu_3\mu_4}$ into irreps which are linear combinations of the components of $T_{\mu_1\mu_2\mu_3\mu_4}$, one may follow the simple rules which we gave at the beginning. This can be done, but the expressions are long. It is simpler to use Young operators. We then get the following decomposition into irreps

$$T_{\mu_1\mu_2\mu_3\mu_4} = (Y^{\begin{array}{c} \boxed{} \\ \boxed{} \\ \boxed{} \\ \boxed{} \end{array}} T)_{\mu_1\mu_2\mu_3\mu_4} + (Y^{\begin{array}{cc} \boxed{} & \boxed{} \\ \boxed{} & \boxed{} \end{array}} T)_{\mu_1\mu_2\mu_3\mu_4} + (Y^{\begin{array}{ccc} \boxed{} & \boxed{} & \boxed{} \\ \boxed{} & & \end{array}} T)_{\mu_1\mu_2\mu_3\mu_4} + \cdots + (Y^{\begin{array}{cccc} \boxed{} & \boxed{} & \boxed{} & \boxed{} \end{array}} T)_{\mu_1\mu_2\mu_3\mu_4}. \quad (28.19)$$

Some explicit formulas are

1) $(Y^{\begin{array}{cccc} \boxed{} & \boxed{} & \boxed{} & \boxed{} \end{array}} T) = \frac{1}{24}(e + (12) + \cdots + (4321))T$. There are 24 terms³ and this is the completely symmetric one-dimensional irrep. The factor $\frac{1}{24}$ is needed in order that $Y^2 = Y$.

$$\begin{aligned}
 2) \quad (Y^{\begin{array}{cc} \boxed{} & \boxed{} \\ \boxed{} & \boxed{} \end{array}} T)_{\mu_1\mu_2\mu_3\mu_4} &= \left(Y \left(\begin{array}{cc} \boxed{1} & \boxed{2} \\ \boxed{3} & \\ \boxed{4} & \end{array} \right) T \right)_{\mu_1\mu_2\mu_3\mu_4} \\
 &= \frac{1}{8}(e - (13) - (14) - (34) + (134) + (143))(e + (12))T_{\mu_1\mu_2\mu_3\mu_4} \\
 &= \frac{1}{8} \begin{pmatrix} e + (12) - (13) - (14) - (34) - (123) - (124) \\ -(12)(34) + (134) + (1234) + (143) + (1243) \end{pmatrix} T_{\mu_1\mu_2\mu_3\mu_4} \\
 &= \frac{1}{8} (T_{\mu_1\mu_2\mu_3\mu_4} + T_{\mu_2\mu_1\mu_3\mu_4} - T_{\mu_3\mu_2\mu_1\mu_4} + \cdots + T_{\mu_2\mu_4\mu_1\mu_3}). \quad (28.20)
 \end{aligned}$$

There are 12 terms in this expression. We used that $(123)T_{\mu_1\mu_2\mu_3\mu_4} = T_{\mu_2\mu_3\mu_1\mu_4}$ (“replace μ_1 by μ_2 , μ_2 by μ_3 and μ_3 by μ_1 in $T_{\mu_1\mu_2\mu_3\mu_4}$ ”) and we read the products of cycles from right to left.

³In expressions such as $e + (12)$, one adds group elements. In groups one can only multiply group elements, but in a “group algebra” one can both multiply and take linear combinations.

As one last example, we construct the irrep corresponding to $\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}$

$$\begin{aligned}
(Y^{\boxplus, I} T)_{\mu_1 \mu_2 \mu_3 \mu_4} &= \left(Y \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix} \right) T \right)_{\mu_1 \mu_2 \mu_3 \mu_4} \\
&= \frac{1}{12} \left[(e - (13))(e - (24))(e + (12))(e + (34))T \right]_{\mu_1 \mu_2 \mu_3 \mu_4} \\
&= \frac{1}{12} \left[(e - (13))(e - (24)) \right] \left[T_{\mu_1 \mu_2 \mu_3 \mu_4} + T_{\mu_2 \mu_1 \mu_3 \mu_4} + T_{\mu_1 \mu_2 \mu_4 \mu_3} + T_{\mu_2 \mu_1 \mu_4 \mu_3} \right] \\
&= \frac{1}{12} \begin{bmatrix} T_{\mu_1 \mu_2 \mu_3 \mu_4} + T_{\mu_2 \mu_1 \mu_3 \mu_4} + T_{\mu_1 \mu_2 \mu_4 \mu_3} + T_{\mu_2 \mu_1 \mu_4 \mu_3} \\ -T_{\mu_3 \mu_2 \mu_1 \mu_4} - T_{\mu_2 \mu_3 \mu_1 \mu_4} - T_{\mu_3 \mu_2 \mu_4 \mu_1} - T_{\mu_2 \mu_3 \mu_4 \mu_1} \\ -T_{\mu_1 \mu_4 \mu_3 \mu_2} - T_{\mu_4 \mu_1 \mu_3 \mu_2} - T_{\mu_1 \mu_4 \mu_2 \mu_3} - T_{\mu_4 \mu_1 \mu_2 \mu_3} \\ +T_{\mu_3 \mu_4 \mu_1 \mu_2} + T_{\mu_4 \mu_3 \mu_1 \mu_2} + T_{\mu_3 \mu_4 \mu_2 \mu_1} + T_{\mu_4 \mu_3 \mu_2 \mu_1} \end{bmatrix}. \tag{28.21}
\end{aligned}$$

The factor $\frac{1}{12}$ is needed to obtain a projection operator ($Y^2 = Y$). (The easiest way to check this factor is to note that there are 16 terms in Y , but one finds the unit element only 12 times in Y^2 .)

The Young operators form a complete set of orthogonal projection operators. Denoting them by Y^J for (in this example) $J = 1, \dots, 10$, they satisfy

$$Y^J Y^K = 0 \text{ if } J \neq K; \quad Y^J Y^J = Y^J; \quad \sum_J Y^J = I. \tag{28.22}$$

They also give the irreps of the finite group $S(n)$ (the “symmetric group” of all $n!$ permutations of n objects), but that subject we do not discuss [2].

Let us now check whether the tensors obtained in (28.6) and (28.7) correspond to the states of an octet of $SU(3)$. We denote the states $|_1\rangle$, $|_2\rangle$ and $|_3\rangle$ of the **3** of $SU(3)$ by u, d and s , the usual notation for the three lightest quarks. Then the highest weight state of the decuplet **10** of $SU(3)$ is uuu , and by using the lowering operators $E_{-\alpha^1} = E_{12}^-$, $E_{-\alpha^2} = E_{23}^-$ and $E_{-\alpha^{13}}$ we get the rest of the decuplet. The lowering operators act as follows on the **3**,

$$\begin{array}{ccc}
& & E_{-\alpha^1} \\
d & \xleftarrow{\quad} & u \\
& \searrow \quad \swarrow & \\
& E_{-\alpha^2} \quad E_{-\alpha^{13}} & \\
& & s
\end{array} \tag{28.23}$$

and using this, beginning with the state uuu one finds for the decuplet

$$\begin{array}{ccc}
 & \xleftarrow{\frac{1}{\sqrt{3}}(udd+dud+ddu)} & \\
 ddd & \xleftarrow{\frac{1}{\sqrt{3}}(uud+udu+duu)} & uuu \\
 & \xleftarrow{\frac{1}{\sqrt{6}}(uds+usd+dus+dsu+sud+sdu)} & \\
 \frac{1}{\sqrt{3}}(dds+dsd+sdd) & \xleftarrow{\frac{1}{\sqrt{3}}(uus+usu+suu)} & \\
 & \xleftarrow{\frac{1}{\sqrt{3}}(uss+sus+ssu)} & \\
 & sss &
 \end{array}
 \quad (28.24)$$

Moving to the left from uuu along the α^1 chain, by acting once with $E_{-\alpha^1}$, we find three states, uud , udu , duu , the totally symmetric combination of which, $\frac{1}{\sqrt{3}}(uud+udu+duu)$, belongs to the decuplet, leaving a two-dimensional space orthogonal to it which should contain the highest-weight states of the two octets. There is a two-parameter freedom to choose these highest weight states of the octets, but a useful way is to start with states which are antisymmetric in 2 quarks⁴, for example $\frac{1}{\sqrt{2}}(ud-du)u$ and $\frac{1}{\sqrt{2}}u(ud-du)$. (The third combination, antisymmetric in the first and last quark, $\frac{1}{\sqrt{2}}(uud-duu)$, is not linearly independent but equal to the sum of the previous two states.) By acting with the lowering operators on the highest weight state of the first octet, we find the complete first octet as follows

$$\begin{array}{ccccc}
 & \frac{1}{\sqrt{2}}(ud-du)d & \xleftarrow{\quad} & \frac{1}{\sqrt{2}}(ud-du)u & \\
 & \frac{1}{\sqrt{2}}(ds-sd)d & \xleftarrow{\frac{1}{2}[(us-su)d+(ds-sd)u]} & \frac{1}{\sqrt{2}}(us-su)u & \\
 & \frac{1}{\sqrt{2}}(sd-ds)s & \xleftarrow{\quad} & \frac{1}{\sqrt{2}}(us-su)s &
 \end{array}
 \quad (28.25)$$

$I = 0$ state

The $I = 0$ state at the center should be annihilated by $E_{-\alpha^1}$ (or equivalently, be orthogonal to the $I = 1$ state in the center), and should also be orthogonal to the state in the center of the decuplet;

⁴The state $\frac{1}{\sqrt{2}}(ud-du)u$ is a linear combination of the states $T_{\mu_1\mu_2\mu_3}^{\square\square I}$ in (28.6) and $T_{\mu_1\mu_2\mu_3}^{\square\square II}$ in (28.7), namely $\frac{1}{3}(T_{112}^{\square\square I} + 2T_{112}^{\square\square II})$. Similarly, $u(ud-du)$ is equal to $\frac{1}{3}(T_{112}^{\square\square I} - T_{112}^{\square\square II})$.

this fixes it only as far as

$$\alpha(ud - du)s + \beta(us - su)d - \beta(ds - sd)u \quad (28.26)$$

However, using $E_{-\alpha^2}$ or $E_{-\alpha^{13}}$ one should find a linear combination of the two states in the center. For example, $E_{-\alpha^{13}}$ acting on $\frac{1}{\sqrt{2}}(ud - du)u$ produces a state proportional to $\frac{1}{\sqrt{2}}(sd - ds)u + \frac{1}{\sqrt{2}}(ud - du)s$, and this is only a linear combination of the state with $I = 1$ in the center and the state with $I = 0$ given in (28.26), if $\alpha = 2\beta$.

$$I = 0 \text{ state} = \frac{1}{\sqrt{12}}[2(ud - du)s + (us - su)d - (ds - sd)u] \quad (28.27)$$

Then $E_{-\alpha^2}$ acting on $\frac{1}{\sqrt{2}}(ud - du)d$ produces a linear combination of the state with $I = 0$ and $I = 1$ in the center. This fixes all states in the first octet.

We defined the states of the octet by acting with $E_{-\alpha^1}$ and $E_{-\alpha^2}$, starting with the highest weight state⁵. This yields a representation of $E_{-\alpha^1}$ and $E_{-\alpha^2}$ in terms of 8×8 matrices. However, one can also act with $E_{-\alpha^{13}}$ on states, for example acting on the state $\frac{1}{\sqrt{2}}(ud - du)d$ one obtains minus the state $\frac{1}{\sqrt{2}}(ds - sd)d$. This is not an inconsistency; it only means that in the 8×8 matrix representation of $E_{-\alpha^{13}}$ one finds an entry with -1 .

There is also the second octet which starts with the highest weight state $\frac{1}{\sqrt{2}}u(ud - du)$, and using the 3 lowering operators we can find the complete second octet as well. In all there are 27 states which fall into 4 multiplets of dimension **10**, **8**, **8**, **1**. This agrees with the number of 4 tensor multiplets in (28.8) with the same number of components as the number of corresponding states. We come now to the point we wanted to reach: to explicitly check that the tensor multiplets agree with the state multiplets. The tensors $T^{\square\square I}$ and $T^{\square\square II}$ have each 27 components, but most components vanish or are equal to other components, such that there are only 8 independent components for each. For example, starting with (28.7), and using the notation $\mu = 1 = u$, $\mu = 2 = d$ and $\mu = 3 = u$, we find from (28.7) an expression for T_{udu} , and similar expressions for T_{duu} and T_{uud}

$$T_{udu}^{\square\square I} = udu + udu - duu - duu \sim (ud - du)u; \quad T_{duu}^{\square\square I} = T_{udu}^{\square\square I}; \quad T_{uud}^{\square\square I} = 0. \quad (28.28)$$

So T_{udu} corresponds to the highest weight state of the first octet. Let us now see whether taking other values of μ^1, μ^2, μ^3 we do find the other states of the octet. For example, first T_{udd}

$$T_{udd}^{\square\square I} = udd + dud - 2ddu; \quad T_{dud}^{\square\square I} = 0; \quad T_{ddu}^{\square\square I} = -T_{udd}^{\square\square I}. \quad (28.29)$$

⁵Except the state at the lower right in (28.25) which was obtained by acting with $E_{-\alpha^{13}}$, but we obtain this same state if we act with $E_{-\alpha^2}$ on the two states in the center.

This is **not** a state of the multiplet. The same problem arises for $T^{\square II}$. Now one finds

$$\begin{aligned} T_{uud}^{\square II} &= 0; & T_{udu}^{\square II} &= 2udu - duu - uud; & T_{duu}^{\square II} &= -T_{udu}^{\square II} \\ T_{ddu}^{\square II} &= 0; & T_{dud}^{\square II} &= 2dud - udd - ddu; & T_{udd}^{\square II} &= -T_{dud}^{\square II} \end{aligned} \quad (28.30)$$

What is the solution of this conundrum?

The resolution is that one has constructed the wrong tensor multiplets. Rather than using the Young operator QP one should use⁶ PQ . So one should first antisymmetrize and then symmetrize the tensors. We leave as an exercise to start all over with these different multiplets, and check that they correspond precisely to the state multiplets.

Bibliography

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- [3] H. Georgi, *Lie Algebras in Particle Physics*, Avalon Publishing 1999.

⁶I thank Howard Georgi (private communication) who ran into the same problem and found the solution given in the text.

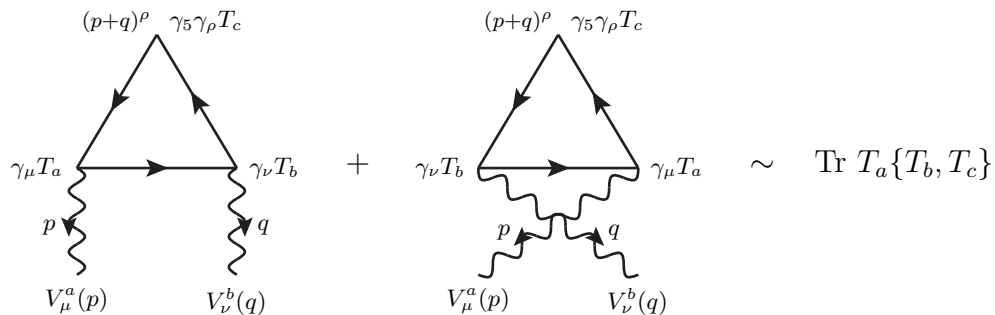
Chapter 29

When do anomalies cancel, and when should they cancel?

The study of anomalies in quantum field theory is intimately linked to group theory, more precisely to the properties of Lie algebras. In this section we consider chiral anomalies in four dimensions. We do not assume knowledge of quantum field theory.

The nonabelian triangle anomalies for the AVV and AAA case are proportional to the d -symbol, defined by $d_{abc}^{(R)} = -\frac{i}{2} \text{Tr} (T_a^{(R)} \{T_b^{(R)}, T_c^{(R)}\})$, where $T_a^{(R)}$ are the antihermitian matrices of a representation R of the Lie algebra. (The factor i is needed to make d_{abc} real if the generators T_a are antihermitian.) The argument goes as follows: the AVV anomaly of the first graph in the figure has the form

$$An \sim \epsilon^{\mu\nu\rho\sigma} p_\rho q_\sigma \text{Tr} T_a T_b T_c \quad (\text{omitting } V_\mu^a(p) \text{ and } V_\nu^b(q)) \quad (29.1)$$



and since $\epsilon^{\mu\nu\rho\sigma} p_\rho q_\sigma$ is symmetric under simultaneous exchange of $\mu \leftrightarrow \nu$ and $p \leftrightarrow q$, Bose symmetry requires to add the contribution of the second graph: the same factor $\epsilon^{\mu\nu\rho\sigma} p_\rho q_\sigma$ times $\text{Tr} T_a T_c T_b$. Hence we get as group factor the **anticommutator** $\text{tr} T_a \{T_b, T_c\}$.

There are two important general statements one can make about axial or chiral anomalies

- if the one-loop triangle anomalies cancel, also all other one-loop anomalies cancel (box and pentagon anomalies). This follows from the uniqueness of the Bardeen or the chiral consistent anomaly up to an overall constant.

- if the one loop anomalies cancel, all higher-loop anomalies must also cancel. The reason is that one can use a higher-derivative regularization scheme (adding $\mathcal{L} = \bar{\psi} D^\mu D_\mu \psi$ to the action) which makes 2-loop and higher-loop graphs finite, and finite graphs carry no anomalies.

We now turn to the chiral basis with left-handed fermions ψ_L and right-handed fermions ψ_R . The nonabelian triangle anomalies for the left-handed fermions are proportional to

$$d_{abc}^L = \frac{-i}{2} \text{Tr}(T_a^L \{T_b^L, T_c^L\}); \quad [T_a^L, T_b^L] = f_{ab}^L T_c^L \quad (29.2)$$

When do these triangle anomalies cancel? This is a purely group-theoretical problem [1]. One can distinguish between anomalies in the local gauge symmetry which in general lead to an inconsistent (nonunitary and/or nonrenormalizable) quantum field theory, and anomalies in axial vector currents which are the Noether currents of rigid symmetries. The latter are in some cases welcome because they explain certain experimental facts, such as the fact that neutral pions decay into two photons, without rendering the quantum field theory inconsistent.

If there are, in addition to left-handed fermions ψ_L with a gauge algebra with structure constants f_{ab}^L , also right-handed fermions ψ_R with a gauge algebra with structure constants $f_{ab}^{\hat{R}}$, there are also anomalies due to triangle loops with ψ_R , proportional to

$$d_{abc}^R = \frac{-i}{2} \text{Tr} \left(T_a^{\hat{R}} \{T_b^{\hat{R}}, T_c^{\hat{R}}\} \right); \quad [T_a^{\hat{R}}, T_b^{\hat{R}}] = f_{ab}^{\hat{R}} T_c^{\hat{R}} \quad (29.3)$$

(We denote the representation of the right-handed fermion by $\hat{R}$ in order to avoid confusion with the R of right-handedness.) One can always rewrite right-handed fermions as charge-conjugate left-handed fermions

$$\begin{aligned} \psi_L^c &= C^{-1} \bar{\psi}_R^T; \quad \left(\text{or } \psi_L^{c,\alpha} = (C^{-1})^{\alpha\beta} \bar{\psi}_{R,\beta} \text{ with } \bar{\psi}_{R,\beta} = (\psi^\dagger)_{\beta'} (i\gamma^0)^{\beta'\beta} \right) \\ \bar{\psi}_L^c &\equiv \psi_L^{c\dagger} i\gamma^0 = \bar{\psi}_R^* C i\gamma^0 = \psi_R^T (i\gamma^0)^* C i\gamma^0 = -\psi_R^T C \end{aligned} \quad (29.4)$$

(Sometimes one defines $\bar{\psi}_L^c = \psi_R^T C$, without minus sign; then $\psi_L^c = -C^{-1} \bar{\psi}_R^T$.) If the right-handed fermions belong to a representation $\hat{R}$ of the gauge group, these charge-conjugate fermions belong to the complex conjugate representation $\hat{R}^*$.¹ Because we can always take the generators of a compact Lie algebra to be antihermitian (if need be, by making a similarity transformation), it is clear that the anomaly of the charge-conjugate fermions in the representation $\hat{R}^*$ is minus the anomaly for left-handed fermions in the representation $\hat{R}$. That is obvious, because anomalies

¹The current $\bar{\psi}_R \gamma^\mu \psi_R$ can be rewritten as $-\psi_R^T (-C_- \gamma^\mu C_-^{-1}) \bar{\psi}_R^T = -\bar{\psi}_L^c \gamma^\mu \psi_L^c$ where we transposed the spinors and used that $C_- \gamma^\mu C_-^{-1} = -\gamma^{\mu,T}$. So the current of the conjugate spinors has an extra minus sign. The ψ_L^c couple to color matrices $T_a^T = -T_a^*$ instead of T_a due to the transposition of the spinors. The two minus signs cancel, so the charge-conjugate fermions couple the same way as the original fermions, but with matrices T_a^* . Because traces are invariant under transposition, the d symbol of the complex conjugate representation $\hat{R}^*$ is equal to minus the d symbol of the representation $\hat{R}$.

of left-handed and right-handed fermions in the same representation have opposite sign, as is clear from the fact that in the V-A basis the anomaly involves a trace with γ_5 , and the left-handed fermions are proportional to $\frac{1}{2}(1+\gamma_5)$ while the right-handed fermions are proportional to $\frac{1}{2}(1-\gamma_5)$.

Hence the total anomaly due to triangle graphs of left-handed fermions in representations $\hat{R}_L$, and right-handed fermions in representations $\hat{R}_R$, is proportional to

$$A_{abc} = d_{abc}^L - d_{abc}^R = \sum_{\hat{R}_L} \frac{-i}{2} \text{Tr} \left(T(\hat{R}_L)_a \left\{ T(\hat{R}_L)_b, T(\hat{R}_L)_c \right\} \right) - \text{idem for } \hat{R}_R \quad (29.5)$$

From this simple formula it is clear that anomalies will cancel either if $d_{abc}^L = d_{abc}^R$, or (a special but important case) if both d_{abc}^L and d_{abc}^R vanish. If we rewrite the right-handed fermions as charge-conjugate left-handed fermions, then anomalies cancel if the contributions from the various representations cancel in the sum

$$A_{abc} = \sum_{\hat{R}=\hat{R}_L, \hat{R}_R^*} \frac{-i}{2} \text{Tr} \left(T_a^{(\hat{R})}, \left\{ T_b^{(\hat{R})}, T_c^{(\hat{R})} \right\} \right) \quad (29.6)$$

It is clear that the symbols d_{abc} are totally symmetric. Since the trace is invariant under similarity transformations with group elements (recall that $g^{-1}T_a g$ lies again in the Lie algebra), the d_{abc} are also invariant tensors in the adjoint representation. We shall normalize the generators of simple Lie algebras by the condition $\text{Tr } T_a T_b = -\frac{1}{2}\delta_{ab}$ for the defining $N \times N$ matrix representation for $SO(N)$, $SU(N)$ and $USp(N)$.

Before we go on, let us check that there exist groups G and matrix representations R for which $d_{abc}(R)$ does not vanish. We repeat that we normalize the anti-hermitian generators T_a such that for the fundamental (defining) representation one has $\text{Tr } T_a(\text{def})T_b(\text{def}) = -\frac{1}{2}\delta_{ab}$.

- **$SU(2)$.** For $SU(2)$ one has $T_a(\text{def}) = -\frac{i}{2}\sigma_a$ where σ_a are the Pauli matrices, and since $\{\sigma_a, \sigma_b\} = 2\delta_{ab}$, d_{abc} vanishes for the **2** of $SU(2)$ and hence, as will become clear from the discussion how to compute d symbols for representations that are composed from several fundamental ones, all representations of $SU(2)$ are anomaly free.
- **$SU(3)$.** Consider the generators

$$T_3 = -\frac{i}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad T_8 = \frac{-i}{\sqrt{12}} \begin{pmatrix} 1 & & \\ & 1 & \\ & & -2 \end{pmatrix}$$

Since $d_{338} = -i \text{tr } T_3 T_3 T_8 \neq 0$, the **3** of $SU(3)$ is anomalous. Since one can always embed these generators T_3 and T_8 in the fundamental representation **N** of any $SU(N)$, it is clear that the fundamental representations of all $SU(N)$ groups for $N \geq 3$ are anomalous.

The d symbols are closely related to Casimir operators, which are operators which commute

with every generator. For any simple Lie algebra with rank r there are r Casimir operators. A Casimir operator which is built out of k generators is said to be of rank k . There is always the quadratic Casimir operator $C_2(G) = -g^{ab}T_aT_b$ where g^{ab} is the inverse of the Killing metric $g_{ab} \equiv f_{pa}{}^q f_{qb}{}^p$. (For semisimple groups, the Killing metric has an inverse.) So for $SU(2)$ which has rank 1, this is the only Casimir operator. Other Casimir operators are of the form

$$C_k(G) = d^{a_1 \dots a_k} T_{a_1} \dots T_{a_k} \quad (29.7)$$

where $d^{a_1 \dots a_k}$ are totally symmetric real irreducible invariant tensors. (By irreducible we mean that it is not the product of invariant tensors of lower rank or that it is not the direct sum of two or more Casimir operators.) Thus, our d symbols d_{abc} (or d^{abc} after raising indices which is possible in semisimple Lie algebras) are nonvanishing if and only if there exists a third rank Casimir operator.

If a Lie algebra has no cubic Casimir operator $C_3(G)$, then none of its representations can give rise to an anomaly. The corresponding groups are called safe groups. **Safe groups never lead to anomalies.** It is easy to find the safe groups. The ranks of the Casimir operators $C_k(G)$ for the various Lie algebras are given by the following list

$$\begin{aligned} A(n-1) &= SU(n) : C_2, C_3, \dots, C_n \\ B(n) &= SO(2n+1) : C_2, C_4, \dots, C_{2n} \\ C(n) &= Sp(2n) : C_2, C_4, \dots, C_{2n} \\ D(n) &= SO(2n) : C_2, C_4, \dots, C_{2n-2}, C_n \\ G_2 &: C_2, C_6 \\ F_4 &: C_2, C_6, C_8, C_{12} \\ E_6 &: C_2, C_5, C_6, C_8, C_9, C_{12} \\ E_7 &: C_2, C_6, C_8, C_{10}, C_{12}, C_{14}, C_{18} \\ E_8 &: C_2, C_8, C_{12}, C_{14}, C_{18}, C_{20}, C_{24}, C_{30} \end{aligned} \quad (29.8)$$

Hence, the only Lie algebras with anomalies are $SU(n)$ with $n \geq 3$ and $SO(6)$. But since the algebra of $SO(6)$ is isomorphic to the algebra of $SU(4)$, the appearance of $SO(6)$ in the list of unsafe algebras is already included in the list of $SU(n)$ algebras. Also since $SU(4)$ has a cubic Casimir operator C_3 , also $SO(6)$ has one, which explains the C_n in the list for $SO(2n)$.

In higher dimensions the nonabelian anomalies are proportional to higher-dimensional Casimir invariants. For example, in 10 dimensions the simplest Feynman diagram with an anomaly is the hexagon graph, and the corresponding group factor is the rank 6 Casimir operator. We shall soon show that in 4 dimensions only complex representations can carry anomalies, but the argument is based on the fact that in 4 dimensions one needs d_{abc} which is a trace over an odd number of generators. In 10 dimensions one can also have anomalies in real and pseudoreal representations.

Since invariant tensors are Clebsch-Gordan coupling constants for the product of irreducible representations, a nonvanishing tensor d_{abc} means that in the symmetric product of two adjoint

representations one finds the adjoint representation back. Since there is only one C_3 in the list of Casimir operators of $SU(n)$, **the symbol d_{abc} is unique** (up to an overall scale). One may prove this by using Young-tableau methods that show that one finds the adjoint representation indeed only once if one decomposes the symmetric product of two adjoint representations of $SU(n)$ into irreducible representations. For example, for $SU(3)$ one finds

$$\mathbf{8} \times \mathbf{8} = \mathbf{1} + \mathbf{8}_a + \mathbf{8}_s + \mathbf{10} + \mathbf{10}^* + \mathbf{27}$$

$$\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} \otimes \begin{array}{|c|c|} \hline a & a \\ \hline b & \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline & & a \\ \hline & a & \\ \hline & b & \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline & & a \\ \hline & b & \\ \hline a & & \\ \hline \end{array} \oplus \dots$$

The $\mathbf{27}$, $\mathbf{8}_s$ and $\mathbf{1}$ are symmetric, while the $\mathbf{10}$, $\mathbf{10}^*$ and $\mathbf{8}_a$ are antisymmetric. (The $\mathbf{8}_s$ corresponds to d_{abc} and the $\mathbf{8}_a$ to f_{abc} .)²

Even though the $SU(N)$ groups are not safe, some of their representations may be safe. Namely, d_{abc} may vanish for some representations while for other representations it may be nonvanishing. In fact, the following representations are safe:

(i) **Real representations.** More precisely representations for which all matrices T_a are real (or can be made real by a similarity transformation) and thus antisymmetric. It is clear that for these representations d_{abc} vanishes because $T_a^* = T_a$ but also $T_a^* = -T_a^T$ and the trace goes over into minus itself after transposition. (The matrix used in the similarity transformation need not itself be an element of the group.)

(ii) **Pseudoreal representations.** These are representations for which $T_a^* = ST_a S^{-1}$ for some matrix S (not necessarily in the Lie algebra or group), but one cannot make the T_a real by a similarity transformation (if one could, one would be dealing with a real representation).³ It is clear that the matrices T_a^* satisfy the same commutation relations as the T_a , and when $T_a^* = ST_a S^{-1}$ this is obviously the case (the relation $T_a^* = -ST_a S^{-1}$ is inconsistent with the commutation relations). Since T_a are antihermitian, $\text{Tr } T_a \{T_b, T_c\} = -\text{Tr} \{T_c^*, T_b^*\} T_a^*$, so when $T_a^* = ST_a S^{-1}$, the d_{abc} will vanish.

If a representation R is neither real nor pseudoreal, it is called complex. The unsafe groups are $SU(N)$ with $N \geq 3$; they have thus complex representations, for example the $\mathbf{N}$ which is unsafe. Not every complex representation has anomalies (for example the two irreducible spinor representations of $SO(10)$ are complex, but $SO(10)$ is safe because it has no third-order Casimir operator), but all complex representations of the unsafe groups $SU(N)$ carry anomalies. So, safe groups and safe representations of nonsafe groups are without anomalies. However, a third possibility may occur: the nonvanishing anomalies of several anomalous representations may sum

²Of course, the structure constants are a totally antisymmetric invariant tensor, hence one also always finds the adjoint representation at least once in the antisymmetric product of two adjoint representations of any Lie algebra.

³The classic example is the two-dimensional (defining) representation $\mathbf{2}$ of $SU(2)$. The generators are $-\frac{i}{2}\tau_a$ where τ_a are the Pauli matrices, and $S = \tau_2$ (or $\alpha\tau_2$ for any α). One cannot find a matrix S such that $S\tau_a S^{-1}$ is real for $a = 1, 2, 3$.

up to zero. To study this further, we must first determine how the d_{abc} of different representations are related to each other.

Since for simple groups d_{abc} is unique (if it exists) up to an overall scale, the d symbol of representation R is related to the d symbol of the defining representation R_{def} by

$$d_{abc}^{(R)} = d(R) d_{abc}^{(\text{def})} \quad (29.9)$$

where $d(R)$ is an integer. To compute the coefficients $d(R)$ for $SU(N)$, we take a diagonal traceless purely imaginary generator A of $SU(N)$ in the $N \times N$ fundamental representation (for example, the diagonal matrix with entries $(i, i, \dots, (1-N)i)$ but we shall not need its explicit form). Since every irreducible representation of $SU(N)$ can be obtained from the fundamental representation by tensoring and reducing, we see that the $d(R)$ are positive or negative **integers** (or vanish). The following rules are obvious but useful

- (1) For a direct sum of two representations R and R' one has $d(R \oplus R') = d(R) + d(R')$. To prove this note that $A^{R \oplus R'} = \begin{pmatrix} A^R & 0 \\ 0 & A^{R'} \end{pmatrix}$ and take the trace.
- (2) For a direct product of two traceless representations R and R' one has $d(R \otimes R') = \dim R' d(R) + \dim R d(R')$. This follows from $A^{R \otimes R'} = A^R \otimes I^{R'} + I^R \otimes A^{R'}$. Cross terms in the trace $\text{tr}(A^{R \otimes R'})^3$ cancel because $\text{tr} A^R = \text{tr} A^{R'} = 0$.
- (3) Under complex conjugation one has $d(R^*) = -d(R)$. This follows from $T_a^\dagger = -T_a^T$ as we already discussed.

Let us now compute the $d(R)$ for some frequently used irreducible representations of $SU(N)$.

Antisymmetric tensors t^{ij} : The components t^{ij} with $1 \leq i < j \leq N$ form a $\frac{1}{2}N(N-1)$ dimensional linear vector space in which the generators T_a act according to

$$\delta t^{ij} = (T_a)^i_{i'} t^{i'j} + (T_a)^j_{j'} t^{ij'} \quad (29.10)$$

where $(T_a)^i_j$ are the generators of the fundamental representation. The generators $\hat{A}$ are diagonal on the states t^{ij} with eigenvalues $(A_i^f + A_j^f)$. Hence, denoting the irreducible representation $t^{ij} = -t^{ji}$ by $f \wedge f$, we find for the generator A (we drop from now on the factor $-i$ in (29.1))

$$d_{AAA}^{f \wedge f} = \sum_{i < j} (A_i^f + A_j^f)^3 = \frac{1}{2} \left(\sum_{i,j} - \sum_{i=j} \right) (A_i^f + A_j^f)^3 \quad (29.11)$$

where A_i^f denotes the i -th diagonal entry of the generator A in the fundamental (defining) representation. Since $\sum_i A_i^{(f)} = 0$, one finds

$$d_{AAA}^{f \wedge f} = N \sum_i (A_i^f)^3 - \frac{1}{2} \sum_i (2A_i^f)^3 = (N-4) \sum_i (A_i^f)^3 \quad (29.12)$$

In other words

$$d^{(f \wedge f)} = (N - 4)d^{(f)} \quad (29.13)$$

As a check, note that for $SU(4)$ the anomaly cancels because the **6** representation $t^{ij} = -t^{ji}$ is real⁴ (it is the vector representation of $SO(6)$). Note also that d is an integer, but negative for $N = 3$. For $N = 3$, an antisymmetric tensor $t^{ij} = -t^{ji}$ transforms like $(v_i)^*$ since $t^{ij} = \epsilon^{ijk}(v_k)^*$. Thus the anomaly for t^{ij} should be $-d(f)$, which agrees with (29.13). For $N = 2$ $d_{abc}^{(f)}$ vanishes because $SU(2)$ is pseudoreal,⁵ and $\sum_i (A_i^f)^3$ in (29.11) indeed vanishes in that case because $A_1^f = -A_2^f$.

Similarly, for $R = f \wedge f \wedge f$ the value of $d(R)$ follows from a sum over $i < j < k$, and one finds

$$\begin{aligned} d_{AAA}^{f \wedge f \wedge f} &= \sum_{i < j < k} (A_i^f + A_j^f + A_k^f)^3 = \frac{1}{6} \left(\sum_{i,j,k} -3 \sum_{i=j,k} + 2 \sum_{i=j=k} \right) (A_i^f + A_j^f + A_k^f)^3 \\ &= \left(\frac{1}{2}N^2 - \frac{1}{2}(2^3 + 1)N + \frac{1}{3}3^3 \right) d(f) \\ d(f \wedge f \wedge f) &= \frac{1}{2}(N - 3)(N - 6)d(f) \end{aligned} \quad (29.14)$$

As a check note that the totally antisymmetric representation t^{ijk} is pseudoreal for $N = 6$ and a singlet for $N = 3$. (We prove the pseudoreality of t^{ijk} below, and for $SU(3)$ one has $t^{ijk} \sim \epsilon^{ijk}$, which is a singlet).

Symmetric tensors: The symmetric tensor $t^{ij} = t^{ji}$ forms a representation R which we denote by $f \vee f$. Then

$$\begin{aligned} d_{AAA}^{f \vee f} &= \sum_{i \leq j} (A_i^f + A_j^f)^3 = \frac{1}{2} \left(\sum_{i,j} + \sum_{i=j} \right) (A_i^f + A_j^f)^3 \\ &= N \sum_i (A_i^f)^3 + \frac{1}{2} \sum_i (2A_i^f)^3 = (N + 4) \sum_i (A_i^f)^3 \end{aligned} \quad (29.15)$$

Hence

$$d(f \vee f) = (N + 4)d(f) \quad (29.16)$$

As a check, note that

$$\begin{aligned} d(f \otimes f) &= d(f \wedge f) + d(f \vee f) = 2Nd(f) \\ \square \otimes \square &= \begin{array}{|c|} \hline \square \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \end{aligned} \quad (29.17)$$

⁴Actually, the situation is a bit more subtle for $SU(4)$. The complex representation t^{ij} is reducible, and can be decomposed into a selfdual and an anti selfdual part, $t^{ij} \pm \frac{1}{2}\epsilon^{ijkl}t_{kl}^*$, where $t_{kl}^* \equiv (t^{kl})^*$. For a selfdual representation satisfying $t^{ij} = \frac{1}{2}\epsilon^{ijkl}t_{kl}^*$, one can express t_{k4}^* in terms of t^{ij} with $1 \leq i, j \leq 3$, and the solution for t_{k4}^* transforms under $SU(4)$ the same way as t^{ij} (because for example $(A_3^f + A_4^f)^* = A_1^f + A_2^f$). Thus the irreducible representation of a selfdual tensor has only 3 complex components (the t^{ij}) and hence has 6 real components, just as the vector of $SO(6)$. However, t^{ij} with $1 \leq i, j \leq 3$ transform under $SU(4)$ into t^{k4} , and the selfdual representation is still 6-dimensional. The contributions of t^{ij} to the anomaly cancel those of t^{k4} , in agreement with (29.13).

⁵More precisely, all representations of $SU(2)$ are either real or pseudoreal, namely real if they are made from an even number of fundamental representations, and pseudoreal otherwise.

Similarly one finds for a totally symmetric t^{ijk}

$$d(f \vee f \vee f) = \frac{1}{2}(N+3)(N+6)d(f) \quad (29.18)$$

Mixed tensor products: For mixed tensors, the rules for direct sums and direct products of representations can be used. For example, for $N \geq 3$ the representation $t_k^{ij} = -t_k^{ji}$ is given by $(f \wedge f) \otimes f^* - f$. (Since t_k^{ij} should be traceless if it is to be irreducible, $t_k^{ij} \delta_i^k = 0$, one must subtract for $N \geq 3$ the fundamental representation once). One finds then

$$\begin{aligned} d(t_k^{ij}) &= d(f \wedge f) \dim f^* + \dim(f \wedge f) d(f^*) - d(f) \\ &= [(N-4)N + \frac{1}{2}N(N-1)(-1) - 1] d(f) \\ &= \frac{1}{2}(N^2 - 7N - 2)d(f) \text{ for } N \geq 3 \end{aligned} \quad (29.19)$$

which is, of course, always an integer. As a check note that for $N = 3$ the t_k^{ij} transform like the $t_{(kl)}$ (because $\epsilon^{ijl} t_{(kl)}$ transforms like t_k^{ij}). The results of (29.16) and (29.19) indeed agree with opposite signs for $N = 3$.

As another application, consider the representation in the form of a gun, $\begin{smallmatrix} \square & \square \\ \square & \end{smallmatrix}$. Using Young tableaux rules for the decomposition of a product of irreducible tensors into a sum of irreducible tensors, we find

$$\begin{aligned} d(f \otimes f \otimes f) &= d(f \vee f \vee f) + d(f \wedge f \wedge f) + 2d\left(\begin{smallmatrix} \square & \square \\ \square & \end{smallmatrix}\right) = 3N^2 d(f) \\ \square \otimes \square \otimes \square &= \begin{smallmatrix} \square & \square & \square \end{smallmatrix} \oplus \begin{smallmatrix} \square \\ \square \\ \square \end{smallmatrix} \oplus \begin{smallmatrix} \square & \square \\ \square & \end{smallmatrix} \end{aligned} \quad (29.20)$$

Using (29.14) and (29.18), it follows that $d\left(\begin{smallmatrix} \square & \square \\ \square & \end{smallmatrix}\right) = (N^2 - 9)d(f)$ which makes sense because for $N = 3$ this is the adjoint representation, which is of course real and thus has no anomaly.

We can also organize the calculation of $d\left(\begin{smallmatrix} \square & \square \\ \square & \end{smallmatrix}\right)$ differently by noting that $\begin{smallmatrix} \square & \square \\ \square & \end{smallmatrix}$ is equal to $t^{[ij],k}$ minus the totally antisymmetric part: $\begin{smallmatrix} \square & \square \\ \square & \end{smallmatrix} = (f \wedge f) \otimes f - f \wedge f \wedge f = \begin{smallmatrix} \square & \square \\ \square & \end{smallmatrix} \otimes \square - \begin{smallmatrix} \square \\ \square \\ \square \end{smallmatrix}$. Then

$$\begin{aligned} d\left(\begin{smallmatrix} \square & \square \\ \square & \end{smallmatrix}\right) &= d(f \wedge f) \dim f + \dim(f \wedge f) d(f) - d(f \wedge f \wedge f) \\ &= (N-4)Nd(f) + \frac{1}{2}N(N-1)d(f) - \frac{1}{2}(N-3)(N-6)d(f) \\ &= (N^2 - 9)d(f). \end{aligned} \quad (29.21)$$

We now study in more detail when representations are real, pseudoreal or complex. (Recall that a representation which is not real nor pseudoreal is called complex: it is not equivalent to its complex conjugate).

A real or pseudoreal representation R satisfies by definition $T_a^* = S T_a S^{-1}$. We always assume that R is irreducible. It follows from Schur's lemma that if such a matrix S exists, it is unique (up

to an overall scale). Furthermore, by taking the complex conjugate of this relation, we see that S^*S is proportional to the unit matrix. For compact groups we can without loss of generality take T_a to be unitary. For a unitary representation the condition can then be rewritten as

$$ST + T_a^T S = 0 \quad (29.22)$$

This shows that S is an invariant tensor in the representation $R \times R$.

$$\delta S_{ij} = S_{i'j}(T_a)^{i'}_i + S_{ij'}(T_a)^{j'}_j = 0 \quad (29.23)$$

Taking once more the transpose of $-(T_a)^T = ST_a S^{-1}$, we find that $S^{-1,T}S$ is proportional to the unit matrix. Hence S^T is proportional to S , and thus **S is either a symmetric or an antisymmetric invariant tensor**. Together with the information that $S^*S = \alpha I$, we conclude that $S^\dagger S = \beta I$, hence (after rescaling) **S is unitary**. We claim that **if and only if S is symmetric, then R is similar to a real representation, whereas if and only if S is antisymmetric, then R is pseudoreal** but not similar to a real representation. To prove these statements, first assume that R is equivalent to a real antisymmetric representation R'

$$T'_a = VT_a V^{-1}, (T'_a)^T = -T'_a; \quad T'_a \text{ real} \quad (29.24)$$

Then

$$\begin{aligned} (T'_a)^T &= V^{-1,T} T_a^T V^T = -V^{-1,T} T_a^* V^T \\ &= -(T'_a) = -VT_a V^{-1} \end{aligned} \quad (29.25)$$

It follows that $T_a^* = V^T V T_a V^{-1} V^{-1,T}$. Hence, $V^T V = S$ and S is symmetric. Conversely, if $T_a^* = ST_a S^{-1}$ and S is symmetric (and unitary, see above) we can take⁶ a unitary symmetric square root $\sqrt{S}$. Using $(\sqrt{S})^{-1} = (\sqrt{S})^*$ it follows that $T'_a = \sqrt{S} T_a (\sqrt{S})^{-1}$ is real.

Applying this knowledge to the 3-index antisymmetric tensor representation of $SU(6)$ denoted by the Young tableau $\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array}$, we conclude from the existence of an antisymmetric invariant tensor $S^{IJ} = \epsilon^{\mu\nu\rho\alpha\beta\gamma}$ (with $I = (\mu, \nu, \rho)$ and $J = (\alpha, \beta, \gamma)$) that this representation is pseudoreal.⁷ This also proves our earlier claim that the **6** of $SU(4)$ is real because $S^{IJ} = \epsilon^{\mu\nu\rho\sigma}$ (with $I = \mu\nu$ and $J = \rho\sigma$) is a symmetric invariant tensor.

The representations of $SU(2)$ are pseudoreal for odd spin, and real for even spin. This is clear for the fundamental **2** representation (because the Pauli matrices τ_a satisfy $(i\tau_a)^* = S(i\tau_a)S^{-1}$ where $S = \tau_2$ is indeed antisymmetric) and for the adjoint representation (because the latter is

⁶Decompose $S = A + iB$ where $A = \frac{1}{2}(S + S^*)$ and $B = \frac{-i}{2}(S - S^*)$. Since $S^* = S^{-1}$ (S is unitary), S^* commutes with S , so A and B are real, commuting, symmetric matrices which can be simultaneously diagonalized by an orthogonal matrix O as $S = Oe^{i\phi}O^T$ where ϕ is a real diagonal matrix. Then the square root of S is $Oe^{\frac{i}{2}\phi}O^T$ which is again symmetric and unitary.

⁷Read (29.22) backwards: $-T_a^T = T_a^* = ST_a S^{-1}$.

always real, namely $(T_a)^b{}_c = f^b{}_{ac}$. All representations of $Sp(2n)$ are pseudoreal or real because the fundamental representation is pseudoreal (it satisfies $\mathcal{O}T_a + T_a^T\mathcal{O} = 0$, hence $T_a^* = ST_aS^{-1}$ with $S = \mathcal{O}$, where the symplectic metric $\mathcal{O}$ is antisymmetric). The same holds for $SO(5)$ because it is equivalent to $Sp(4)$.

In fact, all tensor representations of $SO(n)$ are real, but the spinor representations need not be real. For even n there are two inequivalent fundamental spinor representations, but in odd dimensions only one. The reality properties of the fundamental spinor representations of $SO(n)$ are as follows

$$\begin{aligned} SO(8n) & : \text{both real} \\ SO(8n+1) & : \text{real} \\ SO(8n+2) & : \text{both complex} \\ SO(8n+3) & : \text{pseudoreal} \\ SO(8n+4) & : \text{both pseudoreal} \\ SO(8n+5) & : \text{pseudoreal} \\ SO(8n+6) & : \text{both complex} \\ SO(8n+7) & : \text{real} \end{aligned}$$

The properties of $SO(8n+p)$ do not depend on n , the so-called Bott-periodicity. For noncompact groups such as the Lorentz groups $SO(n-1, 1)$ the reality properties are different. We now explain these results by constructing simple explicit representations.

Since $SO(3) \sim SU(2)$ we understand the pseudoreality for $SO(8n+3)$. Further, $SO(4) = SO(3) \otimes SO(3)$, so also here the pseudoreality is clear. For $SO(5)$ we use that $SO(5) \sim Sp(4)$ where the fundamental representation $\mathbf{4}$ of $Sp(4)$ is pseudoreal.⁸ For $SO(6)$ the spinor representations are complex, as one may note from the isomorphy of $SO(6)$ and $SU(4)$. (The generators are the 15 antihermitian matrices $[\gamma^\mu, \gamma^\nu], i\gamma^\mu, \gamma^\mu\gamma^5, i\gamma^5$ of 4 Euclidean space dimensions). Thus $SO(8n+6)$ has complex spinor representations. For $SO(8n+7)$ with $n=0$ we use that there exists a purely imaginary 8×8 representation of the Dirac matrices $\Gamma_{(7)}^\mu$ in 7 space dimensions, hence the spinor representation of $SO(7)$ is real. Taking for 8 space dimensions $\Gamma_{(7)}^\mu \otimes \tau_2$ and $\mathbf{1} \otimes \tau_1$ as real symmetric Dirac matrices, it is also clear that both spinor representations of $SO(8)$ are real (and inequivalent. One denotes them by $\mathbf{8}_s$ and $\mathbf{8}_c$, and together with the vector representation $\mathbf{8}_v$ they satisfy triality relations). For 9 space dimensions one may then take as Dirac matrices $\Gamma_{(8)}^\mu$ and $\Gamma_{(8)}^9$ where $\Gamma_{(8)}^9 \equiv \Gamma_{(8)}^1 \dots \Gamma_{(8)}^8$ is real and symmetric and satisfies $(\Gamma_{(8)}^9)^2 = I$. Thus the spinor representation of $SO(9)$ is also real.⁹

The situation is different for $SO(10)$. As Dirac matrices in 10 Euclidean space dimensions we

⁸The generators of the spinor representation of $SO(5)$ are $[\Gamma^M, \Gamma^N]$ with hermitian $\Gamma^M = \{i\gamma^\mu\gamma^5, \gamma^5\}$ where γ^μ, γ^5 are hermitian Dirac matrices in 4 dimensions. The Lorentz generators are then $[\gamma^\mu, \gamma^\nu]$ and $i\gamma^\mu$. They satisfy $C\Gamma^{MN}C^{-1} = -(\Gamma^{MN})^T$ and $(\Gamma^{MN})^T = -(\Gamma^{MN})^*$. Thus $(\Gamma^{MN})^* = C\Gamma^{MN}C^{-1}$. Since the charge conjugation matrix C is antisymmetric in $d=5$, the spinor representation of $SO(5)$ is pseudoreal.

⁹The charge conjugation matrix C in $d=9$ satisfies $C\Gamma^\mu C^{-1} = \sigma\Gamma^{\mu,T}$ and is given by $C = I$ and $\sigma = +1$. In 10 dimensions the charge conjugation matrix is either $C_+ = I \otimes \tau_1$ or $C_- = I \otimes \tau_2$, where $C_\pm \Gamma^M C_\pm^{-1} = \pm(\Gamma^M)^T$ with $M = 1, \dots, 10$.

take $\Gamma_{(9)}^\mu \otimes \tau_1$ and $I_{8 \times 8} \otimes \tau_2$. The generators of the two spinor representations in $d = 10$ are then block-diagonal and given by $[\Gamma_{(9)}^\mu, \Gamma_{(9)}^\nu]$ and $\pm i\Gamma_{(9)}^\mu$. We denote them by L_I^{MN} and L_{II}^{MN} . They are complex conjugates of each other and antihermitian. The charge conjugation matrices C_+ and C_- , defined by $C_\pm \Gamma^M C_\pm^{-1} = \pm \Gamma^{M,T}$ with $M = 1, \dots, 10$ are block off-diagonal, and thus cannot play the role of S in the reality condition for the block-diagonal Lorentz generators. So, the charge conjugation matrices are not invariant tensors of either of the spinor representation, but, rather, C_+ and C_- couple the two spinor representations to a scalar. In fact, there is no matrix S such that $SL_I^{MN} S^{-1} = (L_I^{MN})^* = L_{II}^{MN}$ because for $L_I^{MN} = i\Gamma_{(9)}^\mu$ this would imply that $S\Gamma_{(9)}^\mu S^{-1} = -\Gamma_{(9)}^\mu$ but if this were true for $\mu = 1, \dots, 8$, it would be false for $\mu = 9$ (recall $\Gamma_{(9)}^9 = \Gamma_{(9)}^1 \dots \Gamma_{(9)}^8$). Thus the spinor representations of $SO(10)$ are complex. (For $SO(9, 1)$ the reality properties are, of course, very different. The two spinor representations are both real, as is clear by taking $I_{8 \times 8} \otimes i\tau_2$ instead of $I_{8 \times 8} \otimes \tau_2$ as tenth Dirac matrix.)

We recapitulate that if a group is unsafe, it must have at least one complex representation. However, not every complex representation leads to an anomaly (is unsafe): $SO(10)$ is safe but has complex spinor representations. The unsafe groups are $SU(N)$ with $N \geq 3$; the representations with Dynkin labels $(m_1, \dots, m_{N-1})$ are real if the sequence $m_1, \dots, m_{N-1}$ is equal to the sequence $m_{N-1}, \dots, m_1$ (if the corresponding Young tableaux is self-conjugate). The other representations are complex, and all complex representations of $SU(N)$ with $N \geq 3$ are unsafe.

As a first and very important application of these general results, we discuss the cancellation of chiral anomalies in the Standard Model. The cancellation takes place within each of the three families. The color group $SU(3)$ does not couple to chiral fermions, so it does not lead to anomalies. Triangle graphs with three $SU(2)$ gauge fields do not contain anomalies either since the fermions are in the **2** of $SU(2)$ which is pseudoreal. The only triangle graphs which could possibly contain anomalies are either graphs with three $U(1)$ gauge fields, or graphs with one $U(1)$ gauge field and two $SU(2)$ gauge fields. Since all fermions which couple to $SU(2)$ form doublets, the anomalies in triangles with one $U(1)$ gauge field and two $SU(2)$ gauge fields are proportional to $\sum e_j \text{Tr } T_a T_b \sim \sum e_j$, and these anomalies cancel because the sum of the hypercharges vanishes. (It is equal to the sum of the electric charges because the average electric charge in a multiplet is the hypercharge of that multiplet). In fact the sum of the (hyper)charges of quarks cancels separately, as does the sum of the (hyper)charges of leptons because there are as many charged left-handed particles as right-handed particles. The triangle graphs with three $U(1)$ gauge fields do not contain anomalies either because the electromagnetic group $U(1)$ only couples to nonchiral fermions. (The $U(1)$ gauge field of the $U(1)$ in $SU(2) \otimes U(1)$ is the hypercharge gauge field. It couples with the hypercharge e_Y , but also $\sum_j (e_Y^j)^3 = 0$ for each family: from the left-handed quarks $6(\frac{1}{6})^3$, from the right-handed quarks $-3(\frac{2}{3})^3 - 3(-\frac{1}{3})^3$, from left-handed leptons $2(-\frac{1}{2})^3$ and from the right-handed lepton $-(-1)^3$, adding up to zero.)

For another application we consider the grand unified $SU(5)$ theory. The fundamental representation of the generators of $SU(5)$ consists of 5×5 antihermitian matrices. One can embed the

group $SU(3) \times SU(2) \times U(1)$ of the Standard Model into $SU(5)$ in the following way.

$$SU(3) : \left(\begin{array}{c|c} -\frac{i}{2}\lambda_\alpha & 0 \\ \hline 0 & 0 \end{array} \right), \quad SU(2) : \left(\begin{array}{c|c} 0 & 0 \\ \hline 0 & -\frac{i}{2}\sigma_a \end{array} \right), \quad U(1) : \frac{i}{\sqrt{15}} \begin{pmatrix} 1 & & & & 0 \\ & 1 & & & \\ & & 1 & & \\ & & & -\frac{3}{2} & \\ 0 & & & & -\frac{3}{2} \end{pmatrix} \quad (29.26)$$

where λ_α ($\alpha = 1, \dots, 8$) are the eight hermitian 3×3 ‘‘Gell-Mann’’ matrices for $SU(3)$, σ_a ($a = 1, 2, 3$) are the three Pauli matrices, and the last matrix represents the hypercharge $U(1)$ generator. Clearly $SU(3)$, $SU(2)$ and $U(1)$ commute with each other. Consider now the 15 fermions of the first generation ($3 u_L, 3 d_L, 3 u_R, 3 d_R, e^-, \nu_e, e^+$). They form $SU(3) \times SU(2)$ multiplets of the following dimensions

$$\begin{pmatrix} u^a \\ d^a \end{pmatrix}_L = (3, 2); \quad u_R^a = (3, 1); \quad d_R^a = (3, 1); \quad \begin{pmatrix} \nu \\ e^- \end{pmatrix}_L = (1, 2) \quad (29.27)$$

The lowest-dimensional irreps (irreducible representations) of $SU(5)$ are

$$v^I = \mathbf{5}, \quad t^{IJ} = -t^{JI} = \mathbf{10}, \quad t^{IJ} = t^{JI} = \mathbf{15} \quad (I, J = 1, \dots, 5) \quad (29.28)$$

One cannot put all 15 fermions into the $\mathbf{15}$ because for example in the sector with $I, J = 1, \dots, 3$ in t^{IJ} one would find the $SU(3)$ representations $\mathbf{3} \times \mathbf{3} = \mathbf{6} + \mathbf{3}^*$, and there are no $\mathbf{6}$ in Standard Model. But it is possible to use the other two irreps. Writing the right-handed u_R^a and d_R^a as left-handed $u_{L,a}^c$ and $d_{L,a}^c$, consider

$$\mathbf{5}^* = \begin{pmatrix} d_1^c \\ d_2^c \\ d_3^c \\ e^- \\ -\nu_e \end{pmatrix}, \quad \mathbf{10} = \left(\begin{array}{c|cc} \epsilon^{ijk} u_k^c & u^a & d^a \\ \hline -u^{a,T} & 0 & e^c \\ -d^{a,T} & -e^c & 0 \end{array} \right) \quad (29.29)$$

It is clear that the d_a^c in the $\mathbf{5}^*$ transform as $(\mathbf{3}^*, \mathbf{1})$. The doublet $\begin{pmatrix} \nu_e^- \\ -\nu_e \end{pmatrix}$ transforms as $(\mathbf{1}, \mathbf{2}^*)$ because $\begin{pmatrix} \nu_e \\ e^- \end{pmatrix}$ transforms under $SU(2)$ as a vector $v^i = \mathbf{2}$, and $\epsilon_{ij} v^j$ transforms as $v_i^* = (v^i)^*$. (The $\mathbf{2}$ of $SU(2)$ is pseudoreal.) The $\mathbf{10}$ contains the following building blocks. In the upper-left corner we find $\epsilon^{ijk} u_k^c$ which transforms as an $SU(3)$ tensor $t^{ij} \sim t_k^* = \mathbf{3}^*$ and as an $SU(2)$ singlet. Clearly we can take $t_k^* = u_k^c$. The entries t^{i4} and t^{i5} transform as $(\mathbf{3}, \mathbf{2})$, so they correspond to (u^a, d^a) . And finally we have the 2×2 block in the lower-right part of t^{IJ} . It should transform as a singlet under $SU(3)$ and as $t^{ij} \sim \epsilon^{ij}$ under $SU(2)$, hence it should also be an $SU(2)$ singlet. This leads to the identification $t^{45} = -t^{54} = e_L^c$.

Because the electromagnetic field is part of $SU(2) \times U(1)$, one of the generators of $SU(5)$ should generate the electric charges of the particles. Denoting this charge in the defining representation by Q with diagonal entries q^I we have

$$\delta v_I^* = -(Q^T)_I{}^{I'} v_{I'}^*, \quad \delta t^{IJ} = (Q)^I{}_{I'} t^{I'J} + t^{IJ'} (Q^T)_{J'}{}^J \quad (29.30)$$

It is clear that

$$Q^I{}_J = \begin{pmatrix} -\frac{1}{3} & & & 0 \\ & -\frac{1}{3} & & \\ & & -\frac{1}{3} & \\ 0 & & & 1 \\ & & & & 0 \end{pmatrix} \quad (29.31)$$

yields the correct charges of the $\mathbf{5}^*$ and the $\mathbf{10}$. For example $Qt^{14} = (Q^1 + Q^4)t^{14} = \frac{2}{3}t^{14}$ which is the correct charge of the up-quark, and $Qv_4^* = -v_4^*$ which is correct for the electron.¹⁰

Consider now in the full $SU(5)$ theory the gauge couplings of the $SU(5)$ gauge fields $(W_\mu)^I{}_J$ to these 15 left-handed fermions. There are two triangle graphs which are potentially anomalous, one with the fermions from the $\mathbf{10}$ in the loop, and the other with the fermions from the $\mathbf{5}^*$ in the loop. Since for $SU(5)$ according to (29.13)

$$\begin{aligned} d(\mathbf{10}) &= (N-4)d(\mathbf{5}) = d(\mathbf{5}) \\ d(\mathbf{5}^*) &= -d(\mathbf{5}) \end{aligned} \quad (29.32)$$

it is clear that the anomalies cancel.

Another interesting grand unified model is $SO(10)$. The 15 fermions of one family, plus an extra hypothetical right-handed neutrino, form the $\mathbf{16}$ dimensional spinor representation of $SO(10)$. But because $SO(10)$ is safe there should be no anomaly due to these fermions. In fact, one can decompose $SO(10)$ representations with respect to the subgroup $SU(5)$, and then $\mathbf{16} \rightarrow \mathbf{10} + \mathbf{5}^* + \mathbf{1}$. (The spinor generators of $SO(10)$ are the 16×16 matrices $M^{\mu\nu} = \{\gamma^\mu \gamma^\nu, i\gamma^\mu\}$ with $1 \leq \mu < \nu \leq 10$, where γ^μ are the nine real 16×16 Dirac matrices for $SO(9)$.) This demonstrates that the $\mathbf{16}$ of $SO(10)$ has no $SU(5)$ anomalies; in fact, it does not even have $SO(10)$ anomalies as we already explained.

¹⁰If one considers the grand unified theory $SO(10)$, there is a natural subgroup $SU(5) \times U(1)$. If one allows the generator for the electromagnetic field to lie in $SO(10)$ but not in $SU(5)$, one obtains a theory called flipped $SU(5)$. In that theory $Q = Q(SU(5)) + X = c_3 T_3 + c_8 T_8 + c_{15} T_{15}$ and the particle assignments are [2]

$$\mathbf{10} = \begin{pmatrix} & & & u & d \\ & \epsilon^{ijk} d_k^c & & u & d \\ & & & u & d \\ -u & -u & -u & 0 & e \\ -d & -d & -d & e & 0 \end{pmatrix}, \quad \mathbf{5}^* = \begin{pmatrix} u \\ -e^c \\ \nu^c \end{pmatrix}$$

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Chapter 30

The Wigner-Eckart theorem

The Wigner-Eckart theorem deals with matrix elements of tensor operators between states which transform according to representations of a Lie algebra (LA). A tensor operator $V(\Lambda, \lambda)$ is a set of operators that transform as an irreducible representation of the LA

$$[T_a, V(\Lambda, \lambda)] = \sum_{\lambda'} \langle \Lambda \lambda' | T_a | \Lambda \lambda \rangle V(\Lambda, \lambda') \quad (30.1)$$

Here the T_a are the generators of the LA, Λ labels the irrep (for example, by giving its highest weight), and λ, λ' label the various states of the multiplet Λ . The states $|\Lambda \lambda\rangle$ and $\langle \Lambda \lambda|$ themselves transform similarly

$$\begin{aligned} T_a |\Lambda \lambda\rangle &= \sum_{\lambda'} \langle \Lambda \lambda' | T_a | \Lambda \lambda \rangle |\Lambda \lambda'\rangle \\ \langle \Lambda \lambda | T_a &= \sum_{\lambda'} \langle \Lambda \lambda | T_a | \Lambda \lambda' \rangle \langle \Lambda \lambda' | \end{aligned} \quad (30.2)$$

These latter two relations follow from the completeness relation $\sum_{\lambda'} |\Lambda \lambda'\rangle \langle \Lambda \lambda'| = I$.

In their pioneering work in Quantum Mechanics, Eckart (1930) considered the rotation group $SO(3)$, and Wigner (1940) considered general simple compact Lie groups. The theory for arbitrary compact groups proceeds along similar lines, as does the theory for finite groups, but the Wigner-Eckart theorem for noncompact groups is rather complicated and will not be discussed. We shall restrict ourselves to general compact simple LA's. Even for them, there are fundamental differences with $SO(3)$. Using angular momentum theory, we know that if one couples two spins j_1 and j_2 , the resulting spin j runs over the values $|j_1 - j_2| \leq j \leq j_1 + j_2$, but each spin j occurs only once. Already for $SU(3)$ this is in general no longer true. For example,

$$\mathbf{8} \times \mathbf{8} = \mathbf{27} + \mathbf{10} + \mathbf{10}^* + \mathbf{8} + \mathbf{8} + \mathbf{1} \quad \text{in } SU(3) \quad (30.3)$$

We see that the $\mathbf{8}$ occurs twice in the Clebsch-Gordan (CG) series.

The Wigner-Eckart (WE) theorem states that matrix elements of tensor operators factorize into a kinematical factor which does not depend on the dynamics (does not depend on the details

of V and the details of the bra and ket states, for example the radial dependence of wave functions) and a dynamical factor which does not depend on the kinematics (does not depend on $\lambda_1, \lambda_2, \lambda$, but only on $\Lambda_1, \Lambda_2, \Lambda$). The WE theorem for general tensor operators of compact simple LA's reads

Theorem: $\langle \Lambda_2, \lambda_2, \beta | V(\Lambda, \lambda) | \Lambda_1, \lambda_1, \alpha \rangle = \langle \Lambda_2 \lambda_2 | \Lambda \lambda; \Lambda_1 \lambda_1 \rangle \langle \Lambda_2 \beta | | V(\Lambda, \lambda) | | \Lambda_1 \alpha \rangle$.

Here α and β contain information about the states in addition to their transformation properties. The first factor on the right-hand side is independent of the physical system considered (independent of α and β); it is a CG coefficient for the coupling of two irreps to a third irrep and contains only information from group theory. The second factor, often called **reduced matrix element**, does depend on the physical details of the bra and ket state (α and β), and on the particular operator V . This second factor is a (complex) number that cannot be obtained from group theory alone. However, by picking a particular value for λ_2, λ and λ_1 , the evaluation of the left-hand side and the CG coefficients is often easy, and then the value of the reduced matrix element follows. Once it is known, all other matrix elements (all other choices of λ_2, λ and λ_1) follow from group theory.

We shall first discuss the WE theorem for $SO(3)$ since this is still the case where it is most often used. Then we shall generalize to general compact simple LA's.

$SO(3)$. We begin by recalling some angular momentum theory. The angular momentum operators are hermitian operators J_i with $i = 1, 2, 3$, satisfying $[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$. We set $\hbar = 1$ (or work with $\frac{J_i}{\hbar}$). Defining

$$J_{\pm} = J_1 \pm iJ_2 \quad (30.4)$$

we have for eigenstates $|j, m\rangle$ of J^2 and $J_z = J_3$

$$\begin{aligned} J^2|j, m\rangle &= j(j+1)|j, m\rangle; & J_z|j, m\rangle &= m|j, m\rangle; \\ J_+|j, m\rangle &= \sqrt{(j+m+1)(j-m)}|j, m+1\rangle; \\ J_-|j, m\rangle &= \sqrt{(j+m)(j-m+1)}|j, m-1\rangle. \end{aligned} \quad (30.5)$$

Tensor operators $V(J, M)$ with spin J satisfy by definition the following commutation relations

$$\begin{aligned} [J_z, V(J, M)] &= MV(J, M) \\ [J_+, V(J, M)] &= \sqrt{(J+M+1)(J-M)}V(J, M+1) \\ [J_-, V(J, M)] &= \sqrt{(J+M)(J-M+1)}V(J, M-1). \end{aligned} \quad (30.6)$$

Typical examples of tensor operators with $J = 1$ (vector operators) are the coordinates $\vec{r}$, the momenta $\vec{p}$, and the angular momenta $\vec{J}$. Using $\vec{J} = \vec{r} \times \vec{p}$ and $[r_i, p_j] = i\delta_{ij}$ one may check that they all satisfy (30.6). For example, for r^j one may start from $V(1, 0) = V_z = r^3 = z$, and then

use (30.6) to obtain $V(1, 1) = V_+ = -\frac{1}{\sqrt{2}}(x + iy)$ and $V(1, -1) = V_- = \frac{1}{\sqrt{2}}(x - iy)$. Other tensor operators are $r_i r_j - \frac{1}{3}\delta_{ij}r^2$ ($J = 2$) and the spin $\frac{1}{2}$ operators $\vec{\sigma}$ ($J = \frac{1}{2}$).

To obtain explicit expressions for the matrix elements of vector operators $V(1, m)$ with $-1 \leq m \leq 1$

$$\langle j_2, m_2, \beta | V(j, m) | j_1, m_1, \alpha \rangle, \quad (30.7)$$

we observe that $[J_+, V_+] = 0$. Consider the case with $j_1 = j$ and $j_2 = j + 1$ and the following matrix element

$$\langle j + 1, m + 1, \beta | J_+ V_+ | j, m - 1, \alpha \rangle - \langle j + 1, m + 1, \beta | V_+ J_+ | j, m - 1, \alpha \rangle = 0. \quad (30.8)$$

Using (30.5) we obtain

$$\begin{aligned} \sqrt{(j + m + 2)(j - m + 1)} \langle j + 1, m, \beta | V_+ | j, m - 1, \alpha \rangle \\ = \sqrt{(j + m)(j - m + 1)} \langle j + 1, m + 1, \beta | V_+ | j, m, \alpha \rangle \end{aligned} \quad (30.9)$$

Since both sides contain the same factor $\sqrt{j - m + 1}$, we may replace it by $\sqrt{j + m + 1}$, and this yields

$$\frac{\langle j + 1, m, \beta | V_+ | j, m - 1, \alpha \rangle}{\sqrt{(j + m)(j + m + 1)}} = \frac{\langle j + 1, m + 1, \beta | V_+ | j, m, \alpha \rangle}{\sqrt{(j + m + 2)(j + m + 1)}} = -\sigma(j, m, V, \alpha, \beta) \quad (30.10)$$

Observe now that $\sigma(j, m, V, \alpha, \beta)$ does not depend on m ! (Replacing m by $m - 1$ in the second factor yields the first factor.) Hence

$$\langle j + 1, m + 1, \beta | V_+ | j, m, \alpha \rangle = -\sigma(j, V, \alpha, \beta) \sqrt{(j + m + 2)(j + m + 1)} \quad (30.11)$$

The factor $\sqrt{(j + m + 2)(j + m + 1)}$ is proportional to the CG coefficient for the coupling of the angular momentum states $|1, 1\rangle$ and $|j, m\rangle$ to the state $|j + 1, m + 1\rangle$

$$\langle j + 1, m + 1 | 1, 1; j, m \rangle = a \sqrt{(j + m + 2)(j + m + 1)}. \quad (30.12)$$

To fix the coefficient a , one may take the case $m = j$ which yields $a^{-1} = \sqrt{(2j + 2)(2j + 1)}$. Absorbing this factor a^{-1} into σ yields the reduced matrix element, and we find the WE formula

$$\langle j + 1, m + 1, \beta | V_+ | j, m, \alpha \rangle = \langle j + 1, m + 1 | 1, 1; j, m \rangle \langle j + 1, \beta || V_+ || j, \alpha \rangle. \quad (30.13)$$

These matrix elements have $j_1 = j$ and $j_2 = j + 1$. Given the knowledge of these matrix elements, we can also obtain the same matrix elements of the operator V_z between states with $j_1 = j$ and $j_2 = j + 1$ by using $[J_-, V_+] = \sqrt{2}V_z$. Subsequently the same matrix elements for V_- can be obtained from $[J_-, V_z] = \sqrt{2}V_-$.

To obtain the matrix elements $\langle j, m_1, \beta | V(1, k) | j, m_1, \alpha \rangle$ with $j_1 = j_2 = j$, one may begin again

with $[J_+, V_+] = 0$, but the reduced matrix element will introduce a new constant. (In this case it turns out that one does not even need to replace a factor like $\sqrt{j - m + 1}$ by $\sqrt{j + m + 1}$.) Similarly $\langle j, m_2, \beta | V_- | j, m_1, \alpha \rangle$ yields yet another constant.

The final conclusion is that all matrix elements of a vector operator are parametrized by only 3 constants. For example, all matrix elements $\langle j_2, m_2, \beta | x_k | j_1, m_1, \alpha \rangle$ of the operators x_k are equal to the matrix elements $\langle j_2, m_2, \beta | J_k | j_1, m_1, \alpha \rangle$ of the operators J_k up to 3 constants.

Another application of the WE theorem is the selection rules $\Delta j = \pm 1, 0$ and no $0 \rightarrow 0$ transitions for the electric dipole radiation of atoms ($j_1 - j_2 = \pm 1$ and no transitions with $j_1 = j_2 = 0$. For atoms with a single valence electron, transitions with $j_1 = j_2 \neq 0$ are also forbidden by parity). The matrix element for electric dipole radiation is $\langle j_2, m_2 | \vec{x} | j_1, m_1 \rangle \cdot \vec{E}$ which is proportional to the CG coefficient $\langle j_2, m_2 | 1, k; j_1, m_1 \rangle$. The latter vanishes if j_2 is not equal to $j_1 \pm 1$ or j_1 , or if $j_1 = j_2 = 0$. This proves the selection rules.

As final application we discuss the Landé g -factor for an atom with a total orbital angular momentum $\vec{L}$ and total spin $\vec{S}$. We use eigenstates of $\vec{J} = \vec{L} + \vec{S}$. The energy in an external magnetic field is equal to $\vec{\mu} \cdot \vec{B}$ where $\vec{\mu} = \vec{\mu}_L + \vec{\mu}_S$ is the total magnetic moment of the atom, and $\vec{\mu}_L$ is the contribution of the orbital momenta of the electrons while $\vec{\mu}_S$ is the contribution of the spins. One has

$$\vec{\mu}_L = \frac{e}{2mc} \vec{L} \quad \text{but} \quad \vec{\mu}_S = \frac{e}{mc} \vec{S}. \quad (30.14)$$

Hence we must evaluate

$$\langle j, m | L_3 + 2S_3 | j, m \rangle = \langle j, m | J_3 + S_3 | j, m \rangle = gm \quad (30.15)$$

which defines g . The expectation value of $\vec{J}_3$ is easy since we have eigenstates of $\vec{J}$, but to evaluate the expectation value of S_3 we use the following chain of equalities

$$\frac{\langle j, m | \vec{S} \cdot \vec{J} | j, m \rangle}{\langle j, m | J^2 | j, m \rangle} = \frac{\langle j | |S \cdot J| | j \rangle}{\langle j | |J^2| | j \rangle} = \frac{\langle j | |S| | j \rangle}{\langle j | |J| | j \rangle} = \frac{\langle j, m | S_3 | j, m \rangle}{\langle j, m | J_3 | j, m \rangle}. \quad (30.16)$$

We used the WE theorem twice, and shall prove the second equality in a moment. From these equations we obtain

$$\langle j, m | S_3 | j, m \rangle = \frac{\langle j | |S \cdot J| | j \rangle}{\langle j | |J^2| | j \rangle} \times m, \quad (30.17)$$

since $\langle j, m | J_3 | j, m \rangle = m$. The remaining steps are standard in Quantum Mechanics: we square $\vec{L} = \vec{J} - \vec{S}$ to obtain $\vec{S} \cdot \vec{J} = \frac{1}{2} (S^2 + J^2 - L^2)$, which brings us to

$$\frac{\langle j | |S \cdot J| | j \rangle}{\langle j | |J^2| | j \rangle} = \frac{S(S+1) + J(J+1) - L(L+1)}{2J(J+1)} \quad (30.18)$$

and from here we obtain

$$g = 1 + \frac{S(S+1) + J(J+1) - L(L+1)}{2J(J+1)} \quad (30.19)$$

where the 1 came from $\langle j, m | J_3 | j, m \rangle$ in (30.15).

So let us prove that $\frac{\langle j || S \cdot J || j \rangle}{\langle j || J^2 || j \rangle}$ is equal to $\frac{\langle j || S || j \rangle}{\langle j || J || j \rangle}$. We begin by considering $\langle j, m | \vec{S} \cdot \vec{J} | j, m \rangle$ and insert a complete set of states. Writing $\vec{S} \cdot \vec{J} = \sum_j S_j J_j = -S_+ \hat{J}_- + S_3 \hat{J}_3 - S_- \hat{J}_+$ with $S_+ = -\frac{1}{\sqrt{2}}(S_1 + iS_2)$ and $\hat{J}_+ = -\frac{1}{\sqrt{2}}(J_1 + iJ_2)$, we obtain for the term $S_+ \hat{J}_-$

$$\begin{aligned} \langle j, m | S_+ \hat{J}_- | j, m \rangle &= \sum_{j', m'} \langle j, m | S_+ | j', m' \rangle \langle j', m' | \hat{J}_- | j, m \rangle \\ &= \langle j, m | S_+ | j, m-1 \rangle \langle j, m-1 | \hat{J}_- | j, m \rangle \end{aligned} \quad (30.20)$$

because $\hat{J}_-$ does not change j but only m . Using the WE theorem once more, we obtain

$$\langle j, m | 1, m=1 | j, m-1 \rangle \langle j || S || j \rangle \langle j, m-1 | 1, m=-1 | j, m \rangle \langle j || \hat{J} || j \rangle \quad (30.21)$$

The ratio $\frac{\langle j, m | S \cdot J | j, m \rangle}{\langle j, m | J^2 | j, m \rangle}$ becomes then a ratio of sum of 3 CG factors which yields unity, times the ratio of the reduced matrix elements where only $\frac{\langle j || S || j \rangle}{\langle j || J || j \rangle}$ remains. This proves the second step in (30.16).

General compact simple LA's. We now consider the general case of compact simple LA's. Their unitary irreps are all finite-dimensional ...

Chapter 31

Non-semisimple Lie algebras

Any finite-dimensional Lie algebra $\mathcal{A}$ is a semidirect sum of a solvable Lie algebra P and a semisimple Lie algebra S .

$$\begin{aligned} \mathcal{A} &= S \ltimes P \quad (\text{the Levi decomposition}) \\ [S, S] &\subset S; \quad [P, P] \subset P; \quad [S, P] \subset P. \end{aligned} \quad (31.1)$$

An example is the Poincaré algebra in $d = 3+1$ with generators P_μ (P_0, P_1, P_2, P_3 , the translations) and $M_{\mu\nu} = -M_{\nu\mu}$ ($M_{01}, \dots, M_{23}$, the Lorentz generators). Here P_μ form the solvable part of the algebra

$$\begin{aligned} [M_{\mu\nu}, M_{\rho\sigma}] &= \eta_{\nu\rho} M_{\mu\sigma} - \eta_{\mu\rho} M_{\nu\sigma} - \eta_{\nu\sigma} M_{\mu\rho} + \eta_{\mu\sigma} M_{\nu\rho} \\ [M_{\mu\nu}, P_\rho] &= \eta_{\nu\rho} P_\mu - \eta_{\mu\rho} P_\nu; \quad [P_\mu, P_\nu] = 0. \end{aligned} \quad (31.2)$$

A semisimple Lie algebra is a direct sum of simple Lie algebras

$$S = S_1 \oplus S_2 \oplus \dots \oplus S_n \quad (31.3)$$

A simple Lie algebra $\mathcal{A}$ contains no other ideals than $\{0\}$ and $\mathcal{A}$ and by definition $U(1)$ is not simple.

To define a **solvable Lie algebra** P we need the series of derived subalgebras $P, P^{(1)}, P^{(2)}, \dots$

$$[P, P] = P^{(1)}, \quad [P^{(1)}, P^{(1)}] = P^{(2)}, \dots [P^{(k)}, P^{(k)}] = P^{(k+1)}, \dots \quad (31.4)$$

If for some k we have $P^{(k+1)} = 0$ we have a solvable Lie algebra.

Example 1. The translation generators of the Poincaré algebra form a solvable Lie algebra ($P^{(1)} = 0$).

Example 2. The Euclidean group E_2 (the Poincaré group in two Euclidean dimensions) is the semidirect product $E_2 = T_2 \rtimes R_2$ where $T_2 = \{P_1, P_2\}$ contains the translation generators and

$R_2 = L_{12}$ is the rotation generator.

$$[P_1, P_2] = 0; \text{ and } [L_{12}, P_2] = P_1 \text{ but } [L_{12}, P_1] = -P_2. \quad (31.5)$$

The whole algebra of E_2 is solvable.

Example 3. The Euclidean group $E_3 = T_3 \ltimes R_3$ is not solvable by itself, but T_3 is solvable and $R_3 = SO(3)$ is semisimple (in fact, simple).

Example 4. The set of all upper triangular $n \times n$ matrices form a solvable Lie algebra. (In fact, Lie proved that any solvable Lie algebra is isomorphic to an upper triangular algebra.)

Theorem: A Lie subalgebra of a solvable algebra is solvable (obvious) and a homomorphism maps a solvable algebra to a solvable algebra.

The **radical** R of a Lie algebra is the maximal solvable ideal. There exists such a subalgebra; it is unique and contains all other solvable ideals. Thus Levi's theorem can be sharpened to: every Lie algebra is the direct sum of its radical and a semisimple Lie algebra. For example, the group

$$SU(3) \otimes SU(2) \otimes U(1) \quad (31.6)$$

of the standard model decomposes into the semisimple Lie algebra $SU(3) \otimes SU(2)$ and its radical $U(1)$.

The semisimple Lie algebras are completely classified, but much less is known about non-semisimple Lie algebras. We focus on a few examples.

Chapter 32

Infinite-dimensional Lie algebras

The finite-dimensional Lie algebras which we have been discussing so far have been used with great success to describe the rigid (*i.e.* constant parameter) internal and spacetime symmetries in physics. Even more important are the local (gauge) symmetries in physics, and they can be viewed as infinite-dimensional Lie algebras. A case in-between are the conformal (and superconformal) symmetries in 2-dimensional field theories (which describe strings and condensed matter systems); in these cases, most of the gauge symmetries have been fixed, but a residual infinite-dimensional rigid symmetry group remains. (It can also be interpreted as a one-dimensional gauge symmetry, namely diffeomorphisms of the unit circle in the complex plane.) It may be that there exist other infinite-dimensional Lie algebras which play a role in physics (in string theory they are ubiquitous) and for this reason it is important to understand the structure of infinite-dimensional extensions of the finite-dimensional semi-simple (and even non-semi-simple) Lie algebras we have discussed.

We present two ways to introduce these extensions. The first one is a mathematical approach, which starts from the Cartan matrix, the Chevalley basis for simple Lie algebras, and the Serre relations which restrict the multiple commutators of the generators in the Chevalley basis [1]. By relaxing the criteria which the Cartan matrices for simple finite-dimensional Lie algebras fulfill, one finds new algebras, which one then may try to classify and find representations thereof. The other approach to infinite-dimensional Lie algebras starts from particular models which have a clear physical and geometrical meaning, and then through studies of these models one is led to more general Lie algebras. String theory and two-dimensional conformal field theories provide models which can be viewed as maps from the circle S^1 in the complex plane into a (finite-dimensional, usually simple) Lie algebra, and this yields what physicists call current algebras and what mathematicians call loop algebras. One can add to these algebras an extra term which physicists call an anomaly and which mathematicians call a central charge.

Kac-Moody algebras [2] are a class of Lie algebras that can be described by generalized Cartan matrices. The Kac-Moody algebra can be reconstructed from the data in the generalized Cartan matrix uniquely, and different generalized Cartan matrices yield non-isomorphic Lie algebras. We define generalized Cartan matrices as certain square matrices with integer entries; diagonal

elements are required to be equal to two, off-diagonal zeros are symmetric with respect to the diagonal, and off-diagonal elements are negative or zero. It turns out that the Kac-Moody algebras fall into three classes:

Finite. The Cartan matrix is nonsingular and its symmetrization¹ is Euclidean (the Cartan matrix is positive definite). In this case, one obtains the Cartan matrices of finite-dimensional simple Lie algebras, *i.e.* the series $A-D$, F_4 , G_2 and E_6 , E_7 , E_8 . In commonly accepted (mathematical) terminology, these form a class of Kac-Moody algebras.

Affine. They have a Cartan matrix with precisely one zero eigenvalue (the Cartan matrix is positive semi-definite). This yields the twisted and untwisted affine Lie algebras appearing in conformal field theory. Conversely, all affine Lie algebras, twisted and untwisted, are Kac-Moody algebras.

Indefinite. This class contains the Lorentzian Kac-Moody algebras for which the Cartan matrix has negative eigenvalues (whether or not $\det A = 0$). Hyperbolic Lie algebras are a subclass of Lorentzian Kac-Moody algebras.

Any vector space of maps from the circle to a Lie algebra is again a Lie algebra g . It can be possibly extended by a central element, as is the case for affine Lie algebras to yield a Lie algebra $\hat{g}$. However, there are definitely (many) examples when $\hat{g}$ is not a Kac-Moody algebra. $\hat{g}$ is Kac-Moody, if and only if g is a finite-dimensional simple Lie algebra. For example, the maps from the circle into an affine Lie algebra (called toroidal Lie algebras) are not Kac-Moody algebras. On the other hand, the maps from the circle into a finite Lie algebra are Kac-Moody algebras, namely affine Lie algebras. Hyperbolic Lie algebras (which are indefinite Kac-Moody algebras) cannot be represented as central extensions of maps from circles.

There are many more Lie algebras than only affine Lie algebras, but we shall only briefly discuss these more general algebras.

32.1 The Cartan matrix, the Chevalley basis and Serre relations for simple LA

If $\vec{\alpha}^i$ for $i = 1, \dots, r$ denote the simple roots of a simple finite-dimensional Lie algebra (with r the rank of the Lie algebra), the Cartan matrix is defined by

$$A^{ij} = 2 \frac{\vec{\alpha}^i \cdot \vec{\alpha}^j}{\vec{\alpha}^i \cdot \vec{\alpha}^i} . \quad (32.1)$$

¹Any Cartan matrix A can be written as DS , where D is a nonsingular real diagonal matrix and S is a real symmetric matrix [3]. This decomposition is unique up to an overall scale of D . If $\det A \neq 0$ and S has Euclidean signature, one obtains one of the finite Lie algebras but if S has Lorentzian signature (whether or not $\det A = 0$), one obtains an infinite-dimensional Lie algebra.

From the properties of α -chains we know that the off-diagonal entries of A^{ij} are zero or negative integers. Let us consider 3 examples.

$SU(3)$. The simple roots are $\vec{\alpha}^1 = (1, 0)$ and $\vec{\alpha}^2 = \left(-\frac{1}{2}, \frac{1}{2}\sqrt{3}\right)$ if $H_1 = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ & 0 \end{pmatrix}$ and $H_2 = \frac{1}{\sqrt{12}} \begin{pmatrix} 1 & 1 \\ & -2 \end{pmatrix}$. Then the Cartan matrix and the Dynkin diagram are given by

$$A_{SU(3)}^{ij} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \quad \bigcirc \text{---} \bigcirc \quad (32.2)$$

$SO(5)$. The simple roots are $\vec{\alpha}^1 = (1, -1)$ and $\vec{\alpha}^2 = (0, 1)$. Of course, one gets the same Cartan matrix for $USp(4)$ whose simple roots are $(1, 0)$ and $(-1, 1)$.

$$A_{SO(5)}^{ij} = \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix} \quad \bigcirc \text{---} \bullet \quad (32.3)$$

G_2 . The roots of G_2 are the roots of $SU(3)$ and the weights $\vec{\mu}^i$ of $\mathbf{3}$ and $\vec{\mu}_i = -\vec{\mu}^i$ of $\mathbf{3}^*$ of $SU(3)$. Then the simple roots are $\vec{\alpha}^1 = \vec{\mu}^1 - \vec{\mu}^2 = (1, 0)$ and $\vec{\alpha}^2 = \vec{\mu}^2 = \left(-\frac{1}{2}, \frac{1}{2\sqrt{3}}\right)$. (With the convention that a root is positive if its last entry is positive and $\alpha^1 < \alpha^2$, the weight $\vec{\mu}^2$ is the second simple root.)

$$A_{G_2}^{ij} = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix} \quad \bigcirc \text{---} \bullet \quad (32.4)$$

From these three examples and from the definition in (32.1) we see that a Cartan matrix for simple finite-dimensional Lie algebras has the following properties

- Its entries are integers (follows from α -chains).
- Its diagonal entries are 2 (obvious from (32.1)).
- If $A^{ij} = 0$, then also $A^{ji} = 0$ (obvious from (32.1)).
- If $A^{ij} = -1$, then A^{ji} can be $-1, -2, -3$, but if $A^{ij} = -2$ then $A^{ji} = -1$, and if $A^{ij} = -3$ then $A^{ji} = -1$ (follows from α -chains; in particular $A^{ij}A^{ji}(\text{no sums}) = 0, 1, 2, 3, 4$).
- Because the simple roots are linearly independent, one always has $\det A^{ij} \neq 0$. Moreover, A can always be written as $A = DS$, with D a real diagonal matrix and S a real symmetric matrix. Then for finite-dimensional Lie algebras S is positive definite. (Proof: $D_{ii} = \frac{2}{\alpha^i \cdot \alpha^i}$ and $S = \alpha^i \cdot \alpha^j$ and $x_i(\alpha^i \cdot \alpha^j)x_j = (x_i\alpha^i) \cdot (\alpha^j x_j) > 0$.) **To obtain new (and infinite-dimensional) Lie algebras, the properties $\det A \neq 0$ and/or $S > 0$ will be relaxed.**

In 1955 Chevalley constructed a new basis for simple Lie algebras in which all structure constants are integers. This was merely an increase in elegance, and as Boltzmann has said, “Elegance

should be left to tailors and shoemakers". (However, Chevalley used this reformulation of Lie algebras to construct new finite groups, "groups of Lie type", in which the entries of the matrix elements of a representation of a Lie algebra are not real or complex numbers, but elements of a finite field, of the kind that arises in Galois theory. We shall not discuss this any further, but note that in 1983 the classification of all simple finite groups² was completed, with an error corrected in 2004.)

To obtain the Chevalley basis, let us recall the general form of a simple Lie algebra on the Cartan-Weyl basis

$$\begin{aligned} [H_I, E_{\pm\alpha}] &= \pm\alpha_I E_{\pm\alpha}; \quad [H_I, H_J] = 0 \\ [E_\alpha, E_{-\alpha}] &= \left(\alpha_I g^{IJ} H_J \right) g_{\alpha, -\alpha} \\ [E_\alpha, E_\beta] &= f_{\alpha\beta}^\gamma E_\gamma \text{ with } f_{\alpha\beta}^\gamma \neq 0 \text{ if and only if } \vec{\gamma} = \vec{\alpha} + \vec{\beta}. \end{aligned} \quad (32.5)$$

Here α, β and γ are any roots.

Chevalley took the following linear combinations in terms of the simple roots $\vec{\alpha}^i$

$$h^i = \frac{2\alpha_I^i g^{IJ} H_J}{\vec{\alpha}^i \cdot \vec{\alpha}^i}; \quad e_i = \frac{\sqrt{2} E_{\alpha^i}}{\sqrt{\vec{\alpha}^i \cdot \vec{\alpha}^i} \sqrt{g_{\alpha, -\alpha}}}; \quad f_i = \frac{\sqrt{2} E_{-\alpha^i}}{\sqrt{\vec{\alpha}^i \cdot \vec{\alpha}^i} \sqrt{g_{\alpha, -\alpha}}}. \quad (32.6)$$

These are only $3r$ of the generators of the Lie algebra, but we shall see how to construct the rest of the Lie algebra from them. One finds then the following commutation relations for these $3r$ generators

$$\begin{aligned} [h^i, e_j] &= A^{ij} e_j; \quad [h^i, f_j] = -A^{ij} f_j; \\ [e_i, f_j] &= \delta_{ij} h^j; \quad [h^i, h^j] = 0. \end{aligned} \quad (32.7)$$

The reasons that $[e_i, f_j]$ vanishes if $i \neq j$ is that it can only be nonvanishing if $\vec{\alpha}^i - \vec{\alpha}^j$ is a root, but for simple roots this is impossible. So far, the Chevalley basis looks somewhat simpler than the Cartan-Weyl basis, but this has been achieved by rescalings, and by itself this is not very impressive. But now comes the punchline: for any positive root β Chevalley defines

$$h_\beta = \frac{2\beta_I g^{IJ} H_J}{\vec{\beta} \cdot \vec{\beta}}; \quad e_\beta = \frac{\sqrt{2} E_\beta}{\sqrt{\vec{\beta} \cdot \vec{\beta}} \sqrt{g_{\beta, -\beta}}}; \quad f_\beta = \frac{\sqrt{2} E_{-\beta}}{\sqrt{\vec{\beta} \cdot \vec{\beta}} \sqrt{g_{\beta, -\beta}}}. \quad (32.8)$$

and finds then

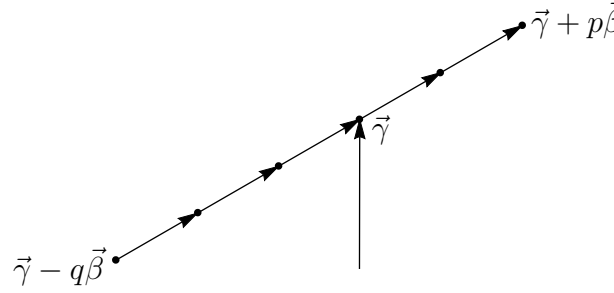
$$[h_\beta, e_\gamma] = \frac{2\vec{\beta} \cdot \vec{\gamma}}{\vec{\beta} \cdot \vec{\beta}} e_\gamma; \quad [h_\beta, f_\gamma] = -\frac{2\vec{\beta} \cdot \vec{\gamma}}{\vec{\beta} \cdot \vec{\beta}} f_\gamma \quad (32.9)$$

²The complete set of simple finite groups consists of 1) cyclic groups of prime order, 2) the alternating groups A_n for $n \geq 5$, 3) simple groups of Lie type (which contains the groups Chevalley found), and 4) the 26 sporadic groups (the largest of which is called the monster group).

which generalize (32.7), and the remarkable formulas

$$\begin{aligned} [e_\beta, e_\gamma] &= \pm(q+1)e_{\beta+\gamma} & \text{if } \vec{\beta} + \vec{\gamma} \text{ is a root} \\ [f_\beta, f_\gamma] &= \pm(q+1)f_{\beta+\gamma} & \text{if } \vec{\beta} + \vec{\gamma} \text{ is a root} \end{aligned} \quad (32.10)$$

where $\frac{2\beta \cdot \gamma}{\beta \cdot \beta}$ are integers, and q is the greatest positive integer such that $\vec{\gamma} - q\vec{\beta}$ is a root (the lower end of the β -chain containing γ).

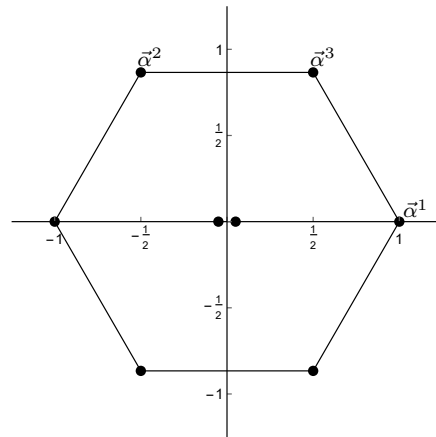


(32.11)

The signs in front of the factor $(q+1)$ must be fixed by some suitable convention, and we shall not go into the consistency conditions such sign conventions must satisfy. We fix them in the example of $SU(3)$ which we now consider.

$SU(3)$. For $SU(3)$ the commutation relations in the Chevalley basis read as follows. The simple roots α^i are $\vec{\alpha}^1 = (1, 0)$ and $\vec{\alpha}^2 = (-\frac{1}{2}, \frac{1}{2}\sqrt{3})$. Denote e_i and f_i by $e_{\pm i}$ to simplify the notation. Then from (32.7) we obtain

$$\begin{aligned} [h^1, e_{\pm 1}] &= \pm 2e_{\pm 1}; & [h^2, e_{\pm 2}] &= \pm 2e_{\pm 2}; \\ [h^1, e_{\pm 2}] &= \mp e_{\pm 2} \quad (A^{12} = -1); & [h^2, e_{\pm 1}] &= \mp e_{\pm 1} \quad (A^{21} = -1); \\ [e_1, e_{-1}] &= h^1; & [e_2, e_{-2}] &= h^2; & [e_1, e_{-2}] &= [e_2, e_{-1}] = 0. \end{aligned} \quad (32.12)$$



In the defining representation one finds from (32.6) that $h^1 = \begin{pmatrix} 1 & & \\ & -1 & \\ & & 0 \end{pmatrix}$; $h^2 = \begin{pmatrix} 0 & & \\ & 1 & \\ & & -1 \end{pmatrix}$; $e_1 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $e_{-1} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and $e_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, $e_{-2} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. (Recall that $\alpha^i \cdot \alpha^i = \alpha_I^i g^{IJ} \alpha_J^i = 2$ since

$g_{IJ} = \frac{1}{2}\delta_{IJ}$ while $g_{\alpha,-\alpha} = \text{tr } E_{\alpha}E_{-\alpha} = 1$.) Further $[e_{\pm 1}, e_{\pm 2}]$ yields a new generator which we call $\pm e_{\pm 3}$

$$[e_{\pm 1}, e_{\pm 2}] = \pm e_{\pm 3} \quad (q = 0 \text{ since } \vec{\alpha}^1 \text{ and } \vec{\alpha}^2 \text{ are simple roots}) \quad (32.13)$$

The signs in $[e_{\pm 1}, e_{\pm 2}] = \pm e_{\pm 3}$ must be fixed by a convention, and we have chosen them such that in the defining representation $e_3 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $e_{-3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$. So at this point we have found that the commutator $[e_{\pm 1}, e_{\pm 2}]$ does not vanish, and it yields a new generator which we call $e_{\pm 3}$. We shall now investigate whether there are any further commutators of e 's that do not vanish. Each time we find such a nonvanishing commutator we view it as a new generator.

One finds the following commutators of $e_{\mp 3}$ with $e_{\pm 1}$ and $e_{\pm 2}$ (the commutators $[e_{\pm 1}, e_{\pm 3}]$ and $[e_{\pm 2}, e_{\pm 3}]$ will be analyzed later)

$[e_{\pm 1}, e_{\mp 3}] = \mp e_{\mp 2}$

The e_1 chain acting on the weight $-\vec{\alpha}_3$ yields minus the weight $-\alpha_2$, which agrees with $-(1+q) = -1$ in (32.10).

(32.14)

$[e_{\pm 2}, e_{\mp 3}] = \pm e_{\mp 1}$

The e_2 chain acting on the weight $-\vec{\alpha}_3$ yields plus the weight $-\alpha_1$, which agrees with (32.10) if one uses $-(1+q) = -1$.

(32.15)

(The signs in the last two commutators follow from $[e_1, e_2] = e_3$ and can be checked by the Jacobi identities,³ or directly from the 3×3 matrix representation.) If one next evaluates $[e_{\pm 1}, e_{\pm 3}] = [e_{\pm 1}, [e_{\pm 1}, e_{\pm 2}]]$ and $[e_{\pm 2}, e_{\pm 3}] = -[e_{\pm 2}, [e_{\pm 2}, e_{\pm 1}]]$, by using the 3×3 matrices given before, one finds that they vanish. Thus the commutators in (32.14) and (32.15) are the only two nontrivial double commutators with simple roots and we see that in $SU(3)$ all triple and higher commutators vanish.

One would like a relation that only depends on the Cartan matrix A^{ij} to prove that the double commutators $[e_{\pm 1}, [e_{\pm 1}, e_{\pm 2}]]$ and $[e_{\pm 2}, [e_{\pm 2}, e_{\pm 1}]]$ vanish. Such a rule was found by Serre who proved in 1966 the following relations for multiple commutators of the e_i and f_i in (32.6)

$$\underbrace{[e_i, [e_i, \dots [e_i, [e_i, e_j]]]]}_{1-A^{ij} \text{ brackets}} \equiv (\text{adj } e_i)^{1-A^{ij}} e_j = 0 \quad i \neq j. \quad (32.16)$$

$$\underbrace{[f_i, [f_i, \dots [f_i, [f_i, f_j]]]]}_{1-A^{ij} \text{ brackets}} = (\text{adj } f_i)^{1-A^{ij}} = 0 \quad (32.17)$$

³The Jacobi identities applied to $[e_{+1}, e_{-3}] = [e_{+1}, -[e_{-1}, e_{-2}]]$ yield $[e_{-2}, [e_{+1}, e_{-1}]] + [e_{-1}, [e_{-2}, e_{+1}]]$. The first term yields $[e_{-2}, h^1] = A^{12}e_{-2} = -e_{-2}$, while the second term vanishes since $[e_{-2}, e_1] = 0$ ($[e_i, f_j] = \delta_{ij}h^j$).

For example, for $SU(3)$, $A^{12} = -1$, hence $[e_1, [e_1, e_2]]$ and $[e_2, [e_2, e_1]]$ vanish, as we indeed found above. Thus the whole Lie algebra of $SU(3)$ has been constructed using only the information supplied by the Cartan matrix.

G_2 . As a second example we consider G_2 with $\vec{\alpha}^1 = (1, 0)$ and $\vec{\alpha}^2 = (-\frac{1}{2}, \frac{1}{2\sqrt{3}})$, and $A^{ij} = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$. We get

$$[e_2, [e_2, [e_2, [e_2, e_1]]]] = 0 \quad \text{and} \quad [e_1, [e_1, e_2]] = 0. \quad (32.18)$$

Of the $\frac{1}{2}(14-2) = 6$ positive roots of G_2 , the two simple roots correspond to e_1 and e_2 . The other four come from the multiple commutators of e_1 and e_2 that are still allowed by (32.18)

$$[e_1, e_2]; \quad [e_2, [e_2, e_1]]; \quad [e_2, [e_2, [e_2, e_1]]]; \quad (32.19)$$

$$[e_1, [e_2, [e_1, e_2]]]. \quad (32.20)$$

Those in (32.19) are nonvanishing according to the Serre relations, and then the only nonvanishing mixed triple commutator is the one in (32.20). (The other mixed triple commutator $[e_2, [e_1, [e_1, e_2]]]$, vanishes according to (32.18).) They are all linearly independent because their weights (sums of $\vec{\alpha}^1$ and $\vec{\alpha}^2$) are different and nonzero. All fourfold commutators in G_2 vanish

$$[e_1, [e_2, [e_2, [e_2, e_1]]]] = 0; \quad [e_2, [e_1, [e_2, [e_1, e_2]]]] = 0; \quad [e_1, [e_1, [e_2, [e_1, e_2]]]] = 0. \quad (32.21)$$

This one may show by using (repeatedly) the Jacobi identities. So, also in the case of G_2 we can construct the complete Lie algebra by using only the Cartan matrix.

32.2 Affine Kac-Moody algebras from conformal field theories

We could now drop some of the restrictions on the Cartan matrix and in this way start the investigation of Kac-Moody algebras. Instead, we shall follow a more physical approach, and study currents in the conformal field theory for string theory and condensed matter theory. The currents of (super)conformal field theories in two Euclidean dimensions with coordinates $z = x + iy$ and $\bar{z} = x - iy$, give rise to infinite-dimensional Lie algebras which are of affine Kac-Moody type [1]. Consider the current

$$j_a(z) = \bar{\psi}_i(z) (T_a)^i{}_j \psi^j(z). \quad (32.22)$$

The $\psi^j(z)$ and $\bar{\psi}_i(z)$ are one-component spinors and the indices j and i belong to the $\mathbf{N}$ and $\mathbf{N}^*$ representation of $SU(N)$. The Dirac equation for ψ and $\bar{\psi}$ reads $\partial_{\bar{z}}\psi^i = 0$, $\partial_z\bar{\psi}_j = 0$ so that on-shell both ψ^i and $\bar{\psi}_j$ depend only on z . From canonical quantization one finds for the operator product expansion (OPE) of $\psi^j(z)$ and $\bar{\psi}_i(w)$

$$\psi^j(z)\bar{\psi}_i(w) = \delta_j^i \frac{1}{z-w} + \cdots \quad (32.23)$$

Then it is straightforward to derive the OPE (operator product expansion) for the currents

$$\begin{aligned}\mathcal{T}j_a(z)j_b(w) &= -\frac{\text{tr } T_a T_b}{(z-w)^2} + \frac{\bar{\psi}_i(z) (T_a T_b)^i{}_l \psi^l(w)}{z-w} - \frac{\bar{\psi}_k(w) (T_b T_a)^k{}_j \psi^j(z)}{z-w} + \dots \\ &= \frac{\frac{1}{2}\delta_{ab}}{(z-w)^2} + f_{ab}{}^c \frac{j_c(w)}{z-w} + \dots\end{aligned}\quad (32.24)$$

where $\mathcal{T}$ denotes radial ordering (a kind of time ordering in Euclidean space). We normalized the generators T_a in the defining representation of $SU(N)$ by $\text{tr } T_a T_b = -\frac{1}{2}\delta_{ab}$. Expanding $j_a(z) = \sum_{n=-\infty}^{\infty} j_{an} z^{-n-1}$, and using contour integrals in a way that is standard in conformal field theory, one finds then the following infinite-dimensional Lie algebra

$$[j_{am}, j_{bn}] = f_{ab}{}^c j_{c,m+n} + \delta_{ab} \delta_{m+n,0} m k I. \quad (32.25)$$

The real number k is the value of the central charge, and the central charge is proportional to the unit operator I which is to be considered as one of the generators. For the current in terms of fermions $k = \frac{1}{2}$, but for other models other values are possible.

In addition to the internal currents $j_a(z)$, also the stress tensor $T(z)$ plays an important role in (super)conformal field theory. It is natural to add a stress tensor to the currents $j_a(z)$ because one can construct a particular stress tensor from the currents (the so-called Sugawara construction [4]). Expanding $T(z) = \sum_{m=-\infty}^{\infty} L_m z^{-m-2}$, one finds the Virasoro algebra

$$[L_m, L_n] = (m-n)L_{m+n} + \delta_{m+n,0} (m^3 - m) \frac{c}{12} I, \quad (32.26)$$

where c depends on the fields from which $T(z)$ is constructed. Physicists call c an anomaly and then c is proportional to $\hbar$ (although one usually sets $\hbar = 1$), whereas mathematicians call $\frac{c}{12} I$ a central extension.

The currents $j^a(z)$ and $T(z)$ are analytic in z except at $z = 0$ and $|z| = \infty$. Thus we do not lose information if we view them as defined on the circle $|z| = 1$. Then the algebra of the j_{am} is a centrally extended loop algebra of a finite Lie algebra. We shall soon see that it contains simple roots such that there is precisely one null vector of the generalized Cartan matrix. This proves that the centrally extended loop algebra of a simple Lie algebra is an affine Kac-Moody algebra.

In mathematics, one defines loop groups as maps from the circle S^1 into an ordinary Lie group G

$$S^1 \xrightarrow[g(z) \in \Gamma]{} G. \quad (32.27)$$

The points of S^1 are defined by $\{z \in \mathbb{C} : |z| = 1\}$, and every map $g(z)$ from S^1 into G is an element of the Lie group G . The infinite set of all maps $g(z)$ is called the loop group, and it is denoted by Γ . The product of two group elements (two maps) in Γ is defined to be a product at the same point z for all z

$$g_1 g_2 \in \Gamma \text{ where } (g_1 g_2)(z) = g_1(z) g_2(z). \quad (32.28)$$

One can write group elements of a compact Lie group as exponents of the generators: $g = \exp(\lambda^a T_a)$. For loop groups λ^a depends on z , and expanding $\lambda^a(z)$ into a Laurent series as $\lambda^a(z) = \sum_{n=-\infty}^{\infty} \lambda^{an} z^{-n}$, we can write $g(z)$ in two ways

$$g(z) = e^{\lambda^a(z) T_a} = e^{\lambda^{an} T_{an}}, \quad T_{an} = T_a z^{-n}, \quad (32.29)$$

where $T_{an} = T_a z^{-n}$ is the coordinate representation of the generators T_{an} of the loop group. We can also define a coordinate representation of the Virasoro generators

$$L_m = -z^{m+1} \partial_z \quad (32.30)$$

and the classical Lie algebra (*i.e.* without central extension) of the L_m is sometimes called the “Witt algebra” (not de Wit, or DeWitt).

From these coordinate representations one can abstract the following commutation relations

$$\hat{\mu}_0 : [T_{am}, T_{bn}] = f_{ab}^c T_{c,m+n} \quad (32.31)$$

$$\hat{\nu}_0 : [L_m, L_n] = (m - n) L_{m+n} \quad (32.32)$$

$$[L_m, T_{an}] = n T_{a,m+n}. \quad (32.33)$$

The infinite-dimensional Lie algebra $\mathcal{A}_0$ is then a semidirect sum of the loop algebra $\hat{\mu}_0$ and the Witt algebra $\hat{\nu}_0$

$$\mathcal{A}_0 = \hat{\mu}_0 \oplus \hat{\nu}_0.$$

At the quantum level an ordinary semisimple Lie algebra does not admit a central extension, but infinite-dimensional Lie algebras often do have such extensions. The most general central extension of a loop algebra contains only one central element, kI , just as we found above, and the central extension of the Witt algebra, c , was also given above.

For each simple finite-dimensional Lie algebra one can construct an extended Dynkin diagram by adding to the set of simple roots $\vec{\alpha}^1, \dots, \vec{\alpha}^r$ a new “root” $\vec{\alpha}^0 = -\vec{\alpha}_{\text{highest}}$, where $\vec{\alpha}_{\text{highest}}$ is the highest (most positive) root of the Lie algebra. If one then applies the rules for constructing a Cartan matrix, one finds a singular matrix because the set $\vec{\alpha}^0, \vec{\alpha}^1, \dots, \vec{\alpha}^r$ is linearly dependent. It is also clear that this extended Cartan matrix has only one null vector because without $\vec{\alpha}^0$ the matrix A^{ij} is non-singular. Hence, we enter the area of affine Lie algebras. We shall work out this approach, although one can also consider the most general extended Cartan matrix, only requiring that its rank is still r . For the general case one may consult [5], whereas our approach follows [6] where however only the case for $SU(2)$ is discussed. Let us start with $SU(2)$ and $SU(3)$.

For $SU(2)$ there is only one root, hence $\vec{\alpha}^1 = \vec{\alpha} = 1$ and $\vec{\alpha}^0 = -\vec{\alpha} = -1$. This is a degenerate case. For $SU(3)$ we have instead $\vec{\alpha}^1 = (1, 0)$; $\vec{\alpha}^2 = \left(-\frac{1}{2}, \frac{1}{2}\sqrt{3}\right)$; $\vec{\alpha}^0 = -\left(\frac{1}{2}, \frac{1}{2}\sqrt{3}\right)$. The extended

Cartan matrix becomes

$$A_{\text{ext. } SU(2)}^{ij} = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}; \quad A_{\text{ext. } SU(3)}^{ij} = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}. \quad (32.34)$$

Clearly, $\det A = 0$ in both cases. Also $\det A_k > 0$ for $k = 1, 2$ for $SU(2)$, and $k = 1, 2, 3$ for $SU(3)$, where A_k is the matrix obtained from A^{ij} by deleting the k th row and column.

32.3 Dynkin diagrams for Kac-Moody algebras

The Kac-Moody algebra in (32.25) with generators T_{am} and kI has a Cartan subalgebra consisting of H_{i0} and kI . The roots $(\vec{\alpha}, 0)$ are $(r+1)$ -dimensional vectors, and correspond to $[H_{i0}, E_{\alpha m}] = \alpha_i E_{\alpha m}$ and $[kI, E_{\alpha m}] = 0$. Thus each root is infinitely degenerate. To lift this degeneracy we extend the affine Kac-Moody algebra by adding a “derivative generator” d , satisfying $[d, T_{am}] = mT_{am}$. In practice one takes $d = L_0$, which is not really a restriction since L_0 can be constructed from the T_{am} according to the Sugawara construction. Then, on the Cartan-Weyl basis with generators H_i and E_α for the finite-dimensional Lie algebra, the extended affine Kac-Moody algebra can be written as follows⁴

$$\begin{aligned} [H_m^i, H_n^j] &= mk\delta^{ij}\delta_{m+n,0} \\ [H_m^i, E_n^\alpha] &= \alpha^i E_{m+n}^\alpha \\ [E_m^\alpha, E_n^\beta] &= \epsilon(\alpha, \beta) E_{m+n}^{\alpha+\beta} \quad \text{if } \alpha + \beta \text{ is a root} \\ &= \frac{2(\vec{\alpha} \cdot \vec{H}_{m+n} + km\delta_{m+n,0})}{\vec{\alpha} \cdot \vec{\alpha}} \quad \text{if } \alpha = -\beta \\ &= 0 \quad \text{otherwise.} \\ [k, E_m^\alpha] &= [k, H_m^i] = [k, k] = [d, k] = 0 \\ [d, E_m^\alpha] &= mE_m^\alpha; \quad [d, H_m^i] = mH_m^i. \end{aligned} \quad (32.35)$$

Since $(H_m^i)^\dagger = H_{-m}^i$ and $(E_m^\alpha)^\dagger = E_{-m}^{-\alpha}$, the operators k and d are hermitian

$$k = k^\dagger, \quad d = d^\dagger. \quad (32.36)$$

The Cartan subalgebra now consists of the following $r+2$ operators

$$H_0^i (1 \leq i \leq r), k, d \quad (32.37)$$

⁴To simplify the notation, we write the indices $i, j, \dots$ and $\alpha, \beta, \dots$ as contravariant indices.

and the roots are given by

$$\begin{aligned} E_m^\alpha &\text{ gives } (\vec{\alpha}, 0, m) \\ H_m^i &\text{ gives } (\vec{0}, 0, m) \equiv m\vec{\delta}, \end{aligned} \quad (32.38)$$

where $\vec{\delta} = (\vec{0}, 0, 1)$. So, the roots form an $(r+1)$ -dimensional space (not $(r+2)$ -dimensional) and are given by $m\vec{\delta}$ and $\vec{r} = (\vec{\alpha}, 0, m)$ with $\vec{\alpha}$ a root of the ordinary Lie algebra, and $m \in \mathbb{Z}$. The roots $(\vec{\alpha}, 0, m)$ with $\vec{\alpha}$ nonzero are nondegenerate, but the roots $(\vec{0}, 0, m)$ for given $m \neq 0$ are all r -fold degenerate because they correspond to $H_m^i = 0$ for all i . This degeneracy is a first deviation from ordinary Lie algebras.

If we define that a root is positive if its last nonvanishing entry is positive, then the positive roots are by definition $(\vec{\alpha}, 0, m)$ with $m > 0$, and $(\vec{\alpha}, 0, 0)$ with $\vec{\alpha}$ positive, and $(\vec{0}, 0, m)$ with $m > 0$. A basis for the simple roots is given by

$$\vec{\alpha}_i = (\vec{\alpha}_i, 0, 0) \text{ with } 1 \leq i \leq r, \quad \vec{\alpha}_0 = (-\vec{\psi}, 0, 1), \quad (32.39)$$

where $\vec{\psi}$ is the highest root of the ordinary Lie algebra. To avoid confusion with the indices i of H_m^i we write the indices i which label the simple roots $\vec{\alpha}_i$ as covariant indices. Any root $\vec{\alpha}$ can be then written as

$$\vec{\alpha} = \sum_{i=0}^r n_i \vec{\alpha}_i \quad (32.40)$$

with either all $n_i \geq 0$ (positive roots), or all $n_i \leq 0$ (negative roots). These $r+1$ positive roots are the **simple** roots of the affine KM algebra.

To summarize: The Cartan subalgebra is $(r+2)$ -dimensional, but the root space is $(r+1)$ -dimensional, and there are $r+1$ simple roots.

To construct Dynkin diagrams, we need an inner product between the simple roots. For ordinary Lie algebras, one uses $(T^a, T^b) \sim \text{tr } T^a T^b$ to define the Killing metric. This inner product is invariant under similarity transformations, in particular under the action of the group

$$(gT^a g^{-1}, gT^b g^{-1}) = (T^a, T^b). \quad (32.41)$$

Infinitesimally setting $g \simeq I + Z$, one finds then

$$([Z, X], Y) + (X, [Z, Y]) = 0. \quad (32.42)$$

For affine KM algebras we require that the inner product be again symmetric and have the same invariance of the inner product. By taking $Z = d$ or T^a_0 we find that (T^a_m, T^b_n) is nonzero only if $m+n=0$ and then it is proportional to γ^{ab} . Consider now $Z = T^a_1$, $X = T^b_m$, $Y = T^c_{-m-1}$. One finds then

$$f^{ab}_{b'}(T^{b'}_{m+1}, T^c_{-m-1}) + f^{ac}_{c'}(T^b_m, T^{-c'}_{-m}) = 0, \quad (32.43)$$

which shows that (T_m^a, T_{-m}^b) is independent of m . So

$$(T_m^a, T_n^b) = \gamma^{ab} \delta_{m+n,0} \quad (32.44)$$

by a suitable overall normalization of the inner product. By taking $Z = T_m^a$, $X = T_n^b$ and $Y = k$ one finds that

$$(T_m^a, k) = 0; \quad (k, k) = 0. \quad (32.45)$$

Finally, with $X = T_m^a$, $Y = d$, $Z = T_n^b$ we find

$$f^{ba}_c(T_{m+n}^c, d) + n\gamma^{ba} \delta_{m+n,0}(k, d) = (T_m^a, nT_n^b) = n\gamma^{ab} \delta_{m+n,0} \quad (32.46)$$

from which we read off

$$(d, T_m^a) = 0; \quad (k, d) = 1. \quad (32.47)$$

We are left with (d, d) . If we redefine $d \rightarrow d' = d - \eta k$ with η a constant, then d' satisfies the same relations as d , except that $(d', d') = (d, d) - 2\eta$. So, we can always achieve that $(d', d') = 0$. Therefore, we define

$$(d, d) = 0. \quad (32.48)$$

We have now defined an indefinite inner product in root space. The basis

$$\left\{ T_0^a, \frac{(T_m^a + T_{-m}^a)}{\sqrt{2}}, \frac{(T_m^a - T_{-m}^a)}{\sqrt{2}}, \frac{k+d}{\sqrt{2}}, \frac{k-d}{\sqrt{2}} \right\} \quad (32.49)$$

is then orthonormal with Lorentzian signature, meaning that all generators in this basis are orthogonal and have norm $+1$, except the last one which has norm -1 .

It is now clear that the roots $(\vec{\alpha}, 0, n)$ have length $(\vec{\alpha}, \vec{\alpha})$ while the roots $(0, 0, n)$ belonging to H_n^i have length zero, and are orthogonal to all other roots. (In the vertex operator representation of affine KM algebras, one represents H_m^i by using photon operators α_{-m}^i where the photons are massless in $r+2$ dimensions, so with r polarizations.)

The Cartan matrix is now formed as usual from the simple roots

$$A_{ij} = \frac{2(a_i, a_j)}{(a_j, a_j)}, \quad 0 \leq i, j \leq r. \quad (32.50)$$

The entries A_{00} , A_{0i} and A_{i0} are obtained by evaluating $(-\vec{\psi}, -\vec{\psi})$ and $(-\vec{\psi}, \vec{\alpha}_i)$, where $\vec{\psi}$ is the highest root. Since $\vec{\psi}$ is the longest root, $(\vec{\psi}, \vec{\psi}) \geq (\vec{\alpha}_i, \vec{\alpha}_i)$ and since $\vec{\psi}$ is not itself a simple root (except for $SU(2)$) $|(a_0, a_i)| < |(\vec{a}_0, \vec{a}_0)|$. On the other hand, $\frac{2(\vec{r}_i, \vec{r}_j)}{(\vec{r}_j, \vec{r}_j)}$ is always an integer. Hence

$$\begin{aligned} A_{i0} &= -1 \quad \text{if } A_{0i} \neq 0 \\ A_{i0} &= 0 \quad \text{if } A_{0i} = 0. \end{aligned} \quad (32.51)$$

We now construct Dynkin diagrams by

- 1) adding a point corresponding to $\vec{a}_0$.
- 2) joining it by $-A_{0i}$ lines to the points corresponding to $\vec{a}_i$.
- 3) putting an arrow towards $\vec{a}_0$ if $-A_{01} > 1$.

Example 1: $s\hat{u}(r+1)$ with $r \geq 2$. Here $\vec{\psi} = \vec{\lambda}_1 + \vec{\lambda}_r$ where $\vec{\lambda}_1$ and $\vec{\lambda}_r$ are the first and last fundamental weights. Then $A_{10} = -1$, $A_{s0} = 0$ for $2 \leq s \leq r-1$ while $A_{r0} = 1$. Thus

Example 2: $s\hat{o}(n)$ with $n \geq 5$. Here $\vec{\psi} = \vec{\lambda}_2$, so

Example 3: $\hat{e}(8)$. One finds $\vec{\psi} = \vec{\lambda}_8$, hence

Example 4: $s\hat{u}(2)$. Now there are two simple roots: $\vec{a}_1 = (\vec{\alpha}, 0, 0)$ and $\vec{a}_0 = (-\vec{\alpha}, 0, 1)$. Hence $A_{01} = A_{10} = -2$. The matrix A thus reads

$$A = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}. \quad (32.52)$$

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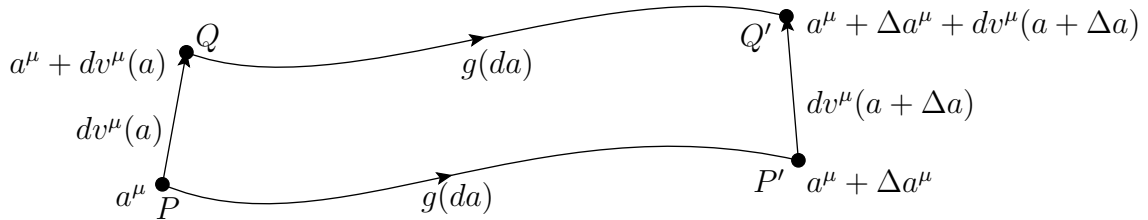
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Chapter 33

Geometry of Group Manifolds

33.1 Vielbeins and connections with either torsion or curvature

Consider a Lie group G with group elements $g(a), g(b), g(c), g(x)$, etc, where $a^\mu, b^\mu, c^\mu, x^\mu$, etc are the group coordinates. We denote the composition rule of the group coordinates by $\phi^\mu(a, b) = c^\mu$ if $g(a)g(b) = g(c)$. We choose the origin of the group coordinates such that the unit element of the group has $a^\mu = 0$, so $\phi^\mu(0, 0) = 0$ and $\phi^\mu(0, b) = \phi^\mu(b, 0) = b^\mu$. If the inverse of $g(x)$ is $g(x)^{-1}$, we denote its group coordinates by $(x^{-1})^\mu$. So $\phi^m(x, x^{-1}) = \phi^m(x^{-1}, x) = 0$.¹ The associativity of group multiplication $(g(a)g(b))g(c) = g(a)(g(b)g(c))$ reads in the notation with ϕ^μ as follows $\phi^\mu(\phi(a, b), c) = \phi^\mu(a, \phi(b, c))$. The group coordinates a^μ form a manifold, the group manifold, and each point on the group manifold corresponds to an element of the group. Let two points P and Q near each other have group coordinates a^μ and $a^\mu + \epsilon^\mu$. Then the difference ϵ^μ can be viewed as a little vector $dv^\mu(a)$ which runs from P to Q . Multiplying a point P (group element $g(a)$) by $g(b)$, either on the left or on the right, the point P moves to another point P' . We have **motion on the group manifold**.



Let us now move the point P corresponding to group element $g(a)$ with group parameters a^μ , to a point P' by left multiplication with group element $g(da)$ where da^m are small

$$g(da)g(a) = g(a + \Delta a); \quad \phi^\mu(da, a) = a^\mu + \Delta a^\mu. \quad (33.1)$$

¹We use the following notation. Objects near the origin carry latin indices $(m, n, \dots)$, while objects far away from the origin carry greek indices $(\mu, \nu, \dots)$. Thus $g(da)$ has group coordinates da^m but $g(a)$ has a^μ , and $g(a + \Delta a)$ has $a^\mu + \Delta a^\mu$. Since $g(x)g(x)^{-1}$ is near (at) the origin we write $\phi^m(x, x^{-1})$ and not $\phi^\mu(x, x^{-1})$.

A Taylor expansion² introduces the right vielbein field³ $f_m^\mu(a)$

$$\phi^\mu(da, a) - \phi^\mu(0, a) \simeq da^m \frac{\partial}{\partial b^m} \phi^\mu(b, a)|_{b=0} = da^m {}_m\phi^\mu(0, a) = da^m f_m^\mu(a) = \Delta a^\mu. \quad (33.2)$$

We denote its matrix inverse by $f_\mu^m(a)$. Similarly a point Q near P is moved to a point Q' near P' , and thus the vector $dv^\mu(a)$ between P and Q is moved to a vector $dv^\mu(a + \Delta a)$ between P' and Q' , see the figure above. This defines a connection $\bar{\Gamma}_{\mu\nu}^\rho(a)$ where the minus sign on top of $\bar{\Gamma}$ indicates that it is due to left multiplication with $g(da)$

$$dv^\mu(a + \Delta a) - dv^\mu(a) = -\Delta a^\sigma \bar{\Gamma}_{\sigma\rho}^\mu(a) dv^\rho(a) \quad (33.3)$$

with $dv^\mu(a + \Delta a) = \phi^\mu(da, a + dv) - \phi^\mu(da, a)$. Expanding to first order first in da and then in dv yields an expression for $\bar{\Gamma}$

$$\text{From } g(da)g(a) : \bar{\Gamma}_{\sigma\rho}^\mu(a) = -(\partial_\rho f_m^\mu) f_\sigma^m = f_m^\mu \partial_\rho f_\sigma^m. \quad (33.4)$$

Similarly one can use right multiplication $g(a)dg(a) = g(a + \Delta a)$ to define left vielbeins $e_m^\mu(a)$ and a connection $\bar{\Gamma}_{\sigma\rho}^\mu$ as follows

$$\begin{aligned} \phi^\mu(a, da) &= a^\mu + \Delta a^\mu; \quad \Delta a^\mu = da^m e_m^\mu(a); \quad e_m^\mu(a) = \frac{\partial}{\partial b^m} \phi^\mu(a, b)|_{b=0} = \phi_m^\mu(a, 0) \\ \text{From } g(a)g(da) : \bar{\Gamma}_{\sigma\rho}^\mu &= e_m^\mu \partial_\rho e_\sigma^m. \end{aligned} \quad (33.5)$$

We define the inverse of e_m^μ by e_μ^m as usual. We consider a matrix representation of g and T_m where T_m are the group generators.

Lie group elements $g(a)$ can be written as $e^{a^\mu T_\mu}$. Then the equation $g(da)g(a) = g(a^\mu + da^m f_m^\mu(a)) = g(a^\mu + \Delta a^\mu)$ can be rewritten as $g(\Delta a^\mu f_\mu^m(a)) = g(a^\mu + \Delta a^\mu)g^{-1}(a)$. Subtracting unity we find $g(\Delta a^\mu f_\mu^m(a)) - I = (g(a + \Delta a) - g(a))g^{-1}(a)$, so for small Δa we find

$$\Delta a^\mu f_\mu^m T_m = f^m T_m = f = \Delta a^\rho (\partial_\rho g(a)) g^{-1}(a) \quad (33.6)$$

so in form language $f^m T_m = dg(a)g^{-1}(a)$. Similarly $e^m T_m = g^{-1}(a)dg(a)$. Taking the exterior derivative d of these equations yields the Cartan-Maurer equations

$$\begin{aligned} de^m + \frac{1}{2} c_{np}^m e^n \wedge e^p &= 0 \\ df^m - \frac{1}{2} c_{np}^m f^n \wedge f^p &= 0 \end{aligned} \quad (\text{Cartan-Maurer}) \quad (33.7)$$

where c_{np}^m are the structure constants of the group G , defined by $[T_m, T_n] = c_{mn}^p T_p$, and $c_{mn}^p =$

²Here we have used the notation that differentiation of $\phi^\mu(a, b)$ w.r.t. its first argument a^ρ is denoted by ${}_\rho\phi^\mu(a, b)$, but differentiation w.r.t. its second argument b^σ is written as $\phi_\sigma^\mu(a, b)$. But since da in $\phi^\mu(da, a)$ is near zero, we write da^m and ${}_m\phi^\mu(0, a)$.

³The proper name is right-invariant vielbein field, for reasons to be explained shortly.

c^p_{mn} . The relative minus sign in (33.7) is due to moving d past $dg(a)$ in $f^m T_m = dg(a)g^{-1}$, but for $e^m T_m = g^{-1}dg(a)$ there is no such minus sign.

Torsion is defined as twice the antisymmetric part of the Γ connection, $\Gamma_{\sigma\rho}^\mu - \Gamma_{\rho\sigma}^\mu \equiv T_{\sigma\rho}^\mu$. Using the Maurer-Cartan equations in appendix I, eq. (33.53), yields

$$\bar{T}_{\sigma\rho}^\mu(a) = f_m^\mu(\partial_\rho f_\sigma^m - \partial_\sigma f_\rho^m) = f_m^\mu c_{rs}^m f_\rho^r f_\sigma^s \quad (33.8)$$

$$T_{\sigma\rho}^\mu(a) = e_m^\mu(\partial_\rho e_\sigma^m - \partial_\sigma e_\rho^m) = -e_m^\mu c_{rs}^m e_\rho^r e_\sigma^s. \quad (33.9)$$

where c_{rs}^m are the structure constants of the group. At the origin these two torsions are each other's opposite, $\bar{T}_{\sigma\rho}^\mu(0) = -T_{\sigma\rho}^\mu(0)$, because $f_m^\mu(0) = e_m^\mu(0) = \delta_m^\mu$, see (33.2) and (33.5), but also at other points with coordinates a^μ : $\bar{T}_{\sigma\rho}^\mu(a) = -T_{\sigma\rho}^\mu(a)$. To prove this we need the matrix $f_m^\mu(a)e_\mu^n(a) \equiv A_m^n(a)$. We now first show that A_m^n is a matrix in the adjoint representation of the Lie algebra, and then we shall come back to the torsion.

Consider the following group element

$$\mathcal{A} = e^{-a \cdot T} e^{da \cdot T} e^{a \cdot T}; \quad a \cdot T = a^\mu T_\mu; \quad da \cdot T = da^m T_m. \quad (33.10)$$

(We use here a representation of group elements $g(a)$ in terms of exponentials; this is not always possible for noncompact groups, but everything can be phrased in terms of $\phi^\mu(a, da)$ etc. Manipulations with exponentials are visually easier to follow.) Using that the product of these exponentials can be written as the exponent of exponentials we get

$$\begin{aligned} \mathcal{A} &= \exp \left[e^{-a \cdot T} da \cdot T e^{a \cdot T} \right] = \exp \left[da \cdot T + [da \cdot T, a \cdot T] + \frac{1}{2!} [[da \cdot T, a \cdot T], a \cdot T] + \dots \right] \\ &= \exp \left[da^m T_m + da^m (a^\nu c_{m\nu}^n) T_n + \frac{1}{2!} da^m (a^\nu c_{m\nu}^p) (a^\rho c_{p\rho}^n) T_n + \dots \right] \\ &= \exp \left[da^m A_m^n(a) T_n \right] \end{aligned} \quad (33.11)$$

where the matrix A_m^n is given in terms of the structure constants c_{mk}^n by

$$A_m^n(a) = \left(e^{a^\nu c_{\bullet\nu}^\bullet} \right)_m^n. \quad (33.12)$$

This is the adjoint representation of the group G . We can also obtain another expression for A by evaluating $\mathcal{A}$ as $e^{-a \cdot T} (e^{da \cdot T} e^{a \cdot T})$, and this yields

$$\mathcal{A} = e^{-a \cdot T} e^{(a^\mu + da^m f_m^\mu) T_\mu} = e^{da^m f_m^\mu e_\mu^n T_n}. \quad (33.13)$$

(We used $g^{-1}(a)g(a + \Delta a) = g(\Delta a^\mu e_\mu^n(a))$ which follows from $g(a)g(\Delta a^\mu e_\mu^n) = g(a + \Delta a)$.) Thus

$$\boxed{A_m^n(a) = \left(e^{a^\nu c_{\bullet\nu}^\bullet} \right)_m^n = f_m^\mu(a) e_\mu^n(a)} \quad (33.14)$$

This shows first of all how left vielbeins (e_μ^m) are related to right vielbeins (f_μ^m)

$$f_m^\mu = A_m^n e_n^\mu; \quad e_\mu^n = f_\mu^m A_m^n. \quad (33.15)$$

Secondly, it proves that the $\overset{+}{T}$ torsion and the $\bar{T}$ torsion are opposite at all points.⁴ In fact, the $\pm$ torsion with flat indices is equal to $\pm$ the structure constants, see (33.8) and (33.9),

$$-e_m^\mu e_n^\nu \overset{+}{T}_{\mu\nu}^\rho e_\rho^r = f_m^\mu f_n^\nu \bar{T}_{\mu\nu}^\rho f_\rho^r = c_{mn}^r \quad (33.16)$$

It should be clear that if one parallel transports a small vector (two nearby points) with the $\overset{+}{\Gamma}$ or the $\bar{\Gamma}$ connection along a closed loop, the vector does not change because the product of the group elements which produce the round transport of the top and the bottom of the little vectors is in both cases the unit element. Hence

The $\overset{+}{\Gamma}$ and $\bar{\Gamma}$ connections have opposite torsion but no curvature.

One can sweep any small vector out from any point on the group manifold to all other points, and since this sweeping out is unambiguous (independent of the choice of path along which one does the sweeping out), one speaks of “parallelizability” of group manifolds. Let us discuss this important property in more detail.

The left vielbeins e_m^μ and the right vielbeins f_m^μ are actually called left-invariant and right-invariant vielbeins for a geometrical reason which we now explain. If the group manifold is n -dimensional, one can choose n points $Q_1, \dots, Q_n$ near the origin with coordinates $da_{(1)}^\mu, da_{(2)}^\mu, \dots, da_{(n)}^\mu$. The little vectors $dv_{(j)}^\mu(0)$ from the origin to these points form then a basis into which all little vectors at the origin can be expanded. Using a group element $g(a)$, the origin is moved to a point P with coordinates a^μ , and using left-(right-)multiplication the points $Q_1, \dots, Q_n$ (group elements $g(da_{(1)}), \dots, g(da_{(n)})$ near the origin) are moved to points $Q'_1, \dots, Q'_n$ near P , where they yield n little vectors $dv_{(j)}^\mu(a)$ which form a “frame”. In this way, given a frame at the origin, one can sweep this frame out to create corresponding frames at all other points of the group manifold. Using left-multiplication to do the sweeping out, we obtain a different set of frames from using right-multiplication, but the complete set of left frames (and similarly the complete set of right frames) has a property which allows them to be called left-invariant frames or left-invariant vielbeins (right-invariant frames or right-invariant vielbeins). Namely, given frames at points P and Q and a group element $g(b)$ which moves the point P to a point Q on the group manifold, **the frames obtained at Q by moving the frames at P to Q coincide with the frames which were already at Q** . In other words, the frames are independent of the way they were swept out, they are “invariant”. If one chooses the little vectors at the origin to lie along the coordinate axes,

⁴Move the e -vielbeins from $\overset{+}{T}$ in (33.9) to $\bar{T}$ in (33.8); one then finds that the torsions are opposite provided $c_{rs}^m = (A^{-1})_{m'}^m c_{r's'}^{m'} A_r^{r'} A_s^{s'}$ (the structure constants are “invariant tensors”). This can be proven by inserting $e^{-a \cdot T} da \cdot T e^{a \cdot T}$ instead of $da \cdot T$ into the definition of a Lie algebra.

the frames swept out by left-multiplication with $g(a)$ are the left-invariant vielbein fields $e_m^\mu(a)$, and the frames created by right-multiplication are the right-invariant vielbein fields $f_m^\mu(a)$. Note that the roles of $g(da)$ and $g(a)$ have been interchanged in this process of sweeping out: to construct connections we moved points $g(a)$ by group elements $g(da)$, but to sweep out frames (little vectors at the origin) we multiplied by $g(a)$.

Given a connection $\Gamma_{\mu\nu}^\rho$, and any set of frames E_m^μ (for example the e_m^μ or f_m^μ we have constructed), one can introduce a new set of “tangent space connections” ω as follows. Given a vector field $v^\mu(x)$ with “curved index” μ and a set of frames E_m^μ , one can construct a vector field $v^m(x)$ with “flat index” m by contraction with the vielbein field

$$v^m(x) = v^\mu(x) E_\mu^m(x) \quad (33.17)$$

Conversely, given a vector field $v^m(x)$ with “flat index” m , one can define $v^\mu(x) = v^m(x) E_m^\mu(x)$ where E_m^μ is the matrix inverse of E_μ^m . Parallel transport of v^μ is defined in terms of an **arbitrary** real connection $\Gamma_{\rho\sigma}^\mu$ (not necessarily symmetric)

$$\tilde{v}^\mu(x + \Delta x) = v^\mu(x) - \Delta x^\rho \Gamma_{\rho\sigma}^\mu(x) v^\sigma(x) \quad (33.18)$$

whereas parallel transport of v^m is defined in terms of an arbitrary connection $\omega_\rho^m{}_n(x)$

$$\tilde{v}^m(x + \Delta x) = v^m(x) - \Delta x^\rho \omega_\rho^m{}_n(x) v^n(x) \quad (33.19)$$

The minus signs are only a convention, and the terms with Δx^ρ are the most general expressions which are linear in Δx^ρ and v so that all indices match. For vectors $v_\mu(x)$ and $v_m(x)$ we define parallel transport such that in $\tilde{v}_\mu \tilde{v}^\mu$ and $\tilde{v}_m \tilde{v}^m$ the connections cancel

$$\begin{aligned} \tilde{v}_\mu(x + \Delta x) &= v_\mu(x) + \Delta x^\rho \Gamma_{\rho\mu}^\sigma(x) v_\sigma(x) \\ \tilde{v}_m(x + \Delta x) &= v_m(x) + v_n(x) \Delta x^\rho \omega_\rho^m{}_n(x). \end{aligned} \quad (33.20)$$

The geometrical meaning of the vielbein field E_μ^m is that it gives the orientation of the inertial frames (with Cartesian coordinates $\xi^m(x)$) with respect to the coordinates x^μ of curved space at a point P : we have $E_\mu^m(x_P) = \left. \frac{\partial}{\partial x^\mu} \xi^m(x) \right|_{x=x_P}$. Then $v^\mu(x)$ are the components of the vector v with respect to x^μ , and v^m are the components of v with respect to the inertial frame. In other words v^μ and v^m describe the same vector, just in different frames, and then parallel transport of v^μ and v^m should give the same result. The operations of parallel transport and projection onto flat or curved indices should commute. If we first transport $v^\mu(x)$ and then convert at $x + \Delta x$ to flat indices, we should get the same result as when we first convert $v^\mu(x)$ to flat indices at x and then transport to $x + \Delta x$

$$[\tilde{v}^\mu(x + \Delta x)] E_\mu^m(x + \Delta x) = \tilde{v}^m(x + \Delta x) \quad (33.21)$$

Substituting (33.17), (33.18) and (33.19) yields

$$[v^\mu(x) - \Delta x^\rho \Gamma_{\rho\sigma}{}^\mu v^\sigma(x)] E_\mu{}^m(x + \Delta x) = v^m(x) - \Delta x^\rho \omega_\rho{}^m{}_n(x) v^n(x) \quad (33.22)$$

Expanding $E_\mu{}^m(x + \Delta x)$ in terms of Δx , the terms without Δx yield (33.17), and to first order in Δx one finds

$$v^\mu(x) \Delta x^\rho \partial_\rho E_\mu{}^m(x) - \Delta x^\rho \Gamma_{\rho\sigma}{}^\mu v^\sigma(x) E_\mu{}^m(x) = -\Delta x^\rho \omega_\rho{}^m{}_n(x) v^n(x) \quad (33.23)$$

Using (33.17) and extracting Δx^ρ and $v^\mu(x)$ one arrives at what is called the vielbein postulate

$$\mathcal{D}_\rho E_\mu{}^m \equiv \partial_\rho E_\mu{}^m(x) - \Gamma_{\rho\mu}{}^\nu E_\nu{}^m + \omega_\rho{}^m{}_n E_\mu{}^n = 0 \quad (33.24)$$

The connection $\Gamma_{\rho\mu}{}^\nu$ “sees” the curved index μ of $E_\mu{}^m$, and the connection $\omega_\rho{}^m{}_n$ “sees” the flat index m . Given $\Gamma_{\rho\mu}{}^\nu$, one can calculate $\omega_\rho{}^m{}_n$ from (33.24); conversely, given $\omega_\rho{}^m{}_n$, this equation gives $\Gamma_{\rho\mu}{}^\nu$. We now apply these ideas to the $\overset{+}{\Gamma}$ and $\bar{\Gamma}$ connection.

Given either the $\overset{+}{\Gamma}$ or the $\bar{\Gamma}$ connection, we can choose either the left-invariant vielbein frames $e_\mu{}^m$ or the right-invariant vielbein frames $f_\mu{}^m$ to construct ω connections by means of the vielbein postulate which act on vectors in these frames: ${}^L\omega$ connections for $\overset{+}{\Gamma}$ acting on vectors in the frames $e_m{}^\mu$, ${}^R\omega$ connections for $\overset{+}{\Gamma}$ acting on vectors in the frames $f_m{}^\mu$, and further ${}^L\bar{\omega}$ connections for $\bar{\Gamma}$ acting on vectors in the frames $e_m{}^\mu$ and finally ${}^R\bar{\omega}$ connections acting on vectors in the frames $f_m{}^\mu$

$$\begin{aligned} \partial_\rho e_\sigma{}^m - \overset{+}{\Gamma}_{\rho\sigma}{}^\tau e_\tau{}^m + {}^L\omega_\rho{}^m{}_n e_\sigma{}^n &= 0 \\ \partial_\rho f_\sigma{}^m - \overset{+}{\Gamma}_{\rho\sigma}{}^\tau f_\tau{}^m + {}^R\omega_\rho{}^m{}_n f_\sigma{}^n &= 0. \\ \partial_\rho e_\sigma{}^m - \bar{\Gamma}_{\rho\sigma}{}^\tau e_\tau{}^m + {}^L\bar{\omega}_\rho{}^m{}_n e_\sigma{}^n &= 0 \\ \partial_\rho f_\sigma{}^m - \bar{\Gamma}_{\rho\sigma}{}^\tau f_\tau{}^m + {}^R\bar{\omega}_\rho{}^m{}_n f_\sigma{}^n &= 0. \end{aligned} \quad (33.25)$$

Since the left-invariant frames $e_m{}^\mu$ were obtained by left multiplication of $g(da)$ by $g(a)$, the left-invariant frames at x when parallel transported to $x + dx$ will coincide with the left-invariant frames at $x + dx$. So a vector v^m at x when parallel transported to $x + dx$ by left-multiplication will have the same components at $x + dx$ as at x . This implies that ${}^L\bar{\omega}_\rho{}^m{}_n = 0$.

Recall that $e_m{}^\mu(a)$ and $\bar{\Gamma}(a)$ were both obtained from left multiplication by $g(a)$.⁵ It follows that the components of a vector $v^n(a)$ in the frame $e_m{}^\mu(a)$ do not change under parallel transport with the $-$ connection; the connection for this parallel transport is ${}^L\bar{\omega}$, and thus ${}^L\bar{\omega}$

⁵Note that in (33.1) the action of $g(da)$ on $g(a)$ by left-multiplication as $g(da)g(a)$ defined the connection $\bar{\Gamma}$, while also the sweeping out of frames $d\tilde{a}^m$ from the origin by left-multiplication with $g(a)$ produced the left-invariant vielbein fields $e_m{}^\mu$. So the word left in left-invariant vielbeins $e_m{}^\mu$ refers to the left-multiplication of the little vectors $d\tilde{a}$ of a frame at the origin by finite group elements g .

vanishes. Similarly right multiplication produces f_m^μ and $\bar{\Gamma}^+$, and thus also $R\omega^+$ vanishes

$$\boxed{R\omega_\rho^{+m}{}_n(a) = 0; \quad L\bar{\omega}_\rho^{-m}{}_n(a) = 0.} \quad (33.26)$$

The vielbein postulates in (33.25) yield then another expression for $\bar{\Gamma}^+$ and $\bar{\Gamma}^-$

$$\begin{aligned} \bar{\Gamma}_{\rho\sigma}^{+\mu} &= f_m^\mu \partial_\rho f_\sigma^m \quad (\text{previously} = e_m^\mu \partial_\sigma e_\rho^m) \\ \bar{\Gamma}_{\rho\sigma}^{-\mu} &= e_m^\mu \partial_\rho e_\sigma^m \quad (\text{previously} = f_m^\mu \partial_\sigma f_\rho^m) \end{aligned} \quad (33.27)$$

These two expressions for $\bar{\Gamma}^+$ are equal since, as we prove in the appendix, the left-invariant tangent vectors $e_m \equiv e_m^\mu \partial_\mu$ commute with the right-invariant tangent vectors $f_n \equiv f_n^\nu \partial_\nu$, $[e_m, f_n] = 0$. Similarly the two expressions for $\bar{\Gamma}^-$ are equal.

Let us now analyze the nonvanishing connections $L\omega^+$ and $R\bar{\omega}^-$. If we consider a vector v in the right-invariant frames,⁶ $v^\mu f_\mu^m = Rv^m$, it will not change under parallel transport with the $+$ connection (because $R\omega^+ = 0$)

$$Rv^m(a + \Delta a) = Rv^m(a) \quad \text{for the } + \text{ connection.} \quad (33.28)$$

But we can use this $+$ connection in the left-invariant frames, $L\omega^+$, to parallel transport the vector in the left frames, $Lv^m = v^\mu e_\mu^m$, and afterwards we can use the relation (see (33.15))

$$Lv^m = Rv^n A_n^m \quad (33.29)$$

both for $Lv^m(a + \Delta a)$ and $Lv^m(a)$ to determine $L\omega^+$. The details are as follows

$$\begin{aligned} Lv(a + \Delta a)^m - Lv(a)^m &= -\Delta a^\mu (L\omega_\mu^{+m}{}_n) Lv(a)^n \\ &= Rv(a + \Delta a)^n A_n^m(a + \Delta a) - Rv(a)^n A_n^m(a) \\ &= Rv(a)^n [A_n^m(a + \Delta a) - A_n^m(a)] \\ &= Lv(a)^k (A^{-1})_k{}^n(a) [A_n^m(a + \Delta a) - A_n^m(a)] \\ &= Lv(a)^k [(A^{-1})_k{}^n(a) A_n^m(a + \Delta a) - \delta_k^m] \\ &= Lv(a)^k [A(\Delta a^\mu e_\mu^t)_k{}^m - \delta_k^m] \\ &= Lv(a)^k [\Delta a^\mu e_\mu^l c_{kl}^m]. \end{aligned} \quad (33.30)$$

⁶Given a vector v^μ we can expand it in terms of the left-invariant vielbein fields e_m^μ as $v^\mu = (v^\nu e_\nu^m) e_m^\mu = Lv^n e_n^\mu$. The components Lv^n of the vector define then a vector Lv in the left-invariant frames.

This yields ${}^L\omega_\mu^m{}_n$. In a similar manner one finds ${}^R\bar{\omega}_\mu^m{}_n$.

$$\boxed{{}^L\omega_\mu^m{}_n(a) = -e_\mu^l c_{nl}^m; \quad {}^R\bar{\omega}_\mu^m{}_n(a) = f_\mu^l c_{nl}^m.} \quad (33.31)$$

(The reason for the different sign is that for ${}^R\bar{\omega}$ we need the inverse relation ${}^Rv^m = {}^Lv^n(A^{-1})_n{}^m$.)

Suppose one has an arbitrary connection Γ , which gives rise in the left-invariant frames $e_m{}^\mu$ to a connection ${}^L\omega_\mu^m{}_n$, and in the right-invariant frames $f_m{}^\mu$ to a connection ${}^R\omega_\mu^m{}_n$. Using the relation ${}^Lv^m = {}^Rv^n A_n{}^m$ we can then first perform parallel transport of ${}^Lv^m(a)$ by ${}^L\omega_\mu^m{}_n$, then replace ${}^Lv^m(a + da)$ and ${}^Lv^m(a)$ by ${}^Rv^n(a + da)A_n{}^m(a + da)$ and ${}^Rv^n(a)A_n{}^m(a)$, next use the definition of parallel transport of ${}^Rv^n(a + da)$ by ${}^R\omega_\mu^m{}_n$, and finally convert the ${}^Rv^k(a)$ back into ${}^Lv^k(a)$. This gives a relation between ${}^L\omega$ and ${}^R\omega$

$$\boxed{{}^L\omega_\mu^m{}_n(a) = -e_\mu^l(a)c_{nl}^m + {}^R\omega_\mu^{m'}{}_{n'}(a)A_{m'}{}^m(a)(A^{-1})_n{}^{n'}(a).} \quad (33.32)$$

The structure of the last term is clear: the indices m and n of ${}^L\omega_\mu^m{}_n$ are transformed like ${}^Lv^m = {}^Rv^n A_n{}^m$. We can also start with ${}^R\omega_\mu^m{}_n$, and find in a similar manner an expression for ${}^R\omega_\mu^m{}_n$ in terms of ${}^L\omega_\mu^m{}_n$, but it is more interesting to directly invert (33.32). This yields

$${}^R\omega_\mu^m{}_n = e_\mu^l c_{n'l}^{m'}(A^{-1})_{m'}{}^m A_n{}^{n'} + {}^L\omega_\mu^{m'}{}_{n'}(A^{-1})_{m'}{}^m(a)A_n{}^{n'}(a). \quad (33.33)$$

Using that c_{nl}^m is an invariant tensor and $e_\mu^{l'}(A^{-1})_{l'}{}^l = f_\mu^l$, we obtain

$$\boxed{{}^R\omega_\mu^m{}_n(a) = f_\mu^l(a)c_{nl}^m + {}^L\omega_{\mu n'}^{m'}(a)(A^{-1})_{m'}{}^m(a)A_n{}^{n'}(a).} \quad (33.34)$$

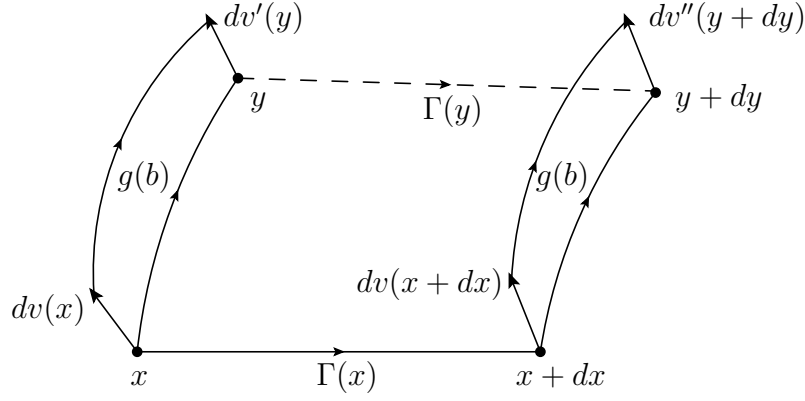
Again the structure of the last term agrees with ${}^Rv^m = {}^Lv^n(A^{-1})_n{}^m$.

As a check one may apply these general relations to the particular cases with ${}^R\omega^+ = 0$ and ${}^L\bar{\omega} = 0$, and recover in this way the expressions for ${}^L\omega^+$ and ${}^R\bar{\omega}$ which we obtained in (33.31).

33.2 General invariant connections on group manifolds

We have found the very particular connections ${}^L\omega^+$ and ${}^R\bar{\omega}$ which are uniquely determined by the group multiplication, and the most general connections ${}^L\omega$ and ${}^R\omega$ which are only related to each other by (33.32) and (33.34). This cannot be the whole story because the Christoffel connections (which have vanishing torsion) have not been found. We now discuss a class of connections, general invariant connections, which are less general than the most general connections, but more general than the ${}^L\omega^+$ and ${}^R\bar{\omega}$, and which include the Christoffel connections.

Suppose one has a connection Γ at all points of the manifold. We parallel transport with $\Gamma(x)$ a little vector at x to $x + dx$. Then we move the two little vectors $dv(x)$ and $dv(x + dx)$ by left-multiplication with $g(b)$ to little vectors dv' and dv'' at points y and $y + dy$, respectively. (The



little vectors consist really of the two extreme points, and points are group elements which we can move by group multiplication.) If we now parallel transport $dv'(y)$ with $\Gamma(y)$ over a distance dy , we get a vector $dv'_{PT}(y + dy)$ at $y + dy$. If for all points of the group manifold this new vector coincides with the vector that was already there

$$dv'_{PT}(y + dy) = dv''(y + dy) \quad (33.35)$$

the connection Γ is called an invariant connection. In other words, if we first use $\Gamma(x)$ to parallel transport a little vector from x to $x + dx$ and then use the group action to sweep out the little vector to $y + dy$, we get the same result as when we first use the group action to sweep out the little vector at x to a point y , and then use $\Gamma(y)$ to parallel transport the little vector at y to $y + dy$. So for an invariant connection, parallel transport (PT) and group action (GA) commute

$$[\text{PT}, \text{GA}] = 0. \quad (33.36)$$

A general left-invariant connection (an invariant connection in the e -frames, using left-multiplication) satisfies (proof later)

$$\boxed{{}^L\omega_\mu{}^m{}_n(x) = e_\mu{}^k(x){}^L\omega_k{}^m{}_n(0)} \quad (33.37)$$

A general right-invariant connection satisfies

$$\boxed{{}^R\omega_\mu{}^m{}_n(x) = f_\mu{}^k(x){}^R\omega_k{}^m{}_n(0)} \quad (33.38)$$

Any connection Γ produces connections ${}^L\omega_\mu{}^m{}_n(x)$ and ${}^R\omega_\mu{}^m{}_n(x)$ which are related by (see (33.32))

$${}^L\omega_\mu{}^m{}_n(x) = -e_\mu{}^l(x)c_{nl}{}^m + {}^R\omega_\mu{}^{m'}{}_{n'}(x)A_{m'}{}^m(x)A_n{}^{-1n'} \quad (33.39)$$

Substituting (33.37) and (33.38) into (33.39), and using (33.32) once more, now at $x = 0$, and using that the structure constants constitute an invariant tensor, yields

$${}^L\omega_k{}^m{}_n(0) = A^{-1}{}_k{}^{k'} \left({}^L\omega_{k'}{}^{m'}{}_{n'}(0) \right) A_{m'}{}^m A_n{}^{-1n'} \quad (33.40)$$

Thus the most general left-invariant and right-invariant connection at the origin is an invariant tensor $t_k^m{}_n$ in the adjoint representation of the group. For general groups there is only one such invariant tensor: the structure constants.⁷ Hence we conclude:

$$L\omega_k^m{}_n(0) = \lambda c_{kn}^m, \quad \lambda \text{ free!} \quad (33.41)$$

Substituting this result into (33.37) and (33.38) determines completely the most general left- and right- invariant connection: it is a one parameter class of connections.

The torsion and curvature follow from

$$L\omega_\mu^m{}_n = e_\mu^k(x) \lambda c_{kn}^m \quad (33.42)$$

and the Maurer-Cartan equations:

$$\begin{aligned} \Gamma_{\mu\nu}^\rho &= e_m^\rho \left[\partial_\mu e_\nu^m + \lambda e_\mu^k e_\nu^l c_{kl}^m \right] = e_m^\rho \left[\partial_\nu e_\mu^m + (\lambda - 1) e_\mu^k e_\nu^l c_{kl}^m \right] \\ T_{\mu\nu}^\rho &= (2\lambda - 1) c_{mn}^r e_\mu^m e_\nu^n e_r^\rho \\ R_{\mu\nu}^m{}_n &= \lambda(\lambda - 1) e_\mu^r e_\nu^s c_{rs}^k c_{kn}^m \end{aligned} \quad (33.43)$$

For $\lambda = 0$ we obtain the $\bar{\Gamma}$ connection (see (33.27)), for $\lambda = 1$ we obtain the $\bar{\Gamma}^+$ connection, and for $\lambda = \frac{1}{2}$ we obtain the connection $\Gamma_{\rho\sigma}^{(0)\mu} = \frac{1}{2} e_m^\mu (\partial_\rho e_\sigma^m + \partial_\sigma e_\rho^m)$, which clearly is torsionless.⁸

As a check we can also start from $R\omega_\mu^m{}_n$ and compute $\Gamma_{\mu\nu}^\rho$ using the vielbein postulate. We should obtain the same result for $\Gamma_{\mu\nu}^\rho$ because $\Gamma_{\mu\nu}^\rho$ produced $L\omega_\mu^m{}_n$ in the e_m^μ frames and $R\omega_\mu^m{}_n$ in the f_m^μ frames. Using (33.39) and (33.41) one easily finds

$$R\omega_\mu^m{}_n = (\lambda - 1) f_\mu^k c_{kn}^m \quad (33.44)$$

The vielbein postulate yields

$$\Gamma_{\mu\nu}^\rho = f_m^\rho \left[\partial_\mu f_\nu^m + (\lambda - 1) f_\mu^k c_{kl}^m f_\nu^l \right] \quad (33.45)$$

and using

$$\begin{aligned} f_m^\rho \partial_\mu f_\nu^k &= e_m^\rho \partial_\nu e_\mu^k \\ f_\mu^k f_\nu^l c_{kl}^m f_m^\rho &= e_\mu^k e_\nu^l c_{kl}^m e_m^\rho \end{aligned} \quad (33.46)$$

we see that (33.45) agrees with (33.42).

⁷For $SU(N)$ there are two such tensors: the structure constants and the d -symbols. The first is antisymmetric, the second is symmetric. Up to now the connections Γ and $L\omega$ and $R\omega$ were the most general, only related by the vielbein postulate. In applications we shall view the $\omega_\mu^m{}_n$ connection as antisymmetric in m and n . Then (33.41) is unique.

⁸If c_{kn}^m is antisymmetric in m and n , $L\omega_\mu^m{}_n$ is the spin connection. If this is not the case, one can add terms to c_{kn}^m to obtain a modified $\bar{c}_{kn}^m$, and then $L\omega_{\mu m}^n = \gamma e_\mu^k \frac{1}{2} \bar{c}_{km}^n$ is the spin connection. We discuss this further in the next chapter.

So we see that on group manifolds we have “preferred connections”: left- and right-invariant connections. They are parametrized by a real parameter λ , and for $\lambda = 0, 1$ there is no curvature but (opposite) torsion, while for $\lambda = \frac{1}{2}$ there is no torsion, but curvature. Both the torsion and the curvature are expressed in terms of the structure constants, and with flat indices both the torsion and the curvature are constant.

$$T_{mn}{}^r = (2\lambda - 1)c_{mn}{}^r; \quad R_{rs}{}^m{}_n = \lambda(\lambda - 1)c_{rs}{}^k c_{kn}{}^m \quad (33.47)$$

Sweeping out left-invariant connections from the origin. We still have to prove (33.37). Consider ${}^L\omega_\mu{}^m{}_n(0)$. A vector ${}^Lv^m(0)$ is parallel-transported to dx^k to give

$${}^Lv^m(dx) = {}^Lv^m(0) - dx^k {}^L\omega_k{}^m{}_n {}^Lv^n(0) \quad (33.48)$$

Now act with $g(x)$ on ${}^Lv^m(0)$ and ${}^Lv^m(dx)$ to give

$${}^Lv^m(x) = {}^Lv^m(0); \quad {}^Lv^m(x + \Delta x) = {}^Lv^m(dx). \quad (33.49)$$

If Γ is an invariant connection for left-multiplication, the connection ${}^L\omega_\mu{}^m{}_n(x)$ follows from

$$\begin{aligned} {}^Lv^m(x + \Delta x) - {}^Lv^m(x) &= -\Delta x^\sigma \left({}^L\omega_\sigma{}^m{}_n(x) \right) \left({}^Lv^n(x) \right) \\ &= {}^Lv^m(dx) - {}^Lv^m(0) = -dx^k \left({}^L\omega_k{}^m{}_n(0) \right) \left({}^Lv^n(0) \right) \end{aligned} \quad (33.50)$$

So $\Delta x^\sigma {}^L\omega_\sigma{}^m{}_n(x) = dx^k {}^L\omega_k{}^m{}_n(0)$. But $\phi^\mu(x, dx) = x^\mu + dx^m e_m{}^\mu(x) = x^\mu + \Delta x^\mu$. Hence

$$\boxed{{}^L\omega_\sigma{}^m{}_n(x) = e_\sigma{}^k(x) {}^L\omega_k{}^m{}_n(0)} \quad (33.51)$$

It follows that any left-invariant connection with flat indices is constant. The same holds for any right-invariant connection.

33.I The Maurer-Cartan equations

In this appendix we derive the Maurer-Cartan (MC) equations for $e^m = da^\mu e_\mu{}^m(a)$ and $f^m = da^\mu f_\mu{}^m(a)$, and the fundamental relation $[e_m, f_n] = 0$ where $e_m = e_m{}^\mu \partial_\mu$ and $f_n = f_n{}^\nu \partial_\nu$. Define⁹ $g^{-1}dg = e^m T_m$ and act with d , using $dd = 0$ and $dg^{-1} = -g^{-1}dgg^{-1}$. One obtains

$$de^m T_m = -g^{-1}dgg^{-1}dg = -e^n T_n e^p T_p = -\frac{1}{2}e^n e^p [T_n, T_p] = -\frac{1}{2}c_{np}{}^m e^n e^p T_m, \quad (33.52)$$

⁹This definition of e^m agrees with the definition of $e_\mu{}^m$ in (33.5) since $g(a + \Delta a) = g(a)g(da)$ with $\Delta a^\mu = da^\mu e_m{}^\mu(a)$. Then $g(a)^{-1}(g(a + \Delta a) - g(a)) = g(da) - I$, or $g^{-1}dg = e^m T_m$.

where c_{np}^m are the structure constants.¹⁰ Hence $de^m + \frac{1}{2}c_{np}^m e^n e^p = 0$, or with $e = e^m T_m$, one finds $de + ee = 0$. Similarly, defining $dgg^{-1} = f^m T_m$ we find $df^m - \frac{1}{2}c_{np}^m f^n f^p = 0$ and $df - ff = 0$. Now consider $[e_m, e_n] = e_m^\mu (\partial_\mu e_n^\nu) \partial_\nu - e_n^\mu (\partial_\mu e_m^\nu) \partial_\nu$. Using $\partial_\mu e_m^\nu = -e_m^{\nu'} (\partial_\mu e_{\nu'}^{m'}) e_{m'}^{\nu'}$ one finds curls of e_μ^m which we can eliminate with the equation $de^m + \frac{1}{2}c_{np}^m e^n e^p = 0$. This yields $[e_m, e_n] = -c_{mn}^p e_p$. In total

$$\left. \begin{aligned} \partial_\mu e_\nu^m - \partial_\nu e_\mu^m + c_{rs}^m e_\mu^r e_\nu^s &= 0 \\ \partial_\mu f_\nu^m - \partial_\nu f_\mu^m - c_{rs}^m f_\mu^r f_\nu^s &= 0 \\ de^m + \frac{1}{2}c_{rs}^m e^r e^s &= 0; [e_m, e_n] - c_{mn}^c e_c = 0 \\ df^m - \frac{1}{2}c_{rs}^m f^r f^s &= 0; [f_m, f_n] + c_{mn}^c f_c = 0. \end{aligned} \right\} \text{ the MC equations} \quad (33.53)$$

We now derive the relation $[e_m, f_n] = 0$. As we shall discuss, it states that f_n^ν is a Killing vector of e_m^μ : if one moves in the direction of f_n^ν **and** changes the coordinates in a suitable way, the vielbein e_m^μ does not change. We begin by writing down the law of associativity for group multiplication with small a and c , but finite b

$$\phi^\mu(\phi(da, b), dc) = \phi^\mu(da, \phi(b, dc)) \quad (33.54)$$

Next we expand the left-hand-side (LHS) and right-hand-side (RHS) to first order in da and dc

$$\begin{aligned} \text{LHS} &= \phi^\mu(da, b) + dc^m e_m^\mu (\phi(da, b)) \\ &= b^\mu + da^n f_n^\mu + dc^m e_m^\mu (b + da^n f_n) \\ &= b^\mu + da^n f_n^\mu + dc^m e_m^\mu + dc^m (\partial_\rho e_m^\mu) da^n f_n^\rho \end{aligned} \quad (33.55)$$

$$\begin{aligned} \text{RHS} &= b^\mu + dc^m e_m^\mu + da^n f_n^\mu (b + dc^m e_m) \\ &= b^\mu + dc^m e_m^\mu + da^n f_n^\mu + da^n (\partial_\rho f_n^\mu) dc^m e_m^\rho \end{aligned} \quad (33.56)$$

Equating the LHS to RHS yields

$$f_n^\rho (\partial_\rho e_m^\mu) = e_m^\rho \partial_\rho f_n^\mu \quad (33.57)$$

Contracting with the derivatives ∂_μ yields the commutator of two tangent vectors

$$f_n e_m - e_m f_n = [f_n, e_m] = 0. \quad (33.58)$$

The meaning of this equation is that the f_n^μ are Killing vectors for e_m^μ as we now explain.

¹⁰Because f_μ^m denotes the right-invariant vielbeins, we do not employ the usual notation f_{mn}^p for the structure constants, but use instead c_{mn}^p .

Killing vectors. Killing vectors are usually introduced to describe symmetries (the more precise term is isometries) of the metric. The equation

$$g'_{\mu\nu}(x) = g_{\mu\nu}(x) \quad (33.59)$$

states that the metric $g'_{\mu\nu}$ in a new coordinate frame $\bar{x}^\rho$ is component-by-component equal to the metric $g_{\mu\nu}(x)$ in the old coordinates at points x which are in different frames (the $\bar{x}$ -frame and the x -frame) but with the same numerical values. So the points with coordinates x in $g'_{\mu\nu}(x)$ are different **physical** points from the points with coordinates x in $g_{\mu\nu}(x)$. (Note that the equation $g'_{\mu\nu}(x) = g_{\mu\nu}(x)$ is not the equation for a coordinate scalar; that equation reads $S'(x') = S(x)$.) One can now make the following set of steps

$$\begin{aligned} g'^{\mu\nu}(x') &= \frac{\partial x'^\mu}{\partial x^\alpha} \frac{\partial x'^\nu}{\partial x^\beta} g^{\alpha\beta}(x) \\ g^{\alpha\beta}(x) &= \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} g'^{\mu\nu}(x') \\ &= \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} g^{\mu\nu}(x') \end{aligned} \quad \begin{array}{l} \updownarrow \text{inverting the transformation} \\ \updownarrow \text{isometry} \end{array} \quad (33.60)$$

Setting $x'^\mu = x^\mu - \xi^\mu(x)$, we find for infinitesimal coordinate transformations

$$\frac{\partial \xi^\alpha}{\partial x^\mu} g^{\mu\beta} + \frac{\partial \xi^\beta}{\partial x^\nu} g^{\alpha\nu} - \xi^\alpha \partial_\alpha g^{\mu\nu}(x) = 0 \quad (33.61)$$

This equation contains the “Lie derivative of the (inverse) metric along ξ ”

$$\mathcal{L}_\xi g^{\mu\nu} = \xi^\alpha \partial_\alpha g^{\mu\nu} - \partial_\mu \xi^\alpha g^{\mu\beta} - \partial_\nu \xi^\beta g^{\alpha\mu}. \quad (33.62)$$

The equation $\mathcal{L}_\xi g^{\mu\nu} = 0$ can be rewritten as a covariant equation

$$-\mathcal{L}_\xi g^{\mu\nu} = D^\alpha \xi^\beta + D^\beta \xi^\alpha \quad (\text{Killing vector}) \quad (33.63)$$

For the vielbein fields $e_m{}^\mu$ we found the equation

$$(\partial_\rho f_n{}^\mu) e_m{}^\rho - f_n{}^\rho \partial_\rho e_m{}^\mu = 0 \quad (33.64)$$

which is the Lie derivative of $e_m{}^\mu$ along $\xi = f_n$

$$\mathcal{L}_\xi e_m{}^\mu = \xi^\rho \partial_\rho e_m{}^\mu - (\partial_\rho \xi^\mu) e_m{}^\rho. \quad (33.65)$$

The equation $\mathcal{L}_\xi e_m{}^\mu = 0$ can be rewritten as the following covariant equation

$$\mathcal{L}_\xi e_m = [\xi, e_m] \text{ with } \xi = \xi^\rho \partial_\rho \text{ and } e_m = e_m{}^\sigma \partial_\sigma. \quad (33.66)$$

The geometrical meaning of this equation is as follows: if one moves on the group manifold in the direction of f_m^μ , and at the same time changes coordinates as $\bar{x}^\mu = x^\mu - f_m^\mu \delta x^m$, then the components of the vielbein field in the new frame look exactly the same as in the old frame: we have a symmetry.

(We shall see that on coset manifolds, the equation $\mathcal{L}_\xi e_m^\mu = 0$ will be modified into

$$\mathcal{L}_\xi e_m^\mu = e_n^\mu \lambda_n^m$$

where λ_n^m are the parameters of the subgroup H of the coset manifold.)

We now get a deeper understanding of the symmetries of group manifolds. Each f_n^μ (for each n) corresponds to an isometry of the e_m^μ . It follows that if one f_m^μ leaves e_m^μ unchanged, and a second f_n^μ leaves it unchanged, then the commutator $[f_m, f_n]$ will leave it unchanged. This means that $[f_m, f_n]$ must be a linear combination of Killing vectors: $[f_m, f_n] = -c_{mn}^p f_p$. (The minus sign does not follow from this simple argument, but was explained before.) This is one of the MC equations (33.53), whose geometrical meaning we now have uncovered. But there is more. The equation $[f_n, e_m] = 0$ can also be written as $[e_m, f_n] = 0$ and this means that one can also view the left-invariant vielbein fields e_m^μ as Killing vectors of f_n^μ . Again the commutator $[e_m, e_n]$ must then be equal to a linear combination of the e_p , and this explains geometrically the MC equation $[e_m, e_n] = c_{mn}^p e_p$.

33.II The shadow of a group

Not all elements of a noncompact group can be written in exponential form as $g = e^{\lambda^a T_a}$ where λ^a are the real group parameters and T_a the group generators. The set of elements that cannot be written in exponential form is called the shadow of a group. We consider here an example, the group $Sl(2, R)$ of real 2×2 matrices with determinant one.

Any group element can be written as

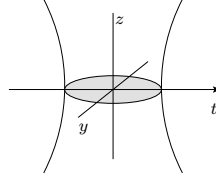
$$g = \begin{pmatrix} t+z & x-y \\ x+y & t-z \end{pmatrix} \quad \text{with } t^2 - z^2 - x^2 + y^2 = 1. \quad (33.67)$$

We choose as group parameters x, y, z , hence t is given by $t = \pm(1 + z^2 + x^2 - y^2)^{\frac{1}{2}}$. The group generators are $T_x = \sigma_1$, $T_y = -i\sigma_2$, $T_z = \sigma_3$, hence the subset of group elements in exponential form is given by

$$g = e^{a\sigma_1 - b i\sigma_2 + c\sigma_3} \quad \text{for real } a, b, c. \quad (33.68)$$

For $a^2 - b^2 + c^2 = -k^2 \leq 0$, these group elements can be written as $\cos k + \frac{1}{k}(a\sigma_1 - b i\sigma_2 + c\sigma_3) \sin k$. For $a^2 - b^2 + c^2 = k^2 \geq 0$, these group elements can be written as $\cosh k + \frac{1}{k}(a\sigma_1 - b i\sigma_2 + c\sigma_3) \sinh k$. For the first set $t = \cos k$ lies between -1 and $+1$. For the second set $t = \cosh k$ is always positive. Hence the shadow consists of all group elements with $t < -1$.

One can cover this group with two patches of elements, the first patch containing the elements in (33.68), and the second patch containing for example $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ times the elements in (33.68). These patches overlap in the region $-1 \leq t \leq 1$.



To get an idea about the shape of this group manifold, consider first the case of $x = 0$. Then $t^2 + y^2 - z^2 = 1$ forms a hyperboloid which is rotationally symmetric about the z -axis, with a “throat” which is narrowest at $z = 0$. If one adds $-x^2$, one fills in the area outside the hyperboloid. The shadow consists of all points with $t < -1$. This group manifold is “connected” (any two points can be connected by a curve) but not “simply-connected” (not every closed curve can be contracted to a point).

Chapter 34

Geometry of Coset Manifolds

Coset manifolds are a generalization of group manifolds. Their geometry has more structure because we have now two Lie groups, a group G and a subgroup H . Group manifolds correspond to the case $H = e$. On group manifolds we found the (inverse of) left-**in**variant vielbein fields e_m^μ and right-invariant vielbein fields f_m^μ , but on coset manifolds we shall find left-**c**ovariant vielbein fields e_m^μ (again square matrices) and Killing vectors k_M^μ (rectangular matrices with $M > m$). The Lie derivatives along these Killing vectors generate symmetry transformations. In addition there are an H -connection ω_μ^i and an H -compensator Ω_M^i which can be used to define covariant external derivatives D_m and covariant Lie derivatives $\mathcal{L}_{k_M}$. These two derivatives commute (because left-multiplication and right-multiplication commute in groups), and can be used to construct the transformation rules of matter fields on coset manifolds using the theory of induced representations.

In physics, one of the main applications of coset manifolds is to Kaluza-Klein dimensional reduction, where usually a $D + d$ dimensional spacetime is decomposed into a compact internal d dimensional space and a noncompact D dimensional spacetime. As internal space one often considers coset manifolds. Other important applications are the description of Minkowski space as the coset manifold Poincaré/Lorentz, superspace for rigid supersymmetry and supergravity as the coset manifold superPoincaré/Lorentz, and the relations anti-de Sitter (AdS) space = $SO(3, 2)/SO(3, 1)$ and de Sitter (dS) space = $SO(4, 1)/SO(3, 1)$, as well as superspaces for AdS and dS.

Coset manifolds arise as follows. Consider a Lie group G with group elements g that act on points x of a given carrier space Γ such that if $\tau(g)x$ denotes the action of g on x , then

$$\tau(g_1)\tau(g_2) = \tau(g_1g_2) \quad (34.1)$$

So the group G acts on the carrier space Γ , and the carrier space forms (“carries”) a representation of G . The set of all points x in Γ that can be transformed into each other by the group elements g of G forms an equivalence class¹, called a transitivity class. It follows that Γ splits up into

¹If $x \rightarrow \tau(g)x = y$ then x and y are said to be equivalent. An equivalence relation $x \sim y$ must satisfy the following three properties: 1) $x \sim x$, 2) if $x \sim y$ then $y \sim x$, 3) if $x \sim y$ and $y \sim z$ then $x \sim z$. Clearly the map $\tau(g)$ is an equivalence relation.

transitivity classes. The maps $\tau(g)$ map each transitivity class into itself in 1 – 1 fashion, hence each $\tau(g)$ effects a permutation of the points of each transitivity class. We restrict the discussion to one such transitivity class. So Γ is from now on a transitive carrier space for a group G .

Choosing a particular point y in Γ , the set of all group elements which leave y fixed form a group called the stability group (or isotropy group) H_y of the point y . That they form a group is clear: if $\tau(g_1)y = y$ and $\tau(g_2)y = y$ then also $\tau(g_1g_2)y = \tau(g_1)\tau(g_2)y = y$, so closure holds. Consider another point z in Γ , and let y be transformed to z by the group element g_z , so $\tau(g_z)y = z$. The stability subgroup H_z of z is then related to the stability subgroup H_y of y by a similarity transformation

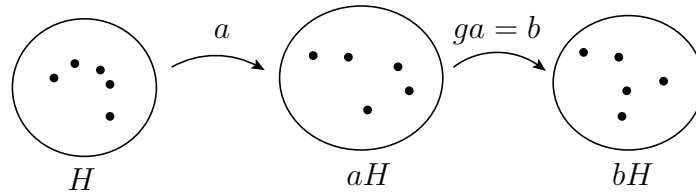
$$H_z = g_z H_y g_z^{-1} \quad (34.2)$$

because g_z^{-1} maps z back to y , and all group elements that keep y fixed form H_y , while finally g_z maps y back to z . The stability groups $H_y, H_z, \dots$ contain in general different group elements, but they are related by similarity transformations: they are equivalent (if $H_y \sim H_z$ and $H_z \sim H_w$ then $H_y \sim H_w$).

Next pick a particular point y_0 in Γ which we shall call the origin, and denote the stability subgroup of y_0 by H (instead of H_{y_0} to simplify the notation). Then we can construct the left-coset elements $H, aH, bH, \dots$ of G/H all with the same $H = H_{y_0}$. (We shall never consider right-coset elements $H, Hp, Hq, \dots$.) We claim that

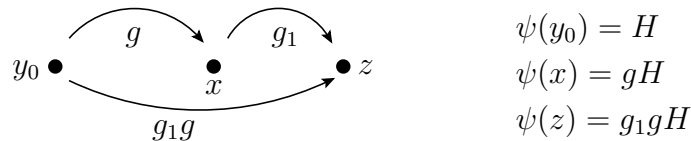
any transitive carrier space is (isomorphic to) a corresponding coset manifold G/H .

First notice that entire coset elements are moved to other entire coset elements: if $ga = b$, then $gaH = bH$ for all elements h in H . The action of G on the “points” $H, aH, bH, \dots$ is also transitive, so the coset manifold (the set of all points $H, aH, bH, \dots$) also forms a carrier space which is a transitivity class of G .



To each point x in Γ we associate a coset element $g_x H$ by requiring that if $\tau(g_x)y_0 = x$ for some group element g_x , then the coset element associated to x is $g_x H$. One may check this establishes a 1 – 1 map ψ from Γ onto G/H .

We now construct a map ψ that maps the action of G on Γ into action of G on G/H . Let $\tau(g)y_0 = x$, and $\tau(g_1)x = \tau(g_1)\tau(g)y_0 = \tau(g_1g)y_0 = z$. Then the corresponding coset elements are as follows



But then we have

$$\psi(\tau(g_1)x) = \psi(\tau(g_1g))y_0 = g_1gH = g_1\psi(x). \quad (34.3)$$

So the group action commutes with the map ψ , and thus we can identify points x in Γ with coset elements gH in G/H .

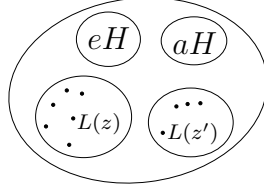


Figure 34.1: Cosets of G .

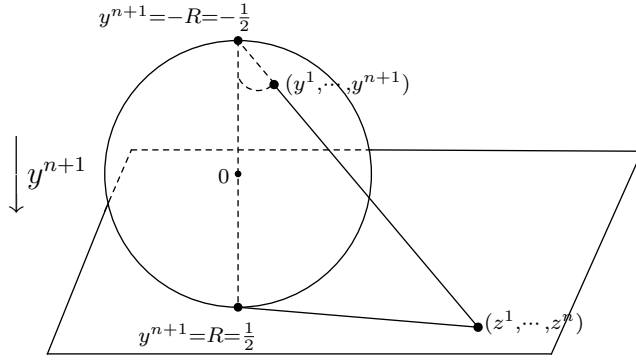
Cosets are disjoint, and every element of G lies in one and only one coset. To characterize cosets, one chooses in each coset a particular group element, and calls this group element the **coset representative**. For Lie groups, the coset representatives are denoted by $L(z)$ where $z = (z^1, \dots, z^d)$ are real coordinates in R^d which label the coset representatives, and hence the coset elements, in a 1 – 1 way. Thus $L(z)$ lies in G , and if $z \neq z'$, then $L(z) \neq L(z')$, and $L(z)$ and $L(z')$ lie in different cosets. Since one can always find a group element g that maps a coset representative $L(z)$ to any other coset representative, $gL(z) = L(z')$, one can also map any coset element $L(z)H$ to any other coset element $L(z')H$. Since all points of G/H are in this sense equivalent, one calls coset manifolds also **homogeneous spaces**. Homogeneous spaces need not be isotropic (invariant under rotations). If they are invariant under all possible rotations, such spaces are called maximally symmetric. We shall consider general homogeneous spaces and not make any further restrictions except reductivity (to be defined).

We will never need explicit expressions for $L(z)$ in a particular representation of G . Yet, in order to be able to deal abstractly with these $L(z)$, it is useful to have seen a few examples of how to obtain coset representatives for a given manifold G/H . Having gone through these examples, we shall then forget about them, and work afterwards with abstract objects $L(z)$.

34.1 First example of a coset manifold: the sphere S_n

As our first example of a coset manifold we consider S_n . As a particular set of coordinates on S_n we use stereographic coordinates: real z^a with $a = 1, \dots, n$. We view S_n as embedded in a Euclidean Cartesian space R^{n+1} with coordinates y^M with $M = 1, \dots, n+1$. In this space we consider the sphere S_n with radius R . Sometimes we shall put $R = \frac{1}{2}$. (This will give nice formulas; for example, we shall get a vielbein $e_\alpha^a(z) = \delta_\alpha^a(1+z^2)^{-1}$ instead of $\delta_\alpha^a(4R^2+z^2)^{-1}$.) We summarize the stereographic coordinates in the figure 34.2.

The relations between y^M and z^α follow from $\frac{y^\alpha}{z^\alpha} = \frac{1}{2R}(R+y^{n+1}) = \frac{1}{2}+y^{n+1}$, and squaring this relation. Note that for reasons to be explained the coordinate y^{n+1} runs in the opposite direction



$$\begin{aligned}
y^\alpha &= z^\alpha \frac{4R^2}{4R^2 + z^2} = \frac{z^\alpha}{1 + z^2}; \quad \alpha = 1, \dots, n \\
y^{n+1} &= R \frac{4R^2 - z^2}{4R^2 + z^2} = \frac{1}{2} \frac{1 - z^2}{1 + z^2} \\
y^\Pi y^\Pi &= R^2 = \frac{1}{4}; \quad \Pi = 1, \dots, n+1 \\
z^\alpha &= y^\alpha \frac{2R}{R + y^{n+1}}.
\end{aligned}$$

Figure 34.2: Stereographic coordinates on S_n .

from usual: $y^{n+1} = -R = -\frac{1}{2}$ at the north pole, and $y^{n+1} = R = \frac{1}{2}$ at the south pole. (One could, of course, turn the figure upside down; then the plane touches the sphere at the north pole where as usual $y^{n+1} = R = +\frac{1}{2}$.)

Before we begin our discussion of coset models, we want to compute the metric and corresponding vielbein on S_n by direct means. Evaluating $(ds)^2 = \sum_{M=1}^{n+1} (dy^M)^2$ after substituting the expressions for y^M , one finds after some algebra a very simple result²: $(ds)^2 = (dz^\alpha)^2 \frac{16R^4}{(4R^2 + z^2)^2}$. Hence the vielbein one-forms are $e^a = dz^\alpha \frac{4R^2}{4R^2 + z^2}$, or $e_\alpha^a = \delta_\alpha^a \frac{4R^2}{4R^2 + z^2}$ ($= \delta_\alpha^a \frac{1}{1 + z^2}$ if $R = \frac{1}{2}$), and the vielbein fields are the dimensionless fields e_α^a . We now turn to coset models.

"The sphere S_n is the coset manifold $SO(n+1)/SO(n)$." By this statement one means that one can decompose the set of all group elements of $SO(n+1)$ into nonoverlapping cosets $g_1 H, g_2 H, \dots$ with $H = SO(n)$ and $g_1 H, g_2 H, \dots$ corresponding to points on S_n , and any point on S_n corresponds to one of the $g_1 H, g_2 H, \dots$. We shall begin with the vector representation of $SO(n+1)$ which are $(n+1) \times (n+1)$ orthogonal matrices, and choose the following coset representatives

$$L_B^A(z) = \left(\begin{array}{c|c} \delta_b^a - \frac{y_b y^a}{R^2} \frac{1}{1 + \frac{y^{n+1}}{R}} & \frac{y_B}{R} \\ \hline -\frac{y^a}{R} & \end{array} \right) = \left(\begin{array}{cc} \delta_b^a - \frac{2z^b z^a}{4R^2 + z^2} & \frac{4Rz^b}{4R^2 + z^2} \\ -\frac{4Rz^a}{4R^2 + z^2} & \frac{4R^2 - z^2}{4R^2 + z^2} \end{array} \right) = \left(\begin{array}{cc} \delta_b^a - \frac{2z^b z^a}{1 + z^2} & \frac{2z^b}{1 + z^2} \\ -\frac{2z^a}{1 + z^2} & \frac{1 - z^2}{1 + z^2} \end{array} \right), \quad (34.4)$$

where $A, B = 1, \dots, n+1$ and $a, b = 1, \dots, n$. For $z = 0$ we get the unit element, and to linear order in z the matrix $L - I$ is antisymmetric. This suggests that these real matrices are orthogonal which one can check with some work

$$L^T = L^{-1} \quad (34.5)$$

Let us now check explicitly that any group element g in $SO(n+1)$ can always be written as a coset representative $L(z)$ as in (34.4) for some z , times an element h in $H = SO(n)$. We use the vector representation, so $g, L(z)$ and h are $(n+1) \times (n+1)$ matrices. Given g , we now show how

²Another way to obtain coordinates on S_n is to solve for y^{n+1} from $\sum_{j=1}^{n+1} (y^j)^2 = R^2$. If one then evaluates $\sum_{M=1}^{n+1} (dy^M)^2$, one obtains a metric that is not diagonal. For that reason stereographic coordinates are preferred.

to construct $L(z)$ and h . We begin by assuming that g, L and h have the following form

$$g = \begin{pmatrix} A & B \\ C & D \end{pmatrix}; \quad L = \begin{pmatrix} M & 2y^1 \\ & \vdots \\ W & 2y^{n+1} \end{pmatrix}; \quad h = \begin{pmatrix} P & 0 \\ 0 & 1 \end{pmatrix} \quad (34.6)$$

where $2y^k = \frac{y^k}{R}$ for $R = \frac{1}{2}$, and A, M and P are $n \times n$ matrices. In order that $g = Lh$, we select a matrix L which has the same last column as g . Then we fix h by requiring that $L^{-1}g$ be equal to h . Since $L^{-1} = L^T$, the expression for $L^{-1}g$ is given by $L^T g = \begin{pmatrix} M^T & W^T \\ B^T & D \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix}$. Because the columns (AC) in g are orthogonal to the column BD and normalized, the bottom row of h is as in (34.6). Similarly, because the columns of L are orthonormal, the last column of h is as in (34.6). Since $L^T g$ is orthogonal, this proves that h is of the form in (34.6) with P in $SO(n)$. The matrix M and the row W form $n - 1$ columns which must be chosen to be orthonormal and orthogonal to the last column of L . This can be done in many ways, for example as in (34.4). Then g has indeed been decomposed into $L(z)h$.

The south pole $y^A = (0, 0, \dots, R)$ is mapped by L in (34.4) to the point $(y^1, \dots, y^{n+1})$ on S_n , as is clear from $\frac{z^b}{1+z^2} = y^b$ and $\frac{1}{2} \frac{1-z^2}{1+z^2} = y^{n+1}$. (If we had chosen $y^{n+1} = -R = -\frac{1}{2}$ at the south pole, we would have obtained $\frac{z^2-1}{z^2+1}$ in the lower right corner, and then we would not have gotten $L = I$ for $z = 0$.) Thus every matrix L corresponds to a point on S_n . Conversely, given a point $(y^1, \dots, y^{n+1})$ on S_n , the corresponding matrix L is the matrix in (34.4).³ Furthermore, all group elements of $SO(n+1)$ which keep the south pole fixed form the subgroup $SO(n)$. Thus S_n corresponds to the coset manifold $SO(n+1)/SO(n)$.

Matrix representations of compact groups such as $SO(n+1)$ can always be written as the exponent of an element of the Lie algebra. In particular the coset representative L in (34.4) can be written in exponential form as

$$L = e^{(\frac{1}{2}v^a K_a)}; \quad v^a = \frac{z^a}{z} \arcsin \left(\frac{2z}{1+z^2} \right) \quad (34.7)$$

where K_a are antisymmetric $(n+1) \times (n+1)$ matrices with only an entry $+2(-2)$ in the last column (row), see (34.11).⁴ To show this, diagonalize the antisymmetric matrix $v^a K_a$ and exponentiate

³This is not quite true. There is no matrix L which corresponds to the north pole ($|z| \rightarrow \infty$). One can, however, introduce two patches which cover the sphere, one without the north pole and the other without the south pole. The second patch corresponds to the stereographic projection onto a plane which touches S_n at the north pole. Then the 1-1 correspondence between points on S_n and coset representatives is restored.

⁴We have $L = e^{Rv^a K_a}$ and K_a has entries $\frac{1}{R}$ and $-\frac{1}{R}$, so that v^a are dimensionless. To check (34.7) it is useful to write $\frac{1}{2}v^a K_a$ as $\begin{pmatrix} 0 & \vec{v} \\ -\vec{v}^T & 0 \end{pmatrix}$. Then $(\frac{1}{2}v^a K_a)(\frac{1}{2}v^b K_b)$ is the matrix $\begin{pmatrix} v v^T & 0 \\ 0 & v^2 \end{pmatrix}$. For simplicity of notation we have set $R = \frac{1}{2}$.

(or, more easily, determine the first few terms in the expansion of $e^{\frac{1}{2}v^a K_a}$). One finds then

$$\left(e^{\frac{1}{2}v^a K_a}\right)_A^B = \begin{pmatrix} \cos \sqrt{vv^T} & v^a \frac{\sin v}{v} \\ -v^a \frac{\sin v}{v} & \cos v \end{pmatrix}; \quad v^2 = \sum_{n=1}^n (v^a)^2; \quad v = \sqrt{v^2}. \quad (34.8)$$

(The expression vv^T denotes the matrix obtained by multiplying the vector and its transpose v^T . The indices Π of y^Π are curved vector indices and A, B or M, N are flat vector indices in R^{n+1} . Further a, b or m, n are flat vector indices on the coset, and α, β or μ, ν are curved coset indices.) Setting $\frac{v^a \sin v}{v} = \frac{y^a}{R} = \frac{2z^a}{1+z^2}$ yields then $\sin v = \frac{2z}{1+z^2}$, while $\frac{v^a}{v} = \frac{z^a}{z}$ so $v^a = \frac{z^a}{z} \arcsin\left(\frac{2z}{1+z^2}\right)$. Also $\cos v = \frac{(1-z^2)}{(1+z^2)}$, and $\cos \sqrt{vv^T} = I + (\cos v - 1) \frac{vv^T}{v^2} = I - \frac{2zz^T}{1+z^2}$, proving the exponentiation. We shall not use this exponential form but directly work with (34.4).

The coset vielbein $e_\alpha^a(z)$ and coset H -connection $\omega_\alpha^{ab}(z)$ are by definition given in terms of L by⁵

$$L^{-1}dL = e^a K_a + \frac{1}{2}\omega^{ab} H_{ab}, \quad (34.9)$$

where K_a are the coset generators and H_{ab} the generators of the subgroup $SO(n)$ in the vector representation. (In general one writes $\omega^i H_i$, but for $SO(n)$ the generators H_i can be labeled by H_{ab} with $H_{ab} = -H_{ba}$, and then one needs the factor $\frac{1}{2}$ in $\frac{1}{2}\omega^{ab} H_{ab}$ to avoid double counting.) We shall show that in this example the H -connection is equal to the spin connection, but this is not true in general. Direct but tedious evaluation of $L^{-1}dL$, using $L^{-1} = L^T$, yields

$$(L^{-1}dL)_A^B = \begin{pmatrix} \frac{2z^a dz^b - 2z^b dz^a}{4R^2 + z^2} & \frac{4R dz^a}{4R^2 + z^2} \\ -\frac{4R dz^b}{4R^2 + z^2} & 0 \end{pmatrix} \quad (34.10)$$

We choose the $(n+1) \times (n+1)$ generators K_a and H_{ab} in the vector representation as follows: as we already mentioned, $K_a^{(v)}$ has only one entry in the last row and column

$$(K_a^{(v)})_B^C = 2(\delta_{aB}\delta_{n+1}^C - \delta_a^C \delta_{n+1,B}); \quad K_a = \left(\begin{array}{cccc|c} & & & & 0 \\ & & & & \vdots \\ & & & & 0 \\ & & 0 & & 2 \\ & & & & 0 \\ & & & & \vdots \\ \hline 0 & \dots & 0 & -2 & 0 & \dots & 0 \end{array} \right) \quad (34.11)$$

⁵It is well-known that in Yang-Mills theory $g^{-1}dg$ lies in the Lie algebra. We shall anyhow prove this later.

whereas $H_{ab}^{(v)}$ has only two entries in the first $n \times n$ block

$$(H_{ab}^{(v)})_B^C = \delta_{aB}\delta_b^C - \delta_a^C\delta_{bB}; \quad H_{ab} = \left(\begin{array}{ccc|cc} 0 & \cdots & 0 & 0 & 0 \\ & & 1 & & \\ & & \vdots & & \vdots \\ & -1 & & & \\ \hline 0 & \cdots & 0 & 0 & 0 \\ 0 & \cdots & 0 & 0 & 0 \end{array} \right) \quad (34.12)$$

These matrices satisfy the following commutation relations

$$\begin{aligned} [K_a, K_b] &= -4H_{ab} \left(= -\frac{1}{R^2}H_{ab} \right) \\ [H_{ab}, H_{cd}] &= \delta_{bc}H_{ad} + 3 \text{ terms} \\ [H_{ab}, K_c] &= \delta_{bc}K_a - \delta_{ac}K_b \end{aligned} \quad (34.13)$$

These are the commutation relations of $SO(n+1)$ with $K_a = 2H_{a,n+1}$. It is clear that the coset manifold $SO(n+1)/SO(n)$ is “reductive” ($[H, K] \subset K$, which means that the coset generators K form a representation of the subgroup H), and is “symmetric” (meaning that there are no terms with K in the $[K, K]$ commutators).

Equating $(L^{-1}dL)_a^{n+1} = \frac{4Rdz^a}{4R^2+z^2}$ to $e^b(K_b)_a^{n+1} = \frac{e^a}{R}$ yields the group vielbein in stereographic coordinates

$$e_\alpha^a = \delta_\alpha^a \frac{4R^2}{4R^2+z^2} \left(= \frac{\delta_\alpha^a}{1+z^2} \right) \quad (34.14)$$

Similarly, equating $(L^{-1}dL)_a^b = \frac{2(z^a dz^b - z^b dz^a)}{(1+z^2)}$ to $(L^{-1}dL)_a^b = \frac{1}{2}\omega^{cd}(H_{cd})_a^b = \omega_a^b$ yields the H -connection on S_n

$$\omega_\alpha^{ab} = \frac{-2\delta_\alpha^a z^b + 2\delta_\alpha^b z^a}{4R^2+z^2} \left(= \frac{-2\delta_\alpha^a z^b + 2\delta_\alpha^b z^a}{1+z^2} \right) \quad (34.15)$$

(We chose the normalization and signs of K_a and H_{ab} as in (34.11) and (34.12) to get simple formulas for $R = \frac{1}{2}$. Note that from their definition in (34.9) the coset vielbeins e_α^a are dimensionless and ω_α^{ab} have the dimension of an inverse length. The vielbein we obtained earlier by direct computation of $(ds)^2$ is also dimensionless, and agrees with (34.14).

One may check that

$$D_\alpha e_\beta^a - D_\beta e_\alpha^a = 0; \quad D_\alpha e_\beta^a = \partial_\alpha e_\beta^a + \omega_\alpha^a{}_b e_\beta^b. \quad (34.16)$$

This confirms that in this example the H -connection $\omega_\alpha^{ab} = -\omega_\alpha^{ba}$ is the spin connection. (The spin connection has no torsion; it can be expressed in terms of the vielbeins and curls of vielbeins.)

The Riemann curvature on S_n is found to be

$$\begin{aligned}
R_{\alpha\beta}{}^{ab}(\omega) &= \partial_\alpha \omega_\beta{}^{ab} + \omega_\alpha{}^a \omega_\beta{}^c \omega_\gamma{}^b - \alpha \leftrightarrow \beta = 4(\delta_\alpha{}^a \delta_\beta{}^b - \delta_\alpha{}^b \delta_\beta{}^a) \frac{4R^2}{(4R^2 + z^2)^2} \\
&= \frac{1}{R^2} (e_\alpha{}^a e_\beta{}^b - e_\alpha{}^b e_\beta{}^a) \quad (= 4(\delta_\alpha{}^a \delta_\beta{}^b - \delta_\alpha{}^b \delta_\beta{}^a) \frac{1}{(1+z^2)^2}) \\
R_{abcd}(\omega) &= \frac{1}{R^2} (\delta_{ac} \delta_{bd} - \delta_{ad} \delta_{bc}) \quad (= 4(\delta_{ac} \delta_{bd} - \delta_{ad} \delta_{bc})) .
\end{aligned} \tag{34.17}$$

The Ricci tensor and scalar curvature are defined and given by⁶

$$R_{ab} = R_{ca}{}^c{}_b(\omega) = \frac{1}{R^2} (n-1) \delta_{ab}; \quad R = \delta^{ab} R_{ab} = \frac{1}{R^2} n(n-1). \tag{34.18}$$

So, for S_2 the scalar curvature is equal to $\frac{2}{R^2}$.

34.2 Example 2: Complex projective space CP^n

Consider points in complex $(n+1)$ dimensional space C^{n+1} (not the origin). Points in CP^n are per definition the set of points in $C^{n+1}/(0)$ which are equivalent by complex rescalings. Thus, “complex rays” in C^{n+1} correspond to points in CP^n . Or, equivalently, the points $(z^1, \dots, z^{n+1})y$ for all complex $y \neq 0$ are identified as one point in CP^n . The space $C^{n+1}/(0)$ splits into points of CP^n .

The $(n+1) \times (n+1)$ matrices of $SU(n+1)$ act not only in C^{n+1} , but also in CP^n as follows. Let s be in $SU(n+1)$. Then, if s maps z into sz , it maps yz into $y(sz)$. Hence, s maps a point of CP^n (namely, the collection of all points yz with all possible y) into another point of CP^n (namely, the collection of all $y'(sz)$ with all possible y'). The action of s is transitive: one can go from any point in CP^n to any other point by a suitable $SU(n+1)$ matrix. Thus, CP^n is a carrier space on which $SU(n+1)$ acts transitively.

To identify points of CP^n with a coset manifold, we again first determine the isotropy group H of a point p in CP^n . We thus ask: which $SU(n+1)$ matrices keep a point p in C^{n+1} fixed up to an overall complex scale factor?

Let us, for convenience, consider the point $(0, 0, \dots, 0, 1)$ in C^{n+1} . Then the collection of $SU(n+1)$ matrices which maps this point into itself times any phase factor, is given by⁷

$$\begin{pmatrix} “SU(n)” e^{i\phi} & 0 \\ 0 & e^{-ni\phi} \end{pmatrix} = \begin{pmatrix} “SU(n)” & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} e^{i\phi} \mathbb{1}_n & 0 \\ 0 & e^{-ni\phi} \end{pmatrix} \tag{34.19}$$

⁶In the supergravity community one usually defines the Ricci tensor by $R_{ab} = R_{ac}{}^c{}_b(\omega) = R_{acb}{}^c(\Gamma)$, but in General Relativity one defines R_{ab} with opposite sign. Then the scalar curvature on spheres is positive in General Relativity but negative in supergravity. We follow the conventions of supergravity.

⁷Repeating the construction of (34.6) for $SU(n+1)$, we take the last column of L again equal to the last column of g , but now $D = 2y^{n+1}$ acquires a phase, $D = \frac{1-|z|^2}{1+|z|^2} e^{-in\phi}$. In order that $\det L = 1$, M and W acquire a phase $e^{i\phi}$. Extracting these phase factors yields (34.19).

where “ $SU(n)$ ” denotes any $SU(n)$ matrix. Hence, the isotropy group of any point in CP^n is $SU(n) \times U(1)$. It follows that CP^n is the coset manifold $SU(n+1)/SU(n) \times U(1)$.

$$CP^n = SU(n+1)/SU(n) \times U(1) \quad (34.20)$$

The next problem to be solved is how to parametrize these coset elements. Since an element of $SU(n+1)/SU(n) \times U(1)$ is described by $2n$ real parameters, it is natural to look for a complex vector $(z^1, \dots, z^n)$. Since antipodal points $(y^1, \dots, y^{n+1})$ in C^{n+1} correspond to the same point in CP^n , we could pick as coset representative a point on the complex n -sphere in C^{n+1} whose $(n+1)$ -th coordinate is real and positive (or negative). We can also use complex stereographic coordinates as follows. Write a complex ray in C^{n+1} as

$$(w^1, w^2, \dots, w^n, \frac{1}{2}(1 - |w|^2)) (1 + |w|^2)^{-1} 2y \quad (34.21)$$

where $|w|^2 \equiv |w^1|^2 + \dots + |w^n|^2 \leq 1$. By rescaling, this can always be achieved. For $y = 1$, this is a point on the complex n -sphere “above the equator”. The point in CP^n which corresponds to this ray is now parametrized by $(w^1, \dots, w^n)$. Conversely, given $(w^1, \dots, w^n)$ with $|w|^2 \leq 1$, we can determine the corresponding point in CP^n by the above formula. Hence, points in CP^n are in a 1-1 way parametrized by $(w^1, \dots, w^n)$. (For points with $|w|^2 = 1$, one must make an additional convention which select one of the two antipodal points.) Thus, unlike in the case of S_n , one patch covers the whole of CP^n , but not in a continuous way at $|w|^2 = 1$. (The reason this is possible is that in CP^n antipodal points are identified.)

Finally, we must find coset representatives. As in the case of the real sphere, we look for an $SU(n+1)$ matrix which maps the north=south pole in CP^n to the point p in CP^n . Hence, in particular, we need an $SU(n+1)$ matrix which maps the south pole in C^{n+1} with coordinates $(0, \dots, 0, \frac{1}{2})$ into the point $(w^1, \dots, w^n, \frac{1}{2}(1 - |w|^2)) (1 + |w|^2)^{-1}$. Clearly, one such $SU(n+1)$ matrix is:

$$L(w^1, \dots, w^n) = \begin{pmatrix} \delta^{\alpha\beta} - \frac{2w^\alpha(w^\beta)^*}{1+|w|^2} & \frac{2w^\alpha}{1+|w|^2} \\ -\frac{2(w^\beta)^*}{1+|w|^2} & \frac{1-|w|^2}{1+|w|^2} \end{pmatrix}. \quad (34.22)$$

Comment: The reader may wonder why we did not simply choose coset representatives elements in G of the form $e^{z^\alpha K_\alpha}$, where the K_α form a basis for those generators in G which do not lie in H . The answer is that, as in the case of $SO(n+1)/SO(n)$, it is very hard to find the explicit form of such matrices. We will sometimes use the exponential parametrization of such coset representations for general discussions, but if one needs explicit expressions for $L(z)$ (for example, if one wants to evaluate the derivative of $L(z)$), the exponential representations are not very useful.

34.3 Reductivity and the freedom in choosing generators

We shall only consider reductive cosets G/H . Given a set of generators H_i which span the algebra of H , we call G/H reductive if one can find a basis of the remaining generators of G , called K_α , such that $[H, K] \subset K$. Thus the algebras we consider have the structure

$$\begin{aligned} [H_i, H_j] &= c_{ij}^k H_k & (H \text{ is subgroup}) \\ [H_i, K_\alpha] &= c_{i\alpha}^\beta K_\beta & (\text{reductivity}) \\ [K_\alpha, K_\beta] &= c_{\alpha\beta}^i H_i + c_{\alpha\beta}^\gamma K_\gamma & (\text{general}) \end{aligned} \quad (34.23)$$

If $c_{\alpha\beta}^\gamma$ would vanish, we would have a symmetric algebra, but we shall not restrict our attention to symmetric algebras.

Given a reductive split H_i, K_α of the generators of G , one can choose new bases **inside** the linear vector spaces spanned by H_i and K_α , respectively; the new split $\{H'_i = M_i^j H_j, K'_\alpha = N_\alpha^\beta K_\beta\}$ will again be reductive. Thus reductivity does not depend on the choice of basis inside $\{H_i\}$ and $\{K_\alpha\}$. We do **not** require that the H_i and K_α are orthogonal with respect to the Killing metric $(\gamma^{-1})^{AB}$

$$\gamma_{AB} = -c_{DA}^E c_{EB}^D \quad (34.24)$$

For one reason, γ^{-1} may not exist; furthermore rather than restrict the cases covered, it will be our strategy to keep as much generality as possible.

If one wants compact manifolds G/H (meaning that one can find a set of coordinates z parametrizing the coset representatives which belong to a compact space), both G and H must be compact groups. For the K -generators, it is intuitively clear that they must be compact: if one would replace K_α by iK_α , the corresponding coordinates z^α would range over an infinite domain. (cf. Fourier series e^{iz} with $0 \leq z \leq 2\pi$ and e^z with $-\infty \leq z \leq \infty$). However, also H must be compact; one can thus not “divide out” the noncompactness such that G/H is compact although G and H are noncompact. Also this is clear: if $[H, K] \subset K$, with compact H_i and K_α , then by multiplying some H -generators by i , also some K generators must become noncompact (structure constants are always real).

It is a theorem that whenever H is compact or semi-simple (even if G is noncompact) one can always find a set of $\{K_\alpha\}$ such that $\{H_i, K_\alpha\}$ is reductive.

Not all interesting algebras are reductive, for example the conformal group with H spanned by all generators except the translations P_m , is not reductive. (The commutator of P_m with conformal boosts is not proportional to P_m , but to dilations and Lorentz generators.)

Given a reductive split $\{H_i, K_\alpha\}$, what freedom does one have in choosing K_α (in addition to basis changes inside $\{K_\alpha\}$)? The new coset generators are

$$\hat{K}_\alpha = K_\alpha + d_\alpha^j H_j \quad (34.25)$$

and requiring reductivity one finds

$$[H_i, \hat{K}_\alpha] = c_{i\alpha}{}^\beta \hat{K}_\beta = c_{i\alpha}{}^\beta (K_\beta + d_\beta{}^j H_j) = c_{i\alpha}{}^\beta K_\beta + d_\alpha{}^j c_{ij}{}^k H_k \quad (34.26)$$

Hence, the matrix d must be an interpolating matrix between the adjoint representation of H , given by $(H_i^{\text{adj}})_j{}^k = -c_{ij}{}^k$, and the coset representation of H , given by $(H_i^{\text{cos}})_\alpha{}^\beta = -c_{i\alpha}{}^\beta$. In matrix notation: $dH_i^{\text{adj}} = H_i^{\text{cos}}d$ for all i . According to Schur's lemma, d can only be nonzero if the adjoint and coset representations of H have one or more irreducible representations of H in common.

Example 1: $SO(n+1)/SO(n)$. Taking as coset generators those matrices with nonzero entries in only the $(n+1)$ row and column, the $\{K_\alpha\}$ transform under H as vectors. Hence, the coset representation is irreducible (“Isotropy irreducible”). Since the adjoint representation of any simple group is irreducible, also the adjoint representation of $SO(n)$ is irreducible. Since these two irreps are inequivalent, $d = 0$ and the choice of $\{K_a\}$ is unique.

Example 2: $SO(5)/SO(3)$ From the root diagram for $SO(5)$, one sees that there are two ways to embed $SO(3)$ into $SO(5)$: along the diagonal or horizontally. In the first case the K_α are unique, in the second case there is a freedom.

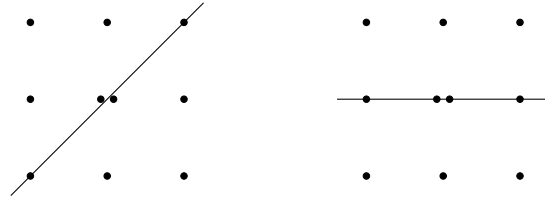


Figure 34.3: Two ways to embed $SO(3)$ into $SO(5)$.

34.4 Motion along Killing vectors and H -compensators

Given coset representatives $L(z)$ of a coset manifold G/H , one can multiply a given $L(z)$ by a group element g from the left. Since $L(z)$ is an element of G , the product lies again in G , hence in some coset, whose representative we denote by $L(z')$. Hence

$$gL(z) = L(z')h(z, g). \quad (34.27)$$

Any other element in the same coset as $L(z)$ is of the form $L(z)h$, and is moved by g to $L(z')h(z, g)h$. Hence, all points of a given coset move to only one other coset: **we have a motion on coset space**. In particular, given two coset elements aH and bH , one can move from one to the other by $g = ba^{-1}$. Hence, the action of G on G/H is transitive and the coset manifold is **homogeneous**.

If g is near the identity, also z' will be near z , and h will be near the identity element of H . Let us expand as follows

$$g = 1 + \epsilon^M T_M, \quad (z')^\alpha = z^\alpha + \epsilon^M k_M^\alpha(z), \quad h = 1 - \epsilon^M \Omega_M^i(z) H_i \quad (34.28)$$

where T_M are the generators of G , hence $T_M = \{H_i, K_\alpha\}$. **We will only consider constant ϵ^M .** In this case we find from (34.27) and (34.28)

$$L(z') - L(z) = \epsilon^M k_M^\alpha \partial_\alpha L(z) = \epsilon^M (T_M L(z) + L(z) \Omega_M^i H_i). \quad (34.29)$$

The contravariant vectors $k_M^\alpha(z)$ are called **Killing vectors**, and will be shown later to be indeed Killing vectors. The $\Omega_M^i(z)$ will be called **H -compensators** for a reason to be explained later.

From associativity, $g_1(g_2 L) = (g_1 g_2) L$, we obtain two results⁸ because the decomposition of a group element into $L(z)h$ is unique. From the terms with $L(z)$ one finds

$$[k_M, k_N] = -c_{MN}^P k_P, \quad k_M \equiv k_M^\alpha \partial_\alpha \quad (34.30)$$

where c_{MN}^P are the structure constants of G

$$[T_M, T_N] = c_{MN}^P T_P. \quad (34.31)$$

From the terms with H elements we find the relation

$$k_M^\alpha \partial_\alpha \Omega_N^i - k_N^\alpha \partial_\alpha \Omega_M^i = -c_{MN}^P \Omega_P^i - c_{jk}^i \Omega_M^j \Omega_N^k. \quad (34.32)$$

Equation (34.30) is nothing but the Cartan-Maurer equations for the right-vielbeins of the group G (the left-vielbeins have no minus sign in (34.30)), while, as we shall see, (34.32) is the integrability condition for the H -covariant Lie derivatives.

If $L(z)$ is given by $e^{z^\alpha K_\alpha}$ (a very particular and not very convenient parametrization for explicit calculations, as we discussed before) one may rewrite (34.27) as

$$e^{\epsilon^M T_M} e^{z^\alpha K_\alpha} = e^{(z^\alpha + \epsilon^M k_M^\alpha(z)) K_\alpha} e^{-\epsilon^M \Omega_M^i(z) H_i}. \quad (34.33)$$

We shall often use this notation for general discussions.

⁸Use that $g_1(g_2 L(z))$ is equal to $(g_1(L(z + k_2(z))))h(z, g_2)$, which is equal to $L(z + k_z(2) + k_1(z + k_2(z)))$ times $h(z + k_z(z), g_1)h(z, g_2)$. Then expand and antisymmetrize.

34.5 Vielbeins, H -connections and Lie derivatives

Consider the matrix $L^{-1}(z)\partial_\alpha L(z)$. Since $L(z)$ lies in G , $L^{-1}\partial_\alpha L(z)$ lies in the algebra of G . Hence, at each point z we can decompose $L^{-1}\partial_\alpha L$ into part along H and a part along K .

$$L^{-1}(z)\partial_\alpha L(z) = e_\alpha^a(z)K_a + \omega_\alpha^i(z)H_i. \quad (34.34)$$

The $e_\alpha^a(z)$ are called **vielbeins** and the $\omega_\alpha^i(z)$ are called **H -connections**.

If $L(z)$ is of the form $e^{z^\alpha K_\alpha}$, then the vielbeins and H -connections are obtained by left multiplications on an infinitesimal element

$$e^{z^\alpha K_\alpha} e^{dz^a K_a} = e^{z^\alpha K_\alpha + dz^a e_\alpha^a(z) K_a} e^{-dz^a e_\alpha^a(z) \omega_\alpha^i(z) H_i} \quad (34.35)$$

Indeed, denoting $dz^a e_\alpha^a = \Delta z^\alpha$, we rewrite this equation as

$$e^{\Delta z^\alpha e_\alpha^a K_a} = L^{-1}(z) L(z + \Delta z) e^{-\Delta z^\alpha \omega_\alpha^i H_i}. \quad (34.36)$$

Subtracting the unit element, the terms to first order in Δz yields

$$\Delta z^\alpha e_\alpha^a K_a + \Delta z^\alpha \omega_\alpha^i H_i = L(z)^{-1} (L(z + \Delta z) - L(z)) \quad (34.37)$$

This proves (34.34).

From (34.34) and $gL(z) = L(z')h(z, g)$ we find

$$\begin{aligned} L^{-1}(z')\partial'_\alpha L(z') - L^{-1}(z)\partial_\alpha L(z) &= (hL^{-1}(z)) \partial'_\alpha (L(z)h^{-1}) - L^{-1}(z)\partial_\alpha L(z) \\ &= [e_\alpha^a(z')K_a + \omega_\alpha^i(z')H_i] - [e_\alpha^a(z)K_a + \omega_\alpha^i(z)H_i]. \end{aligned} \quad (34.38)$$

Note that the constant g and g^{-1} have cancelled.

Since $\partial'_\alpha = \frac{\partial}{\partial z'^\alpha} = \partial_\alpha - (\partial_\alpha k^\beta(z)) \partial_\beta$ where $z'^\alpha = z^\alpha + k^\alpha(z)$, we find

$$l_{k_M} e_\alpha^a(z) = e_\alpha^b(z) c_{bi}^a \Omega_M^i(z) = \delta_H(\Omega_M^i) e_\alpha^a \quad (34.39)$$

$$l_{k_M} \omega_\alpha^i(z) = \omega_\alpha^j(z) c_{jk}^i \Omega_M^k(z) + \partial_\alpha \Omega_M^i(z) = \delta_H(\Omega_M^i) \omega_\alpha^i \quad (34.40)$$

where l_k are the **Lie derivatives along k** , defined by

$$\begin{aligned} l_{k_M} e_\alpha^a &= k_M^\beta \partial_\beta e_\alpha^a + (\partial_\alpha k_M^\beta) e_\beta^a \\ l_{k_M} \omega_\alpha^i &= k_M^\beta \partial_\beta \omega_\alpha^i + (\partial_\alpha k_M^\beta) \omega_\beta^i. \end{aligned} \quad (34.41)$$

In general, we define the Lie derivative of a scalar by $l_k s = k^\alpha \partial_\alpha s$, and for tensors l_k has the form of a general coordinate transformation (with transport term) with parameter k^α .

We can introduce differential forms. The exterior derivative $d = dz^\alpha \partial_\alpha$ satisfies $dd = 0$, and

the one forms $e^a \equiv e_\alpha^a dz^\alpha$ and $\omega^i \equiv \omega_\alpha^i dz^\alpha$ satisfy

$$L^{-1}dL = e^a K_a + \omega^i H_i. \quad (34.42)$$

We can also consider $L^{-1}(z')dL(z')$ but we must relate the dz'^α to dz^α in agreement with $z'^\alpha = z^\alpha + k^\alpha$, namely

$$dz'^\alpha = dz^\alpha + dk^\alpha(z) = dz^\alpha + dz^\beta \partial_\beta k^\alpha. \quad (34.43)$$

In that case one has

$$L^{-1}(z')dL(z') - L^{-1}(z)dL(z) = (e^a(z') - e^a(z)) K_a + (\omega^i(z') - \omega^i(z)) H_i. \quad (34.44)$$

It is convenient to introduce also **the Lie derivative of forms**. For a covariant vector v_α , the Lie derivative can be written as the sum of a curl and a gradient

$$k^\beta \partial_\beta v_\alpha + (\partial_\alpha k^\beta) v_\beta = k^\beta (\partial_\beta v_\alpha - \partial_\alpha v_\beta) + \partial_\alpha (k^\beta v_\beta). \quad (34.45)$$

For a one-form $v \equiv v_\alpha dz^\alpha$, the Lie derivative l_k can be written as, $\iota_k d + d\iota_k$

$$l_k v = \iota_k dv + d\iota_k v \quad (34.46)$$

where the symbol ι_k replaces each dz^α in a p -form by k^α , taking into account minus signs which are caused by bringing a particular dz^α to the left. Further $d = dz^\alpha \partial_\alpha$ acts per definition on everything to its right. Note that for a one-form v according to (34.41) and (34.46)

$$l_k v = (l_k v_\alpha) dz^\alpha. \quad (34.47)$$

In other words, if we define $l_k dz^\alpha = 0$, then the Leibniz rule holds. For p -forms we define the Lie derivatives by (34.46), too. Note that $e^a(z') - e^a(z) = l_k e^a$ for infinitesimal $z' - z = k$. The change in vielbeins and H -connections by motions along Killing vectors is thus given by

$$l_{k_M} e^a(z) = e^b(z) c_{bi}^a \Omega_M^i(z) \quad (34.48)$$

$$l_{k_M} \omega^i(z) = d\Omega_M^i(z) + \omega^j(z) c_{jk}^i \Omega_M^k(z). \quad (34.49)$$

We see that the vielbein is not invariant, but transforms by an H -transformation with parameter $\Omega_M^k(z)$. Thus the vielbein is invariant under motion along Killing vectors provided one adds to this motion a compensating H -transformation. This is the reason we call $\Omega_M^k(z)$ an H -compensator.

The H -connection transforms under motion along k_M as a Yang-Mills gauge field. Note, however, that the functions $\Omega_M^i(z)$ are fixed by the group multiplication rules and are not arbitrary functions of z .

The Lie derivatives commute with the exterior derivative. To prove this, it suffices to write out

the definition of Lie derivatives on p forms ω and on $d\omega$

$$\begin{aligned} l_k d\omega &= \iota_k d\omega + d\iota_k \omega = d\iota_k \omega \\ dl_k \omega &= d(\iota_k d\omega + d\iota_k \omega) = d\iota_k d\omega. \end{aligned} \quad (34.50)$$

Hence, indeed

$$l_k d\omega = dl_k \omega. \quad (34.51)$$

Let us now derive the integrability conditions for (34.48) and (34.49). The Lie derivatives satisfy the same algebra as (34.30) because they generate general coordinate transformations with composite parameter $k_M^\beta \partial_\beta k_N^\alpha - M \leftrightarrow N$. Hence

$$[l_{k_M}, l_{k_N}] = -c_{MN}^P l_{k_P}. \quad (34.52)$$

Differentiating (34.48) w.r.t. l_{k_N} and antisymmetrizing in (M, N) thus yields the following integrability condition

$$l_{k_M} W_N - l_{k_N} W_M + [W_M, W_N] = -c_{MN}^P W_P \quad (34.53)$$

where for later use we have defined (see also next section)

$$W_{M,b}^a = c_{bi}^a \Omega_M^i. \quad (34.54)$$

This is the same equation as (34.32), so (34.32) is the integrability condition of (34.48).

One can prove (34.52) directly, but it is interesting to prove it using form language as follows. In general the following identity holds for p -forms

$$\iota_{k^A} l_{k^B} \omega - l_{k^B} \iota_{k^A} \omega = \iota_{[k^A, k^B]} \omega \quad (34.55)$$

(It is easy to verify this relation for scalars and one-forms.) Hence

$$\begin{aligned} l_{k_M} l_{k_N} \omega &= l_{k_M} (d\iota_{k_N} + \iota_{k_N} d) \omega = dl_{k_M} \iota_{k_N} \omega + l_{k_M} \iota_{k_N} d\omega \quad (\text{using (34.51)}) \\ &= d(\iota_{k_N} l_{k_M} + \iota_{[k_M, k_N]}) \omega + (\iota_{k_N} l_{k_M} + \iota_{[k_M, k_N]}) d\omega \quad \text{using (34.55)} \end{aligned} \quad (34.56)$$

The second and fourth terms yield the right hand side of (34.52), whereas the first and third term yield

$$dl_{k_N} l_{k_M} + \iota_{k_N} l_{k_M} d = l_{k_N} l_{k_M} \quad (34.57)$$

since $[l, d] = 0$ according to (34.51). Combining (34.56) and (34.57), one indeed recovers (34.52). (See R. D'Auria, P. Fré, T. Regge, *Rivista Nuov. Cim.* **3** (1980) 12.)

The integrability condition for (34.40) is again (34.53). We leave the proof as an exercise.

We have at this point four geometrical objects: vielbeins and H -connections (obtained by right multiplications with coset representative near unity, see (34.35)), and Killing vectors and H -

compensators (obtained by left multiplication with an element in G near unity). They are related by the adjoint matrix $D(g)$ of G , defined by

$$g^{-1}T_M g = D(g)_M{}^N T_N. \quad (34.58)$$

[If $g = e^{\epsilon^M T_M}$, then $D(g) = e^{-\epsilon^P c_P \bullet \bullet}$.] Indeed

$$L^{-1}k_M^\alpha \partial_\alpha L = k_M^\alpha \left(e_\alpha{}^a K_a + \omega_\alpha{}^i H_i \right) \quad (34.59)$$

while from (34.29)

$$L^{-1}k_M^\alpha \partial_\alpha L = L^{-1}T_M L + \Omega_M{}^i H_i. \quad (34.60)$$

Hence, taking terms proportional to K_a

$$k_M^\alpha e_\alpha{}^a = D(L)_M{}^a \quad (34.61)$$

while from the terms proportional to H_i

$$k_M^\alpha \omega_\alpha{}^i = D(L)_M{}^i + \Omega_M{}^i. \quad (34.62)$$

These two relations generalize the equation $f_m{}^\mu(a)e_\mu{}^n(a) = A_m{}^n(a) = (e^{a^\nu c_\nu \bullet \bullet})_m{}^n$ which we obtained in (33.15) for group manifolds.

This result shows that $\Omega_M{}^i$ is a good H -connection for the ordinary derivative ∂_α along k_M^α i.e., for the ordinary Lie derivative. We will use this in later section.

The matrix elements of $D(L)$ can be expressed in terms of Killing vectors and vielbeins only, see (34.61) and the lemma below (34.202).

Finally a word about notations. There is no distinction between curved and flat indices on coset manifolds because one cannot locally rotate vielbeins (globally one can do so). Nevertheless, we will follow the same rule as for the group manifolds, namely objects near the origin have flat indices (like dz^a) while objects far away have curved indices (like z^α).

34.6 Antisymmetry of the $H - K - K$ structure constants

So far we considered any set of generators $\{H_i\}$ and $\{K_\alpha\}$ which was reductive. We now show that **by a change of basis inside the space spanned by K_α , one can make the $c_{ia}{}^b$ antisymmetric if H is compact.**

$$c_{ia}{}^{b'} \delta_{b'b} = -c_{ib}{}^{a'} \delta_{a'a}. \quad (34.63)$$

As we have discussed, this change of basis keeps reductivity.

Theorem: By redefining $\hat{K}_\alpha = N_\alpha^\beta K_\beta$ for a suitable N_α^β one can achieve that the new structure constants are antisymmetric as in (34.63).

Proof: Since H is compact, any representation of H can be made unitary by a suitable change of basis (just as in the case of finite groups). Since the $-c_{ia}^b$ generate a real representation of H (namely, the coset representation), this representation can be made orthogonal, and the c_{ia}^b antisymmetric. q.e.d.

For readers less familiar with properties of compact groups, we give a few more details. Consider the matrix

$$D = \int_H d\mu(h) h h^\dagger, \quad h \text{ generated by } c_{ai}^b \quad (34.64)$$

where $d\mu(h) = dy \det e_\alpha^a(y)$ is the Haar measure, $e_\alpha^a(y)$ the H -vielbein, y parametrizes the elements of H , and h is the coset representation of elements of H (generated by c_{ai}^b). This integral over all elements of H is the analogue of the sum over all elements of a finite group, and makes sense because H is compact. Since H is a group, the definition of vielbein is simpler than in the case of coset manifolds, namely the subgroup factor is absent.

$$h(y)h(dy) = h(y^\alpha + dy^\alpha e_a^\alpha(y)) . \quad (34.65)$$

Let us recall that the Haar measure is translationally invariant

$$\int_M d\mu(h) = \int_{h_0 M} d\mu(h_0 h) \quad (34.66)$$

as follows from (34.65). We shall need the case $M = H$, in which case also $h_0 M = H$.

Clearly D is hermitian and real (and thus symmetric) and positive semi-definite. If $Dy = 0$, then $hy = 0$ for all $h \in H$. Clearly, D maps the unitary complement of the null-space into itself because D is hermitian.

$$(x, Dy) = (Dx, y) = 0 \quad \text{for } y \in N \text{ and } x \in N^\perp . \quad (34.67)$$

In the space $N^\perp$, D is again real and symmetric, but now positive definite. Hence it can be made diagonal and positive by an orthogonal matrix O . We form $S = OD^{-\frac{1}{2}}$ which is a real matrix. Then ShS^{-1} is an orthogonal matrix for all $h \in H$. Indeed

$$(ShS^{-1})(ShS^{-1})^\dagger = ShD^{\frac{1}{2}}O^{-1}OD^{\frac{1}{2}}h^\dagger S^\dagger = ShDh^\dagger S^\dagger = SDS^\dagger = OO^\dagger = I . \quad (34.68)$$

We used $hDh^\dagger = D$, which is true due to (34.64) and (34.66).

Thus, **given** a reductive set of generators, they span two linear vector spaces $\{H_i\}$ and $\{K_\alpha\}$ and by changing basis in the K -space, we can achieve that (34.63) holds. From now on we shall always assume that (34.63) holds. Note that we do not go outside the linear vector space spanned by the K_α , and thus we have not lost any generality.

Comment 1. Although the $-c_{ia}{}^b$ form the coset representation of H , and are generators of orthogonal matrices, this does not imply that H is a subgroup of $SO(n)$ (where n = number of independent K -generators = dimension of coset space). A simple counter example suffices:

$$G/H = U(1) \times U(1) \times U(1)/U(1) \times U(1). \quad (34.69)$$

The dimension of this coset space is $3 - 2 = 1$, but evidently $U(1) \times U(1)$ is not a subgroup of $SO(1)$. However, when H acts faithfully on the coset (i.e., for no H generator $\lambda^i H_i$ is it true that $[\lambda^i H_i, K_\alpha] = 0$ for all K_α , then H is indeed a subgroup of $SO(n)$.

Comment 2. We can bring the coset representation of H on block diagonal form by a further orthogonal basis transformation in the space of K_α generators. On this basis one has

$$c_{ia}{}^b = \begin{pmatrix} \square & & \emptyset & \\ & \square & & \\ \emptyset & & 0 & \\ & & & \ddots \\ & & & & 0 \end{pmatrix} \quad (34.70)$$

The blocks and the zeros along the diagonal correspond to the decomposition of the coset representation of H into irreducible representations of H .

For an elementary discussion of Haar measures, see, for example, “Group Theory” by M. Hamermesh.

34.7 Cartan-Maurer equations, spin connection and Ricci tensor

We recall the definition of vielbein and H -connection in terms of the one-form $L^{-1}dL$ in (34.42). Using the Poincaré identity $dd = 0$, we thus have

$$d(L^{-1}dL) = -(L^{-1}dL) \wedge (L^{-1}dL). \quad (34.71)$$

Substituting $L^{-1}dL = e^a K_a + \omega^i H_i$, and taking separately the terms proportional to K_a and H_i , we obtain the celebrated Cartan-Maurer equations

$$de^a + \frac{1}{2}c_{bc}{}^a e^b \wedge e^c + c_{ib}{}^a \omega^i \wedge e^b = 0 \quad (34.72)$$

$$d\omega^i + \frac{1}{2}c_{jk}{}^i \omega^j \wedge \omega^k + \frac{1}{2}c_{ab}{}^i e^a \wedge e^b = 0. \quad (34.73)$$

Because of reductivity there is no $e \wedge \omega$ term in (34.73).

At this point, mathematicians and physicists part ways. Namely, **physicists do not use the H -connection but the spin connection**. In fact, in a given action, one can always solve the spin connection from its non-propagating field equation (Palatini formalism), and the result is $\omega = \omega(e) + \text{more}$, where $\omega(e)$ is the spin connection, and “more” usually denotes terms bilinear in fermion fields. Sometimes physicists do not use the spin connection $\omega(e)$ but impose by hand a connection ω' with desirable properties. However, such ω' do not satisfy the field equations. One determines in (super)gravity the spin connection from the Einstein action

$$\begin{aligned} R_{ab} &= R_{\alpha\beta}{}^c{}_b e_a{}^\beta e_c{}^\alpha; \quad R = R_{ab}\delta^{ab} \text{ (or } \eta^{ab}) \\ R_{\alpha\beta}{}^a{}_b &= \partial_\alpha \omega_\beta{}^a{}_b + \omega_\alpha{}^a{}_c \omega_\beta{}^c{}_b - \alpha \leftrightarrow \beta. \end{aligned} \quad (34.74)$$

In the action one thus has an explicit, Lorentz or $SO(n)$ invariant tensor, namely η^{ab} or δ^{ab} . In general relativity, one also requires that $\omega_\alpha{}^a{}_b$ be antisymmetric

$$\omega_\alpha{}^{a'}{}_b \delta_{a'a} = -\omega_\alpha{}^{b'}{}_a \delta_{b'b} \quad (34.75)$$

and thus one selects a priori the Lorentz or $SO(n)$ group as the tangent group. One might set up a different formalism, with $\omega_\alpha{}^a{}_b$ valued in some other Lie algebra, and δ^{ab} replaced by an invariant tensor of this other Lie algebra. We shall not do so here. Thus, for us (34.75) and (34.74) will hold. We summarize: **the spin connection is always antisymmetric**. On the compact spaces we consider, it is $SO(n)$ -valued.

If one solves $\omega_\alpha{}^a{}_b$ from its field equation due to (34.74), the result is a complicated expression in terms of vielbeins (see below). However, this complicated expression happens to satisfy a simple identity.

$$\partial_\alpha e_\beta{}^a + \omega_\alpha{}^a{}_b(e) e_\beta{}^b - \alpha \leftrightarrow \beta = 0. \quad (34.76)$$

Moreover, the general solution of (34.76) which satisfies also (34.75), is equal to the spin connection. Thus, given an arbitrary vielbein on a given manifold, we can solve for the spin connection by using (34.76). Details will be given shortly.

Let us now return to the Cartan-Maurer equations. If we define

$$\tilde{\omega}_b{}^a = \omega^i c_{ib}{}^a + \frac{1}{2} e^d c_{db}{}^a \quad (34.77)$$

then it would seem that $\tilde{\omega}_b{}^a$ is the spin connection, since according to (34.72)

$$de^a + \tilde{\omega}_b{}^a e^b = 0. \quad (34.78)$$

However, although $c_{ib}{}^a$ is antisymmetric according to (34.63), the $K - K - K$ structure constants $c_{db}{}^a$ are in general not antisymmetric in (b, a) . Hence $\tilde{\omega}$ is in general not antisymmetric, and therefore in general not equal to the spin connection. One can find the spin connection from (34.78) by adding further terms to $\tilde{\omega}$ which make $\tilde{\omega}$ antisymmetric but which cancel in (34.78).

We claim that the general result is

$$\begin{aligned}
\omega(\text{spin})^a_b &= \tilde{\omega}_b^a + \frac{1}{2}e^d\delta^{aa'}\left(c_{a'd}{}^{b'}\delta_{b'b} + b \leftrightarrow d\right) \\
&= \omega^i c_{ib}^a + \frac{1}{2}e^d\bar{c}_{db}^a \\
\bar{c}_{db}^a &= c_{db}^a + \delta^{aa'}\left(c_{a'd}{}^{b'}\delta_{b'b} + b \leftrightarrow d\right).
\end{aligned} \tag{34.79}$$

Indeed, the extra terms cancel because $e^d \wedge e^b$ is anti-symmetric in (b, d) , and thus $\bar{c}_{db}^e \delta_{ea}$ is antisymmetric in (a, b) .

The Kronecker deltas are not as noncovariant as they may seem. Since c_{ia}^b is antisymmetric, $\delta^{aa'}$ is an H -invariant tensor, and so is therefore c_{db} .

One can also directly determine the spin connection from (34.76). (For details, see, for example section 7 of the Physics Report of the author.) The result contains curls of vielbeins, and using the Cartan-Maurer equations to eliminate these curls, one recovers (34.79).

Let us now compute the Riemann tensor

$$R_{\alpha\beta}{}^b{}_a = \partial_\alpha \left(\omega_\beta^i c_{ia}^b + \frac{1}{2}e_\beta^d \bar{c}_{da}^b \right) + \left(\omega_\alpha^i c_{ic}^b + \frac{1}{2}e_\alpha^d \bar{c}_{dc}^b \right) \left(\omega_\beta^j c_{ja}^c + \frac{1}{2}e_\beta^t \bar{c}_{ta}^c \right) - \alpha \leftrightarrow \beta. \tag{34.80}$$

Eliminating the curls of ω^i and e^d by their Cartan-Maurer equations, one finds that the terms with $\omega\omega$ cancel, due to the Jacobi identities. For the remainder one finds in terms of flat indices

$$R_{st}{}^b{}_a = c_{ai}^b c_{st}^i - \frac{1}{2}\bar{c}_{ra}^b c_{st}^r + \frac{1}{4} \left(\bar{c}_{ta}^r \bar{c}_{sr}^b - \bar{c}_{sa}^r \bar{c}_{tr}^b \right) + \omega_s^j \left(-\bar{c}_{ra}^b c_{jt}^r + c_{jr}^b \bar{c}_{ta}^r + \bar{c}_{tr}^b c_{aj}^r \right). \tag{34.81}$$

The last term should separately be antisymmetric in (ab) but it vanishes, due to the identity

$$c_{tj}^r \bar{c}_{ra}^b + \bar{c}_{ta}^r c_{jr}^b + c_{aj}^r \bar{c}_{tr}^b = 0. \tag{34.82}$$

This equation states that $\bar{c}$ is an H -invariant tensor. If $\bar{c}$ were equal to c , this would be the usual Jacobi identity, but the terms proportional to $\bar{c} - c$ cancel separately. Indeed, these extra terms are

$$c_{tj}^r \left(c_{ra}^b + c_{ar}^b \right) + c_{jr}^b \left(c_{ta}^r + c_{at}^r \right) + c_{aj}^r \left(c_{tr}^b + c_{rt}^b \right) \tag{34.83}$$

and cancel, because the first, third and fifth term combine to a Jacobi identity, and so do the remaining three terms.

Hence, **the Riemann tensor with flat indices is constant, and given by**

$$R_{st}{}^b{}_a = c_{ai}^b c_{st}^i - \frac{1}{2} \left(\bar{c}_{ra}^b c_{st}^r \right) + \frac{1}{4} \left(\bar{c}_{ta}^r \bar{c}_{sr}^b - \bar{c}_{sa}^r \bar{c}_{tr}^b \right). \tag{34.84}$$

It is easy to check that $R_{st}{}^b{}_a$ is antisymmetric in (st) and in (ab) , using that $\bar{c}$ is antisymmetric.

For the Ricci tensor one finds

$$R_{ab} = R_{at}{}^t{}_b = c_{at}{}^i c_{bi}{}^t - \frac{1}{2} c_{at}{}^r \bar{c}_{rb}{}^t + \frac{1}{4} \bar{c}_{ar}{}^t \bar{c}_{tb}{}^r - \frac{1}{4} \bar{c}_{ab}{}^r \bar{c}_{tr}{}^t. \quad (34.85)$$

Or, writing out the $\bar{c}$ terms in full

$$R_{ab} = c_{at}{}^i c_{bi}{}^t + \frac{1}{2} c_{ar}{}^t (c_{tb}{}^r + c_{rb}{}^t) + \frac{1}{4} c_{rb}{}^t c_{ta}{}^r - \frac{1}{2} c_{tr}{}^t \bar{c}_{ab}{}^r. \quad (34.86)$$

Since G is compact, $c_{Mr}{}^M = 0$ while $c_{ir}{}^i = 0$ due to reductivity, hence the trace $c_{tr}{}^t$ vanishes, and thus the last term in R_{ab} is zero. As a check one may verify that R_{ab} is symmetric in (ab) , as it should be since the spin connection has no torsion. The proof follows from the following Jacobi identity

$$c_{at}{}^i c_{bi}{}^t - c_{bt}{}^i c_{ai}{}^t = c_{ki}{}^i c_{ab}{}^k = 0. \quad (34.87)$$

Because H is compact, the trace $c_{ki}{}^i$ vanishes. (We used that H is compact because we used that $c_{ia}{}^b$ is antisymmetric.)

If $c = \bar{c}$, one finds for the Ricci tensor

$$R_{ab} = c_{at}{}^i c_{bi}{}^t + \frac{1}{4} c_{at}{}^r c_{br}{}^t. \quad (34.88)$$

For symmetric algebras the last term vanishes while the first term is proportional to the Killing metric times the Casimir operator of G .

The result that the components of the Riemann tensor with flat indices are constant, can be understood as follows. First note that the ordinary Lie derivative of the spin connection is an H -gauge transformation with parameter

$$(W_M)_b{}^a = \Omega_M{}^k c_{bk}{}^a. \quad (34.89)$$

This follows from $[l_k, d] = 0$ in (34.51) and

$$l_{kM} (de^a + \omega(\text{spin})^{ab} \wedge e^b) = 0. \quad (34.90)$$

Next note that as a consequence the ordinary Lie derivative of the Riemann curvature is equal to an H -gauge rotation of the Riemann curvature with parameter (34.89). Now the Riemann tensor in (34.84) is in fact H -invariant, because both the structure constants and the Kronecker deltas are H -invariant tensors (section 5 and Jacobi identities). Hence, the ordinary Lie derivative of the Riemann tensor vanishes, and since on coset manifolds there are as many Killing vectors as dimensions (coset manifolds are homogeneous), the Riemann tensor is constant.

One may also verify by direct computation that the Riemann tensor is invariant under trans-

formation with $c_{ia}{}^b$, and hence the Ricci tensor satisfies

$$R_{ab}c_{ib}{}^d + R_{db}c_{ia}{}^d = 0. \quad (34.91)$$

Let us finally recall that for group manifolds the Riemann tensor reads

$$R_{st}{}^b{}_a = -\frac{1}{4}c_{st}{}^r c_{ra}{}^b. \quad (34.92)$$

The results involving $\bar{c}$ -constants for Riemann tensors seem new. However, the $\bar{c}$ constants were introduced in connection with rescalings of vielbeins (section 8) by L. Castellani, L. J. Romans and N. P. Warner, *Annals of Physics* **157** (1984) 394.

34.8 H -covariant Lie derivatives and their properties

We saw in section 34.5 that the Lie derivative of the vielbein is equal to an H -rotation of the vielbein. In fact, since $c_{ia}{}^b$ is antisymmetric, the H -rotation is an $SO(n)$ rotation. By bringing the right hand side of (34.48) to the left hand side, we obtain what will be called the H -covariant Lie derivative

$$\mathcal{L}_{k_M} e^a = l_{k_M} e^a + \Omega_M^i c^a{}_{ib} e^b = 0. \quad (34.93)$$

Thus, **the H -covariant Lie derivative of the vielbein vanishes.**

For a field ϕ^I which forms a representation of $SO(n)$, transforming according to $SO(n)$ matrices $(t_{ab})^I{}_J$ we define the H -covariant derivative by

$$\mathcal{L}_{k_M} \phi^I = l_{k_M} \phi^I + \Omega_M^i (-\frac{1}{2}c_{iab})(t^{ab})^I{}_J \phi^J. \quad (34.94)$$

Since $c_{ia}{}^b = c_{iab}$ is antisymmetric, this definition makes sense.

For a vector, $(t^{ab})^I{}_J = (\delta^{aI}\delta_J^b - \delta^{bI}\delta_J^a)$ and one recovers (34.93). For spinors, $t_{ab} = \frac{1}{2}\tau_{ab}$ where τ_{ab} are Dirac matrices, satisfying $[\tau_a, \tau_b] = 2\delta_{ab}$.

The mass operators in Kaluza-Klein theories commute with the H -covariant Lie derivative, and this will be the starting point for an analysis of spherical harmonics. In order to set the stage, we begin by deriving a number of theorems concerning the H -covariant Lie derivative.

Theorem I. The H -covariant Lie derivative satisfies the same algebra as the ordinary Lie derivative

$$[\mathcal{L}_{k_M}, \mathcal{L}_{k_N}] = -c_{MN}{}^P \mathcal{L}_{k_P}. \quad (34.95)$$

Proof: Since l_{k_M} satisfied (34.95) according to (34.52) one is left with the same equation as (34.53) but with W_M given by $\Omega_M^i (-\frac{1}{2}c_{iab}) t^{ab}$ instead of $\Omega_M^i (-c_{iab})$. Since (34.95) is obtained by contracting (34.53) with $\frac{1}{2}t^{ab}$, it holds in any representation of the Lorentz of $SO(n)$ group.

Theorem II. The H -covariant derivative commutes with the gravitationally covariant exterior derivative

$$[\mathcal{L}_{k_M}, D] = 0, \quad D \equiv dz^\alpha D_\alpha(\omega(\text{spin})). \quad (34.96)$$

Proof: Recall $\mathcal{L} = l - W^{ab}t_{ab}$ and $D = d + \omega(\text{spin})^{ab}t_{ab}$. Since $[l, d] = 0$ according to (34.51), there are only three contributions. As we shall show, the Lie derivative of the spin connection is equal to the covariant derivative D (with spin connection) acting on W . This is sufficient to prove (34.96) because the two other terms involve dW and $[W, \omega(e)]$ and cancel the first term.

To evaluate $l\omega(\text{spin})$, use (34.79) and (34.48), (34.49). Then use (34.82).

Theorem III. The H -covariant Lie derivative commutes with the Dirac operator and the D'Alembertian.

Proof: First note that $[\mathcal{L}, \tau^a] = 0$, because the Dirac matrices are $SO(n)$ invariant tensors. Indeed, since the H -covariant Lie derivative acts both on the vector index and on the spinor index of a spinor

$$[\mathcal{L}_{k_M}, \tau^a] = \frac{1}{4}\Omega_M^i(-c_i^{de})([\tau_{de}, \tau^a] + 4\delta_{[d}^a\delta_{e]}^b\tau_b). \quad (34.97)$$

Since $\mathcal{L}$ is distributive, and $[\mathcal{L}, dz^\alpha] = 0$, we get according to (34.47)

$$[\mathcal{L}, \tau^a D_a] = 0. \quad (34.98)$$

Hence also $[\mathcal{L}, D_\alpha^2] = 0$.

The H -covariant Lie derivatives were introduced by R. D'Auria and P. Fré, Ann. of Phys., to be published.

34.9 Rescalings of vielbeins preserving G invariance Einstein metrics

In the previous chapters we discussed how, once an algebra with generators H and K is given, one can define vielbeins on the coset manifold. In general, however, these vielbeins are not useful for applications because they do not lead to Einstein spaces. (An Einstein space satisfies $R_{\alpha\beta} = \Lambda g_{\alpha\beta}$.) One can, however, construct from the vielbeins new vielbeins which do lead to Einstein spaces in all known cases; this may be a general theorem.

The vielbeins were G -invariant; under motion along a Killing vector, they changed only by an H -rotation. Hence they are physically invariant. The mathematical statement is that the H -covariant Lie derivative of the vielbein vanishes,

$$\mathcal{L}_{k_M} e^a = 0. \quad (34.99)$$

Writing this result out

$$l_{k_M} e^a = \Omega_M^i c_i^{ab} e^b \quad (34.100)$$

we recognize that if we rescale

$$e^a = \lambda^a \bar{e}^a \quad (\text{no sum}) \quad (34.101)$$

then

$$l_{k_M} \bar{e}^a = \left(\Omega_M^i c_i^{ab} \right) \bar{e}^b \left(\frac{\lambda^b}{\lambda^a} \right) \quad (34.102)$$

Hence, as long as $\lambda^a/\lambda^b = 1$ whenever c_i^{ab} is non-vanishing, the new $\bar{e}^a$ are again G -invariant. Recalling the structure of c

$$c_{iab} = \begin{pmatrix} \boxed{} & & \emptyset \\ \emptyset & \boxed{} & \\ & & \ddots \end{pmatrix} \quad (34.103)$$

this means that the λ^a 's must be constant inside each block.

The Cartan-Maurer equation for the new $\bar{e}^a$ read

$$\begin{aligned} d\bar{e}^a + \omega^i c_{ib}^a \bar{e}^b + \frac{1}{2} \hat{c}_{bd}^a \bar{e}^b \wedge \bar{e}^d &= 0 \\ \hat{c}_{bd}^a &= c_{bd}^a (\lambda^a)^{-1} \lambda^b \lambda^d. \end{aligned} \quad (34.104)$$

The structure constants c_{ij}^k , c_{ib}^a and $\hat{c}_{ab}^d$ are clearly those of the rescaled generators (H_i , $\lambda^a K_a$ no sum). Hence, the Ricci tensor for the new vielbeins is still constant, but contains now the rescaled structure constants. Thus, the requirement that one have an Einstein space, reduces to a set of algebraic equations for the λ^a .

If the coset representation of H splits into irrepses $R_1, R_2, \dots$, and some of the R_i are equivalent, then one can take as new vielbeins linear combinations of the vielbeins which belong to equivalent representations

$$\bar{e}^a = \lambda (p e^{a_1} + q e^{a_2} + r e^{a_3} + \dots) \quad (34.105)$$

because, again $\mathcal{L}\bar{e}^a = 0$. However, the constants p, q and r must then be integers in order that the coset be compact. In these cases one finds for the Einstein spaces algebraic equations involving both the λ 's and the $p, q, r, \dots$.

A typical example are the spaces

$$SU(3) \times SU(2) \times U(1) / SU(2) \times U(1) \times U(1) \quad (34.106)$$

where the $SU(2)$ in the denominator is the $SU(2)$ in $SU(3)$, but where the $U(1)$'s in the denominator are linear combinations of the three $U(1)$'s in $SU(3)$ and $SU(2)$ and the $U(1)$ in the numerator. The original set of e^a consists then of the remaining quartet in $SU(3)$, a doublet in $SU(2)$ and a particular linear combination of the three $U(1)$'s which is linearly independent from (but not necessarily orthogonal to) the two $U(1)$'s in H . Thus the matrix c_{ia}^b splits up into a quartet, a

doublet and a singlet. Thus there are three scaling constants.

Given a vielbein e_α^a how should one construct a metric out of it?

We require that the metric $g_{\alpha\beta}$ be G -invariant. We call a metric G -invariant when $g_{\alpha\beta}(z)dz^\alpha dz^\beta$ is equal to $g_{\alpha\beta}(\bar{z})d\bar{z}^\alpha d\bar{z}^\beta$ where $\bar{z}$ is a point on the coset manifold obtained from the point z by left multiplication with a group element g . Since $\bar{z} = z + k$, the condition of G -invariant is equal to

$$g_{\alpha\beta}(\bar{z}) = g_{\alpha\beta}(z) - (\partial_\alpha k^\gamma) g_{\gamma\beta} - (\partial_\beta k^\gamma) g_{\gamma\alpha} \quad (34.107)$$

or, expanding $g_{\alpha\beta}(\bar{z})$

$$l_{k_M} g_{\alpha\beta} = 0. \quad (34.108)$$

Thus we must find a suitable matrix m_{ab} such that

$$g_{\alpha\beta} = e_\alpha^a m_{ab} e_\beta^b, \quad l_{k_M} g_{\alpha\beta} = 0. \quad (34.109)$$

Since $g_{\alpha\beta}$ has no flat indices, $\mathcal{L}g$ should be equal to l_g . However le^a is equal to an H -rotation, hence m_{ab} must be an H -invariant tensor

$$\Omega_M^i \left(c_{ia}^{a'} m_{a'b} + c_{ib}^{b'} m_{ab'} \right) = 0. \quad (34.110)$$

Or, in other words, $e^a m_{ab} e^b$ must be an H -scalar. The only possible way to construct a scalar from two irrepses R_1 and R_2 of a Lie group is if R_2 is similar to $(R_1)^*$ and in that case $m_{ab} = \delta_{ab}$. Indeed, δ_{ab} is an invariant tensor in the coset representation of H , since c_{iab} were antisymmetric matrices.

Hence, the general G -invariant metric is given by

$$e_\alpha^a m_{ab} e_\beta^b, \quad m_{ab} = \left(\begin{array}{cc|c} a\mathbb{1} & b\mathbb{1} & \emptyset \\ b\mathbb{1} & c\mathbb{1} & \emptyset \\ \hline \emptyset & \emptyset & d\mathbb{1} \end{array} \right) \quad (34.111)$$

where a, b, c, d are real constants. If $R_1 = R_2$ one can also contract the e_α^a with a in R_1 , with the e_β^b with b in R_2 , by means of a δ_{ab} (if $R_1 = SR_2S^{-1}$ but not $S = 1$, one must use not δ_{ab} but S to contract).

We are now in a position to prove that the k_M^α defined by (34.28) are indeed Killing vectors. A Killing vector satisfies

$$D_\alpha k_\beta + D_\mu k_a = 0 \quad (34.112)$$

where D_α is the gravitationally covariant derivative with spin connection. Since

$$\begin{aligned} l_{k_M} g_{\alpha\beta} &= (\partial_\alpha k_M^\gamma) g_{\gamma\beta} + (\partial_\beta k_M^\gamma) g_{\alpha\gamma} + k_M^\gamma \partial_\gamma g_{\alpha\beta} \\ &= \partial_\alpha k_{M\beta} + \partial_\beta k_{M\alpha} + k_M^\gamma (\partial_\gamma g_{\alpha\beta} - \partial_\alpha g_{\gamma\beta} - \partial_\beta g_{\alpha\gamma}) \\ &= D_\alpha k_\beta + D_\beta k_\alpha. \end{aligned} \quad (34.113)$$

it follows that the k_M^α in (34.107) are indeed Killing vectors.

The freedom one has in rescaling vielbeins is well-known also outside supergravity. See, for example, papers by R. Cocqueraux, Szczyrk lectures (1983), and Marseille preprint CPT-84/p. 1585 (1984).

For an example, see example 34.20 at the end of these lectures.

34.10 Extra Killing vectors due to right multiplication

On group manifolds, left and right cosets are equivalent, and one can define not only the left-vielbeins e_α^a , but also right-vielbeins f^α_a by the reverse of (34.37). On coset manifolds the f^α_a become Killing vectors k_M^α . From the Cartan-Maurer equations one has for the vielbeins of group manifolds

$$[e_a, e_b] = c_{ab}^d e_d \quad (34.114)$$

while the algebra of Killing vectors shows that

$$[f_a, f_b] = -c_{ab}^d f_d \quad (34.115)$$

Now $[f_a, e_b]$ is equal to the Lie derivative of e_b

$$[f_a, e_b] = \left(f_a^\alpha \partial_\alpha e_b^\beta - e_b^\alpha \partial_\alpha f_a^\beta \right) \partial_\beta = (l_{f_a} e_b^\alpha) \partial_\alpha. \quad (34.116)$$

Since the H -covariant Lie derivative of e^a vanishes, it follows that on group manifolds

$$[f_a, e_b] = 0. \quad (34.117)$$

This last result expresses the fact that group multiplication is associative, hence one can first multiply only on the right (yielding e^a) and then on the left (yielding f^a), or first on the left and then on the right.

On group manifolds, the right vielbeins are Killing vectors of the metric $e^a_\alpha m_{ab} e^b_\beta$ for any m_{ab} , but one can also consider a linear combination of left vielbeins $k_M^\alpha \epsilon^a_\alpha$ and ask whether it, too, can be a Killing vector. For certain m_{ab} , certain k^α can indeed be Killing vectors. Rather than discuss the case of group manifolds further, let us jump directly to coset manifolds, and ask: which left vielbeins are Killing vectors? We saw that if the ordinary Lie derivative along a certain k is an H -rotation, then these k are Killing vectors. Let us now repeat the steps of section 34.4, but

start with g on the right.

Consider

$$L(z)g = L(z')h. \quad (34.118)$$

(Since we deal with left cosets, we must write h on the right.) In order that one obtains a motion on the coset, all points of a given coset must move to the same other coset. Hence, in $L(z)\hat{h}g = L(z')\hat{h}\tilde{h}$ the z' must be independent of h . In other words $hg^{-1} = \hat{h}\tilde{h}g^{-1}h^{-1}$. Hence,

$$g^{-1}\hat{h}h \in H \quad \text{for all } \hat{h} \in H \quad (34.119)$$

Clearly, for $g \in H$ there is no motion. Let us consider $g = g(\epsilon)$. In this case the requirement for motions on the coset becomes

$$g^{-1}(\epsilon)hg(\epsilon) \in H \quad (\text{the } g(\epsilon) \text{ are the normalizer of } H \text{ in } G) \quad (34.120)$$

(Normalizer means: the largest subgroup of G in which H is a normal subgroup.) In that case, the motion is $z^\alpha \rightarrow z^\alpha + \epsilon^a e_a^\alpha$ since

$$\begin{aligned} L(z)g(\epsilon) &= L(z^\alpha + \epsilon^g k_g^\alpha) h, & g(\alpha) &= \Lambda + \epsilon^M X_M \\ (\Lambda + \epsilon^a k_a + \epsilon^i H_i) &= \epsilon^M k_M^\alpha (e_a^\alpha k_a + \omega_a^i H_i) - \epsilon^M \Omega_M^i H_i \\ k^a &= \epsilon^g k_g^\alpha = \epsilon^a e_a^\alpha, & k^\alpha \omega_\alpha^i &= e^i + \epsilon^M \Omega_M^i. \end{aligned} \quad (34.121)$$

When are the $k^\alpha = \epsilon^a e_a^\alpha$ indeed Killing vectors? In that case the Lie derivative along k of e must be equal to an H -rotation. Since $l_k e^b = [k, e^b]$ we get from the Cartan-Maurer equations the condition

$$\epsilon^a (c_{ab}^d + e_a^\alpha \omega_\alpha^i c_{ib}^d - e_b^\alpha \omega_\alpha^i c_{ia}^d) e_d^\beta = H \text{ rotation of } e_b^\beta. \quad (34.122)$$

The last term vanishes because (34.120) says that the commutator of $l + \epsilon^M X_M$ and H_i must lie in H whereas reductivity says that it must lie in K . Hence

$$[\epsilon^a K_a, H_i] = \epsilon^a c_{ai}^b K_b = 0. \quad (34.123)$$

The second term in (34.122) is indeed an H -rotation of the vielbein. If the first term in (34.122) is also an H -rotation, then $\epsilon^a e_a^\alpha$ is a Killing vector. Thus

$$\epsilon^a e_a^\alpha \text{ is Killing vector} \quad \Leftrightarrow \quad \epsilon^a c_{ab}^d \in H. \quad (34.124)$$

In section 34.15 we shall see how such extra Killing vectors often arise: from the Killing spinors η^A ($A = 1, \dots, N$) one can build Killing vectors $\bar{\eta}^A \tau_A \eta^B$ which generate a group $SO(N)$ which may commute with the Killing vectors of the coset manifold G/H . Of course, if one takes as G the group of **all** Killing vectors, and subsequently determines its isotropy subgroup, one does never find extra Killing vectors.

Let us summarize which isometry groups (= groups of Killing vectors) a group manifold and a coset manifold can have.

Coset Manifolds G/H have at least an isometry group $S = G \times N(H)/H$ where $N(H)/H$ is the normalizer of H in G . It can happen that some left Killing vectors are also right-Killing vectors. $UG = G' \times U(1)$, the $U(1)$ gives both left and right. In this case they must correspond to $U(1)$ generators, and S is smaller than $G \times N(H)/H$. There can be extra Killing vectors, for example, from Killing spinors. Sometimes a vielbein of G/H can have more symmetries by accident. A familiar example is $SO(7)/G_2$ whose vielbein equal to $SO(8)/SO(7)$ and has thus an $SO(8)$ isometry group. Or a given vielbein can get more symmetries by rescaling. Example: the vielbein of $SO(5)/SO(3)$ becomes equal to the vielbein of $SO(5) \times SO(3)/SO(3) \times SO(3)$ after rescaling. The same holds for the vielbeins of $SO(7)/SO(6)$ and $G_2/SU(3)$.

Group Manifolds: The isometry group is $G \times G$, namely left- and right-Killing vectors. However, the group can be smaller, namely if some Killing vectors are in both groups G . In that case they must generate a $U(1)$ and $G = G' \times U(1)$.

From $L(y)g = L(y')h$ we find for the right Killing vectors

$$\text{if } L(\epsilon) = 1 + \epsilon^a K_a \text{ then } k_R^\alpha \partial_\alpha L = L(y) (\epsilon^a K_a + \epsilon^a \Omega_{R,a}^i H_i) \quad (34.125)$$

where we follow the definition in (34.28). Hence, after multiplying by L^{-1}

$$k_R^\alpha e_\alpha^a = \epsilon^a; \quad k_R^\alpha \omega_a^i = \Omega_{R,\alpha}^i \epsilon^a. \quad (34.126)$$

The right- and left-Killing vectors commute, because of associativity. One can show this directly as follows

$$[k_M, \epsilon^a e_a] = \epsilon^a \Omega_M^i c_{ia}^b e_b = 0 \quad (34.127)$$

because $\epsilon^a k_a$ commutes with all H_i , see (34.123).

An extra $SO(2)$ Killing vector was found in R. D'Auria, P. Fré and the author, *Phys. Lett. B* **136** (1984) 347. The treatment using normalizers is due to L. Castellani, L. J. Romans and N. P. Warner, *Annals of Physics* **157** (1984) 394.

34.11 Killing spinors and their integrability conditions

A Killing spinor $\eta(z)$ on a n -dimensional compact manifold with coordinates z^α is defined by

$$D_\alpha \eta = c \tau_\alpha \eta. \quad (34.128)$$

Here c is a constant, to be fixed presently, τ_α are Dirac matrices of the compact space, satisfying $\{\tau_\alpha, \tau_\beta\} = 2\delta_{\alpha\beta}$ and D_α is the gravitationally covariant derivative with spin connection.

Iterating, and using $[D_\beta, \tau_\alpha] = 0$, one finds for the commutator of two covariant derivatives a curvature. Hence

$$[D_\alpha, D_\beta]\eta = \frac{1}{4}R_{\alpha\beta}{}^{ab}\tau_{ab}\eta = -2c^2\tau_{\alpha\beta}\eta. \quad (34.129)$$

After multiplication by τ_α one finds on the left hand side the following Dirac matrices

$$\tau^\alpha\tau_{ab} = \tau^\alpha{}_{ab} + e_a{}^\alpha\tau_b - e_b{}^\alpha\tau_a. \quad (34.130)$$

The terms with $\tau^{\alpha ab}$ cancel due to the cyclic identity, and one finds

$$(R_{\alpha\beta} - 4c^2(n-1)g_{\alpha\beta})\tau^\beta\eta = 0. \quad (34.131)$$

Multiplying by $\bar{\eta}\tau^\gamma$ and using that $\bar{\eta}\tau^\gamma\tau^\beta\eta = g^{\gamma\beta}\bar{\eta}\eta$ in $d = 7$ dimensions (in other dimensions one may need modified Majorana spinors), one obtains

$$R_{\alpha\beta} = 4c^2(n-1)g_{\alpha\beta}. \quad (34.132)$$

Hence, **only Einstein spaces admit Killing spinors and $4(n-1)c^2$ is equal to the cosmological constant.**

In fact, even the sign of c is fixed (except on spheres). Suppose there were two Killing spinors.

$$D_\alpha\eta^+ - c\tau_\alpha\eta^+, \quad D_\alpha\eta^- = -c\tau_\alpha\eta^-, \quad c \geq 0. \quad (34.133)$$

Then $\phi = \bar{a}\eta_+\eta_-$ would have the property that $D_\alpha\phi$ is a conformal Killing vector $k_\alpha{}^c$ (a conformal Killing vector satisfies $D_\alpha k_\beta + D_\beta k_\alpha \sim g_{\alpha\beta}$). Now conformal Killing vectors only exist on spheres S_n , hence either η^+ or η^- does not exist (what we want to prove) or otherwise $\phi = 0$. However, if $\phi = 0$ then the vector $V_\alpha = \bar{\eta}^+\tau_\alpha\eta^-$ is nonvanishing, since

$$\eta^+\bar{\eta}^+ + \tau_\alpha\eta^+\bar{\eta}^+\tau^\alpha = \mathbb{1} \quad (\text{completeness in } d = 7) \quad (34.134)$$

so that

$$\eta^+\phi + \tau^\alpha\eta^+V_\alpha = \eta^-. \quad (34.135)$$

This vector V_α would be covariantly constant because $D_\alpha V_\beta = -2cg_{\alpha\beta}\phi = 0$. Hence,

$$D_{[\gamma}D_{\alpha]}V_\beta = 0 = -R^\tau{}_{\beta\gamma\alpha} \Rightarrow R^\tau{}_\alpha V_\tau = \Lambda V_\alpha = 0 \quad (34.136)$$

which would imply $\Lambda = c = 0$. Thus, in general, Ricci curved manifolds have only η^+ or η^- Killing spinors, except spheres which have both. For Ricci flat manifolds there is no distinction between η^+ and η^- because $c = 0$, see (34.132). M. J. Duff, B. Nilsson and C. N. Pope, *Phys. Lett. B* **139** (1984) 154.

The above argument is valid only for $d = 7$ because of (34.134) but may be generalized to other

d.

Returning to (34.129), we decompose the Riemann tensor for the Einstein space (34.132) as

$$R_{\alpha\beta}{}^{ab} = C_{\alpha\beta}{}^{ab} + \delta_{[\alpha}^a \delta_{\beta]}^b \frac{2}{n(n-1)} R. \quad (34.137)$$

Since $R = 4c^2 n(n-1)$ according to (34.132), we see that (34.129) reduces to

$$C_{\alpha\beta}{}^{ab} \tau_{ab} \eta = 0. \quad (34.138)$$

On arbitrary coset manifolds, the anholonomic (= with flat indices) components of the Riemann tensor, and hence of the Weyl tensor, are constant. Hence

$$C_{ab}{}^{cd} \tau_{cd} = \text{holonomy generators} \quad (34.139)$$

generate a subgroup of $SO(n)$, which is called the holonomy group. It is **not** the holonomy group of relativity which is the group of linear rotations of a vector generated by round transport. It is rather the holonomy group for the “de Sitter” connection which contains the $c\tau_\alpha\eta$ terms. The connection $\frac{1}{4}\omega^{ab}\tau_{ab} - ce^a\tau_a$ is $SO(n+1)$ valued. Since $SO(n+1)$ cannot act linearly on an n -dimensional vector, one must consider a vector plus a scalar in general relativity before one can discuss the holonomy group in (34.139) acting on vectors instead of spinors. For spinors there is no such problem in $d=7$, because spinors of $SO(7)$ and chiral spinors of $SO(8)$ have the same number of components, namely 8.

Further integrability conditions of (34.128) follow by differentiating (34.138) and using (34.128) to eliminate derivatives of n . The next condition is

$$(D_\lambda C_{\mu\nu\rho\sigma}) \tau^{\rho\sigma} \eta = 2c C_{\mu\nu\lambda\rho} \tau^\rho \eta. \quad (34.140)$$

One may show that if the $(n+1)$ th condition is satisfied as a consequence of the first n conditions, then no further integrability conditions will be found beyond the first n conditions. Thus on spheres, where the Weyl tensor vanishes, there are twice as Killing spinors as a spinor has components, namely, spinors with positive and spinors with negative constant c .

One may build Killing vectors from Killing spinors η_A ($A = 1, \dots, N$)

$$k_{AB}{}^\alpha = \bar{\eta}_A \tau^\alpha \eta_B = -\bar{\eta}_B \tau^\alpha \eta_A. \quad (34.141)$$

Such k_{AB}^α may vanish. For example, on K_3 , which is a $d=4$ compact $R_{\alpha\beta}=0$ manifold without Killing vectors but with two chiral Killing spinors, the expression (34.141) vanishes. It may also be that some of the k_{AB}^α are linearly related. An example is the n -sphere, where the Killing vectors constructed from $+$ spinors ($D_\alpha \eta^+ = +C\tau_\alpha \eta^+$) are linearly related to the Killing vectors constructed from η^- . That this should be the case is a priori clear, since S_n has an isometry group $SO(n+1)$ and should have $\frac{1}{2}n(n+1)$ Killing vectors, whereas (34.141) would give twice as

many if k_{AB}^+ and k_{AB}^- were independent.

To show that on S_n the $k_{AB}^{+\alpha}$ and $k_{AB}^{-\alpha}$ are linearly dependent, we use an explicit representation. Namely

$$e_\alpha^a = \frac{\delta_\alpha^a}{1+z^2}, \quad \omega_\alpha^{ab} = \frac{-2\delta_\alpha^a z^b + 2\delta_\alpha^b z^a}{1+z^2}, \quad \eta^\pm(z) = \frac{1 \pm i\tau_a z^\alpha \delta_\alpha^a}{\sqrt{1+z^2}} \eta(0). \quad (34.142)$$

One may show by computing $\partial_\alpha e_\beta^a - \partial_\beta e_\alpha^a$ that ω_α^{ab} is indeed the spin connection for e_α^a , and from ω_α^{ab} one can show that the Riemann tensor is equal to $g_{\alpha\gamma}g_{\beta\delta} - g_{\alpha\delta}g_{\beta\gamma}$ which proves that e_α^a is a vielbein on S_n . Finally, one shows by direct computation that

$$D_\alpha \eta^\pm(z) = \pm i\tau_\alpha \eta^\pm(z); \quad \tau_\alpha = e_\alpha^a \tau_a. \quad (34.143)$$

Assume now $\eta_A^\pm(0)$ to be spinors which have zero entries, except one in the A th entry. Then, on S_7 with purely imaginary and hence antisymmetric τ^α ,

$$(1+z^2)k_{AB}^{\pm\alpha}(z) = [(1 \mp i\tau)^\alpha (1 \pm i\tau)]_{AB}. \quad (34.144)$$

Note that this expression is antisymmetric in AB . Because the trace of Dirac matrices is invariant under $\tau_\alpha \rightarrow -\tau_\alpha$ we have the identity

$$[(1-i\tau)^\alpha (1+i\tau)]_{AB} \begin{pmatrix} \tau^\beta \\ \tau^{\beta\gamma} \end{pmatrix}_{BA} = [(1+i\tau)^\alpha (1-i\tau)] \begin{pmatrix} \tau^\beta \\ -\tau^{\beta\gamma} \end{pmatrix}_{DC}. \quad (34.145)$$

From the Fierz rearrangement formula

$$\bar{\eta}^{[B} \eta_L \bar{\eta}_K \eta^{A]} = \frac{1}{8} \bar{\eta}^B \tau^\beta \eta^A \bar{\eta}_K \tau_\beta \eta_L - \frac{1}{16} \bar{\eta}^B \tau^{\beta\gamma} \eta^A \bar{\eta}_K \tau_{\beta\gamma} \eta_L \quad (34.146)$$

we derive the identity

$$\delta_K^{[A} \delta_L^{B]} = \frac{1}{8} (\tau^B)_{BA} (\tau^\beta)_{KL} - \frac{1}{16} (\tau^{\beta\gamma})_{BA} (\tau^{\beta\gamma})_{KL}. \quad (34.147)$$

Thus, by multiplying (34.145) by $\frac{1}{8}(\tau^\beta)_{KL}$ and $-\frac{1}{16}(\tau^{\beta\gamma})_{KL}$ respectively, and adding both terms, one arrives with (34.147) at

$$k_{KL}^{+\alpha} = k_{CD}^{-\alpha} \left(\frac{1}{8} \tau^\beta{}_{DC} \tau^\beta{}_{KL} + \frac{1}{16} \tau^{\beta\gamma}{}_{DC} \tau^{\beta\gamma}{}_{KL} \right). \quad (34.148)$$

Hence, indeed, on S_n the $k^{+\alpha}$ and $k^{-\alpha}$ are linearly related.

In a similar way one can prove that all $k_{AB}^{+\alpha}$ are linearly independent. Namely suppose $C^{AB} k_{AB}^{+\alpha} = 0$ for some constants C^{AB} . Since C^{AB} is antisymmetric, one can expand it into $(c^a \tau_a + c^{ab} \tau_{ab})^{AB}$. Contraction of this expression for C^{AB} with (34.144) then shows that for no c^a and c^{ab} one can obtain zero, except $c^{AB} = 0$.

34.12 Holonomy Group

The results of this section are from F. Englert, M. Romain and P. Spindel, *Phys. Lett. B* **127** (1983) 47. P. van Nieuwenhuizen and N. J. Warner, *Comm. Math. Phys.* **93** (1984) 277.

Since the Riemann curvature of the spin connection with flat indices R^{ab}_{cd} is constant, we can replace the Cartan-Maurer equations by what is sometimes called the structure equations of a manifold. They read

$$de^a + \omega(\text{spin})^{ab} \wedge e^b = 0 \quad (34.149)$$

$$d\omega(\text{spin})^{ab} + \omega(\text{spin})^{ac} \wedge \omega(\text{spin})^{cb} = \frac{1}{2} R^{ab}_{cd} e^c \wedge e^d. \quad (34.150)$$

Since the vielbein belongs to a coset G/H , R^{ab}_{cd} is constant, see section 34.7.

One can combine the spin connection and the vielbein into an $SO(8)$ connection

$$\Omega^{AB} = \{\omega(\text{spin})^{ab}, \omega^{a8} = \rho e^a\} \quad (34.151)$$

where ρ is a constant. With this $SO(8)$ connection one can define $SO(8)$ covariant derivatives. For example, for an 8-component spinor one can define $SO(8)$ generators $\{\tau^{ab}, \tau^{a8} = i\tau^a\}$ and their $SO(8)$ covariant derivative becomes

$$\left(\partial_\alpha + \frac{1}{4} \omega(\text{spin})_\alpha^{ab} \tau_{ab} - c e_\alpha^a \tau_a \right) \eta = 0 \quad (34.152)$$

where c is a constant. These η 's are Killing spinors.

As a second example, consider the curl F_{abcd} of the 3-index photon A_{abc} . Defining $F_{abc8} = \sigma A_{abc}$ with σ a constant, one can define an $SO(8)$ -covariantly constant tensor by

$$D^{SO(8)} F_{ABCD} = \alpha F_{ABCD} + 4 \Omega_{[D}^{D'} F_{ABC]D'} = 0. \quad (34.153)$$

Decomposition of this equation shows that it contains two parts: a part which repeats that F is the curl of A , and another part which says that the derivative of F is proportional to A . Contraction then leads to: the divergence of F is proportional to A . Hence, if F_{ABCD} is selfdual, we have a solution of the internal Maxwell equation (but not the reverse: not every solution need be $SO(8)$ -covariantly constant).

Such $SO(8)$ -covariantly constant tensors T as in (34.152) and (34.153) lead to integrability conditions on the $SO(8)$ curvature, namely

$$\left(R^{AB} J_{AB} \right) T = 0 \quad (34.154)$$

where J_{AB} are $SO(8)$ generators for T . The $(a8)$ components of R^{AB} vanish, because they vanish according to (34.149). Furthermore, the (ab) components of R^{AB} are equal to the Weyl tensor (see

next section). Hence, the integrability conditions become

$$\left(C^{ab}{}_{cd}J^{cd}\right)T(z)=0 \quad (34.155)$$

where J^{cd} are $SO(7)$ generators, and $C^{ab}{}_{cd}$ **constants**.

The group generated by $C^{ab}{}_{cd}J^{cd}$ is called the holonomy group. It was first introduced into supergravity by M. A. Awada, M. J. Duff and C. N. Pope, *Phys. Rev. Lett.* **50** (1983) 294. The discussion below follows L. Castellani, R. D'Auria, P. Fré and the author, *J. Math. Phys.* (1984).

Let us briefly discuss the relation between our definition of holonomy group and the holonomy group one is familiar with in general relativity, that is, the holonomy group of Riemannian geometry. In general relativity one parallel-transport a vector v^a around a closed curve and finds a holonomy group generated by ordinary Riemann tensors while in our case it is generated by Weyl tensors. Thus the maximally symmetric n -sphere has in general relativity the holonomy group $SO(8)$ while in our case S_n has as holonomy group the unit group. Said differently, in general relativity one considers the holonomy group of the metric connection (= spin connection) while we consider here the holonomy group of the de Sitter connection ($SO(8)$ in our case) which is the sum of the spin connection plus the vielbein term. The vector representation of the de Sitter group splits into a vector plus a scalar under the Lorentz subgroup, and since one restricts one's attention in general relativity to Lorentz vectors, one cannot consider the de Sitter holonomy group. However, for such representations as spinors and anti-selfdual F_{ABCD} which remain irreducible under the Lorentz subgroup, one can define both holonomy groups.

The dependence of $SO(8)$ -covariantly constant tensors T on z can be related to the holonomy group as follows.

Consider an $SO(8)$ -covariantly constant $SO(8)$ representation, denoted generically by $F(z)$. Consistence of the covariantly constancy requires that the $SO(8)$ Lie algebra valued curvatures $R^{AB}J_{AB}$ annihilate $F(z)$ at all z . Hence $F(z)$ must belong to the null space N of the matrix $R^{AB}J_{AB}$.

$$R^{AB}J_{AB}v=0 \quad \Leftrightarrow \quad v \in N, F(z) \in N \text{ for all } z. \quad (34.156)$$

The 21 $SO(8)$ Lie algebra valued matrices $R^{AB}{}_{cd}J_{AB}$ lie actually in $SO(7)$ due to (34.149). In general $R_{cd} = C_{cd}$ do not span already a Lie algebra; however their commutators generate the holonomy algebra C of the given manifold, which may coincide with $SO(7)$ or be a proper subgroup thereof. Given the null space N of C , we denote by Σ the set of generators of $SO(8)$ which annihilate N . Clearly Σ is a (proper or improper) subalgebra of $SO(8)$ and contains C

$$C = SO(7) \cap \Sigma. \quad (34.157)$$

Since $CF(z) = 0$ for all z , and $F(z) \in N$, also $\Sigma F(z) = 0$ for all z . Let us denote by $g(z)$ the $SO(8)$ matrix which sweeps $F(z)$ out from the origin, and the group generated by Σ by S . Then

$gS \in SO(8)$ and $(gS - Sg)v = 0$ for $v \in N$. Hence

$$F(z) = g(z)F(0), \quad g(z) \in SO(8) \text{ for all } z. \quad (34.158)$$

Clearly, $g(z)$ maps null vectors of Σ into null vectors of Σ . Let us denote the generators of $g(z)$ by Z . Then we must have

$$[Z, \Sigma] \subset \Sigma. \quad (34.159)$$

These generators Z define a new subalgebra of $SO(8)$ because of the Jacobi identities. Z is the normalizer of Σ in $SO(8)$, i.e., Z is the largest subalgebra of $SO(8)$ in which Σ is an invariant subalgebra (an ideal). The matrix $g(z)$ can now be written as $e^{\epsilon^I(z)Z_I}$ where I labels the generators of Z which are not in Σ . The relation between $\epsilon^I(z)$ and the $SO(8)$ connection $\Omega_{AB}(z)$ is dictated by the requirement that F be covariantly constant, namely by

$$\left(dg(z) + \Omega^{AB}(z)J_{AB} \right) F(0) = 0. \quad (34.160)$$

Example: Suppose $\Sigma = SO(p)$. Writing $SO(8)$ as $SO(p+q)$, clearly, $[SO(p), SO(q)] = 0$, hence Z contains $SO(p) \times SO(q)$. Since the Grassman manifold $SO(p+q)/SO(p)$ is symmetric, no other generators of $SO(p+q)$, when commuted with $SO(p)$, produce $SO(p)$, hence $Z = SO(p) \times SO(q)$.

Let us give two further examples.

- (i) **Round S_7 with torsion.** Since R^{ab}_{cd} is maximally symmetric, the holonomy group C is either $SO(7)$ or the unit group, depending on the value of the cosmological constant Λ . By choosing Λ such that the $SO(8)$ curvature vanishes, $C = 0$. This value of Λ is indeed the one which follows from the Maxwell equations. Since the null space is the whole space, there are no $SO(8)$ generators which leave the null space invariant. Hence also $\Sigma = 0$. It follows that the normalizer of Σ in $SO(8)$ coincides with $SO(8)$: $Z = SO(8)$. Hence, in this case a general representation F depends on z as $F(z) = g(z)F(0)$ where $g(z)$ is any $SO(8)$ matrix whose coefficients depend on z . For an example see (34.142).
- (ii) **The squashed S_7 with torsion.** In this case C equals G_2 (recall: C is a subgroup of $SO(7)$ and representation independent). From the explicit form of the Riemann curvatures, C can be identified as G_2 . Hence the subalgebra of $SO(8)$ which leaves for example (recall that Σ is representation independent), an $SO(8)$ spinor invariant, is $SO(7)$. The normalizer Z of $SO(7)$ in $SO(8)$ is this $SO(7)$ itself (see the example). Hence, the spinor is invariant under Z , and thus the spinor is actually constant. Thus the solution is $F_{ABCD}(z) = F_{ABCD}(0) = \bar{\eta}\tau_{ABCD}\tau$.

34.13 Killing spinors as Killing vectors in superspace

It is possible to give a more geometrical characterization of Killing spinors than the statement that they are a square root of Killing vectors (in the sense that $\bar{\eta}^A \tau^\alpha \eta^B$ is a Killing vector). Namely,

Killing spinors are Killing vectors in fermionic directions in superspace.

To show this deep property, we must first show how one constructs a superspace, once one has given an ordinary manifold with coordinates z^α . We shall use the method of gauge completion (“a principal fibre bundle over the manifold M ”). This method “lifts” the z -space fields from the $\theta = 0$ section of superspace into the full super space, without the need of introducing more fields than there are in z -space. It is a purely kinematical procedure; nothing is said about field equations. (When the ordinary space theory has a closed algebra and thus one knows a set of auxiliary fields, then also the superspace fields will be off-shell.) When we ask about the isometries of a given field configuration, we do not need field equations, of course. What we must find are Killing vectors k in superspace such that the change in a field if one moves in the direction of k , is equal to a local symmetry of the theory. Thus, for the supervielbein to be constructed, we need to find k^Λ such that

$$k^\Lambda \partial_\Lambda E_\Pi^M + (\partial_\Pi k^\Lambda) E_\Lambda^M = W(k)^M{}_N E_\Lambda^N \quad (34.161)$$

where $\Lambda = \{\alpha, \dot{A}\}$ are curved (bosonic, fermionic) indices. The superscripts M are flat superindices, decomposing into bosonic a and fermionic A , and W is a Lorentz matrix.

The method to lift fields into superspace, is to require that the local symmetries of ordinary space (general coordinate transformation, local supersymmetry transformations, local Lorentz transformations, local Maxwell transformations, ...) are compatible with the local symmetries of superspace (general supercoordinate transformations and local Lorentz transformation). In this way one obtains a kind of **minimal superspace**: it contains only the super-vielbein, but the various ordinary space fields can appear at various θ -levels in the supervielbein.

When we discuss local supersymmetry, we cannot restrict our attention to the compact coset manifold above, but we must consider the full $d = 11$ spacetime manifold, which factorizes into a $d = 4$ anti-de Sitter spacetime, and a compact coset manifold. In this background the gravitino vanishes: $\psi_\Lambda(x, z, \theta) = 0$ at $\theta = 0$. The superspace we consider contains, in addition to the bosonic coordinates x^μ and z^α , a 32-component Majorana spinor coordinate θ^α .

We identify the $\theta = 0$ bose-bose part of the supervielbein with the vielbein of $x - z$ space, and the $\theta = 0$ bose-fermi part with the gravitino. We will need also the fermi-bose and fermi-fermi parts at $\theta = 0$, and fix these as follows. We identify the $\theta = 0$ bose part of the superparameter of general supercoordinate transformations with the parameter of general $x - z$ coordinate transformations, and we identify the $\theta = 0$ fermi part of superparameter with the parameter of local supersymmetry transformation in $d = 11$ x, z spacetime.

Compatibility of superspace and ordinary space transformations thus requires for the $\theta = 0$ parts

$$(\delta E_\Pi^M)_{\theta=0} = (\xi^{\dot{A}} \partial_{\dot{A}} E_\Pi^M + (\partial_\Pi \xi^{\dot{A}}) E_{\dot{A}}^M) = \begin{cases} \frac{1}{2} \bar{\epsilon} \gamma^{\hat{m}} \psi_{\hat{m}} & \text{if } E_\Pi^M = e_{\hat{\mu}}^{\hat{m}} = \text{vielbein in } d = 11 \\ \partial_{\hat{\mu}} \epsilon + \text{“more”} & \text{if } E_\Pi^M = \psi_{\hat{\mu}}^A = \text{gravitino in } d = 11 \end{cases} \quad (34.162)$$

(hatted indices refer to $d = 11$ bosonic spacetime, and $\dot{A}$, A are curved, flat spinor indices). Thus, $E_{\hat{\mu}}^{\hat{m}}$ at order θ is equal to $\frac{1}{2}\bar{\theta}\gamma^{\hat{m}}\psi_{\hat{\mu}}$ while $E_{\dot{A}}^A(\theta = 0) = 0$. Further, from $\delta\psi_{\hat{\mu}}^A$ we deduce that $E_{\dot{A}}^A(\theta = 0) = \delta_{\dot{A}}^A$ while the order θ -part of $E_{\hat{\mu}}^A$ is equal to

$$\text{“more”} = \delta\psi_{\hat{\mu}}^A(x, z; \epsilon \rightarrow \theta) - \partial_{\hat{\mu}}\theta^A \quad (34.163)$$

(namely to the variation of the gravitino, but with ϵ replaced by θ and without the leading term $\partial_{\hat{\mu}}\epsilon$). We conclude that the supervielbein in minimal superspace has the form

$$E_{\Lambda}^M(x, z, \theta) = \begin{pmatrix} e_{\hat{\mu}}^{\hat{m}}(x, z) & \psi_{\hat{\mu}}^A(x, z) \\ 0 & \delta_{\dot{A}}^A \end{pmatrix} + \begin{pmatrix} \frac{1}{2}\bar{\theta}\gamma^{\hat{m}}\psi_{\hat{\mu}} & \text{“more”} \\ \dots & \dots \end{pmatrix}. \quad (34.164)$$

We are now in a position to investigate the isometries of the supervielbein. Let us return to (34.161) and restrict ourselves to the case

$$k^A(\theta = 0) = (0, \epsilon^{\dot{A}}), \quad E_{\Pi}^M = E_{\hat{\mu}}^A. \quad (34.165)$$

In that case, the Killing equation (34.161) at $\theta = 0$ reduces to

$$\epsilon^{\dot{A}}(\partial_{\dot{A}}E_{\hat{\mu}}^A)_{\theta=0} + (\partial_{\hat{\mu}}\epsilon^{\dot{A}})E_{\dot{A}}^A(\theta = 0) = 0. \quad (34.166)$$

We find zero on the right hand side because in the Kaluza-Klein vacuum the gravitino field vanishes. Substitution of (34.164) shows that (34.166) is equivalent to

$$\delta\psi_{\hat{\mu}}^A = 0. \quad (34.167)$$

Hence, those spinors $\epsilon^A(x, z)$ for which the $d = 11$ gravitino variation vanishes, are Killing spinors in the sense of (34.161). On the other hand, substituting the Freund-Rubin and the Englert ansatz into the gravitino law

$$F_{\mu\nu\rho\sigma} = e\epsilon_{\mu\nu\rho\sigma}, \quad F_{\alpha\beta\gamma\delta} = a\bar{\eta}\tau_{\alpha\beta\gamma}\delta\eta \quad (34.168)$$

we arrive at

$$\begin{aligned} \hat{\mu} = \mu : \quad & \left(D_{\mu} + \frac{ie\sqrt{2}}{6}\gamma_5\gamma_{\mu}\right)\epsilon + \frac{\sqrt{2}}{288}\gamma_{\mu}F_{\alpha\beta\gamma\delta}\tau^{\alpha\beta\gamma\delta}\epsilon = 0 \\ \hat{\mu} = \alpha : \quad & \left(D_{\alpha} + \frac{ie}{6\sqrt{2}}\tau_{\alpha}\right)\epsilon + \frac{\sqrt{2}}{288}F_{\beta\gamma\delta\epsilon}\left(\tau_{\alpha}^{\beta\gamma\delta\epsilon} - 8\delta_{\alpha}^{\beta}\tau^{\gamma\delta\epsilon}\right)\gamma_5\epsilon = 0. \end{aligned} \quad (34.169)$$

If we write $\epsilon = \epsilon(x, z) = \sum \epsilon^I(x)\eta^I(z)$, possible solutions exist iff $a = 0$, namely

$$\begin{aligned} (D_{\mu} - 2c\gamma_5\gamma_{\mu})\epsilon^I(x) &= 0 \\ (D_{\alpha} - c\tau_{\alpha})\eta^I(z) &= 0. \end{aligned} \quad (34.170)$$

Thus the $\eta^I(z)$ are Killing spinors.

Conclusion: a Killing spinor $\eta(z)$ on a compact $d = 7$ manifold is a Killing vector in the minimal superspace into which fields are lifted from $d = 11$ spacetime by gauge completion.

For an introduction and further literature concerning gauge completion, see P. van Nieuwenhuizen, *Physics Reports* **68** (1981) 189.

34.14 The relation between ordinary and conformal Killing spinor-tensors

Conformal Killing spinor-tensors are spinor-tensors of the Lorentz or $SO(n)$ group, whose derivatives do not contain the highest irreducible representation possible. For example, the derivative of a conformal Killing vector $D_\alpha k_\beta$ contains a symmetric traceless piece $D_{(\alpha} k_{\beta)}$, a trace $D^\alpha k_\alpha$ and an antisymmetric piece, of which the highest piece, $D_{[\alpha} k_{\beta]}$ is absent. Hence a conformal Killing vector satisfies

$$D_\alpha k_\beta + D_\beta k_\alpha = g_{\alpha\beta} \phi. \quad (34.171)$$

It immediately follows that $\phi = \frac{2}{n} D^\gamma k_\gamma$.

As a second example, consider a conformal Killing spinor χ . Its derivative contains a gamma-traceless part and a gamma-trace, $(D_\alpha - \frac{1}{n} \tau_\alpha \not{D})\chi$ and $\tau_a \not{D}\chi$. Thus a conformal Killing spinor satisfies

$$D_\alpha \chi = \tau_\alpha \lambda. \quad (34.172)$$

It immediately follows that $\lambda = \frac{1}{n} \not{D}\chi$.

As a third example, consider a conformal Killing vector-spinor. It is defined by

$$D_\mu \chi_\nu = A_{\mu\nu} + (\gamma_\mu \phi_\nu + \gamma_\nu \phi_\mu) + g_{\mu\nu} \lambda \quad (34.173)$$

where $A_{\mu\nu} = -A_{\nu\mu}$ and A, ϕ and λ are arbitrary. Again one can express A, ϕ and λ in terms of χ .

Let us find the general solution of (34.172). Its divergence yields

$$\square \chi = \frac{1}{n} \not{D} \not{D} \chi = \frac{1}{n} (\square \chi + \frac{1}{4} R \chi) \quad (34.174)$$

from which we deduce for constant R (for example on Einstein spaces) that

$$\square \chi = \frac{1}{4(n-1)} R \chi \equiv \lambda^2 \chi. \quad (34.175)$$

Defining

$$\psi_\pm = \chi^\pm \lambda^{-1} \not{D} \chi \quad (34.176)$$

we see that

$$\chi = \frac{1}{2}(\psi_+ + \psi_-), \quad \not{D} \psi_\pm = \pm \lambda \psi_\pm \quad (34.177)$$

as follows from (34.175). If we have an Einstein space, we can prove that both ψ_+ and ψ_- are ordinary Killing spinors. The proof proceeds as follows

$$D_\alpha \psi_+ = D_\alpha \chi + \lambda^{-1} [D_\alpha, \not{D}] \chi + \lambda^{-1} \not{D} \left(\frac{1}{n} \tau_\alpha \not{D} \chi \right) \quad (34.178)$$

where we used (34.172) in the last term. Thus

$$D_\alpha \psi_+ = D_\alpha \chi + \lambda^{-1} \left(R^\beta_{\gamma\alpha\beta} \tau^\gamma \chi + \frac{1}{4} R_{\alpha\beta}{}^{ab} \tau_{ab} \tau^\beta \chi \right) + \frac{1}{n} \lambda^{-1} \left(-\tau_\alpha n \lambda^2 + 2 D_\alpha \not{D} \right) \chi. \quad (34.179)$$

where we used (34.174) and (34.175). Since we have an Einstein space, the terms with curvatures yield, due to the cyclic identity,

$$\lambda^{-1} \left(\frac{1}{n} \tau_\alpha \chi \right) \left(1 - \frac{2}{4} \right) \left(\lambda^2 4(n-1) \right). \quad (34.180)$$

We see that in (34.179) $D_\alpha \not{D} \chi$ appears on the left hand side, while the right hand side only contains $\tau_\alpha \chi$ terms (the $D_\alpha \chi$ terms cancel clearly). Thus we find

$$\left(1 - \frac{2}{n} \right) D_\alpha \not{D} \chi = \frac{\lambda^2}{n} \tau_\alpha \chi \quad (34.181)$$

and with this result it follows that

$$D_\alpha \psi_\pm = \pm \frac{\lambda}{n} \tau_\alpha \psi_\pm. \quad (34.182)$$

Hence, a conformal Killing spinor is equal to an ordinary Killing spinor (see below (34.136)) except on spheres where it can be a sum of two ordinary Killing spinors.

Conformal Killing vectors only exist on spheres, not on other compact spaces. Let us prove this for symmetric Einstein spaces. Splitting

$$k_\alpha = k_\alpha^T + D_\alpha \phi, \quad D^\alpha k_\alpha^T = 0 \quad (34.183)$$

and squaring and integrating $(D_\alpha k_\beta + D_\beta k_\alpha)^2$, the cross terms cancel because

$$\begin{aligned} \int -k_\beta^T D_\alpha \left(D^\alpha D^\beta \phi - \frac{1}{n} g^{\alpha\beta} \square \phi \right) &= \int -k_\beta^T \left(\square D^\beta \phi - \frac{1}{n} D^\beta \square \phi \right) \\ &= \int -k_\beta^T \left(\left(1 - \frac{1}{n} \right) D^\beta \square \phi - R^{\beta\gamma} D_\gamma \phi \right) = 0 \end{aligned} \quad (34.184)$$

which vanishes since $D^\gamma k_\gamma^T = 0$. Hence $D_\alpha k_\beta^T + D_\beta k_\alpha^T = 0$ and $(D_\alpha D_\beta - \frac{1}{n} g_{\alpha\beta} \square) \phi = 0$. Taking the D_α derivative of the last equation we get

$$\left(\frac{n-1}{n} \right) D^\beta (\square \phi + n \phi) = 0 \quad (34.185)$$

from which we deduce that $\square \phi + n \phi = 0$ (possibly after absorbing a constant into ϕ).

Hence, $D_\alpha D_\beta \phi + g_{\alpha\beta} \phi = 0$, and taking the D_γ derivative

$$D_\alpha D_\gamma D_\beta \phi - R_{\beta\gamma\alpha} D_\delta \phi + g_{\alpha\beta} D_\gamma \phi = 0 \quad (34.186)$$

from which it follows, using $D_\gamma D_\beta \phi + g_{\gamma\beta} \phi = 0$, that

$$C^\delta_{\beta\gamma\alpha} D_\delta \phi = 0. \quad (34.187)$$

On symmetric manifolds, the Riemann tensor and hence the Weyl tensor are covariantly constant; hence, taking the D_ϵ derivative and using $D_\epsilon D_\delta \phi + g_{\epsilon\delta} \phi = 0$,

$$C^\delta_{\beta\gamma\alpha} = 0. \quad (34.188)$$

Hence, conformal Killing vectors are a sum of an ordinary Killing vector and a gradient of a scalar; they can only exist on spheres, and the scalar is the lowest nontrivial eigenmode of the D'alembertian.

Reference, N. P. Warner and P. van Nieuwenhuizen, *Comm. Math. Phys.* **93** (1984) 277; M. Walker and R. Penrose, *Comm. Math. Phys.* **18** (1970) 265.

As an application, we consider the algebra of ordinary and conformal Killing tangent vectors. (M. Gunaydin and N. P. Warner, unpublished.) The lowest nontrivial scalar modes on S_n can be written as $\bar{\eta}^A \eta^B$ with η_+ and η_- defined in (34.133) and $A, B = 1, \dots, k$ (where k is the number of components of a spinor). There are only $(n+1)$ independent scalars

$$\phi^{\dot{\gamma}} = \bar{\eta}_+^A \eta_-^B (i\tau^{\dot{\gamma}})_{AB}, \quad \phi^{n+1} = (\bar{\eta}_+^A \eta_-^B) \delta_{AB} \quad (34.189)$$

as follows from (34.142) and (34.144). The $n+1$ scalars $\phi^I = (\phi^i, \phi^{n+1})$ satisfy

$$D_\alpha D_\beta \phi^I = -g_{\alpha\beta} \phi^I, \quad \phi^I \phi_I = k \quad \text{sum over } I. \quad (34.190)$$

Hence $D_\alpha \phi^I$ are conformal Killing vectors and

$$c_\beta^I \equiv D_\beta \phi^I, \quad \phi^I D_\alpha \phi^I = 0, \quad D_\alpha \phi^I D_\beta \phi_I = g_{\alpha\beta}. \quad (34.191)$$

The $(n+1) \times (n+1)$ matrix $(D_\alpha \phi^I, D_\beta \phi_I)$ has orthogonal rows, hence also its columns are orthogonal

$$D_\alpha \phi^I D^\alpha \phi^J + \phi^I \phi^J = \delta^{IJ}. \quad (34.192)$$

The ordinary Killing vectors on S_n can be written as

$$k_\alpha^{IJ} = \phi^I D_\alpha \phi^J - \phi^J D_\alpha \phi^I \quad (34.193)$$

and from (34.192) one may prove the identity

$$k_\alpha^{IJ} K_{KL}^\alpha + \frac{1}{2} D_\alpha k_\beta^{IJ} D^\alpha k_{KL}^B = \delta_K^I \delta_L^J - \delta_L^I \delta_K^J \quad (34.194)$$

and from this result one may prove that the tangent vectors C^I and k^{IJ} generate $SO(n+1, 1)$.

34.15 The isometries of manifolds with Killing spinors

Given a manifold with N Killing spinors η_A , they yield Killing vectors k_{AB}^α . If the manifold is a coset manifold G/H , there is certainly an isometry group G . There may be a larger isometry group, due to right multiplication, if the normalizer $N(H)$ of H in G is nontrivial.

Theorem I: The k_{AB}^α generate the group $SO(N)$.

Proof:

$$(2c)^{-1} [k_{AB}, k_{CD}] = \left(k_{AB}^\alpha \bar{\eta}_C \tau^{\beta\alpha} \eta_D - k_{CD}^\alpha \bar{\eta}_A \tau^{\beta\alpha} \eta_B \right) \partial_\beta. \quad (34.195)$$

After a Fierz rearrangement, the right hand side yields

$$R^\beta = \frac{3}{2} (\bar{\eta}_A \eta_D) \left(\bar{\eta}_C \tau^\beta \eta_B \right) - \frac{3}{2} \left(\bar{\eta}_A \tau^\beta \eta_D \right) (\bar{\eta}_C \eta_B) + \text{“more”} \quad (34.196)$$

where “more” includes terms with two and more Dirac matrices. The first terms are of the form we want, and we will call them “good terms”. Hence

$$\text{“more”} = R^\beta - \text{“good terms”} \quad (34.197)$$

Fierzing the two “good terms” once more, such that AB and CD are taken together, and antisymmetrizing in AB and CD because R^β has this antisymmetry, one finds

$$\text{“good terms”} = \frac{3}{8} R^\beta. \quad (34.198)$$

Hence, “more” equals $\frac{5}{8} R^\beta$, and thus, with (34.196), we find that $\frac{3}{8} R^\beta$ is equal to the “good terms”. Normalizing the η_A such that $\bar{\eta}_A \eta_B = \delta_{AB}$, we arrive at the desired result

$$[k_{AB}, k_{CD}] = 4c (\delta_{BC} k_{AD} + 3 \text{ more terms}). \quad (34.199)$$

Theorem II: For any Killing vector k^α , the H -covariant Lie derivative is given by

$$\mathcal{L}_k = l_k + \left(D(\omega(e))^a k^b + k^\alpha \omega(\text{spin})_\alpha^{ab} \right) t_{ab}. \quad (34.200)$$

Proof: The H -invariant Lie derivative of the vielbein vanishes per definition. From

$$l_k e^a = \iota_k de^a + d\iota_k e^a, \quad de^a + \omega(\text{spin})^{ab} \wedge e^b = 0 \quad (34.201)$$

where $\omega(\text{spin})(e)$ is the spin connection, we find

$$l_k e^a = -k^\alpha \omega_\alpha^{ab}(e) e^b + \omega^{ab} k^b + dk^a. \quad (34.202)$$

Since $D_\alpha k^a = -(D^a k^b) e^b_\alpha$, the proof of the theorem is completed.

Lemma: The adjoint matrix in (34.58) can be expressed in terms of Killing vectors by

$$D(L)_M^i c_i^{ab} = D(\omega(e))^a k^b - \frac{1}{2} k^d \bar{c}_d^{ab}.$$

Proof: Compare (34.62) with (34.200).

Theorem III:

$$\mathcal{L}_{k_{AB}} \eta_C = 4c (\delta - AC \eta_B - \delta_{BC} \eta_A). \quad (34.203)$$

The H -covariant Lie derivative of a Killing spinor is equal to an $SO(N)$ rotation.

Proof: We follow the same Fierz strategy as in Theorem I. The ordinary Lie derivative of η yields

$$k_{AB}^\alpha \partial_\alpha \eta_C = (\bar{\eta}_A \tau^\alpha \eta_B) \left(-\frac{1}{4} \omega(e)_\alpha^{ab} \tau_{ab} \eta_C \right) + c (\bar{\eta}_A \tau^\alpha \eta_B) (\tau_\alpha \eta_C). \quad (34.204)$$

The last term yields after Fierzing

$$R = \frac{7c}{8} (\bar{\eta}_A \eta_C) \eta_B + \text{“more”} = \text{“good terms”} + \text{“more”} \quad (34.205)$$

Fierzing once more the “good terms” one finds

$$\text{“good terms”} = \frac{7c}{64} (\bar{\eta}_A \tau^\alpha \eta_B) (\tau_\alpha \eta_C) - \frac{7c}{128} (\bar{\eta}_A \tau^{\alpha\beta} \eta_B) (\tau_{\alpha\beta} \eta_C) \quad (34.206)$$

where we used that R is antisymmetric in (AB) . Hence “more” = $R - \text{“good terms”}$, and the

$$\text{“more”} = ck_{AB}^\alpha (\tau_\alpha \eta_C) - \frac{7c}{64} k_{AB}^\alpha (\tau_\alpha \eta_C) + \frac{7c}{128} (\bar{\eta}_A \tau^{\alpha\beta} \eta_B) (\tau_{\alpha\beta} \eta_C). \quad (34.207)$$

On the other hand

$$R = ck_{AB}^\alpha \tau_\alpha \eta_C = \text{“good terms”} + \text{“more”} \quad (34.208)$$

from which we conclude that

$$\frac{7c}{64} k_{AB}^\alpha \tau_\alpha \eta_C = \frac{7c}{128} (\bar{\eta}_A \tau^{\alpha\beta} \eta_B) (\tau_{\alpha\beta} \eta_C) + \text{“good terms”}. \quad (34.209)$$

Since “good terms” is equal to $\frac{7c}{8} \delta_{[A}^C \eta_{B]}$, we arrive at

$$l_{k_{AB}} \eta_C = \left(-k_{AB}^\alpha \omega_\alpha^{ab} \right) (\tau_{ab} \eta_C) + \frac{c}{2} (\bar{\eta}_A \tau_{\alpha\beta} \eta_B) (\tau^{\alpha\beta} \eta_C) + 4c (\delta_{AC} \eta_B - \delta_{BC} \eta_A). \quad (34.210)$$

The proof is completed by noting that

$$\frac{c}{2}\bar{\eta}_A\tau^{\alpha\beta}\eta_B = D^\beta k_{AB}^\alpha = -D^\alpha k_{AB}^\beta \quad (34.211)$$

and by using the definition of $\mathcal{L}_{k_{AB}}$ in Theorem II.

Theorem IV: $\mathcal{L}_{k_G}\eta_A = SO(N)$ rotation where G is the maximal isometry group of the coset manifold. We see thus that Killing vectors of G/H preserve the length of a Killing spinor.

Proof: Since $\mathcal{L}_{k_G}$ commutes with D_α ,

$$D_\alpha (\mathcal{L}_{k_G}\eta_A) = c\tau_a (\mathcal{L}_{k_G}\eta_A). \quad (34.212)$$

Hence, the $\mathcal{L}\eta$ of the group G are a linear contribution of the η 's with constant coefficients

$$\mathcal{L}_{k_G}\eta_A = M_A^B \eta_B, \quad M = \text{constant}. \quad (34.213)$$

Since $\bar{\eta}_A\eta_A$ (no sum) is constant (because $0 = D_\alpha (\bar{\eta}_A\eta_A) = \partial_\alpha (\bar{\eta}_A\eta_A)$) also $l_{k_G} (\bar{\eta}_A\eta_A) = \mathcal{L}_{k_G} (\bar{\eta}_A\eta_A) = 0$. Hence, M_A^B is antisymmetric. Thus M_A^B is an $SO(N)$ generator.

However, the $\mathcal{L}_{k_{IJ}}$ already rotated η_A according to the full $SO(N)$ group. Hence, we can divide the Killing vectors of G into two classes: the k_{IJ}^α , and a remaining set $k_{G'}^\alpha$ which leaves the spinors η_A invariant.

One can also directly prove that (34.213) holds and that M is antisymmetric, by using Theorem II, and eliminating the derivative of η_A from the definition of a Killing spinor. Using that $\sum \eta_B \bar{\eta}_B = 1$, the matrix M is found to be equal to

$$M_A^B = \frac{1}{4} D^a k^b \bar{\eta}_B \tau_{ab} \eta_A - k^\alpha c \bar{\eta}_B \tau_\alpha \eta_A \quad (34.214)$$

which is clearly antisymmetric in (A, B) .

The commutator of $\mathcal{L}_{k_{AB}}$ and $\mathcal{L}_{G'}$ on η_C vanishes. This suggests that the isometry group is a direct product $G = G' \otimes SO(N)$, which is indeed the case as we now prove.

Theorem V: $G = G' \otimes SO(N)$. **Any** isometry group of a coset manifold with Killing spinors has this form. On S_n , G' is trivial.

Proof: We evaluate the commutator of a Killing vector in G' with one in $SO(N)$. One finds

$$\left[k_{g'}^\alpha \partial_\alpha, \bar{\eta}_A \tau^\beta \eta_B \partial_\beta \right] = \left[l_{g'} \left(\bar{\eta}_A \tau^\beta \eta_B \right) \right] \partial_\beta. \quad (34.215)$$

Since $\bar{\eta}_B \tau^\beta \eta_B$ has no flat indices, the Lie derivative can be replaced by the H -covariant Lie derivative, and since the latter is distributive, one finds with Theorem IV that the right hand side of (34.215) vanishes.

Theorem VI: If the Einstein manifold has a nonvanishing cosmological constant, none of the Killing vectors k_{AB}^α vanishes according to (34.203). If the manifold is, however, Ricci flat, all k_{AB}^α vanish; in fact, there are no Killing vectors whatsoever.

Proof: Suppose $R_{\alpha\beta} = 0$ and a Killing vector k^α did exist. Then

$$0 = \int (D^\alpha k^\beta) (D_\alpha k_\beta + D_\beta k_\alpha) = \int (D^\alpha k^\beta)^2. \quad (34.216)$$

Hence k^β would be covariantly constant. This would mean that $R_{\alpha\beta\gamma\delta} k^\delta = C_{\alpha\beta\gamma\delta} k^\delta = 0$ and it can be shown that no compact manifold can satisfy this equation.

Conclusions: $G = G' \otimes SO(N)$ and

$$\mathcal{L}_{k_{G'}} \eta_A = 0, \quad \mathcal{L}_{k_{AB}} \eta_C = 4c (\delta_{AC} \eta_B - \delta_{BC} \eta_A). \quad (34.217)$$

In other words

$$\mathcal{L}_{k_G} \eta_A = -\eta_B (X_G)^B{}_A \quad (34.218)$$

where X_G are generators of G , namely in the identity representation of G' , and in the vector representation of $SO(N)$. The Killing vectors of $SO(N)$ belong in general to the superalgebra $OSp(N|4)$, while those of G belong to matter multiplets.

34.16 Spherical Harmonics

The expansion of an angular scalar function in Y_{lm} 's in quantum mechanics, or the expansion of the electromagnetic vector potential into vector spherical harmonics, are special cases of an expansion of any field which transforms according to a particular representation of the group $SO(n)$. In Kaluza-Klein supergravity, one needs this general expansion to obtain the mass matrices.

Consider an arbitrary representation of the coset representative $L(z)$. Defining

$$Y(z) = L(z)^{-1} \quad (34.219)$$

we find from (34.27), (34.28) and (34.94)

$$\mathcal{L}_{k_M} Y = -Y X_M \quad (34.220)$$

where X_M are the generators of G in the same representation as $L(z)$. Note that due to the minus sign in (34.220), the equations (34.31) and (34.95) are compatible.

Any object Y which satisfies (34.220) will be called a spherical harmonic. We shall explain below why. Let us note that we have already encountered two examples of spherical harmonics. Killing spinors η_A (with $\cdot$ a spinor index and A a label of the various Killing spinors) are spherical

harmonics according to (34.218). But also Killing vectors are spherical harmonics because

$$\mathcal{L}_{k_M} k^{\alpha N} = l_{k_N} k_{\alpha N} = [k_M, k_N]_{\alpha} = k_{\alpha P C M N}^P. \quad (34.221)$$

Note that the matrix X_M of (34.220) is in this case given by $(X_M)^P_N = c_{MN}^P$.

Any representation of $SO(n)$ is also a representation of H when H is represented by the matrices $-c_{iab}t^{ab}$. Because H is compact, we can decompose any $SO(n)$ representations into H -irreducible representations. In each such subspace, one thus has:

$$l_{k_M} Y_s^N - \Omega_M^i c_{iab} (t^{ab})_s^{s'} Y_{s'}^N = -Y_s^{N'} (X_M)_{N'}^N \quad (34.222)$$

where s runs over a basis in the H -irreducible subspace.

This equation tells us what the derivative of a spherical harmonic along a Killing vector is. Because coset manifolds are homogeneous, there are as many Killing vectors as dimensions and thus one should be able to obtain also formulae for a free derivative of a spherical harmonic. These equations follow easily from the Cartan-Maurer equations. Using that

$$L^{-1} dL = Y dY^{-1} = -Y^{-1} dY \quad (34.223)$$

one finds from (34.42), the derivative of Y

$$dY = -(e^a H_a + \omega^i H_i) Y. \quad (34.224)$$

In application, one needs rather the gravitationally covariant derivative with spin connection. Using (34.79), one obtains for this derivative

$$DY \equiv (d + \omega(\text{spin})^{ab} t_{ab}) Y = -e^d (K_d + \frac{1}{2} \bar{c}_d^{ab} t_{ab}) Y. \quad (34.225)$$

From this result, one can compute the eigenvalues of such invariant operators (operators which commute with $\mathcal{L}$) as the D'Alembertian.

Let us first discuss the expansion of a field in Kaluza-Klein supergravity in terms of spherical harmonics. Consider a field $\phi_{\mu\alpha}(x, z)$ in an $SO(3, 1)$ representation T with index $1 \leq \alpha \leq \dim R$. The $SO(n)$ representation splits into H -irrepses

$$\phi_{R\alpha} = \phi_{R_1\alpha_1} + \cdots + \phi_{R_k\alpha_k}. \quad (34.226)$$

Let a complete set of functions for $\phi_{R_i\alpha_i}$ be given by $Y_{R_i\alpha_i}^I(z)$ with $I = 1, \dots, \infty$. Thus,

$$\phi_{R_i\alpha_i}(z) = \sum_I c_I Y_{R_i\alpha_i}^I. \quad (34.227)$$

Consider now $\mathcal{L}_{k_M} Y_{R_i\alpha_i}^I$ for a given i . It is again a function which transforms in the H -irrepses

R_i , since $\mathcal{L}_{k_M}$ is H -covariant. Thus,

$$\mathcal{L}_{k_M} Y_{R_i \alpha_i}^I = Y_{R_i \alpha_i}^J D(M)_J^I. \quad (34.228)$$

Acting once more with $\mathcal{L}_{k_N}$, shows that the matrices $-D(M)$ form a representation of G . Since G is compact, we can decompose it into G -irreps S . Hence,

$$\mathcal{L}_{k_M} Y_{R_i \alpha_i}^{Sj} = -Y_{R_i \alpha_i}^{S'j'} (S_M)_{j'}^j. \quad (34.229)$$

We repeat that S denotes a G -irrep with index $1 \leq j \leq \dim S$. It not follows that

$$\phi_{T_\mu, R\alpha}(x, z) = \sum_i \sum_{Sj} \varphi_{T_\mu}^{Sj}(x) Y_{T_i \alpha_i}^{Sj}(z). \quad (34.230)$$

It follows from Schur's lemma that only those G -irreps S contribute to the Y which contain the H -irreps R_i . To prove this, we consider the relation $gL(z) = L(z')h$ and take for g an element in the stability subgroup of $L(z)$: $g = L(z)hL(z)^{-1}$ for any $h \in H$. Then for h near unity:

$$g = L(z)(1 + \epsilon^i H_i)L(z)^{-1} = 1 + \epsilon^i D_i^{-1}(z)^M X_M \quad (34.231)$$

according to (34.58). Hence

$$D_i^{-1}(z)^M k_M^\alpha = 0. \quad (34.232)$$

It follows from (34.62) that

$$D_i^{-1M} \mathcal{L}_{k_M} Y = D_i^{-1M} \Omega_M^j (-c_{jab} t^{ab}) Y = (c_{iab} t^{ab}) Y. \quad (34.233)$$

Now c_{MN}^P is invariant under adjoint transformations with $(D^{-1})_M^N$. Hence, the matrices $D_i^{-1M} S_M$ form a representation $\hat{H}_i$ of H . Thus (34.229) leads to

$$(c_{iab} t^{ab}) Y = Y (D_i^{-1M} S_M). \quad (34.234)$$

This result states that Y is an interpolating matrix between the H -irreps R_i , α_i and another set of H -irreps, namely the G -irreps S . Schur's lemma tells us that Y can only be nonzero if S contains R_i . These G -irreps are denoted by S in (34.230). It follows that a given G -irrep appears as many times in the expansion as it contains H -fragments R_i . (The set of G -irreps contributing to a given $Y_{R_i \alpha_i}$ depends in general on i , but to simplify the notation somewhat, we have suppressed a subscript i in $(S_i j_i)$).

34.17 Beyond one-forms: antisymmetric tensors

Antisymmetric tensors play a crucial role in supergravity. They appear in the $d = 11$ model as a 3-index antisymmetric tensor, and by dimensional reduction also in models in lower dimensions. One can transform antisymmetric tensors into other antisymmetric tensors by a duality relation.

Consider as an elementary example an action with a field $A_{\nu_1 \dots \nu_k}$ which appears only as its curl F . For example, a Maxwell action plus Pauli couplings of the form $\bar{\psi}\psi F$. Often one finds also terms of the form ϵ -tensor times one F times other fields. Go now to a first order formalism in which the curl

$$F_{\nu_1 \dots \nu_{k+1}} = \partial_{[\nu_1} A_{\nu_2 \dots \nu_{k+1}]} \quad (34.235)$$

is considered an independent field. Adding a Lagrange multiplier field which tells us that F is a curl, we get

$$I = \int dx \left[\mathcal{L}(F) + \epsilon^{\nu_1 \dots \nu_d} \partial_{\nu_1} F_{\nu_2 \dots \nu_{k+2}} M_{\nu_{k+3} \dots \nu_d} \right]. \quad (34.236)$$

The field equation for M says that F is a curl, and inserting this result: in (34.236) we find back the original second order action. However, eliminating instead the field F by solving its algebraic field equation ($\mathcal{L}(F)$ should be of the form $F^2 + FX$), we get a new theory in which the field A has been replaced by the field M .

If the original theory had a local or global symmetry, then the variation of $\mathcal{L}(F)$ in (34.236) (with the variation of the independent field F equal to what it was when F was the curl of A) is proportional to the curl of F (because the variation of $\mathcal{L}(F)$ vanishes when F is the curl of A). Clearly, one can choose δM in (34.236) such that all curl- F variations in (34.236) cancel. The remaining variations, namely that of F in the Lagrange-multiplier term, vanishes, too, since δF is equal to the curl of the variation of the old field A . Hence, the intermediate action in (34.236) is still invariant. Finally, when one solves F and replaces F by its field equation

$$F \sim \epsilon \partial M + \text{terms without } F \text{ or } M \quad (34.237)$$

the variation of F should be replaced by the variation of the right-hand side of (34.237). The action remains invariant since the difference is proportional to the F field equation which is **identically** satisfied **after** one has replaced F according to (34.237). This is the so-called 1.5 order formalism. Hence the theory with M has the same symmetries as the theory with F . The transformation rules of all fields other than A are unchanged, while the variation of M follows from the variation of $\mathcal{L}(F)$. We shall give an example in a moment.

Hence we have the following

Theorem: In d dimensions a massless k -index photon is dual to a massless $d - k - 2$ index photon.

Well-known examples are

$$\begin{aligned}
d = 4 : A_{\mu\nu} &\sim \phi; & A_\mu &\sim V_\mu \\
d = 5 : A_{\mu\nu} &\sim A_\mu \\
d = 7 : A_{\mu\nu\rho} &\sim A_{\mu\nu} \\
d = 10 : A_{\mu\nu\rho} &\sim A_{\mu\nu\rho\sigma\tau}
\end{aligned} \tag{34.238}$$

One can extend these duality relations to massive photons. Consider, for example, photons A and B and the action (Curtright and Freund)

$$\mathcal{L} = A^2 + \epsilon F(A)B + B^2. \tag{34.239}$$

We often omit indices to improve readability. The symbol ϵ is the ϵ -tensor in d dimensions and $F(A)$ is the curl of A . Elimination of **both** A or B leads to a massive antisymmetric tensor theory. Hence

Theorem: In d dimensions a massive k -index photon is dual to a massive $(d - k - 1)$ index photon.

Examples are

$$\begin{aligned}
d = 3 : A_\mu &\sim A_\mu \\
d = 4 : A_{\mu\nu} &\sim A_\mu \\
d = 5 : A_{\mu\nu} &\sim A_{\mu\nu}; & A_\mu &\sim A_{\mu\nu\rho} \\
d = 7 : A_{\mu\nu\rho} &\sim A_{\mu\nu\rho}
\end{aligned} \tag{34.240}$$

Note that in odd dimensions a massive photon can be dual to another massive photon with the same number of indices.

The equivalence of duality related theories at the free-field level may no longer hold at the level of interactions. The classical example is the $d = 11$ model with a 3-index photon, which, at the free field level is equivalent to a model with a 6-index photon, but the latter version cannot be extended to the nonlinear level. In fact, there exists an even simpler example, namely the Wess-Zumino model in $d = 4$ dimensions. It is usually formulated in terms of a scalar A and a pseudoscalar B , but one can use duality transformations to transform A into an antisymmetric tensor $B_{\mu\nu}$, and B into $A_{\mu\nu}$. As we shall now show, the supersymmetry algebra on the bosonic fields $A_{\mu\nu}$ and $B_{\mu\nu}$ no longer closes, and this is a signal that the model cannot survive interactions. So, consider

$$\mathcal{L}(F_\mu, G_\mu, \chi) = -\frac{1}{2}F_\mu^2 - \frac{1}{2}G_\mu^2 - \frac{1}{2}\bar{\chi}\not{\partial}\chi \tag{34.241}$$

where $F_\mu = \partial_\mu A$, $G_\mu = \partial_\mu B$ and

$$\begin{aligned}\delta F_\mu &= \partial_\mu \left(\frac{1}{2} \bar{\epsilon} \chi \right), & \delta G_\mu &= \partial_\mu \left(-\frac{i}{2} \bar{\epsilon} \gamma_5 \chi \right) \\ \delta \chi &= \frac{1}{2} \not{\partial} (A - i \gamma_5 B) \epsilon = \frac{1}{2} \gamma^\mu (F_\mu - i \gamma_5 G_\mu) \epsilon\end{aligned}\tag{34.242}$$

The equivalent of (34.236) becomes

$$I = \int d^4x \left[\mathcal{L} + \epsilon^{\mu\nu\rho\sigma} (\partial_\mu F_\nu B_{\rho\sigma} + \partial_\mu G_\nu A_{\rho\sigma}) \right].\tag{34.243}$$

If F and G are independent fields

$$\delta \mathcal{L} = -\frac{1}{2} \bar{\chi} \gamma^{\rho\sigma} (\partial_\rho F_\sigma - i \gamma_5 \partial_\rho G_\sigma) \epsilon\tag{34.244}$$

and with $\gamma^{\rho\sigma} = \frac{1}{2} \epsilon^{\rho\sigma\tau\nu} \gamma_5 \gamma_{\tau\nu}$ one finds

$$\delta A_{\mu\nu} = -\frac{i}{4} \bar{\chi} \gamma_{\mu\nu} \epsilon, \quad \delta B_{\mu\nu} = \frac{1}{4} \bar{\chi} \gamma_5 \gamma_{\mu\nu} \epsilon.\tag{34.245}$$

So at this point (34.243) is invariant under (34.242) and (34.245). The F_μ , G_μ field equations read

$$F^\mu = \epsilon^{\mu\nu\rho\sigma} \partial_\nu B_{\rho\sigma}, \quad G^\mu = \epsilon^{\mu\nu\rho\sigma} \partial_\nu A_{\rho\sigma}\tag{34.246}$$

and substituting (34.246) into (34.243) one must replace δF_μ in (34.242) by what (34.246) and (34.245) give. The difference is

$$\frac{\delta \mathcal{L}}{\delta F_\mu} [\delta F_\mu \text{ (in (34.246))} - \delta F_\mu \text{ (in (34.242))}]\tag{34.247}$$

but

$$\frac{\delta \mathcal{L}}{\delta F_\mu} = -F^\mu + \epsilon^{\mu\nu\rho\sigma} \partial_\nu B_{\rho\sigma}\tag{34.248}$$

which vanishes identically **after** one has used (34.246). Thus the new action

$$\mathcal{L} = 3 (\partial_{[\mu} B_{\nu\rho]})^2 + 3 (\partial_{[\mu} A_{\nu\rho]})^2 - \frac{1}{2} \bar{\lambda} \not{\partial} \lambda\tag{34.249}$$

is invariant under (34.245) and

$$\delta \chi = \frac{1}{2} \gamma_\mu \epsilon^{\mu\nu\rho\sigma} \partial_\nu (B_{\rho\sigma} - i \gamma_5 A_{\rho\sigma}) \epsilon\tag{34.250}$$

as one may check explicitly. Thus we have checked the general theory of (34.235)-(34.237) in a particular example.

One can now compute the supersymmetry commutator using (34.245) and (34.250). One finds

with $\xi_\mu = \frac{1}{2}\bar{\epsilon}_2\gamma_\mu\epsilon_1$

$$[\delta(\epsilon_1), \delta(\epsilon_2)]A_{\mu\nu} = \xi^\sigma \partial_\sigma A_{\mu\nu} + \partial_\mu (A_{\nu\sigma}\xi^\sigma) - \partial_\nu (A_{\mu\sigma}\xi^\sigma) + \xi_{[\mu} {}^*B_{\nu]} , \quad (34.251)$$

where ${}^*B_\nu = \epsilon_\nu{}^{\rho\sigma\tau}\partial_\rho B_{\sigma\tau}$. The first term is the usual shift, the next two terms are a gauge transformation of $A_{\mu\nu}$ but the last term is strange. On-shell, where $\partial^\mu\partial_{[\mu}B_{\nu\rho]} = 0$

$$\partial_\mu {}^*B_\nu - \partial_\nu {}^*B_\mu = 0 \quad (\text{on-shell}) \quad (34.252)$$

and thus (locally) one has on-shell that ${}^*B_\mu = \partial_\mu B$, i.e., another gauge transformation in (34.251). But off-shell, (34.251) is not closed.

The model in (34.249) cannot be coupled to supergravity, whereas (34.241) can be coupled. **Thus duality related theories need not be equivalent at the level of interactions.** One can show that (34.249) cannot be coupled either directly, by the Noether method, or by the following argument. Suppose the coupling would be possible. Then, in (34.251) the ξ_μ would become local, and, on-shell at least, the last term in (34.251) would read $D_\mu(\xi_\nu\beta) + (D_\mu\xi_\nu)\beta - \mu \leftrightarrow \nu$. Usually the $D_\mu\xi_\nu\beta$ terms would be cancelled by adding terms with ψ_μ , but here these ψ_μ terms cannot do the job because they would be of the form $\epsilon\psi\beta$, whereas there are no bare β terms in the theory but only $B_{\mu\nu}$ fields.

Reference : P. K. Townsend and P. van Nieuwenhuizen, *Phys. Lett. B* **109** (1982) 179.

The S -matrices for duality related fields which are coupled to gravity are still equivalent (E. Sezgin and the author, *Phys. Rev. D* **22** (1980) 301) but the trace anomalies are different (M. J. Duff and the author, *Phys. Lett. B* **94** (1980) 179).

A geometrical framework for antisymmetric tensors is given by the so-called free graded differential super algebras. These are differential algebras (expressions of the form $da = b$) without constraints (“free”, not, for example, $b^2 = c$), which are graded (meaning there are p -forms) and “super” (meaning there are bosonic and fermionic p -forms). These free differential algebras are the natural extensions of (34.72) and (34.73) to p -forms. Although (34.72) and (34.73) are equivalent to (34.114), the extensions of (34.114) to p -forms is much more difficult.

There exists a general recipe to associate to a free differential algebra an action. Several known supergravity actions can be derived in a uniform manner in this way. This method is called “the group manifold approach”. However, the transformation rules in this approach are obtained on-shell to begin with. To find the correct off-shell transformation rules one must do what the component approach has done for years: an order-by-order Noether analysis. (However: the action is already known to all orders.)

An interesting new result is the action of simple $d = 6$ supergravity, which is based on the

following free differential algebra

$$\begin{aligned}
d\omega^{ab} + \omega^{ac} \wedge \omega^{cb} &= 0 \\
de^a + \omega^{ab} \wedge e^b + \bar{\psi} \tau^a \wedge \psi &= 0 \\
d\psi + \frac{1}{4} \omega^{ab} \wedge \tau_{ab} \psi &= 0 \\
dB + \bar{\psi} \wedge \tau_a \psi \wedge e^a &= 0,
\end{aligned} \tag{34.253}$$

where B is a 2-form. The first three equations define the Poincaré superalgebra, whereas the last line is consistent since $dd = 0$ while

$$\tau^a \psi \bar{\psi} \tau_a \psi = 0 \quad \text{for chiral } \psi \text{ in } d = 6. \tag{34.254}$$

One now finds that one of the extra field equations (extra meaning: more than one would have if one were in Minkowski space rather than in the group manifold) yields that $B_{\mu\nu}$ is selfdual. Thus the usual duality constraint on $B_{\mu\nu}$ in simple $d = 6$ supergravity, is, in fact, a field equation in the larger group manifold. The action is on-shell invariant, but cannot be made off-shell invariant. (R. D'Auria, P. Fré, T. Regge, *Phys. Lett. B* **128** (1983) 44.)

Is it possible to write down an action **in Minkowski spacetime** which yields such selfduality conditions as a field equation? This is indeed possible. For example, in the $d = 6$ model one considers the action

$$I = \int d^6x \left[-\frac{1}{12} F_{abc}^2 + \frac{1}{4} \lambda^{ab} F_{acd}^- F_b^{-cd} \right] \tag{34.255}$$

where $F_{abc}^\pm = \frac{1}{2} (F_{abc} \pm *F_{abc})$ and λ^{ab} is a Lagrange multiplier. The λ field equation yields

$$F_{acd}^- F_b^{-cd} = 0 \tag{34.256}$$

and if $a = b = \text{time index}$, all terms are positive definite, hence $F_{acd}^- = 0$. Because $F_{abc}^- = -*F_{abc}$, one concludes that also all other components of F_{abc}^- vanish. One can also include interactions (W. Siegel, Berkeley preprint, Spring 1984). However, so far at least, it has not been possible to combine this dynamical selfduality with a supersymmetric action. The problem is that one must find the variation of λ^{ab} under supersymmetry (H. Kim and the author). The action in (34.255) has to first order in λ^{ab} a local invariance, namely

$$\delta\lambda_{ab} = \partial_{(a}\epsilon_{b)} + \epsilon^c \partial_c \lambda_{ab} + \frac{1}{2} \lambda_{ac} (\partial_b \epsilon^c - \partial^c \epsilon_b). \tag{34.257}$$

The extension to all orders in λ_{ab} seems to give problems (J. H. Schwarz, private communication).

34.18 Selfduality in $d = 4k - 1$ and topological masses

An unexpected aspect of antisymmetric tensors has been revealed by considering the Kaluza-Klein compactification of the $d = 11$ model on S_4 instead of S_7 . In $d = 7$ Minkowski spacetime one expects five 3-index antisymmetric tensors $A_{\mu\nu\rho}^I(x)$. They get **two** kinds of mass terms

- (i) an explicit mass, coming from $\square(d = 11) = \square(d = 7) + \square(d = 4)$ in the Maxwell action for the field $A_{\mu\nu\rho}$ in $d = 11$.
- (ii) a topological mass term, coming from the FFA term in $d = 11$, using the Freund-Rubin ansatz $F_{\mu\nu\rho\sigma} \sim \epsilon_{\mu\nu\rho\sigma}$

$$\epsilon^{\mu_1 \dots \mu_7} \partial_{\mu_1} A_{\mu_2 \dots \mu_4} A_{\mu_5 \dots \mu_7} . \quad (34.258)$$

Counting degrees of freedom in $d = 7$ shows that the five antisymmetric tensors must have as few degrees of freedom as if they were gauge invariant (K. Pilch, P. K. Townsend and the author, Nucl. Phys.). However, the mass in (i) (but not that in (ii)) breaks the gauge invariance under $\delta A_{\mu\nu\rho} = \partial_\mu \Lambda_{\nu\rho}$. How are these seemingly contradictory statements compatible?

Consider the Lagrangian (omitting obvious indices)

$$\mathcal{L} = \frac{1}{2(c_2 - c_1)} A [c_1 \mathbb{1} + \epsilon D] [c_2 \mathbb{1} + \epsilon D] A . \quad (34.259)$$

There is the usual kinetic term from $\epsilon D \epsilon D$, and a mass term (i) from $c_1 c_2$ and a mass term (ii) from $(c_1 + c_2)$. Add now an auxiliary field term $L = B^2$, where $B = B_{\mu\nu\rho}$ has as many indices as $A_{\mu\nu\rho}$. By redefining

$$A = A' + B', \quad B = B' + \frac{1}{c_2 - c_1} (c_1 + *D) (A' + B') \quad (34.260)$$

with Jacobian one, we get

$$\mathcal{L} = \frac{1}{2} A' (c_1 + *D) A' - \frac{1}{2} B' (c_2 + *D) B' . \quad (34.261)$$

The second term turns out to describe fields which are in the same multiplet as the graviton: these fields we call massless. The fields A' are massive. (In more detail: the fermion field equations rotate into the B' field equation under supersymmetry.)

The fields A' (or B') have as few degrees of freedom as gauge invariant antisymmetric tensors. To prove this, we use again duality. Consider the parent action (S. Deser and R. Jackiw, *Phys. Lett. B*)

$$\mathcal{L} = \epsilon F(A) A + \epsilon F(A) B + B^2 . \quad (34.262)$$

Elimination of B gives: $B \sim \epsilon F(A)$, and substitutions into (34.262) yields a topological mass action (see W. Siegel, *Nucl. Phys. B* **156** (1975) 135).

$$\mathcal{L}(\text{top}) = F(A)^2 + \epsilon F(A) A . \quad (34.263)$$

Elimination of A from (34.262) yields $F(A) = F(B)$, hence $A = B + \partial\Lambda$. Substitution into (34.262) yields a selfdual action as in (34.261) (the $\partial\Lambda$ terms cancel)

$$\mathcal{L}(\text{self}) = \epsilon F(B)B + B^2 \quad (34.264)$$

(P. K. Townsend, K. Pilch and the author, *Phys. Lett. B* **136** (1984) 30.) The action in (34.263) is gauge-invariant, that in (34.264) is not, but both are equivalent at the linearized level.

At this point it is not clear whether the equivalence of the duality related (34.263) and (34.264) extends beyond the free field level. My personal opinion is that it does not (see under (34.251) for a reason); however, a clear counterexample has not yet been constructed.

In the $d = 7$ model obtained from the $d = 11$ model by compactification on S_4 , a number of other interesting aspects are found. **The scalars propagate nonconformally.** In $d = 4$, their field equation is conformal namely, the well known $(\square + \frac{1}{6}R)\phi = 0$, but in $d = 7$ the coefficient is not what one needs for conformal invariance. Since these scalars sit in the same multiplet as the graviton, one might call them massless; in $d = 7$, scalars exist with a conformal field equation.

Another interesting aspect is the gauging of noncompact groups, first found by C. Hull in the $d = 4$ $\mathcal{N} = 8$ model. One can also gauge noncompact groups in $d = 7$ or analyze the small fluctuation and large fluctuation stability of the compact or noncompact gauging. For details, see N. P. Warner, M. Pernici, K. Pilch and the author, *Nucl. Phys. B*.

34.19 Example A: $SO(4)/SO(3)$

Since $SO(4) = SO(3) \times SO(3)$, we have two $SO(3)$ algebras with generators E_a and F_a , satisfying $[E_a, E_b] = \epsilon_{abc}E_c$ and idem for F_a . The group vielbeins satisfy thus $SO(3)$ Cartan-Maurer equations

$$de^a + \frac{1}{2}\epsilon^{abc}e^b \wedge e^c = 0, \quad \text{idem } f^a. \quad (34.265)$$

The only subgroups $SO(3)$ are generated by $H_a = E_a$ (or F_a) or $H_a = E_a + F_a$. In the first case the coset becomes the group manifold $SO(3)$. We shall return to this case later. In the second case we find for the most general reductive algebra

$$K_\alpha = \alpha E_\alpha \beta F_\alpha; \quad [K_a, K_b] = \epsilon_{abc}\{-\alpha\beta H_c + (\alpha + \beta)K_c\}. \quad (34.266)$$

The new vielbeins follow from

$$e^a E_a + f^a F_a = \omega^i H_i + B^a K_a \quad (34.267)$$

and one finds

$$\omega^i = (-\beta e^i + \alpha f^i)(\alpha - \beta)^{-1}, \quad B^a = (e^a - f^a)(\alpha - \beta)^{-1}. \quad (34.268)$$

Their Cartan-Maurer equations read

$$\begin{aligned} d\omega^i + \frac{1}{2}\epsilon^{ijk}\omega^j \wedge \omega^k - \frac{1}{2}\alpha\beta\epsilon^{ibc}B^b \wedge B^c &= 0 \\ dB^a + \epsilon^{aic}\omega^i \wedge B^c + \frac{1}{2}(\alpha + \beta)\epsilon^{abc}B^b \wedge B^c &= 0. \end{aligned} \quad (34.269)$$

Only for $\alpha + \beta = 0$ does one have a symmetric algebra. Since both $c_{ia}{}^b = -\alpha\beta\epsilon_{ia}{}^b$ and $c_{da}{}^b = (\alpha + \beta)\epsilon_{da}{}^b$ are antisymmetric in (ab) , the Cartan-connection is equal to the spin connection. We find the Riemann curvature from (34.84) with $\bar{c} = c$.

$$R_{st,ba} = (\delta_{sb}\delta_{ta} - \delta_{sa}\delta_{tb}) \frac{1}{4}(\alpha - \beta)^2. \quad (34.270)$$

This is a maximally symmetric space, and thus equal to S_3 .

Returning to the case $H_a = E_a$, i.e., the group manifold $SO(3)$, we note that its curvature follows from (34.17) and is again equal to (34.270). Thus S_3 and $SO(3)$ have the same Riemann-curvature, and since the vielbein for a maximally symmetric space is unique up to local Lorentz and general coordinate transformations (Weinberg's book), **S_3 and $SO(3)$ have the same vielbein.** S_3 is the only S_n which is a group manifold.

34.20 Example B: $SU(2) \times SU(2)/U(1)$

As in the previous example, we start with generators E_a , F_a and vielbeins e^a , f^a , but now we begin by choosing the coset vielbeins, instead of first choosing the subgroup H . We claim that the Riemann, and hence Ricci, tensors are independent of the choice of H : **it is sufficient to specify the coset vielbeins.** We prove this by actually computing these tensors without ever having to commit ourselves to the choice of H .

The coset vielbeins are chosen to be

$$B^A = \{e^m, f^m, pe^3 + qf^3\} \equiv \{B^m, B^{m'}, B^3\} \quad (34.271)$$

with $m, m' = 1, 2$. The corresponding generators, and the subgroup generators as well as the H -connection ω follow from

$$\begin{aligned} e^a E_a + f^a F_a &= \omega H + B^m E_m + B^{m'} F_{m'} + B^3 K_3 \\ \omega &= \frac{1-p\alpha}{q}e_3 - \alpha f_3; \quad H = qE_3 - pF_3; \quad K_3 = \alpha E_3 + \frac{1-p\alpha}{q}F_3. \end{aligned} \quad (34.272)$$

Since the coset representation of H (for $p \neq 0$, $q \neq 0$) splits into two doublets and one singlet, we can introduce 3 scale parameters without losing G -invariance of the rescaled vielbeins. (One could choose the new vielbeins proportional to $e^m + sf^{m'}$, $f^{m'} + te^m$ and B^3 , but for simplicity we shall

only consider the case $s = t = 0$.) Thus

$$\hat{B}^m = \frac{1}{a}e^m; \quad \hat{B}^{m'} = \frac{1}{b}f^{m'}; \quad \hat{B}^3 = \frac{1}{d}(pe_3 + qf_3). \quad (34.273)$$

We begin by computing the spin connections. From dB^3 we can read off ω^{3m} and $\omega^{3m'}$, and subsequently dB^m and $dB^{m'}$ give the rest (using at this point known results for ω^{3m} and $\omega^{3m'}$).

One finds

$$\begin{aligned} \omega(\text{spin})^{3m} &= \frac{ap}{2d}\epsilon^{mn}e^n, & \omega(\text{spin})^{3m'} &= -\frac{bq}{2d}\epsilon^{m'n'}f^{n'} \\ \omega^{mn} &= \epsilon^{mn} \left[\left(\frac{a^2p^2}{2d^2} - 1 \right) e^3 + \frac{a^2pq}{2d^2} f^3 \right] \\ \omega^{m'n'} &= \epsilon^{m'n'} \left[\left(\frac{b^2q^2}{2d^2} - 1 \right) f^3 + \frac{b^2pq}{2d^2} e^3 \right]. \end{aligned} \quad (34.274)$$

Note that the spin connection depends also on the H -connection.

Let us, as an instructive intermezzo, show that Cartan's H -connection is not equal to the spin connection. Then we shall show that the replacement of $c_{AB}{}^C$ by $\bar{c}_{AB}{}^C$ in the former will yield the latter. The structure constants of the rescaled generators are determined by

$$\begin{aligned} [aE_1, aE_2] &= a^2 \left\{ \frac{p}{d}(dK_3) + \frac{1-p\alpha}{q}H \right\} \\ [aE_2, dK_3] &= \alpha d(aE_1); & [aE_1, dK_3] &= -\alpha d(aE_2) \\ [H, aE_1] &= q(aE_2); & [H, aE_2] &= -q(aE_1). \end{aligned} \quad (34.275)$$

Hence, $\hat{c}_{12}{}^3 = \frac{a^2p}{d}$ and $c_{13}{}^2 = -\alpha d$, so that the coset-coset-coset structure constants are not antisymmetric. Thus we should be able to obtain the spin connection by using the $\bar{c}$ constants. Indeed,

$$\begin{aligned} \omega(\text{spin})^1{}_2 &= \frac{1}{2}\hat{B}^d (\hat{c}_{d2}{}^1 + \hat{c}^1{}_{d2} + \hat{c}^1{}_{2d}) + \omega_{CH^2}{}^1 \\ &= \frac{1}{2d} (pe^3 + qf^3) \left(-2\alpha d + \frac{a^2p}{d} \right) - q \left(\frac{1-p\alpha}{q}e^3 - \alpha f^3 \right) \\ &= \left(\frac{a^2p^2}{2d^2} - 1 \right) e^3 + \frac{a^2pq}{2d^2} f^3 \end{aligned} \quad (34.276)$$

which clearly reproduces the result in (34.274).

One next computes the curvature two-forms

$$R^{AB} = d\omega^{AB} + \omega^{AC} \wedge \omega^{CB} = \frac{1}{2}R_{CD}{}^{AB}\hat{B}^C \wedge \hat{B}^D. \quad (34.277)$$

One finds for the nonvanishing curvatures

$$\begin{aligned}
R^{3m} &= \frac{a^4 p^2}{4d^2} \hat{B}^3 \wedge \hat{B}^m; & R^{3m'} &= \frac{b^4 q^2}{4d^2} \hat{B}^3 \wedge \hat{B}^{m'} \\
R^{mn} &= \left(1 - \frac{3a^2 p^2}{4d^2}\right) a^2 \hat{B}^m \wedge \hat{B}^n - \frac{a^2 b^2 pq}{4d^2} \epsilon^{mn} \left(\epsilon^{m'n'} \hat{B}^{m'} \wedge \hat{B}^{n'}\right) \\
R^{m'n'} &= \left(1 - \frac{3b^2 q^2}{4d^2}\right) b^2 \hat{B}^{m'} \wedge \hat{B}^{n'} - \frac{a^2 b^2 pq}{4d^2} \epsilon^{m'n'} \left(\epsilon^{mn} \hat{B}^m \wedge \hat{B}^n\right) \\
R^{mm'} &= -\frac{a^2 b^2 pq}{4d^2} \epsilon^{mn} \epsilon^{m'n'} \hat{B}^n \wedge \hat{B}^{n'}.
\end{aligned} \tag{34.278}$$

In agreement with our discussion in section 34.7, the components with flat indices are constant, although the structure constants are not antisymmetric.

Next we compute the Ricci tensors $R_A{}^B = R_{CA}{}^{BC}$. One finds for the nonzero components

$$\begin{aligned}
R_3{}^3 &= -\frac{1}{2d^2} (a^4 p^2 + b^4 q^2) \\
R_m{}^n &= \delta_m^n a^2 \left(\frac{a^2 p^2}{2d^2} - 1 \right) \\
R_{m'}{}^{n'} &= \delta_{m'}^{n'} b^2 \left(\frac{b^2 q^2}{2d^2} - 1 \right).
\end{aligned} \tag{34.279}$$

The necessary and sufficient conditions that one have an Einstein space thus read

$$\alpha(1 - \alpha p^2) = \beta(1 - \beta q^2) = \alpha^2 p^2 + \beta^2 q^2; \quad \alpha = \frac{a^2}{2d^2}; \quad \beta = \frac{b^2}{2d^2}. \tag{34.280}$$

The general solution of these algebraic equations is

$$\beta = 2\alpha - 3\alpha^2 p^2, \quad \beta^2 q^2 = \alpha(1 - 2\alpha p^2). \tag{34.281}$$

Thus we have obtained a $d = 5$ Einstein space with $G = SU(2) \times SU(2)$ isometry, without ever having had to specify the embedding of H into G . There are two free parameters, α and p , but since β is positive we have the inequality

$$0 < \alpha p^2 < \frac{1}{2}. \tag{34.282}$$

Let us now go on and compute the holonomy group. The Weyl tensor for an n -dimensional Einstein space reads

$$C_{ABCD} = R_{ABCD} + \frac{1}{n(n-1)} (\delta_{AC}\delta_{BD} - \delta_{AD}\delta_{BC}). \tag{34.283}$$

It is easiest to compute $C_{AB} = C_{ABCD} \hat{B}^C \hat{B}^D$. For C_{31} we find

$$C_{31} = \frac{a^4 p^2}{2d^2} \hat{B}^3 \wedge \hat{B}^1 + \frac{1}{10} \hat{B}^3 \wedge \hat{B}^1 R. \quad (34.284)$$

Taking into account (34.279), one finds

$$\begin{aligned} C_{3m} &= (3\alpha^2 p^2 d^2 - \alpha d^2) \hat{B}^3 - \hat{B}^m \\ C_{3m'} &= (3\beta^2 q^2 d^2 - \beta d^2) \hat{B}^3 - \hat{B}^{m'}. \end{aligned} \quad (34.285)$$

If C_{3m} and $C_{3m'}$ would both be nonvanishing, the holonomy generators τ_{3m} and $\tau_{3m'}$ would generate the whole of $SO(5)$ and no Killing spinor would exist. If either C_{3m} or $C_{3m'}$ vanishes, so does the other, and in this case one is left with a one-parameter solution

$$\alpha p^2 = \frac{1}{3}, \quad \alpha = \beta, \quad p^2 = q^2, \quad R = -\frac{20}{3} \alpha d^2. \quad (34.286)$$

This solution is compatible with (34.271).

The nonvanishing holonomy generators $C_{ABCD} \tau^{CD}$ are

$$\begin{aligned} C_{12} &= -\frac{8\alpha d^2}{3} (\tau^{12} - \tfrac{1}{2} \sigma \tau^{45}), \quad \sigma = \text{sign}(pq) \\ C_{45} &= -\frac{8\alpha d^2}{3} (\tau^{45} - \tfrac{1}{2} \sigma \tau^{12}) \\ C_{14} &= -\frac{2\alpha d^2}{3} (\tau^{14} + \sigma \tau^{25}) = \sigma C_{25} \\ C_{15} &= -\frac{2\alpha d^2}{3} (\tau^{15} - \sigma \tau^{24}) = -\sigma C_{23}. \end{aligned} \quad (34.287)$$

Commuting C_{14} with C_{15} we find that τ^{12} and τ^{45} are separately generators, and hence **no Killing spinors exist** ($\tau^{12} = 0$ has no solution).

It is instructive to compare this example with the analogous case of $SU(2) \times SU(2) \times SU(2)/U(1) \times U(1)$ (R. D'Auria, P. Fré and the author, *Phys. Lett. B* **136** (1984) 347). There are now constants α, β, γ and p, q, r . Again one must require that C_{3A} with $A = 1, 2, 4, 5, 6, 7$ vanish in order that Killing spinors exist. Also $\alpha = \beta = \gamma$ and $p^2 = q^2 = r^2$ (but $\alpha p^2 = \frac{1}{4}$ instead of $\frac{1}{3}$). The holonomy generators $C_{14}, C_{24}, C_{15}, C_{25}$ (and a further set $C_{16}, \dots, C_{57}$) have the same form as in (34.286) (but are now 8×8 matrices). However

$$\begin{aligned} C_{12} &\sim (\tau_{12} - \tfrac{1}{2} \sigma \tau_{45} - \tfrac{1}{2} \rho \tau_{67}), \quad \rho = \text{sign}(pr) \\ C_{45} &\sim (\tau_{45} - \tfrac{1}{2} \sigma \tau_{12} - \tfrac{1}{2} \lambda \tau_{67}), \quad \lambda = \text{sign}(qr) \\ C_{67} &\sim (\tau_{67} - \tfrac{1}{2} \rho \tau_{12} - \tfrac{1}{2} \lambda \tau_{45}), \quad \sigma \rho \lambda = +1 \end{aligned} \quad (34.288)$$

and now the commutator of C_{14} and C_{15} does not generate new generators, but is proportional to $\tau^{12} - \sigma \tau^{45}$ $\rho C_{12} - \lambda C_{45}$. The holonomy group is in this case $SU(3)$, and has two null eigenspinors

η_1 and η_2 . They are both Killing spinors, satisfying $D_\alpha \eta = c\tau_\alpha \eta$ with the same c , in agreement with our discussion of section 34.11. They are actually **constant**, because both spinors satisfy

$$\frac{1}{4}\omega_\alpha{}^{ab}(e)\tau_{ab}\eta = c\tau_\alpha\eta. \quad (34.289)$$

The Killing vector $\bar{\eta}_1\tau^\alpha\eta_2$ is nonvanishing, and generates an extra $SO(2)$ isometry. Thus, the full isometry group is

$$SU(2) \times SU(2) \times SU(2) \times SO(2). \quad (34.290)$$

These results illustrate several of the theorems we derived in section 34.15. The $SO(2)$ is part of $OSp(2|4)$ while the $SU(2)$ groups belong to $\mathcal{N} = 2$ matter multiplets.

34.21 Example C: Superspace

Superspace is the coset manifold of the super Poincaré algebra. The subgroup is the Lorentz group. Thus $K_\alpha = \{P_m, Q_\alpha\}$ and $H_i = \{M_{mn}\}$.

We begin by computing the vielbein from its definition

$$e^{z^\alpha K_\alpha} e^{dz^a K_a} = e^{(z^\alpha + dz^a e_a^\alpha) K_\alpha} e^{-dz^a \omega_a^i H_i} \quad (34.291)$$

Using the composition rule

$$e^A e^{\epsilon B} = \exp \left[A + \epsilon B + \epsilon \sum_{n=1}^{\infty} \frac{1}{(n+1)!} \underbrace{[A, \dots, [A, B] \dots]}_{n\text{-times}} + \mathcal{O}(\epsilon^2) \right] \quad (34.292)$$

one finds for the super vielbein

$$e_\alpha{}^a = \begin{pmatrix} \delta_\mu^m & 0 \\ -\frac{1}{2}(\bar{\theta}\gamma^m)_A & \delta_A^a \end{pmatrix} \quad (34.293)$$

where $A, A = \text{flat, curved spinor indices}$ and $\mu, m = \text{flat, curved vector indices}$. Since commutators of P and Q remain in $\{P, Q\}$, the ω^i **vanish**. Because $\{Q, Q\} \sim P$ but $[Q, P] = 0$, the $K - K - K$ structure constants are not antisymmetric. **The spin connection is non-vanishing**. It is equal to $-\frac{1}{2}e^d \bar{c}_d{}^{ab}$, and can also be obtained directly from $D_\alpha e_\beta{}^b - \alpha \leftrightarrow \beta = 0$ with $e_\alpha{}^a$ given in (34.293). One sometimes introduces a derivative with a connection ω^i ; in that case, superspace has torsion (proportional to $\bar{c}_d{}^{ab}$) but vanishing curvature ($d\omega^i + c_{jk}{}^i \omega^j \omega^k = 0$).

The Killing vectors follow from

$$e^{dg^M X_M} e^{z^\alpha K_\alpha} = e^{z^\alpha + dg^M k_M^\alpha} K_\alpha e^{-dg^M \Omega_M^i H_i}. \quad (34.294)$$

Using

$$e^{\epsilon A} e^B = \exp \left[B + \epsilon A + \epsilon \sum_{n=1}^{\infty} \frac{1}{(n+1)!} [\cdots [A, \underbrace{B, \cdots B}_{n\text{-times}}] + \mathcal{O}(\epsilon^2) \right] \quad (34.295)$$

one finds

$$\begin{aligned} l_m &= \delta_m^\mu \partial_\mu, & l_A &= \delta_A^{\dot{A}} \left(\frac{\partial}{\partial \theta} + \not{\partial} \theta \right)_{\dot{A}}, & l_{mn} &= 0 \\ \Omega_m^{ab} &= \Omega_A^{ab} = 0, & \Omega_{mn}^{kl} &= \delta_{[mn]}^{kl}. \end{aligned} \quad (34.296)$$

Hence, the ordinary and H -covariant Lie derivatives with coset indices coincide, and thus the ordinary Lie derivative commutes with the covariant derivative.

(Minus) the H -covariant Lie derivatives are the (supersymmetry) generators. Since they always commute with the covariant derivatives, one can impose constraints on superfields (fields depending on $z^\alpha = (x^\mu, \theta^A)$) without changing the transformation properties of the superfields. We have shown in these lectures that this is a general property of any reductive coset manifold, and not only of that particular coset manifold which is due to dividing the super Poincaré group by the Lorentz group.

One sometimes reads statements that the supersymmetry generators (34.296) and covariant derivatives commute in superspace because of associativity (see the discussion below (34.117)). This is not correct in general: in general (34.117) does not hold but rather (34.48)

$$[l_{k_M}, e_a] = \Omega_M^i c_{iab} e^b. \quad (34.297)$$

However, since $\Omega_M^i = 0$ in (34.296), it is still correct in this case. This is due to the structure of the algebra: any algebra for which the reductive split $A = (H, K)$ is at the same time a semi-direct sum ($[K, K] \subset K$), has the property that the Ω_M^i vanishes.

For a deeper and even more beautiful geometrical interpretation of superspace following Cartan's original ideas, see the forthcoming book by P. C. West and the author.